Real Analysis

An Undergraduate Notebook

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Chapter 1

Introductory Set Theory

In this chapter, we lay the groundwork for our exposition of real analysis. We will introduce the fundamental notions of **sets**, **relations**, and **maps**, which underlie not only all of analysis but also all of modern mathematics as well.

1.1 What Is a Set?

Since the early 20th century, the practice of mathematics has undergone a radical transformation, which grew out of the idea that the whole of the mathematical canon can and should be rephrased in the language of set theory. Informally, a **set** is a collection of objects, which are called its **elements**. If an object x is an element of a set S, then we write $x \in S$; otherwise, if x is not an element of S, then we write $x \notin S$.

1.1.1 Definitions of Sets

The careful reader may complain that the terms "collection" and "object" are vague and undefined, and they would be right to object; the above characterization of sets is definitely not as rigorous as we might desire, and is in fact internally inconsistent. However, the technical difficulties associated with building a rigorous and internally consistent set theory are outside the scope of this course and in this context, far more trouble than they are worth. In fact, it's entirely possible that such an internally consistent set theory does not exist!

As such, we will leave this ambiguity in favor of moving on to more salient topics. Even though we will not encounter any such issues in this text, the "definition" above does lead to some interesting inconsistencies. We will explore one such contradiction, called Russell's paradox, shortly. First, however, we introduce some shorthand notation which will be useful in specifying sets we wish to discuss.

Convention 1.1.1 Roster notation. It's easy to completely describe a set which contains only a finite number of elements; we can simply write a list of those elements down! Concretely, if X is the set whose elements are precisely the objects $x_1, \ldots, x_n$, then we may write

$$X = \{x_1, \ldots, x_n\}.$$

The above notation for specifying sets is called **roster notation**.

Not all sets are finite, however, and it is obviously impossible to write down a complete roster of all of the elements of an infinite set. We can carefully use ellipses to indicate a set with an infinite roster whose elements follow some discernible pattern. For example, we will see in Section 1.3 Peano Arithmetic and the Natural Numbers [Skip] that the set $\mathbb{N}$ of natural numbers can be written

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\}.$$

Unfortunately, this extension of roster notation still does not suffice for all sets. For example, we will see in Section 1.6 Cardinality and Countability [SKIP] that there is no such roster for the set $\mathbb{R}$ of real numbers.

Definition 1.1.2 Definable collection. Let Φ be a **unary Boolean predicate** (a statement whose truth value depends on a single variable x). The collection of objects x which satisfy Φ is said to be **defined** by Φ, and is denoted

$$\{x \mid \Phi(x)\}.$$

Such a collection of objects which satisfy such a Boolean predicate is called a **definable collection**. We will see that the collection of objects that satisfy a given Boolean predicate is not necessarily a set; this is the content of Russell's paradox. ◆

Example 1.1.3 Sets. The following are examples of sets:

- The **empty set**, denoted $\emptyset$ or $\{\}$, is the set with no elements; that is, $x \notin \emptyset$ for all objects x.
- The set $\{1, 2, \text{red}, \text{blue}\}$ is a **non-empty** set with only finitely many elements. Note that this set contains different types of objects. In general, we will not place any explicit restrictions on the types of objects allowed inside a set, although some formalizations of axiomatic set theories rely on this to avoid inconsistencies such as Russell's paradox.
- The set $\mathbb{N} = \{0, 1, 2, \ldots\}$ of **natural numbers** (also sometimes called the **whole numbers** or **counting numbers**) is a non-empty set with infinitely many elements. We will characterize the natural numbers $\mathbb{N}$ in the usual manner of Peano arithmetic later in this chapter.

▲

As we mentioned before, the set theory based on the "definition" above, which is called **naïve set theory**, is in fact inconsistent; taking this "definition" as the basis for an axiomatic system, we can derive a contradiction, which implies that we may in fact derive all possible statements, including those which are false! This inconsistency was first demonstrated by Bertrand Russell at the beginning of the 20th century, and bears his name.

Proposition 1.1.4 Russell's paradox. (Bertrand Russell) *Not all definable collections are sets.*

Proof. By contradiction. Let

$$\mathfrak{R} = \{X \mid X \notin X\}$$

be the definable collection of sets which do not contain themselves, and suppose for a contradiction that $\mathfrak{R}$ is a set. The question of whether or not $\mathfrak{R}$ contains itself is a natural one; however, we will see that both the positive and negative answers lead to contradictions.

This is because, if $\mathfrak{R} \in \mathfrak{R}$, then $\mathfrak{R} \notin \mathfrak{R}$ by the definition of $\mathfrak{R}$. On the other hand, if $\mathfrak{R} \notin \mathfrak{R}$, then $\mathfrak{R} \in \mathfrak{R}$ by the same reasoning. Therefore, $\mathfrak{R} \in \mathfrak{R}$ if and only if $\mathfrak{R} \notin \mathfrak{R}$. This contradiction implies that $\mathfrak{R}$ is not a set, as desired. ■

Russell's paradox tells us that naïve set theory is inconsistent. However, as we stated before, it is no easy task to construct a consistent set theory which is strong enough to describe the sophisticated structures we wish to investigate while also weak enough to be internally consistent. However, since the question of whether modern formal set theories are consistent is beyond the scope of this text. With this in mind, we will continue formulating definitions and results in the language of naïve set theory, keeping in mind that rigorous systems of formal set theory (which do not suffer from Russell's paradox) exist in which they can be expressed and proven.

1.1.2 Set Containment

As we have just seen, not all definable collections of mathematical objects are sets. However, any collection of objects which themselves belong to a set is also a set in its own right; such a collection of elements of a given set is called a **subset**.

Definition 1.1.5 Set containment. A set U is a **subset** of a set X if for all objects x, if $x \in U$, then $x \in X$. In this case, we say that U is **contained** in X, and we write $U \subseteq X$. Less commonly, we may also call X a **superset** of U, say that X contains U, and write $X \supseteq U$.

A set U is a **proper subset** of a set X if $U \subseteq X$ but $X \not\subseteq U$. In this case, we say that U is **strictly contained** in X, and we write $U \subsetneq X$. Less commonly, we may also call X a **proper superset** of U, say that X strictly contains U, and write $X \supsetneq U$. ◆

Convention 1.1.6 Set containment notation. Some authors use the symbols $\subset$ and $\supset$ in place of $\subseteq$ and $\supseteq$ to indicate set containment, and some authors use them in place of $\subsetneq$ and $\supsetneq$ to indicate strict set containment. We will avoid this ambiguity by opting not to write $\subset$ and $\supset$ at all.

Definition 1.1.7 Set equality. Two sets are considered **equal** if they contain exactly the same elements; that is, one set X is equal to another set Y when objects are elements of X if and only if they are elements of Y. This occurs precisely when both $X \subseteq Y$ and $Y \subseteq X$. ◆

Example 1.1.8 Subsets. The following are examples of subsets of sets:

- For any set X, the empty set $\emptyset$ and X itself are subsets of X.
- The set $\{1, 3, 5, \ldots\}$ of odd natural numbers is a proper subset of the set $\mathbb{N}$ of natural numbers. Written more concisely,

$$\{1, 3, 5, \ldots\} \subsetneq \mathbb{N}.$$

▲

An important property of the set containment relationship is its transitivity, which is the content of the following lemma:

Lemma 1.1.9 Transitivity of set containment. *For any sets U, V, and X, if $U \subseteq V$ and $V \subseteq X$, then $U \subseteq X$.*

Proof. Let $u \in U$. Since $u \in U$ and $U \subseteq V$, $u \in V$. Moreover, since $u \in V$ and $V \subseteq X$, $u \in X$. In summary, we have shown that for $u \in X$ for all elements $u \in U$, and so $U \subseteq X$. ■

Set containment is also useful for constructing new notations for specifying sets as definable collections. Crucially, although not all definable collections are sets, any definable subcollection of a set is itself a set.

Convention 1.1.10 Set-builder notation. Given a set X and a unary Boolean predicate Φ, the definable collection

$$\{x \mid x \in X \text{ and } \Phi(x)\}$$

is a set (and in fact a subset of X). This subset will also be denoted in **set-builder notation** by

$$\{x \in X \mid \Phi(x)\} = \{x \mid x \in X \text{ and } \Phi(x)\}.$$

After learning about Russell's paradox, the careful reader may have some trepidation about sets whose elements are themselves sets. However, not all such definable collections lead to contradictions. For example, the collection of all subsets of a given set also form a set.

Definition 1.1.11 Power set. The **power set** of a set X is the set of all subsets of X; this set is denoted $\mathcal{P}(X)$, so that

$$\mathcal{P}(X) = \{U \mid U \subseteq X\}.$$

◆

Lemma 1.1.12 Power sets are non-empty. *The power set $\mathcal{P}(X)$ of any set X is non-empty; that is, $\mathcal{P}(X) \neq \emptyset$.*

Proof. Fix a set X, and note that the statement $x \in X$ for all $x \in \emptyset$ is vacuously true, so that $\emptyset \subseteq X$. Thus $\emptyset \in \mathcal{P}(X)$, and so $\mathcal{P}(X) \neq \emptyset$ is non-empty. ■

Proof. Fix a set X, and note that the statement $x \in X$ for all $x \in X$ is a tautology, so that $X \subseteq X$. Thus $X \in \mathcal{P}(X)$, and so $\mathcal{P}(X) \neq \emptyset$ is non-empty. ■

Both of the above proofs of Lemma 1.1.12 Power sets are non-empty are correct, although they are nonequivalent. Both are valuable for different reasons; the first proof shows that the power set of a set is nonempty since the empty set $\emptyset$ is a subset of any set, and the second proof shows the same conclusion since any set is a subset of itself. These proofs are called **constructive**, in that they explicitly exhibited an object satisfied the desired properties (being an element of the power set of a given set). Proofs which are not constructive are called **non-constructive**.

In general, sets are likely to have many subsets, so that power sets of sets are likely to have a great many elements. We will investigate the notion of the number of elements of a set, called its **cardinality**, more precisely and in far greater depth in Section 1.6 Cardinality and Countability [SKIP] at the end of this chapter.

1.1.3 Set Operations

We now introduce several ways to construct new sets from old ones which loosely correspond to the logical operators of **conjunction** $\wedge$, **disjunction** $\vee$, and **negation** $\neg$, pronounced "and", "or" and "not", respectively.

Definition 1.1.13 Set operations. Let S and T be sets.

- *Intersection.*

  The **intersection** $S \cap T$ of two sets S and T is the set whose elements are *both* elements of S and elements of T; that is,

  $$S \cap T = \{x \mid x \in S \wedge x \in T\} = \{x \mid x \in S \text{ and } x \in T\}.$$

- *Union.*

  The **union** $S \cup T$ of two sets S and T is the set whose elements are *either* elements of S and elements of T (or both!); that is,

  $$S \cup T = \{x \mid x \in S \vee x \in T\} = \{x \mid x \in S \text{ or } x \in T\}.$$

- *Complement.*

  The **complement** $S \setminus T$ of T in S is the set whose elements are elements of S but not elements of T; that is,

  $$S \setminus T = \{x \mid x \in S \wedge \neg(x \in T)\} = \{x \mid x \in S \text{ and } x \notin T\}.$$

◆

Informally, the intersection $S \cap T$ of two sets S and T is the largest set which is contained in both S and T; the union of S and T is the smallest set which contains both S and T; and the complement $S \setminus T$ is the largest set which is contained in S but disjoint from T. These informal characterizations are formalized in the following result:

Proposition 1.1.14 Set operations and containment. *Let S, T, and U be sets.*

1. *$U \subseteq S \cap T$ if and only if both $U \subseteq S$ and $U \subseteq T$.*
2. *$S \cup T \subseteq U$ if and only if both $S \subseteq U$ and $T \subseteq U$.*
3. *$U \subseteq S \setminus T$ if and only if both $U \subseteq S$ and $U \cap T = \emptyset$.*

Proof.

1. First suppose that $U \subseteq S \cap T$. Since $S \cap T \subseteq S$, Lemma 1.1.9 Transitivity of set containment implies that $U \subseteq S$. Similarly, since $S \cap T \subseteq T$, Lemma 1.1.9 Transitivity of set containment implies that $U \subseteq T$.

   Conversely, now suppose that $U \subseteq S$ and $U \subseteq T$, and let $x \in U$. Since $x \in U$ and $U \subseteq S$, $x \in S$. Similarly, since $x \in U$ and $U \subseteq T$, $x \in T$. Since $x \in S$ and $x \in T$, we conclude that $x \in S \cap T$. In summary, we have shown that $x \in S \cap T$ for all $x \in U$, so that $U \subseteq S \cap T$.

2. First suppose that $S \cup T \subseteq U$. Since $S \subseteq S \cup T$, Lemma 1.1.9 Transitivity of set containment implies that $S \subseteq U$. Similarly, since $T \subseteq S \cup T$, Lemma 1.1.9 Transitivity of set containment implies that $T \subseteq U$.

   Conversely, now suppose that $S \subseteq U$ and $T \subseteq U$. Let $x \in S \cup T$, so that $x \in S$ or $x \in T$. If $x \in S$, then $x \in U$, since $S \subseteq U$. Similarly, if $x \in T$, then $x \in U$, since $T \subseteq U$. We conclude that $x \in U$. In summary, we have shown that $x \in U$ for all $x \in S \cup T$, so that $S \cup T \subseteq U$.

3. First suppose that $U \subseteq S \setminus T$, and let $x \in U$. Since $x \in U$ and $U \subseteq S \setminus T$, $x \in S \setminus T$, so that $x \in S$ and $x \notin T$. In summary, we have shown that $x \in S$ for all $x \in U$, so that $U \subseteq S$. Moreover, we have shown that $x \notin T$ for all $x \in U$, so that no element of U is an element of T; that is, $U \cap T = \emptyset$.

   Conversely, now suppose that $U \subseteq S$ and $U \cap T = \emptyset$, and let $x \in U$. Since $x \in U$ and $U \subseteq S$, $x \in S$. Moreover, since $x \in U$ and $U \cap T = \emptyset$, $x \notin T$. Since $x \in S$ and $x \notin T$, we conclude that $x \in S \setminus T$. In summary, we have shown that $x \in S \setminus T$ for all $x \in U$, and so $U \subseteq S \setminus T$.

■

Corollary 1.1.15 Properties of the set operations.

1. *$S \cap S = S \cup S = S \cup \emptyset = S \setminus \emptyset = S$ and $S \cap \emptyset = S \setminus S = \emptyset \setminus S = \emptyset$ for all sets S.*
2. *Commutativity.*

   $S \cap T = T \cap S$ and $S \cup T = T \cup S$ for all sets S and T.
3. *Associativity.*

   $(S \cap T) \cap U = S \cap (T \cap U)$ and $(S \cup T) \cup U = S \cup (T \cup U)$ for all sets S, T, and U.

Proof.

1. Since $S \subseteq S$, $S \subseteq S \cap S$ and $S \cup S \subseteq S$ by (1) and (2) of Proposition 1.1.14 Set operations and containment, respectively. Of course $S \cap S \subseteq S \subseteq S \cup S$, so that $S \cap S = S = S \cup S$.

   Similarly, since $S \subseteq S$ and $\emptyset \subseteq S$, $S \cup \emptyset \subseteq S$ by (2) of Proposition 1.1.14 Set operations and containment. Of course $S \subseteq S \cup \emptyset$, and so $S \cup \emptyset = S$.

   $\emptyset \subseteq S \cap \emptyset \subseteq \emptyset$, and so $S \cap \emptyset = \emptyset$. Moreover, together with the observation that $S \subseteq S$, this implies by (1) of Proposition 1.1.14 Set operations and containment that $S \subseteq S \setminus \emptyset$. Of course, $S \setminus \emptyset \subseteq S$, and so $S \setminus \emptyset = S$.

   Similarly, $\emptyset \subseteq \emptyset \setminus S \subseteq \emptyset$, and so $\emptyset \setminus S = \emptyset$.

   Finally, note that any element of $S \setminus S$ is both an element of S and not an element of S, so that no such element can exist. Thus $S \setminus S = \emptyset$.

2. Since $S \cap T \subseteq T$ and $S \cap T \subseteq S$, (1) of Proposition 1.1.14 Set operations and containment implies that $S \cap T \subseteq T \cap S$. By symmetry, $T \cap S \subseteq S \cap T$, so that $S \cap T = T \cap S$.

   Similarly, since $S \subseteq T \cup S$ and $T \subseteq T \cup S$, (2) of Proposition 1.1.14 Set operations and containment implies that $S \cup T \subseteq T \cup S$. By symmetry, $T \cup S \subseteq S \cup T$, so that $S \cup T = T \cup S$.

3. We note that

   $$\begin{aligned} &(S \cap T) \cap U \subseteq S \cap T \subseteq T \\ &(S \cap T) \cap U \subseteq U, \end{aligned}$$

   so that $(S \cap T) \cap U \subseteq T \cap U$ by (1) of Proposition 1.1.14 Set operations and containment. Similarly,

   $$(S \cap T) \cap U \subseteq S \cap T \subseteq T,$$

   so that $(S \cap T) \cap U \subseteq S \cap (T \cap U)$ by another application of (1) of Proposition 1.1.14 Set operations and containment.

   Conversely, we note that

   $$\begin{aligned} &S \cap (T \cap U) \subseteq S \\ &S \cap (T \cap U) \subseteq T \cap U \subseteq T, \end{aligned}$$

   so that $S \cap (T \cap U) \subseteq S \cap T$ by (1) of Proposition 1.1.14 Set operations and containment. Similarly,

   $$S \cap (T \cap U) \subseteq T \cap U \subseteq U,$$

so that $S \cap (T \cap U) \subseteq (S \cap T) \cap U$ by another application of (1) of Proposition 1.1.14 Set operations and containment. Thus

$$(S \cap T) \cap U = S \cap (T \cap U).$$

■

In Subsection 1.2.3 Indexing in the next section, we will generalize the first two ways (**intersection** and **union**) of constructing sets to arbitrary sets of sets; this generalization will mirror the similarities between conjunction $\wedge$ and universal quantification $\forall$ and between disjunction $\vee$ and existential quantification $\exists$, respectively.

While sets and set theory form the foundations on which we will build our understanding of analysis, these topics are not the focus of this text. As such, we will leave behind the question of precise definitions of sets in favor of developing set-theoretic ideas which are more relevant to the world of analysis.

1.1.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Elements of sets. Determine whether or not the given object is an element of the given set. Write your answer using the notation $\in$ or $\notin$.

1. Is 3 an element of the set $\{1, 2, 3\}$?

2. Is 5 an element of the set $\{3, 9, 12\}$?

3. Is 8 an element of the set $\{0, 2, 4, \ldots, 12\}$?

4. Is 50 an element of the set $\{0, 1, 4, 9, 16, \ldots\}$?

Roster notation. Write down the specified set in roster notation.

5. Write down the set E of even numbers between 1 and 15 in roster notation.

6. Write down the set L of letters in your full name in roster notation.

7. Write down the set C of colors in the rainbow in roster notation.

Set containment. Let S, T, and U be the sets defined by

$$S = \{0, 1, 2\} \qquad T = \{3, 4\} \qquad U = \{1, 2, 3\}.$$

Determine whether or not the stated set containment is true or false.

8. Is $S \subseteq U$?

9. Is $S \cap T \subsetneq U$?

10. Is $U \subseteq S \cup T$?

Power sets. Write down the power sets of the given sets in roster notation.

11. Write down the power set $\mathcal{P}(S)$ of the set $S = \{1\}$ in roster notation.

12. Write down the power set $\mathcal{P}(T)$ of the set $T = \{1, 2\}$ in roster notation.

13. Write down the power set $\mathcal{P}(U)$ of the set $U = \{1, 2, 3\}$ in roster notation.

Set Operations. Compute the given set operations. Write your answer in roster or set-builder notation, or as a well-known set.

14. Compute $S \cap T$, where

$$S = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \ldots\}$$

is the set of even natural numbers and

$$T = \{n \in \mathbb{N} \mid n = 2k + 1 \text{ for some } k \in \mathbb{N}\} = \{1, 3, 5, 7, \ldots\}$$

is the set of odd integers.

15. Compute $S \cup T$, where $S = \{0, 3, 4, 9\}$ and $T = \{1, 3, 6\}$.

16. Compute $S \setminus T$, where

$$S = \{n \in \mathbb{N} \mid n = 5k \text{ for some } k \in \mathbb{N}\} = \{0, 5, 10, 15, \ldots\}$$

is the set of natural numbers divisible by 5, and

$$T = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \ldots\}$$

is the set of even natural numbers.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.1.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Distributivity of set operations.

(a) Prove that $S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$ for all sets S, T, and U.

(b) Prove that $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$ for all sets S, T, and U.

Taken together, these results are called the **distributive properties** of the set operations.

2. De Morgan's laws.

(a) Prove that $S \setminus (T \cap U) = (S \setminus T) \cup (S \setminus U)$ for all sets S, T, and U.

(b) Prove that $S \setminus (T \cup U) = (S \setminus T) \cap (S \setminus U)$.

Taken together, these results are called **De Morgan's laws**.

3. **Power sets.** Determine whether the following statement is true or false:

The power set of a set contains at least two distinct elements.

If the statement is true, prove it. If the statement is false, disprove it by providing a counterexample.

Hint. Revisit the proofs of Lemma 1.1.12 Power sets are non-empty.

4. **Set operations and containment.** Let S and T be sets.

(a) Prove that $S \cap T = S$ if and only if $S \subseteq T$.

(b) Prove that $S \cup T = S$ if and only if $T \subseteq S$.

(c) Prove that $S \setminus T = S$ if and only if $S \cap T = \emptyset$.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.2 Maps and Functions

In this section, we introduce the fundamental notion of a **map** between sets. Informally, a map is an assignment to each element of one set an element of another. You likely have been exposed to maps in previous math courses in the form of real-valued functions of one real variable. We will treat maps in far greater generality, which will allow us to perform mathematical analysis in a wider array of spaces.

1.2.1 Introduction to Maps

First, however, we define the **Cartesian product** of two sets. This relies on a notion of an ordered pair of objects x and y, which we denote (x, y). Like formalized set theory, rigorous constructions of ordered pairs are, in this context, unnecessarily complicated and annoying to work with. As such, we will take them to be well-defined. The salient properties of an ordered pair (x, y) are its **entries** x and y, which are called its first and second entries, respectively.

Definition 1.2.1 Cartesian product. The **Cartesian product** $X \times Y$ of two sets X and Y is the set which contains all ordered pairs whose first entry is an element of X and whose second entry is an element of Y; that is,

$$X \times Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}.$$

◆

Example 1.2.2 Cartesian products. The sets $X = \{1, 2\}$ and $Y = \{3, 4, 5\}$ have Cartesian product

$$X \times Y = \{(1,3), (1,4), (1,5), (2,3), (2,4), (2,5)\}.$$

Note that $X \times Y \neq X \cup Y$, since the elements of these sets are totally different kinds of objects. Moreover, if $X \neq Y$, then $X \times Y \neq Y \times X$; in the language of Subsection 1.4.2 Finitary Operations, the Cartesian product $\times$ is **non-commutative**. ▲

Recall that we would like a map between two sets to be an association of the elements of one set to the elements of the other. We will model this association via ordered pairs. To that end, a map between two sets will just be a subset of their Cartesian product which satisfies certain existence and uniqueness properties.

Definition 1.2.3 Map. A subset $f \subseteq X \times Y$ of the Cartesian product $X \times Y$ of two sets X and Y is called a **map** from X (which is called the **domain** of f and whose elements are called **inputs**) to Y (which is called the **codomain** of f and whose elements are called **outputs**) if it satisfies the following conditions:

1. *Uniqueness.*

   For all inputs $x \in X$, there is an output $y \in Y$ so that $(x, y) \in f$.

2. *Uniqueness (vertical line test).*

   For all inputs $x \in X$ and outputs $y_1, y_2 \in Y$, if $(x, y_1), (x, y_2) \in f$, then $y_1 = y_2$.

So f is a map from X to Y if and only if for each input $x \in X$, there is a unique output $y \in Y$ so that $(x, y) \in f$. We will write $f\colon X \to Y$ to indicate that f is a map from X to Y, and we will write $f\colon x \mapsto y$ to indicate that $(x, y) \in f$. The set of maps from X to Y is denoted $Y^X \subseteq X \times Y$.

We will call a map $f\colon X \to Y$ whose codomain Y is a number system like the natural numbers $Y = \mathbb{N}$, the integers $Y = \mathbb{Z}$, the rational numbers $Y = \mathbb{Q}$, the real numbers $Y = \mathbb{R}$, or the complex numbers $Y = \mathbb{C}$ a Y-valued **function** on X. We will investigate the natural numbers $\mathbb{N}$ in Section 1.3 Peano Arithmetic and the Natural Numbers [Skip], and Chapter 2 Constructing the Number Systems will be dedicated to constructing the other number systems with which we will become acquainted throughout the course of this text. ◆

So a map between sets is just a subset of their Cartesian product such that each element of the domain appears in precisely one pair with an element of the codomain. The above existence and uniqueness properties of a map implies that there is an unambiguous association between the elements of the domain and the elements of the codomain.

Example 1.2.4 Maps. The following are examples of maps between sets:

- Let $X = \{1, 2\}$ and $Y = \{\text{red}, \text{blue}\}$, and consider the following subsets of $X \times Y$:

  $$\begin{aligned} f_1 &= \{(1, \text{red}), (2, \text{red})\} \\ f_2 &= \{(1, \text{red}), (2, \text{blue})\} \\ f_3 &= \{(1, \text{blue}), (2, \text{red})\} \\ f_4 &= \{(1, \text{blue}), (2, \text{blue})\}. \end{aligned}$$

  As defined above, f_1, f_2, f_3, and f_4 are maps from X to Y. In fact, these are the only such maps; that is,

  $$Y^X = \{f_1, f_2, f_3, f_4\}.$$

- The empty set $\emptyset\colon \emptyset \to Y$ is the unique map from the empty set $\emptyset$ to any set Y; that is,

  $$Y^\emptyset = \{\emptyset\}.$$

  Note that none of the elements of the codomain Y are associated with elements of the empty domain $\emptyset$. However, this does not imply that $\emptyset$ is not a map; in general, not all elements of the codomain of a map need be associated with elements of the domain, since ***the existence property applies only to the domain***.

- Let $U \subseteq X$ be a subset of a set X. The **inclusion map** $\iota_U^X : U \to X$ of U in X is the map from U to X defined by

$$\iota_U^X = \{(u, x) \in U \times X \mid u = x\}.$$

That is, $\iota_U^X : u \mapsto u$ for all $u \in U$.

The inclusion map ι_X^X of X in itself is called the **identity map** on X, and is denoted $\mathrm{id}_X : X \to X$.

▲

Remark 1.2.5 Passive and active characterizations of maps. It may be difficult to see how the above definition of a map is a generalization of the notion of a real-valued function of a real variable. In fact, the above definition might seem more closely related to the graph of such a function, since the uniqueness property of maps is just a generalization of the vertical line test for functions. We will refer to this perspective on maps as subsets of a Cartesian product satisfying certain abstract properties as the **passive characterization** of what maps *are*. In contrast, we often like to think of a map as an active process, which takes an input and transforms it into the unique output with which it is associated. We will refer to this perspective as the **active characterization** of what maps *do*. When it benefits us to favor one perspective over another, we will emphasize either the active or passive characterization over the other. The following definitions allow us to convert between these two equivalent characterizations of maps.

Definition 1.2.6 Image; pre-image. Let $f : X \to Y$ be a map from a set X to a set Y.

1. The **image** $f(x)$ of an input $x \in X$ under f is the unique output so that $(x, f(x)) \in Y$. The **image** $f(U)$ of a subset $U \subseteq X$ is the subset of the codomain Y defined by

$$f(U) = \{f(x) \in Y \mid x \in U\}.$$

The image $f(X)$ of X itself is called the **image** of f, and is denoted $\mathrm{im}(f) = f(X)$.

2. The **pre-image** $f^{-1}(V)$ of a subset $V \subseteq Y$ of the codomain Y under f is the set of all inputs whose image lies in V; that is,

$$f^{-1}(V) = \{x \in X \mid f(x) \in V\}.$$

The **pre-image** $f^{-1}(y)$ of an output $y \in Y$ under f is just the pre-image of the singleton set $\{y\}$ under f; that is,

$$f^{-1}(y) = f^{-1}(\{y\}) = \{x \in X \mid f(x) = y\}$$

◆

Any map between sets induces a map from any subset of its domain to its codomain via a process known as **restriction**. In light of the active characterization of what maps do, this may seem almost trivial. However, we will formalize this notion of restriction in terms of the passive characterization of what maps are in order to prove rigorously our intuitions are correct. You may be better served by conceptualizing the restriction of a map to be the same active process of transforming inputs into outputs but with a smaller domain, rather than the passive definition below.

Definition 1.2.7 Restriction. The **restriction** $f|_U$ of a map $f\colon X \to Y$ between sets X and Y to a subset $U \subseteq X$ of its domain X is the subset of the Cartesian product $U \times Y$ defined by

$$\begin{aligned} f|_U &= \{(u, y) \in U \times Y \mid f(u) = y\} = \{(u, y) \in U \times Y \mid (u, y) \in f\} \\ &= (U \times Y) \cap f. \end{aligned}$$

◆

As our intuition about the active characterization of what maps do tells us, the restriction of a map to a subset of its domain is itself a map from the chosen subset to the domain of the original map.

Proposition 1.2.8 Restrictions are maps. *Let $f\colon X \to Y$ be a map from a set X to a set Y. For any subset $U \subseteq X$, the restriction $f|_U$ of f to U is a map from U to Y.*

Proof. Let $u \in U$. Then $u \in X$ is an input to f, and so $(u, y) \in f$ for some output $y \in Y$. We conclude that

$$(x, y) \in \{(u, y) \in U \times Y \mid (u, y) \in f\} = f|_U\,,$$

so that $f|_U$ satisfies the existence condition.

Let $u \in U$ and $y_1, y_2 \in U$, and suppose that $(u, y_1), (u, y_2) \in f|_U$. Since

$$f|_U = (U \times Y) \cap f \subseteq f,$$

we have that $(u, y_1), (u, y_2) \in f$, and so $y_1 = y_2$; that is, $f|_U$ satisfies the uniqueness condition. Since $f|_U$ satisfies the existence and uniqueness conditions, it is a map from U to Y. ■

We will shortly see another way to construct restrictions as compositions with inclusion maps.

1.2.2 Composition and Invertibility

Viewed through the perspective of the active characterization of what maps do, it may seem obvious that we can combine two maps between sets by first acting on it with one map and then acting on its image by the other map. This new process yields the **composition** of the two maps, and it may be intuitively clear that this composition is itself a map. However, we will formalize this notion of composition in the passive characterization, so that we may prove this fact rigorously. Again, it may better serve you to conceptualize the composition of maps in the context of the active characterization of what maps do rather than the passive characterization of what maps are in order to better inform your intuition about composition.

Definition 1.2.9 Composition. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between sets X, Y, and Z. The **composition** $g \circ f$ of g with f is the map from X to Y defined by the formula

$$(g \circ f)(x) = g(f(x)).$$

◆

We aren't restricted to the composition of just two maps. In fact, we can compose any finite number of maps with compatible domains and codomains by iterating the above construction.

Proposition 1.2.10 Associativity of composition. *Let $f\colon W \to X$, $g\colon X \to Y$, and $h\colon Y \to Z$ be maps between sets W, X, Y, and Z. Then*

$$(h \circ g) \circ f = h \circ (g \circ f)\,.$$

Proof. Note that

$$\begin{aligned} ((h \circ g) \circ f)(w) &= (h \circ g)(f(w)) = h(g(f(w))) \\ &= h((g \circ f)(w)) = (h \circ (g \circ f))(w) \end{aligned}$$

for all $w \in W$. Since $(h \circ g) \circ f$ and $h \circ (g \circ f)$ agree on all inputs $w \in W$,

$$(h \circ g) \circ f = h \circ (g \circ f)\,.$$

■

The above notion of composition introduces a natural question about maps between sets:

To what degree can the action of a map on its domain be undone?

This question of map **invertibility** is fundamental, and will recur throughout this text.

Definition 1.2.11 Invertibility. Let $f\colon X \to Y$ be a map from a set X to a set Y.

1. *Left-invertibility.*

   A map $g\colon Y \to X$ from Y to X is a **left inverse** to f if $g \circ f = \mathrm{id}_X$. f is called **left-invertible** if such a left inverse exists.

2. *Right-invertibility.*

   A map $g\colon Y \to X$ from Y to X is a **right inverse** to f if $f \circ g = \mathrm{id}_Y$. f is called **right-invertible** if such a right inverse exists.

3. *Invertibility.*

   A map $g\colon Y \to X$ from Y to X is an **inverse** to f if it is both a left inverse and a right inverse to f; that is, g is an inverse to f if both $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. f is called **invertible** if such an inverse exists.

◆

Example 1.2.12 Inverses. The following are examples of left inverses, right inverses, and inverses of maps.

- Let $\emptyset \subsetneq U \subseteq X$ be a non-empty subset of a set X, and fix an element $u \in U$. Consider the map $c\colon X \to U$ from X to U defined piecewise by the formula

  $$c(x) = \begin{cases} x & x \in U \\ u & x \notin U. \end{cases}$$

  Note that

  $$\left(c \circ \iota_U^X\right)(u) = c\left(\iota_U^X(u)\right) = c(u) = u = \mathrm{id}_U(u)$$

  for all $u \in U$, and so $c \circ \iota_U^X = \mathrm{id}_U$; that is, c is a left inverse to the inclusion map $\iota_U^X\colon U \to X$ of U in X.

- Fix sets X and Y, and suppose that $Y \neq \emptyset$ is non-empty. Let $\pi\colon X \times Y \to X$ be the **projection map** defined by the formula $\pi(x, y) = x$.

  Fix an element $y \in Y$, and consider the map $c\colon X \to X \times Y$ from X to $X \times Y$ defined by the formula $c(x) = (x, y)$. Then

  $$(\pi \circ c)(x) = \pi(c(x)) = \pi(x, y) = x = \mathrm{id}_X(x)$$

  for all $x \in X$, and so $\pi \circ c = \mathrm{id}_X$; that is, c is a right inverse to the projection map π.

- The identity map $\mathrm{id}_X\colon X \to X$ on a set X is its own inverse, since $\mathrm{id}_X \circ \mathrm{id}_X = \mathrm{id}_X$. This is actually a special case of the first example above, where we take $U = X$. More generally, $f \circ \mathrm{id}_X = f$ for all maps f with domain X, and $\mathrm{id}_X \circ g = g$ for all maps g with codomain X.

▲

We now give alternative characterizations of left-invertibility, right-invertibility, and invertibility. Intuitively, a map is left-invertible if it does not associate two distinct inputs with the same output, and a map is right-invertible if each output is associated with at least one input. These properties mirror the totality and uniqueness conditions for maps, and will turn out to together imply the existence of a two-sided inverse, which is both a left and right inverse.

Definition 1.2.13 Injectivity; surjectivity; bijectivity. Let $f\colon X \to Y$ be a map from a set X to a set Y.

1. *Injectivity.*

   f is called **injective** (or **one-to-one**) if for all inputs $x_1, x_2 \in X$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$. In this case, f is also called an **injection** of X into Y.

2. *Surjectivity.*

   f is called **surjective** (or **onto**) if for all outputs $y \in Y$ there is an input $x \in X$ so that $y = f(x)$. In this case, f is also called a **surjection** of X onto Y.

3. *Bijectivity.*

   f is called **bijective** if it is both injective and surjective. In this case, f is also called a **bijection** or a **one-to-one correspondence** between X and Y.

◆

Example 1.2.14 Injections, surjections, and bijections. The following are examples of injective, surjective, and bijective functions.

1. Let $U \subseteq X$ be a subset of a set X, and consider $u_1, u_2 \in U$. If $\iota_U^X(u_1) = \iota_U^X(u_2)$, then
   $$u_1 = \iota_U^X(u_1) = \iota_U^X(u_2) = u_2.$$
   So the inclusion map $\iota_U^X\colon U \to X$ is injective.

2. Fix sets X and Y, and suppose that $Y \neq \emptyset$ is non-empty. Let $\pi\colon X \times Y \to X$ be the **projection map** defined by the formula $\pi(x, y) = x$.

   Let $x \in X$, and note that since $Y \neq \emptyset$ is non-empty, there is some element $y \in Y$. We note that $\pi(x, y) = x$. Since each output $x \in X$ is the image of some input under π, π is surjective.

3. The identity map $\mathrm{id}_X \colon X \to X$ on a set X is injective by virtue of being an inclusion map. id_X is also surjective, since $\mathrm{id}_X(x) = x$ for all $x \in X$. Thus id_X is bijective.

▲

It is no accident that the examples for left- and right- invertibility coincided with the examples for injectivity and surjectivity. It turns out that injectivity is essentially equivalent to left-invertibility, and surjectivity is equivalent to right-invertibility. This is the content of the following theorem, whose proof relies heavily on Problem 1.2.5.4 Composition and injectivity/surjectivity/bijectivity.

Theorem 1.2.15 Invertibility. *Let $f \colon X \to Y$ be a map from a set X to a set Y.*

1. *If f is left-invertible, then f is injective.*
2. *If $X \neq \emptyset$ is non-empty and f is injective, then f is left-invertible.*
3. *If f is right-invertible, then f is surjective.*

Proof.

1. Suppose that f is left-invertible, so that $g \circ f = \mathrm{id}_X$ for some map $g \colon Y \to X$. That f is injective follows from the injectivity of the identity map id_X and (d) of Problem 1.2.5.4 Composition and injectivity/surjectivity/bijectivity.

2. Conversely, now suppose that $X \neq \emptyset$ is non-empty and f is injective. Then for each output $y \in Y$, there is at most one input $x \in X$ so that $y = f(x)$. Specifically, for each $y \in \mathrm{im}(f)$, there is a unique input $x \in X$ so that $y = f(x)$. Denote by $g(y) = x$ this unique input.

   Since $X \neq \emptyset$ is non-empty, it contains an element $x \in X$. Define $g(y) = x$ for all $y \in Y \setminus \mathrm{im}(f)$. We have defined a map $g \colon Y \to X$. Note that

   $$(g \circ f)(x) = g(f(x)) = x = \mathrm{id}_X(x)$$

   for all inputs $x \in X$, and so $g \circ f = \mathrm{id}_X$; that is, g is a left inverse to f, and so f is left-invertible.

3. Finally, suppose that f is right-invertible, so that $f \circ g = \mathrm{id}_Y$ for some map $g \colon Y \to X$. That f is surjective follows from the surjectivity of the identity map id_Y and (e) of Problem 1.2.5.4 Composition and injectivity/surjectivity/bijectivity.

■

Proving the remaining direction of the equivalence between right invertibility and surjectivity requires a strong assumption about set theory called the Axiom of choice, which we do not yet have the language to express in its simplest form. We will discuss about the Axiom of choice in Subsection 1.4.1 Products and Choice, when we generalize the Cartesian product of two sets using indexing maps.

Corollary 1.2.16 Invertibility. *A map is invertible if and only if it is bijective.*

Proof. Let $f \colon X \to Y$ be a map from a set X to a set Y. If f is invertible, it is both left- and right-invertible, and so f is both injective and surjective by (1) and (3) of Theorem 1.2.15 Invertibility. Hence f is bijective.

Conversely, now suppose that f is bijective, so that f is both injective and surjective. If $X = \emptyset$, then

$$\begin{aligned} Y &= \operatorname{im}(f) = f(X) = f(\emptyset) \\ &= \{f(x) \in Y \mid x \in \emptyset\} = \emptyset \end{aligned}$$

by the surjectivity of f. Thus $f\colon \emptyset \to \emptyset$ is the unique map from $\emptyset$ to itself, the identity map $\mathrm{id}_\emptyset\colon \emptyset \to \emptyset$. In particular, f is invertible.

So we may assume that $X \neq \emptyset$ is non-empty. Since f is injective, it is left-invertible by (2) of Theorem 1.2.15 Invertibility. So $g \circ f = \mathrm{id}_X$ for some map $g\colon Y \to X$; that is, $g(f(x)) = x$ for all inputs $x \in X$. It suffices to show that g is also a right inverse to f.

To that end, consider an output $y \in Y$. Since f is surjective, $y = f(x)$ for some input $x \in X$. Then

$$(f \circ g)(y) = f(g(x)) = f(g(f(x))) = f(x) = y = \mathrm{id}_Y(y).$$

Hence $f \circ g = \mathrm{id}_Y$, and so g is an inverse to f; that is, f is invertible. ■

We will investigate the invertibility and bijectivity of maps and the consequences these properties have on the domains and codomains of such maps more in Section 1.6 Cardinality and Countability [SKIP] at the end of this chapter, when we study **cardinality** and the size of sets. For now, however, we will move on to an important convention which is ubiquitous throughout mathematics: the notions of **indexed sets** and **indexed families**.

1.2.3 Indexing

Considered as an association between sets, a map can be thought to index its image by the elements of its domain. That is, a map can transfer information from its domain to its image. Just what this transferred information looks like can vary tremendously, but suffice it to say that this notion of indexing is fundamental and will recur throughout this text.

Definition 1.2.17 Indexed family/set. Consider a map $x\colon \mathcal{J} \to X$ from a set $\mathcal{J}$, called the **indexing set** and whose elements are called **indices**, to a set X. The image of such an index $j \in \mathcal{J}$ under such an **(indexed) family** x is often denoted $x_j = x(j)$, and x is often denoted $(x_j)_{j \in \mathcal{J}}$. We say that $\operatorname{im}(x)$ is **indexed** by $\mathcal{J}$, and we write

$$\{x_j\}_{j \in \mathcal{J}} = \{x_j \in X \mid j \in \mathcal{J}\} = \operatorname{im}(x).$$

◆

We can also use this notion of indexing to generalize the intersection and union of sets to act on arbitrary collections of sets. This generalization of the intersection and union of two sets relies on the natural similarities between logical conjunction $\wedge$ and universal quantification $\forall$ and between logical disjunction $\vee$ and existential quantification $\exists$, respectively.

Definition 1.2.18 Indexed set operations. Let $(U_j)_{j \in \mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$.

1. *Intersection.*

   The **intersection** $\bigcap_{j \in \mathcal{J}} U_j$ of this indexed family of subsets is the subset of X defined by

   $$\bigcap_{j \in \mathcal{J}} U_j = \{x \in X \mid x \in U_j \text{ for all } j \in \mathcal{J}\}.$$

2. *Union.*

   The **union** $\bigcup_{j\in\mathcal{J}} U_j$ of this indexed family of subsets is the subset of X defined by

$$\bigcup_{j\in\mathcal{J}} U_j = \{x \in X \mid x \in U_j \text{ for some } j \in \mathcal{J}\}\,.$$

◆

These generalizations of the intersection and union of two sets extend naturally to generalizations of each of the results proven in the previous section.

Theorem 1.2.19 Analogous properties of indexed intersections and unions. *Let $(U_j)_{j\in\mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$.*

1. *Analogously to Proposition 1.1.14 Set operations and containment, for any subset $V \subseteq X$,*

$$V \subseteq \bigcap_{j\in\mathcal{J}} U_j$$

   if and only if $V \subseteq U_j$ for all indices $j \in \mathcal{J}$. Moreover,

$$\bigcup_{j\in\mathcal{J}} U_j \subseteq V$$

   if and only if $U_j \subseteq V$ for all indices $j \in \mathcal{J}$.

2. *Reindexing.*

   Analogously to (2) of Corollary 1.1.15 Properties of the set operations,

$$\bigcap_{k\in\mathcal{K}} U_{f(k)} = \bigcap_{j\in\mathcal{J}} U_j$$

   and

$$\bigcup_{k\in\mathcal{K}} U_{f(k)} = \bigcup_{j\in\mathcal{J}} U_j$$

   for any set $\mathcal{K}$ and surjection $f\colon \mathcal{K} \to \mathcal{J}$.

3. *Distributivity.*

   Analogously to Problem 1.1.5.1 Distributivity of set operations,

$$V \cup \bigcap_{j\in\mathcal{J}} U_j = \bigcap_{j\in\mathcal{J}} (V \cup U_j)$$

   and

$$V \cap \bigcup_{j\in\mathcal{J}} U_j = \bigcup_{j\in\mathcal{J}} (V \cap U_j)$$

   for any subset $V \subseteq X$.

4. *De Morgan's laws.*

   Analogously to Problem 1.1.5.2 De Morgan's laws,

$$V \setminus \bigcap_{j\in\mathcal{J}} U_j = \bigcup_{j\in\mathcal{J}} (V \setminus U_j)$$

   and

$$V \setminus \bigcup_{j\in\mathcal{J}} U_j = \bigcap_{j\in\mathcal{J}} (V \setminus U_j)$$

   for any subset $V \subseteq X$.

You will have an opportunity to prove the above results in the problems for this section.

The topics introduced in this section are central not only to the study of real analysis but to all of modern mathematics. In the next section, we will introduce another central object of study, the set $\mathbb{N}$ of **natural numbers**.

1.2.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Cartesian products. Write down the Cartesian products of the given sets in roster notation.

1. Write down the Cartesian product $S \times T$ of the sets $S = \{0,1\}$ and $T = \{0,1,2\}$ in roster notation.

2. Write down the Cartesian product $U \times V$ of the sets $U = \{0\}$ and $V = \{1,2,3,4\}$ in roster notation.

3. Write down the Cartesian product $X \times Y$ of the sets $X = \{0,1,2,3,4,5\}$ and $Y = \emptyset$ in roster notation.

Is it a map? Consider the sets $X = \{0,2,4,6,8\}$ and $Y = \{0,1,2,3,4\}$. Determine whether or not the given object is a map from X to Y.

4. Determine whether or not

$$f = \{(0,0), (2,1), (4,2), (6,3), (8,4)\}$$

is a map from X to Y.

5. Determine whether or not

$$g = \{(0,1), (2,1), (4,4), (8,1)\}$$

is a map from X to Y.

6. Determine whether or not

$$h = \{(0,1), (2,1), (4,4), (4,3), (6,2), (8,1)\}$$

is a map from X to Y.

7. Determine whether or not

$$j = \{(0,0), (2,0), (4,0), (6,0), (8,0)\}$$

is a map from X to Y.

8. **Images and pre-images.** Let $f_4 \colon \{1,2\} \to \{\text{red}, \text{blue}\}$ be the map from $\{1,2\}$ to $\{\text{red}, \text{blue}\}$ defined in Example 1.2.4 Maps; that is,

$$f_4 = \{(1, \text{blue}), (2, \text{blue})\}.$$

Compute the given images and pre-images. Express your answer in roster notation if necessary.

(a) Compute the images $f_4(1)$ and $f_4(2)$.

(b) Compute the image $\mathrm{im}(f_4)$.

(c) Compute the pre-images f_4^{-1} (red) and f_4^{-1} (blue).

Determining injectivity/surjectivity/bijectivity. Determine whether or not the given map is injective, surjective, and/or bijective.

9. Let $X = \{0, 1\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f \colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 0)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

10. Let $X = \{0, 1\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f \colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 2)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

11. Let $X = \{0, 1, 2\}$ and $Y = \{0, 1\}$, and consider the following map $f \colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 1), (2, 1)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

12. Let $X = \{0, 1, 2\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f \colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 2), (2, 1)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.2.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Non-commutativity of Cartesian products.** Give examples of sets X and Y so that $X \times Y \neq Y \times X$.

2. **Existence of maps.** Determine whether the following statement is true or false:

For any sets X and Y, there is a map from X to Y.

If the statement is true, prove it. If the statement is false, disprove it by providing a counterexample.

Hint. Consider your answers to Exercise Group 1.2.4.1–3 Cartesian products.

3. Non-commutativity of composition.

(a) Give examples of maps f and g so that the composition $g \circ f$ is well-defined but the composition $f \circ g$ is not.

(b) Give examples of maps f and g so that the compositions $g \circ f$ and $f \circ g$ are well-defined but $g \circ f \neq f \circ g$.

4. Composition and injectivity/surjectivity/bijectivity. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between sets X, Y, and Z.

(a) Prove that if f and g are injective, then so too is their composition $g \circ f\colon X \to Z$.

(b) Prove that if f and g are surjective, then so too is their composition $g \circ f\colon X \to Z$.

(c) Prove that if f and g are bijective, then so too is their composition $g \circ f\colon X \to Z$.

(d) Prove that if the composition $g \circ f\colon X \to Z$ is injective, then so too is f.

Hint. Prove the desired result by contraposition.

(e) Prove that if the composition $g \circ f\colon X \to Z$ is surjective, then so too is g.

Hint. Prove the desired result by contraposition.

Note that the above results are used in the proof of Theorem 1.2.15 Invertibility. In order to avoid circular reasoning, you should prove the above statements by using only the definitions of injectivity, surjectivity, and bijectivity (and not the alternative characterizations given by Theorem 1.2.15 Invertibility and Corollary 1.2.16 Invertibility).

5. Indexed intersections and unions. Let $(U_j)_{j\in\mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$.

(a) Prove that for any subset $V \subseteq X$,

$$V \subseteq \bigcap_{j\in\mathcal{J}} U_j$$

if and only if $V \subseteq U_j$ for all indices $j \in \mathcal{J}$, and

$$\bigcup_{j\in\mathcal{J}} U_j \subseteq V$$

if and only if $U_j \subseteq V$ for all indices $j \in \mathcal{J}$.

(b) Prove that

$$\bigcap_{k\in\mathcal{K}} U_{f(k)} = \bigcap_{j\in\mathcal{J}} U_j$$

and

$$\bigcup_{k\in\mathcal{K}} U_{f(k)} = \bigcup_{j\in\mathcal{J}} U_j$$

for any set $\mathcal{K}$ and surjection $f\colon \mathcal{K} \to \mathcal{J}$.

(c) Prove that

$$V \cup \bigcap_{j \in \mathcal{J}} U_j = \bigcap_{j \in \mathcal{J}} (V \cup U_j)$$

and

$$V \cap \bigcup_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \cap U_j)$$

for any subset $V \subseteq X$.

(d) Prove that

$$V \setminus \bigcap_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \setminus U_j)$$

and

$$V \setminus \bigcup_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \setminus U_j)$$

for any subset $V \subseteq X$.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.3 Peano Arithmetic and the Natural Numbers [Skip]

In this section, we formally introduce the **natural number system** $\mathbb{N}$. Also called the **counting numbers** and the **whole numbers**, the natural numbers form the simplest of the number systems; in fact, many people would date the advent of mathematics itself to the invention/discovery of these numbers.

1.3.1 Axiomatizing the Natural Numbers

There are many nonequivalent formalizations and models of the natural number system, but the standard axiomatization is due to Giuseppe Peano in the 19th century. **Peano arithmetic** is **non-constructive**; it gives a description of the natural number system but not a model. We won't introduce any specific model of the natural number system, but, rest assured, there are several. For our purposes, $\mathbb{N}$ itself can be any set which conforms to the following characterization:

Definition 1.3.1 Natural numbers. The **natural number system** $\mathbb{N}$ is a nonempty set with an element $0 \in \mathbb{N}$, called **zero**, along with functions $\operatorname{succ}\colon \mathbb{N} \to \mathbb{N}$ and $\operatorname{add}, \operatorname{mult}\colon \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, called the **successor function** and the **addition** and **multiplication operations** and a subset $L \subseteq \mathbb{N} \times \mathbb{N}$ of the Cartesian product $\mathbb{N} \times \mathbb{N}$ of $\mathbb{N}$ with itself, called the **standard ordering**, which satisfy the following properties:

1. The **successor** $\operatorname{succ}(n)$ of any natural number $n \in \mathbb{N}$ is non-zero; that is $\operatorname{succ}(n) \neq 0$ for all natural numbers $n \in \mathbb{N}$.

2. The successor function $\operatorname{succ}\colon \mathbb{N} \to \mathbb{N}$ is injective; that is, for all natural numbers $m, n \in \mathbb{N}$, if $\operatorname{succ}(m) = \operatorname{succ}(n)$, then $m = n$.

3. $\operatorname{add}(n, 0) = n$ for all natural numbers $n \in \mathbb{N}$.

4. $\operatorname{add}(m, \operatorname{succ}(n)) = \operatorname{succ}(\operatorname{add}(m,n))$ for all natural numbers $m, n \in \mathbb{N}$.

5. $\operatorname{mult}(n, 0) = 0$ for all natural numbers $n \in \mathbb{N}$.

6. $\operatorname{mult}(m, \operatorname{succ}(n)) = \operatorname{add}(\operatorname{mult}(m,n), m)$ for all natural numbers $n \in \mathbb{N}$.

7. For all natural numbers $l, m \in \mathbb{N}$, $(l, m) \in L$ if and only if $\operatorname{add}(l, n) = m$ for some natural number $n \in \mathbb{N}$.

8. *Axiom of induction.*

   For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:

   (a) $0 \in U$.

   (b) For all natural numbers $n \in \mathbb{N}$, if $n \in U$, then $\operatorname{succ}(n) \in U$.

Typically, for all natural numbers $m, n \in \mathbb{N}$ we write $m + n = \operatorname{add}(m, n)$ and $mn = m \cdot n = \operatorname{mult}(m, n)$, and we write $m \leq n$ if and only if $(m, n) \in L$. The successor of 0 is denoted $1 = \operatorname{succ}(0)$. Likewise, the successor of 1 is denoted $2 = \operatorname{succ}(1)$, the successor of 2 is denoted $3 = \operatorname{succ}(2)$, and so on. We see that

$$\operatorname{succ}(n) = \operatorname{succ}(\operatorname{add}(n, 0)) = \operatorname{add}(n, \operatorname{succ}(0)) = n + 1$$

for all natural numbers $n \in \mathbb{N}$. With these new notations, we may restate the Peano axioms as follows:

1. $n + 1 \neq 0$ for all $n \in \mathbb{N}$.

2. For all natural numbers $m, n \in \mathbb{N}$, if $m + 1 = n + 1$, then $m = n$.

3. $n + 0 = n$ for all natural numbers $n \in \mathbb{N}$.

4. $m + (n + 1) = (m + n) + 1$ for all natural numbers $m, n \in \mathbb{N}$.

5. $n \cdot 0 = 0$ for all natural numbers $n \in \mathbb{N}$.

6. $m(n + 1) = mn + m$ for all natural numbers $n \in \mathbb{N}$.

7. For all natural numbers $l, m \in \mathbb{N}$, $l \leq m$ if and only if $l + n = m$ for some natural number $n \in \mathbb{N}$.

8. For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:

   (a) $0 \in U$.

   (b) *Axiom of induction.*
   For all natural numbers $n \in \mathbb{N}$, if $n \in U$, then $n + 1 \in U$.

◆

Under the usual (and original) interpretation of the natural number system $\mathbb{N}$ as characterizing quantities of discrete physical objects, the above rules (with the possible exception of the axiom of induction) may seem obviously true but also somewhat arbitrary. It is a fact, though we will not prove it here, that these axioms are independent of each other; it is impossible to derive one of the above statements from only the others. We hope that this, along with the results in this section about arithmetic, convince you that the above axiomatization is a good choice, in that it is efficient in the above sense and that it can be used to prove many arithmetical truths.

First, however, we introduce some ubiquitous terminology and notation.

A family $(x_n)_{n\in\mathcal{N}}$ of objects x_n indexed by some subset $\mathcal{N} \subseteq \mathbb{N}$ is called a **sequence**. If the index set $\mathcal{N}$ is of the form

$$\mathcal{N} = \{n \in \mathbb{N} \mid a \leq n \text{ and } n \leq b\}$$

for some natural numbers $a, b \in \mathbb{N}$, then we denote this family by $(x_n)_{n=a}^{b}$. Similarly, if the index set $\mathcal{N}$ is of the form

$$\mathcal{N} = \{n \in \mathbb{N} \mid a \leq n\}$$

for some natural number $a \in \mathbb{N}$, then we denote this family by $(x_n)_{n=a}^{\infty}$. These notations also apply to intersections and unions of sequences of sets indexed by subsets of $\mathbb{N}$ of the above form.

Lemma 1.3.2 0 is minimal. *For all natural numbers $n \in \mathbb{N}$, if $n \leq 0$, then $n = 0$.*

Proof. We first observe that $\{0\} \cup \operatorname{im}(\operatorname{succ}) = \mathbb{N}$. Indeed, note that $0 \in \{0\} \subseteq \{0\} \cup \operatorname{im}(\operatorname{succ})$. Moreover, if $n \in \{0\} \cup \operatorname{im}(\operatorname{succ})$, then

$$\operatorname{succ}(n) \in \operatorname{im}(\operatorname{succ}) \subseteq \{0\} \cup \operatorname{im}(\operatorname{succ}).$$

Thus $\{0\} \cup \operatorname{im}(\operatorname{succ}) = \mathbb{N}$ by the axiom of induction.

Let $n \in \mathbb{N}$ be a natural number, and suppose that $n \leq 0$ for some natural number $n \in \mathbb{N}$. Then $n + m = 0$ for some natural number $m \in \mathbb{N}$. Suppose for a contradiction that $m = p + 1$ for some natural number $p \in \mathbb{N}$. Then

$$0 = n + m = n + (p + 1) = (n + p) + 1,$$

which contradicts the first Peano axiom. Thus $m \notin \operatorname{im}(\operatorname{succ})$. However, $m \in \mathbb{N} = \{0\} \cup \operatorname{im}(\operatorname{succ})$, and so $m \in \{0\}$; that is, $m = 0$, and so

$$n = n + 0 = n + m = 0.$$

■

Proposition 1.3.3 Principle of strong induction. *For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:*

1. $0 \in U$.
2. *For all natural numbers $n \in \mathbb{N}$, if $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n$, then $n + 1 \in U$.*

Proof. Clearly $\mathbb{N}$ satisfies the above conditions, and so it suffices to show that any subsets which satisfies these conditions is equal to $\mathbb{N}$. To that end, let $U \subseteq \mathbb{N}$ be a subset which satisfies the above conditions, and consider the subset

$$V = \{n \in \mathbb{N} \mid k \in U \text{ for all } k \in \mathbb{N} \text{ so that } k \leq n\}.$$

Since the only $k \in \mathbb{N}$ so that $k \leq 0$ is $k = 0$ itself, and $0 \in U$, we have that $0 \in V$. Moreover if $n \in V$, then $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n$, and so $n + 1 \in U$ by hypothesis. It remains to conclude that $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n + 1$.

To that end, let $k \in \mathbb{N}$, and suppose that $k \leq n + 1$. Then $k + m = n + 1$ for some natural number $m \in \mathbb{N}$. If $m = 0$, then $k = n + 1 \in U$. On the other hand, if $m \neq 0$, then the argument in the proof of Lemma 1.3.2 0 is minimal implies that $m = p + 1$ for some natural number $p \in \mathbb{N}$. Thus

$$(k + p) + 1 = k + (p + 1) = k + m = n + 1,$$

and so the injectivity of the successor function implies that $k + p = n$; that is, $k \leq n$ and so $k \in U$. Since $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n + 1$, we conclude that $n + 1 \in V$.

In summary, we have shown that $0 \in V$ and that for all natural numbers $n \in \mathbb{N}$, if $n \in V$, then $n + 1 \in V$. Thus $V = \mathbb{N}$ by the axiom of induction. One can check that $\mathbb{N} = V \subseteq U \subseteq \mathbb{N}$, and so $U = \mathbb{N}$. ■

The axiom of induction and Proposition 1.3.3 Principle of strong induction form the basis for ubiquitous methods of proof called **proof by induction** and **proof by strong induction**, respectively. You will have the opportunity to practice constructing such proofs in the problems for this section.

1.3.2 Properties of Arithmetic

The axioms of Peano arithmetic actually imply a great deal about the properties of the natural number system. We will now state and prove a great deal of results about the successor function and the addition and multiplication operators. Many of these results will give us practice in the fundamental skill of proof by induction, and we will use these results implicitly for the rest of this text.

Proposition 1.3.4 Associativity of addition. $(l + m) + n = l + (m + n)$ *for all natural numbers* $l, m, n \in \mathbb{N}$.

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$ to show that $(l + m) + n = l + (m + n)$.

Base case. If $n = 0$, then

$$(l + m) + n = (l + m) + 0 = l + m = l + (m + 0) = l + (m + n)\,.$$

Inductive step. Now suppose for the inductive hypothesis that $(l + m) + n = l + (m + n)$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned}(l + m) + (n + 1) &= ((l + m) + n) + 1 = (\ell + (m + n)) + 1\\ &= \ell + ((m + n) + 1) = \ell + (m + (n + 1))\,.\end{aligned}$$

We conclude that $(l + m) + n = l + (m + n)$ for all natural numbers $n \in \mathbb{N}$. ■

One consequence of Proposition 1.3.4 Associativity of addition is that we may unambiguously parse expressions with multiple sums without the need of parentheses; for example,if $l, m, n \in \mathbb{N}$ are natural numbers, then the expression $l + m + n$ is well-defined, because

$$(l + m) + n = l + (m + n)\,.$$

Going forward, we will make use of these expressions wherever it is possible to do so without obscuring the relevant arguments.

Proposition 1.3.4 Associativity of addition implies that the result of adding a finite number of natural numbers depends only on the order of the numbers from left to right, and not on the iterative structure of the summands induced by parentheses. In fact, the order of the summands does not matter either. In order to show this, it helps to notice the following two results about addition.

Lemma 1.3.5 Addition by 0. $n + 0 = 0 + n = n$ *for all natural numbers* $n \in \mathbb{N}$.

Proof. $n + 0 = n$ for all natural numbers $n \in \mathbb{N}$ by the Peano axioms, and so it suffices to show $0 + n = n$. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$0 + n = 0 + 0 = 0 = n.$$

Inductive step. Now suppose for the inductive hypothesis that $0 + n = n$ for some natural number $n \in \mathbb{N}$, and note that

$$0 + (n + 1) = (0 + n) + 1 = n + 1.$$

We conclude that $0 + n = n$ for all natural numbers $n \in \mathbb{N}$. ■

Lemma 1.3.6 Towards commutativity of addition. $1 + n = n + 1$ *for all natural numbers* $n \in \mathbb{N}$.

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$1 + n = 1 + 0 = 0 + 1 = n + 1$$

by Lemma 1.3.5 Addition by 0.

Inductive step. Now suppose for the inductive hypothesis that $1 + n = n + 1$ for some natural number $n \in \mathbb{N}$, and note that

$$1 + (n + 1) = (1 + n) + 1 = (n + 1) + 1$$

by Proposition 1.3.4 Associativity of addition. We conclude that $1 + n = n + 1$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.7 Commutativity of addition. $m + n = n + m$ *for all natural numbers* $m, n \in \mathbb{N}$.

Proof. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$m + n = m + 0 = m = 0 + m = n + m$$

by Lemma 1.3.5 Addition by 0.

Inductive step. Now suppose for the inductive hypothesis that $m + n = n + m$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} m + (n + 1) &= (m + n) + 1 = (n + m) + 1 = 1 + (n + m) \\ &= (1 + n) + m = (n + 1) + m \end{aligned}$$

by Proposition 1.3.4 Associativity of addition and Lemma 1.3.6 Towards commutativity of addition. We conclude that $m + n = n + m$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.4 Associativity of addition and Proposition 1.3.7 Commutativity of addition together imply that, when adding natural numbers, neither the order of the summands nor the order of the addition operations performed matters. In order to see that analogous properties also hold for multiplication, we will require the following supporting results. Along the way, we will also prove the distributivity of multiplication over addition.

Lemma 1.3.8 Multiplication by 1**.** *$n \cdot 1 = n$ and $1n = n$ for all natural numbers $n \in \mathbb{N}$.*

Proof. For all natural numbers $n \in \mathbb{N}$,

$$n \cdot 1 = n\,(0+1) = (n \cdot 0) + n = 0 + n = 0$$

by Lemma 1.3.5 Addition by 0. Thus it suffices to prove that $1n = n$ for all natural numbers $n \in \mathbb{N}$. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$1n = 1 \cdot 0 = 0 = n.$$

Inductive step. Now suppose for the inductive hypothesis that $1n = n$ for some natural number $n \in \mathbb{N}$, and note that

$$1\,(n+1) = 1n + 1 = n + 1.$$

We conclude that $1n = n$ for all natural numbers $n \in \mathbb{N}$. ∎

Proposition 1.3.9 Left distributivity of multiplication over addition. *$l\,(m+n) = lm + ln$ for all natural numbers $l, m, n \in \mathbb{N}$.*

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$l\,(m+n) = l\,(m+0) = lm = lm + 0 = lm + l \cdot 0 = lm + ln.$$

Inductive step. Now suppose for the inductive hypothesis that $l\,(m+n) = lm + ln$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} l\,(m + (n+1)) &= l\,((m+n)+1) = l\,(m+n) + l \\ &= (lm + ln) + l = lm + (ln + l) = lm + l\,(n+1) \end{aligned}$$

by Proposition 1.3.4 Associativity of addition. We conclude that $l\,(m+n) = lm + ln$ for all natural numbers $n \in \mathbb{N}$. ∎

Proposition 1.3.10 Associativity of multiplication. *$(lm)\,n = l\,(mn)$ for all natural numbers $l, m, n \in \mathbb{N}$.*

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$(lm)\,n = (lm)\,0 = 0 = l \cdot 0 = l\,(m \cdot 0) = l\,(mn)\,.$$

Inductive step. Now suppose for the inductive hypothesis that $(lm)\,n = l\,(mn)$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} (lm)\,(n+1) &= (lm)\,n + lm = l\,(mn) + lm = \\ &= l\,(mn + m) = l\,(m\,(n+1)) \end{aligned}$$

by Proposition 1.3.9 Left distributivity of multiplication over addition. We conclude that $(lm)\,n = l\,(mn)$ for all natural numbers $n \in \mathbb{N}$. ∎

Just as with addition, one consequence of Proposition 1.3.10 Associativity of multiplication is that we may unambiguously parse expressions with multiple products without the need of parentheses; for example,if $l, m, n \in \mathbb{N}$ are natural numbers, then the expression lmn is well-defined, because

$$(lm)\,n = l\,(mn)\,.$$

Going forward, we will make use of these expressions wherever it is possible to do so without obscuring the relevant arguments.

Proposition 1.3.11 Right distributivity of multiplication over addition. *$(l+m)\,n = ln + mn$ for all natural numbers $l, m, n \in \mathbb{N}$.*

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$(l+m)\,n = (l+m)\cdot 0 = 0 = 0 + 0 = l \cdot 0 + m \cdot 0 = ln + mn.$$

Inductive step. Now suppose for the inductive hypothesis that $(l+m)\,n = ln + mn$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned}(l+m)\,(n+1) = (l+m)\,n + (l+m) = ln + l + mn + m \\ = l\,(n+1) + m\,(n+1)\end{aligned}$$

by repeated applications of Proposition 1.3.4 Associativity of addition and Proposition 1.3.7 Commutativity of addition. We conclude that $(l+m)\,n = ln + mn$ for all natural numbers $n \in \mathbb{N}$. ■

Lemma 1.3.12 Multiplication by 0. *$0n = 0$ for all natural numbers $n \in \mathbb{N}$.*

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$0n = 0 \cdot 0 = 0.$$

Inductive step. Now suppose for the inductive hypothesis that $0n = 0$ for some natural number $n \in \mathbb{N}$, and note that

$$0\,(n+1) = 0n + 0 = 0 + 0 = 0.$$

We conclude that $0n = 0$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.13 Commutativity of multiplication. *$mn = nm$ for all natural numbers $m, n \in \mathbb{N}$.*

Proof. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$mn = m \cdot 0 = 0 = 0 \cdot m = nm$$

by Lemma 1.3.12 Multiplication by 0.

Inductive step. Now suppose for the inductive hypothesis that $mn = nm$ for some natural number $n \in \mathbb{N}$, and note that

$$m\,(n+1) = mn + m = nm + 1m = (n+1)\,m$$

by Lemma 1.3.8 Multiplication by 1 We conclude that $mn = nm$ for all natural numbers $n \in \mathbb{N}$. ■

Lemma 1.3.14 Cancellation of addition. *For all natural numbers $n \in \mathbb{N}$, addition by n is injective; that is, for all natural numbers $l, m, n \in \mathbb{N}$, if $l + n = m + n$, then $l = m$.*

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$ and $l + n = m + n$, then

$$l = l + 0 = l + n = m + n = m + 0 = m$$

Inductive step. Now suppose for the inductive hypothesis that there is some natural number $n \in \mathbb{N}$ so that if $l + n = m + n$, then $l = m$. Note that if $l + (n + 1) = m + (n + 1)$, then

$$(l + n) + 1 = l + (n + 1) = m + (n + 1) = (m + n) + 1,$$

and so $l + n = m + n$ by the injectivity of the successor function. Thus $l = m$ by the inductive hypothesis. In summary, we have shown that if $l + (n + 1) = m + (n + 1)$, then $l = m$. We conclude that for all natural numbers $n \in \mathbb{N}$, if $l + n = m + n$, then $l = m$. ■

A similar cancellation law exists for multiplication, but its proof will require some properties of the comparison of natural numbers via its standard ordering.

1.3.3 Properties of Comparison

The final component of Peano's axiomatization of the natural numbers $\mathbb{N}$ is that of comparison. The standard ordering $\leq$ on the natural numbers has several important and familiar properties which we will now derive from first principles.

Proposition 1.3.15 Comparison is a total order. *Comparison of natural numbers has the following properties:*

1. *Reflexivity.*

   $n \leq n$ *for all natural numbers* $n \in \mathbb{N}$*.*

2. *Antisymmetry.*

   For all natural numbers $m, n \in \mathbb{N}$*, if* $m \leq n$ *and* $n \geq m$*, then* $m = n$*.*

3. *Transitivity.*

   For all natural numbers $l, m, n \in \mathbb{N}$*, if* $l \leq m$ *and* $m \leq n$*, then* $l \leq n$*.*

4. *Totality.*

   For all natural numbers $m, n \in \mathbb{N}$*, either* $m \leq n$ *or* $n \leq m$*.*

Proof.

1. For all natural numbers $n \in \mathbb{N}$, $n + 0 = n$, so that $n \leq n$.

2. Let $m, n \in \mathbb{N}$ be natural numbers, and suppose that $m \leq n$ and $n \leq m$. Then $n = m + p$ and $m = n + q$ for some natural numbers $p, q \in \mathbb{N}$. Note that

   $$m + 0 = m = n + q = m + p + q,$$

   and so $0 = p + q$ by Lemma 1.3.14 Cancellation of addition; that is, $q \leq 0$. Lemma 1.3.2 0 is minimal now implies that $q = 0$, so that

   $$m = n + q = n + 0 = n.$$

3. Let $l, m, n \in \mathbb{N}$ be natural numbers, and suppose that $l \leq m$ and $m \leq n$. Then $l + p = m$ and $m + q = n$ for some natural numbers $p, q \in \mathbb{N}$, and

so

$$l + (p + q) = (l + p) + q = m + q = n.$$

We conclude that $l \leq n$.

4. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

   Base case. If $n = 0$, then $0 + m = m$ by Lemma 1.3.5 Addition by 0, and so $n = 0 \leq m$.

   Inductive step. Now suppose for the inductive hypothesis that there is a natural number $n \in \mathbb{N}$ so that either $m \leq n$ or $n \leq m$. If $m \leq n$, then $m + p = n$ for some natural number $p \in \mathbb{N}$, and so

   $$m + (p + 1) = (m + p) + 1 = n + 1;$$

   that is, $m \leq n + 1$.

   Now suppose that $n \leq m$, so that $n + p = m$ for some natural number $p \in \mathbb{N}$. If $p = 0$, then

   $$m + 1 = (n + p) + 1 = (n + 0) + 1 = n + 1,$$

   and so $m \leq n + 1$. On the other hand, if $p \neq 0$, then $p = q + 1$ for some natural number $q \in \mathbb{N}$ by an argument in the proof of Lemma 1.3.2 0 is minimal. Then

   $$(n + 1) + q = n + 1 + q = n + q + 1 = n + p = m,$$

   and so $n + 1 \leq m$. We conclude that for all natural numbers $n \in \mathbb{N}$, either $m \leq n$ or $n \leq m$.

■

Corollary 1.3.16 Cancellation of multiplication. *For all natural numbers $n \in \mathbb{N}$, if $n \neq 0$, then addition by n is injective; that is, for all natural numbers $l, m, n \in \mathbb{N}$, if $ln = mn$, then either $l = m$ or $n = 0$.*

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then $n = 0$, and so the desired conclusion holds for all natural numbers $l, m \in \mathbb{N}$.

Inductive step. Now suppose for the inductive hypothesis that there is a natural number $n \in \mathbb{N}$ so that for all natural numbers $l, m \in \mathbb{N}$, if $ln = mn$, then either $n = 0$ or $l = m$. Let $l, m \in \mathbb{N}$ be natural numbers, and suppose that $l\,(n + 1) = m\,(n + 1)$. Then $n + 1 \neq 0$, and so it suffices to show that $l = m$.

To that end, note that (4) of Proposition 1.3.15 Comparison is a total order implies that either $l \leq m$ or $m \leq l$. Without loss of generality, we may suppose that $l \leq m$, so that $m = l + p$ for some natural number $p \in \mathbb{N}$. We observe that

$$\begin{aligned} 0 + (ln + l) &= ln + l = l\,(n + 1) = m\,(n + 1) \\ &= (l + p)\,(n + 1) = ln + l + pn + p = (pn + p) + (ln + l) \end{aligned}$$

by Lemma 1.3.5 Addition by 0, Proposition 1.3.9 Left distributivity of multiplication over addition, Proposition 1.3.11 Right distributivity of multiplication over addition, Proposition 1.3.4 Associativity of addition, and Proposition 1.3.7 Commutativity of addition.

In particular, Lemma 1.3.14 Cancellation of addition implies that $0 = pn + p$, so that $p \leq 0$. We conclude by Lemma 1.3.2 0 is minimal that $p = 0$, so that

$$l = m + p = m + 0 = m.$$

We conclude that for all natural numbers $l, m, n \in \mathbb{N}$, either if $ln = mn$, then either $l = m$ or $n = 0$. ■

Lemma 1.3.17 Comparison and arithmetic. *Let $l, m, n, p \in \mathbb{N}$ be natural numbers.*

1. *If $l \leq n$ and $m \leq p$, then $l + m \leq n + p$.*
2. *If $l \leq n$ and $m \leq p$, then $lm \leq np$.*

Proof.

1. If $l \leq n$ and $m \leq p$, then $n = l + q$ and $p = m + r$ for some natural numbers $q, r \in \mathbb{N}$, and so

$$(l + m) + (q + r) = l + m + q + r = l + q + m + r = n + p.$$

   We conclude that $l + m \leq n + p$.

2. If $l \leq n$ and $m \leq p$, then $n = l + q$ and $p = m + r$ for some natural numbers $q, r \in \mathbb{N}$, and so

$$\begin{aligned} lm + (lr + qm + qr) &= lm + lr + qm + qr = l\,(m + r) + q\,(m + r) \\ &= (l + q)\,(m + r) = np. \end{aligned}$$

   We conclude that $lm \leq np$.

■

For ease of reference, we collect the above results (along with some immediate corollaries) in the following theorem.

Theorem 1.3.18 Properties of Peano arithmetic. *Addition, multiplication, and comparison of natural numbers have the following properties:*

1. *Associativity of addition.*

   $(l + m) + n = l + (m + n)$ *for all natural numbers* $l, m, n \in \mathbb{N}$.

2. *Commutativity of addition.*

   $m + n = n + m$ *for all natural numbers* $m, n \in \mathbb{N}$.

3. *Additive identity.*

   $n + 0 = 0 + n = 0$ *for all natural numbers* $n \in \mathbb{N}$.

4. *Associativity of multiplication.*

   $(lm)\,n = l\,(mn)$ *for all natural numbers* $l, m, n \in \mathbb{N}$.

5. *Commutativity of multiplication.*

   $mn = nm$ *for all natural numbers* $m, n \in \mathbb{N}$.

6. *Multiplicative annihilation.*

   $n \cdot 0 = 0n = 0$ *for all natural numbers* $n \in \mathbb{N}$.

7. *Multiplicative identity.*

   $n \cdot 1 = 1n = n$ *for all natural numbers* $n \in \mathbb{N}$.

8. *Distributivity of multiplication over addition.*

   $l(m+n) = lm + ln$ *and* $(l+m)n = ln + mn$ *for all natural numbers* $l, m, n \in \mathbb{N}$.

9. *Reflexivity of comparison.*

   $n \leq n$ *for all natural numbers* $n \in \mathbb{N}$.

10. *Antisymmetry of comparison.*

    For all natural numbers $m, n \in \mathbb{N}$, *if* $m \leq n$ *and* $n \leq m$, *then* $m = n$.

11. *Transitivity of comparison.*

    For all natural numbers $l, m, n \in \mathbb{N}$, *if* $l \leq m$ *and* $m \leq n$, *then* $l \leq n$.

12. *Totality of comparison.*

    For all natural numbers $m, n \in \mathbb{N}$, *either* $m \leq n$ *or* $n \leq m$.

13. *Compatibility of comparison and addition/multiplication.*

    For all natural numbers $l, m, n, p \in \mathbb{N}$, *if* $l \leq n$ *and* $m \leq p$, *then*

    $$l + m \leq n + p$$

    and

    $$lm \leq np$$

Many of these properties of the arithmetic and comparison of natural numbers also extend to the arithmetic of the other classes of numbers we will consider, namely the integers, rational numbers, real numbers, and complex numbers.

In the next sections, we will generalize the properties of addition and multiplication and the order-theoretic properties of comparison of natural numbers and arrive at fundamental notions of **finitary operations** and **relations**, **partial** and **total orders**, and the Well-ordering principle.

1.3.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.3.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be

less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.4 Finitary Operations and Relations

In this section, we will develop the notion of (finitary) operations and relations on sets. Intuitively, a **finitary operation** on a set is an assignment of an element of that set to some finite number of other elements, and a **finitary relation** on a set is a designation of certain groups of some finite number of elements as associated.

1.4.1 Products and Choice

Previously, we have used the mathematical convention of indexing to generalize the intersection and union of two sets. We may also use indexing to generalize the Cartesian product of two sets to construct products of arbitrary indexed families of sets. In this section, we will develop this **generalized Cartesian product**, and use it to introduce operations and relations on a set. Instead of containing ordered pairs of elements, our new Cartesian product consists of indexed families of elements.

Definition 1.4.1 Cartesian product. Let $(U_j)_{j\in\mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X, called **(Cartesian) factors**, indexed by a set $\mathcal{J}$.

1. *Cartesian product.*

   The **Cartesian product** $\prod_{j\in\mathcal{J}} U_j$ of this indexed family is the set

   $$\prod_{j\in\mathcal{J}} U_j = \left\{ (u_j)_{j\in\mathcal{J}} \in X^{\mathcal{J}} \;\middle|\; u_j \in U_j \text{ for all } j \in \mathcal{J} \right\}$$

   of families $(u_j)_{j\in\mathcal{J}}$ of elements $u_j \in X$ with the property that $u_j \in U_j$ for all indices $j \in \mathcal{J}$.

2. *Projection.*

   For each index $k \in \mathcal{J}$, **projection onto the kth factor** is the map $\pi_k \colon \prod_{j\in\mathcal{J}} U_j \to U_k$ is the map from the Cartesian product $\prod_{j\in\mathcal{J}} U_j$ to the kth factor U_k defined by the formula

   $$\pi^{U_k}_{\prod_{j\in\mathcal{J}} U_j} (u_j)_{j\in\mathcal{J}} = u_k.$$

◆

Convention 1.4.2 Empty and singleton products.

- *Empty product.*

  What does a family indexed by the empty set look like? There is a unique map from the empty set $\emptyset$ to a set X, and so there is only one such family

  $$(x_j)_{j\in\emptyset}$$

of elements x_j of X indexed by the empty set $\emptyset$. This unique indexed family is usually denoted $()$. By the above remark,

$$\prod_{j\in\emptyset} U_j = \{()\}\,.$$

- *Singleton product.*

  Similarly, a family $(x_j)_{j\in\{j\}}$ of elements $x_j \in X$ of a set S indexed by a singleton set $\{j\}$ is uniquely determined by the image x_j of the index j. Such an indexed family is denoted

  $$(x_j) = (x_j)_{j\in\{j\}}\,.$$

  We usually act under the convention $(x_j) = x_j$, so that

  $$\prod_{j\in\{j\}} U_j = U_j.$$

It may seem intuitive or obvious that the Cartesian product of an indexed family of nonempty sets is itself nonempty. After all, one can specify an indexed family of elements of these sets just by choosing an element of each nonempty set. This notion of a choice of elements of nonempty sets gives the following axiom its name.

Axiom 1.4.3 Axiom of choice. *Let $(U_j)_{j\in\mathcal{J}}$ be an indexed family of subsets $U_j \subseteq X$ of a set X. If $U_j \neq \emptyset$ for all indices $j \in \mathcal{J}$, then the Cartesian product*

$$\prod_{j\in\mathcal{J}} U_j \neq \emptyset$$

is also non-empty.

We will take the Axiom of choice as true for the remainder of this text. While seemingly too specific to be of much use, this axiom can be used to prove many useful results. In particular, we can use it to complete the characterization of invertibility given in Theorem 1.2.15 Invertibility.

Corollary 1.4.4 Invertibility. *Let $f\colon X \to Y$ be a map from a set X to a set Y.*

1. *Left-invertibility.*

   f is left-invertible if and only if f is injective or $X = \emptyset$ is empty.

2. *Right-invertibility.*

   f is right-invertible if and only if f is surjective.

3. *Invertibility.*

   f is invertible if and only if f is bijective.

Proof.

1. This follows directly from (1) and (2) of Theorem 1.2.15 Invertibility.

2. The forward implication is (3) of Theorem 1.2.15 Invertibility. Thus we may suppose that f is surjective. Then for each output $y \in Y$, the pre-image

   $$f^{-1}(y) = \{x \in X \mid f(x) = y\} \neq \emptyset$$

   is a non-empty subset of X, and so the Axiom of choice implies that the

Cartesian product

$$\prod_{y \in Y} f^{-1}(y) \neq \emptyset$$

is a non-empty subset of X^Y. Choose an element $g = (x_y)_{y \in Y} \in \prod_{y \in Y} f^{-1}(y)$, and note that for all $y \in Y$, $g(y) = x_y \in f^{-1}(y)$, so that

$$(f \circ g)(y) = f(g(y)) = y = \mathrm{id}_Y(y);$$

that is, $f \circ g = \mathrm{id}_Y$ so that g is a right inverse to f. Hence f is right-invertible.

3. This is (3) of Theorem 1.2.15 Invertibility.

■

One particularly useful kind of Cartesian product is obtained by taking all factors to be the same set. The resulting **Cartesian powers** will be used to define both finitary operations and finitary relations.

Definition 1.4.5 Cartesian power. Let X be a set and $n \in \mathbb{N}$ be a natural number. The nth **Cartesian power** X^n of X is the Cartesian product

$$X^n = \prod_{k=1}^{n} X = \{(x_k)_{k=1}^{n} \mid x_k \in X \text{ for all } 1 \leq k \leq n\}.$$

◆

We will now move on to study the Cartesian powers of sets. Maps from the Cartesian power of a set to the set itself will be called **finitary operations** on that set, and subsets of the Cartesian power of a set will be called **finitary relations** on that set.

1.4.2 Finitary Operations

Informally, a **finitary operation** on a set is an assignment of its elements to all finite sequence of a certain length in that set. That is, a finitary operation on a set is a map from some finite Cartesian power of that set to the set itself. We will view a finitary operation (through the active characterization of what maps do) as a process which combines some fixed number of elements of a set into a new element.

Definition 1.4.6 Finitary operation. Let X be a set and $n \in \mathbb{N}$ be a natural number. A **finitary operation** of **arity** n (or more simply a **n-ary operation**) is a map $f \colon X^n \to X$. ◆

Example 1.4.7 Finitary operations. The following are examples of finitary operations:

- *Nullary operation.*

  A 0-ary operation (or more simply, a **nullary operation**) on a set X is just a map $f \colon \{\emptyset\} \to X$ from $X^0 = \{\emptyset\}$ to X. Such a map $f \colon \{\emptyset\} \to X$ is uniquely determined by the image $f(\emptyset) \in X$ of the unique element $\emptyset \in X^0$. In this way, we may view the nullary operations on X as synonymous with the elements of X.

- *Unary operation.*

  A 1-ary operation (or more simply, a **unary operation**) on a set X is just a map $f \colon X \to X$. One example of such a unary operation on the natural numbers $\mathbb{N}$ is the successor function $\mathrm{succ} \colon \mathbb{N} \to \mathbb{N}$.

- *Binary operation.*

  A 2-ary operation (or more simply, a **binary operation**) on a set X is a map $f\colon X^2 \to X$. Two examples of such binary operations on the natural numbers $\mathbb{N}$ are the addition and multiplication operations $\operatorname{add}, \operatorname{mult}\colon \mathbb{N}^2 \to \mathbb{N}$.

  Binary operations are usually denoted with symbols such as $+$, $\times$, $\cdot$, $*$, and $\star$. For a given binary operation $*\colon X \times X \to X$, we write

  $$x_1 * x_2 = *\left(x_1, x_2\right)$$

  for all elements $x_1, x_2 \in X$.

▲

In many ways, algebra is the study of finitary operations on sets. Particularly emphasized in this study is that of binary operation on sets. We won't delve too deep into the study of binary operations, but we will introduce some nice properties which some binary operations may satisfy.

Definition 1.4.8 Commutativity; associativity. Let $*\colon X^2 \to X$ be a binary operation on a set X.

1. *Commutativity.*

   $*$ is called **commutative** if

   $$x_1 * x_2 = x_2 * x_1$$

   for all elements $x_1, x_2 \in X$.

2. *Associativity.*

   $*$ is called **associative** if

   $$\left(x_1 * x_2\right) * x_3 = x_1 * \left(x_2 * x_3\right)$$

   for all elements $x_1, x_2, x_3 \in X$.

◆

Example 1.4.9 Commutative/associative binary operations. The following are examples of commutativity and associativity in binary operations:

- *Arithmetic of natural numbers.*

  Addition $+\colon \mathbb{N}^2 \to \mathbb{N}$ and multiplication $\cdot\colon \mathbb{N}^2 \to \mathbb{N}$ are commutative and associative binary operations on $\mathbb{N}$ by (1), (2), (4), and (5) of Theorem 1.3.18 Properties of Peano arithmetic.

- *Set operations.*

  Intersection $\cap\colon \mathcal{P}\left(X\right)^2 \to \mathcal{P}\left(X\right)$ and union $\cup\colon \mathcal{P}\left(X\right)^2 \to \mathcal{P}\left(X\right)$ are commutative and associative binary operations on the power set $\mathcal{P}\left(X\right)$ of a set X by (2) and (3) of Corollary 1.1.15 Properties of the set operations.

- *Composition of maps.*

  Composition $\circ\colon \left(X^X\right)^2 \to X^X$ is an associative but non-commutative binary operation on the set X^X of maps from a set X to itself by Proposition 1.2.10 Associativity of composition and Problem 1.2.5.3 Non-commutativity of composition.

- *Exponentiation of natural numbers.*

Let $e\colon \mathbb{N}^2 \to \mathbb{N}$ be the **exponentiation** function defined by the formula $e(m,n) = m^n$; that is, e is defined recursively by the formulae $e(m,0) = 1$ and $e(m,n+1) = e^{m,n} \cdot m$. Then e is a binary operation on $\mathbb{N}$, but e is neither commutative nor associative.

▲

Associative binary operations can be extended recursively to act on finite sequences of any length. For example, given a finite sequence $(n_j)_{j=1}^m$ of natural numbers $n_j \in \mathbb{N}$, we write

$$\sum_{j=1}^m n_j = n_1 + n_2 + \cdots + n_m \qquad\qquad \prod_{j=1}^m n_j = n_1 n_2 \cdots n_m.$$

As in the case of finitary operations, we will introduce finitary relations of all arities, but we will emphasize binary relations in order to study congruences and order relations, which are ubiquitous throughout both classical and modern mathematics and which will recur throughout this text.

1.4.3 Finitary Relations

A **finitary relation** on a set is just a subset of some finite Cartesian power of the set. This subset is usually interpreted as distinguishing its elements certain finite sequences of elements of the set as important in some way.

Definition 1.4.10 Finitary relation. Let X be a set and $n \in \mathbb{N}$ be a natural number. A **finitary relation** of **arity** n (or more simply a n**-ary relation**) is a subset $R \subseteq X^n$. ◆

We will view a finitary relation on a set as elevating specific finite sequences of elements of that set as distinguished.

Example 1.4.11 Finitary relations. The following are examples of finitary relations:

- *Unary relation.*

  In light of Convention 1.4.2 Empty and singleton products that $X^1 = X$, a 1-ary relation (or more simply, a **unary relation**) on a set X is just a subset of X.

- *Binary relation.*

  A 2-ary relation (or more simply, a **binary relation**) on a set X is a subset $R \subseteq X^2$. Two examples of such binary relations are the containment relation $\subseteq$ on the power set $\mathcal{P}(X)$ of a set X and the standard ordering $\leq$ of the natural numbers $\mathbb{N}$.

  Just as there is a special notation for binary operations, there is a special relation for binary relations as well. Specifically, for a given binary relation $R \subseteq X^2$ on a set X, we write $x_1 \mathrel{R} x_2$ to mean that $(x_1, x_2) \in R$.

▲

Definition 1.4.12 Pullback of a finitary relation. Let X and Y be sets and $n \in \mathbb{N}$ be a natural number. The **pullback** R_f of an n-ary relation R on Y along a map $f\colon X \to Y$ is the n-ary relation on X defined by

$$R_f = \left\{ (x_k)_{k=1}^n \in X^n \mid (f(x_k))_{k=1}^n \in R \right\}.$$

◆

Example 1.4.13 Restriction of a finitary relation. Let X be a set and $n \in \mathbb{N}$ be a natural number. The **restriction** $R|_U$ of an n-ary relation R on X to a subset $U \subseteq X$ is the pullback of R along the inclusion map $\iota_U^X : U \to X$; that is,

$$\begin{aligned} R|_U &= R_{\iota_U^X} = \left\{ (u_k)_{k=1}^n \in U^n \;\middle|\; \left(\iota_U^X (u_k)\right)_{k=1}^n \in R \right\} = R \cap U^n \\ &= \{(x_k)_{k=1}^n \in X^n \mid x_1, \ldots, x_n \in U \text{ and } (x_k)_{k=1}^n \in R\} = R \cap U^n. \end{aligned}$$

▲

One particularly nice type of binary relation is the **equivalence relation**. This is a generalization of the notion of equality, which has many of the salient properties of the equality relation. We will see that an equivalence relation partitions a set into disjoint subsets known as **equivalence classes**. When we consider more sophisticated structures than just sets, we will use equivalence relations to construct **quotient** objects, as is done in algebraic settings such as group theory or ring theory.

Definition 1.4.14 Equivalence relation. Let $\equiv$ be a binary relation on a set X.

1. *Reflexivity.*

   The binary relation $\equiv$ is called **reflexive** if $x \equiv x$ for all elements $x \in X$.

2. *Symmetry.*

   The binary relation $\equiv$ is called **symmetric** if for all elements $x_1, x_2 \in X$, if $x_1 \equiv x_2$, then $x_2 \equiv x_1$.

3. *Transitivity.*

   The binary relation $\equiv$ is called **transitive** if for all elements $x_1, x_2, x_3 \in X$, if $x_1 \equiv x_2$ and $x_2 \equiv x_3$, then $x_1 \equiv x_3$.

A binary relation $\equiv$ is called an **equivalence relation** if it is reflexive, symmetric, and transitive. ◆

Lemma 1.4.15 Pullbacks of equivalence relations. *Let $f : X \to Y$ be a map from a set X to a set Y, and consider a binary relation $\equiv$ on Y.*

1. *Reflexivity.*

   If $\equiv$ is reflexive, then its pullback $\equiv_f$ along f is reflexive.

2. *Symmetry.*

   If $\equiv$ is symmetric, then its pullback $\equiv_f$ along f is symmetric.

3. *Transitivity.*

   If $\equiv$ is transitive, then its pullback $\equiv_f$ along f is transitive.

In particular, if $\equiv$ is an equivalence relation on Y, then its pullback $\equiv_f$ along f is an equivalence relation on X.

Proof.

1. Let $x \in X$, and note that $f(x) \equiv f(x)$ by the reflexivity of $\equiv$, and so $x \equiv_f x$; that is, $\equiv_f$ is reflexive.

2. Let $x_1, x_2 \in X$, and suppose that $x_1 \equiv_f x_2$. Then $f(x_1) \equiv f(x_2)$, and so $f(x_2) \equiv f(x_1)$ by the symmetry of $\equiv$. In particular, $x_2 \equiv_f x_1$, and so $\equiv_f$ is symmetric.

3. Let $x_1, x_2, x_3 \in X$, and suppose that $x_1 \equiv_f x_2$ and $x_2 \equiv_f x_3$. Then $f(x_1) \equiv f(x_2)$ and $f(x_2) \equiv f(x_3)$, and so $f(x_1) \equiv f(x_3)$ by the transitivity of $\equiv$. In particular, $x_1 \equiv_f x_3$, and so $\equiv_f$ is transitive.

■

One of the most important properties of equivalence relations is that they partition sets into disjoint sets called **equivalence classes**.

Definition 1.4.16 Equivalence class; quotient set. Let $\equiv$ be an equivalence relation on a set X.

1. *Equivalence class.*

   The **equivalence class** $[x]_\equiv$ of an element $x \in X$ is the subset of X defined by
   $$[x]_\equiv = \{y \in X \mid x \equiv y\};$$
   that is, $[x]_\equiv$ is the set of elements which are equivalent to x under the relation $\equiv$.

2. *Quotient.*

   The **quotient** $X/\equiv$ of X by $\equiv$ is the set of equivalence classes; that is,
   $$X/\equiv = \{[x]_\equiv \in \mathcal{P}(X) \mid x \in X\}.$$
   The **quotient map** $\pi_\equiv : X \to X/\equiv$ is the map from X to the quotient $X/\sim$ defined by the formula $\pi_\equiv(x) = [x]_\equiv$.

   If the set X is equipped with additional structure, then this structure might also descend to the quotient $X/\equiv$. In settings in which this is possible, we might distinguish between the **quotient set** $X/\equiv$ and more structured quotient objects.

◆

As mentioned above, an equivalence relation on a given set partitions that set into pairwise disjoint equivalence classes.

Proposition 1.4.17 Equivalence partition. *The equivalence classes of an equivalence relation $\equiv$ on a set X form a **partition** of X; that is, two equivalence classes are either equal or disjoint, and*
$$X = \bigcup_{c \in X/\equiv} c.$$

Proof. Each equivalence class $c \in X/\equiv$ is a subset of X, and so
$$\bigcup_{c \in X/\equiv} c \subseteq X.$$
In order to show the reverse containment, let $x \in X$, and note that $x \equiv x$ by reflexivity; that is, $x \in [x]_\equiv$, and so
$$x \in [x]_\equiv = \bigcup_{c \in X/\equiv} c.$$
We conclude that
$$X = \bigcup_{c \in X/\equiv} c.$$

It remains to show that two equivalence classes are either equal or disjoint. To that end, consider elements $x_1, x_2 \in X$. First suppose that $x_1 \equiv x_2$, and let $x \in [x_2]_{\equiv}$. Then $x_2 \equiv x$, and so transitivity implies that $x_1 \equiv x$; that is, $x \in [x_1]_{\equiv}$.

In summary, we have shown that if $x_1 \equiv x_2$, then $[x_2]_{\equiv} \subseteq [x_1]_{\equiv}$. Symmetry implies that if $x_1 \equiv x_2$, then $x_2 \equiv x_1$, and so $[x_1]_{\equiv} \subseteq [x_2]_{\equiv}$; that is, if $x_1 \equiv x_2$, then $[x_1]_{\equiv} = [x_2]_{\equiv}$.

Conversely, now suppose that the equivalence classes $[x_1]_{\equiv}$ and $[x_2]_{\equiv}$ are not disjoint, so that $x \in [x_1]_{\equiv} \cap [x_2]_{\equiv}$ for some element $x \in X$. Since $x \in [x_1]_{\equiv}$ and $x \in [x_2]_{\equiv}$, $x \equiv x_1$ and $x \equiv x_2$. We conclude by symmetry and transitivity that $x_1 \equiv x_2$. ■

Equivalence relations are ubiquitous throughout mathematics, and we will work with them for the remainder of this text.

While this section has dealt with finitary operations and relations in broad generality, the vast majority of mathematics relevant to this text deals with *binary* operations and relations. In the next section, we will see another use for binary relations; that of **ordering** sets.

1.4.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

1. **Indexed Cartesian products.** Let $U_1 = \{1, 2\}$, $U_2 = \{1, 2, 3\}$, and $U_3 = \{0, 1\}$, and consider the Cartesian product $U = \prod_{k=1}^{3} U_k$.

   (a) Write down the Cartesian product $U = \prod_{k=1}^{3} U_k$ in roster notation.

   (b) For each index $k \in \{1, 2, 3\}$, let $\pi_k \colon U \to U_k$ be the projection map onto the kth factor. Compute $\pi_1(2, 3, 0)$, $\pi_2(1, 1, 1)$, and $\pi_3(2, 2, 0)$.

2. **Exponentiation of natural numbers.** Let $e \colon \mathbb{N}^2 \to \mathbb{N}$ be the **exponentiation** operation defined by the formula

$$e(m, n) = m^n;$$

   that is, e is defined recursively by the formulae $e(m, 0) = 1$ and $e(m, n+1) = e^{m,n} \cdot m$.

   (a) Compute $e(1, 2)$ and $e(2, 1)$. Is exponentiation a commutative binary operation?

   (b) Compute $e(e(2, 1), 2)$ and $e(2, e(1, 2))$. Is exponentiation an associative binary operation?

Equivalence classes. Let $X = \{0, 1, 2, 3, \ldots, 10\}$. Write down all the equivalence classes for the given equivalence relation on X in roster notation.

3. Consider the equivalence relation F on X defined by the condition that for all $x, y \in X$, $x \mathrel{F} y$ if and only if the English words for x and y start with the same letter. Write down the equivalence classes in roster notation.

4. Consider the equivalence relation L on X defined by the condition that for all $x, y \in X$, $x \; L \; y$ if and only if the English words for x and y end with the same letter. Write down all the equivalence classes in roster notation.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.4.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Revisiting commutativity/associativity of Cartesian products.** Recall from Problem 1.2.5.1 Non-commutativity of Cartesian products that there are sets X and Y so that $X \times Y \neq Y \times X$. Similarly, there are sets X, Y, and Z so that $(X \times Y) \times Z \neq X \times (Y \times Z)$.

   (a) Prove that there is a bijection between $X \times Y$ and $Y \times X$.

   (b) Prove that there is a bijection between $(X \times Y) \times Z$ and $X \times (Y \times Z)$.

   While the sets $X \times Y$ and $Y \times X$ and the sets $(X \times Y) \times Z$ and $X \times (Y \times Z)$ are not equal, they may safely be identified via these bijections.

2. **Cross products.** The **cross product** $\times \colon \left(\mathbb{R}^3\right)^2 \to \mathbb{R}^3$ is the binary operation on 3-dimensional real coordinate space

$$\mathbb{R}^3 = \{(x_1, x_2, x_3) \mid x_1, x_2, x_3 \in \mathbb{R}\}$$

   defined by the formula

$$(x_1, x_2, x_3) \times (y_1, y_2, y_3) = (x_2 y_3 - x_3 y_2, x_3 y_1 - x_1 y_3, x_1 y_2 - x_2 y_1)$$

   For this problem, you may assume all standard properties of the arithmetic of real numbers, even though we will not formally define $\mathbb{R}$ until Section 2.2 Constructing the Real Numbers.

   (a) Determine whether or not the cross product $\times$ on $\mathbb{R}^3$ is commutative. Justify your answer.

   (b) Determine whether or not the cross product $\times$ on $\mathbb{R}^3$ is associative. Justify your answer.

3. **Generating equivalence relations.** Let X be a set.

   (a) Prove that the intersection $R = \bigcap_{j \in \mathcal{J}} R_j$ of any indexed family $(R_j)_{j \in \mathcal{J}}$ of equivalence relations R_j on X is an equivalence relation on X.

   (b) Prove that for any binary relation R on X, there is a unique equivalence relation S on X which satisfies the following conditions:

(a) S is a **refinement** of R; that is, $R \subseteq S$.

(b) S is **coarser** than any equivalence relation which **refines** R; that is, $S \subseteq T$ for any equivalence relation T on X so that $R \subseteq T$.

The unique equivalence relation on X constructed in (b) above is called the equivalence relation **generated** by R.

4. **Universal property of quotients.** Let $\equiv$ be an equivalence relation on a set X. A map $f\colon X \to Y$ from X to a set Y is called **$\equiv$-invariant** if for all elements $x_1, x_2 \in X$, if $x_1 \equiv x_2$, then $f(x_1) = f(x_2)$.

Prove that for every $\equiv$-invariant map $f\colon X \to Y$, there is a map $\overline{f}\colon X/{\equiv} \to Y$ so that $\overline{f} \circ \pi_{\equiv} = f$.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.5 Introduction to Order Theory

In this section, we will investigate another class of binary relations, which can be used to establish a notion of an **ordering** on a set. Touchstone examples will be the standard ordering $\leq$ of the natural numbers, which is a **total order**, and set containment $\subseteq$, which generally is only a **partial order**.

1.5.1 Preorders and Partial Orders

We have seen that a finitary relation on a set can be viewed as elevating its elements as distinguished finite sequences in that set. Some binary relations on a set may be interpreted in this way as establishing an **ordering**, with the first entries of each distinguished ordered pair preceding the second entries in this ordering.

What are the salient properties of the standard ordering of the natural numbers? We will see that, in order to establish a similar notion of ordering on a set, a given binary operation should satisfy similar properties. However, even if a binary operation satisfies some, but not all, of the desired properties, we will see that we can still interpret it as a weaker type of ordering relation. However, all ordering relations should, at the very least, satisfy reflexivity and transitivity. Such a reflexive and transitive binary operation is called a **preorder**.

Definition 1.5.1 Preorder. A binary relation $\precsim$ on a set X is called a **preorder** if it is reflexive and transitive; that is, $\precsim$ is a preorder if it satisfies the following conditions:

- *Reflexivity.*

  $x \precsim x$ for all elements $x \in X$.

- *Transitivity.*

  For all elements $x_1, x_2, x_3 \in X$, if $x_1 \precsim x_2$ and $x_2 \precsim x_3$, then $x_1 \precsim x_3$.

If $\precsim$ is a preorder on a set X, then the ordered pair $(X, \precsim)$ is called a **preordered set** (or a **proset**). In this case, one may refer to X as the **ground set** of the preordered set $(X, \precsim)$. ◆

Lemma 1.5.2 Preorders from pullbacks. *The pullback $\leq_f$ of the standard ordering $\leq$ on the natural numbers $\mathbb{N}$ along a function $f \colon X \to \mathbb{N}$ on a set X is a preorder on X.*

Proof. First let $x \in X$, and note that $f(x) \leq f(x)$ by (1) of Proposition 1.3.15 Comparison is a total order. In particular, $x \leq_f x$, and so $\leq_f$ is reflexive.

Now let $x_1, x_2, x_3 \in X$, and suppose that $x_1 \leq_f x_2$ and $x_2 \leq_f x_3$. Then $f(x_1) \leq f(x_2)$ and $f(x_2) \leq f(x_3)$, and so $f(x_1) \leq f(x_3)$ by (3) of Proposition 1.3.15 Comparison is a total order. In particular, $x_1 \leq_f x_3$, and so $\leq_f$ is transitive. Since $\leq_f$ is both reflexive and transitive, it is a preorder on X. ■

It turns out that, at least relative to the standard ordering $\leq$ of the natural numbers $\mathbb{N}$, the defining properties of a preorder are still too weak to be of much use in the general case. After all, an equivalence relation is just a symmetric preorder, and there is no way of meaningfully assigning a notion of an ordering based on an equivalence relation. This is because an equivalence relation is symmetric; in order to consider an ordering based on a binary operation, we would want to avoid any symmetry at all. Therefore, we must require that an ordering relation satisfy stronger assumptions.

Definition 1.5.3 Partial order. A binary relation $\preceq$ on a set X is called a **partial order** if it satisfies the following conditions:

- *Reflexivity.*

  $x \preceq x$ for all elements $x \in X$.

- *Anti-symmetry.*

  For all elements $x_1, x_2 \in X$, if $x_1 \preceq x_2$ and $x_2 \preceq x_1$, then $x_1 = x_2$.

- *Transitivity.*

  For all elements $x_1, x_2, x_3 \in X$, if $x_1 \preceq x_2$ and $x_2 \preceq x_3$, then $x_1 \preceq x_3$.

If $\preceq$ is a partial order on a set X, then the ordered pair $(X, \preceq)$ is called a **partially ordered set** (or a **poset**). In this case, one may refer to X as the **ground set** of the partially ordered set $(X, \preceq)$. ◆

Example 1.5.4 Partial orders. The following are examples of partial orders on sets:

- *Standard ordering on the natural numbers.*

  The standard ordering $\leq$ on the natural numbers $\mathbb{N}$ is a partial order on $\mathbb{N}$ by (1), (2), and (3) of Proposition 1.3.15 Comparison is a total order.

- *Set containment.*

  Containment $\subseteq$ is a partial order on the power set $\mathcal{P}(X)$ of a set X by Definition 1.1.7 Set equality and Lemma 1.1.9 Transitivity of set containment.

▲

At the end of this section, we will explore a property of partial orders called **totality** which distinguishes between our touchstone examples of the standard ordering $\leq$ of natural numbers and the containment $\subseteq$ of sets.

1.5.2 Boundedness and Extrema

Partial orders satisfy many of the properties of the standard ordering of the natural numbers. We may use these properties for inspiration in generalizing the definitions of many of the usual notions associated with an ordering, such as upper and lower bounds and maximal and minimal elements, to preordered sets. However, we will see that anti-symmetry (and sometimes even stronger assumptions) is needed in order to recover many of these desired results.

Definition 1.5.5 Boundedness. Let $U \subseteq X$ be a subset of a preordered ordered set $(X, \precsim)$.

- *Upper bound.*

  An element $x \in X$ is called an **upper bound** for U if $u \precsim x$ for all elements $u \in U$. If such an upper bound for U exists, then U is called **bounded from above** in X. Otherwise, if no such upper bound for U exists, then U is called **unbounded from above**.

- *Lower bound.*

  An element $x \in X$ is called a **lower bound** for U if $x \precsim u$ for all elements $u \in U$. If such a lower bound for U exists, then U is called **bounded from below** in X. Otherwise, if no such upper bound for U exists, then U is called **unbounded from below**.

A subset $U \subseteq X$ is called **bounded** in X if it is both bounded from above in X and bounded from below in X, and U is called **unbounded** in X otherwise if it is either unbounded from above in X or unbounded from below in X. ◆

Lemma 1.5.6 Unique extrema. *Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$. At most one element of U is an upper bound for U in X, and at most one element of U is a lower bound for U in X.*

Proof. Let $u_1, u_2 \in U$, and suppose first that u_1 and u_2 are upper bounds for U in X, so that $u \preceq u_1$ and $u \preceq u_2$ for all elements $u \in U$. In particular, $u_1 \preceq u_2$ and $u_2 \preceq u_1$, and so $u_1 = u_2$ by anti-symmetry.

Similarly, now suppose that u_1 and u_2 are lower bounds for U in X, so that $u_1 \preceq u$ and $u_2 \preceq u$ for all elements $u \in U$. In particular, $u_1 \preceq u_2$ and $u_2 \preceq u_1$, and so $u_1 = u_2$ by anti-symmetry. ■

We observe that the above proof of Lemma 1.5.6 Unique extrema is strongly dependent on the anti-symmetry of the partial order in question. Indeed, there is no analogous result for preordered sets.

Definition 1.5.7 Extrema. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

- *Maximum.*

  If there is an element of U which is an upper bound for U in X, then this unique element is called the **maximum** of U and is denoted $\max(U)$.

- *Minimum.*

  If there is an element of U which is a lower bound for U in X, then this unique element is called the **minimum** of U and is denoted $\min(U)$.

Together, maxima and minima are called **extrema**. ◆

The uniqueness of extrema in partially ordered sets is one way in which partial orders resemble the standard ordering of the natural number system. However, unlike in the standard ordering of the natural number system, there is in general a distinction between extrema, which are defined above, and **extremal elements**, which are defined below.

Definition 1.5.8 Extremal element. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

- *Maximal element.*

  An element $u \in U$ is called **maximal** in U if for all elements $v \in U$, if $u \preceq v$, then $v = u$.

- *Minimal element.*

  An element $u \in U$ is called **minimal** in U if for all elements $v \in U$, if $v \preceq u$, then $v = u$.

Together, maximal and minimal elements are called **extremal elements**. ◆

Example 1.5.9 Extrema and extremal elements. The following are examples of extrema and extremal elements:

- *Standard ordering on the natural numbers.*

  The minimum of the natural numbers $\mathbb{N}$ is $\min(\mathbb{N}) = 0$, which is the unique minimal element. However, $\mathbb{N}$ has no maximum or maximal element.

- *Set containment.*

  Consider set containment $\subseteq$ on the power set $\mathcal{P}(X)$ of a set X. $\mathcal{P}(X)$ has minimum and maximum $\min(\mathcal{P}(X)) = \emptyset$ and $\max(\mathcal{P}(X)) = X$, which are also the unique minimal and maximal elements, respectively.

- Now consider set containment $\subseteq$ on the set X of proper non-empty subsets of the set $\{1, 2, 3\}$; that is,

  $$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}.$$

  Then X has no maximum or minimum, but $\{1\}$, $\{2\}$, and $\{3\}$ are minimal elements, and $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3\}$ are maximal elements.

▲

Proposition 1.5.10 Extrema are extremal elements. *Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$. If U has a maximum, then $\max(U)$ is the unique maximal element of U. If U has a minimum, then $\min(U)$ is the unique minimal element of U.*

Proof. First suppose that U has a maximum. Let $u \in U$, and suppose that $\max(U) \preceq u$. Of course $u \preceq \max(U)$, and so anti-symmetry implies that $u = \max(U)$; that is, $\max(U)$ is maximal in U.

On the other hand, now consider a maximal element $u \in U$. Since $u \preceq \max(U)$, the maximality of u implies that $u = \max(U)$.

Now suppose that U has a minimum. Let $u \in U$, and suppose that $u \preceq \min(U)$. Of course $\min(U) \preceq u$, and so anti-symmetry implies that $u = \min(U)$; that is, $\min(U)$ is minimal in U.

On the other hand, now consider a minimal element $u \in U$. Since $\min(U) \preceq u$, the minimality of u implies that $u = \min(U)$. ■

The distinction between extrema and extremal elements is one way in which more general partial orders differ from the standard ordering of the natural numbers. However, this subtlety does not arise in all partial orders. We will shortly explore a class of partial orders called **total orders** for which the aforementioned distinction between extrema and extremal elements does not exist.

1.5.3 Comparability and Total Orders

The fact that the converse of Proposition 1.5.10 Extrema are extremal elements does not hold in general stems from the fact that in a partially ordered set, not all pairs of elements are comparable. That is, given two elements of a partially ordered set, it is not necessarily true that one will precede the other.

Definition 1.5.11 Comparability. Two elements $x_1, x_2 \in X$ in a partially ordered set $(X, \preceq)$ are called **comparable** if either $x_1 \preceq x_2$ or $x_2 \preceq x_1$, and **incomparable** otherwise. ◆

Any two natural numbers are comparable in the standard ordering. However, this is not true for partially ordered sets in general. As an example, consider the subsets $S = \{0, 1\}$ and $T = \{1, 2\}$ of the natural numbers $\mathbb{N}$. One can check that $S \not\subseteq T$ and $T \not\subseteq S$, so that S and T are incomparable in the power set $\mathcal{P}(\mathbb{N})$ of the natural numbers $\mathbb{N}$. So partially ordered sets may have elements which are incomparable. This is the reason why partial orders are called “partial”; partial orders in which all elements are comparable are called **total**.

Definition 1.5.12 Total order. A binary relation $\leq$ on a set X is called a **total order** if it is anti-symmetric and transitive and each pair of elements is comparable; that is, $\leq$ is a total order if it satisfies the following conditions:

- *Antisymmetry.*

  For all elements $x_1, x_2 \in X$, if $x_1 \leq x_2$ and $x_2 \leq x_1$, then $x_1 = x_2$.

- *Transitivity.*

  For all elements $x_1, x_2, x_3 \in X$, if $x_1 \leq x_2$ and $x_2 \leq x_3$, then $x_1 \leq x_3$.

- *Totality.*

  For all elements $x_1, x_2 \in X$, either $x_1 \leq x_2$ or $x_2 \leq x_1$.

If $\leq$ is a total order on a set X, then the ordered pair $(X, \leq)$ is called a **totally ordered set**. ◆

Example 1.5.13 Total order. The standard ordering $\leq$ on the natural numbers $\mathbb{N}$ is a total order by (2), (3), and (4) of Proposition 1.3.15 Comparison is a total order. ▲

Lemma 1.5.14 Total orders are partial orders. *Every total order on a set X is a partial order on X.*

Proof. Let $\leq$ be a total order on a set X. Since $\leq$ is antisymmetric and transitive, it remains to show that $\leq$ is reflexive. To that end, let $x \in X$, and note that totality implies that either $x \leq x$ or $x \leq x$; that is $\leq$ is reflexive (and hence a partial order on X). ■

The careful reader might object that the above definition of total orders neglected to require reflexivity. In fact, this property follows from the other properties of total orders.

Proposition 1.5.15 Extremal elements are extrema in totally ordered sets. *Let $\leq$ be a total order on a set X.*

1. *If X contains a maximal element $e \in X$, then $e = \max(X)$.*

2. *If X contains a minimal element $e \in X$, then $e = \min(X)$.*

Proof.

1. Suppose that $e \in X$ is a maximal element, and let $x \in X$. Totality implies that $x \leq e$ or $e \leq x$. If $e \leq x$, then maximality implies that $x = e$, so that $x \leq e$ by reflexivity. Thus $e \leq x$ for all $x \in X$, and so $\max(X) = e$.

2. Now suppose that $e \in X$ is a minimal element, and let $x \in X$. Totality implies that $x \leq e$ or $e \leq x$. If $x \leq e$, then minimality implies that $x = e$, so that $e \leq x$ by reflexivity. Thus $x \leq e$ for all $x \in X$, and so $\min(X) = e$.

■

We end with an important property of the standard ordering $\leq$ on the natural numbers $\mathbb{N}$ related to the axiom of induction.

Theorem 1.5.16 Well-ordering principle. *Every non-empty subset of the natural numbers $\mathbb{N}$ has a minimum.*

Proof. Let $U \subseteq \mathbb{N}$ be a subset of the natural numbers $\mathbb{N}$, and suppose that U does not have a minimum. We proceed by strong induction on $n \in \mathbb{N}$ to show that $n \notin U$.

Base case. Suppose that $n = 0$, and suppose for a contradiction that $n \in U$. Since $0 \leq n$ for all natural numbers $n \in \mathbb{N}$ by Lemma 1.3.2 0 is minimal and (2) of Proposition 1.5.15 Extremal elements are extrema in totally ordered sets, then $n = \min(U)$. This contradiction implies that $n \notin U$.

Inductive step. Suppose for the inductive hypothesis that there is some natural number $n \in \mathbb{N}$ so that $k \notin U$ for all natural numbers $k \in \mathbb{N}$ so that $0 \leq k \leq n$. Suppose for a contradiction that $n + 1 \in U$, and let $u \in U$. Totality implies that either $u \leq n$ or $n \leq u$. The inductive hypothesis implies that $u \nleq n$, so that $n \leq u$.

So $u = n + p$ for some natural number $p \in \mathbb{N}$. If $p = 0$, then

$$n = n + 0 = n + p = u \in U,$$

and so $p \neq 0$. Thus $p = q + 1$ for some natural number $q \in \mathbb{N}$, and so

$$n + 1 \leq n + q + 1 = n + p = u.$$

In summary, we have shown that $n+1 \leq u$ for all $u \in U$, so that $\min(U) = n+1$. This contradiction implies that $n + 1 \notin U$.

We conclude that $n \notin U$ for all natural numbers $n \in \mathbb{N}$, and so $U = \emptyset$. ■

The Well-ordering principle is sometimes taken to be an axiom and sometimes derived from the axiom of induction as in the above proof. In fact, these results are logically equivalent, as one can derive the axiom of induction from the Well-ordering principle.

We will revisit ordering relations in Chapter 2 Constructing the Number Systems, when we extend the standard ordering on the natural numbers to many of the extensions of the natural number system which we construct.

1.5.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Identifying partial and total orders. Determine whether or not the given preorder is a partial order. If the preorder is a partial order, determine whether or not it is total.

1. Let R be the preorder on the set $\{1, 2, 3, 4\}$ defined by

$$\{(4, 2), (4, 3), (2, 1), (3, 1), (4, 1), (1, 3), (2, 3), (4, 4), (3, 3), (2, 2), (1, 1)\}.$$

Determine whether or not R is a partial order on $\{1, 2, 3, 4\}$. If so, determine whether or not it is total.

2. Let $\precsim$ be the preorder on the set P of all people (living or dead) defined so that for all people $p, q \in P$, $p \precsim q$ if and only if either p is either q or an ancestor of q. Determine whether or not $\precsim$ is a partial order on P. If so, determine whether or not it is total.

3. Let $\precsim$ be the preorder on the set C of all circles in the Cartesian plane so that for all circles $c, d \in C$, $c \precsim d$ if and only if the diameter of c is at most the diameter of d. Determine whether or not $\precsim$ is a partial order on C. If so, determine whether or not it is total.

Bounds. Determine whether or not the following subsets of the partially ordered set $\mathcal{P}(\mathbb{N})$ are bounds for the given element.

4. Which of the following are upper bounds for the set $\{1, 2, 3\}$ in the partially ordered set $\mathcal{P}(\mathbb{N})$?

A. $\emptyset$

B. $\{1, 2\}$

C. $\{1, 2, 3\}$

D. $\mathbb{N} \setminus \{1, 2\}$

E. $\mathbb{N} \setminus \{0\}$

F. $\mathbb{N}$

5. Which of the following are lower bounds for the set $\{1, 2, 3\}$ in the partially ordered set $\mathcal{P}(\mathbb{N})$?

A. $\emptyset$

B. $\{1, 2\}$

C. $\{1, 2, 3\}$

D. $\mathbb{N} \setminus \{1, 2\}$

E. $\mathbb{N} \setminus \{0\}$

F. $\mathbb{N}$

Boundedness. Determine the boundedness of the following subsets of the totally ordered set $\mathbb{N}$.

6. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded from below?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

7. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded from above?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

8. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.5.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Order by divisibility.** Consider the binary relation $\mid$ on the natural numbers $\mathbb{N}$ defined so that for all natural numbers $m, n \in \mathbb{N}$, $m \mid n$ if and only if $mk = n$ for some natural number $k \in \mathbb{N}$; that is, $m \mid n$ if and only if n is a multiple of m.

(a) Prove that $\mid$ is a partial order on the set $\mathbb{N}$ of natural numbers.

(b) Is the binary relation $\mid$ a total order on $\mathbb{N}$? Justify your answer.

(c) What are the extrema of the partially ordered set $(\mathbb{N}, |)$?

(d) We will abuse notation and denote the restriction of $|$ to the set $\mathbb{N} \setminus \{0, 1\}$ by $|$ as well. What are the extremal elements of the partially ordered set $(\mathbb{N} \setminus \{0, 1\}, |)$? Are these extremal elements extrema?

2. Extremal subsets. Find the extremal elements of the partially ordered set $\mathcal{P}(\mathbb{N}) \setminus \{\emptyset, \mathbb{N}\}$.

3. Opposite of an order. The **opposite** R^{op} of a binary relation R on a set X is the binary relation defined by

$$R^{\mathrm{op}} = \left\{(x_2, x_1) \in X^2 \,\middle|\, (x_1, x_2) \in R\right\}.$$

(a) Prove that the opposite of a preorder is a preorder.

(b) Prove that the opposite of a partial order is a partial order.

(c) Prove that the opposite of a total order is a total order.

4. Orders from pullbacks.

(a) Prove that the pullback of a preorder is a preorder.

Hint. You may find it useful to review the proof of Lemma 1.5.2 Preorders from pullbacks

(b) Prove that the pullback $\preceq_f$ of a partial order $\preceq$ on a set Y along a map $f\colon X \to Y$ is a partial order on X if and only if f is injective.

(c) Is the pullback of a total order along an injective map necessarily a total order? If so, prove it. If not, provide a counterexample.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.6 Cardinality and Countability [SKIP]

In this section, we define a notion of the size of a set called **cardinality**. The cardinality of a set is a generalization of the number of elements of a set which contains only finitely many elements based on the notion of bijections. First, we will develop these notions for finite sets and then extend them to infinite sets. At first glance, it may seem that there is essentially only one kind of infinite set. We will show that this is not the case; in fact, there are infinitely many "sizes" of infinite sets with respect to this notion of cardinality.

1.6.1 Finiteness

Before we brave the unintuitive mess which is the realm of infinite sets, we will rigorously develop a notion of size for sets which contain only finitely many elements. Like in the infinite case, these finite cardinalities will be based on bijections.

Proposition 1.6.1 Injections of finite sets. *For any natural numbers $m, n \in \mathbb{N}$, there exists an injective map $\{1, \ldots, m\} \to \{1, \ldots, n\}$ if and only if $m \leq n$.*

Proof. If $m \leq n$, then $\{1, \ldots, m\} \subseteq \{1, \ldots, n\}$, and so the inclusion map $\iota_{\{1,\ldots,m\}}^{\{1,\ldots,n\}}\colon \{1, \ldots, m\} \to \{1, \ldots, n\}$ is an injective map from $\{1, \ldots, m\}$ to $\{1, \ldots, n\}$. So it suffices to show the reverse implication.

To that end, consider the subset

$$S = \{m \in \mathbb{N} \mid \text{There are } n \in \mathbb{N} \text{ and } f\colon \{1, \ldots, m\} \to \{1, \ldots, n\} \text{ so that } m \not\leq n \text{ and } f \text{ is injective}\}$$

of the natural numbers $\mathbb{N}$. If suffices to show that $S = \emptyset$. To that end, we first note that $0 \leq n$ for all natural numbers $n \in \mathbb{N}$, so that $0 \notin S$.

Now let $m \in \mathbb{N}$ be a natural number, and suppose that $m + 1 \in S$. Then there is a natural number $n \in \mathbb{N}$ so that $m + 1 \not\leq n$ and an injective function $f\colon \{1, \ldots, m + 1\} \to \{1, \ldots, n\}$. We note that $n \neq 0$, since there are no functions from a non-empty set $\{1, \ldots, m + 1\}$ to the empty set $\emptyset$. Thus $n = p + 1$ for some natural number $p \in \mathbb{N}$.

If $m \leq p$, then $m + 1 \leq p + 1 = n$, and so $m \not\leq p$. It therefore suffices to construct an injective function $\{1, \ldots, m\} \to \{1, \ldots, p\}$. To that end, fix a bijection $g\colon \{1, \ldots, p + 1\} \to \{1, \ldots, p + 1\}$ so that $g\left(f\left(m + 1\right)\right) = p + 1$, and let $h\colon \{1, \ldots, m\} \to \{1, \ldots, p\}$ be the function defined by the formula

$$h\left(j\right) = g\left(f\left(j\right)\right).$$

Since g and f are injective, h is injective by (a) of `[provisional cross-reference: problem-compositions-of-injections-surjections-and-bijections]`. We conclude that $n \in S$.

In summary, we have shown that $0 \notin S$ and for all natural numbers $n \in \mathbb{N}$, if $n + 1 \in S$, then $n \in S$. These conclusions are equivalent to $0 \in \mathbb{N} \setminus S$ and for all natural numbers $n \in \mathbb{N}$, if $n \in \mathbb{N} \setminus S$, then $n + 1 \in \mathbb{N} \setminus S$. We conclude that $\mathbb{N} \setminus S = \mathbb{N}$ by the axiom of induction, and so $S = \emptyset$. ■

Corollary 1.6.2 Finite cardinality is well-defined. *For any set X, if there are natural numbers $m, n \in \mathbb{N}$ and bijections $f\colon \{1, \ldots, m\} \to X$ and $g\colon \{1, \ldots, n\} \to X$, then $m = n$.*

Proof. Under the given assumptions, $g^{-1} \circ f\colon \{1, \ldots, m\} \to \{1, \ldots, n\}$ and $f^{-1} \circ g\colon \{1, \ldots, n\} \to \{1, \ldots, m\}$ are injective functions by (a) of `[provisional cross-reference: problem-compositions-of-injections-surjections-and-bijections]`. Proposition 1.6.1 Injections of finite sets now implies that $m \leq n$ and $n \leq m$, and so $m = n$ by (2) of Proposition 1.3.15 Comparison is a total order. ■

Definition 1.6.3 Finiteness; cardinality. A set X is called **finite** if there is a natural number $n \in \mathbb{N}$ and a bijection $\{1, \ldots, n\} \to X$, in which case n is uniquely defined by this property and is called the **cardinality** of X and denoted $|X| = n$. A set X is called **infinite** if no such number and bijection exist. ◆

Proposition 1.6.4 Maps and cardinality. *Let $f\colon X \to Y$ be a map from a set X to a set Y.*

1. *If f is injective and Y is finite, then X is finite and $|X| \leq |Y|$.*
2. *If f is surjective and X is finite, then Y is finite and $|Y| \leq |X|$.*
3. *If f is bijective and either X or Y is finite, then both X and Y are finite and $|X| = |Y|$.*

Proof. ■

Text after the last subdivision.

1.6.2 Cardinal Comparison

Text before the first subdivision.

Text after the last subdivision.

1.6.3 Countability

Text before the first subdivision.

Text after the last subdivision.

Text after the last subsection.

1.6.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.6.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these reading questions are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

Armed with a solid understanding of the theory of sets, maps, and relations, we now have much of the necessary background to begin studying real analysis. First, however, we will construct the standard number systems (with a focus on the real numbers $\mathbb{R}$) as extensions of the natural numbers $\mathbb{N}$ and catalog their properties.

Chapter 2

Constructing the Number Systems

In this section, we provide rigorous constructions of well-known number systems including the **integers** $\mathbb{Z}$, the **rational numbers** $\mathbb{Q}$, the **real numbers** $\mathbb{R}$, and the **complex numbers** $\mathbb{C}$. These number systems, particularly the real and complex number systems, provide the backdrop for much of the discipline of analysis.

2.1 The Integers and the Rational Numbers [SKIP]

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Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

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Problems

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less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

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2.2 Constructing the Real Numbers

In this section, we construct the **real number system** $\mathbb{R}$ by **Dedekind cuts**, and we explore the resultant properties of arithmetic and comparison. However, we will find the existence of such a formal construction and its properties more useful than the particular details of the construction itself, and we will shortly leave behind the perspective suggested by our construction.

2.2.1 Cutting the Rational Numbers

So far, our progression through the standard number systems $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q}$ has proceeded in order of containment. Every time we have expanded our perspective to consider larger number systems, it has resulted in only the acquisition of new desirable properties. This is an artefact of our focus on the analytical rather than say the number-theoretical; there are for example incredibly interesting properties of the natural numbers $\mathbb{N}$ and the integers $\mathbb{Z}$ which are lost in the jump to the rational numbers $\mathbb{Q}$.

However, we have yet to see any deficiencies with the rational number system $\mathbb{Q}$ which would motivate seeking an extension. These deficiencies exist, and we will see in Section 2.3 The Complete Ordered Field and later in Section 3.6 Completeness and Compactness the benefits in working in the **real numbers** $\mathbb{R}$. As a brief example, we note that $\mathbb{Q}$ can be written as a particular type of disjoint union $\mathbb{Q} = X \sqcup Y$, where

$$\begin{aligned} X &= \left\{ q \in \mathbb{Q} \,\middle|\, q < 0 \text{ or } q^2 < 2 \right\} \\ Y &= \left\{ q \in \mathbb{Q} \,\middle|\, q > 0 \text{ and } q^2 > 2 \right\}. \end{aligned}$$

are defined by strict inequalities. This decomposition of $\mathbb{Q}$ is an example of a **Dedekind cut**, and we will see that such a decomposition of $\mathbb{R}$ cannot exist.

Definition 2.2.1 Dedekind cut. A proper, non-empty subset $\emptyset \subsetneq X \subsetneq \mathbb{Q}$ of the rational numbers $\mathbb{Q}$ is called a **Dedekind cut** if it satisfies the following conditions:

1. For all rational numbers $q_1, q_2 \in \mathbb{Q}$, if $q_1 \leq q_2$ and $q_2 \in X$, then $q_1 \in X$.
2. X does not have a maximum; that is, for all elements $q_1 \in X$, there is an element $q_2 \in X$ so that $q_1 < q_2$.

The set of Dedekind cuts is denoted $\mathbb{R}$. ◆

Example 2.2.2 Dedekind cuts. For each rational number $q \in \mathbb{Q}$, consider the subset

$$X_q = \{p \in \mathbb{Q} \mid p < q\}$$

of the rational numbers $\mathbb{Q}$. We observe that X_q is a proper and non-empty subset of $\mathbb{Q}$, since it contains $q - 1$ but not q itself. Moreover, for all rational numbers $p_1, p_2 \in \mathbb{Q}$, if $p_1 \le p_2$ and $p_2 \in X_q$, then

$$p_1 \le p_2 < q,$$

and so $p_1 \in X_q$. Finally, for all elements $p_1 \in X$, we observe that

$$p < p + \frac{q-p}{2} < p + (q-p) = q,$$

so that $p + \frac{q-p}{2} \in X_q$. In summary, we have shown that associated to each rational number $q \in \mathbb{Q}$ is a Dedekind cut X_q. ▲

We will see that the above construction realizes the rational numbers $\mathbb{Q}$ as a subset of the set of Dedekind cuts. The Dedekind cuts which arise in this manner will be called **rational**. We will shortly see that not every Dedekind cut is rational.

Lemma 2.2.3 Rational Dedekind cuts. *A Dedekind cut X is rational if and only if its complement $\mathbb{Q} \setminus X$ has a minimum.*

Proof. Let X be a Dedekind cut, and suppose that $X = X_q$ for some rational number $q \in \mathbb{Q}$. Note that $X = X_q$ has complement

$$\mathbb{Q} \setminus X = \mathbb{Q} \setminus X_q = \{p \in \mathbb{Q} \mid p \notin X_q\} = \{p \in \mathbb{Q} \mid p \ge q\}.$$

This set contains q, since $q \ge q$. Moreover, $p \ge q$ for all elements $p \in \mathbb{Q} \setminus X_q$. Thus $\min(\mathbb{Q} \setminus X) = q$.

Conversely, now suppose that $\mathbb{Q} \setminus X$ has a minimum element, say $q = \min(\mathbb{Q} \setminus X) \in \mathbb{Q}$. Let $p \in \mathbb{Q}$, and suppose that $p \ge q$. If $p \in X$, then that would imply that $q \in X$. This contradiction implies that $p \notin X$. On the other hand, if $p < q$, then $p \notin \mathbb{Q} \setminus X$, since $\min(\mathbb{Q} \setminus X) = q$. Thus

$$X = \{p \in \mathbb{Q} \mid p < q\} = X_q$$

is rational. ■

Proposition 2.2.4 Irrational Dedekind cuts. *Not all Dedekind cuts are rational.*

Proof. Consider the subset

$$X = \left\{q \in \mathbb{Q} \;\middle|\; q < 0 \text{ or } q^2 < 2\right\}$$

of the rational numbers $\mathbb{Q}$. We observe that X is a non-empty and proper subset of $\mathbb{Q}$, since it contains 1 but not 2. Let $p_1, p_2 \in \mathbb{Q}$ be rational numbers, and suppose that $p_1 \le p_2$ and $p_2 \in X$. If $p_2 < 0$, then

$$p_1 \le p_2 < 0,$$

and so $p_1 \in X$. On the other hand, if $p_2^2 < 2$, then either $p_1 < 0$ or $0 \le p_1 \le p_2$, in which case $p_1^2 \le p_2^2 < 2$. In either case, we conclude that $p_1 \in X$.

Finally, let $q \in X$. If $q \le 0$, then let $p = 1$, and note that $p \in X$ and $q < p$. On the other hand, suppose that $q > 0$ and $q^2 < 2$, and consider the rational number

$$p = \frac{2q+2}{2+q} \in \mathbb{Q}.$$

Note that

$$p = \frac{2q+2}{2+q} = \frac{2q+q^2}{2+q} + \frac{2-q^2}{2+q} = q + \frac{2-q^2}{2+q} > q,$$

and

$$\begin{aligned} 2-p^2 &= \frac{2\left(2+q\right)^2}{\left(2+q\right)^2} - \frac{\left(2q+2\right)^2}{\left(2+q\right)^2} \\ &= \frac{4-2q^2}{\left(2+q\right)^2} = \frac{2\left(2-q^2\right)}{\left(2+q\right)^2} > 0, \end{aligned}$$

so that $p^2 < 2$; that is, $p \in X$. We conclude that X is a Dedekind cut, and so it suffices to show that X is not rational.

To that end, we note that X has complement

$$\mathbb{Q} \setminus X = \left\{q \in \mathbb{Q} \,\middle|\, q \geq 0 \text{ and } q^2 \geq 2\right\}.$$

In fact, since $q^2 \neq 2$ for all rational numbers $q \in \mathbb{Q}$ (this fact is proven in any introductory number theory course),

$$\mathbb{Q} \setminus X = \left\{q \in \mathbb{Q} \,\middle|\, q \geq 0 \text{ and } q^2 > 2\right\}.$$

We will show that $\mathbb{Q} \setminus X$ has no minimum element. Indeed, let $q \in \mathbb{Q} \setminus X$, so that $q \geq 0$ and $q^2 > 2$. We again consider the rational number

$$p = \frac{2q+2}{2+q} \in \mathbb{Q}.$$

We see that $p \geq 0$. Moreover, now that $q^2 > 2$, we observe that

$$p = \frac{2q+2}{2+q} = \frac{2q+q^2}{2+q} + \frac{2-q^2}{2+q} = q + \frac{2-q^2}{2+q} < q$$

and

$$\begin{aligned} 2-p^2 &= \frac{2\left(2+q\right)^2}{\left(2+q\right)^2} - \frac{\left(2q+2\right)^2}{\left(2+q\right)^2} \\ &= \frac{4-2q^2}{\left(2+q\right)^2} = \frac{2\left(2-q^2\right)}{\left(2+q\right)^2} < 0, \end{aligned}$$

so that $p^2 > 2$. In particular, $p < q$ and $p \in \mathbb{Q} \setminus X$. Thus $\mathbb{Q} \setminus X$ has no minimum element, and so Lemma 2.2.3 Rational Dedekind cuts implies that X is not rational. ∎

Once we define the arithmetic of Dedekind cuts, we will see that the Dedekind cut

$$X = \left\{q \in \mathbb{Q} \,\middle|\, q < 0 \text{ or } q^2 < 2\right\}$$

from the above proof of Proposition 2.2.4 Irrational Dedekind cuts represents the **irrational** real number $\sqrt{2}$. First, however, we observe that the rational numbers $\mathbb{Q}$ inject into $\mathbb{R}$.

Proposition 2.2.5 Rational numbers as real numbers. *The map $\iota\colon \mathbb{Q} \to \mathbb{R}$ defined by the formula $\iota(q) = X_q$ is injective.*

Proof. Let $q_1, q_2 \in \mathbb{Q}$ be rational numbers, and suppose that $\iota(q_1) = \iota(q_2)$. As shown in the proof of Lemma 2.2.3 Rational Dedekind cuts, $\min(X_q) = q$ for all rational numbers $q \in \mathbb{Q}$. Therefore,

$$\begin{aligned} q_1 &= \min\left(X_{q_1}\right) = \min\left(\iota\left(q_1\right)\right) \\ &= \min\left(\iota\left(q_2\right)\right) = \min\left(X_{q_2}\right) = q_2. \end{aligned}$$

Thus $\iota\colon \mathbb{Q} \to \mathbb{R}$ is injective. ∎

Going forward, we will often exploit the injection $\iota\colon \mathbb{Q} \to \mathbb{R}$ to pretend that $\mathbb{Q} \subseteq \mathbb{R}$ by identifying a rational number with its associated rational Dedekind cut. We will now construct a total order $\leq$ on $\mathbb{R}$ which extends the standard order on $\mathbb{Q}$.

2.2.2 Comparison of Dedekind Cuts

Our two perspectives on the set $\mathbb{Q}$ (as equivalence classes of fractions and as rational Dedekind cuts) yield two natural orderings: the standard ordering $\leq$ and the partial order of set containment. In fact, these orderings agree.

Corollary 2.2.6 Ordering Dedekind cuts by containment. *The standard ordering $\leq$ on the rational numbers $\mathbb{Q}$ is the pullback of set containment $\subseteq$ on the set $\mathbb{R}$ of Dedekind cuts.*

Proof. Let $q_1, q_2 \in \mathbb{Q}$ be rational numbers. We note that

$$\begin{aligned} X_{q_1} &= \{p \in \mathbb{Q} \mid p < q_1\} \\ X_{q_2} &= \{p \in \mathbb{Q} \mid p < q_2\}. \end{aligned}$$

First suppose that $q_1 \leq q_2$. If $p \in X_{q_1}$, then

$$p < q_1 \leq q_2$$

and so $p \in X_{q_2}$; that is,

$$\iota(q_1) = X_{q_1} \subseteq X_{q_2} = \iota(q_2),$$

so that $q_1 \subseteq_\iota q_2$.

Conversely, now suppose that $q_1 \subseteq_\iota q_2$. Then

$$X_{q_1} = \iota(q_1) \subseteq \iota(q_2) = X_{q_2}.$$

If $q_1 > q_2$, then

$$q_2 < \frac{q_1 + q_2}{2} < q_1,$$

and so $\frac{q_1+q_2}{2} \in X_{q_1} \setminus X_{q_2}$. This would imply that $X_{q_1} \not\subseteq X_{q_2}$, and so we must have $q_1 \leq q_2$.

In summary, we have shown that $q_1 \leq q_2$ if and only if $q_1 \subseteq_\iota q_2$, and so $\leq$ and $\subseteq_\iota$ are the same binary relation on $\mathbb{Q}$. ∎

Corollary 2.2.6 Ordering Dedekind cuts by containment implies that comparing rational numbers is the same as comparing their associated rational Dedekind cuts by containment. This gives us the template for how to compare Dedekind cuts in general: we compare by set containment.

Theorem 2.2.7 Comparison of Dedekind cuts. *Set containment $\subseteq$ is a total order on the set $\mathbb{R}$ of Dedekind cuts.*

Proof. Since set containment is a partial order, it suffices to show only totality. To that end, let $X, Y \in \mathbb{R}$ be Dedekind cuts, and suppose that $X \not\subseteq Y$. Then there is some rational number $q \in X$ so that $q \notin Y$.

Let $p \in Y$. If $q \leq p$, then we would have $q \in Y$, and so we must have $p \leq q$. But this implies that $p \in X$, and so $Y \subseteq X$.

In summary, we have shown that for all Dedekind cuts $X, Y \in \mathbb{R}$, either $X \subseteq Y$ or $Y \subseteq X$. Thus set containment $\subseteq$ is a total order on the set $\mathbb{R}$ of Dedekind cuts. ∎

Just as with the integers $\mathbb{Z}$ and rational numbers $\mathbb{Q}$, we can use the standard ordering on $\mathbb{R}$ to distinguish Dedekind cuts by their **sign**: either **positive**, **negative**, or **zero**.

Definition 2.2.8 Sign of a Dedekind cut. Recall that

$$X_0 = \{q \in \mathbb{Q} \mid q < 0\}$$

is the rational Dedekind cut associated to the rational number 0.

- *Positivity.*

  A Dedekind cut $X \in \mathbb{R}$ is called **positive** if it strictly contains X_0; that is, X is positive if $X \supsetneq X_0$.

- *Negativity.*

  A Dedekind cut $X \in \mathbb{R}$ is called **negative** if it is strictly contained in X_0; that is, X is negative if $X \subsetneq X_0$.

- *Non-negativity.*

  A Dedekind cut $X \in \mathbb{R}$ is called **non-negative** if it contains X_0; that is, X is non-negative if $X \supseteq X_0$.

- *Non-positivity.*

  A Dedekind cut $X \in \mathbb{R}$ is called **non-positive** if it is contained in X_0; that is, X is positive if $X \subseteq X_0$.

In particular, the **sign** of a Dedekind cut is either positive, negative, or zero. ◆

This classification of Dedekind cuts by sign will be useful in formalizing the multiplication of Dedekind cuts. We now proceed in that direction, extending the usual operations of arithmetic on $\mathbb{Q}$ to binary operations on $\mathbb{R}$ and determining their properties.

2.2.3 Addition of Dedekind Cuts

Finally, we begin to define arithmetic for Dedekind cuts. We emphasize that the utility of what follows is neither the details of the construction nor the methods of proof, but rather the very fact that the operations exist with the same properties we would expect. That is to say, we will eventually lose sight of the particularities of the construction of the real number system by Dedekind cuts and treat real numbers as objects which obey the rules of arithmetic.

Lemma 2.2.9 Addition of Dedekind cuts. *Given two Dedekind cuts $X, Y \in \mathbb{R}$, the set*

$$\{p + q \in \mathbb{Q} \mid p \in X \text{ and } q \in Y\}$$

is a Dedekind cut.

Proof. Denote by Z the set

$$Z = \{p + q \in \mathbb{Q} \mid p \in X \text{ and } q \in Y\}.$$

Since X and Y are Dedekind cuts, they are non-empty subsets of $\mathbb{Q}$, and so there are rational numbers $p \in X$ and $q \in Y$. Thus $p + q \in Z$, and so Z is also non-empty.

On the other hand, X and Y are proper subsets of $\mathbb{Q}$, and so there are rational numbers $r \in \mathbb{Q} \setminus X$ and $s \in \mathbb{Q} \setminus Y$. We observe that that $r > p$ for all

$p \in X$ and $s > q$ for all $q \in Y$. In particular, for all $t \in Z$ there are rational numbers $p \in X$ and $q \in Y$ so that $t = p + q$, and so

$$r + s > p + q = t$$

In particular, $r + s \notin Z$, and so Z is also a proper subset of $\mathbb{Q}$.

Now let $r, s \in \mathbb{Q}$ be rational numbers, and suppose that $s \leq r$ and $r \in Z$. We wish to show that $s \in Z$. To that end, we see that $r = p + q$ for some rational numbers $p \in X$ and $q \in Y$. Since $s - r \leq 0$,

$$q + (s - r) \leq q + 0 = q \in Y,$$

and so $q + (s - r) \in Y$. This implies that

$$s = r + (s - r) = p + (q + (s - r)) \in Z.$$

It remains to show that Z does not have a maximum. To that end, let $r_1 \in Z$, so that $r_1 = p_1 + q_1$ for some rational numbers $p_1 \in X$ and $q_1 \in Y$. Since X and Y are Dedekind cuts, they do not have maxima, and so $p_1 < p_2$ and $q_1 < q_2$ for some rational numbers $p_2 \in X$ and $q_2 \in Y$. We conclude that $r_2 = p_2 + q_2 \in Z$, and that

$$r_1 = p_1 + q_1 < p_2 + q_2 = r_2.$$

Thus Z does not have a maximum, and so is a Dedekind cut. ■

Definition 2.2.10 Addition of Dedekind cuts. The **sum** $X + Y$ of two Dedekind cuts $X, Y \in \mathbb{R}$ is the Dedekind cut defined by

$$X + Y = \{p + q \in \mathbb{Q} \mid p \in X \text{ and } q \in Y\}.$$

This defines a binary operation $+$ on the set $\mathbb{R}$ of Dedekind cuts. ◆

We first observe that addition of rational Dedekind cuts mirrors the addition of the corresponding rational numbers.

Lemma 2.2.11 Addition of rational Dedekind cuts. $X_p + X_q = X_{p+q}$ *for all rational numbers* $p, q \in \mathbb{Q}$.

Proof. Let $r \in \mathbb{Q}$ be a rational number, and suppose first that $r \in X_p + X_q$. Then $r = s + t$ for some rational numbers $s \in X_p$ and $t \in X_q$, so that $s < p$ and $t < q$. We observe that

$$r = s + t < p + q,$$

and so $r \in X_{p+q}$.

Conversely, now suppose that $r \in X_{p+q}$, so that $r < p + q$. Consider the rational numbers

$$s = p + \frac{r - (p + q)}{2} \qquad t = q + \frac{r - (p + q)}{2}.$$

Since $\left(\frac{r-(p+q)}{2}\right) < 0$, $s < p$ and $t < q$, and so $s \in X_p$ and $t \in X_q$. We conclude that

$$\begin{aligned} r &= p + q + (r - (p + q)) = p + \frac{r - (p + q)}{2} + q + \frac{r - (p + q)}{2} \\ &= s + t \in X_p + X_q. \end{aligned}$$

In summary, we have shown that $r \in X_p + X_q$ if and only if $r \in X_{p+q}$, and so $X_p + X_q = X_{p+q}$. ■

Proposition 2.2.12 Addition of Dedekind cuts. *Addition $+$ of Dedekind cuts has the following properties:*

1. *Commutativity.*

   $X + Y = Y + X$ *for all Dedekind cuts* $X, Y \in \mathbb{R}$.

2. *Associativity.*

   $(X + Y) + Z = X + (Y + Z)$ *for all Dedekind cuts* $X, Y, Z \in \mathbb{R}$.

3. *Identity element.*

   The rational Dedekind cut X_0 associated to 0 is an additive identity element; that is, $X + X_0 = X_0 + X = X$ for all Dedekind cuts $X \in \mathbb{R}$.

4. *Inverses.*

   For all Dedekind cuts $X \in \mathbb{R}$, there is a Dedekind cut $Y \in \mathbb{R}$ so that $X + Y = X_0$.

5. *Compatibility with the standard order.*

   For all Dedekind cuts $W, X, Y, Z \in \mathbb{R}$, if $W \subseteq Y$ and $X \subseteq Z$, then $W + X \subseteq Y + Z$.

Proof.

1. We note that

$$\begin{aligned} X + Y &= \{p + q \in \mathbb{Q} \mid p \in X \text{ and } q \in Y\} \\ &= \{q + p \in \mathbb{Q} \mid q \in Y \text{ and } p \in X\} = Y + X \end{aligned}$$

   for all Dedekind cuts $X, Y \in \mathbb{R}$, and so addition $+$ of Dedekind cuts is commutative.

2. We note that

$$\begin{aligned} (X + Y) + Z &= \{s + r \in \mathbb{Q} \mid s \in X + Y \text{ and } r \in Z\} \\ &= \{(p + q) + r \in \mathbb{Q} \mid p \in X \text{ and } q \in Y \text{ and } r \in Z\} \\ &= \{p + (q + r) \in \mathbb{Q} \mid p \in X \text{ and } q \in Y \text{ and } r \in Z\} \\ &= \{p + t \in \mathbb{Q} \mid p \in X \text{ and } t \in Y + Z\} = X + (Y + Z)\,. \end{aligned}$$

   for all Dedekind cuts $X, Y, Z \in \mathbb{R}$, and so addition $+$ of Dedekind cuts is associative.

3. Let $p \in \mathbb{Q}$ be a rational number, and suppose first that $p \in X + X_0$. Then $p = q + r$ for some rational numbers $q \in X$ and $r \in X_0$. Since

$$X_0 = \{s \in \mathbb{Q} \mid s < 0\}$$

   is the set of negative rational numbers,

$$p = q + r < q + 0 = q \in X,$$

   and so $p \in X$.

   Conversely, now suppose that $p \in X$. Since X does not have a maximum, $r \in X$ for some rational number $r > p$. Let $s = p - r \in X_0$, and note that

$$p = r + (p - r) = r + s$$

   is the sum of a rational number $r \in X$ and a rational number $p - r \in X_0$. We conclude that $p \in X + X_0$.

In summary, we have shown that $p \in X + X_0$ if and only if $p \in X$, and so $X + X_0 = X$. That $X_0 + X = X$ now follows from (1) above.

4. There are two cases: either X is rational, or X is irrational. If X is rational, say $X = X_q$ for some rational number $q \in \mathbb{Q}$, then we may choose $Y = X_{-q}$, since Lemma 2.2.11 Addition of rational Dedekind cuts implies that
$$X + Y = X_q + X_{-q} = X_{q+(-q)} = X_0.$$
So we may suppose that X is irrational. Let
$$Y = \{q \in \mathbb{Q} \mid -q \notin X\}.$$
Since X is a non-empty proper subset of $\mathbb{Q}$, so too is its complement $\mathbb{Q} \setminus X$, which implies that Y is also non-empty and proper. We aim to show first that Y is a Dedekind cut.

   To that end, let $p, q \in \mathbb{Q}$ be rational numbers, and suppose that $p < q$ and $q \in Y$. Then $-q \notin X$, and so $-q > r$ for all rational numbers $r \in X$. We observe that
$$-p \geq -q > r$$
for all rational numbers $r \in X$, and so $-p \notin X$; that is, $p \in Y$.

   Now let $q \in Y$, so that $-q \notin X$. Since X is irrational, Lemma 2.2.3 Rational Dedekind cuts implies that $\mathbb{Q} \setminus X$ does not have a minimum. Thus there is some rational number $r \in \mathbb{Q} \setminus X$ so that $r < -q$. We conclude that $-r \in Y$ and $-r > q$, so that Y does not have a maximum.

   We have shown that Y is a Dedekind cut, but it remains to show that $X + Y = X_0$. To that end, let $r \in \mathbb{Q}$ be a rational number, and suppose first that $r \in X + Y$. Then $r = p + q$ for some $p \in X$ and $q \in Y$. Since $-q \notin X$, we must have $p < -q$, so that $r = p + q < 0$; that is, $r \in X_0$.

   Conversely, now suppose that $r \in X_0$, so that $r < 0$. In particular, $-r > 0$. If $p + (-r) \in X$ for all rational numbers $p \in X$, then X is not bounded from above, which would imply that $X = \mathbb{Q}$. Thus $p - r \notin X$ for some rational number $p \in X$. We conclude that
$$r = p + (r - p) \in X + Y$$
is the sum of two rational numbers $p \in X$ and $r - p \in Y$.

   In summary, we have shown that $r \in X + Y$ if and only if $r \in X_0$, and so $X + Y = X_0$.

5. We observe that if $W \subseteq Y$ and $X \subseteq Z$, then
$$\begin{aligned} W + X &= \{p + q \in \mathbb{Q} \mid p \in W \text{ and } q \in X\} \\ &\subseteq \{p + q \in \mathbb{Q} \mid p \in Y \text{ and } q \in Z\} = Y + Z. \end{aligned}$$

■

All that remains is to define the multiplication of Dedekind cuts and derive relevant results about multiplication itself and its relation to addition and comparison.

2.2.4 Multiplication of Dedekind Cuts

Finally, we define multiplication for Dedekind cuts. We will see that this definition depends in a complicated way on the signs of the two factors. As a result of this case-by-case definition of multiplication, we offer the remaining results of this section without proof.

Lemma 2.2.13 Multiplication of positive Dedekind cuts. *Given two positive Dedekind cuts $X, Y \in \mathbb{R}$, the set*

$$\{pq \in \mathbb{Q} \mid p \in X \text{ and } q \in Y \text{ and } (p \geq 0 \text{ or } q \geq 0)\}$$

is a Dedekind cut.

Definition 2.2.14 Multiplication of Dedekind cuts. The **product** $X \cdot Y$ of two Dedekind cuts $X, Y \in \mathbb{R}$ is the Dedekind cut defined by the following formulae:

- *Zero.*

  If either $X = X_0$ or $Y = X_0$, then $X \cdot Y = X_0$.

- *Positive times positive.*

  If both X and Y are positive, then

  $$X \times Y = \{pq \in \mathbb{Q} \mid p \in X \text{ and } q \in Y \text{ and } (p \geq 0 \text{ or } q \geq 0)\}.$$

- *Positive times negative.*

  If X is positive and Y is negative, then

  $$X \times Y = -(X \times (-Y)).$$

- *Negative times positive.*

  If X is negative and Y is positive, then

  $$X \times Y = -((-X) \times Y).$$

- *Negative times negative.*

  If both X and Y are negative, then

  $$X \times Y = (-X) \times (-Y).$$

This defines a binary operation $\cdot$ on the set $\mathbb{R}$ of Dedekind cuts. ◆

We end with a compilation of all relevant results about the arithmetic of Dedekind cuts. While one can verify these results, we omit the proofs of those that involve multiplication in order to simplify the narrative of exposition. Indeed, these proofs rely on casework which, while not difficult, would require more time than we can dedicate to this topic.

Theorem 2.2.15 Arithmetic of Dedekind cuts. *The addition and multiplication of Dedekind cuts has the following properties:*

- *Commutativity of addition.*

  $X + Y = Y + X$ *for all Dedekind cuts* $X, Y \in \mathbb{R}$.

- *Associativity of addition.*

$(X+Y)+Z = X+(Y+Z)$ and $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$ for all Dedekind cuts $X, Y, Z \in \mathbb{R}$.

- *Additive identity element.*

  The rational Dedekind cut X_0 associated to 0 is an additive identity element; that is, $X + X_0 = X_0 + X = X$ for all Dedekind cuts $X \in \mathbb{R}$.

- *Additive inverses.*

  For all Dedekind cuts $X \in \mathbb{R}$, there is a Dedekind cut $-X \in \mathbb{R}$ so that $X + (-X) = (-X) + X = X_0$.

- *Commutativity of multiplication.*

  $X \cdot Y = Y \cdot X$ for all Dedekind cuts $X, Y \in \mathbb{R}$.

- *Associativity of multiplication.*

  $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$ for all Dedekind cuts $X, Y, Z \in \mathbb{R}$.

- *Multiplicative identity element.*

  The rational Dedekind cut X_1 associated to 1 is a multiplicative identity element; that is, $X \cdot X_1 = X_1 \cdot X = X$ for all Dedekind cuts $X \in \mathbb{R}$.

- *Multiplicative inverses.*

  For all Dedekind cuts $X \in \mathbb{R}$, if $X \neq X_0$, then there is a Dedekind cut $X^{-1} \in \mathbb{R}$ so that $X \cdot X^{-1} = X^{-1} \cdot X = X_1$.

- *Distributivity of multiplication over addition.*

  For all Dedekind cuts $X, Y, Z \in \mathbb{R}$, both

  $$X \cdot (Y + Z) = (X \cdot Y) + (X \cdot Z)$$

  and

  $$(X + Y) \cdot Z = (X \cdot Z) + (Y \cdot Z)\,.$$

- *Compatibility of addition and the standard order.*

  For all Dedekind cuts $W, X, Y, Z \in \mathbb{R}$, if $W \subseteq Y$ and $X \subseteq Z$, then $W + X \subseteq Y + Z$.

- *Compatibility of multiplication and the standard order.*

  For all Dedekind cuts $X, Y \in \mathbb{R}$, if X and Y are non-negative, then so too is their product $X \cdot Y$.

All of the above properties of the arithmetic of Dedekind cuts not involving multiplication have already been proven. Again, we highlight that while proofs of the remaining parts are not too difficult to construct, they rely on significant casework involving signs; we unfortunately don't have time to study these proofs in detail.

In the next section, we will continue to explore the properties of the real number system. In particular, we will characterize $\mathbb{R}$ as the unique **Dedekind complete ordered field**.

2.2.5 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

2.2.6 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Cancellation of addition. Prove that for all Dedekind cuts $X, Y, Z \in \mathbb{R}$, if $X + Z \subseteq Y + Z$, then $X \subseteq Y$.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

2.3 The Complete Ordered Field

In the previous section, we gave an explicit and detailed construction of the real number system $\mathbb{R}$. In this section, we give an alternative and non-constructive characterization of $\mathbb{R}$ as the unique **Dedekind complete ordered field**. This will set the stage for abandoning the Dedekind cut perspective in favor of an abstract conception of the properties of the real number system.

2.3.1 The Ordered Field Axioms

We begin by revisiting the properties of the arithmetic of Dedekind cuts collected in Section 2.2 Constructing the Real Numbers. These properties define an algebraic object called a **field**.

Definition 2.3.1 Field. Let $+$ and $\cdot$ be binary operations on a set F, called **addition** and **multiplication**. The ordered triple $(F, +, \cdot)$ is called a **field** if the binary operations satisfy the following conditions:

- *Commutativity of addition.*

  $x + y = y + x$ for all elements $x, y \in F$.

- *Associativity of addition.*

  $(x + y) + z = x + (y + z)$ for all elements $x, y, z \in F$.

- *Additive identity element.*

There is an element $0 \in F$ so that $x + 0 = 0 + x = x$ for all elements $x \in F$.

- *Additive inverses.*

  For all elements $x \in X$, there is an element $-x \in F$ called the **additive inverse** of x so that $x + (-x) = (-x) + x = 0$.

- *Commutativity of multiplication.*

  $x \cdot y = y \cdot x$ for all elements $x, y \in F$.

- *Associativity of addition.*

  $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all elements $x, y, z \in F$.

- *Multiplicative identity element.*

  There is an element $1 \in F \setminus \{0\}$ so that $x \cdot 1 = 1 \cdot x = x$ for all elements $x \in F$.

- *Multiplicative inverses.*

  For all elements $x \in X$, if $x \neq 0$ then there is an element $x^{-1} \in F$ called the **multiplicative inverse** of x so that $x \cdot x^{-1} = x^{-1} \cdot x = 1$.

- *Distributivity of multiplication over addition.*

  $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $(x + y) \cdot z = (x \cdot z) + y \cdot z$ for all elements $x, y, z \in F$.

If $(F, +, \cdot)$ is a field, then the set F is called its **ground set** or **underlying set**. If the binary operations $+$ and $\cdot$ are clear from context, then we will usually abuse notation and refer to a field by the ground set alone. ◆

Example 2.3.2 Fields. The following are examples of fields:

- *The field of rational numbers.*

  Addition $+$ and multiplication $\cdot$ of rational numbers satisfies the above field axioms, and so $\mathbb{Q}$ is a field.

- *The field of real numbers.*

  Addition $+$ and multiplication $\cdot$ of real numbers satisfies the above field axioms by Theorem 2.2.15 Arithmetic of Dedekind cuts, and so $\mathbb{R}$ is a field.

- *The field of rational functions.*

  Let $n \in \mathbb{N}$ be a natural number, and fix real numbers $x_1, \ldots c, x_n \in \mathbb{R}$. A real-valued function $f \colon \mathbb{R} \setminus \{x_1, \ldots, x_n\} \to \mathbb{R}$ is called a **rational function** if it has a formula of the form

  $$f(x) = \frac{p(x)}{q(x)}$$

  for some polynomials $p(x)$ and $q(x)$. Such a rational function $f(x)$ has a **removable discontinuity** at x_k if both $p(x)$ and $q(x)$ have a root at $x = x_k$, and the corresponding factor $(x - x_k)$ appears at least as many times in the factorization of $p(x)$ as it does in the factorization of $q(x)$.

  The set of rational functions without removable discontinuities is denoted $\mathbb{R}(x)$. One can verify that the addition and multiplication of these rational functions satisfies the field axioms, and so $\mathbb{R}(x)$ is a field.

▲

We have seen that the fields $\mathbb{Q}$ and $\mathbb{R}$ of rational and real numbers are equipped with additional structure beyond that of arithmetic: we can compare rational numbers and real numbers by the standard orders $\leq$ on $\mathbb{Q}$ and $\mathbb{R}$, respectively. These are total orders, and we have seen they are compatible with arithmetic in key ways. Generalizing these relationships, we arrive at the notion of an **ordered field**.

Definition 2.3.3 Ordered field. Let $+$ and $\cdot$ be binary operations on a set F, called **addition** and **multiplication**, and consider a binary relation $\leq$ on F, called **comparison**. The ordered quadruple $(F, +, \cdot, \leq)$ is called an **ordered field** if it satisfies the following conditions:

- The ordered triple $(F, +, \cdot)$ is a field.
- The ordered pair $(F, \leq)$ is a totally ordered set.
- *Compatibility of addition and comparison.*

  For all elements $x, y, z \in F$, if $x \leq y$, then $x + z \leq y + z$.
- *Compatibility of multiplication and comparison.*

  For all elements $x, y \in F$, if $0 \leq x$ and $0 \leq y$, then $0 \leq x \cdot y$.

If $(F, +, \cdot, \leq)$ is an ordered field, then the set F is called its **ground set** or **underlying set**. If the binary operations $+$ and $\cdot$ and the binary relation $\leq$ are clear from context, then we will usually refer to an ordered field by the ground set alone. ◆

Example 2.3.4 Ordered fields. The following are examples of ordered fields:

- *The ordered field of rational numbers.*

  Addition $+$, multiplication $\cdot$, and comparison $\leq$ of rational numbers satisfies the above ordered field axioms, and so $\mathbb{Q}$ is an ordered field.
- *The ordered field of real numbers.*

  Addition $+$, multiplication $\cdot$, and comparison $\leq$ of rational numbers satisfies the above ordered field axioms by Theorem 2.2.15 Arithmetic of Dedekind cuts, and so $\mathbb{R}$ is an ordered field.

▲

That the above list of examples includes both the rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$ begs the question: Why bother with the real numbers $\mathbb{R}$, when the rational numbers $\mathbb{Q}$ are already an ordered field?

2.3.2 Dedekind completeness

So far, we have failed to meaningfully distinguish between the properties enjoyed by the rational number system $\mathbb{Q}$ and the real number system $\mathbb{R}$. We now address this issue via the notion of **Dedekind completeness**. Informally, Dedekind completeness is defined in terms of the set of upper bounds of a subset which is bounded from above (or dually the set of lower bounds of a subset which is bounded from below).

Definition 2.3.5 Supremum; infimum. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

- *Supremum.*

If the set of upper bounds for U in X has a minimum, then this minimum is called the **supremum** (or **least upper bound**) of U and is denoted $\sup(U)$; that is,

$$\sup(U) = \min(\{x \in X \mid u \leq x \text{ for all } u \in U\}).$$

- *Infimum.*

  If the set of lower bounds for U in X has a maximum, then this maximum is called the **infimum** (or **greatest lower bound**) of U and is denoted $\inf(U)$; that is,

$$\inf(U) = \max(\{x \in X \mid x \leq u \text{ for all } u \in U\}).$$

◆

Example 2.3.6 Suprema and infima. Let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of subsets $U_j \subseteq X$ of a set X. These subsets U_j are elements of the power set $\mathcal{P}(X)$, which is partially ordered by set containment $\subseteq$.

We may reinterpret (a) of Problem 1.2.5.5 Indexed intersections and unions as saying that the indexed intersection $\bigcap_{j \in \mathcal{J}} U_j$ and the indexed union $\bigcup_{j \in \mathcal{J}} U_j$ are the infimum and supremum of $\{U_j\}_{j \in \mathcal{J}}$. ▲

Remark 2.3.7 Suprema of Dedekind cuts. The complement $\mathbb{Q} \setminus X$ of a Dedekind cut $X \in \mathbb{R}$ is exactly the set of upper bounds of X in $\mathbb{Q}$, and so Lemma 2.2.3 Rational Dedekind cuts implies that X is rational if and only if it has a supremum in $\mathbb{Q}$.

This is the deficiency of the rational numbers $\mathbb{Q}$ from the analytic perspective: Proposition 2.2.4 Irrational Dedekind cuts posits the existence of subsets of $\mathbb{Q}$ which are bounded from above ***and yet do not have suprema***. We will shortly see that this is not the case in the real number system $\mathbb{R}$.

Definition 2.3.8 Dedekind completeness. An ordered field $(F, +, \cdot, \leq)$ is called **Dedekind complete** if every non-empty subset of F which is bounded from above has a supremum. ◆

Theorem 2.3.9 Dedekind completeness of the real numbers. *The real number system* $(\mathbb{R}, +, \cdot, \leq)$ *is Dedekind complete.*

Proof. Let $\emptyset \subsetneq U \subseteq \mathbb{R}$ be a non-empty subset of U which is bounded from above, and consider the indexed union

$$S = \bigcup_{X \in U} X = \{q \in \mathbb{Q} \mid q \in X \text{ for some } X \in U\}.$$

Since U is a non-empty collection of non-empty subsets of $\mathbb{Q}$, S is non-empty. Moreover, since U is bounded from above, there is some Dedekind cut $Y \in \mathbb{R}$ so that $X \subseteq Y$ for all $X \in U$. In particular, if $q \in \mathbb{Q} \setminus Y$, then $q \notin X$ for all $X \in U$, and so $q \notin S$.

Let $p, q \in \mathbb{Q}$ be rational numbers, and suppose that $p \leq q$ and $q \in S$. Since $q \in S$, $q \in X$ for some $X \in U$, and so $p \in X \subseteq S$.

Now let $p \in S$, so that $p \in X$ for some $X \in U$. Since X does not have a maximum, there is some rational number $q \in X \subseteq S$ so that $p < q$. We conclude that S does not have a maximum, and so is a Dedekind cut.

We note that (a) of Problem 1.2.5.5 Indexed intersections and unions implies that for a Dedekind cut $Y \in \mathbb{R}$, $S \subseteq Y$ if and only if $X \subseteq Y$ for all $X \in U$; that is, any upper bound for U in $\mathbb{R}$ contains S, so that

$$S = \min(\{Y \in \mathbb{R} \mid X \subseteq Y \text{ for all } X \in U\}) = \sup(U).$$

■

We will see that Theorem 2.3.9 Dedekind completeness of the real numbers has far-reaching consequences for the discipline of real analysis, most notably in the existence of limits for certain objects constructed out of the real numbers $\mathbb{R}$. First, however, we present the following non-constructive characterization of $\mathbb{R}$ as the *unique* Dedekind complete ordered field.

Theorem 2.3.10 Uniqueness of the real numbers. *If $(F, +, \cdot, \leq)$ is a Dedekind complete ordered field, then there is a bijection $\varphi\colon \mathbb{R} \to F$ so that for all real numbers $x, y \in \mathbb{R}$, $\varphi(x+y) = \varphi(x) + \varphi(y)$, $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$, and if $x \leq y$, then $\varphi(x) \leq \varphi(y)$.*

With Theorem 2.3.10 Uniqueness of the real numbers, we are now ready to shake off the burden of viewing the real number system $\mathbb{R}$ through the perspective of Dedekind cuts. All relevant properties of the analysis of real numbers can be derived abstractly from the axioms of a Dedekind complete ordered field, and as a consequence, we will see that we no longer need to rely on any specific formal model of $\mathbb{R}$.

In the next section, we will construct various extensions of the real number system $\mathbb{R}$. In particular, we will see the **extended real numbers**, the **complex numbers**, and the **quaternions**.

2.3.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Algebraic structures other than fields. Determine which of the ordered field axioms are satisfied by the given algebraic structure.

1. Which of the ordered field axioms are not satisfied by the set $\mathbb{Z}$ of integers?

2. Which of the ordered field axioms are not satisfied by the set $\mathbb{N}$ of natural numbers?

3. Let X be a set. Which of the ordered field axioms are not satisfied by the ordered quadruple $(\mathcal{P}(X), \cup, \cap, \subseteq)$?

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

2.3.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be

less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Uniqueness in a field. Let $(F, +, \cdot)$ be a field

(a) Show that the additive identity element $0 \in F$ is unique.

(b) Show that the multiplicative identity element $1 \in F$ is unique.

(c) Show that additive inverses are unique; that is, given elements $x, y, z \in F$, show that if

$$x + y = x + z = 0,$$

then $y = z$.

(d) Show that multiplicative inverses are unique; that is, given elements $x, y, z \in F$, show that if

$$x \cdot y = x \cdot z = 1,$$

then $y = z$.

2. Sign in an ordered field. Let $(F, +, \cdot, \leq)$ be an ordered field. As in the case of the rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$, we can define the **sign** of an element of F by comparison with the additive identity element 0:

- *Positivity.*

  A non-zero element $x \in F \setminus \{0\}$ is called **positive** if $0 \leq x$.

- *Negativity.*

  A non-zero element $x \in F \setminus \{0\}$ is called **negative** if $x \leq 0$.

- *Zero.*

  The additive identity element $0 \in F$ is not positive or negative.

(a) Let $x \in F$. Show that if $x \neq 0$, then x and its additive inverse $-x$ have opposite sign.

Hint. Add $-x$ to the inequality defining the sign of x.

(b) More generally, show that for all elements $x, y \in F$, if $x \leq y$ then $-y \leq -x$.

(c) Show that $(-1) \cdot x = -x$ for all elements $x \in F$.

(d) Show that the product of two negative elements is positive.

3. Infima and Dedekind completeness. Let $(F, +, \cdot, \leq)$ be an ordered field, and consider the map $n \colon F \to F$ defined by the formula $n(x) = -x$; that is, n sends an input $x \in F$ to its additive inverse $-x$.

(a) Show that for a subset $U \subseteq F$, U is bounded from above if and only if $n(U)$ is bounded from below, and U is bounded from below if and only if $n(U)$ is bounded from above.

(b) Show that a subset $U \subseteq F$ has a supremum if and only if $n(U)$ has an infimum, and in this case

$$\inf(n(U)) = n(\sup(U)).$$

Similarly, show that U has an infimum if and only if $n(U)$ has a supremum, and in this case

$$\sup(n(U)) = n(\inf(U))$$

(c) Show that the ordered field $(F, +, \cdot, \leq)$ is Dedekind complete if and only if every non-empty subset of F which is bounded from below has an infimum.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

2.4 Beyond the Real Number System

In this section, we address several common extensions to the real number system $\mathbb{R}$. Some of these, like the **complex numbers** and the **quaternions**, see extensions to arithmetic (at the cost of some of the ordered field axioms). Others, like the **extended real numbers**, abandon arithmetic altogether in favor of certain order-theoretic properties.

2.4.1 The Complex Numbers

Despite not playing a large role in the field of real analysis, the **complex number system** is ubiquitous throughout mathematics and merits at least a brief mention. The extension of the real numbers to the complex numbers ameliorates an algebraic insufficiency of $\mathbb{R}$, namely that it does not contain the roots of all polynomials. For example, there is no real number $x \in \mathbb{R}$ so that $x^2 + 1 = 0$.

Definition 2.4.1 Complex number. A **complex number** is an ordered pair $z = (x, y)$ of real numbers $x, y \in \mathbb{R}$, the first of which is called the **real part** and is denoted $\Re(z) = x$ and the second of which is called the **imaginary part** and is denoted $\Im(z) = y$. The set of all complex numbers is denoted $\mathbb{C}$.

The **sum** $z_1 + z_2$ and **product** $z_1 z_2$ of two complex numbers $z_1, z_2 \in \mathbb{C}$ are the complex numbers defined by the formulae

$$\begin{aligned} z_1 + z_2 &= (\Re(z_1) + \Re(z_2), \Im(z_1) + \Im(z_2)) \\ z_1 z_2 &= (\Re(z_1) \cdot \Re(z_2) - \Im(z_1) \cdot \Im(z_2), \Re(z_1) \cdot \Im(z_2) + \Im(z_1) \cdot \Re(z_2)). \end{aligned}$$

The **imaginary unit** i is the complex number $(0, 1)$, and we note that

$$i \cdot i = (0, 1) \cdot (0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0).$$

◆

Lemma 2.4.2 Real numbers as complex numbers. *There is an injective map* $\iota \colon \mathbb{R} \to \mathbb{C}$ *so that* $\iota(x_1 + x_2) = \iota(x_1) + \iota(x_2)$ *and* $\iota(x_1 x_2) = \iota(x_1) \cdot \iota(x_2)$ *for all real numbers* $x_1, x_2 \in \mathbb{R}$.

Proof. Let $\iota \colon \mathbb{R} \to \mathbb{C}$ be the function defined by the formula $\iota(x) = (x, 0)$. Since projection $\pi \colon \mathbb{C} \to \mathbb{R}$ onto the first factor is a left inverse to ι, ι is left-invertible and hence injective by (1) of Theorem 1.2.15 Invertibility. Moreover,

$$\iota(x_1 + x_2) = (x_1 + x_2, 0) = (x_1, 0) + (x_2, 0) = \iota(x_1) + \iota(x_2),$$

and

$$\begin{aligned}\iota(x_1x_2) &= (x_1x_2, 0) = (x_1x_2 - 0\cdot 0, x_1\cdot 0 - 0\cdot x_2)\\ &= (x_1, 0)\cdot(x_2, 0) = \iota(x_1)\cdot\iota(x_2)\end{aligned}$$

for all real numbers $x_1, x_2 \in \mathbb{R}$. ■

As in previous extensions of the usual number systems, we will use the injective map $\iota\colon \mathbb{R} \to \mathbb{C}$ defined in Lemma 2.4.2 Real numbers as complex numbers to view the real numbers $\mathbb{R}$ as a subset of the complex numbers $\mathbb{C}$. As a first example of this, we will denote the complex number (x, y) as $x + yi$ going forward.

Definition 2.4.3 Complex conjugate. The **conjugate** $\overline{z}$ of a complex number $x + yi \in \mathbb{C}$ is the complex number $\overline{z} = x - yi$. ◆

Lemma 2.4.4 Complex conjugates. *For all complex numbers $z \in \mathbb{C}$, $z\overline{z}$ is a non-negative real number, and $z\overline{z}$ is positive if and only if $z \neq 0$ is non-zero*

Proof. Let $z = x + yi \in \mathbb{C}$ be a complex number, and note that

$$z\overline{z} = (x + yi)(x - yi) = \left(x^2 + y^2\right) + 0i = x^2 + y^2$$

is a sum of squares of real numbers, and hence is non-negative. Moreover, $z\overline{z} = x^2 + y^2$ is positive unless $x = y = 0$, in which case $z = 0$ as well. ■

Definition 2.4.5 Magnitude of a complex number. The **magnitude** $|z|$ of a complex number $z \in \mathbb{C}$ is the non-negative real number defined by $|z|^2 = z\overline{z}$. ◆

Proposition 2.4.6 The field of complex numbers. *$(\mathbb{C}, +, \cdot\,)$ is a field.*

While the complex numbers $\mathbb{C}$ are a field, they are not an ordered field.

Proposition 2.4.7 Orders on the complex numbers. *There is no total order $\leq$ on the complex numbers $\mathbb{C}$ so that $(\mathbb{C}, +, \cdot, \leq)$ is an ordered field.*

So the addition of the non-real complex numbers to the real number system $\mathbb{R}$ comes at the expense of some of the ordered field axioms. We will see that this is also true of further extensions; for example, multiplication of **quaternions** is not commutative.

2.4.2 Infinity

Definition 2.4.8 Extended real number. An **extended real number** is either a real number or one of two **infinite** elements ∞ or $-\infty$. The set of extended real numbers is denoted $\overline{\mathbb{R}} = \mathbb{R} \cup \{\infty, -\infty\}$.

The standard ordering $\leq$ on the real numbers $\mathbb{R}$ is extended to a total order on $\overline{\mathbb{R}}$ by requiring $-\infty \leq x \leq \infty$ for all real numbers $x \in \mathbb{R}$. ◆

Definition 2.4.9 Interval. Let $a, b \in \overline{\mathbb{R}}$ be extended real numbers so that $a \leq b$.

1. *Closed interval.*

   The **closed interval** $[a, b]$ from a to b is the subset of the extended real numbers $\overline{\mathbb{R}}$ defined by

$$[a, b] = \left\{x \in \overline{\mathbb{R}} \;\middle|\; a \leq x \leq b\right\}.$$

2. *Open interval.*

The **open interval** (a, b) from a to b is the subset of the extended real numbers $\overline{\mathbb{R}}$ defined by

$$(a, b) = \left\{x \in \overline{\mathbb{R}} \,\middle|\, a < x < b\right\}.$$

3. *Half-closed, half-open interval.*

   The **half-closed, half-open interval** $[a, b)$ from a to b is the subset of the extended real numbers $\overline{\mathbb{R}}$ defined by

$$[a, b) = \left\{x \in \overline{\mathbb{R}} \,\middle|\, a \leq x < b\right\}.$$

4. *Closed interval.*

   The **half-open, half-closed interval** $(a, b]$ from a to b is the subset of the extended real numbers $\overline{\mathbb{R}}$ defined by

$$(a, b] = \left\{x \in \overline{\mathbb{R}} \,\middle|\, a < x \leq b\right\}.$$

◆

Example 2.4.10 Intervals. The following are examples of intervals:

- *(Extended) real numbers.*

  The real numbers $\mathbb{R}$ and the extended real numbers $\overline{\mathbb{R}}$ are open and closed intervals, respectively; concretely, $\mathbb{R} = (-\infty, \infty)$ and $\overline{\mathbb{R}} = [-\infty, \infty]$.

- *Dedekind cuts.*

  Given a real number $x \in \mathbb{R}$, the Dedekind cut X corresponding to x is $X = \mathbb{Q} \cap (-\infty, x)$.

▲

Theorem 2.4.11 Suprema and infima. *Any subset of the extended real numbers $\overline{\mathbb{R}}$ has a supremum in $\overline{\mathbb{R}}$.*

Proof. Let $U \subseteq \overline{\mathbb{R}}$ be a subset of the extended real numbers $\mathbb{R}$. If $\infty \in U$, then $\sup(U) = \max(U) = \infty$. If $\infty \notin U$ and $U \cap \mathbb{R}$ is not bounded from above, then

$$\sup(U) = \sup(U \cap \mathbb{R}) = \infty.$$

If $\infty \notin U$ and $U \cap \mathbb{R} \neq \emptyset$ is bounded from above, then

$$\sup(U) = \sup(U \cap \mathbb{R}).$$

Finally, if $\infty \notin U$ and $U \cap \mathbb{R} = \emptyset$, then $\sup(U) = -\infty$. ■

We note that the binary operations of addition and multiplication do not fully extend to the set $\overline{\mathbb{R}}$ of extended real numbers. Indeed, there are several **indeterminate forms** like $\infty + (-\infty)$ and $0 \cdot \infty$ which cannot be consistently resolved.

Text after the last subsection.

2.4.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

2.4.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

Text after the last section.

Part II

Point-Set Topology

Chapter 3

Metric Spaces

In this chapter, we introduce **metric spaces**, which generalize the notion of distance in the physical world, and are the natural setting for beginning our study of analysis. Analysis as a field of mathematics is concerned with the limiting behavior of mathematical objects. In a metric space, it makes sense to talk about the distance between any two points. Armed only with this notion of distance, we will define what it means

- for sequences and functions to **converge** to a **limit**;
- for sets to be **open** and/or closed;
- for maps to be **continuous**; and
- for metric spaces to be **complete**.

However, in spite of the strength and applicability of the results to be proven in this chapter, we will eventually find that we require further generalizations of the above notions, which will occur in the next chapter.

3.1 Introduction to Metric Geometry

To begin, we introduce a generalization of our usual notion of distance which is called a **metric**. The properties of physical distance make it an extremely useful tool for formalizing the analytical notions mentioned above.

3.1.1 Introduction to Metrics

Not all properties of physical distance are generic enough to generalize to other settings in which we might want to measure distance-like quantities. This question of which properties of a specific example (usually Euclidean space) are salient enough to merit requirement in a generalization is ubiquitous throughout mathematical exposition, and will recur frequently throughout this text.

Definition 3.1.1 Metric space. A real-valued function $\rho\colon X^2 \to \mathbb{R}$ on the Cartesian square X^2 of a set X, which is called the **ground set** and whose elements are called **points**, is a **metric** on X if it has the following properties:

1. *Non-degeneracy.*

   For any points $x_1, x_2 \in X$, $\rho(x_1, x_2) = 0$ if and only if $x_1 = x_2$.

2. *Symmetry.*

$\rho(x_1, x_2) = \rho(x_2, x_1)$ for any points $x_1, x_2 \in X$.

3. *Triangle inequality.*

   $\rho(x_1, x_3) \leq \rho(x_1, x_2) + \rho(x_2, x_3)$ for any points $x_1, x_2, x_3 \in X$.

In this case, the ordered pair (X, ρ) is called a **metric space**. ◆

Remark 3.1.2 Implied metrics. We will see that, in general, there are many metrics on a given set, no one of which may truly be considered canonical. In fact, even though some distinct metrics give identical answers to point-set topological concerns such as convergence and continuity, in most cases even a choice one of these classes of metrics is arbitrary in some respects.

However, sometimes, if a metric $\rho\colon X^2 \to \mathbb{R}$ on a set X is clear from context, previously specified, or implicitly present, some mathematicians will refer to the metric space (X, ρ) by only the underlying set X. We will try to avoid such ambiguities, and we will attempt to always explicitly specify the metric with which we will be working.

Example 3.1.3 Metric spaces. The following are examples of metrics and metric spaces:

1. Let $n \in \mathbb{N}$ be a natural number. The **Euclidean metric** $d\colon (\mathbb{R}^n)^2 \to \mathbb{R}$ on n-dimensional real coordinate space $\mathbb{R}^n$ is the function defined by the formula

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}.$$

   Similarly, the **Euclidean metric** $d\colon (\mathbb{C}^n)^2 \to \mathbb{R}$ on n-dimensional real coordinate space $\mathbb{C}^n$ is the function defined by the formula

$$d(\mathbf{w}, \mathbf{z}) = \sqrt{\sum_{k=1}^{n} |w_k - z_k|^2}.$$

   As defined above, the Euclidean metrics are metrics on $\mathbb{R}^n$ and $\mathbb{C}^n$, respectively. The metric spaces $(\mathbb{R}^n, d)$ and $(\mathbb{C}^n, d)$ are called n-dimensional **real Euclidean space** and n-dimensional **complex Euclidean space**, respectively. 1- and 2-dimensional real Euclidean space $(\mathbb{R}, d)$ and $(\mathbb{R}^2, d)$ are called the **real line** and the **Euclidean plane**, respectively, and 1-dimensional complex Euclidean space $(\mathbb{C}, d)$ is called the **complex plane**.

2. Consider the real-valued function $\rho\colon (0, \infty)^2 \to \mathbb{R}$ defined by the formula

$$\rho(x, y) = \left|\ln\left(\frac{x}{y}\right)\right|$$

   As defined above, ρ is a metric on the open interval $(0, \infty)$, and so $((0, \infty), \rho)$ is a metric space.

3. The **discrete metric** $\delta_X : X^2 \to \{0, 1\}$ on a set X is the function defined by the formula

$$\delta_X(x_1, x_2) = \begin{cases} 0 & x_1 = x_2 \\ 1 & x_1 \neq x_2. \end{cases}$$

   As defined above, the discrete metric δ_X is a metric on X, and the metric space (X, δ_X) is called a **discrete metric space**.

▲

The definition of a metric is pretty abstract, and it may be difficult to see how or why requiring only nondegeneracy, symmetry, and the triangle inequality would give a useful generalization of distance, especially given the strange and outlandish properties of the discrete metric described above, which we will explore later in this section. However, requiring the three above properties actually forces a metric to have many of the properties of our usual notion of distance, and so we can often use our knowledge of distance in the physical world to inform our intuition about metric spaces.

As an example, consider for a moment the notion of negative distance. In the real world, it makes no sense to consider negative distances between physical objects, and we would be equally confused by negative distances between points in a metric space. However, the defining axioms of a metric do not a priori explicitly forbid this nonsensical outcome from occurring. At first glance, it might seem like a metric might be allowed to take negative values, even though the three examples of a metric given above are certainly nonnegative. In fact, this is never the case.

Lemma 3.1.4 Metrics are non-negative. *A metric takes only non-negative values.*

Proof. Let ρ be a metric on a set X, and note that

$$\begin{aligned}\rho(x_1, x_2) &= \frac{2\rho(x_1, x_2)}{2} = \frac{\rho(x_1, x_2) + \rho(x_1, x_2)}{2} \\ &= \frac{\rho(x_1, x_2) + \rho(x_2, x_1)}{2} \geq \frac{\rho(x_1, x_1)}{2} = \frac{0}{2} = 0\end{aligned}$$

for all points $x_1, x_2 \in X$. ■

The above argument shows that, just as in the physical world, the distance between any two points in a metric space must be nonnegative. In fact, unlike many of the spaces which are studied by analysts (some of which we will encounter in the following chapters), metric spaces can be surprisingly accessible using only our intuitions about distance in the physical world. For example, the triangle inequality expresses that there are no “shortcuts” between two points in a metric space through a third point; in geometric terms, we require that no leg of a triangle may exceed in length the sum of the lengths of the other two legs. While the triangle inequality establishes an upper bound on the length of a leg of a triangle, the following property of a metric is called the **reverse triangle inequality**, and it establishes a lower bound on the length of a leg of a triangle.

Proposition 3.1.5 Reverse triangle inequality. *Let (X, ρ) be a metric space. Then*

$$\rho(x_1, x_2) \geq |\rho(x_1, x_3) - \rho(x_2, x_3)|$$

for all points $x_1, x_2, x_3 \in X$.

Proof. If $\rho(x_1, x_3) \leq \rho(x_2, x_3)$, then $\rho(x_1, x_3) - \rho(x_2, x_3) \leq 0$, and so

$$\begin{aligned}|\rho(x_1, x_3) - \rho(x_2, x_3)| &= \rho(x_2, x_3) - \rho(x_1, x_3) \\ &\leq \rho(x_2, x_1) + \rho(x_1, x_3) - \rho(x_1, x_3) = \rho(x_1, x_2).\end{aligned}$$

On the other hand, if $\rho(x_1, x_3) \geq \rho(x_2, x_3)$, then $\rho(x_1, x_3) - \rho(x_2, x_3) \geq 0$, and so

$$\begin{aligned}|\rho(x_1, x_3) - \rho(x_2, x_3)| &= \rho(x_1, x_3) - \rho(x_2, x_3) \\ &\leq \rho(x_1, x_2) + \rho(x_2, x_3) - \rho(x_2, x_3) = \rho(x_1, x_2).\end{aligned}$$

■

We have just seen that metric spaces often agree with our geometric intuitions about distance. However, we must be careful not to assume too much about the geometry of metric spaces in general; even though many of the usual properties of distance in the physical world are implied by the defining axioms of a metric space, there is still lots of room for geometric peculiarities. In particular, some of the metrics defined above disagree with our geometric intuitions in confusing ways. This is because they satisfy a stronger form of the triangle inequality than is true in the physical world. We will explore these **ultrametrics** in the reading questions for this section.

3.1.2 Constructing New Metrics

We will now see how to construct new metrics out of given metric spaces. Given a metric space, there is a natural metric on any subset of its ground set. This restriction of the metric structure to a subset is done in the obvious way; the distance between any points in the subset is just their distance when considered as points in the bigger space. In this way, a metric on a set is said to induce a metric on all of its subsets.

Definition 3.1.6 Induced metric. Let $U \subseteq X$ be a subset of a metric space (X, ρ). The **induced metric** on U is the restriction $\rho|_{U^2} : U^2 \to \mathbb{R}$ of ρ to U^2. The metric space $(U, \rho|_U)$ is called a **metric subspace** of the metric space (X, ρ).

In practice, we often write that ρ is a metric on U, even if the domain of ρ is strictly larger than U^2. We will not distinguish between a metric and its restriction to a smaller space. ◆

Example 3.1.7 Discrete metric subspaces. Let $U \subseteq X$ be a subset of a set X. The discrete metric δ_U on U is induced on U by the discrete metric δ_X on X; that is, $\delta_X|_{U^2} = \delta_U$. Thus the discrete metric space (U, δ_U) is a metric subspace of the discrete metric space (X, δ_X). ▲

Thus far, we have been very careful when working in a metric space to specify not only the ground set but also the metric. This is because we can in general find many metrics on a given underlying set. In fact, given one or more metrics on a set, there are several ways of constructing new metrics on that set. In particular, any finite positive pointwise linear combination of metrics is itself a metric.

Proposition 3.1.8 Linear combinations of metrics. *Any positive real linear combination of metrics is a metric.*

Proof. Let X be a set, and consider positive real numbers $\alpha_1, \ldots, \alpha_n \in (0, \infty]$ and metrics $\rho_1, \ldots, \rho_n \colon X^2 \to \mathbb{R}$ on X. Denote by $\rho \colon X^2 \to \mathbb{R}$ the linear combination defined by the formula

$$\rho(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_2).$$

First let $x_1, x_2 \in X$ be points, and suppose that $\rho(x_1, x_2) = 0$. Then

$$\sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_2) = \rho(x_1, x_2) = 0.$$

Since $\alpha_k > 0$ and $\rho_k(x_1, x_2) \geq 0$ for all indices $1 \leq k \leq n$, we conclude that $\rho_k(x_1, x_2) = 0$ for all indices $1 \leq k \leq n$. In particular, $x_1 = x_2$.

Conversely, now suppose that $x_1 = x_2$. Then

$$\rho(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \cdot 0 = 0.$$

Thus ρ is non-degenerate.

Note that

$$\rho(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \rho_k(x_2, x_1) = \rho(x_2, x_1)$$

for all points $x_1, x_2 \in X$, and so ρ is symmetric.

Finally, note that

$$\begin{aligned}\rho(x_1, x_3) &= \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_3) \leq \sum_{k=1}^{n} \alpha_k \left(\rho_k(x_1, x_2) + \rho_k(x_2, x_3)\right) \\ &= \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_2) + \sum_{k=1}^{n} \alpha_k \rho_k(x_2, x_3) = \rho(x_1, x_2) + \rho(x_2, x_3)\end{aligned}$$

for all points $x_1, x_2, x_3 \in X$, and so ρ satisfies the triangle inequality. Since ρ is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on X. ■

In particular, the sum of two metrics is a metric, and any positive real multiple of a metric is a metric.

3.1.3 Normed Vector Spaces

The ground sets of many of the metric spaces which we will investigate over the course of this chapter can also be considered as the ground sets of vector spaces. Moreover, in practically all of these cases, the linear-algebraic structure of the vector space is compatible with the analytical structure of the metric, in a way which we will make precise shortly. We now introduce a mathematical structure called a **normed vector space**, the study of which is called **functional analysis** and lies at the intersection of linear algebra and mathematical analysis. Aptly named, a normed vector space is a vector space equipped with the additional structure of a norm. Just as metrics generalize the notion of the distance between two points, norms on vector spaces generalize the related notion of the length of a vector.

Definition 3.1.9 Normed vector space. A real-valued function $\|\cdot\| : V \to \mathbb{R}$ on a real or complex vector space V is a **norm** on V if it has the following properties:

1. *Non-degeneracy.*

   For any vector $\mathbf{v} \in V$, $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

2. *Absolute homogeneity.*

   $\|\alpha\mathbf{v}\| = |\alpha| \|\mathbf{v}\|$ for any scalar α and vector $\mathbf{v} \in V$.

3. *Sub-additivity.*

   $\|\mathbf{v}_1 + \mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\|$ for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$.

In this case, the ordered pair $(V, \|\cdot\|)$ is called a **normed vector space**. ◆

Normed vector spaces are ubiquitous throughout mathematics. In fact, many of the vector spaces which you will have encountered in any course in linear algebra have canonical norms associated to them; lots of the vector spaces which are familiar to you have been normed vector spaces all along.

Example 3.1.10 Normed vector spaces. The following are examples of normed vector spaces:

1. Let $n \in \mathbb{N}$ be a natural number, and fix a positive real number $p > 0$. The p**-norm** $\|\cdot\|_p : \mathbb{R}^n \to \mathbb{R}$ on n-dimensional real coordinate space $\mathbb{R}^n$ is the real-valued function defined by the formula

$$\|\mathbf{x}\|_p = \left(\sum_{k=1}^{n} x_k^p\right)^{\frac{1}{p}}.$$

   Similarly, the p**-norm** $\|\cdot\|_p : \mathbb{C}^n \to \mathbb{R}$ on n-dimensional complex coordinate space $\mathbb{C}^n$ is the real-valued function defined by the formula

$$\|\mathbf{z}\|_p = \left(\sum_{k=1}^{n} z_k^p\right)^{\frac{1}{p}}.$$

   If $p \geq 1$, then the p-norms are norms on $\mathbb{R}^n$ and $\mathbb{C}^n$, and so $\left(\mathbb{R}^n, \|\cdot\|_p\right)$ and $\left(\mathbb{C}^n, \|\cdot\|_p\right)$ are normed vector spaces.

   In particular, the 2-norms $\|\cdot\|_2$ on $\mathbb{R}^n$ and $\mathbb{C}^n$ are called the **Euclidean norms** on $\mathbb{R}^n$ and $\mathbb{C}^n$. Because the Euclidean norms are so ubiquitous, we will often drop the subscripts in our notation, so that $\|\cdot\| = \|\cdot\|_2$.

2. Let $n \in \mathbb{N}$ be a natural number. The families of p-norms on n-dimensional real and complex coordinate space $\mathbb{R}^n$ and $\mathbb{C}^n$ can be extended to the case $p = \infty$. The ∞**-norm** $\|\cdot\|_\infty : \mathbb{R}^n \to \mathbb{R}$ on $\mathbb{R}^n$ is the real-valued function defined by the formula

$$\|\mathbf{x}\|_\infty = \max_{1 \leq k \leq n} |x_k|.$$

   Similarly, the ∞**-norm** $\|\cdot\|_\infty : \mathbb{C}^n \to \mathbb{R}$ on $\mathbb{C}^n$ is the real-valued function defined by the formula

$$\|\mathbf{z}\|_\infty = \max_{1 \leq k \leq n} |z_k|.$$

   The ∞-norms are norms on $\mathbb{R}^n$ and $\mathbb{C}^n$, and so $(\mathbb{R}^n, \|\cdot\|_\infty)$ and $(\mathbb{C}^n, \|\cdot\|_\infty)$ are normed vector spaces.

3. Any Hermitian inner product $\langle \cdot\,, \cdot \rangle$ on a real or complex vector space V induces a norm $\|\cdot\| : V^2 \to \mathbb{R}$ on V defined by the formula

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

   In fact, for all natural numbers $n \in \mathbb{N}$, the norms produced in this manner by the dot product on n-dimensional real and complex coordinate space $\mathbb{R}^n$ and $\mathbb{C}^n$ are precisely the Euclidean norms. For this reason, the dot product is often called the **Euclidean inner product** on $\mathbb{R}^n$ and $\mathbb{C}^n$.

The above examples show that many of our familiar vector spaces have canonical norms readily available. ▲

Remark 3.1.11 p-norms on 1-dimensional vector spaces. Note that

$$\|x\|_p = (|x|^p)^{\frac{1}{p}} = |x| = \max(\{|x|\}) = \|x\|_\infty$$

for all real numbers $x \in \mathbb{R}$, and similarly

$$\|z\|_p = (|z|^p)^{\frac{1}{p}} = |z| = \max(\{|z|\}) = \|z\|_\infty$$

for all complex numbers $z \in \mathbb{C}$; that is, all of the p norms (including the case $p = \infty$) on the real numbers $\mathbb{R}$ and the complex numbers $\mathbb{C}$ agree. Such statements do not hold in higher dimensions, although we will see that the p-norms remain related for topological concerns.

Just as in the case of metrics, we would prefer to consider only non-negative vector lengths. It makes no sense to consider negative lengths of physical vectors, and it is just as difficult to make sense of negative values of a norm in a normed vector space. Thankfully, the defining axioms of a norm prevent this from occurring.

Lemma 3.1.12 Norms are non-negative. *A norm takes only non-negative values.*

Proof. Let $\|\cdot\| : V \to \mathbb{R}$ be a norm on a real or complex vector space V, and note that

$$\begin{aligned}\|\mathbf{v}\| &= \frac{2\|\mathbf{v}\|}{2} = \frac{\|\mathbf{v}\| + \|\mathbf{v}\|}{2} = \frac{\|\mathbf{v}\| + |-1|\,\|\mathbf{v}\|}{2} \\ &= \frac{\|\mathbf{v}\| + \|-\mathbf{v}\|}{2} \geq \frac{\|\mathbf{v} - \mathbf{v}\|}{2} = \frac{\|\mathbf{0}\|}{2} = \frac{0}{2} = 0\end{aligned}$$

for all vectors $\mathbf{v} \in V$. ■

Theorem 3.1.13 Normed vector spaces are metric spaces. *A norm $\|\cdot\| : V \to \mathbb{R}$ on a real or complex vector space V induces a metric $\rho\colon V^2 \to \mathbb{R}$ defined by the formula*

$$\rho(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1 - \mathbf{v}_2\|.$$

Proof. First, let $\mathbf{v}_1, \mathbf{v}_2 \in V$ be vectors, and suppose that $\rho(\mathbf{v}_1, \mathbf{v}_2) = 0$. Then

$$\|\mathbf{v}_1 - \mathbf{v}_2\| = \rho(\mathbf{v}_1, \mathbf{v}_2) = 0,$$

and so $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$; that is, $\mathbf{v}_1 = \mathbf{v}_2$. Conversely, if $\mathbf{v}_1 = \mathbf{v}_2$, then

$$\rho(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1 - \mathbf{v}_2\| = \|\mathbf{0}\| = 0.$$

Thus ρ is non-degenerate.

Since

$$\begin{aligned}\rho(\mathbf{v}_1, \mathbf{v}_2) &= \|\mathbf{v}_1 - \mathbf{v}_2\| = |-1|\,\|\mathbf{v}_1 - \mathbf{v}_2\| \\ &= \|\mathbf{v}_2 - \mathbf{v}_1\| = \rho(\mathbf{v}_2, \mathbf{v}_1)\end{aligned}$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$, ρ is symmetric.

Finally, since

$$\begin{aligned}\rho(\mathbf{v}_1, \mathbf{v}_3) &= \|\mathbf{v}_1 - \mathbf{v}_3\| = \|(\mathbf{v}_1 - \mathbf{v}_2) + (\mathbf{v}_2 - \mathbf{v}_3)\| \\ &\leq \|\mathbf{v}_1 - \mathbf{v}_2\| + \|\mathbf{v}_2 - \mathbf{v}_3\| = \rho(\mathbf{v}_1, \mathbf{v}_2) + \rho(\mathbf{v}_2, \mathbf{v}_3)\end{aligned}$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in V$, ρ satisfies the triangle inequality. Since ρ is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on V. ■

So, given a normed vector space, we may easily convert it into a metric space by defining the distance between two vectors to be the norm of their difference. Many important metric spaces arise as a result of this construction.

Example 3.1.14 ***p*-metrics.** Let $n \in \mathbb{N}$ be a natural number, and fix an extended real number $1 \leq p \leq \infty$. The metrics d_p on n-dimensional real and complex coordinate spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ by the p-norms $\|\cdot\|_p$ are called ***p*-metrics**. These p-metrics are defined by the formula

$$d_p(\mathbf{w}, \mathbf{z}) = \|\mathbf{w} - \mathbf{z}\|_p = \left(\sum_{k=1}^{n} |w_k - z_k|^p \right)^{\frac{1}{p}}.$$

In particular, we note that the metrics $d_2 = d$ induced by the Euclidean norms $\|\cdot\|_2 = \|\cdot\|$ are the Euclidean metrics. ▲

So any norm structure on a real or complex vector space can be reanalyzed as also providing a metric structure. However, a metric on a real or complex vector space need not be induced by a norm. Moreover, there exist metric spaces which cannot be constructed from a normed vector space. So, while normed vector spaces are very useful for defining specific metric spaces, they do not capture all of what can be said about metric spaces, and so will not merit any more special consideration just yet. With this illuminating example of normed vector spaces in mind, we now resume our study of metric spaces in their full generality.

In the next section, we introduce one of the most important ideas in the field of analysis: that of convergence to a limit.

3.1.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Computing distances in metric spaces. Compute the distance between the given points in the indicated metric space.

1. Let d be the Euclidean metric on the real coordinate plane $\mathbb{R}^2$.

- **(a)** Compute $d(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.
- **(b)** Compute $d(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

2. Let d_3 be the 3-metric on the real coordinate plane $\mathbb{R}^2$.

- **(a)** Compute $d_3(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.

3. Let d_∞ be the ∞-metric on the real coordinate plane $\mathbb{R}^2$.

- **(a)** Compute $d_\infty(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.
- **(b)** Compute $d_\infty(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

4. Let $\delta_{\mathbb{R}^2}$ be the discrete metric on the real coordinate plane $\mathbb{R}^2$.

(a) Compute $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.

(b) Compute $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

Verifying metric properties. Determine whether or not the given function is a metric.

5. Let $d\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be the Euclidean metric on the real numbers $\mathbb{R}$; that is,

$$d(x_1, x_2) = |x_1 - x_2|$$

for all real numbers $x_1, x_2 \in \mathbb{R}$. Verify that d is a metric on $\mathbb{R}$.

6. Let $\delta_X : X \times X \to \mathbb{R}$ be the discrete metric on a set X; that is,

$$\delta_X(x_1, x_2) = \begin{cases} 0 & x_1 = x_2 \\ 1 & x_1 \neq x_2 \end{cases}$$

for all points $x_1, x_2 \in X$. Verify that δ_X is a metric on X.

7. Let $\rho\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be the real-valued function defined by the formula

$$\rho(x_1, x_2) = x_1 - x_2.$$

Determine whether or not ρ is a metric on the real numbers $\mathbb{R}$.

8. Subspaces of discrete metric spaces. Let $U \subseteq X$ be a subset of a discrete metric space (X, δ_X). Verify that the metric $\delta_X|_U$ induced on U by δ_X is the discrete metric δ_U on U.

9. Scalar multiples of metrics. Give an example of a real number $\alpha \in \mathbb{R}$ and a metric $\rho\colon X \times X \to \mathbb{R}$ so that $\alpha\rho$ is not a metric.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.1.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task.*

1. From pseudometrics to metrics. A real-valued function $\rho\colon X^2 \to \mathbb{R}$ on the Cartesian square X^2 of a set X, which is called the **ground set** and whose elements are called **points**, is a **pseudometric** on X if it has the following properties:

(a) *Identity.*

$\rho(x, x) = 0$ for any point $x \in X$.

(b) *Symmetry.*

$\rho(x_1, x_2) = \rho(x_2, x_1)$ for any points $x_1, x_2 \in X$.

(c) *Triangle inequality.*

$\rho(x_1, x_3) \leq \rho(x_1, x_2) + \rho(x_2, x_3)$ for any points $x_1, x_2, x_3 \in X$.

In this case, the ordered pair (X, ρ) is called a **pseudometric space**.

(a) Let ρ be a pseudometric on a set X, and consider the binary relation $\equiv_\rho$ on X defined so that $x_1 \equiv_\rho x_2$ if and only if $\rho(x_1, x_2) = 0$. Prove that $\equiv_\rho$ is an equivalence relation on X.

(b) Prove that a pseudometric ρ descends to a well-defined metric $\overline{\rho}$ on the quotient set $X/{\equiv_\rho}$ defined by the formula

$$\overline{\rho}\left([x_1]_{\equiv_\rho}, [x_2]_{\equiv_\rho}\right) = \rho(x_1, x_2).$$

2. **Norm equivalence.** Two norms $\|\cdot\|_1, \|\cdot\|_2 : V \to \mathbb{R}$ on a real or complex vector space V are called **equivalent** if there are positive real numbers $A, B \in (0, \infty)$ so that

$$A\|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1 \leq B\|\mathbf{v}\|_2$$

for all vectors $\mathbf{v} \in V$.

(a) Prove that norm equivalence is an equivalence relation on the set of norms on V.

(b) Prove that for a given natural number $n \in \mathbb{N}$, any two p-norms $\|\cdot\|_p$ on n-dimensional real coordinate space $\mathbb{R}^n$ for $1 \leq p \leq \infty$ are all equivalent.

3. **Ultrametrics.** A real-valued function $\rho\colon X^2 \to \mathbb{R}$ on the Cartesian square X^2 of a set X is called an **ultrametric** on X, which is called the **ground set** and whose elements are called **points** if it satisfies the following conditions:

- *Non-degeneracy.*

  For any points $x_1, x_2 \in X$, $\rho(x_1, x_2) = 0$ if and only if $x_1 = x_2$.

- *Symmetry.*

  $\rho(x_1, x_2) = \rho(x_2, x_1)$ for any points $x_1, x_2 \in X$.

- *Strong triangle inequality.*

  $\rho(x_1, x_3) \leq \max(\{\rho(x_1, x_2), \rho(x_2, x_3)\})$ for any points $x_1, x_2, x_3 \in X$.

If a real-valued function $\rho\colon X^2 \to \mathbb{R}$ is an ultrametric on X, then the ordered pair (X, ρ) is called an ultrametric space.

(a) Prove that any ultrametric is a metric.

(b) Fix a prime number $p \in \mathbb{N}$. The **p-adic valuation** $|\cdot|_p : \mathbb{Q} \to \mathbb{Q}$ is a rational-valued function defined by the formula

$$|q|_p = \begin{cases} 0 & q = 0 \\ p^{-k_p(q)} & q \neq 0, \end{cases}$$

where $k_p(q) \in \mathbb{Z}$ is the unique integer so that $q = p^{k_p(q)}\left(\frac{a}{b}\right)$ for some integers $a, b \in \mathbb{Z}$ which are not divisible by p.

Prove that the p-adic valuation $|\cdot|_p$ induces an ultrametric $\rho_p\colon \mathbb{Q}^2 \to \mathbb{R}$ defined by the formula

$$\rho_p(q_1, q_2) = |q_1 - q_2|_p.$$

This ultrametric is called the **p-adic metric** on $\mathbb{Q}$.

(c) Prove that for all points $x_1, x_2, x_3 \in X$ in an ultrametric space (X, ρ), at least two of the distances $\rho(x_1, x_2)$, $\rho(x_2, x_3)$, and $\rho(x_1, x_3)$ are equal.

The geometry of non-Archimedean metric spaces differs in several ways from our intuitions about Euclidean geometry informed by our experience of distance in the physical world. We will see several of these differences over the course of this chapter.

4. Reverse triangle inequality for norms. Prove that

$$|\|\mathbf{v}_1\| - \|\mathbf{v}_2\|| \leq \|\mathbf{v}_1 - \mathbf{v}_2\|$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$ in a normed vector space $(V, \|\cdot\|)$.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.2 Convergence and Limits

In this section, we introduce the most fundamental notion of all of mathematical analysis: that of **convergence**. Convergence is a formalization of limiting behavior and closeness in metric spaces and, as we will see in the next chapter, other related analytical settings. In this section we will be concerned with convergence in the setting of abstract metric spaces, which can be interpreted as a rigorous formalization of the informal phrase "eventually close." In the following section, we will gain a more detailed understanding of limiting behavior in less general example of the Euclidean spaces. Note also that in this chapter, we will be concerned only with the convergence of sequences in metric spaces and maps between metric spaces, but in the next chapter, we will make generalizations which in some ways subsume the discussion in this section.

3.2.1 Sequential Convergence

First, we will investigate the limiting behavior of a sequence in a metric space. Informally, a sequence in a metric space **converges** to a point if the values of the sequence eventually grows arbitrarily close to that point; that is, a sequence converges to a point if the values of the sequence are close to that point for large indices. Such a sequence is called **convergent**, and the point to which it converges (which we will see to be uniquely determined by this property) is called its **limit**.

Definition 3.2.1 Sequential convergence/divergence. A sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) **converges** to a point $x \in X$ if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(x_n, x) < \epsilon$$

for all indices $n \geq n_\epsilon$.

A sequence of points in a metric space is called **convergent** if it converges to a point and **divergent** if it does not converge to any point. ◆

So a sequence converges to a point in a metric space if the values of the sequence can be made arbitrarily close to that point solely by choosing large enough indices, and the sequence diverges if this cannot be done.

Example 3.2.2 Sequential convergence/divergence. Consider the sequence $(x_n)_{n=1}^{\infty}$ of real numbers defined by the formula

$$x_n = \frac{1}{n}.$$

We will show that $(x_n)_{n=1}^{\infty}$ converges to 0 in the Euclidean metric d on $\mathbb{R}$.

Indeed, let $\epsilon > 0$ be a positive real number. By the Archimedean principle, there is a natural number $n_\epsilon \in \mathbb{N}$ so that $\epsilon \cdot n_\epsilon > 1$. For all indices $n \geq n_\epsilon$, we observe that

$$d(x_n, 0) = \left|\frac{1}{n} - 0\right| = \frac{1}{|n|} = \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

In summary, we have shown that for all positive real numbers $\epsilon > 0$, there is an index $n_\epsilon \in \mathbb{N}$ so that $d(x_n, 0) < \epsilon$ for all indices $n \geq n_\epsilon$. Thus $(x_n)_{n=1}^{\infty}$ converges to 0.

Note that, in this example, the data of the sequence, the limiting value, the ground set, and the metric are necessary for convergence. By altering any one of these, we may invalidate the convergence we just proved. For example, if we consider $(x_n)_{n=1}^{\infty}$ as a sequence in the metric subspace $(0, \infty)$ of the real line $\mathbb{R}$ consisting of only positive real numbers, then the sequence diverges, since it does not converge in the Euclidean metric to a positive real number.

Similarly, if we change the metric, then the sequence $(x_n)_{n=1}^{\infty}$ may not converge. We have just proven that this sequence is convergent in the Euclidean metric d by finding a point $0 \in \mathbb{R}$ to which it converges, but it is divergent in the discrete metric $\delta_{\mathbb{R}}$. All this is to say that all of the data of a sequence, a metric space, and a purported limiting value are all necessary for claims of convergence or divergence. ▲

Ideally, we would like "the **limit**" of a convergent sequence in a metric space, which is "the point" to which that sequence converges, to be unique with this property. That is, if a sequence in a metric space converges to a point, then we would like this point to be the only such point to which it converges, which a priori need not be the case. Thankfully, the limits of convergent sequences in metric spaces are indeed unique.

Proposition 3.2.3 Uniqueness of sequential limits in a metric space. *Convergent sequences of points in metric spaces converge to unique points.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a metric space (X, ρ), and suppose that $(x_n)_{n=0}^{\infty}$ converges to two points $x, y \in X$. We will show that $x = y$.

To that end, let ϵ be a positive real number, and note that $\frac{\epsilon}{2}$ is a positive real number. There are thus natural numbers $m_{\frac{\epsilon}{2}}, n_{\frac{\epsilon}{2}} \in \mathbb{N}$ so that $\rho(x_n, x) < \frac{\epsilon}{2}$ for all indices $n \geq m_{\frac{\epsilon}{2}}$ and $\rho(x_n, y) < \frac{\epsilon}{2}$ for all indices $n \geq n_{\frac{\epsilon}{2}}$.

Let $p_\epsilon = \max\left(m_{\frac{\epsilon}{2}}, n_{\frac{\epsilon}{2}}\right)$, and note that $p_\epsilon \geq m_{\frac{\epsilon}{2}}$ and $p_\epsilon \geq n_{\frac{\epsilon}{2}}$. We conclude that

$$\rho(x, y) \leq \rho(x, x_{p_\epsilon}) + \rho(x_{p_\epsilon}, y) = \rho(x_{p_\epsilon}, x) + \rho(x_{p_\epsilon}, y) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

In summary, we have shown that $\rho(x, y) < \epsilon$ for all positive real numbers $\epsilon > 0$. Since $\rho(x, y) \geq 0$ by Lemma 3.1.4 Metrics are non-negative, we conclude that $\rho(x, y) = 0$ and hence $x = y$. ■

So if a sequence in a metric space converges to a point, then that point is unique with this property, and is called the limit of the sequence. Many texts in analysis place limits of both sequences and maps (we will define this shortly) in the spotlight; limits are certainly the most important analytical concept introduced in this text. However, we have tried to emphasize the more abstract notion of convergence over that of a limit. We have just proven that the two notions are equivalent in a metric space, but in the next chapter we will explore an analytical setting in which convergent sequences need not converge to a unique limit.

Definition 3.2.4 Sequential limit. The unique point $x \in X$ to which a convergent sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) converges is called the **limit** of said sequence, and it is denoted

$$\lim_{n\to\infty} x_n = x.$$

◆

Example 3.2.5 Sequential limits. Let $x \in X$ be a point in a metric space (X, ρ), and consider the constant sequence $(x_n)_{n=1}^{\infty}$ of points defined by the formula $x_n = x$. For all positive real numbers $\epsilon > 0$ and all indices $n \geq 0$, we observe that

$$\rho(x_n, x) = \rho(x, x) = 0 < \epsilon.$$

Thus $(x_n)_{n=0}^{\infty} = (x)_{n=0}^{\infty}$ converges to x; that is,

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} x = x.$$

▲

Note that the notation introduced in the above definition is somewhat ambiguous; nowhere does it specify with respect to which metric a sequence converges or in which ground set the values of the sequence are considered to lie. We will try to avoid these ambiguities, but (at least in this text) both the ground set and the metric should always be clear from context. If the expression to the right of the limit operator is a real or complex number, then it's a safe bet to assume that convergence is in the standard Euclidean metric d on the real numbers $\mathbb{R}$ or complex numbers $\mathbb{C}$, respectively. Similarly, if the expression to the right of the limit operator is a real or complex coordinate vector, then you should assume that we are working in real or complex coordinate space, respectively, taken with the Euclidean metric. We will properly investigate Euclidean convergence in the section to come.

We next establish a necessary and sufficient condition for sequence convergence in any metric space; that the distances between the points of the sequence and the limiting value converge to zero in the real line. The proof of the following proposition follows directly from the definitions of convergence and limit.

Proposition 3.2.6 Sequential convergence. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a metric space (X, ρ). Then for any point $x \in X$,*

$$\lim_{n\to\infty} x_n = x$$

if and only if

$$\lim_{n\to\infty} \rho(x_n, x) = 0.$$

Proof. We observe that

$$d(\rho(x_n, x), 0) = |\rho(x_n, x) - 0| = |\rho(x_n, x)| = \rho(x_n, x)$$

for all indices $n \in \mathbb{N}$. First suppose that

$$\lim_{n\to\infty} x_n = x,$$

and let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that $\rho(x_n, x) < \epsilon$ for all indices $n \geq n_\epsilon$. In particular,

$$d(\rho(x_n, x), 0) = \rho(x_n, x) < \epsilon$$

for all indices $n \geq n_\epsilon$, and so

$$\lim_{n\to\infty} \rho(x_n, x) = 0.$$

Conversely, now suppose that

$$\lim_{n\to\infty} \rho(x_n, x) = 0,$$

and let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that $d(\rho(x_n, x), 0) < \epsilon$ for all indices $n \geq n_\epsilon$. In particular,

$$\rho(x_n, x) = d(\rho(x_n, x), 0) < \epsilon$$

for all indices $n \geq n_\epsilon$, and so

$$\lim_{n\to\infty} x_n = x.$$

■

The implications of the above result are that ***in order to understand sequential convergence and divergence in any metric space, it suffices to understand the convergence and divergence of sequences of real numbers in the Euclidean metric***. In the next section, we will examine the properties of convergence and divergence in the Euclidean metric, and use the standard ordering on the real numbers to enrich our understanding of limiting behavior in the real numbers.

We have just seen that the question of convergence or divergence in an arbitrary metric space can be reduced to the question of convergence or divergence of a related sequence of real numbers in the Euclidean metric. We will begin our study of these Euclidean limits by generalizing to the metric induced on real coordinate space by an arbitrary p-norm. In real coordinate space, sequential convergence in the Euclidean metric is equivalent to sequential convergence in any of the metrics induced on real coordinate space by a p-norm. In fact, something much stronger and much more useful is true; we can reduce the question of convergence or divergence of any sequence of complex coordinate vectors in such a metric to the question of the convergence or divergence of the component sequences in the Euclidean metric on the real numbers.

Theorem 3.2.7 Sequential convergence in the p-metric. *Let $m \in \mathbb{N}$ be a natural number, and consider a sequence $(\mathbf{x}_n)_{n=0}^\infty$ of vectors $\mathbf{x}_n = (x_{j,n})_{j=1}^m \in \mathbb{R}^m$.*

For any vector $\mathbf{x} = (x_j)_{j=1}^m$ and real number $1 \leq p < \infty$, the sequence $(\mathbf{x}_n)_{n=0}^\infty$ converges to $\mathbf{x}$ in the p-metric d_p on $\mathbb{R}^m$ if and only if for all indices $1 \leq j \leq m$, the sequence $(x_{j,n})_{n=0}^\infty$ converges to x_j in the Euclidean metric d on $\mathbb{R}$.

Proof. First suppose that $(\mathbf{x}_n)_{n=0}^\infty$ converges to $\mathbf{x}$ in the p-metric d_p on $\mathbb{R}^m$. Let $1 \leq j \leq m$, and consider a positive real number $\epsilon > 0$. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$d_p(\mathbf{x}_n, \mathbf{x}) < \epsilon$$

for all indices $n \geq n_\epsilon$. We observe that

$$d(x_{j,n}, x_j)^p = |x_{j,n} - x_j|^p \leq \sum_{k=1}^{m} |x_{k,n} - x_k|^p$$
$$= \|\mathbf{x}_n - \mathbf{x}\|_p^p = d_p(\mathbf{x}_n, \mathbf{x})^p < \epsilon^p,$$

so that $d(x_{j,n}, x_j) < \epsilon$ for all indices $n \geq n_\epsilon$. In particular, $(x_{j,n})_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$.

Conversely, now suppose that for all indices $1 \leq j \leq m$, $(x_{j,n})_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$. Let $\epsilon > 0$ be a positive real number. Then $\frac{\epsilon}{m^{\frac{1}{p}}} > 0$ is a positive real number, and so for each index $1 \leq j \leq m$ there is a natural number $n_{j,\epsilon}$ so that

$$d(x_{j,n}, x_j) < \frac{\epsilon}{m^{\frac{1}{p}}}$$

for all indices $n \geq n_{j,\epsilon}$. Let $n_\epsilon = \max_{1 \leq j \leq m} n_{j,\epsilon}$, and note that

$$d_p(\mathbf{x}_n, \mathbf{x})^p = \sum_{j=1}^{m} |x_{j,n} - x_j|^p = \sum_{j=1}^{m} d(x_{j,n}, x_j)^p$$
$$< \sum_{j=1}^{m} \left(\frac{\epsilon}{m^{\frac{1}{p}}}\right)^p = \sum_{j=1}^{m} \frac{\epsilon^p}{m} = \epsilon^p$$

and therefore $d_p(\mathbf{x}_n, \mathbf{x}) < \epsilon$ for all indices $n \geq n_\epsilon$. We conclude that $(\mathbf{x}_n)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the p-metric d_p on $\mathbb{R}^n$. ■

Theorem 3.2.7 Sequential convergence in the p-metric implies that, in order to know everything about sequential convergence and divergence in the metric induced on real coordinate space by a p-norm, it suffices to know about the convergence and divergence of sequences of real numbers in the Euclidean metric. Moreover, all of the p-metrics induced on real coordinate space by the p-norms are equivalent with respect to the question of sequential convergence and divergence; that is, the convergence of a sequence of real coordinate vectors converges in the metric induced on real coordinate space by a p-norm does not depend on p.

The reduction of problems of sequential convergence in a metric space to that of convergence of one or more sequences of real numbers in the Euclidean metric will be a common theme in this chapter. In some respect, this reduction is possible because

1. given a point, a metric necessarily associates the other points of a space with the positive real numbers $(0, \infty)$; and
2. convergence in the Euclidean metric d can be defined solely in terms of the standard order $\leq$ on the real line $\mathbb{R}$.

In the following section, we will discuss sequential convergence and divergence in the Euclidean metric on the real numbers in greater detail.

3.2.2 Pointwise and Uniform Convergence

Later in this chapter, we will establish metrics on certain collections of maps with codomain a given metric space. However, without yet defining a metric on these sets, there are two related but distinct modes of convergence of a sequence of such maps. A sequence of maps to a metric space is said to **converge**

pointwise to a limit map if the images of each input in the domain converge to their images under the limit map. Stronger than pointwise convergence is the notion of **uniform convergence**, which requires that the speed of pointwise convergence not depend on the given input.

Definition 3.2.8 Pointwise convergence/divergence. Let $(f_n)_{n=0}^{\infty}$ be a sequence of maps $f_n \colon X \to Y$ from a set X to a metric space (Y, ρ).

- *Pointwise convergence.*

  The sequence $(f_n)_{n=0}^{\infty}$ **converges pointwise** to a map $f \colon X \to Y$ from X to Y if for all inputs $x \in X$, the sequence $(f_n(x))_{n=0}^{\infty}$ of corresponding outputs $f_n(x) \in Y$ converges to $f(x)$.

  The sequence $(f_n)_{n=0}^{\infty}$ is called **pointwise convergent** if it converges pointwise to a map from X to Y.

- *Pointwise divergence.*

  The sequence $(f_n)_{n=0}^{\infty}$ is called **pointwise divergent** if it does not converge pointwise to any map from X to Y.

◆

Example 3.2.9 Pointwise convergence. Consider the sequence $(f_n)_{n=0}^{\infty}$ of functions $f_n \colon [0,1] \to [0,1]$ defined by the formulae $f_n(x) = x^n$. Since

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} x^n = \begin{cases} 0 & 0 \le x < 1 \\ 1 & x = 1, \end{cases}$$

the sequence $(f_n)_{n=0}^{\infty}$ converges pointwise to the function $f \colon [0,1] \to [0,1]$ defined piecewise by the formula

$$f(x) = \begin{cases} 0 & 0 \le x < 1 \\ 1 & x = 1. \end{cases}$$

▲

Because of the uniqueness of limits of sequences in metric spaces, if a sequence of maps to a metric space converges pointwise to a map, then this map is unique with this property. In this case, the map is called the **pointwise limit** of the sequence.

Definition 3.2.10 Pointwise limit. For any pointwise convergent sequence $(f_n)_{n=0}^{\infty}$ of maps $f_n \colon X \to Y$ from a set X to a metric space (Y, ρ), the unique map $f \colon X \to Y$ from X to Y to which $(f_n)_{n=0}^{\infty}$ converges pointwise is called the **pointwise limit** of $(f_n)_{n=0}^{\infty}$, and is denoted

$$\lim_{n\to\infty} f_n = f.$$

◆

While some properties of a pointwise convergent sequence of maps to a metric space are shared with its pointwise limit, pointwise convergence does not in general preserve all desirable properties of maps. In particular, Example 3.2.9 Pointwise convergence implies that a pointwise limit of a sequence of continuous maps need not itself be continuous. We will explore continuity in metric spaces and its relationship with convergence later in this chapter. In order to preserve these properties, we will need a stronger notion of convergence. Informally, a sequence of maps to a metric space is said to **converge**

uniformly it converges pointwise at roughly the same rate at each point in the domain.

Definition 3.2.11 Uniform convergence. Let $(f_n)_{n=0}^{\infty}$ be a sequence of maps $f_n : X \to Y$ from a set X to a metric space (Y, ρ).

- *Uniform convergence.*

  The sequence $(f_n)_{n=0}^{\infty}$ **converges uniformly** to a map $f : X \to Y$ from X to Y if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that
  $$\rho\left(f_n\left(x\right), f\left(x\right)\right) < \epsilon$$
  for all inputs $x \in X$ and indices $n \geq n_\epsilon$.

  The sequence $(f_n)_{n=0}^{\infty}$ is called **uniformly convergent** if it converges uniformly to a map from X to Y.

- *Uniform divergence.*

  The sequence $(f_n)_{n=0}^{\infty}$ is called **uniformly divergent** if it does not converge uniformly to any map from X to Y.

◆

Example 3.2.12 Uniform convergence/divergence. The following are examples of uniformly convergent and divergent sequences of maps:

- Consider the sequence $(f_n)_{n=0}^{\infty}$ of functions $f_n\colon\ [0,1] \to [0,1]$ defined by the formulae $f_n\left(x\right) = x^n$, and let $f\colon\ [0,1] \to [0,1]$ be the function defined piecewise by the formula
  $$f\left(x\right) = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1. \end{cases}$$
  We observe that
  $$\begin{aligned} d\left(f_n\left(x\right), f\left(x\right)\right) &= \left|f_n\left(x\right) - f\left(x\right)\right| = \left|x^n - 0\right| = \left|x^n\right| = x^n \\ &> \left(\frac{1}{2^{\frac{1}{n}}}\right)^n = \frac{1}{2} \end{aligned}$$
  for all positive integers $n \geq 1$ and inputs $\frac{1}{\sqrt[n]{2}} \leq x < 1$. We conclude that $(f_n)_{n=0}^{\infty}$ does not converge uniformly to f.

- Consider the sequence $(f_n)_{n=0}^{\infty}$ of functions $f_n : \mathbb{R} \to [-1,1]$ defined by the formulae
  $$f_n\left(x\right) = \frac{\sin\left(nx\right)}{n},$$
  and let $f : \mathbb{R} \to [-1,1]$ be the function defined by the formula $f\left(x\right) = 0$. For all positive real numbers $\epsilon > 0$, the Archimedean principle implies that $n_\epsilon \cdot \epsilon > 1$ for some positive integer $n_\epsilon \geq 1$. We observe that
  $$\begin{aligned} d\left(f_n\left(x\right), f\left(x\right)\right) &= \left|f_n\left(x\right) - f\left(x\right)\right| = \left|\frac{\sin\left(nx\right)}{n} - 0\right| = \left|\frac{\sin\left(nx\right)}{n}\right| \\ &= \frac{\left|\sin\left(nx\right)\right|}{n} \leq \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon \end{aligned}$$
  for all indices $n \geq n_\epsilon$ and inputs $x \in \mathbb{R}$. We conclude that $(f_n)_{n=0}^{\infty}$ converges uniformly to f.

▲

We remark that pointwise and uniform convergence are distinct concepts; that is, there are sequences of maps to a metric space which converge pointwise but not uniformly. However, the opposite relationship is not true: uniform convergence is a stronger condition on sequences of maps to a metric space than pointwise convergence; that is, a sequence of maps to a metric space converges uniformly only if it converges pointwise.

Proposition 3.2.13 Pointwise and uniform convergence. *If a sequence of maps from a set to a metric space converges uniformly, then it converges pointwise.*

Proof. Suppose that a sequence $(f_n)_{n=0}^{\infty}$ of maps $f_n \colon X \to Y$ from a set X to a metric space (Y, ρ) converges uniformly to a map $f \colon X \to Y$ from X to Y. Then for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho\left(f_n(x), f(x)\right) < \epsilon$$

for all indices $n \geq n_\epsilon$ and inputs $x \in X$. In particular, for each input $x \in X$,

$$\rho\left(f_n(x), f(x)\right) < \epsilon$$

for all indices $n \geq n_\epsilon$, and so the sequence $(f_n(x))_{n=0}^{\infty}$ converges to $f(x)$. We conclude that $(f_n)_{n=0}^{\infty}$ converges pointwise to f. ■

Taken with the uniqueness of the pointwise limit of a sequence of maps to a metric space, Proposition 3.2.13 Pointwise and uniform convergence immediately implies that if a sequence of maps to a metric space converges uniformly to a map, then this map is unique with this property. In this case, the map is called the **uniform limit** of the sequence.

Definition 3.2.14 Uniform limit. For any uniformly convergent sequence $(f_n)_{n=0}^{\infty}$ of maps $f_n \colon X \to Y$ from a set X to a metric space (Y, ρ), the unique map $f \colon X \to Y$ from X to Y to which $(f_n)_{n=0}^{\infty}$ converges uniformly is called the **uniform limit** of $(f_n)_{n=0}^{\infty}$, and is denoted

$$\operatorname*{unif\,lim}_{n \to \infty} f_n = f.$$

◆

Example 3.2.15 Uniform limit. Consider the sequence $(f_n)_{n=0}^{\infty}$ of functions $f_n \colon \mathbb{R} \to [-1, 1]$ defined by the formulae

$$f_n(x) = \frac{\sin(nx)}{n},$$

and let $f \colon \mathbb{R} \to [-1, 1]$ be the function defined by the formula $f(x) = 0$. Since $(f_n)_{n=0}^{\infty}$ converges uniformly to f as shown in Example 3.2.12 Uniform convergence/divergence, f is the uniform limit of $(f_n)_{n=0}^{\infty}$; that is,

$$\operatorname*{unif\,lim}_{n \to \infty} f_n = f.$$

We might also write that

$$\operatorname*{unif\,lim}_{n \to \infty} \frac{\sin(nx)}{n} = 0.$$

▲

So a sequence of maps to a metric space converges uniformly only if it converges pointwise. The reverse implication, however, is not in general true. In the following sections, we will see that, unlike in the case of pointwise convergence, many of the desirable properties of maps between metric spaces are preserved under uniform convergence.

3.2.3 Map Convergence

Just as we can use a metric to define the convergence of a sequence of points or maps in a metric space, we can similarly consider the convergence of maps between metric spaces. In our definition of sequence convergence, we considered a sequence to converge to a limit if the points of the sequence could be made arbitrarily close to the limit point solely by choosing large enough indices. We will similarly consider a map between metric spaces to **converge** to a limit in the codomain at a point in the domain if we can make outputs close to the limit in the codomain solely by choosing inputs close to the point in the domain.

In the case of sequential convergence, we used the standard ordering on the natural numbers to induce a notion of limiting behavior by considering arbitrarily large indices. For **map convergence**, however, we use the metric in the domain to consider arbitrarily close points to the point in question.

Definition 3.2.16 Map convergence. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y).

- *Map convergence.*

  f **converges** at a point $c \in X$ to a point $l \in Y$ if for all positive real numbers $\epsilon > 0$, there exists a positive real number $\delta > 0$ so that

  $$\rho_Y\left(f\left(x\right), l\right) < \epsilon$$

  for all inputs $x \in X$ so that $0 < \rho_X\left(x, c\right) < \delta$.

  The map f is called **convergent** at a point $c \in X$ if it converges at c to some point $l \in Y$.

- *Map divergence.*

  The map f is called **divergent** at a point $c \in X$ if it does not converge at c to any point $l \in Y$.

◆

Example 3.2.17 Map convergence. Consider the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined piecewise by the formula

$$f\left(x\right) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

As the graph of $y = f\left(x\right)$ suggests, f converges to 0 at 0 in the Euclidean metric d on $\mathbb{R}$.

Indeed, let $\epsilon > 0$ be a positive real number, and let $\delta = \epsilon > 0$. Then

$$\begin{aligned} d\left(f\left(x\right), 0\right) &= \left|x \sin\left(\frac{1}{x}\right) - 0\right| = \left|x\right| \cdot \left|\sin\left(\frac{1}{x}\right)\right| \\ &\leq \left|x\right| = \left|x - 0\right| = d\left(x, 0\right) < \delta = \epsilon \end{aligned}$$

for all inputs $x \in \mathbb{R}$ so that $0 < d\left(x, 0\right) < \delta$. ▲

In analogy to the case of sequential convergence, in order to study map convergence between metric spaces, it suffices to study the convergence of real-valued functions on metric spaces.

Proposition 3.2.18 Map convergence. *Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). For each point $l \in Y$, let $g_l\colon X \to \mathbb{R}$ be the real-valued function on X defined by the formula*

$$g_l(x) = \rho_Y(f(x), l).$$

Then f converges to l at c if and only if g_l converges to 0 at c.

Proof. We first observe that

$$d(g_l(x), 0) = |\rho_Y(f(x), l) - 0| = |\rho_Y(f(x), l)| = \rho_Y(f(x), l)$$

for all $x \in X$.

Suppose first that f converges to l at c, and let $\epsilon > 0$ be a positive real number. Then there is some positive real number $\delta > 0$ so that

$$\rho_Y(f(x), l) < \epsilon$$

for all inputs $x \in X$ so that $0 < \rho_X(x, c) < \delta$. We observe that

$$d(g_l(x), 0) = \rho_Y(f(x), l) < \epsilon$$

for all inputs $x \in X$ so that $0 < \rho_X(x, c) < \delta$, and so g_l converges to 0 at c.

Conversely, now suppose that g_l converges to 0 at c, and let $\epsilon > 0$ be a positive real number. Then there is some positive real number $\delta > 0$ so that

$$d(g_l(x), 0) < \epsilon$$

for all inputs $x \in X$ so that $0 < \rho_X(x, c) < \delta$. We observe that

$$\rho_Y(f(x), l) = d(g_l(x), 0) < \epsilon$$

for all inputs $x \in X$ so that $0 < \rho_X(x, c) < \delta$, and so f converges to l at c. ■

So we can reduce the question of convergence for maps between metric spaces to that of Euclidean convergence of real-valued functions on metric spaces. We will study the important topic of convergence of such real-valued functions under the Euclidean metric on the real numbers in detail in Section 3.3 Limits in the Euclidean Metric.

The following theorem establishes another necessary and sufficient condition for map convergence; a map between metric spaces converges if and only if all convergent sequences in the domain can be "pushed through" to convergent sequences in the codomain.

Theorem 3.2.19 Sequential characterization of map convergence. *Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). For all points $c \in X$ and $l \in Y$, f converges to l at c if and only if for all sequences $(x_n)_{n=0}^{\infty}$ of points $x_n \in X \setminus \{c\}$, if*

$$\lim_{n\to\infty} x_n = c,$$

then

$$\lim_{n\to\infty} f(x_n) = l.$$

Proof. First suppose that f converges to l at c, and consider a sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X \setminus \{c\}$ which converges to c. We observe that $\rho_X(x_n, c) > 0$

for all indices $n \in \mathbb{N}$.

Let $\epsilon > 0$ be a positive real number. Then there is a positive real number $\delta > 0$ so that

$$\rho_Y (f(x), l) < \epsilon$$

for all points $x \in X$ so that $0 < \rho_X (x, c) < \delta$. Since $\delta > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$0 < \rho_X (x_n, c) < \delta$$

for all indices $n \geq n_\epsilon$. In particular, this implies that

$$\rho_Y (f(x_n), l) < \epsilon$$

for all indices $n \geq n_\epsilon$. We conclude that $(f(x_n))_{n=0}^{\infty}$ converges to l.

Conversely, now suppose that f does not converge to l at c. Then there is a positive real number ϵ so that for all positive real numbers $\delta > 0$, there is a point $x \in X \setminus \{c\}$ so that

$$0 < \rho_X (x, c) < \delta$$

but

$$\rho_Y (f(x), l) \geq \epsilon.$$

In particular, let $(\delta_n)_{n=0}^{\infty}$ be a sequence of positive real numbers $\delta_n \geq 0$ which converges in the Euclidean metric to 0. For each index $n \in \mathbb{N}$, there is a point $x_n \in X \setminus \{c\}$ so that

$$0 < \rho_X (x_n, c) < \delta_n$$

but

$$\rho_Y (f(x_n), l) \geq \epsilon.$$

Since

$$d(\rho_X (x_n, c), 0) = |\rho_X (x_n, c) - 0| = \rho_X (x_n, c) < \delta_n$$

for all indices $n \in \mathbb{N}$, Proposition 3.2.6 Sequential convergence implies that $(x_n)_{n=0}^{\infty}$ converges to c but $(f(x_n))_{n=0}^{\infty}$ does not converge to l. ■

The above result relates map convergence and sequential convergence. Note, however, that this theorem and the uniqueness of sequential limits in metric spaces do not necessarily imply the uniqueness of the limit of a convergent map between metric spaces. In fact, ***map limits need not be uniquely determined in some metric spaces***.

Example 3.2.20 Non-uniqueness of map limits. Consider the discrete metric δ_B on the set $B = \{0, 1\}$, and consider the function $f : B \to B$ defined by the formula $f(b) = b$. Then f converges to 0 at 0, and f converges to 1 at 0.

Indeed, let $c = 0$, and consider $l \in B$. For all positive real numbers $\epsilon > 0$, let $\delta = \frac{1}{2}$, and note that there are no points $b \in B$ so that $0 < \delta_B (b, c) < \frac{1}{2}$. Thus f converges to l at c vacuously. ▲

So the limit of a convergent map between metric spaces is not necessarily unique. However, this undesirable situation may be rectified by restricting our attention to certain points in the domain of such a map. Note that the uniqueness of the limit in Example 3.2.20 Non-uniqueness of map limits failed because the point $c = 0$ is far away from the rest of the set B; that is, there are no points close to but distinct from 0; equivalently, any sequence in B converging to 0 must eventually take only 0 as a value. Such points are called **isolated points**, and other points are called **limit points**.

Definition 3.2.21 Accumulation; isolation. Consider a subset $U \subseteq X$ of a metric space (X, ρ).

- *Limit point.*

  We say that U **accumulates** at a point $c \in X$ if for all positive real numbers $\epsilon > 0$, there is a point $u \in U$ so that $0 < \rho(u, c) < \epsilon$. In this case, c is called a **limit point** (or **accumulation point**) of U in X. The set of such limit points of U is called the **derived set** of U, and is denoted U'.

- *Isolated point.*

  We say that a point $c \in X$ is **isolated** from U if c is not a limit point of U; that is, c is isolated from U if there is a positive real number $\epsilon > 0$ so that $\rho(u, c) \geq \epsilon$ for all points $u \in U$ so that $u \neq c$. If $c \in U$ is isolated from U, then c is called an **isolated point** of U.

◆

Example 3.2.22 Limit points and isolated points. The following are examples of limit points and isolated points in metric spaces:

- *Euclidean space.*

  Let $n \geq 1$ be a positive integer, and consider a real number $1 \leq p < \infty$. If $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, then consider for each positive real number $\epsilon > 0$ the vector $\mathbf{x}_\epsilon = \left(x_1 + \frac{\epsilon}{2}, x_2, \ldots, x_n\right)$. We observe that

$$\begin{aligned} d_p(\mathbf{x}, \mathbf{x}_\epsilon) &= \left(\left|x_1 - \left(x_1 + \frac{\epsilon}{2}\right)\right|^p + |x_2 - x_2|^p + \cdots + |x_n - x_n|^p\right)^{\frac{1}{p}} \\ &= \left|-\frac{\epsilon}{2}\right| = \frac{\epsilon}{2}. \end{aligned}$$

  In particular,

$$0 < d_p(\mathbf{x}, \mathbf{x}_\epsilon) < \epsilon,$$

  and so $\mathbf{x}$ is a limit point of $\mathbb{R}^n$. Since $\mathbf{x}$ was chosen arbitrarily, we conclude that every point of $\mathbb{R}^n$ is a limit point.

- *Discrete space.*

  Let (X, δ_X) be a discrete metric space, and consider a point $x \in X$. Since the values of δ_X are 0 and 1, there is no point $y \in X$ so that $0 < \delta_X(x, y) < 1$. Thus x is an isolated point of X. Since x was chosen arbitrarily, we conclude that every point of (X, δ_X) is an isolated point.

- The only limit point of the set $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ in $\mathbb{R}$ is 0.

- The only limits point of the set $\left\{(-1)^n + \frac{1}{n}\right\}_{n=0}^{\infty}$ in $\mathbb{R}$ are 1 and -1.

▲

A point in a metric space is a limit point of a subset if and only if it is the limit of a sequence of points in that subset which are distinct from that point.

Proposition 3.2.23 Sequential characterization of limit points. *Let $U \subseteq X$ be a subset of a metric space (X, ρ). A point $c \in X$ is a limit point of U if and only if*

$$c = \lim_{n \to \infty} u_n$$

for some sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U \setminus \{c\}$.

Proof. First suppose that c is a limit point of U. Then for each positive integer $n \geq 1$, $\frac{1}{n} > 0$, and so there is a point $u_n \in U \setminus \{c\}$ so that

$$0 < \rho(u_n, c) < \frac{1}{n}.$$

We claim that $(u_n)_{n=0}^{\infty}$ converges to c. Indeed, for any positive real number $\epsilon > 0$, the Archimedean principle implies that there is a natural number $n_\epsilon > \frac{1}{\epsilon}$. We observe that

$$\rho(u_n, c) < \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon$$

for all indices $n \geq n_\epsilon$. We conclude that

$$\lim_{n\to\infty} u_n = c.$$

Conversely, now suppose that

$$\lim_{n\to\infty} u_n = c$$

for some sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U \setminus \{c\}$. Let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon > 0$ so that $\rho(u_n, c) < \epsilon$ for all indices $n \geq n_\epsilon$. In particular,

$$0 < \rho(u_{n_\epsilon}, c) < \epsilon,$$

since $u_{n_\epsilon} \neq c$. We conclude that c is a limit point of U. ■

As we have seen, the limit of a convergent map between metric spaces at an isolated point of the domain need not be unique. However, the limit of such a map at a limit point of the domain is unique. This follows immediately from the above results.

Corollary 3.2.24 Uniqueness of map limits. *Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). If f converges at a limit point $c \in X$ of X, then it converges to a unique point $l \in Y$.*

Proof. Suppose that f converges to two points $l, m \in Y$ at a limit point $c \in X$ of X. Since c is a limit point of X,

$$c = \lim_{n\to\infty} x_n$$

for some sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X \setminus \{c\}$ by Proposition 3.2.23 Sequential characterization of limit points. Theorem 3.2.19 Sequential characterization of map convergence now implies that

$$l = \lim_{n\to\infty} f(x_n) = m.$$

■

Analogously to the case of unique sequential limits, the above corollary leads naturally to the following definition, which is that of the unique **limit** of a map between metric spaces at a limit point.

Definition 3.2.25 Map limit. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). If f converges at a limit point $c \in X$ of X, the unique point $l \in Y$ to which f converges at c is called the **limit** of f at c, and is denoted

$$\lim_{x\to c} f(x) = l.$$

◆

Again, note the ambiguity of the above notation; it is entirely unclear with respect to which metrics this convergence is satisfied. Indeed, as in the case of sequential limits, maps between sets may converge in one pair of metrics and diverge in another, or may converge to different limits in different metrics. We will try to avoid this ambiguity if at all possible. Analogously to the case of sequential limits, if the map inside the limit operator is a map between real coordinate spaces, it will usually be safe to assume that convergence is with respect to the relevant Euclidean metrics.

Example 3.2.26 Map limits. Consider the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = 2x + 1$. We claim that

$$\lim_{x\to c} f(x) = 2c + 1$$

for all points $c \in \mathbb{R}$.

Indeed, let $\epsilon > 0$ be a positive real number, and consider the positive real number $\delta = \frac{\epsilon}{2} > 0$. We observe that

$$\begin{aligned} d(f(x), 2c+1) = |2x + 1 - (2c+1)| = |2(x-c)| \\ = 2|x-c| = 2d(x,c) < 2\delta = \epsilon \end{aligned}$$

for all points $x \in \mathbb{R}$ so that $0 < d(x,c) < \delta$, and so f converges to $2c+1$ at c. Since c is a limit point of $\mathbb{R}$, we conclude that

$$\lim_{x\to c} f(x) = 2c + 1.$$

▲

In this section, we have seen several ways in which the convergence of objects in an arbitrary metric space can be reduced to convergence of related objects in the real line $\mathbb{R}$ taken with the Euclidean metric d. In this way, the topic of convergence in the Euclidean metric d on $\mathbb{R}$ plays a central role in the point-set topology of metric spaces. In the next section, we will investigate the limiting behavior of sequences of real numbers and of real-valued functions on metric spaces in greater depth.

3.2.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Convergence in your own words. Describe *in your own words* the given mode of convergence. ***Do not*** just copy the definition.

1. Describe *in your own words* what it means for a sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) to converge to a limit $x \in X$.
2. Describe *in your own words* what it means for a sequence $(f_n)_{n=0}^{\infty}$ of maps $f\colon X \to Y$ from a set X to a metric space (Y, ρ) to converge pointwise to a limit $f\colon X \to Y$.

3. Describe *in your own words* what it means for a sequence $(f_n)_{n=0}^{\infty}$ of maps $f\colon X \to Y$ from a set X to a metric space (Y, ρ) to converge uniformly to a limit $f\colon X \to Y$.

4. Describe *in your own words* what it means for a map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) to converge at a point $c \in X$ to a limit $l \in Y$.

Sequential convergence. Determine whether the following sequences of real numbers converge or diverge in the Euclidean metric d on the real numbers $\mathbb{R}$. If the given sequence converges, find the limit.

5. Consider the sequence $(a_n)_{n=1}^{\infty}$ of real numbers $a_n \in \mathbb{R}$ defined by the formula

$$a_n = \frac{\cos(n\pi)}{n}.$$

Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

6. Consider the sequence $(b_n)_{n=0}^{\infty}$ of real numbers $b_n \in \mathbb{R}$ defined by the formula

$$b_n = n\cos(n\pi).$$

Determine whether the sequence $(b_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

7. Consider the sequence $(c_n)_{n=0}^{\infty}$ of real numbers $c_n \in \mathbb{R}$ defined by the formula

$$c_n = n\sin(n\pi).$$

Determine whether the sequence $(c_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

8. Consider the sequence $(d_n)_{n=0}^{\infty}$ of real numbers $d_n \in \mathbb{R}$ defined by the formula

$$d_n = \frac{2}{n+1}.$$

Determine whether the sequence $(d_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

9. Consider the sequence $(e_n)_{n=0}^{\infty}$ of real numbers $e_n \in \mathbb{R}$ defined by the formula

$$e_n = \frac{2n}{n+1}.$$

Determine whether the sequence $(e_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

10. Consider the sequence $(f_n)_{n=0}^{\infty}$ of real numbers $f_n \in \mathbb{R}$ defined by the formula

$$f_n = \sqrt{\frac{2n}{n+1}}.$$

Determine whether the sequence $(f_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

11. Consider the sequence $(g_n)_{n=0}^{\infty}$ of real numbers $g_n \in \mathbb{R}$ defined by the formula

$$g_n = \frac{3^n}{n!}.$$

Determine whether the sequence $(g_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Pointwise and uniform convergence. Determine whether the following sequences of real-valued functions converge or diverge both pointwise and uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If the given sequence converges pointwise, find the pointwise limit. Express your answer using limit notation. If the given sequence converges uniformly, find the uniform limit.

12. Consider the sequence $(a_n)_{n=1}^{\infty}$ of real-valued functions $a_n\colon \mathbb{R} \to \mathbb{R}$ defined by the formulae

$$a_n(x) = \left(x - \frac{1}{n}\right)^2.$$

(a) Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

(b) Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

13. Consider the sequence $(b_n)_{n=1}^{\infty}$ of real-valued functions $b_n\colon [-10, 10] \to \mathbb{R}$ defined by the formulae

$$b_n(x) = \frac{x^2}{n}.$$

(a) Determine whether the sequence $(b_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

(b) Determine whether the sequence $(b_n)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

14. Consider the sequence $(c_n)_{n=1}^{\infty}$ of real-valued functions $c_n\colon \mathbb{R} \to \mathbb{R}$ defined by the formulae

$$c_n(x) = \frac{x^2}{n}.$$

(a) Determine whether the sequence $(c_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

(b) Determine whether the sequence $(c_n)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

15. Consider the sequence $(d_n)_{n=0}^{\infty}$ of real-valued functions $d_n\colon [-10,10] \to \mathbb{R}$ defined by the formulae

$$d_n(x) = \frac{1}{3^{x+n}}.$$

(a) Determine whether the sequence $(d_n)_{n=0}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

(b) Determine whether the sequence $(d_n)_{n=0}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

Limit points and isolated points. Find all limit points and isolated points of the given subset of the Euclidean plane $\mathbb{R}^2$.

16. Let U be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$U = \left\{\mathbf{x} \in \mathbb{R}^2 \,\middle|\, 1 \leq \|\mathbf{x}\| < 2\right\}.$$

Find all limit points and isolated points of U in $\mathbb{R}^2$.

17. Let V be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$V = \left\{\mathbf{x} \in \mathbb{R}^2 \,\middle|\, \|\mathbf{x}\| \in \mathbb{N}\right\}.$$

Find all limit points and isolated points of V in $\mathbb{R}^2$.

18. Let W be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$W = \left\{(x,y) \in \mathbb{R}^2 \,\middle|\, x, y \in \mathbb{Z}\right\}.$$

Find all limit points and isolated points of W in $\mathbb{R}^2$.

19. Let X be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$X = \left\{(x,y) \in \mathbb{R}^2 \,\middle|\, x, y \in \mathbb{Q}\right\}.$$

Find all limit points and isolated points of X in $\mathbb{R}^2$.

20. Let Y be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$Y = \left\{(x,y) \in \mathbb{R}^2 \,\middle|\, x, y \notin \mathbb{Q}\right\}.$$

Find all limit points and isolated points of Y in $\mathbb{R}^2$.

21. Let Z be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$Z = \left\{\left(\frac{1}{m}, \frac{1}{n}\right) \in \mathbb{R}^2 \,\middle|\, m, n \in \mathbb{N} \setminus \{0\}\right\}.$$

Find all limit points and isolated points of Z in $\mathbb{R}^2$.

Map convergence. Determine whether or not the given real-valued function converges at the given input. If the function converges, find the limit and express your answer using limit notation.

22. Consider the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined piecewise by the formula

$$f(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0. \end{cases}$$

Determine whether or not f converges at 0. If f converges at 0, find the limit and express your answer using limit notation.

23. Consider the real-valued function $g\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$g(y) = y^2 - 2y + 1.$$

Determine whether or not g converges at 0. If g converges at 0, find the limit and express your answer using limit notation.

24. Consider the real-valued function $h\colon [0, \infty) \to \mathbb{R}$ defined by the formula

$$h(z) = \sqrt{z}.$$

Determine whether or not h converges at 4. If h converges at 4, find the limit and express your answer using limit notation.

Hint. Use the fact that $|\sqrt{x} - 2| \cdot |\sqrt{x} + 2| = |x - 4|$ for all non-negative real numbers $x \geq 0$.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.2.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Eventually constant sequences.** A sequence $(x_n)_{n=0}^{\infty}$ of elements $x_n \in X$ of a set X is called **eventually constant** if there is a point $x \in X$ and a natural number $n_0 \in \mathbb{N}$ so that $x_n = x$ for all indices $n \geq n_0$.

(a) Prove that any eventually constant sequence of points in a metric space is convergent.

(b) Prove that any sequence of points in a discrete metric space is convergent if and only if it is eventually constant.

(c) Give an example of a sequence of real numbers which converges in the Euclidean metric d on $\mathbb{R}$ but diverges in the discrete metric $\delta_{\mathbb{R}}$ on $\mathbb{R}$.

2. **Sequential convergence in the infinity metric.** Let $m \in \mathbb{N}$ be a natural number, and consider a sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ of points $\mathbf{x}_n = (x_{1,n}, \ldots, x_{m,n}) \in \mathbb{R}^m$. Prove that for any vector $\mathbf{x} = (x_1, \ldots, x_m) \in \mathbb{R}$, the following are equivalent:

(a) The sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the ∞-metric d_∞ on $\mathbb{R}^m$.

(b) For all indices $1 \leq j \leq m$, the sequence $(x_{j,n})_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$.

Hint. You may want to revisit the proof of Theorem 3.2.7 Sequential convergence in the p-metric for inspiration.

3. **Alternative characterization of uniform convergence.** Let $(f_n)_{n=0}^{\infty}$ be a sequence of maps $f_n \colon X \to Y$ from a non-empty set $X \neq \emptyset$ to a metric space (Y, ρ). Prove that $(f_n)_{n=0}^{\infty}$ converges to a map $f \colon X \to Y$ from X to Y if and only if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$0 \leq \sup_{x \in X} \rho\left(f_n\left(x\right), f\left(x\right)\right) < \epsilon$$

for all indices $n \geq n_\epsilon$.

4. **Uniqueness of map limits.** Let $f \colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) which has at least two distinct points. Prove the following partial converse to Corollary 3.2.24 Uniqueness of map limits: If f converges at a point $c \in X$ to a unique point $l \in Y$, then c is a limit point of X.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.3 Limits in the Euclidean Metric

In the previous section, we defined and explored the convergence and divergence of sequences and maps in the context of general metric spaces. In this section, however, we turn our attention to the limiting behavior in Euclidean spaces specifically. Euclidean spaces, being (at least in low dimensions) good models of the space in which we live, are the metric spaces in which our intuitions about distance are most accurate and useful.

3.3.1 Limits and Arithmetic

Much of our work in this text will rely on limits taken in Euclidean spaces, and the material introduced in this section is fundamental to the field of real analysis. Euclidean spaces are not just metric objects, however; as normed vector spaces, Euclidean spaces have a vector space structure as well as a norm and an inner product. We will see that Euclidean limits preserve this additional structure; for example, linear combinations of convergent sequences of real coordinate vectors converge in the Euclidean metric to the linear combination of the limits. In fact, this holds true in any normed vector space.

Proposition 3.3.1 Limits and vector arithmetic. *Let $\|\cdot\|$ be a norm on a real or complex vector space V, and consider the induced metric ρ on V.*

1. *Vector addition.*

   For all convergent sequences $(\mathbf{v}_n)_{n=0}^{\infty}$ and $(\mathbf{w}_n)_{n=0}^{\infty}$ of vectors $\mathbf{v}_n, \mathbf{w}_n \in V$,

$$\lim_{n\to\infty} \left(\mathbf{v}_n + \mathbf{w}_n\right) = \lim_{n\to\infty} \mathbf{v}_n + \lim_{n\to\infty} \mathbf{w}_n.$$

2. *Scalar multiplication.*

For all scalars α and convergent sequences $(\mathbf{v}_n)_{n=0}^{\infty}$ of vectors $\mathbf{v}_n \in V$,

$$\lim_{n\to\infty} \alpha\mathbf{v}_n = \alpha \lim_{n\to\infty} \mathbf{v}_n.$$

3. *Norm.*

   For all convergent sequences $(\mathbf{v}_n)_{n=0}^{\infty}$ of vectors $\mathbf{v}_n \in V$,

$$\lim_{n\to\infty} \|\mathbf{v}_n\| = \left\|\lim_{n\to\infty} \mathbf{v}_n\right\|.$$

4. *Inner product.*

   If the norm $\|\cdot\|$ is induced by an inner product $\langle\,\cdot\,,\,\cdot\,\rangle$ on V, then for all convergent sequences $(\mathbf{v}_n)_{n=0}^{\infty}$ and $(\mathbf{w}_n)_{n=0}^{\infty}$ of vectors $\mathbf{v}_n, \mathbf{w}_n \in V$,

$$\lim_{n\to\infty} \langle\mathbf{v}_n, \mathbf{w}_n\rangle = \left\langle \lim_{n\to\infty} \mathbf{v}_n, \lim_{n\to\infty} \mathbf{w}_n \right\rangle.$$

Proof.

1. Let $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$ and $\mathbf{w} = \lim_{n\to\infty} \mathbf{w}_n$. Let $\epsilon > 0$ be a positive real number. Since $\frac{\epsilon}{2} > 0$ is a positive real number, there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that

$$\rho(\mathbf{v}_n, \mathbf{v}) < \frac{\epsilon}{2}$$

   for all indices $n \geq m_\epsilon$ and

$$\rho(\mathbf{w}_n, \mathbf{w}) < \frac{\epsilon}{2}$$

   for all indices $n \geq n_\epsilon$. Let $p_\epsilon = \max(m_\epsilon, n_\epsilon) \in \mathbb{N}$, and note that

$$\begin{aligned}\rho(\mathbf{v}_n + \mathbf{w}_n, \mathbf{v} + \mathbf{w}) &= \|\mathbf{v}_n + \mathbf{w}_n - (\mathbf{v} + \mathbf{w})\| = \|\mathbf{v}_n - \mathbf{v} + \mathbf{w}_n - \mathbf{w}\| \\ &\leq \|\mathbf{v}_n - \mathbf{v}\| + \|\mathbf{w}_n - \mathbf{w}\| = \rho(\mathbf{v}_n, \mathbf{v}) + \rho(\mathbf{w}_n, \mathbf{w}) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon\end{aligned}$$

   for all indices $n \geq p_\epsilon$. We conclude that

$$\lim_{n\to\infty} (\mathbf{v}_n + \mathbf{w}_n) = \mathbf{v} + \mathbf{w}.$$

2. Let $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$. Let $\epsilon > 0$ be a positive real number. Note that

$$\begin{aligned}\rho(\alpha\mathbf{v}_n, \alpha\mathbf{v}) &= \|\alpha\mathbf{v}_n - \alpha\mathbf{v}\| = |\alpha|\,\|\mathbf{v}_n - \mathbf{v}\| \\ &= |\alpha|\,\rho(\mathbf{v}_n, \mathbf{v})\end{aligned}$$

   for all indices $n \in \mathbb{N}$. If $\alpha = 0$, then

$$\rho(\alpha\mathbf{v}_n, \alpha\mathbf{v}) = |\alpha|\,\rho(\mathbf{v}_n, \mathbf{v}) = 0 \cdot \rho(\mathbf{v}_n, \mathbf{v}) = 0 < \epsilon$$

   for all indices $n \in \mathbb{N}$. We conclude that in this case,

$$\lim_{n\to\infty} \alpha\mathbf{v}_n = \alpha\mathbf{v}.$$

   Now suppose that $\alpha \neq 0$. Since $\frac{\epsilon}{|\alpha|} > 0$ is a positive real number, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(\mathbf{v}_n, \mathbf{v}) < \frac{\epsilon}{|\alpha|}$$

for all indices $n \geq m_\epsilon$. In particular,

$$\rho(\alpha \mathbf{v}_n, \alpha \mathbf{v}) = |\alpha| \rho(\mathbf{v}_n, \mathbf{v}) < |\alpha| \left(\frac{\epsilon}{|\alpha|}\right) = \epsilon.$$

We conclude that in this case,

$$\lim_{n\to\infty} \alpha \mathbf{v}_n = \alpha \mathbf{v}.$$

3. Let $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$. Let $\epsilon > 0$ be a positive real number. Then there is a natural number n_ϵ so that

$$\rho(\mathbf{v}_n, \mathbf{v}) < \epsilon$$

for all indices $n \geq n_\epsilon$. We observe that Proposition 3.1.5 Reverse triangle inequality implies that

$$\begin{aligned} d(\|\mathbf{v}_n\|, \|\mathbf{v}\|) &= |\|\mathbf{v}_n\| - \|\mathbf{v}\|| = |\|\mathbf{v}_n - \mathbf{0}\| - \|\mathbf{v} - \mathbf{0}\|| \\ &= |\rho(\mathbf{v}_n, \mathbf{0}) - \rho(\mathbf{v}, \mathbf{0})| \leq \rho(\mathbf{v}_n, \mathbf{v}) < \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$, and so

$$\lim_{n\to\infty} \|\mathbf{v}_n\| = \|\mathbf{v}\|.$$

4. Let $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$ and $\mathbf{w} = \lim_{n\to\infty} \mathbf{w}_n$. Let $\epsilon > 0$ be a positive real number. We observe that

$$\begin{aligned} d(\langle \mathbf{v}_n, \mathbf{w}_n\rangle, \langle \mathbf{v}, \mathbf{w}\rangle) &= |\langle \mathbf{v}_n, \mathbf{w}_n\rangle - \langle \mathbf{v}, \mathbf{w}\rangle| \\ &= |\langle \mathbf{v}_n, \mathbf{w}_n\rangle - \langle \mathbf{v}, \mathbf{w}_n\rangle + \langle \mathbf{v}, \mathbf{w}_n\rangle - \langle \mathbf{v}, \mathbf{w}\rangle| \\ &= |\langle \mathbf{v}_n, \mathbf{w}_n\rangle - \langle \mathbf{v}, \mathbf{w}_n\rangle| + |\langle \mathbf{v}, \mathbf{w}_n\rangle - \langle \mathbf{v}, \mathbf{w}\rangle| \\ &= |\langle \mathbf{v}_n - \mathbf{v}, \mathbf{w}_n\rangle| + |\langle \mathbf{v}, \mathbf{w}_n - \mathbf{w}\rangle| \\ &\leq \|\mathbf{v}_n - \mathbf{v}\| \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \|\mathbf{w}_n - \mathbf{w}\| \\ &= \rho(\mathbf{v}_n, \mathbf{v}) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho(\mathbf{w}_n, \mathbf{w}) \end{aligned}$$

for all indices $n \in \mathbb{N}$ by the Cauchy--Schwarz inequality. There are four cases, depending on whether or not $\mathbf{v}$ and $\mathbf{w}$ are zero or non-zero vectors. If $\mathbf{v} = \mathbf{w} = \mathbf{0}$, then

$$d(\langle \mathbf{v}_n, \mathbf{w}_n\rangle, \langle \mathbf{v}, \mathbf{w}\rangle) \leq \rho(\mathbf{v}_n, \mathbf{v}) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho(\mathbf{w}_n, \mathbf{w}) = 0 + 0 = 0 < \epsilon.$$

If $\mathbf{v} = \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$, then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(\mathbf{v}_n, \mathbf{v}) < \frac{\epsilon}{\|\mathbf{w}\|}$$

for all indices $n \geq n_\epsilon$. We observe that

$$\begin{aligned} d(\langle \mathbf{v}_n, \mathbf{w}_n\rangle, \langle \mathbf{v}, \mathbf{w}\rangle) &\leq \rho(\mathbf{v}_n, \mathbf{v}) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho(\mathbf{w}_n, \mathbf{w}) \\ &< \left(\frac{\epsilon}{\|\mathbf{w}\|}\right) \|\mathbf{w}\| + 0 = \epsilon + 0 = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$.

If $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{w} = \mathbf{0}$, then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(\mathbf{w}_n, \mathbf{w}) < \frac{\epsilon}{\|\mathbf{v}\|}$$

for all indices $n \geq n_\epsilon$. We observe that

$$\begin{aligned} d\left(\langle \mathbf{v}_n, \mathbf{w}_n\rangle, \langle \mathbf{v}, \mathbf{w}\rangle\right) &\leq \rho\left(\mathbf{v}_n, \mathbf{v}\right) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho\left(\mathbf{w}_n, \mathbf{w}\right) \\ &< 0 + \|\mathbf{v}\| \left(\frac{\epsilon}{\|\mathbf{w}\|}\right) \|\mathbf{w}\| = 0 + \epsilon = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$.

Finally, if $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$, then there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that

$$\rho\left(\mathbf{v}_n, \mathbf{v}\right) < \frac{\epsilon}{2\|\mathbf{w}\|}$$

for all indices $n \geq m_\epsilon$ and

$$\rho\left(\mathbf{w}_n, \mathbf{w}\right) < \frac{\epsilon}{2\|\mathbf{v}\|}$$

for all indices $n \geq n_\epsilon$. Let $p_\epsilon = \max\left(\{m_\epsilon, n_\epsilon\}\right)$. We observe that

$$\begin{aligned} d\left(\langle \mathbf{v}_n, \mathbf{w}_n\rangle, \langle \mathbf{v}, \mathbf{w}\rangle\right) &\leq \rho\left(\mathbf{v}_n, \mathbf{v}\right) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho\left(\mathbf{w}_n, \mathbf{w}\right) \\ &< \left(\frac{\epsilon}{2\|\mathbf{w}\|}\right) \|\mathbf{w}\| + \|\mathbf{v}\| \left(\frac{\epsilon}{2\|\mathbf{v}\|}\right) = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

for all indices $n \geq p_\epsilon$.

In all cases, we conclude that

$$\lim_{n\to\infty} \langle \mathbf{v}_n, \mathbf{w}_n\rangle = \langle \mathbf{v}, \mathbf{w}\rangle.$$

■

Proposition 3.3.1 Limits and vector arithmetic details the ways in which the metric structure induced on a vector space by a norm or inner product interacts nicely with the additional structure of the vector space. In the real numbers $\mathbb{R}$, there are similar interactions between the metric structure and the standard ordering $\leq$.

Proposition 3.3.2 Comparison of limits. *Let $(x_n)_{n=0}^\infty$ and $(y_n)_{n=0}^\infty$ be convergent sequences of real numbers $x_n, y_n \in \mathbb{R}$. If $x_n \leq y_n$ for all indices $n \in \mathbb{N}$, then*

$$\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n.$$

Proof. By contraposition. Let $x = \lim_{n\to\infty} x_n$ and $y = \lim_{n\to\infty} y_n$, and suppose that $x > y$. Consider the positive real number

$$\epsilon = \frac{x-y}{2}.$$

Then there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that

$$d\left(x_n, x\right) < \epsilon$$

for all indices $n \geq m_\epsilon$ and

$$d\left(y_n, y\right) < \epsilon$$

for all indices $n \geq n_\epsilon$. Let $p_\epsilon = \max\left(\{m_\epsilon, n_\epsilon\}\right) \in \mathbb{N}$. We observe that

$$x - x_{p_\epsilon} \leq |x_{p_\epsilon} - x| = d\left(x_{p_\epsilon}, x\right) < \epsilon = \frac{x-y}{2}$$

and

$$y_{p_\epsilon} - y \leq |y_{p_\epsilon} - y| = d(y_{p_\epsilon}, y) < \epsilon = \frac{x-y}{2},$$

so that

$$y_{p_\epsilon} < \frac{x+y}{2} < x_{p_\epsilon}.$$

In summary, we have shown that if $x > y$, then $x_n > y_n$ for some index $n \in \mathbb{N}$. Taking the contrapositive, we arrive at the desired result. ■

This is as good a place as any to introduce another useful result concerning the relationship between sequential convergence in the real line $\mathbb{R}$ and the standard ordering $\leq$. Called the Squeeze theorem, this result can be used cleverly to avoid lengthy convergence arguments.

Theorem 3.3.3 Squeeze theorem. *Let $(x_n)_{n=0}^{\infty}$, $(y_n)_{n=0}^{\infty}$, and $(z_n)_{n=0}^{\infty}$ be sequences of real numbers $x_n, y_n, z_n \in \mathbb{R}$. If*

$$x_n \leq y_n \leq z_n$$

for all indices $n \in \mathbb{N}$ and $(x_n)_{n=0}^{\infty}$ and $(z_n)_{n=0}^{\infty}$ converge to the same real number $l \in \mathbb{R}$, then $(y_n)_{n=0}^{\infty}$ converges to l.

Proof. Let $\epsilon > 0$ be a positive real number. Then there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that

$$d(x_n, l) < \epsilon$$

for all indices $n \geq m_\epsilon$ and

$$d(z_n, l) < \epsilon$$

for all indices $n \geq n_\epsilon$. Let $p_\epsilon = \max(\{m_\epsilon, n_\epsilon\}) \in \mathbb{N}$, and note that

$$y_n - l \leq z_n - l \leq |z_n - l| = d(z_n, l) < \epsilon$$

and

$$l - y_n \leq l - x_n \leq |x_n - l| = d(x_n, l) < \epsilon$$

for all indices $n \geq p_\epsilon$. In particular, we conclude that

$$d(y_n, l) = |y_n - l| = \max(\{y_n - l, l - y_n\}) < \epsilon$$

for all indices $n \geq p_\epsilon$, and so $(y_n)_{n=0}^{\infty}$ converges to l. ■

Example 3.3.4 Squeeze theorem. Since $0 \leq \frac{1}{n^2} \leq \frac{1}{n}$ for all positive integers $n \geq 1$ and

$$\lim_{n\to\infty} 0 = \lim_{n\to\infty} \frac{1}{n} = 0,$$

we can conclude by the Squeeze theorem that

$$\lim_{n\to\infty} \frac{1}{n^2} = 0$$

without worrying about a lengthy convergence argument. ▲

Proposition 3.3.1 Limits and vector arithmetic and Proposition 3.3.2 Comparison of limits further justify the choice to emphasize the limiting behavior of objects on the real line $\mathbb{R}$ in the rest of this section; as we have already seen in Section 3.2 Convergence and Limits, these results further emphasize that knowledge of such limiting behavior can be used to make conclusions about the limiting behavior of a wide variety of metric objects.

3.3.2 Divergence to Infinity

In some ways, the binary distinction between convergence and divergence may be viewed as more coarse than an analyst might like. On the real line $\mathbb{R}$, we can make more qualitative statements about the limiting behavior of certain divergent sequences and maps to alleviate this issue somewhat. We now consider the special case of divergence of a sequence or map to infinity. Some sequences increase or decrease without bound. Such sequences can be said to have infinite limits, even though they don't converge to any real number.

Definition 3.3.5 Sequential divergence to infinity. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$.

- *Divergence to* ∞*.*

  The sequence $(x_n)_{n=0}^{\infty}$ is said to **diverge to** ∞ if for each real number $B \in \mathbb{R}$, there is a natural number $n_B \in \mathbb{N}$ so that $x_n > B$ for all indices $n \geq n_B$. In this case, we write

$$\lim_{n\to\infty} x_n = \infty.$$

- *Divergence to* $-\infty$*.*

  The sequence $(x_n)_{n=0}^{\infty}$ is said to **diverge to** $-\infty$ if for each real number $B \in \mathbb{R}$, there is a natural number $n_B \in \mathbb{N}$ so that $x_n < B$ for all indices $n \geq n_B$. In this case, we write

$$\lim_{n\to\infty} x_n = -\infty.$$

◆

Example 3.3.6 Sequential divergence to infinity. The following are examples and non-examples of sequences of real numbers which diverge to infinity.

- The sequence $(n)_{n=0}^{\infty} = (0, 1, 2, 3, \ldots)$ diverges to ∞; that is,

$$\lim_{n\to\infty} n = \infty.$$

- The sequence $\left((-1)^n\, n\right)_{n=0}^{\infty} = (0, -1, 2, -3, \ldots)$ diverges, but it does not diverge to ∞ or $-\infty$.

▲

There are also versions of divergence to infinity for real-valued functions on metric spaces.

Definition 3.3.7 Function divergence to infinity. Let $f\colon X \to \mathbb{R}$ be a real-valued function on a metric space (X, ρ).

- *Divergence to* ∞*.*

  The function f is said to **diverge to** ∞ at a point $c \in X$ if for each real number $B \in \mathbb{R}$, there is a positive real number $\delta > 0$ so that $f(x) > B$ for all points $x \in X$ so that $0 < \rho(x, c) < \delta$.

  In this case, if c is a limit point of X, then we write

$$\lim_{x\to c} f(x) = \infty.$$

- *Divergence to* $-\infty$*.*

The function f is said to **diverge to** $-\infty$ at a point $c \in X$ if for each real number $B \in \mathbb{R}$, there is a positive real number $\delta > 0$ so that $f(x) < B$ for all points $x \in X$ so that $0 < \rho(x, c) < \delta$.

In this case, if c is a limit point of X, then we write

$$\lim_{x \to c} f(x) = -\infty.$$

◆

Example 3.3.8 Function divergence to infinity. The following are examples and non-examples of real-valued functions on metric spaces which diverge to infinity

- Consider the real-valued function $f : \mathbb{R} \setminus \{1\} \to \mathbb{R}$ defined by the formula

  $$f(x) = -\frac{1}{(x-1)^2}.$$

  Then f diverges to $-\infty$ at 1; that is,

  $$\lim_{x \to 1} -\frac{1}{x^2} = -\infty.$$

- Consider the real-valued function $g : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ defined by the formula

  $$g(y) = \frac{1}{y}.$$

  Then g diverges at 0, but it does not diverge to ∞ or $-\infty$ at 0.

▲

Divergence of a sequence of real numbers or a real-valued function to infinity is a strong statement about the limiting behavior of such a sequence or function. However, since there is no metric on the extended real number system which agrees with the Euclidean metric on the real numbers, we cannot view divergence to infinity as convergence in a metric space of which the real line is a metric subspace. We will see in Chapter 4 Topological Spaces a way in which the situation of divergence to infinity can be said to be a form of convergence to an extended real number, namely the infinite quantities ∞ and $-\infty$.

In many ways, the symbols ∞ and $-\infty$ can be viewed as one-sided "reciprocals" of 0. While division by zero is left undefined for very important arithmetical reasons, it turns out that divergence of a sequence of nonzero real numbers to infinity implies the Euclidean convergence of the reciprocals to zero.

Proposition 3.3.9 Reciprocals of infinite limits. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of positive real numbers $x_n > 0$. Then*

$$\lim_{n \to \infty} x_n = \infty$$

if and only if

$$\lim_{n \to \infty} \frac{1}{x_n} = 0.$$

Proof. First suppose that $(x_n)_{n=0}^{\infty}$ diverges to ∞, and let $\epsilon > 0$. Then $\frac{1}{\epsilon} \in \mathbb{R}$

is a real number, and so there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$x_n > \frac{1}{\epsilon}$$

for all indices $n \geq n_\epsilon$. In particular,

$$d\left(\frac{1}{x_n}, 0\right) = \left|\frac{1}{x_n} - 0\right| = \left|\frac{1}{x_n}\right| = \frac{1}{x_n} < \epsilon$$

for all indices $n \geq n_\epsilon$, and so $\left(\frac{1}{x_n}\right)_{n=0}^{\infty}$ converges to 0.

Conversely, now suppose that $\left(\frac{1}{x_n}\right)_{n=0}^{\infty}$ converges to 0, and let $B \in \mathbb{R}$ be a real number. If $B \leq 0$ is non-positive, then

$$x_n > 0 \geq B$$

for all indices $n \in \mathbb{N}$. On the other hand, if $B > 0$ is positive, then $\frac{1}{B} > 0$ is a positive real number, and so there is a natural number n_B so that

$$d\left(\frac{1}{x_n}, 0\right) < \frac{1}{B}$$

for all indices $n \geq n_B$. In particular,

$$x_n = |x_n| = \frac{1}{\left|\frac{1}{x_n} - 0\right|} = \frac{1}{d\left(\frac{1}{x_n}, 0\right)} > B$$

for all indices $n \geq n_B$, and so $(x_n)_{n=0}^{\infty}$ diverges to ∞. ■

So, as far as limiting behavior of sequences of real numbers goes, the infinite quantities ∞ and $-\infty$ behave like one-sided reciprocals of zero. It's incredibly important to avoid assuming anything stronger than what is explicitly stated in the proposition above. Specifically, the hypothesis that the sequence in question have positive real number values is crucial, and the statement is not necessarily true without this requirement.

A sequence of real numbers diverges to ∞ or $-\infty$ if it increases or decreases, respectively, without bound. This notion of **boundedness** is directly related to the one introduced in Section 1.5 Introduction to Order Theory for subsets of partially ordered sets. Informally, a map with codomain a partially ordered set is considered to be bounded if its image is bounded.

Definition 3.3.10 Map boundedness. Let $f\colon X \to Y$ be a map from a set X to a partially ordered set $(Y, \preceq)$.

- *Upper bound.*

  An output $y \in Y$ is an **upper bound** for the map f if it is an upper bound for the image $\operatorname{im}(f) = f(X)$; that is, y is an upper bound for f if $f(x) \preceq y$ for all inputs $x \in X$.

  The map f is called **bounded from above** if there is an upper bound for f, and **unbounded from above** if there is no such upper bound.

- *Lower bound.*

  An output $y \in Y$ is a **lower bound** for the map f if it is a lower bound for the image $\operatorname{im}(f) = f(X)$; that is, y is a lower bound for f if $y \preceq f(x)$ for all inputs $x \in X$.

The map f is called **bounded from below** if there is a lower bound for f, and **unbounded from below** if there is no such lower bound.

- *Boundedness.*

  The map f is called **bounded** if it is both bounded from above and bounded from below, and f is called **unbounded** otherwise if it is either unbounded from above or unbounded from below (or both).

◆

Example 3.3.11 Sequential boundedness in the real line. A sequence $(x_n)_{n=0}^{\infty}$ of real numbers $x_n \in \mathbb{R}$ is bounded if and only if there are real numbers $a, b \in \mathbb{R}$ such that $a \leq x_n \leq b$ for all indices $n \in \mathbb{N}$. For example, the sequence $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ is bounded, because

$$0 < \frac{1}{n} \leq 1$$

for all indices $n \geq 1$. ▲

Clearly, any sequence of real numbers or any real-valued function which diverges to infinity cannot be bounded. However, the converse is not true; unbounded sequences need not diverge to infinity. We have already seen one such example of this in Example 3.3.8 Function divergence to infinity.

Considered as a real-valued function on the set $\mathbb{N}$ of natural numbers, a sequence of real numbers is a map whose codomain is a totally ordered set. However, the domain of such a sequence is also totally ordered. Because the metric structure on the real line $\mathbb{R}$ can be described in terms of the standard ordering $\leq$, sequences which preserve these orders have special convergence properties. Such sequences are called **monotone**.

Definition 3.3.12 Monotonicity. Let $f: X \to Y$ be a map from a partially ordered set $(X, \preceq_X)$ to a partially ordered set $(Y, \preceq_Y)$.

- *Isotonicity.*

  The map f is called **non-decreasing**, **order-preserving**, or **isotone** if for all inputs $x_1, x_2 \in X$, if $x_1 \preceq_X x_2$, then $f(x_1) \preceq_Y f(x_2)$. The map f is called **increasing**, **strictly order-preserving**, or **strictly isotone** if it is non-decreasing and injective.

- *Antitonicity.*

  The map f is called **non-increasing**, **order-reversing**, or **antitone** if for all inputs $x_1, x_2 \in X$, if $x_1 \preceq_X x_2$, then $f(x_2) \preceq_Y f(x_1)$. The map f is called **decreasing**, **strictly order-reversing**, or **strictly antitone** if it is non-increasing and injective.

- *Monotonicity.*

  The map f is called **monotone** if it is either non-decreasing or non-increasing, and f is called **strictly monotone** if it is either increasing or decreasing.

◆

It turns out that all monotone sequences of real numbers have limits, in the sense that a monotone sequence of real numbers either converges to a real number or diverges to infinity. In fact, something slightly stronger is true; the limit of a non-decreasing sequence exists and is its supremum, and the limit of a non-increasing sequence exists and is its infimum. This useful fact is a property of the extended real number system $\mathbb{R}$. Other totally ordered number subsystems, such as the rational numbers $\mathbb{Q}$, do not enjoy this property, since suprema and infima of sets need not exist in these number systems, which are Dedekind incomplete in the sense introduced in Section 2.3 The Complete Ordered Field.

Theorem 3.3.13 Monotone convergence theorem. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$.*

1. *If $(x_n)_{n=0}^{\infty}$ is non-decreasing, then*

$$\lim_{n\to\infty} x_n = \sup_{n\in\mathbb{N}} x_n.$$

2. *If $(x_n)_{n=0}^{\infty}$ is non-increasing, then*

$$\lim_{n\to\infty} x_n = \inf_{n\in\mathbb{N}} x_n.$$

In particular, if $(x_n)_{n=0}^{\infty}$ is bounded and monotone, then it is convergent.

Proof.

1. We observe that $\{x_n\}_{n=0}^{\infty} \neq \emptyset$ is non-empty, and so $\sup_{n\in\mathbb{N}} x_n = \sup\left(\{x_n\}_{n=0}^{\infty}\right) \neq -\infty$. There are two cases; either $\sup_{n\in\mathbb{N}} x_n = \infty$ or $\sup_{n\in\mathbb{N}} x_n \in \mathbb{R}$ is a real number.

   First suppose that $\sup_{n\in\mathbb{N}} x_n = \infty$, and let $B \in \mathbb{R}$ be a real number. Since

$$B < \infty = \sup_{n\in\mathbb{N}} x_n,$$

   B is not an upper bound for $\{x_n\}_{n=0}^{\infty}$; that is, $x_{n_B} > B$ for some natural number $n_B \in \mathbb{N}$. We observe that since $(x_n)_{n=0}^{\infty}$ is non-decreasing,

$$x_n \geq x_{n_B} > B$$

   for all indices $n \geq n_B$, and so

$$\lim_{n\to\infty} x_n = \infty = \sup_{n\in\mathbb{N}} x_n.$$

   On the other hand, now suppose that $S = \sup_{n\in\mathbb{N}} x_n \in \mathbb{R}$ is a real number, and let $\epsilon > 0$ be a positive real number. Since

$$S - \epsilon < S = \sup_{n\in\mathbb{N}} x_n,$$

   $S - \epsilon$ is not an upper bound for $\{x_n\}_{n=0}^{\infty}$; that is, $x_{n_\epsilon} > S - \epsilon$ for some natural number $n_\epsilon \in \mathbb{N}$. Since $(x_n)_{n=0}^{\infty}$ is non-decreasing,

$$S - \epsilon < x_{n_\epsilon} \leq x_n \leq S$$

   for all indices $n \geq n_\epsilon$, and so

$$\begin{aligned} d\left(x_n, S\right) &= |x_n - S| = S - x_n \\ &< S - (S - \epsilon) = \epsilon \end{aligned}$$

   for all indices $n \geq n_\epsilon$. We conclude that

$$\lim_{n\to\infty} x_n = S = \sup_{n\in\mathbb{N}} x_n.$$

2. We observe that $\{x_n\}_{n=0}^{\infty} \neq \emptyset$ is non-empty, and so $\inf_{n\in\mathbb{N}} x_n = \inf\left(\{x_n\}_{n=0}^{\infty}\right) \neq \infty$. There are two cases; either $\inf_{n\in\mathbb{N}} x_n = -\infty$ or $\inf_{n\in\mathbb{N}} x_n \in \mathbb{R}$ is a real number.

   First suppose that $\inf_{n\in\mathbb{N}} x_n = -\infty$, and let $B \in \mathbb{R}$ be a real number. Since
$$B > -\infty = \inf_{n\in\mathbb{N}} x_n,$$
   B is not a lower bound for $\{x_n\}_{n=0}^{\infty}$; that is, $x_{n_B} < B$ for some natural number $n_B \in \mathbb{N}$. We observe that since $(x_n)_{n=0}^{\infty}$ is non-increasing,
$$x_n \leq x_{n_B} < B$$
   for all indices $n \geq n_B$, and so
$$\lim_{n\to\infty} x_n = -\infty = \inf_{n\in\mathbb{N}} x_n.$$

   On the other hand, now suppose that $I = \inf_{n\in\mathbb{N}} x_n \in \mathbb{R}$ is a real number, and let $\epsilon > 0$ be a positive real number. Since
$$I + \epsilon > S = \inf_{n\in\mathbb{N}} x_n,$$
   $I + \epsilon$ is not a lower bound for $\{x_n\}_{n=0}^{\infty}$; that is, $x_{n_\epsilon} < I + \epsilon$ for some natural number $n_\epsilon \in \mathbb{N}$. Since $(x_n)_{n=0}^{\infty}$ is non-increasing,
$$I + \epsilon > x_{n_\epsilon} \geq x_n \geq I$$
   for all indices $n \geq n_\epsilon$, and so
$$\begin{aligned} d(x_n, I) &= |x_n - I| = x_n - I \\ &< I + \epsilon - I = \epsilon \end{aligned}$$
   for all indices $n \geq n_\epsilon$. We conclude that
$$\lim_{n\to\infty} x_n = I = \inf_{n\in\mathbb{N}} x_n.$$

■

What we have just shown will prove incredibly useful; in particular, we will use the fact that all bounded, monotone sequences of real numbers are convergent several times throughout this chapter. The converse is not true; while convergent sequences of real numbers are bounded, not all such sequences need be monotone.

3.3.3 Upper and Lower Limits

If a sequence of points in a metric space is convergent, then its image has exactly one limit point. More generally, studying the limit points of the image of a sequence of points is a fruitful exercise regardless of whether or not the series converges. We will see that the Dedekind completeness of the real numbers $\mathbb{R}$ allows us to characterize these limit points via the **upper** and **lower limits**.

Lemma 3.3.14 Monotonicity of suprema/infima. *Let $U, V \subseteq X$ be subsets of a partially ordered set $(X, \preceq)$, and suppose that $U \subseteq V$.*

1. *If U and V have suprema, then* $\sup(U) \leq \sup(V)$.

2. *If U and V have infima, then* $\inf(U) \geq \inf(V)$.

Proof.

1. If $U \subseteq V$, then every upper bound for V is an upper bound for U, and so

$$\begin{aligned}\sup(U) &= \min(\{x \in X \mid u \preceq x \text{ for all } u \in U\}) \\ &\leq \min(\{x \in X \mid v \preceq x \text{ for all } v \in V\}) = \sup(V).\end{aligned}$$

2. If $U \subseteq V$, then every lower bound for V is a lower bound for U, and so

$$\begin{aligned}\inf(U) &= \max(\{x \in X \mid x \preceq u \text{ for all } u \in U\}) \\ &\geq \min(\{x \in X \mid x \preceq v \text{ for all } v \in V\}) = \inf(V).\end{aligned}$$

■

Definition 3.3.15 Upper/lower limit. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$.

- *Upper limit.*

  The sequence $\left(\sup_{n\geq m} x_n\right)_{m=0}^{\infty}$ is non-increasing, and so by the Monotone convergence theorem has a well-defined limit in the extended real numbers $\mathbb{R}\cup\{\pm\infty\}$. This limit is called the **upper limit** or **limit supremum** of the sequence $(x_n)_{n=0}^{\infty}$, and is denoted

$$\limsup_{n\to\infty} x_n = \lim_{m\to\infty} \sup_{n\geq m} x_n = \inf_{m\in\mathbb{N}} \sup_{n\geq m} x_n.$$

- *Lower limit.*

  The sequence $(\inf_{n\geq m} x_n)_{m=0}^{\infty}$ is non-decreasing, and so by the Monotone convergence theorem has a well-defined limit in the extended real numbers $\mathbb{R}\cup\{\pm\infty\}$. This limit is called the **lower limit** or **limit infimum** of the sequence $(x_n)_{n=0}^{\infty}$, and is denoted

$$\liminf_{n\to\infty} x_n = \lim_{m\to\infty} \inf_{n\geq m} x_n = \sup_{m\in\mathbb{N}} \inf_{n\geq m} x_n.$$

◆

Example 3.3.16 Upper/lower limits. Consider the sequence

$$x_n = (-1)^n \left(1 + \frac{1}{n}\right) = (-1)^n \left(\frac{n+1}{n}\right).$$

Then $\limsup_{n\to\infty} x_n = 1$, and $\liminf_{n\to\infty} x_n = -1$. ▲

Proposition 3.3.17 Properties of upper limits. *Upper limits of sequences of real numbers have the following properties:*

1. *A sequence $(x_n)_{n=0}^{\infty}$ of real numbers $x_n \in \mathbb{R}$ is bounded from above if and only if*

$$\limsup_{n\to\infty} x_n < \infty.$$

2. *For any sequences $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ of real numbers $x_n, y_n \in \mathbb{R}$,*

$$\limsup_{n\to\infty} (x_n + y_n) \leq \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n$$

whenever the right-hand side of the above inequality is well-defined.

3. *For any non-negative real number* $\alpha \in [0, \infty)$ *and sequence* $(x_n)_{n=0}^{\infty}$ *of real numbers* $x_n \in \mathbb{R}$,

$$\limsup_{n \to \infty} \alpha x_n = \alpha \limsup_{n \to \infty} x_n$$

whenever the right-hand side of the above equality is well-defined.

Proof.

1. Suppose that $(x_n)_{n=0}^{\infty}$ is bounded from above. Then there is a real number $B \in \mathbb{R}$ so that $x_n \leq B$ for all indices $n \in \mathbb{N}$. In particular, for all natural numbers $m \in \mathbb{N}$, $x_n \leq B$ for all indices $n \geq m$, and so $\sup_{n \geq m} x_n \leq B$. Proposition 3.3.2 Comparison of limits implies that

$$\limsup_{n \to \infty} x_n = \lim_{n \to \infty} \sup_{n \geq m} x_n \leq B < \infty.$$

Conversely, now suppose that $\limsup_{n \to \infty} x_n < \infty$. For each natural number $m \in \mathbb{N}$, let $S_m = \sup_{n \geq m} x_n$, and let

$$S = \lim_{m \to \infty} S_m = \lim_{m \to \infty} \sup_{n \geq m} x_n = \limsup_{n \to \infty} x_n.$$

Since 1 is a positive real number, there is a natural number $m_1 \in \mathbb{N}$ so that $d(S_m, S) < 1$ for all indices $m \geq m_1$. In particular,

$$S_{m_1} - S \leq |S_{m_1} - S| = d(S_{m_1}, S) < 1,$$

and so $\sup_{n \geq m_1} x_n = S_{m_1} < S + 1$. Let

$$B = \max(\{x_0, x_1, \ldots, x_{m_1 - 1}, S + 1\}),$$

and note that $x_n \leq B$ for all indices $n \in \mathbb{N}$. Since such an upper bound exists, we conclude that $(x_n)_{n=1}^{\infty}$ is bounded from above.

2. For each natural number $m \in \mathbb{N}$, let $S_m = \sup_{n \geq m} x_n$ and $T_m = \sup_{n \geq m} y_n$. We observe that $x_n + y_n \leq S_m + T_m$ for all natural numbers $m \in \mathbb{N}$ and indices $n \geq m$. Thus

$$\sup_{n \geq m} (x_n + y_n) \leq S_m + T_m$$

for all natural numbers $m \in \mathbb{N}$, and so Proposition 3.3.2 Comparison of limits and (1) of Proposition 3.3.1 Limits and vector arithmetic together imply that

$$\begin{aligned} \limsup_{n \to \infty} (x_n + y_n) &= \lim_{n \to \infty} \sup_{n \geq m} (x_n + y_n) \leq \lim_{m \to \infty} (S_m + T_m) \\ &= \lim_{m \to \infty} S_m + \lim_{m \to \infty} T_m = \limsup_{n \to \infty} x_n + \limsup_{n \to \infty} y_n. \end{aligned}$$

3. If $\alpha = 0$, then our hypotheses imply that $\limsup_{n \to \infty} x_n \in \mathbb{R}$ is a real number, and so

$$\limsup_{n \to \infty} \alpha x_n = \limsup_{n \to \infty} 0 = 0 = 0 \limsup_{n \to \infty} x_n.$$

Thus we may suppose that $\alpha > 0$ is positive. In this case, multiplication by α is a strictly order-preserving bijection of $\mathbb{R} \cup \{\pm\infty\}$, and so

$$\sup_{n \geq m} \alpha x_n = \sup(\{\alpha x_n\}_{n=m}^{\infty}) = \alpha \sup(\{x_n\}_{n=m}^{\infty}) = \alpha \sup_{n \geq m} x_n$$

for all natural numbers $m \in \mathbb{N}$. We conclude that

$$\begin{aligned}\limsup_{n\to\infty} \alpha x_n &= \lim_{m\to\infty} \sup_{n\geq m} \alpha x_n = \lim_{m\to\infty} \alpha \sup_{n\geq m} x_n \\ &= \alpha \lim_{m\to\infty} \sup_{n\geq m} x_n = \alpha \limsup_{n\to\infty} x_n.\end{aligned}$$

■

One natural follow-up from Proposition 3.3.17 Properties of upper limits is the question of negative scalings of sequences: If $\alpha < 0$ is a negative real number and $(x_n)_{n=0}^{\infty}$ is a sequence of real numbers $x_n \in \mathbb{R}$, what is the upper limit $\limsup_{n\to\infty} \alpha x_n$?

Lemma 3.3.18 Upper and lower limits. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers. Then*

$$\limsup_{n\to\infty} (-x_n) = -\liminf_{n\to\infty} x_n$$

and

$$\liminf_{n\to\infty} (-x_n) = -\limsup_{n\to\infty} x_n.$$

Here we define $-(\infty) = -\infty$ and $-(-\infty) = \infty$.

Proof. We observe that

$$\begin{aligned}\sup_{n\geq m} (-x_n) &= \min(\{x \in [-\infty,\infty] \mid -x_n \leq x \text{ for all } n \geq m\}) \\ &= \min(\{x \in [-\infty,\infty] \mid -x \leq x_n \text{ for all } n \geq m\}) \\ &= \min(\{-x \in [-\infty,\infty] \mid x \leq x_n \text{ for all } n \geq m\}) \\ &= -\max(\{-x \in [-\infty,\infty] \mid x \leq x_n \text{ for all } n \geq m\}) = -\inf_{n\geq m} x_n\end{aligned}$$

for all natural numbers $m \in \mathbb{N}$, and so

$$\begin{aligned}\limsup_{n\to\infty} (-x_n) &= \lim_{n\to\infty} \sup_{n\geq m} (-x_n) \\ &= \lim_{n\to\infty} -\inf_{n\geq m} x_n \\ &= -\lim_{n\to\infty} \inf_{n\geq m} x_n = -\liminf_{n\to\infty} x_n.\end{aligned}$$

Applying this result to the sequence $(-x_n)_{n=0}^{\infty}$, we see that

$$\begin{aligned}-\limsup_{n\to\infty} x_n &= -\limsup_{n\to\infty} (-(-x_n)) \\ &= -\left(-\liminf_{n\to\infty} (-x_n)\right) = \liminf_{n\to\infty} (-x_n).\end{aligned}$$

■

In addition to answering our question about the behavior of upper limits under scaling by negative real numbers, we can use Lemma 3.3.18 Upper and lower limits to translate results about upper limits into results about lower limits. For example, each of the properties in Proposition 3.3.17 Properties of upper limits has an analogue for lower limits.

Corollary 3.3.19 Properties of lower limits. *Lower limits of sequences of real numbers have the following properties:*

1. *A sequence $(x_n)_{n=0}^{\infty}$ of real numbers $x_n \in \mathbb{R}$ is bounded from below if and*

only if

$$\liminf_{n\to\infty} x_n > -\infty.$$

2. *For any sequences* $(x_n)_{n=0}^\infty$ *and* $(y_n)_{n=0}^\infty$ *of real numbers* $x_n, y_n \in \mathbb{R}$,

$$\liminf_{n\to\infty} (x_n + y_n) \geq \liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n$$

whenever the right-hand side of the above inequality is well-defined.

3. *For any non-negative real number* $\alpha \in [0, \infty)$ *and sequence* $(x_n)_{n=0}^\infty$ *of real numbers* $x_n \in \mathbb{R}$,

$$\liminf_{n\to\infty} \alpha x_n = \alpha \liminf_{n\to\infty} x_n$$

whenever the right-hand side of the above equality is well-defined.

Finally, we end with a characterization of convergence and divergence to infinity in terms of upper and lower limits.

Theorem 3.3.20 Convergence and upper and lower limits. *Let* $(x_n)_{n=0}^\infty$ *be a sequence of real numbers* $x_n \in \mathbb{R}$.

1. *Convergence.*

   For any real number $l \in \mathbb{R}$, *the sequence* $(x_n)_{n=0}^\infty$ *converges to* l *if and only if*

$$\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = l.$$

2. *Divergence to* ∞.

   The sequence $(x_n)_{n=0}^\infty$ *diverges to* ∞ *if and only if*

$$\liminf_{n\to\infty} x_n = \infty.$$

3. *Divergence to* $-\infty$.

   The sequence $(x_n)_{n=0}^\infty$ *diverges to* $-\infty$ *if and only if*

$$\limsup_{n\to\infty} x_n = -\infty.$$

Proof.

1. For each natural number $m \in \mathbb{N}$, let $S_m = \sup_{n\geq m} x_n$ and $I_m = \inf_{n\geq m} x_n$. First suppose that $(x_n)_{n=0}^\infty$ converges to l. Let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$l - \epsilon < x_n < l + \epsilon$$

   for all indices $n \geq n_\epsilon$. In particular,

$$l - \epsilon < I_m \leq S_m < l + \epsilon$$

   for any natural number $m \geq n_\epsilon$, and so

$$l - \epsilon \leq \lim_{m\to\infty} I_m \leq \lim_{m\to\infty} S_m \leq l + \epsilon$$

   by Proposition 3.3.2 Comparison of limits. Since this holds for all positive real numbers $\epsilon > 0$,

$$\limsup_{m\to\infty} x_n = \liminf_{n\to\infty} x_n = l.$$

Conversely, now suppose that

$$\limsup_{m\to\infty} x_n = \liminf_{n\to\infty} x_n = l.$$

Let $\epsilon > 0$ be a positive real number. Then there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that $d(S_m, l) < \epsilon$ for all indices $m \geq m_\epsilon$ and $d(I_m, l) < \epsilon$ for all indices $m \geq n_\epsilon$. Let $p_\epsilon = \max(\{m_\epsilon, n_\epsilon\}) \in \mathbb{N}$, and note that

$$\begin{aligned} x_n - l \leq S_{p_\epsilon} - l &\leq |S_{p_\epsilon} - l| \\ &= d(S_{p_\epsilon}, l) < \epsilon \end{aligned}$$

and

$$\begin{aligned} l - x_n \leq l - I_{p_\epsilon} &\leq |I_{p_\epsilon} - l| \\ &= d(I_{p_\epsilon}, l) < \epsilon \end{aligned}$$

for all indices $n \geq p_\epsilon$. We conclude that

$$\begin{aligned} d(x_n, l) &= |x_n - l| \\ &= \max(\{x_n - l, l - x_n\}) < \epsilon \end{aligned}$$

for all indices $n \geq p_\epsilon$, and so $(x_n)_{n=0}^{\infty}$ converges to l.

■

You will be given the opportunity to prove (2) and (3) of Theorem 3.3.20 Convergence and upper and lower limits in the problems for this section.

In Chapter 4 Topological Spaces, we will reexamine limits, convergence, and divergence to infinity in the extended real number system as an example of the more general context of topological spaces. For now, however, we will return to our study of the point-set topology of metric spaces.

In this section, we have seen that much of the metric structure of the real line $\mathbb{R}$ can be phrased in terms of the standard ordering $\leq$. In particular, we have seen that convergence in the Euclidean metric is deeply related to the properties of small intervals of real numbers. In the next section, we will examine the analogs of these intervals in general metric spaces.

3.3.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Limits and arithmetic. Compute the limits of the given sequences of real numbers.

1. Compute $\lim_{n\to\infty} \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3}$.

2. Compute $\lim_{n\to\infty} \frac{2n+1}{3n}$.

3. Compute $\lim_{n\to\infty} \frac{1-n^2}{n}$.

4. **Limits and strict comparison.** Give examples of convergent sequences $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ of real numbers $x_n, y_n \in \mathbb{R}$ so that $x_n < y_n$ for all indices $n \in \mathbb{N}$ but

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n.$$

Computing upper/lower limits. Compute the upper and lower limits of the given sequences of real numbers.

5. Compute $\limsup_{n\to\infty} 2^n$ and $\liminf_{n\to\infty} 2^n$.

6. Compute $\limsup_{n\to\infty} 2^{-n}$ and $\liminf_{n\to\infty} 2^{-n}$.

7. Compute $\limsup_{n\to\infty} (-2)^n$ and $\liminf_{n\to\infty} (-2)^n$.

8. Compute $\limsup_{n\to\infty} (-2)^{-n}$ and $\liminf_{n\to\infty} (-2)^{-n}$.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.3.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Limits and scalar multiplication.** Let $\|\cdot\|$ be a norm on a real or complex vector space V. Prove that

$$\lim_{n\to\infty} \alpha_n \mathbf{v}_n = \lim_{n\to\infty} \alpha_n \lim_{n\to\infty} \mathbf{v}_n$$

for any convergent sequences $(\alpha_n)_{n=0}^{\infty}$ and $(\mathbf{v}_n)_{n=0}^{\infty}$ of scalars α_n and vectors $\mathbf{v}_n \in V$.

2. **Limits and division.** Prove that

$$\lim_{n\to\infty} \frac{1}{x_n} = \frac{1}{\lim_{n\to\infty} x_n}$$

for every sequence $(x_n)_{n=0}^{\infty}$ of non-zero real numbers $x_n \in \mathbb{R} \setminus \{0\}$ which converges to a non-zero real number limit

$$\lim_{n\to\infty} x_n \neq 0.$$

3. **Squeezing to infinity.** Let $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ be sequences of real numbers $x_n, y_n \in \mathbb{R}$.

(a) Prove that if $x_n \leq y_n$ for all indices $n \in \mathbb{N}$ and $(x_n)_{n=0}^{\infty}$ diverges to ∞, then $(y_n)_{n=0}^{\infty}$ diverges to ∞.

(b) Prove that if $x_n \leq y_n$ for all indices $n \in \mathbb{N}$ and $(y_n)_{n=0}^{\infty}$ diverges to $-\infty$, then $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$.

We can interpret the above results as versions of the Squeeze theorem for divergence to infinity.

4. Reciprocals of infinite limits. Prove that a sequence $(x_n)_{n=0}^{\infty}$ of negative real numbers $x_n < 0$ diverges to $-\infty$ if and only if

$$\lim_{n\to\infty} \frac{1}{x_n} = 0.$$

5. Divergence and upper/lower limits. In this problem, you will complete the proof of Theorem 3.3.20 Convergence and upper and lower limits. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$.

(a) Prove that $(x_n)_{n=0}^{\infty}$ diverges to ∞ if and only if

$$\liminf_{n\to\infty} x_n = \infty.$$

(b) Prove that $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$ if and only if

$$\limsup_{n\to\infty} x_n = -\infty.$$

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.4 Open and Closed Sets

In this section, we generalize the notion of open and closed intervals to subsets of arbitrary metric spaces. Recall that the open interval (a, b) between two real numbers $a, b \in \mathbb{R}$ is the subset of the real numbers $\mathbb{R}$ defined as follows.

$$(a, b) = \{x \in \mathbb{R} : a < x < b\} \subsetneq \mathbb{R}.$$

Similarly, the closed interval $[a, b]$ between two real numbers $a, b \in \mathbb{R}$ is the subset of the real numbers $\mathbb{R}$ defined as follows.

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\} \subsetneq \mathbb{R}.$$

Soon, we will develop the more general notion of the open and closed subsets of a metric space, which will play a crucial role in the remainder of this text.

3.4.1 Neighborhoods in Metric Spaces

Informally, a subset of a metric space is **open** if it contains all points which are close to its elements and **closed** if it contains all of its limit points. These are one of many provably equivalent definitions of openness and closedness. In order to formalize this, we make the following preliminary definitions.

Definition 3.4.1 Open ball; closed disk; sphere. Fix a positive real number $r > 0$ and a point $c \in X$ in a metric space (X, ρ).

- *Open metric ball.*

  The **open (metric) ball** $\mathrm{B}_r(c)$ in X of **radius** r and **center** c is the subset of X defined by

  $$\mathrm{B}_r(c) = \{x \in X \mid \rho(x, c) < r\}.$$

- *Closed metric disk.*

The **closed (metric) disk** $\mathrm{D}_r(c)$ in X of **radius** r and **center** c is the subset of X defined by

$$\mathrm{D}_r(c) = \{x \in X \mid \rho(x, c) \leq r\}.$$

- *Open metric ball.*

  The **(metric) sphere** $\mathrm{S}_r(c)$ in X of **radius** r and **center** c is the subset of X defined by

  $$\mathrm{S}_r(c) = \{x \in X \mid \rho(x, c) = r\}.$$

◆

So an open ball contains precisely the points of distance strictly less than its radius from its center; a closed disk contains precisely the points of distance at most its radius from its center; and a sphere contains precisely the points of distance precisely its radius from its center. Just as the difference between an open and a closed interval is the question of the inclusion of its endpoints, the distinction between open balls and closed disks is that a closed disk contains all points of distance precisely its radius from its center, while an open ball does not.

We will see that in many ways, this section is an investigation of the properties of subsets of metric spaces which contain or are contained in certain open balls. The following definition, that of a **neighborhood**, will be a useful notion for both the rest of this chapter and the next chapter as well. Informally, a subset of a metric space is a neighborhood of a point if it contains all points which are very close to that point.

Definition 3.4.2 Neighborhood. A subset $U \subseteq X$ is a **neighborhood** of a point $x \in X$ in a metric space (X, ρ) if U contains an open ball centered at x; that is, U is a neighborhood of x if $\mathrm{B}_r(x) \subseteq U$ for some positive real number $r > 0$. ◆

Example 3.4.3 Neighborhoods. Every open ball and closed disk in a metric space is a neighborhood of its center. ▲

Remark 3.4.4 Locality. The notion of a neighborhood is our latest exposure to the conceptual metaphor of **locality**. We conceive of metric spaces by spatial reasoning, saying for example that for points $x, y \in X$ in a metric space (X, ρ), if $\rho(x, y)$ is a small positive number, then x and y are "close".

This idea is pervasive throughout point-set topology, and it implicitly informs much of the relevant terminology. Just as your neighborhood contains all of the people closest to where you live, a neighborhood in a metric space contains all nearby points.

We now have the terminology to make the crucial and titular definitions of this section; that of **open** and **closed subsets** of a metric space. A subset of a metric space is open if it is a neighborhood of all of its points, and is closed if its complement is open.

Definition 3.4.5 Open/closed subset. Let $U \subseteq X$ be a subset of a metric space (X, ρ).

- *Open subset.*

  U is **open** in X if it is a neighborhood of each of its points; that is, U is open if for each point $u \in U$, there is a positive real number $r > 0$ so that $\mathrm{B}_r(u) \subseteq U$.

- *Closed subset.*

  U is **closed** in X if its complement $X \setminus U$ is open in X; that is, U is closed if each point $x \in X \setminus U$ has a neighborhood in X which is disjoint from U.

◆

Lemma 3.4.6 Open balls and closed disks. *Open balls in metric spaces are open, and closed disks are closed.*

Proof. Fix a positive real number $r > 0$ and a point $c \in X$ in a metric space (X, ρ). Let $U = \mathrm{B}_r(c)$ and $V = \mathrm{D}_r(c)$ be the open ball and closed disk of radius r and center c, respectively. We will show that U and V are open and closed in X, respectively.

To that end, first let $u \in U$, so that $\rho(u, c) < r$. Let

$$s = r - \rho(u, c).$$

We note that for a point $x \in X$, if $\rho(x, u) < s$, then

$$\begin{aligned}\rho(x, c) &\leq \rho(x, u) + \rho(u, c) < s + r \\ &= r - \rho(u, c) + \rho(u, c) = r,\end{aligned}$$

and so $x \in U$. Thus $\mathrm{B}_s(u) \subseteq \mathrm{B}_r(c) = U$. Since U contains an open ball centered at each of its points, it is a neighborhood of each of its points, and so U is open in X.

On the other hand, now suppose that $x \in X \setminus V$, so that $\rho(x, c) > r$. Let

$$s = \rho(x, c) - r.$$

We note that for a point $y \in X$, if $\rho(y, x) < s$, then

$$\begin{aligned}r + s = \rho(x, c) &\leq \rho(x, y) + \rho(y, c) \\ &= \rho(y, x) + \rho(y, c) < s + \rho(y, c).\end{aligned}$$

We conclude that $\rho(y, c) > r$, and so $y \notin \mathrm{D}_r(c) = V$. Since $X \setminus V$ contains an open ball centered at each of its points, it is a neighborhood of each of its points, and so $X \setminus V$ is open in X; that is, V is closed in X. ■

Lemma 3.4.6 Open balls and closed disks illustrates that the names "open" and "closed" for balls and disks are well-deserved; open balls and closed disks in a metric space are open and closed, respectively. This immediately implies that open intervals and closed intervals of real numbers (with finite endpoints) are open and closed, respectively, in the Euclidean metric, since any such interval is either a open ball or a closed disk (depending on whether or not it includes its endpoints) around its center, which is the arithmetic mean of its endpoints.

We next develop a series of point-set topological definitions based on the notion of neighborhoods which will play an important role in both this and the next chapter. Given a subset of a metric space, we may consider open or closed sets which approximate the set as well as possible from the inside or outside, respectively.

Definition 3.4.7 Interior; closure; boundary. Fix a subset $U \subseteq X$ of a metric space (X, ρ).

- *Interior.*

  A point $x \in X$ is an **interior point** of U if U is a neighborhood of x. The set of interior points of U is called the **interior** of U and is denoted $\mathrm{Int}(U)$ or U°.

- *Closure.*

A point $x \in X$ is called a **point of closure** of U in X if $N \cap U \neq \emptyset$ for all neighborhoods N of x in X. The set of points of closure of U in X is called the **closure** of U in X and is denoted $\mathrm{Cl}(U)$ or $\overline{U}$.

- *Boundary.*

  A point $x \in X$ is called a **boundary point** of U in X if $N \cap U \neq \emptyset$ and $N \setminus U \neq \emptyset$ for all neighborhoods N of x in X. The set of all boundary points of U in X is called the **boundary** of U in X and is denoted $\mathrm{Bd}(U)$ or ∂U.

◆

The above definitions are equivalent if we quantify over open balls centered at the point in question rather than over neighborhoods of said point. However, the notion of neighborhood will extend into our discussion in Chapter 4 Topological Spaces, so we will continue in this spirit of generality, which we hope will ease our generalization of the above notions.

Example 3.4.8 Interior, closure, and boundary. Consider the half-closed, half-open interval

$$[0,1) = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$$

of real numbers between 0 and 1.

- *Interior.*

  $[0,1)$ has interior the open interval

  $$\mathrm{Int}([0,1)) = (0,1) = \{x \in \mathbb{R} \mid 0 < x < 1\}.$$

- *Closure.*

  $[0,1)$ has closure the closed interval

  $$\mathrm{Cl}([0,1)) = [0,1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}.$$

- *Boundary.*

  $[0,1)$ has boundary the set $\mathrm{Bd}([0,1)) = \{0,1\}$.

▲

Clearly, a subset of a metric space is open if and only if it is its own interior. Perhaps less obvious, however, is the second half of the following result; a subset of a metric space is closed if and only if it is its own closure.

Proposition 3.4.9 Interior/closure. *Let $U \subseteq X$ be a subset of a metric space (X,ρ).*

1. *Interior.*

   U is open in X if and only if $U = \mathrm{Int}(U)$.

2. *Closure.*

   U is closed in X if and only if $U = \mathrm{Cl}(U)$.

Proof.

1. We first note that for any point $x \in \mathrm{Int}(U)$, there is some neighborhood N of x so that $N \subseteq U$. Since N is a neighborhood of x, it contains some open ball $\mathrm{B}_r(x) \subseteq N$. We conclude that

   $$x \in \mathrm{B}_r(x) \subseteq N \subseteq U,$$

and so $x \in U$. We conclude that $\mathrm{Int}(U) \subseteq U$. Therefore, it suffices to show that U is open in X if and only if $U \subseteq \mathrm{Int}(U)$.

To that end, we first suppose that U is open, and let $u \in U$. Then U is a neighborhood of u in X, and of course $U \subseteq U$. Thus $u \in \mathrm{Int}(U)$ is an interior point of U. We conclude that $U \subseteq \mathrm{Int}(U)$.

Conversely, now suppose that $U \subseteq \mathrm{Int}(U)$, and let $u \in U$. Then $u \in \mathrm{Int}(U)$, and so $N \subseteq U$ for some neighborhood N of U in X. Since N is a neighborhood of u, it contains some open ball $\mathrm{B}_r(u) \subseteq N$. We conclude that

$$\mathrm{B}_r(u) \subseteq N \subseteq U,$$

and so U is a neighborhood of u in X. Since this holds for all $u \in U$, U is open in X.

2. Let N be a neighborhood of a point $u \in U$ in X. Then N contains some open ball $\mathrm{B}_r(u)$, and so

$$u \in \mathrm{B}_r(u) \subseteq N.$$

In particular, $u \in N \cap U$, and so $N \cap U \neq \emptyset$ is non-empty. Thus $u \in \mathrm{Cl}(U)$ is a point of closure of U in X. We conclude that $U \subseteq \mathrm{Cl}(U)$. Therefore, it suffices to show that U is closed in X if and only if $\mathrm{Cl}(U) \subseteq U$.

To that end, we first suppose that U is closed, and let $x \in X$. If $x \in X \setminus U$, then our hypothesis that U is closed implies that x has a neighborhood N in X so that $N \cap U = \emptyset$. This implies that $x \in X \setminus \mathrm{Cl}(U)$. In summary, we have shown that $X \setminus U \subseteq X \setminus \mathrm{Cl}(U)$, and so $\mathrm{Cl}(U) \subseteq U$.

Conversely, now suppose that $\mathrm{Cl}(U) \subseteq U$, and let $x \in X \setminus U$. Then $x \notin \mathrm{Cl}(U)$ is not a point of closure of U in X, and so $N \cap U = \emptyset$ for some neighborhood N of x in X. This implies that $N \subseteq X \setminus U$, and so $X \setminus U$ is a neighborhood of x in X. Since this holds for all points $x \in X \setminus U$, we conclude that $X \setminus U$ is open in X; that is, U is closed in X.

■

So a subset of a metric space is open if and only if it is its own interior, and it is closed if and only if it is its own closure. One consequence of this is that the interior of a subset is open, and the closure of a set is closed. In order to show this, we make the following observation about the behavior of the openness and closedness conditions under the set operations.

Proposition 3.4.10 Topology of a metric space. *Let (X, ρ) be a metric space.*

1. *The empty set $\emptyset$ and the ground set X are both open and closed in X.*
2. *The union of any collection of open subsets of X is itself open in X.*
3. *The intersection of any collection of closed subsets of X is itself closed in X.*
4. *The intersection of finitely many open subsets of X is itself open in X.*
5. *The union of finitely many closed subsets of X is itself closed in X.*

Proof.

1. The empty set $\emptyset$ is vacuously a neighborhood of each of its points, and

so $\emptyset$ is open in X. On the other hand, let $x \in X$. Then

$$\mathrm{B}_1(x) = \{y \in X \mid \rho(y, x) < 1\} \subseteq X,$$

and so X is a neighborhood of x in X. We conclude that X is open in X.

Since the empty set $\emptyset$ and the ground set X are both open in X, their complements $X \setminus \emptyset = X$ and $X \setminus X = \emptyset$ are closed in X.

2. Let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of open sets $U_j \subseteq X$ in X, and consider the union
$$U = \bigcup_{j \in \mathcal{J}} U_j.$$
Let $u \in U$. Then $u \in U_k$ for some index $k \in \mathcal{J}$. Since U_k is open in X, it is a neighborhood of x in X, and so $\mathrm{B}_r(x) \subseteq U_k$ for some positive real number $r > 0$.

   We observe that
$$\mathrm{B}_r(x) \subseteq U_k \subseteq \bigcup_{j \in \mathcal{J}} U_j = U,$$
and so U is a neighborhood of u in X. Since this holds for all points $u \in U$, U is open in X.

3. Let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of closed sets $U_j \subseteq X$ in X, and consider the intersection
$$U = \bigcap_{j \in \mathcal{J}} U_j.$$
Since for each index $j \in \mathcal{J}$, $X \setminus U_j$ is open in X, (b) above implies that
$$X \setminus U = X \setminus \bigcap_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (X \setminus U_j)$$
is open in X. We conclude that $U = X \setminus (X \setminus U)$ is closed in X.

4. Let $(U_j)_{j=1}^m$ be a finite sequence of open sets $U_j \subseteq X$ in X, and consider the intersection
$$U = \bigcap_{j=1}^{m} U_j.$$
Let $u \in U$, so that $u \in U_j$ for all indices $1 \leq j \leq m$. For each index $1 \leq j \leq m$, U_j is open in X, and so U_j is a neighborhood of u in X; that is, $\mathrm{B}_{r_j}(u) \subseteq U_j$.

   Let $r = \min_{1 \leq j \leq m} r_j > 0$, and note that
$$\mathrm{B}_r(u) \subseteq \mathrm{B}_{r_j}(u) \subseteq U_j$$
for all indices $1 \leq j \leq m$. We conclude that
$$\mathrm{B}_r(u) \subseteq \bigcap_{j=1}^{m} U_j = U,$$
so that U is a neighborhood of u in X. Since this holds for all points $u \in U$, U is open in X.

5. Let $(U_j)_{j=1}^m$ be a finite sequence of closed sets $U_j \subseteq X$ in X, and consider the union
$$U = \bigcup_{j=1}^m U_j.$$
Since for each index $1 \leq j \leq m$, $X \setminus U_j$ is open in X, (d) above implies that
$$X \setminus U = X \setminus \bigcup_{j=1}^m U_j = \bigcap_{j=1}^m (X \setminus U_j)$$
is open in X. We conclude that $U = X \setminus (X \setminus U)$ is closed in X.

■

Along with Proposition 3.4.10 Topology of a metric space, the following result implies that the interior of a subset of a metric space is the largest open set which is contained in it, the closure of such a subset is the smallest closed set in which it is contained.

Theorem 3.4.11 Interior/closure. *Let $U \subseteq X$ be a subset of a metric space (X, ρ).*

1. *Interior.*

   U has interior
$$\mathrm{Int}\,(U) = \bigcup_{V \in \mathcal{O}_U} V,$$
   where $\mathcal{O}_U$ is the collection of open subsets of X contained in U.

2. *Closure.*

   U has closure
$$\mathrm{Cl}\,(U) = \bigcap_{V \in \mathcal{C}_U} V,$$
   where $\mathcal{C}_U$ is the collection of closed subsets of X which contain U.

Proof.

1. Let $x \in X$, and first suppose that $x \in \mathrm{Int}\,(U)$ is an interior point to U in X. Then U is a neighborhood of x in X, and so $\mathrm{B}_r\,(x) \subseteq U$ for some positive real number $r > 0$. Lemma 3.4.6 Open balls and closed disks implies that $\mathrm{B}_r\,(x)$ is open in X, and so $\mathrm{B}_r\,(x) \in \mathcal{O}_U$. We conclude that
$$x \in \mathrm{B}_r\,(x) \subseteq \bigcup_{V \in \mathcal{O}_U} V.$$
   Conversely, now suppose that $x \in \bigcup_{V \in \mathcal{O}_U} V$. Then $u \in V$ for some $V \in \mathcal{O}_U$. Since V is open in X, it is a neighborhood of x in X, and so $\mathrm{B}_r\,(x) \subseteq V$ for some positive real number $r > 0$. Since
$$\mathrm{B}_r\,(x) \subseteq V \subseteq U,$$
   U is a neighborhood of x in X; that is, $x \in \mathrm{Int}\,(U)$ is an interior point of U in X.

   In summary, we have shown that $x \in \mathrm{Int}\,(U)$ if and only if $x \in \bigcup_{V \in \mathcal{O}_U} V$, and so
$$\mathrm{Int}\,(U) = \bigcup_{V \in \mathcal{O}_U} V.$$

2. Consider an element $V \in \mathcal{C}_U$, and let $x \in X \setminus V$. Since V is closed in X, $X \setminus V$ is open in X, and so $X \setminus V$ is a neighborhood of each of its points. Moreover, since $U \subseteq V$, $(X \setminus V) \cap U = \emptyset$, and so none of the points of $X \setminus V$ are points of closure of U in X; that is, $X \setminus V \subseteq X \setminus \mathrm{Cl}\,(U)$. We conclude that $\mathrm{Cl}\,(U) \subseteq V$. Since this holds for all $V \in C$,

$$\mathrm{Cl}\,(U) \subseteq \bigcap_{V \in \mathcal{C}_U} V.$$

Conversely, now let $x \in X \setminus \mathrm{Cl}\,(U)$. Then $N \cap U = \emptyset$ for some neighborhood N of x in X. In particular, $\mathrm{B}_r\,(x) \cap U = \emptyset$ for some positive real number $r > 0$. Note that $X \setminus \mathrm{B}_r\,(x) \in \mathcal{C}_U$. However, $x \notin X \setminus \mathrm{B}_r\,(x)$, and so $x \notin \bigcap_{V \in \mathcal{C}_U} V$. We conclude that

$$\bigcap_{V \in \mathcal{C}_U} V \subseteq \mathrm{Cl}\,(U),$$

and so

$$\mathrm{Cl}\,(U) = \bigcap_{V \in \mathcal{C}_U} V..$$

■

Corollary 3.4.12 Idempotence of interior/closure. *Let (X, ρ) be a metric space.*

1. *Idempotence of interior.*

   $\mathrm{Int}\,(\mathrm{Int}\,(U)) = \mathrm{Int}\,(U)$ *for all subsets* $U \subseteq X$*.*

2. *Idempotence of closure.*

   $\mathrm{Cl}\,(\mathrm{Cl}\,(U)) = \mathrm{Cl}\,(U)$ *for all subsets* $U \subseteq X$*.*

Proof.

1. (1) of Theorem 3.4.11 Interior/closure implies that $\mathrm{Int}\,(U)$ is a union of open sets in X, and so $\mathrm{Int}\,(U)$ is itself open in X by (2) of Proposition 3.4.10 Topology of a metric space. Thus

$$\mathrm{Int}\,(\mathrm{Int}\,(U)) = \mathrm{Int}\,(U)$$

   by (1) of Proposition 3.4.9 Interior/closure.

2. (2) of Theorem 3.4.11 Interior/closure implies that $\mathrm{Cl}\,(U)$ is an intersection of closed sets in X, and so $\mathrm{Cl}\,(U)$ is itself closed in X by (3) of Proposition 3.4.10 Topology of a metric space. Thus

$$\mathrm{Cl}\,(\mathrm{Cl}\,(U)) = \mathrm{Cl}\,(U)$$

   by (2) of Proposition 3.4.9 Interior/closure.

■

For a subset $U \subseteq X$ of a metric space (X, ρ), the property of being a neighborhood of a point $x \in X$ is the existence of a certain kind lower bound on U in the power set $\mathcal{P}\,(X)$ of X. We now examine the dual condition, that of **metric boundedness**.

3.4.2 Metrics and Boundedness

In this section, we have introduced open balls and closed disks of positive radius. While the notion of radius does not easily generalize to arbitrary subsets of metric spaces (even for just open balls and closed disks), the related concept of **diameter** is easily generalized from its meaning for Euclidean circles and spheres of higher dimension. Informally, the diameter of a subset of a metric space is the largest possible value taken by the metric when considering only points in that subset.

Definition 3.4.13 Diameter. The diameter $\operatorname{diam}(U)$ of a subset $U \subseteq X$ of a metric space (X, ρ) is the extended real number defined by

$$\operatorname{diam}(U) = \begin{cases} 0 & U = \emptyset \\ \sup_{u,v \in U} \rho(u, v) & U \neq \emptyset. \end{cases}$$

◆

Lemma 3.4.14 Diameter and radius. *Let $r > 0$ be a positive real number and $c \in X$ be a point in a metric space (X, ρ). The open ball $\mathrm{B}_r(c)$, closed disk $\mathrm{D}_r(c)$, and sphere $\mathrm{S}_r(c)$ of radius r and center c have diameter at most $2r$.*

Proof. Since $c \in \mathrm{B}_r(c)$ the open ball $\mathrm{B}_r(c) \neq \emptyset$ is non-empty. Since

$$\rho(u, v) \leq \rho(u, c) + \rho(c, v) = \rho(u, c) + \rho(v, c) < r + r = 2r$$

for all points $u, v \in \mathrm{B}_r(c)$,

$$\operatorname{diam}(\mathrm{B}_r(c)) = \sup_{u,v \in \mathrm{B}_r(c)} \rho(u, v) \leq 2r.$$

Similarly, since $c \in \mathrm{D}_r(c)$ the closed disk $\mathrm{D}_r(c) \neq \emptyset$ is non-empty. Since

$$\rho(u, v) \leq \rho(u, c) + \rho(c, v) = \rho(u, c) + \rho(v, c) \leq r + r = 2r$$

for all points $u, v \in \mathrm{D}_r(c)$,

$$\operatorname{diam}(\mathrm{D}_r(c)) = \sup_{u,v \in \mathrm{D}_r(c)} \rho(u, v) \leq 2r.$$

If the sphere $\mathrm{S}_r(c) = \emptyset$ is empty, then $\operatorname{diam}(\mathrm{S}_r(c)) = \operatorname{diam}(\emptyset) = 0 < 2r$. On the other hand, if the sphere $\mathrm{S}_r(c) \neq \emptyset$ is non-empty, then we observe that

$$\rho(u, v) \leq \rho(u, c) + \rho(c, v) = \rho(u, c) + \rho(v, c) \leq r + r = 2r$$

for all points $u, v \in \mathrm{S}_r(c)$,

$$\operatorname{diam}(\mathrm{S}_r(c)) = \sup_{u,v \in \mathrm{D}_r(c)} \rho(u, v) \leq 2r.$$

■

So the diameter of an open ball, closed disk, or sphere in a metric space is bounded above by twice its radius. Note that, unlike in Euclidean spaces, the diameter of an open ball, closed disk, or sphere in arbitrary metric space need not actually reach this upper bound. This is because, unlike in Euclidean space, in which a given open ball or closed disk has a unique center and radius, balls and disks of different centers and radii may be equal. In particular, the geometry of open balls, closed disks, and spheres may be quite different in a given metric space than our intuition from Euclidean space might suggest.

Lemma 3.4.15 Monotonicity of the diameter. *Let $S, T \subseteq X$ be subsets of a metric space (X, ρ). If $S \subseteq T$, then* $\operatorname{diam}(S) \leq \operatorname{diam}(T)$.

Proof. If $T = \emptyset$, then $S = \emptyset$ as well, and so

$$\operatorname{diam}(S) = \operatorname{diam}(\emptyset) = \operatorname{diam}(T).$$

So we may suppose that $T \neq \emptyset$ is non-empty. We observe that $\rho(x, y) \geq 0$ for all points $x, y \in T$, and so

$$\operatorname{diam}(S) = 0 \leq \sup_{x,y \in T} \rho(x, y) = \operatorname{diam}(T).$$

Thus we may additionally suppose that $S \neq \emptyset$ is non-empty. In this case,

$$\begin{aligned} \operatorname{diam}(S) &= \sup(\{\rho(x, y) \in \mathbb{R} \mid x, y \in S\}) \\ &\leq \sup(\{\rho(x, y) \in \mathbb{R} \mid x, y \in T\}) = \operatorname{diam}(T) \end{aligned}$$

■

Much of this section thus far has been almost entirely concerned with the properties of subsets of metric spaces which contain certain open balls. We now consider the opposite case; what can we say about subsets which are themselves contained in open balls of finite radius? Such subsets of metric spaces are called **bounded**.

Definition 3.4.16 Boundedness. A subset $U \subseteq X$ of a metric space (X, ρ) is called **bounded** (resp. **unbounded**) if $\operatorname{diam}(U) < \infty$ (resp. $\operatorname{diam}(U) = \infty$). A metric space (X, ρ) is called **bounded** (resp. **unbounded**) if its ground set X is bounded (resp. unbounded). ◆

A priori, the above definition of boundedness does not appear to be related to the definition introduced in Section 1.5 Introduction to Order Theory. Namely, that kind of boundedness requires a partial order, and this one requires a metric. However, there is a way to reconcile these perspectives, and the two notions coincide in the real numbers, taken with their standard ordering $\leq$ and the Euclidean metric d.

Example 3.4.17 Boundedness. The following are examples of bounded and unbounded subsets of metric spaces:

- The empty set $\emptyset$ is a bounded subset of every metric space.
- Any discrete metric space is bounded, since its diameter is either 0 or 1.
- The n-dimensional Euclidean space $\mathbb{R}^n$ is bounded when $n = 0$ and unbounded when $n \geq 1$.

▲

Any subset of a bounded subset of a metric space is itself bounded, and any set which contains an unbounded set is itself unbounded. Moreover, any subset of a nonempty metric space is bounded if and only if it is contained in an open ball or closed disk of finite radius.

Proposition 3.4.18 Metric characterization of boundedness. *A subset of a nonempty metric space is bounded if and only if it is contained in an open ball of finite radius.*

Proof. Let $U \subseteq X$ be a subset of a non-empty metric space (X, ρ). First suppose that U is bounded, so that $\operatorname{diam}(U) < \infty$. Since $X \neq \emptyset$ is non-empty, it contains a point $x \in X$. If $U = \emptyset$ is empty, then $U = \emptyset \subseteq \mathrm{B}_r(x)$ for any positive real number $r > 0$.

On the other hand, if $U \neq \emptyset$ is non-empty, then it contains a point $u \in U$. Let $r = \operatorname{diam}(U) + 1$, and note that

$$\rho(u, v) \leq \sup_{s,t \in U} \rho(s, t) = \operatorname{diam}(U) < r$$

for all points $v \in U$, and so $U \subseteq \mathrm{B}_r(u)$.

Conversely, now suppose that $U \subseteq \mathrm{B}_r(x)$ for some point $x \in X$. Then Lemma 3.4.15 Monotonicity of the diameter and Lemma 3.4.14 Diameter and radius imply that

$$\operatorname{diam}(U) \leq \operatorname{diam}(\mathrm{B}_r(x)) \leq 2r < \infty,$$

so that U is bounded. ■

In fact, something stronger is true for normed vector spaces; a subset of a normed vector space is bounded if and only if it is contained in some open ball of finite radius centered at the origin. This kind of result is typical of analysis done on normed vector spaces; the interplay between the norm and the vector space structure implies that most point-set topological concerns can be reduced to questions which are answerable in a neighborhood of the origin.

Corollary 3.4.19 Norm characterization of boundedness. *A subset of a normed vector space is bounded if and only if it is contained in an open ball of finite radius centered at the origin.*

Proof. Let $(V, \|\cdot\|)$ be a normed vector space. By Proposition 3.4.18 Metric characterization of boundedness, it suffices to show that any bounded subset of V is contained in an open ball centered at $\mathbf{0}$. To that end, let $U \subseteq V$ be a bounded subset of V. Proposition 3.4.18 Metric characterization of boundedness implies that $U \subseteq \mathrm{B}_r(\mathbf{v})$ for some positive real number $r > 0$ and some vector $\mathbf{v}$.

Let $s = r + \|\mathbf{v}\|$, and observe that

$$\begin{aligned} \|\mathbf{u} - \mathbf{0}\| &= \|\mathbf{u} - \mathbf{v} + \mathbf{v}\| \leq \|\mathbf{u} - \mathbf{v}\| + \|\mathbf{v}\| \\ &< r + \|\mathbf{v}\| = s \end{aligned}$$

for all $\mathbf{u} \in U$. Thus $U \subseteq \mathrm{B}_s(\mathbf{0})$. ■

Any finite subset of a metric space is bounded. In fact, any finite union of bounded subsets of a metric space is itself bounded. In order to see this, it suffices to show that the binary union of two bounded subsets of a metric space is itself bounded.

Proposition 3.4.20 Unions of bounded sets. *The union of two bounded sets in a metric space is itself bounded.*

Proof. Let $U, V \subseteq X$ be bounded subsets of a metric space (X, ρ), and consider the union $U \cup V$. If $X = \emptyset$ is empty, then $U = V = \emptyset$, and so

$$\operatorname{diam}(U \cup V) = \operatorname{diam}(\emptyset \cup \emptyset) \operatorname{diam}(\emptyset) = 0 < \infty,$$

so that U and V are bounded. On the other hand, if $X \neq \emptyset$ is non-empty, then Proposition 3.4.18 Metric characterization of boundedness implies that $U \subseteq \mathrm{B}_r(x)$ and $V \subseteq \mathrm{B}_s(y)$ for some positive real numbers $r, s > 0$ and points $x, y \in X$. Let

$$t = \max(\{r, s + \rho(y, x)\}) > 0,$$

and consider a point $z \in U \cup V$. If $z \in U$, then

$$\rho(z, x) < r \leq t,$$

and if $z \in V$, then

$$\rho(z, x) \leq \rho(z, y) + \rho(y, x) < s + \rho(y, x) \leq t.$$

In either case, we conclude that $z \in \mathrm{B}_t(x)$, and so $U \cup V \subseteq \mathrm{B}_t(x)$. Proposition 3.4.18 Metric characterization of boundedness now implies that $U \cup V$ is bounded. ■

The following corollary follows immediately from the above result by induction. Essentially, it asserts that the property of boundedness is stable under finite unions.

Corollary 3.4.21 Unions of bounded sets. *The union of any finite collection of bounded subsets of a metric space is itself bounded.*

Before we move on to discuss the boundedness of maps between metric spaces, we will briefly mention one stronger kind of boundedness for subsets of metric spaces. A subset of a metric space is called **totally bounded** if it can be covered by finitely many open balls of arbitrarily small radius.

Definition 3.4.22 Total boundedness. A subset $U \subseteq X$ of a metric space (X, ρ) is called **totally bounded** if for all positive real numbers $r > 0$, there is a finite list of points $x_1, \ldots, x_n \in X$ so that

$$U \subseteq \bigcup_{k=1}^{n} \mathrm{B}_r(x_k).$$

A metric space (X, ρ) is called **totally bounded** if its ground set X is totally bounded. ◆

Example 3.4.23 Total boundedness. The following are examples of totally bounded and not totally bounded subsets of metric spaces:

- The empty set $\emptyset$ is a totally bounded subset of any metric space. More generally, any finite set of a metric space is totally bounded.
- Any infinite subset of a discrete metric space is bounded but not totally bounded.
- A subset of n-dimensional Euclidean space $\mathbb{R}^n$ is totally bounded if and only if it is bounded.

▲

The above examples illustrate that while boundedness and total boundedness are not in general equivalent for subsets of metric spaces, the two notions are also not necessarily entirely unrelated. In fact, total boundedness is a stronger condition than boundedness for subsets of a metric space.

Proposition 3.4.24 Total boundedness implies boundedness. *If a subset of a metric space is totally bounded, then it is bounded.*

Proof. Let $U \subseteq X$ be a totally bounded subset of a metric space (X, ρ). Since $1 > 0$ is a positive real number,

$$U \subseteq \bigcup_{k=1}^{n} \mathrm{B}_1(x_k)$$

for some finite list of points $x_1, \ldots, x_n \in X$. Lemma 3.4.14 Diameter and radius and Corollary 3.4.21 Unions of bounded sets together imply that U is bounded. ■

We have seen that infinite discrete metric spaces provide a counterexample to the converse of Proposition 3.4.24 Total boundedness implies boundedness. However, in Euclidean space, total boundedness and boundedness are equivalent. We offer this result without proof.

Proposition 3.4.25 Total boundedness in Euclidean space. *A subset of Euclidean space is totally bounded if and only if it is bounded.*

One way to see why Proposition 3.4.25 Total boundedness in Euclidean space is true is that any open ball or closed disk of finite radius in Euclidean space is a subset of a cube, which may be subdivided into finitely many appropriately small cubes. While boundedness and total boundedness are equivalent in Euclidean spaces, they are not the same for metric spaces in general. In some of the generalizations of theorems about Euclidean space to results about metric spaces in general, total boundedness is a more natural notion to discuss. We will further discuss totally bounded sets in the Section 3.6 Completeness and Compactness, in which we investigate **completeness** and **compactness** for metric spaces. For now, however, we move on in favor of exploring boundedness for maps between metric spaces.

3.4.3 Bounded maps between metric spaces

In Section 1.5 Introduction to Order Theory, we defined boundedness for subsets of partially ordered sets, and then we defined a map from a set to a partially ordered set to be bounded if and only if its image was a bounded subset of the codomain. In this section, we have introduced a different notion of boundedness for subsets of a metric space, but an there is an analogous way to extend this notion to maps to metric spaces. A map to a metric space is **bounded** if its image is bounded in the codomain, and unbounded otherwise.

We note that this is a different notion than a bounded linear transformation between normed vector spaces, which is not bounded in this sense. The homogeneity of a norm on a vector space implies that no nonzero linear transformation to a normed vector space can be bounded in the above sense.}

Definition 3.4.26 Map boundedness. Let $f\colon X \to Y$ be a map from a set X to a metric space (Y, ρ).

- *Boundedness.*

  The map f is called **bounded** if its image $\operatorname{im}(f) = f(X)$ is a bounded subset of Y.

- *Unboundedness.*

  The map f is called **unbounded** if its image $\operatorname{im}(f) = f(X)$ is an unbounded subset of Y.

The set of bounded maps from a set X to a metric space (Y, ρ) is denoted $\mathrm{B}(X, Y)$. In the special case $(Y, \rho) = (\mathbb{R}, d)$, the set of bounded real-valued functions on X is denoted $\mathrm{B}(X) = \mathrm{B}(X, \mathbb{R})$. ◆

Remark 3.4.27 Boundedness for linear transformations. We remark that the above definition of boundedness is different than the usual definition of boundedness for linear maps between normed vector spaces. Indeed, in the setting of normed vector spaces, the absolute homogeneity of the norm implies that no non-zero linear transformation to a normed vector space can be bounded in the above sense. Instead, a linear transformation is called **bounded** if its restriction to the unit sphere around the origin is bounded in the above sense.

Example 3.4.28 Bounded functions. Let $f\colon \mathbb{R} \to \mathbb{R}$ be the real-valued function defined by the formula

$$f(x) = e^{-x^2}.$$

Then f is a bounded function on the real line $\mathbb{R}$. ▲

Just as in the cases of sequential and map convergence, in attempting to understand boundedness in arbitrary metric spaces, it suffices to investigate boundedness in the real line $\mathbb{R}$. These links between arbitrary metric spaces and the metric space of the real numbers taken with the Euclidean metric stem from the definition of a metric; the structure of a real-valued function such as a metric can be seen to pull back some of the properties of its codomain $\mathbb{R}$ to its domain. In order to check that a metric space is bounded, it is both necessary and sufficient for the metric itself to be a bounded function.

Lemma 3.4.29 Map boundedness. *A metric space (X, ρ) is bounded if and only if the metric $\rho\colon X \times X \to \mathbb{R}$ is a bounded function.*

Proof. First suppose that (X, ρ) is bounded, so that $\operatorname{diam}(X) < \infty$. If $X = \emptyset$, then

$$\begin{aligned}\operatorname{diam}(\operatorname{im}(\rho)) &= \operatorname{diam}(\rho(\emptyset \times \emptyset)) = \operatorname{diam}(\rho(\emptyset)) \\ &= \operatorname{diam}(\emptyset) = 0 < \infty,\end{aligned}$$

and so ρ is bounded. Thus we may suppose that $X \neq \emptyset$ is non-empty. In particular,

$$0 \leq \rho(x, y) \leq \operatorname{diam}(X)$$

for all points $x, y \in X$, which implies that

$$d(\rho(x, y), 0) = |\rho(x, y) - 0| = |\rho(x, y)| = \rho(x, y) \leq \operatorname{diam}(X)$$

for all points $x, y \in X$. In particular, this implies that $\operatorname{im}(\rho) \subseteq \mathrm{D}_{\operatorname{diam}(X)}(0)$. Now Lemma 3.4.15 Monotonicity of the diameter and Lemma 3.4.14 Diameter and radius imply that

$$\operatorname{diam}(\operatorname{im}(\rho)) \leq \operatorname{diam}\left(\mathrm{D}_{\operatorname{diam}(X)}(0)\right) \leq 2\operatorname{diam}(X) < \infty,$$

so that $\operatorname{im}(\rho)$ is bounded.

Conversely, now suppose that ρ is bounded. Then Corollary 3.4.19 Norm characterization of boundedness implies that $\operatorname{im}(\rho) \subseteq \mathrm{B}_r(0)$ for some positive real number $r > 0$. This implies that

$$\rho(x, y) = |\rho(x, y) - 0| = d(\rho(x, y), 0) < r$$

for all $x, y \in X$, so that $\operatorname{diam}(X) \leq r < \infty$; that is, (X, ρ) is bounded. ■

The set of bounded maps to a metric space may inherit some of the structure of the codomain. We will see some examples of this shortly. First, however, a useful observation about the relationship between the boundedness of maps and the boundedness of their codomains.

Remark 3.4.30 Maps to bounded metric spaces. Let (Y, ρ) be a metric space. If Y is bounded, then Lemma 3.4.15 Monotonicity of the diameter implies that for any set X, any map $f\colon X \to Y$ is bounded; that is, $\mathrm{B}(X, Y) =$

Y^X.

There is a canonical way of constructing a metric on the set of bounded maps to a metric space.

Definition 3.4.31 Uniform metric. Let X be a set and (Y, ρ) be a metric space. The **uniform metric** ρ_∞ on the set $\mathrm{B}(X, Y)$ of bounded maps from X to Y is the map $\rho_\infty \colon \mathrm{B}(X, Y) \times \mathrm{B}(X, Y) \to \mathbb{R}$ defined by the formula

$$\rho_\infty(f, g) = \begin{cases} 0 & X = \emptyset \\ \sup_{x \in X} \rho(f(x), g(x)) & X \neq \emptyset. \end{cases}$$

◆

Proposition 3.4.32 Uniform metric. *The uniform metric is a metric.*

Proof. Let X be a set and (Y, ρ) be a metric space. If $X = \emptyset$, then there is exactly one map from X to Y, and this map is bounded. The uniform metric ρ_∞ is identically zero in this case, and so is a metric on $\mathrm{B}(X, Y)$.

Thus we may suppose that $X \neq \emptyset$ is non-empty. Let $f, g \in \mathrm{B}(X, Y)$ be a bounded map from X to Y, and note that if $f = g$, then

$$\begin{aligned} \rho_\infty(f, g) &= \sup_{x \in X} \rho(f(x), g(x)) = \sup_{x \in X} \rho(f(x), f(x)) \\ &= \sup_{x \in X} 0 = 0. \end{aligned}$$

Conversely, if $\rho_\infty(f, g) = 0$, then

$$0 \leq \rho(f(x), g(x)) \leq \rho_\infty(f, g) = 0$$

for all inputs $x \in X$, so that $\rho(f(x), g(x)) = 0$. The non-degeneracy of the metric ρ implies that $f(x) = g(x)$ for all inputs $x \in X$, and so $f = g$. We conclude that ρ_∞ is non-degenerate.

We observe that the symmetry of the metric ρ implies that

$$\rho_\infty(f, g) = \sup_{x \in X} \rho(f(x), g(x)) = \sup_{x \in X} \rho(g(x), f(x)) = \rho_\infty(g, f)$$

for all bounded maps $f, g \in \mathrm{B}(X, Y)$, and so ρ_∞ is symmetric.

Finally, we observe that the triangle inequality for the metric ρ implies that

$$\begin{aligned} \rho_\infty(f, h) &= \sup_{x \in X} \rho(f(x), h(x)) \leq \sup_{x \in X} (\rho(f(x), g(x)) + \rho(g(x), h(x))) \\ &\leq \sup_{x \in X} \rho(f(x), g(x)) + \sup_{x \in X} \rho(g(x), h(x)) = \rho_\infty(f, g) + \rho_\infty(g, h) \end{aligned}$$

for all bounded maps $f, g, h \in \mathrm{B}(X, Y)$, and so ρ_∞ is satisfies the triangle inequality.

Since ρ_∞ is reflexive, symmetric, and satisfies the triangle inequality, it is a metric on $\mathrm{B}(X, Y)$. ■

You will have an opportunity to explore the uniform metric in the setting of normed vector spaces in the problems for this section. In particular, you will see that in this setting, the uniform metric is induced by a norm.

The similarity in the names of the uniform metric and uniformly convergent sequences of functions is no coincidence. In fact, it turns out that for bounded maps to a metric space, these two notions coincide.

Theorem 3.4.33 Convergence in the uniform metric. *Let $(f_n)_{n=0}^\infty$ be a sequence of bounded maps $f_n \in \mathrm{B}(X, Y)$ from a set X to a metric space (Y, ρ).*

For a bounded map $f \in \mathrm{B}(X, Y)$, $(f_n)_{n=0}^{\infty}$ converges uniformly to f if and only if it converges to f in the uniform metric.

In the next section, we reintroduce one of the fundamental ideas in analysis and topology: that of **continuity**.

3.4.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Neighborhoods. Determine whether or not the given set is a neighborhood of the given point.

1. Let
$$A = (0, 2) = \{x \in \mathbb{R} \mid 0 < x < 2\}$$
be the open interval between 0 and 2. Determine whether or not A is a neighborhood of the point 1 in the metric space $(\mathbb{R}, d)$.

2. Let
$$B = [0, 2] = \{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$$
be the closed interval between 0 and 2. Determine whether or not B is a neighborhood of the point 1 in the metric space $(\mathbb{R}, d)$.

3. Let
$$C = [0, 2) = \{x \in \mathbb{R} \mid 0 \leq x < 2\}$$
be the half-closed, half-open interval between 0 and 2. Determine whether or not C is a neighborhood of the point 0 in the metric space $(\mathbb{R}, d)$.

4. Let
$$D = 2\mathbb{Z} = \{2k \in \mathbb{Z} \mid k \in \mathbb{Z}\} = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$$
be the set of even integers. Determine whether or not D is a neighborhood of the point 0 in the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

5. Let
$$E = [0, 1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$
be the closed interval between 0 and 1. Determine whether or not E is a neighborhood of the point 0 in the metric subspace $([0, 2], d)$ of the real line $(\mathbb{R}, d)$.

Open and closed sets. Determine whether the given subset of a metric space is open, closed, both open and closed, or neither open nor closed.

6. Let
$$A = (0, 2) = \{x \in \mathbb{R} \mid 0 < x < 2\}$$
be the open interval between 0 and 2. Determine whether or not A is open or closed in the metric space $(\mathbb{R}, d)$.

7. Let
$$B = [0, 2] = \{x \in \mathbb{R} \mid 0 \le x \le 2\}$$
be the closed interval between 0 and 2. Determine whether or not B is open or closed in the metric space $(\mathbb{R}, d)$.

8. Let
$$C = [0, 2) = \{x \in \mathbb{R} \mid 0 \le x < 2\}$$
be the half-closed, half-open interval between 0 and 2. Determine whether or not C is open or closed in the metric space $(\mathbb{R}, d)$.

9. Let
$$D = 2\mathbb{Z} = \{2k \in \mathbb{Z} \mid k \in \mathbb{Z}\} = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$$
be the set of even integers. Determine whether or not D is open or closed in the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

10. Let
$$E = [0, 1] = \{x \in \mathbb{R} \mid 0 \le x \le 1\}$$
be the closed interval between 0 and 1. Determine whether or not E is open or closed in the metric subspace $([0, 2], d)$ of the real line $(\mathbb{R}, d)$.

11. Let
$$F = [0, 1] = \{x \in \mathbb{R} \mid 0 \le x \le 1\}$$
be the closed interval between 0 and 1. Determine whether or not F is open or closed in the metric subspace $([0, 1] \cup [2, 3], d)$ of the real line $(\mathbb{R}, d)$.

Diameter. Compute the diameter of the given subsets of metric spaces.

12. Let $A = \{0, 1, 2\}$. Compute the diameter of A as a subset of the metric space $(\mathbb{R}, d)$.

13. Let $B = \{0, 1, 2, 5\}$. Compute the diameter of B as a subset of the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

14. Let
$$C = \{0, 1, 2\} \cup \left\{3 - \frac{1}{n}\right\}_{n=1}^{\infty}.$$
Compute the diameter of C as a subset of the metric space $(\mathbb{R}, d)$.

15. Let
$$D = 3\mathbb{Z} = \{3k \in \mathbb{Z}\}\, k \in \mathbb{Z}$$
be the set of integers which are divisible by 3. Compute the diameter of D as a subset of the metric space $(\mathbb{R}, d)$.

16. Let $E = \{\pi\}$. Compute the diameter of E as a subset of the metric space $(\mathbb{R}, d)$.

17. Let $F = \{\pi\}$. Compute the diameter of F as a subset of the discrete metric space $(\mathbb{R}, \delta_X)$.

Bounded and unbounded subsets of metric spaces. Let
$$U = (0, 1) = \{x \in \mathbb{R} \mid 0 < x < 1\}$$
be the open interval between 0 and 1. Determine whether U is a bounded or unbounded subset of the given metric space.

18. Determine whether U is bounded or unbounded in the metric space $(\mathbb{R}, d)$.

19. Determine whether U is bounded or unbounded in the subspace $((0, \infty), d)$ of the metric space $(\mathbb{R}, d)$.

20. Determine whether U is bounded or unbounded in the metric space $((0, \infty), \rho)$, where ρ is the metric on $(0, \infty)$ defined by the formula

$$\rho(x, y) = \left| \ln \left(\frac{x}{y} \right) \right|.$$

21. Determine whether U is bounded or unbounded in the discrete metric space $\left((0, \infty), \delta_{(0,\infty)}\right)$.

The uniform metric. Compute the distance between the given bounded maps in the uniform metric.

22. Let $f, g \colon \mathbb{R} \to \mathbb{R}$ be the constant real-valued functions defined by the formulae $f(x) = 2$ and $g(x) = \pi$. Compute the uniform distance $d_\infty(f, g)$.

23. Let $f, g \colon [-1, 1] \to \mathbb{R}$ be the real-valued functions defined by the formulae $f(x) = 2$ and $g(x) = x^2$. Compute the uniform distance $d_\infty(f, g)$.

24. Let $f, g \colon (0, 2) \to \mathbb{R}$ be the real-valued functions defined by the formulae $f(x) = 2 - x$ and $g(x) = 1 + 3x$. Compute the uniform distance $d_\infty(f, g)$.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.4.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Open and closed sets in metric subspaces.** Let $U \subseteq X$ be a subset of a metric space (X, ρ).

(a) Prove that a subset $V \subseteq U$ is open in the metric subspace (U, ρ) if and only if $V = U \cap W$ for some open set $W \subseteq X$ in X.

(b) Prove that if a subset $V \subseteq U$ is open in X, then it is open in U.

(c) Construct an explicit counterexample to the converse of (b) above.

2. **Interior, closure, and boundary of complements.** Let $U \subseteq X$ be a subset of a metric space (X, ρ).

(a) Prove that $\operatorname{Int}(X \setminus U) = X \setminus \operatorname{Cl}(U)$.

(b) Prove that $\operatorname{Cl}(X \setminus U) = X \setminus \operatorname{Int}(U)$.

(c) Prove that $\operatorname{Bd}(X \setminus U) = \operatorname{Bd}(U) = \operatorname{Cl}(U) \cap \operatorname{Cl}(X \setminus U)$.

(d) Prove that $\operatorname{Bd}(U) = \operatorname{Cl}(U) \setminus \operatorname{Int}(U)$.

3. **Closure and limit points.** Let $U \subseteq X$ be a subset of a metric space (X, ρ). Prove that $\operatorname{Cl}(U) = U \cup U'$.

Hint. Recall that the **derived set** U' of a subset $U \subseteq X$ of a metric space (X, ρ) is the set of all limit points of U in X.

4. **Open balls in ultrametric spaces.** Prove that any point in an open ball in an ultrametric space is its center.

Hint. You may find (c) of Problem 3.1.5.3 Ultrametrics to be of use.

5. **Diameter in the real line.** Let $\emptyset \subsetneq U \subseteq \mathbb{R}$ be a non-empty subset of the real line $(\mathbb{R}, d)$. Prove that U has diameter

$$\operatorname{diam}(U) = \sup(U) - \inf(U)$$

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.5 Notions of Metric Continuity

In this section, we present three distinct but related notions of continuity on a metric space. Informally, these forms of continuity are conditions related to the nature of the convergence of a function to its value at a point. We will in this section make this precise, and investigate the structure of continuous maps between metric spaces.

3.5.1 Continuous Maps and Functions

In calculus, you may have learned that a function of the real numbers is continuous if the function commutes with limits, if limits of the function agree with the values of the function. We will see that this is also the case for maps of metric spaces. More precisely, map of metric spaces is **continuous** at a point in its domain if its values can be made arbitrarily close to the image of that point just by choosing points close to that point.

Proposition 3.5.1 Characterizations of metric continuity. *Let $f : X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). For a point $c \in X$, the following are equivalent:*

1. *Convergence characterization of continuity.*

   The map f converges at c to $f(c)$.

2. *ϵ-δ characterization of continuity.*

   For all positive real numbers $\epsilon > 0$, there is a positive real number $\delta > 0$ so that for all points $x \in X$, if $\rho_X(x, c) < \delta$, then $\rho_Y(f(x), f(c)) < \epsilon$.

3. *Neighborhood characterization of continuity.*

   The pre-image $f^{-1}(N)$ of any neighborhood N of $f(c)$ in Y is a neighborhood of c in X.

4. *Sequential characterization of continuity.*

   For every sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$, if

$$\lim_{n \to \infty} x_n = c,$$

then

$$\lim_{n\to\infty} f(x_n) = f(c).$$

Proof. First suppose that f converges at c to $f(c)$, and let $\epsilon > 0$. Then there exists a positive real number $\delta > 0$ so that for all points $x \in X$, if $0 < \rho_X(x,c) < \delta$, then $\rho_Y(f(x), f(c)) < \epsilon$. We note that if $\rho_X(x,c) = 0$, then the non-degeneracy of ρ_X implies that $x = c$. In this case, $f(x) = f(c)$, and so $\rho_Y(f(x), f(c)) = 0$.

Now suppose that for all positive real numbers $\epsilon > 0$, there is a positive real number $\delta > 0$ so that for all points $x \in X$, if $\rho_X(x,c) < \delta$, then $\rho_Y(f(x), f(c)) < \epsilon$, and let $N \subseteq Y$ be a neighborhood of $f(c)$ in Y. Then $\mathrm{B}_\epsilon(f(c)) \subseteq N$ for some positive real number $\epsilon > 0$.

Since $\epsilon > 0$ is positive, there is a positive real number $\delta > 0$ so that for all points $x \in X$, if $\rho_X(x,c) < \delta$, then $\rho_Y(f(x), f(c)) < \epsilon$. In particular,

$$f(\mathrm{B}_\delta(c)) \subseteq \mathrm{B}_\epsilon(f(c)) \subseteq N,$$

and so $\mathrm{B}_\delta(c) \subseteq f^{-1}(N)$; that is, $f^{-1}(N)$ is a neighborhood of c in X.

Now suppose that the pre-image $f^{-1}(N)$ of any neighborhood N of $f(c)$ in Y is a neighborhood of c in X, and let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ so that

$$\lim_{n\to\infty} x_n = c.$$

Let $\epsilon > 0$ be a positive real number. Then $\mathrm{B}_\epsilon(f(c))$ is a neighborhood of $f(c)$ in Y, and so $f^{-1}(\mathrm{B}_\epsilon(f(c)))$ is a neighborhood of c in X. Thus

$$\mathrm{B}_\delta(c) \subseteq f^{-1}(\mathrm{B}_\epsilon(f(c)))$$

for some positive real number $\delta > 0$. Since $(x_n)_{n=0}^{\infty}$ converges to c, there is some natural number $n_\epsilon \in \mathbb{N}$ so that $\rho_X(x_n, c) < \delta$ for all indices $n \geq n_\epsilon$. This implies that

$$f(x_n) \in f(\mathrm{B}_\delta(c)) \subseteq \mathrm{B}_\epsilon(f(c))$$

for all indices $n \geq n_\epsilon$, and so $\rho_Y(f(x_n), f(c)) < \epsilon$ for all such indices. We conclude that

$$\lim_{n\to\infty} f(x_n) = f(c).$$

Finally, suppose that for every sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$, if

$$\lim_{n\to\infty} x_n = c,$$

then

$$\lim_{n\to\infty} f(x_n) = f(c).$$

Then Theorem 3.2.19 Sequential characterization of map convergence implies that f converges at c to $f(c)$. ■

Definition 3.5.2 Continuity. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y).

- *Continuity.*

  f is **continuous** at a point $c \in X$ if it satisfies any (and therefore all) of the conditions in Proposition 3.5.1 Characterizations of metric continuity. f is **continuous** if it is continuous at every point $c \in X$.

- *Discontinuity.*

  f is **discontinuous** at a point $c \in X$ if it does not satisfies any of the conditions in Proposition 3.5.1 Characterizations of metric continuity. f is **discontinuous** if it is discontinuous at some point $c \in X$.

The set of continuous maps from X to Y is denoted $\mathrm{C}(X, Y)$. In the special case $(Y, \rho_Y) = (\mathbb{R}, d)$, the set of continuous, real-valued functions on a metric space (X, ρ_X) is denoted $\mathrm{C}(X) = \mathrm{C}(X, \mathbb{R})$. ◆

Example 3.5.3 Continuity. The following are examples of continuous and discontinuous maps.

- Any constant map between metric spaces is continuous.
- The identity map $\mathrm{id}_X : X \to X$ on a metric space (X, ρ) is continuous. More generally, the inclusion map $\iota : U \to X$ of a metric subspace $U \subseteq X$ is continuous.
- Any map $f : X \to Y$ from a discrete metric space (X, δ_X) to a metric space (Y, ρ_Y) is continuous.
- Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be the real-valued function defined by the formula

  $$f(x) = \frac{1}{x}.$$

  Then f is continuous. In particular, we note that f is neither continuous nor discontinuous at 0, since 0 is not in the domain of f.

  You may have been taught that f has a discontinuity at the point 0, but we will require more specific language. We can describe the behavior of f near 0 by saying that f does not have a continuous extension to $\mathbb{R}$.

▲

A map of metric spaces is continuous at a point if and only if the outputs of the map can be made close to the image of that point solely by choosing inputs close to the point. Perhaps unsurprising in light of Theorem 3.2.19 Sequential characterization of map convergence is the fact that continuity is totally determined by the behavior of the map on convergent sequences.In fact, unlike the more general spaces we will investigate in Chapter 4 Topological Spaces, the answers to all point-set topological questions about metric spaces are captured by the behavior of sequences. Perhaps more surprising is the following characterization of continuity in terms of open sets.

Corollary 3.5.4 Open set characterization of metric continuity. *A map $f : X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is continuous if and only if the pre-image of any open subset of Y is an open subset of X.*

Proof. Suppose first that f is continuous, and let $V \subseteq Y$ be an open subset of Y. Let $c \in f^{-1}(V)$. Since V is open, it is a neighborhood of $f(c)$, and so the neighborhood characterization of continuity in Proposition 3.5.1 Characterizations of metric continuity implies that $f^{-1}(V)$ is a neighborhood of c. Since this is true of every point $c \in f^{-1}(V)$, $f^{-1}(V)$ is open in X.

Conversely, now suppose that the pre-image of every open subset of Y is an open subset of X, and consider a point $c \in X$ and a neighborhood $N \subseteq Y$ of $f(c)$ in Y. Then $\mathrm{B}_r(f(c)) \subseteq N$ for some positive real number $r > 0$. Since $\mathrm{B}_r(c)$ is open in Y, $f^{-1}(\mathrm{B}_r(f(c)))$ is open in X and is therefore a neighborhood of each of its points.

In particular, $f(c) \in \mathrm{B}_r(f(c))$, and so $c \in f^{-1}(\mathrm{B}_r(f(c)))$. We conclude that $f^{-1}(\mathrm{B}_r(f(c)))$ is a neighborhood of c. The neighborhood characterization of continuity in Proposition 3.5.1 Characterizations of metric continuity now implies that f is continuous at c. Since this holds for all points $c \in X$, f is continuous. ■

In general, proving that a given map between metric spaces is continuous can be annoyingly difficult. For example, if you want to use the ϵ-δ characterization of continuity, then finding a correct positive real number $\delta > 0$ for each positive real number $\epsilon > 0$ is not necessarily an easy task. The following results allow us to decompose such a problem into smaller parts, which may be much easier to solve than taking on the entire problem at once.

Proposition 3.5.5 Composition of continuous maps. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between metric spaces (X, ρ_X), (Y, ρ_Y), and (Z, ρ_Z). If f is continuous at a point $c \in X$ and g is continuous at $f(c)$, then the composition $g \circ f\colon X \to Z$ is continuous at c.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$, and suppose that

$$\lim_{n\to\infty} x_n = c.$$

Since f is continuous at c,

$$\lim_{n\to\infty} f(x_n) = f(c).$$

Since g is continuous at $f(c)$,

$$\lim_{n\to\infty} (g \circ f)(x_n) = \lim_{n\to\infty} g(f(x_n)) = g(f(c)) = (g \circ f)(c).$$

Thus $g \circ f$ is continuous at c. ■

Proposition 3.5.6 Vector arithmetic of continuous maps. *Let (X, ρ) be a metric space and $(V, \|\cdot\|)$ be a normed vector space.*

1. *Vector addition.*

   Let $f, g\colon X \to V$ be maps from X to V. If f and g are continuous at a point $c \in X$, then their pointwise sum $f + g$ is continuous at c.

2. *Scalar multiplication.*

   Let α be a scalar and $f\colon X \to V$ be a map from X to V. If f is continuous at a point $c \in X$, then the scalar multiplication αf is continuous at c.

3. *Norm.*

   Let $f\colon X \to V$ be a map from X to V. If f is continuous at a point $c \in X$, then the norm $\|f\| = \|\cdot\| \circ f$ is continuous at c.

4. *Inner product.*

   Let $f, g\colon X \to V$ be maps from X to V. If the norm $\|\cdot\|$ on V is induced by an inner product $\langle\,\cdot\,,\,\cdot\,\rangle$ and f and g are continuous at a point c, then the pointwise inner product $\langle f, g\rangle$ is continuous at c.

Proof.

1. Let $\epsilon > 0$ be a positive real number. Since f is continuous at c, there is a positive real number $\delta_f > 0$ so that for all points $x \in X$, if $\rho_X(x, c) < \delta_f$, then $\|f(x) - f(c)\| < \frac{\epsilon}{2}$. Similarly, since g is continuous at c, there is a positive real number $\delta_g > 0$ so that for all points $x \in X$, if $\rho_X(x, c) < \delta_g$, then $\|g(x) - g(c)\| < \frac{\epsilon}{2}$.

   Let $\delta = \min(\{\delta_f, \delta_g\}) > 0$, and note that for all points $x \in X$, if

$\rho_X(x, c) < \delta$, then

$$\begin{aligned}\|(f+g)(x) - (f+g)(c)\| &= \|f(x) + g(x) - f(c) - g(c)\| \\ &\leq \|f(x) - f(c)\| + \|g(x) - g(c)\| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.\end{aligned}$$

We conclude that $f + g$ is continuous at c.

2. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$, and suppose that

$$\lim_{n\to\infty} x_n = c.$$

Since f is continuous at c, we observe that

$$\begin{aligned}\lim_{n\to\infty} (\alpha f)(x_n) = \lim_{n\to\infty} \alpha f(x_n) &= \alpha \lim_{n\to\infty} f(x_n) \\ &= \alpha f(c) = (\alpha f)(c),\end{aligned}$$

and so αf is continuous at c.

3. By Proposition 3.5.5 Composition of continuous maps, it suffices to show that $\|\cdot\| : V \to \mathbb{R}$ is a continuous real-valued function on V. This follows directly from (3) of Proposition 3.3.1 Limits and vector arithmetic.

4. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$, and suppose that

$$\lim_{n\to\infty} x_n = c.$$

Since f and g are continuous at c, we observe that

$$\begin{aligned}\lim_{n\to\infty} \langle f, g\rangle(x_n) = \lim_{n\to\infty} \langle f(x_n), g(x_n)\rangle &= \left\langle \lim_{n\to\infty} f(x_n), \lim_{n\to\infty} g(x_n)\right\rangle \\ &= \langle f(c), g(c)\rangle = \langle f, g\rangle(c),\end{aligned}$$

and so $\langle f, g\rangle$ is continuous at c.

■

Corollary 3.5.7 Arithmetic of continuous functions. *Let $f, g \colon X \to \mathbb{R}$ be real-valued functions on a metric space (X, ρ). If f and g are continuous at a point $c \in X$, then $f + g$, $f - g$, and fg are continuous at c.*

Moreover, if f and g are continuous at at point $c \in X$ and $g(x) \neq 0$ for all points $x \in X$, then $\frac{f}{g}$ is continuous at c.

Having introduced the notion of continuity for maps between metric spaces, one natural question is to describe the relationship between continuity and the limiting behavior of sequences of maps. At first glance, it may seem plausible that the limit of a sequence of continuous functions which converges pointwise must also be continuous. This, however, is not the case.

Example 3.5.8 Discontinuous limits of sequences of continuous functions. Consider the sequence $(f_n)_{n=0}^{\infty}$ of continuous real-valued functions $f_n \colon [0, 1] \to \mathbb{R}$ defined by the formulae

$$f_n(x) = x^n.$$

We recall that $(f_n)_{n=0}^{\infty}$ converges pointwise to the function $f \colon [0, 1] \to \mathbb{R}$ defined piecewise by the formula

$$f(x) = 1 \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1. \end{cases}$$

Even though each function f_n is continuous, the pointwise limit f is not continuous at 1. ▲

So the pointwise limit of a sequence of continuous maps between metric spaces need not be continuous. However, if such a sequence of continuous maps converges uniformly, then the limit is a continuous map.

Theorem 3.5.9 Uniform limits of continuous maps. *The limit of a uniformly convergent sequence of continuous maps between metric spaces is continuous.*

Proof. Let $(f_n)_{n=0}^{\infty}$ be a sequence of continuous maps $f_n \colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y), and suppose that $(f_n)_{n=0}^{\infty}$ converges uniformly to a map $f \colon X \to Y$.

Fix a point $c \in X$, and let $\epsilon > 0$. Since $(f_n)_{n=0}^{\infty}$ converges uniformly to f, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho_Y(f_n(x), f(x)) < \frac{\epsilon}{3}$$

for all indices $n \geq n_\epsilon$ and points $x \in X$. Fix one such index $n \geq n_\epsilon$. Since f_n is continuous, there is a positive real number $\delta > 0$ so that

$$\rho_Y(f_n(x), f_n(c)) < \frac{\epsilon}{3}$$

for all points $x \in X$ so that $\rho_X(x, c) < \delta$. We observe that

$$\begin{aligned} \rho_Y(f(x), f(c)) &\leq \rho_Y(f(x), f_n(x)) + \rho_Y(f_n(c), f_n(c)) + \rho_Y(f_n(c), f(c)) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon \end{aligned}$$

for all points $x \in X$ so that $\rho_X(x, c) < \delta$, and so f is continuous at c. Since this holds for all points $c \in X$, we conclude that f is continuous. ■

So, while the pointwise limit of a sequence of continuous maps between metric spaces need not in general be continuous, any such sequence which converges uniformly must converge to a continuous limit. Uniformity, or the relative independence of the "size" of neighborhoods of points in a space from position, is a useful property which will allow us to reconcile complex limiting behavior where several limits interact. We now turn to a strengthening of the notion of continuity based on this idea of uniformity.

3.5.2 Stronger Forms of Metric Continuity

Continuity is a fundamental and useful tool for mathematical analysis of maps between metric spaces. However, there are many pathological examples of maps which are continuous and yet do not behave as well as we might like. Therefore, we now turn our attention to a stronger notion of continuity, that of uniform continuity. At the beginning of this section, we saw a definition for metric continuity which posited the existence of a positive real number δ for each point in the domain and each positive real number ϵ. Informally, a map between metric spaces is **uniformly continuous** if the δ associated to a given ϵ may be chosen independent of position in the domain.

Definition 3.5.10 Uniform continuity. A map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is called **uniformly continuous** if for all positive real numbers $\epsilon > 0$, there is a positive real number $\delta > 0$ so that

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) < \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho_X\left(x_1, x_2\right) < \delta$.

The set of uniformly continuous maps from X to Y is denoted $\mathrm{UC}\left(X, Y\right)$. In the special case $(Y, \rho_Y) = (\mathbb{R}, d)$, the set of uniformly continuous real-valued functions on X is denoted $\mathrm{UC}\left(X\right) = \mathrm{UC}\left(X, \mathbb{R}\right)$. ◆

Note that uniform continuity does not apply at a specific point, but rather over the entire domain of a map. ***It does not make sense to say that a map between metric spaces is uniformly continuous at a point.***

Example 3.5.11 Uniform continuity. The following are examples of uniformly continuous and not uniformly continuous maps between metric spaces:

- Any constant map between metric spaces is uniformly continuous. Indeed, fix metric spaces (X, ρ_X) and (Y, ρ_Y), and consider a point $y \in Y$. The constant map $f\colon X \to Y$ from X to Y defined by the formula $f\left(x\right) = y$ satisfies the condition

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) = \rho_Y\left(y, y\right) = 0$$

  for all points $x_1, x_2 \in X$. Thus for a given positive real number $\epsilon > 0$, we may choose any positive real number $\delta > 0$ and check that this choice verifies the uniform continuity condition.

- The inclusion map of a metric subspace is uniformly continuous. Indeed, fix a subset $U \subseteq X$ of a metric space (X, ρ), and note that the inclusion map $\iota\colon U \to X$ satisfies the condition

$$\rho\left(\iota\left(u_1\right), \iota\left(u_2\right)\right) = \rho\left(u_1, u_2\right)$$

  for all points $u_1, u_2 \in U$. Thus for a given positive real number $\epsilon > 0$, we may choose $\delta = \epsilon > 0$ and check that this choice verifies the uniform continuity condition.

- The real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f\left(x\right) = x^2$ is continuous but not uniformly continuous. Indeed, for any positive real number $\delta > 0$, let

$$x_1 = \frac{1}{2\delta} \qquad\qquad x_2 = \frac{1 + \delta^2}{2\delta}.$$

  We observe that

$$|x_1 - x_2| = \left|\frac{1}{2\delta} - \frac{1 + \delta^2}{2\delta}\right| = \left|\frac{\delta^2}{2\delta}\right| = \frac{\delta}{2} < \delta,$$

  but

$$|x_1 + x_2| = \left|\frac{1}{2\delta} + \frac{1 + \delta^2}{2\delta}\right| = \left|\frac{2 + \delta^2}{2\delta}\right| = \frac{2 + \delta^2}{2\delta} = \frac{1}{\delta} + \frac{\delta}{2},$$

  which implies that

$$\begin{aligned} \left|f\left(x_1\right) - f\left(x_2\right)\right| &= \left|x_1^2 - x_2^2\right| = \left|\left(x_1 - x_2\right)\left(x_1 + x_2\right)\right| = |x_1 - x_2| \cdot |x_1 + x_2| \\ &= \left(\frac{\delta}{2}\right)\left(\frac{1}{\delta} + \frac{\delta}{2}\right) = \frac{1}{2} + \frac{\delta^2}{4} > \frac{1}{2}. \end{aligned}$$

  In particular, for the positive real number $\epsilon = \frac{1}{2}$, there is no positive real number $\delta > 0$ for which the uniform continuity criterion is satisfied.

▲

The definitions of continuity and uniform continuity are very similar at first glance. In fact, despite the above example of a continuous but not uniformly continuous function, it's not difficult to see that uniform continuity implies continuity.

Proposition 3.5.12 Uniform continuity implies continuity. *Any uniformly continuous map between metric spaces is continuous.*

Proof. Let $f\colon X \to Y$ be a uniformly continuous map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). Consider a point $x \in X$, and let $\epsilon > 0$ be a positive real number. Since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) < \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho_X\left(x_1, x_2\right) < \delta$. In particular, we observe that

$$\rho_Y\left(f\left(x\right), f\left(x\right)\right) < \epsilon$$

for all points $x \in X$ so that $\rho_X\left(x, c\right) < \delta$, and so f is continuous at c. Since c was chosen to be an arbitrary point in X, f is continuous. ■

Continuous maps between metric spaces (or other more general spaces) are fundamental objects in point-set topology. This is because lots of topological properties of spaces, which we will examine in the following section and chapter, are preserved under continuous maps. However, some of the more geometric properties, which start to approach notions of size, are not preserved under all continuous maps in general, but only under uniformly continuous maps. In the previous section, we introduced a notion of boundedness for subsets. While the image of a bounded subset of a metric space under a continuous map is not necessarily bounded, the image of a totally bounded subset of a metric space under a uniformly continuous map is totally bounded.

Proposition 3.5.13 Uniform continuity and total boundedness. *The image of a totally bounded subset of a metric space under a uniformly continuous map between metric spaces is totally bounded.*

Proof. Let $f\colon X \to Y$ be a uniformly continuous map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y), and consider a totally bounded subset $U \subseteq X$. We wish to show that $f\left(U\right)$ is totally bounded in Y.

To that end, let $\epsilon > 0$ be a positive real number. Since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) < \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho_X\left(x_1, x_2\right) < \delta$. In particular, this implies that

$$f\left(\mathrm{B}_\delta\left(x\right)\right) \subseteq \mathrm{B}_\epsilon\left(f\left(x\right)\right)$$

for all points $x \in X$.

Since U is totally bounded in X, there is a finite list $x_1, \ldots, x_n \in X$ so that

$$U \subseteq \bigcup_{k=1}^{n} \mathrm{B}_\delta\left(x_k\right).$$

We observe that

$$f\left(U\right) \subseteq f\left(\bigcup_{k=1}^{n} \mathrm{B}_\delta\left(x_k\right)\right) = \bigcup_{k=1}^{n} f\left(\mathrm{B}_\delta\left(x_k\right)\right)$$

$$\subseteq \bigcup_{k=1}^{n} \mathrm{B}_\epsilon\left(f\left(x_k\right)\right),$$

and so $f(U)$ is totally bounded in Y. ■

So the uniformly continuous image of a totally bounded subset of a metric space is totally bounded. Just as in the case of plain continuity, the set of uniformly continuous maps from a metric space to a normed vector space itself forms a vector space under pointwise addition and scalar multiplication.

Proposition 3.5.14 Vector arithmetic and uniform continuity. *Let (X, ρ) be a metric space and $(V, \|\cdot\|)$ be a normed vector space.*

1. *Vector addition.*

   If two maps $f, g\colon X \to V$ from X to Y are uniformly continuous, then their pointwise sum $f + g\colon X \to V$ is also uniformly continuous.

2. *Scalar multiplication.*

   If a map $f\colon X \to V$ is uniformly continuous, then any pointwise scalar multiple αf is also uniformly continuous.

3. *Norm.*

   If a map $f\colon X \to V$ is uniformly continuous, then its pointwise norm $\|f\| = \|\cdot\| \circ f$ is also uniformly continuous.

Proof.

1. Let $\epsilon > 0$ be a positive real number. Since f and g are continuous, there are positive real numbers $\delta_f, \delta_g > 0$ so that

   $$\|f(x_1) - f(x_2)\| < \frac{\epsilon}{2}$$

   for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta_f$ and

   $$\|g(x_1) - g(x_2)\| < \frac{\epsilon}{2}$$

   for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta_g$. Let $\delta = \min(\{\delta_f, \delta_g\}) > 0$. We observe that

   $$\begin{aligned}\|(f+g)(x_1) - (f+g)(x_2)\| &= \|f(x_1) + g(x_1) - f(x_2) - g(x_2)\| \\ &\leq \|f(x_1) - f(x_2)\| + \|g(x_1) - g(x_2)\| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon\end{aligned}$$

   for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$, and so $f + g$ is uniformly continuous.

2. Let $\epsilon > 0$ be a positive real number. If $\alpha = 0$, then $\alpha f = 0f = \mathbf{0}$ is a constant map and therefore is uniformly continuous. We may therefore suppose that $\alpha \neq 0$ is non-zero.

   Let $\epsilon > 0$ be a positive real number. Since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

   $$\|f(x_1) - f(x_2)\| < \frac{\epsilon}{|\alpha|}$$

   for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$. We observe that

   $$\|(\alpha f)(x_1) - (\alpha f)(x_2)\| = \|\alpha f(x_1) - \alpha f(x_2)\| = \|\alpha(f(x_1) - f(x_2))\|$$

$$= |\alpha| \, \|f(x_1) - f(x_2)\| < |\alpha| \left(\frac{\epsilon}{|\alpha|}\right) = \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$, and so αf is uniformly continuous.

3. Let $\epsilon > 0$ be a positive real number. Since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

$$\|f(x_1) - f(x_2)\| < \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$. We observe that Proposition 3.1.5 Reverse triangle inequality implies that

$$\begin{aligned} |\|f\|(x_1) - \|f\|(x_2)| &= |\|f(x_1)\| - \|f(x_2)\|| = |\|f(x_1)\| - \|f(x_2)\|| \\ &= |\|f(x_1) - \mathbf{0}\| - \|f(x_2) - \mathbf{0}\|| \\ &\leq \|f(x_1) - f(x_2)\| < \epsilon \end{aligned}$$

for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$, so that $\|f\|$ is uniformly continuous.

■

A natural question to ask is whether the same holds for the pointwise product of uniformly continuous real-valued functions on a metric space. In fact, we have seen that this is not necessarily the case: the real-valued function $f: \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x$ is uniformly continuous, but the pointwise product $(f \cdot f)(x) = f(x) \cdot f(x) = x^2$ is not uniformly continuous. However, since uniformly continuous maps are continuous and because the product of two continuous, real-valued functions is continuous, the pointwise product of two uniformly continuous, real-valued functions on a metric space is continuous.

Just as in the case of continuous maps, the composition of uniformly continuous maps is uniformly continuous. This allows us to simplify the proofs of uniform continuity for a vast class of maps.

Proposition 3.5.15 Composition of uniformly continuous maps. *The composition of uniformly continuous maps between metric spaces is uniformly continuous.*

Proof. Let $f: X \to Y$ and $g: Y \to Z$ be uniformly continuous maps between metric spaces (X, ρ_X), (Y, ρ_Y), and (Z, ρ_Z). Let $\epsilon > 0$ be a positive real number. Since g is uniformly continuous, there is a positive real number $\eta > 0$ so that

$$\rho_Z(g(y_1), g(y_2)) < \epsilon$$

for all points $y_1, y_2 \in Y$ so that $\rho_Y(y_1, y_2) < \eta$. Moreover, since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

$$\rho_Y(f(x_1), f(x_2)) < \eta$$

for all points $x_1, x_2 \in X$ so that $\rho_X(x_1, x_2) < \delta$.

We observe that for all points $x_1, x_2 \in X$ if $\rho_X(x_1, x_2) < \delta$, then $\rho_Y(f(x_1), f(x_2)) < \eta$, and so

$$\rho_Z((g \circ f)(x_1), (g \circ f)(x_2)) = \rho_Z(g(f(x_1)), g(f(x_2))) < \epsilon.$$

We conclude that $g \circ f$ is uniformly continuous. ■

Uniform continuity of a map has strong implications for its properties. However, there are even stronger notions of continuity.

3.5.3 Even Stronger Forms of Metric Continuity

We conclude this section with a look at a stronger form of continuity than uniform continuity. A map between metric spaces is called **Lipschitz continuous** if it changes the distance between points by at most some positive scalar multiple.

Definition 3.5.16 Lipschitz continuity. A map f from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is called **Lipschitz continuous** if there is a non-negative real number $L \geq 0$ so that

$$d_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L d_X\left(x_1, x_2\right)$$

for all points $x_1, x_2 \in X$.

The set of Lipschitz continuous maps from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is denoted $\mathrm{LC}\left(X, Y\right)$, and the set of Lipschitz continuous real-valued functions on (X, ρ_X) is denoted $\mathrm{LC}\left(X\right) = \mathrm{LC}\left(X, \mathbb{R}\right)$. ◆

As with uniform continuity, Lipschitz continuity does not apply at a specific point, but rather over the entire domain of a map. ***It does not make sense to say that a map between metric spaces is Lipschitz continuous at a point.***

Example 3.5.17 Norms are Lipschitz continuous. Let $(V, \|\cdot\|)$ be a normed vector space. The Reverse triangle inequality implies that

$$d\left(\left\|\mathbf{v}_1\right\|, \left\|\mathbf{v}_2\right\|\right) = \left|\left\|\mathbf{v}_1\right\| - \left\|\mathbf{v}_2\right\|\right| \leq \left\|\mathbf{v}_1 - \mathbf{v}_2\right\| = 1\left\|\mathbf{v}_1 - \mathbf{v}_2\right\|$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$, and so the norm $\|\cdot\| : V \to \mathbb{R}$ is Lipschitz continuous. ▲

For any Lipschitz continuous map between metric spaces, there is a unique smallest non-negative real number such that the above condition is satisfied. This unique smallest constant is called the **Lipschitz constant** for the map.

Definition 3.5.18 Lipschitz constant. The **Lipschitz constant** L_f of a map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is the infimum

$$L_f = \inf\left(\left\{L \in [0, \infty) \mid \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L\rho_X\left(x_1, x_2\right) \text{ for all } x_1, x_2 \in X\right\}\right).$$

◆

Proposition 3.5.19 Lipschitz constants and Lipschitz continuity. *A map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is Lipschitz continuous if and only if $L_f < \infty$, in which case*

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L_f \rho_X\left(x_1, x_2\right)$$

for all points $x_1, x_2 \in X$.

Proof. We observe that

$$C = \left\{L \in [0, \infty) \mid \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L\rho_X\left(x_1, x_2\right) \text{ for all } x_1, x_2 \in X\right\}$$

is a subset of the real numbers, and so $L_f = \inf\left(C\right) < \infty$ if and only if $C \neq \emptyset$ is non-empty. That $C \neq \emptyset$ is non-empty is precisely the Lipschitz continuity condition for f, and so f is Lipschitz continuous if and only if $L_f < \infty$.

If $C \neq \emptyset$ is non-empty, let $(l_n)_{n=0}^{\infty}$ be a sequence of elements $l_n \in L$ which converge to $L_f = \inf(C)$. We observe that

$$\begin{aligned}\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) &= \lim_{n\to\infty} \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right)\\ &\leq \lim_{n\to\infty} l_n \rho_X\left(x_1, x_2\right) = L_f \rho\left(x_1, x_2\right)\end{aligned}$$

for all points $x_1, x_2 \in X$. ∎

A map is Lipschitz continuous if it increases the distance between points by at most some nonnegative constant multiplier. This implies that Lipschitz continuity is a stronger condition than uniform continuity, and therefore is also a stronger condition than continuity.

Proposition 3.5.20 Lipschitz continuity implies uniform continuity. *If a map between metric spaces is Lipschitz continuous, then it is uniformly continuous and continuous.*

Proof. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y), and consider a positive real number $\epsilon > 0$. There are two cases; either $L_f = 0$, or $0 < L_f < \infty$.

If $L_f = 0$, then Proposition 3.5.19 Lipschitz constants and Lipschitz continuity implies that

$$0 \leq \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L_f \rho_X\left(x_1, x_2\right) = 0\rho_X\left(x_1, x_2\right) = 0 < \epsilon$$

for all points $x_1, x_2 \in X$, and so $f(x_1) = f(x_2)$ for all points $x_1, x_2 \in X$. In particular, this implies that f is constant and therefore uniformly continuous.

If on the other hand $0 < L_f < \infty$, let $\delta = \frac{\epsilon}{L_f}$, and note that Proposition 3.5.19 Lipschitz constants and Lipschitz continuity implies that

$$0 \leq \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L_f \rho_X\left(x_1, x_2\right) < L_f\left(\frac{\epsilon}{L_f}\right) = \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho_X(x_1, x_2) < \delta$. Thus f is uniformly continuous. We conclude that f is continuous by Proposition 3.5.12 Uniform continuity implies continuity. ∎

Example 3.5.21 Uniform continuity does not imply Lipschitz continuity. Consider the real-valued function $f\colon [0,\infty) \to \mathbb{R}$ defined by the formula $f(x) = \sqrt{x}$. For each positive real number $\epsilon > 0$, let $\delta = \epsilon^2 > 0$. We observe that

$$|\sqrt{x_1} + \sqrt{x_2}| \leq |\sqrt{x_1}| + |\sqrt{x_2}| = \sqrt{x_1} + \sqrt{x_2} = |\sqrt{x_1} + \sqrt{x_2}|$$

for all points $x_1, x_2 \in [0, \infty)$, and so

$$\begin{aligned}d\left(f\left(x_1\right), f\left(x_2\right)\right)^2 &= |\sqrt{x_1} - \sqrt{x_2}|^2 \leq |\sqrt{x_1} + \sqrt{x_2}| \cdot |\sqrt{x_1} - \sqrt{x_2}| = |x_1 - x_2|\\ &= d\left(x_1, x_2\right) < \delta = \epsilon^2\end{aligned}$$

for all points $x_1, x_2 \in [0, \infty)$ so that $d(x_1, x_2) < \delta$. In particular, $d(f(x_1), f(x_2)) < \epsilon$ for all points $x_1, x_2 \in [0, \infty)$ so that $d(x_1, x_2) < \delta$, and so f is uniformly continuous.

On the other hand, we observe that

$$\frac{d\left(f\left(\frac{1}{(L+1)^2}\right), f(0)\right)}{d\left(\frac{1}{(L+1)^2}, 0\right)} = \frac{\left|\sqrt{\frac{1}{(L+1)^2}} - \sqrt{0}\right|}{\left|\frac{1}{(L+1)^2} - 0\right|} = \left|\frac{\left(\frac{1}{L+1}\right)}{\left(\frac{1}{(L+1)^2}\right)}\right|$$

$$= \left| \frac{(L+1)^2}{L+1} \right| = |L+1| = L+1 > L$$

for any non-negative real number $0 \leq L < \infty$, so that

$$d\left(f\left(\frac{1}{(L+1)^2}\right), f(0)\right) > Ld\left(\frac{1}{(L+1)^2}, 0\right).$$

Thus $L_f > L$. Since this holds for each non-negative real number $0 \leq L < \infty$, $L_f = \infty$ and so f is not Lipschitz continuous by Proposition 3.5.19 Lipschitz constants and Lipschitz continuity. ▲

Just as in the case of continuity and uniform continuity, Lipschitz continuity is preserved by composition, although possibly at the cost of increasing the Lipschitz constant.

Lemma 3.5.22 Lipschitz continuity and composition. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between metric spaces (X, ρ_X), (Y, ρ_Y), and (Z, ρ_Z). If f and g are Lipschitz continuous, then so is their composition $g \circ f\colon X \to Z$, and $L_{g\circ f} \leq L_g L_f$.*

Proof. We observe that

$$\begin{aligned}\rho_Z\left((g\circ f)(x_1), (g\circ f)(x_2)\right) &= \rho_Z\left(g\left(f\left(x_1\right)\right), g\left(f\left(x_2\right)\right)\right) \\ &\leq L_g \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \\ &\leq L_g L_f \rho_X\left(x_1, x_2\right)\end{aligned}$$

for all points $x_1, x_2 \in X$. We conclude that the composition $g \circ f$ is Lipschitz continuous and $L_{g\circ f} \leq L_g L_f$. ■

Just as in the case of continuity and uniform continuity, Lipschitz continuous maps from a metric space to a normed vector space form a vector space under the pointwise operations of addition and scalar multiplication, although possibly at the cost of increasing the Lipschitz constant.

Proposition 3.5.23 Vector arithmetic and Lipschitz continuity. *Let (X, ρ) be a metric space and $(V, \|\cdot\|)$ be a normed vector space.*

1. *Vector addition.*

   If two maps $f, g\colon X \to V$ from X to Y are Lipschitz continuous, then their pointwise sum $f + g\colon X \to V$ is also Lipschitz continuous, and $L_{f+g} \leq L_f + L_g$.

2. *Scalar multiplication.*

   If a map $f\colon X \to V$ is Lipschitz continuous, then any pointwise scalar multiple αf is also Lipschitz continuous, and $L_{\alpha f} \leq |\alpha| L_f$.

3. *Norm.*

   If a map $f\colon X \to V$ is uniformly continuous, then its pointwise norm $\|f\| = \|\cdot\| \circ f$ is also uniformly continuous, and $L_{\|f\|} \leq L_f$.

Proof.

1. We observe that

$$\begin{aligned}\left\|(f+g)(x_1) - (f+g)(x_2)\right\| &= \left\|f(x_1) + g(x_1) - f(x_2) - g(x_2)\right\| \\ &\leq \left\|f(x_1) - f(x_2)\right\| + \left\|g(x_1) - g(x_2)\right\|\end{aligned}$$

$$\begin{aligned} &\leq L_f \rho(x_1, x_2) + L_g \rho(x_1, x_2) \\ &= (L_f + L_g)\, \rho(x_1, x_2) \end{aligned}$$

for all points $x_1, x_2 \in X$. We conclude that the composition $g \circ f$ is Lipschitz continuous and $L_{f+g} \leq L_f + L_g$.

2. We observe that

$$\begin{aligned} \|(\alpha f)(x_1) - (\alpha f)(x_2)\| = \|\alpha f(x_1) - \alpha f(x_2)\| &= \|\alpha (f(x_1) - f(x_2))\| \\ &\leq |\alpha|\, \|f(x_1) - f(x_2)\| \leq |\alpha|\, L_f \rho(x_1, x_2) \end{aligned}$$

for all points $x_1, x_2 \in X$. We conclude that the pointwise scalar multiple αf is Lipschitz continuous and $L_{\alpha f} \leq |\alpha|\, L_f$.

3. Recall from Example 3.5.17 Norms are Lipschitz continuous that the norm $\|\cdot\| : V \to \mathbb{R}$ is Lipschitz continuous with Lipschitz constant $L_{\|\,\cdot\,\|} \leq 1$. Thus Lemma 3.5.22 Lipschitz continuity and composition implies that the composition $\|f\| = \|\cdot\| \circ f$ is Lipschitz continuous, with Lipschitz constant

$$L_{\|f\|} = L_{\|\,\cdot\,\| \circ f} \leq L_{\|\,\cdot\,\|} L_f \leq 1 L_f = L_f.$$

■

You will have several opportunities to explore applications of Lipschitz continuity in the problems for both this section and Section 3.6 Completeness and Compactness.

In this section, we have examined several nonequivalent notions of continuity for maps between metric spaces. In the next and final section of this chapter, we will introduce the notions of **completeness** and **compactness** for metric spaces, which both have to do with the convergence of sequences which morally should have a limit, but might not.

3.5.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Continuity at a point. Determine whether or not the given map between metric spaces is continuous at the given point.

1. Let $f : \mathbb{R} \to \mathbb{R}$ be the linear function defined by the formula $f(x) = 3x - 1$. Determine whether or not f is continuous at 2.

2. Let $g : \mathbb{R} \to \mathbb{R}$ be the linear function defined by the formula $f(y) = 2y^2 - 1$. Determine whether or not g is continuous at 4.

3. Let $h : [0, \infty) \to \mathbb{R}$ be the linear function defined by the formula $h(z) = \sqrt{z}$. Determine whether or not h is continuous at 1.

4. **Uniform and Lipschitz continuity.** Let $-\infty < a < b < \infty$ be real numbers, and consider the real-valued function $f : [a, b] \to \mathbb{R}$ defined by the formula

$$f(x) = x^3.$$

(a) Verify that f is not uniformly continuous.

(b) Without appealing to Proposition 3.5.20 Lipschitz continuity implies uniform continuity, verify that f is not Lipschitz continuous.

(c) Verify that the restriction of f to a closed interval $[a,b] \subsetneq \mathbb{R}$ is uniformly continuous.

(d) Verify that the restriction of f to a closed interval $[a,b] \subsetneq \mathbb{R}$ is Lipschitz continuous.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.5.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Continuity of rational functions.

(a) Prove using one or more characterizations of continuity that the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x$ is continuous.

(b) Argue that any polynomial function is continuous.

(c) Argue that any rational function is continuous on its implicit domain.

2. Uniform limits of uniformly continuous maps. Let $(f_n)_{n=0}^{\infty}$ be a sequence of uniformly continuous maps $f_n\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). Prove that if $(f_n)_{n=0}^{\infty}$ converges uniformly, the uniform limit is also uniformly continuous.

3. Continuous pointwise limits of discontinuous maps. We have seen in Theorem 3.5.9 Uniform limits of continuous maps and Problem 3.5.5.2 Uniform limits of uniformly continuous maps that uniform limits preserve continuity and uniform continuity. In this problem, we will construct counterexamples to the converses of these results.

(a) Give an example of a sequence $(f_n)_{n=0}^{\infty}$ of continuous maps $f_n\colon X \to Y$ from a metric space X to a metric space Y which converges pointwise but not uniformly to a continuous map $f\colon X \to Y$ from X to Y.

(b) Give an example of a sequence $(g_n)_{n=0}^{\infty}$ of discontinuous maps $g_n\colon X \to Y$ from a metric space X to a metric space Y which converges uniformly to a continuous map $g\colon X \to Y$ from X to Y.

4. **Uniform limits of Lipschitz continuous maps.** Consider the sequence $(f_n)_{n=1}^{\infty}$ of real-valued functions $f_n\colon [0,\infty) \to \mathbb{R}$ defined by the formula

$$f_n(x) = \sqrt{x + \frac{1}{n}}.$$

   (a) Show that f_n is Lipschitz continuous for each index $n \geq 1$.

   (b) Show that the sequence $(f_n)_{n=0}^{\infty}$ converges uniformly, but the uniform limit is not Lipschitz continuous.

   (c) Prove that the limit of a uniformly convergent sequence $(g_n)_{n=0}^{\infty}$ of Lipschitz continuous maps $g_n\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is Lipschitz continuous if the sequence $(L_{g_n})_{n=0}^{\infty}$ of Lipschitz constants L_{g_n} is bounded from above.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.6 Completeness and Compactness

In this section, we will investigate two related notions which, like much of the material of this chapter, have to do with the convergence of sequences in metric spaces. First, we will reexamine the limiting behavior of sequences from a more intrinsic perspective, drawing a distinction between convergent sequences and **Cauchy sequences**. We will then examine the properties of **complete metric spaces**, in which this distinction does not exist. Next, we will discuss **compact metric spaces**, which are those in which, even if a sequence diverges, it has a convergent **subsequence**. Finally, to conclude this chapter, we will use both of these notions to prove several powerful results about metric spaces.

3.6.1 Cauchy Sequences and Metric Completeness

Thus far, we have examined the limiting behavior of sequences through an extrinsic perspective, focusing on the distance between the values of a given sequence and the points of the metric space. We now turn our attention to a more broad class of sequences, which are called **Cauchy sequences**, and which are viewed in a more intrinsic way. While a sequence in a metric space is convergent if its values eventually grow close to a specific point, it is Cauchy if its values eventually grow close to each other.

Definition 3.6.1 Cauchy sequence. A sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) is called **Cauchy** if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that $\rho(x_m, x_n) < \epsilon$ for all indices $m, n \geq n_\epsilon$. ◆

The above requirement is sometimes called the **Cauchy criterion**, so a sequence in a metric space satisfies the Cauchy criterion if the distances between its values eventually becomes negligible. Note the distinction between this notion and that of convergence: The entries of a convergent sequence must approach a specific point but the entries of a Cauchy sequence must approach each other (while not necessarily approaching a point of the metric space). At first glance, these notions might seem to be equivalent characterizations of sequences. However, we will see that in some metric spaces, this is not the case.

Example 3.6.2 Cauchy sequence. For each natural number $k \in \mathbb{N}$, let $d_k \in \{0, 1, 2, \ldots, 9\}$ denote the kth digit in the decimal expansion of $\sqrt{2} \approx 1.414213562$, so that

$$d_0 = 1 \qquad d_1 = 4 \qquad d_2 = 1 \qquad d_3 = 4 \qquad d_4 = 2 \qquad \ldots.$$

Consider the sequence $(x_n)_{n=0}^{\infty}$ of rational numbers $x_n \in \mathbb{Q}$ defined by the formula

$$x_n = \sum_{k=0}^{n} \frac{d_k}{10^k} = d_0 + \frac{d_1}{10} + \frac{d_2}{100} + \cdots + \frac{d_n}{10^n}.$$

Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \log\left(\frac{1}{\epsilon}\right)$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d(x_m, x_n) = |x_m - x_n| &= x_{\max(\{m,n\})} - x_{\min(\{m,n\})} \\ &= \sum_{k=0}^{\max(\{m,n\})} \frac{d_k}{10^k} - \sum_{k=0}^{\min(\{m,n\})} \frac{d_k}{10^k} = \sum_{k=\min(\{m,n\})+1}^{\max(\{m,n\})} \frac{d_k}{10^k} \\ &\leq \frac{1}{10^{\min(\{m,n\})}} \leq \frac{1}{10^{n_\epsilon}} < \epsilon \end{aligned}$$

for all indices $m, n \geq n_\epsilon$. Thus $(x_n)_{n=0}^{\infty}$ is Cauchy. ▲

The above example of Cauchy sequences shows that Cauchy sequences in a metric space need not converge; the given Cauchy sequence of rational numbers converges to $\sqrt{2}$ and thus diverges in the metric space $(\mathbb{Q}, d)$. However, any convergent sequence is Cauchy.

Proposition 3.6.3 Convergent sequences are Cauchy. *Any convergent sequence of points in a metric space is Cauchy.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a metric space (X, ρ), and suppose that $(x_n)_{n=0}^{\infty}$ converges to a limit $x \in X$.

Let $\epsilon > 0$ be a positive real number. Since $(x_n)_{n=0}^{\infty}$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(x_n, x) < \frac{\epsilon}{2}$$

for all indices $n \geq n_\epsilon$. In particular, we observe that

$$\rho(x_m, x_n) \leq \rho(x_m, x) + \rho(x, x_n) = \rho(x_m, x) + \rho(x_n, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all indices $m, n \geq n_\epsilon$. We conclude that the sequence $(x_n)_{n=0}^{\infty}$ is Cauchy. ■

So the convergence of a sequence implies that it satisfies the Cauchy criterion. A natural follow-up question to the above result is whether or not it is possible for the reverse implication holds. The answer to this question actually depends on the particulars of the metric space. Such a metric space in which Cauchy sequences converge is called **complete**.

Definition 3.6.4 Metric completeness. Let $U \subseteq X$ be a subset of a metric space (X, ρ).

- *Completeness.*

  The subset U is called **complete** in X if every Cauchy sequence of points in U converges in X to a limit in U.

- *Incompleteness.*

  The subset U is called **incomplete** in X if some Cauchy sequence of points in U does not converge in X to a limit in U.

A metric space (X, ρ) is called **complete** (resp. **incomplete**) if X is complete (resp. incomplete) in X. A complete normed vector space is called a **Banach space**. ◆

Complete metric spaces are ideal for performing mathematical analysis, as it is often much easier to verify that a given sequence is Cauchy (and therefore has a limit) than to actually compute its limiting value.

Example 3.6.5 Metric completeness. The following are examples of complete and incomplete metric spaces.

- Any discrete metric space or finite metric space is complete, since any Cauchy sequence in a discrete metric space is eventually constant.
- The rational numbers $\mathbb{Q}$ are not complete in the real line $(\mathbb{R}, d)$. Indeed, we have already seen an example of a Cauchy sequence of rational numbers which converges in $\mathbb{R}$ to an irrational number $\sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$.

▲

The above fact about the rational number system is one of the reasons why it doesn't serve as a good context in which to perform mathematical analysis. However, as we will shortly see, the real number system (and, more generally, Euclidean space of any dimension) is a complete metric space, and so there is a rich theory of real analysis. These facts are themselves intimately related to the fact that the rational number system is not Dedekind complete, while the real number system is. This relationship between Dedekind completeness and completeness as a metric space is true for subspaces of the real numbers because of the relationship between the standard ordering of the real numbers and the point-set topology of the Euclidean metric, which we will examine in more detail in Chapter 4 Topological Spaces. We will prove that the real numbers are complete in the Euclidean metric shortly.

First, however, let us examine some of the properties of complete subsets of metric spaces in general. The following two results will be useful in determining whether or not a subset of a given metric space is complete. First, notice that any complete subset of a metric space needs to contain all of its limit points, and so must be closed.

Proposition 3.6.6 Complete subsets are closed. *A complete subset of a metric space is closed.*

Proof. By contraposition. Let $U \subseteq X$ be a subset of a metric space (X, ρ), and suppose that U is not closed in X. Then $U \subsetneq \mathrm{Cl}(U) = U \cup U'$ by (2) of Proposition 3.4.9 Interior/closure and Problem 3.4.5.3 Closure and limit points, and so $U' \nsubseteq U$; that is, there is some limit point $x \in U'$ which is not an element of U.

Since $x \in U'$ is a limit point of U in X,

$$x = \lim_{n\to\infty} u_n$$

for some sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U$. Since $(u_n)_{n=0}^{\infty}$ converges in X, it is Cauchy. However, the limit of this Cauchy sequence of points in U does not lie in U. Thus U is not complete in X. ■

So any complete subset of a metric space is closed. The converse to this statement is not, in general, true, since any metric space is closed in the metric, but not every metric space is complete. However, there is a partial converse to the above result; any closed subset of a complete metric space is also complete.

Proposition 3.6.7 Closed subsets of complete metric spaces. *A subset of a complete metric space is complete if and only if it is closed.*

Proof. Let $U \subseteq X$ be a subset of a complete metric space (X, ρ). We have already seen in Proposition 3.6.6 Complete subsets are closed that if U is complete in X, then it is closed in X. It remains to prove the converse implication. To that end, suppose that U is not complete in X, so that some Cauchy sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U$ does not converge to a limit in U.

However, since $(u_n)_{n=0}^{\infty}$ is a Cauchy sequence in a complete metric space (X, ρ), it converges to a limit

$$x = \lim_{n \to \infty} u_n \in X.$$

Since $u_n \neq x$ for all indices $n \in \mathbb{N}$, we conclude that $x \in U'$ is a limit point of U in X which is not contained in U. Problem 3.4.5.3 Closure and limit points implies that $x \in \mathrm{Cl}\,(U)$ is a point of closure of U in X, and so U is a proper subset of its closure $\mathrm{Cl}\,(U)$. (2) of Proposition 3.4.9 Interior/closure now implies that U is not closed in X. ■

We now would like to prove the completeness of various metric spaces associated with the real number system $\mathbb{R}$. First, however, let us consider the notion of a **subsequence**. Informally, a subsequence of a sequence is a sequence which takes some, but not necessarily all, of the same entries, and these entries appear in the same relative order. We can form a subsequence of a sequence by selecting some infinite subset of indices to include in the subsequence. In fact, any subsequence can be formed in this manner. In order to formalize this notion, recall that a sequence is formally a map whose domain is the natural numbers or the positive integers, and that an increasing function is an injective and non-decreasing function.

Definition 3.6.8 Subsequence. A sequence $y \colon \mathbb{N} \to X$ of points in a metric space (X, ρ) is a **subsequence** of a sequence $x \colon \mathbb{N} \to X$ of points in X if $y = x \circ f$ for some increasing function $f \colon \mathbb{N} \to \mathbb{N}$. In this case, if $x = (x_n)_{n=0}^{\infty}$ and $f = (n_k)_{k=0}^{\infty}$, then we write $y = (x_{n_k})_{k=0}^{\infty}$. ◆

Example 3.6.9 Subsequences. The following are examples of subsequences:

- The sequences

  $$(n)_{n=3}^{\infty} = (n+3)_{n=0}^{\infty} = (3, 4, 5, 6, \ldots)$$

  and

  $$(2n+1)_{n=0}^{\infty} = (1, 3, 5, 7, \ldots)$$

  are subsequences of the sequence $(n)_{n=0}^{\infty} = (0, 1, 2, 3, 4, \ldots)$.

- The sequence

  $$\left(\frac{1}{n^2}\right)_{n=1}^{\infty} = \left(1, \frac{1}{4}, \frac{1}{9}, \ldots\right)$$

  is a subsequence of the sequence

  $$\left(\frac{1}{n}\right)_{n=1}^{\infty} = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right).$$

- The convergent sequence

$$(1)_{n=0}^{\infty} = \left((-1)^{2n}\right)_{n=0}^{\infty} = (1, 1, 1, \ldots)$$

is a subsequence of the divergent sequence $((-1)^n)_{n=0}^{\infty} = (1, -1, 1, -1, 1, \ldots)$.

▲

The above example shows that divergent sequences may have convergent subsequences. In fact, much of this section will be devoted to examining the properties of metric spaces in which this is the only possible situation. First, however, note that any subsequence of a convergent sequence in a metric space converges to the same limit. In order to show this, it helps to see the following fact about strictly increasing functions on the natural numbers.

Lemma 3.6.10 Increasing functions. *$f(n) \geq n$ for any increasing function $f \colon \mathbb{N} \to \mathbb{N}$.*

Proof. By induction on $n \in \mathbb{N}$. For the base case $n = 0$, we observe that $f(0) \in \mathbb{N}$ is a natural number, and so

$$f(n) = f(0) \geq 0 = n.$$

For the inductive step, suppose for the inductive hypothesis that $f(n) \geq n$ for some natural number $n \in \mathbb{N}$. We observe that $n \leq n + 1$, and so $f(n) \leq f(n+1)$, since f is non-decreasing. Moreover, $n \neq n + 1$, and so $f(n) \neq f(n+1)$, since f is injective. In particular,

$$f(n+1) > f(n) \geq n$$

by the inductive hypothesis, and so $f(n+1) \geq n + 1$. ■

We are now ready to show that a subsequence of a convergent sequence in a metric space is itself convergent, and converges to the same limit as the larger sequence. So while a subsequence of a divergent sequence might converge, the reverse situation cannot occur.

Proposition 3.6.11 Subsequences of convergent sequences. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a metric space (X, ρ). If $(x_n)_{n=0}^{\infty}$ converges in X, then any subsequence $(x_{n_k})_{k=0}^{\infty}$ also converges in X, and*

$$\lim_{k\to\infty} x_{n_k} = \lim_{n\to\infty} x_n.$$

Proof. Let

$$x = \lim_{n\to\infty} x_n,$$

and consider a positive real number $\epsilon > 0$. There is some natural number $n_\epsilon \in \mathbb{N}$ so that $\rho(x_n, x) < \epsilon$ for all indices $n \geq n_\epsilon$.

We observe that Lemma 3.6.10 Increasing functions implies that $n_k \geq n_\epsilon$ for all indices $k \geq n_\epsilon$, and so

$$\rho(x_{n_k}, x) < \epsilon$$

for all such indices. Thus

$$\lim_{k\to\infty} x_{n_k} = x = \lim_{n\to\infty} x_n.$$

■

An analogous argument can be used to show that any subsequence of a sequence of real numbers which diverges to infinity must itself diverge to infinity. Taken together with Proposition 3.6.11 Subsequences of convergent sequences, these results state that a subsequence of a sequence in a metric space with a limit shares that limit. We have seen that the opposite is not in general true; divergent sequences which do not approach a limit may have subsequences which do. However, this cannot happen if the original sequence was Cauchy; a Cauchy sequence in a metric space with a convergent subsequence converges to the limit of this subsequence.

Proposition 3.6.12 Cauchy sequences with convergent subsequences. *If a Cauchy sequence of points in a metric space has a convergent subsequence, then it converges.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a Cauchy sequence of points $x_n \in X$ in a metric space X, and suppose that some subsequence $(x_{n_k})_{k=0}^{\infty}$ converges to a point $x \in X$. We will show that the entire sequence $(x_n)_{n=0}^{\infty}$ also converges to x.

To that end, let $\epsilon > 0$ be a positive real number. Since $(x_n)_{n=0}^{\infty}$ is Cauchy, there is a natural number $m_\epsilon \in \mathbb{N}$ so that

$$\rho(x_m, x_n) < \frac{\epsilon}{2}$$

for all indices $m, n \geq m_\epsilon$. Since the subsequence $(x_{n_k})_{k=0}^{\infty}$ converges to x, there is a natural number $k_\epsilon \in \mathbb{N}$ so that

$$\rho(x_{n_k}, x) < \frac{\epsilon}{2}$$

for all indices $k \geq k_\epsilon$. Let $n_\epsilon = \max(\{m_\epsilon, k_\epsilon\})$. We observe that Lemma 3.6.10 Increasing functions implies that $n_k \geq n_\epsilon$ for all indices $k \geq n_\epsilon$. In particular,

$$\rho(x_n, x) \leq \rho(x_n, x_{n_n}) + \rho(x_{n_n}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(x_n)_{n=0}^{\infty}$ converges to x. ■

The above result will come in handy in proving the completeness of the real numbers in the Euclidean metric. So will the following foundational theorem, the content of which is that any bounded sequence of real numbers has a subsequence which is convergent in the Euclidean metric.

Theorem 3.6.13 Bolzano-Weierstrass theorem. (Bernard Bolzano; Karl Weierstrass) *Any bounded sequence of real numbers has a convergent subsequence.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$ which is bounded in the Euclidean metric on $\mathbb{R}$. For each natural number $m \in \mathbb{N}$, consider the set

$$T_m = \{x_n \in \mathbb{R} \mid n \geq m\}.$$

There are two cases: either T_m has a maximum for every index $m \in \mathbb{N}$ or T_m does not have a maximum for some index $m \in \mathbb{N}$. We will show that in either case, $(x_n)_{n=0}^{\infty}$ has a monotone subsequence.

First suppose that T_m has a maximum for each index $m \in \mathbb{N}$. Consider the sequence $(n_k)_{k=0}^{\infty}$ of natural numbers $n_k \in \mathbb{N}$ defined recursively by

$$\begin{aligned} n_0 &= \min(\{n \in \mathbb{N} \mid x_n = \max(T_0)\}) \\ n_{k+1} &= \min\left(\left\{n \in \mathbb{N} \,\middle|\, n \geq n_k + 1 \text{ and } x_n = \max\left(T_{n_{k+1}}\right)\right\}\right). \end{aligned}$$

Since $n_{m+1} \geq n_m + 1 > n_m$, the sequence $(n_k)_{n=0}^{\infty}$ is increasing, and so $(x_{n_k})_{k=0}^{\infty}$ is a subsequence of $(x_n)_{n=0}^{\infty}$. Moreover, we observe that

$$x_{n_{k+1}} = \max\left(T_{n_{k+1}}\right) \leq \min\left(T_{n_k}\right) = x_{n_k}$$

for all indices $k \in \mathbb{N}$, and so $(x_{n_k})_{k=0}^{\infty}$ is non-increasing.

On the other hand, now suppose that T_m has no maximum for some index $m \in \mathbb{N}$. Consider the sequence $(n_k)_{k=0}^{\infty}$ of natural numbers $n_k \in \mathbb{N}$ defined recursively by

$$\begin{aligned} n_0 &= m \\ n_{k+1} &= \min\left(\{n \in \mathbb{N} \mid n \geq n_k + 1 \text{ and } x_n > x_{n_k}\}\right). \end{aligned}$$

Since $n_{m+1} \geq n_m + 1 > n_m$, the sequence $(n_k)_{n=0}^{\infty}$ is increasing, and so $(x_{n_k})_{k=0}^{\infty}$ is a subsequence of $(x_n)_{n=0}^{\infty}$. Moreover, we observe that

$$x_{n_{k+1}} > x_{n_k}$$

for all indices $k \in \mathbb{N}$, and so $(x_{n_k})_{k=0}^{\infty}$ is increasing.

In either case, $(x_n)_{n=0}^{\infty}$ has a monotone subsequence $(x_{n_k})_{k=0}^{\infty}$. Since $(x_n)_{n=0}^{\infty}$ is bounded, so too is $(x_{n_k})_{k=0}^{\infty}$, and so Theorem 3.3.13 Monotone convergence theorem implies that $(x_{n_k})_{k=0}^{\infty}$ converges. ■

So any bounded sequence of real numbers has a subsequence which converges in the Euclidean metric. We will see that this implies that the real line is complete, and so is a Banach space under the Euclidean norm.

Lemma 3.6.14 Cauchy sequences are bounded. *Any Cauchy sequence of points in a metric space is bounded.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a Cauchy sequence of points in a metric space (X, ρ). Then there is a natural number $n_1 \in \mathbb{N}$ so that $\rho(x_m, x_n) < 1$ for all indices $n \geq n_1$. Let

$$r = \max\left(\{\rho(x_1, x_n), \ldots, \rho(x_{n_1-1}, x_0), \rho(x_{n_1}, x_0) + 1\}\right) > 0.$$

For all indices $n \in \mathbb{N}$, if $n < n_1$, then $\rho(x_n, x_0) \leq r$. On the other hand, if $n \geq n_1$, then

$$\rho(x_n, x_0) \leq \rho(x_n, x_{n_1}) + \rho(x_{n_1}, x_0) \leq \rho(x_n, x_{n_1}) + 1 \leq r.$$

Thus $\{x_n\}_{n=0}^{\infty} \subseteq \mathrm{D}_r(x_0)$, and so the sequence $(x_n)_{n=0}^{\infty}$ is bounded. ■

Corollary 3.6.15 Completeness of the real line. *The real line* $(\mathbb{R}, d)$ *is complete, and so* $(\mathbb{R}, \|\cdot\|)$ *is a Banach space.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a Cauchy sequence of real numbers $x_n \in \mathbb{R}$. Then $(x_n)_{n=0}^{\infty}$ is bounded by Lemma 3.6.14 Cauchy sequences are bounded, and so it has a convergent subsequence by the Bolzano-Weierstrass theorem. Since $(x_n)_{n=0}^{\infty}$ is a Cauchy sequence with a convergent subsequence, it is itself convergent by Proposition 3.6.12 Cauchy sequences with convergent subsequences. Since every Cauchy sequence in $(\mathbb{R}, d)$ converges, this metric space is complete. ■

So the real line is a complete metric space. As the Euclidean metric on $\mathbb{R}$ is induced by the Euclidean norm, this implies that $\mathbb{R}$ is a Banach space. That this completeness result is dependent on the Dedekind completeness of the real number system is not immediately obvious. However, in our proof of the Bolzano-Weierstrass theorem, we used the fact that a bounded monotone sequence of real numbers converges to a real number, which requires the existence of upper and lower limits and is not true in ordered number systems such at the rational numbers $\mathbb{Q}$ in which these upper and lower limits do not necessarily exist.

In fact, the completeness of the real line implies that the normed vector spaces associated to the p-norms are all also Banach spaces.

Corollary 3.6.16 Completeness of the p-norm. $\left(\mathbb{R}^n, \|\cdot\|_p\right)$ *is a Banach space for all natural numbers $n \in \mathbb{N}$ and extended real numbers $1 \leq p \leq \infty$.*

Proof. Let $(\mathbf{x}_m)_{m=0}^{\infty}$ be a Cauchy sequence of points $\mathbf{x}_m = (x_{m,k})_{k=1}^{n} \in \mathbb{R}^n$ in the metric space $(\mathbb{R}^n, d_p)$. Let $\epsilon > 0$ be a positive real number. Then there is a natural number $m_\epsilon > 0$ so that

$$d\left(\mathbf{x}_m, \mathbf{x}_p\right) < \epsilon$$

for all indices $m, p \geq m_\epsilon$. We observe that for each index $1 \leq k \leq n$,

$$d\left(x_{m,k}, x_{p,k}\right) = |x_{m,k} - x_{p,k}| \leq \|\mathbf{x}_m - \mathbf{x}_p\|_p = d_p\left(\mathbf{x}_m, \mathbf{x}_p\right) < \epsilon$$

for all indices $m, p \geq m_\epsilon$, and so $(x_{m,k})_{m=0}^{\infty}$ is a Cauchy sequence of real numbers.

Corollary 3.6.15 Completeness of the real line now implies that this sequence converges to some real number

$$x_k = \lim_{m \to \infty} x_{m,k}.$$

Let $\mathbf{x} = (x_k)_{k=1}^{n} \in \mathbb{R}^n$. Theorem 3.2.7 Sequential convergence in the p-metric and Problem 3.2.5.2 Sequential convergence in the infinity metric now imply that $(\mathbf{x}_m)_{m=0}^{\infty}$ converges to $\mathbf{x}$.

Since every Cauchy sequence in $(\mathbb{R}^n, d_p)$ converges, this metric space is complete. We conclude that $\left(\mathbb{R}^n, \|\cdot\|_p\right)$ is a Banach space. ∎

It may appear from the above result that almost all commonly appearing metric spaces are complete, but this is certainly not the case. The rational numbers, $\mathbb{Q}$, for example, form a metric subspace of the real line $\mathbb{R}$ which are not complete. We showed this by exhibiting a Cauchy sequence of rational numbers which converge to an irrational number, but this could also be observed by noting that the rational numbers are not a closed subset of the real numbers.

As the properties of complete metric spaces are incredibly useful for performing analysis, it is almost always preferable to work in a complete metric space. Shortly, we will see that this is usually possible, by passing to the **completion** of an incomplete metric space.

3.6.2 Completion of a Metric Space [SKIP]

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3.6.3 Metric Compactness

When is it the case that a sequence in a metric space has a convergent subsequence? This is a natural question to ask about metric spaces. We have seen so far that this occurs when the original sequence converges, and when the sequence is a bounded sequence of real numbers, but this phenomenon is not limited to the case of convergent sequences. In some metric spaces, all sequences have convergent subsequences. These metric spaces are called **(sequentially) compact**, and we will devote the rest of this section to investigating the properties of sequentially compact metric spaces.

Definition 3.6.17 Sequential compactness. A subset $U \subseteq X$ of a metric space (X, ρ) is called **(sequentially) compact** if every sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U$ has a subsequence which converges to a limit in U, and **(sequentially) non-compact** otherwise. A metric space (X, ρ) is called **compact** (resp. **non-compact**) if X is compact (resp. non-compact). ◆

Example 3.6.18 Sequential compactness. The following are examples of compact and non-compact metric spaces:

- A discrete metric space (X, δ_X) is compact if and only if the ground set X is finite. More generally, any finite metric space is compact.
- The n-dimensional Euclidean space $\mathbb{R}^n$ is compact if $n = 0$ and non-compact if $n \geq 1$. In particular, the real line $\mathbb{R}$ is non-compact.

▲

The compactness of a metric space has profound implications for the behavior of continuous maps on that space. We will see in the next chapter that much can be said about continuous maps with compact (in the more general sense which we will introduce later) domains. For example, a continuous map with domain a compact metric space is uniformly continuous.

Theorem 3.6.19 Heine-Cantor theorem. (Eduard Heine; Georg Cantor) *A continuous map from a compact metric space to a metric space is uniformly continuous.*

Proof. Let $f\colon X \to Y$ be a map from a compact metric space (X, ρ_X) to a metric space (Y, ρ_Y), and suppose that f is not uniformly continuous. Then there is a positive real number $\epsilon > 0$ so that for all positive real numbers $\delta > 0$,

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \geq \epsilon$$

for some points $x_1, x_2 \in X$ with $\rho_X\left(x_1, x_2\right) < \delta$.

Let $(\delta_n)_{n=0}^{\infty}$ be a sequence of positive real numbers $\delta_n > 0$ so that

$$\lim_{n\to\infty} \delta_n = 0.$$

For each index $n \in \mathbb{N}$, there are points $x_{1,n}, x_{2,n} \in X$ so that $\rho_X\left(x_{1,n}, x_{2,n}\right) < \delta_n$ but $\rho_Y\left(f\left(x_{1,n}\right), f\left(x_{2,n}\right)\right) \geq \epsilon$. Since X is compact, the sequence $(x_{2,n})_{n=0}^{\infty}$ has a convergent subsequence $(x_{2,n_k})_{k=0}^{\infty}$; that is,

$$\lim_{k\to\infty} x_{2,n_k} = x$$

for some point $x \in X$. We will argue that $(x_{1,n_k})_{k=0}^{\infty}$ also converges to x. Indeed, let $\epsilon > 0$ be a positive real number. Since $(\delta_n)_{n=0}^{\infty}$ converges to 0, there is some natural number $n_\epsilon \in \mathbb{N}$ so that

$$\delta_n < \frac{\epsilon}{2}$$

for all indices $n \geq n_\epsilon$. Moreover, since $(x_{2,n_k})_{k=0}^{\infty}$ converges to x, there is a natural number $k_\epsilon \in \mathbb{N}$ so that

$$\rho_X\left(x_{2,n_k}, x\right) < \frac{\epsilon}{2}$$

for all indices $k \geq k_\epsilon$. Let $m_\epsilon = \max(\{n_\epsilon, k_\epsilon\}) \in \mathbb{N}$, and note that Lemma 3.6.10 Increasing functions implies that $n_k \geq m_\epsilon$ whenever $k \geq m_\epsilon$. In particular, this implies that

$$\rho_X(x_{1,n_k}, x) \leq \rho_X(x_{1,n_k}, x_{2,n_k}) + \rho_X(x_{2,n_k}, x) < \delta_n + \rho_X(x_{2,n_k}, x)$$
$$< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all indices $k \geq m_\epsilon$. Thus $(x_{1,n_k})_{k=0}^\infty$ also converges to x.

If f were continuous at x, then both $(f(x_{1,n_k}))_{k=0}^\infty$ and $(f(x_{2,n_k}))_{k=0}^\infty$ would converge to $f(x)$. In this case, we would see that

$$\epsilon \leq \lim_{k\to\infty} \rho_Y(f(x_{1,n_k}), f(x_{2,n_k})) \leq \lim_{k\to\infty} (\rho_Y(f(x_{1,n_k}), f(x)) + \rho_Y(f(x_{2,n_k}), x))$$
$$= \lim_{k\to\infty} \rho_Y(f(x_{1,n_k}), f(x)) + \lim_{k\to\infty} \rho_Y(f(x_{2,n_k}), f(x)) = 0 + 0 = 0$$

This contradiction implies that f is not continuous at x.

In summary, we have shown that if f is not uniformly continuous, then it is not continuous. Equivalently, if f is continuous, then it is uniformly continuous. ■

We observe that there is no version of the Heine-Cantor theorem for Lipschitz continuity; indeed, the function $f\colon [0,1] \to [0,1]$ defined by the formula $f(x) = \sqrt{x}$ is continuous but not Lipschitz continuous, even though we will see that the closed interval $[0,1]$ is compact.

Continuity of a map between metric spaces is a local property which places bounds on the amount the map can increase the distance between close points. Compactness allows us to port this local property of continuity to a global property of uniform continuity, so that any continuous map is uniformly continuous on a compact subset of the domain. It is for this reason (that we may transfer many local geometric and topological properties to global properties) that compact subsets of metric spaces are so useful. We will see that compact metric spaces are also complete, and so it is often the case that the most salient information about a metric space are its compact subspaces.

In fact, more can be said about this situation. The image of a compact subset of a metric space under a continuous map is a compact subset of the codomain.

Proposition 3.6.20 Continuous images of compact subsets. *Let $f\colon X \to Y$ be a continuous map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). If a subset $U \subseteq X$ is compact, then its image $f(U) \subseteq Y$ is also compact.*

Proof. Let $(v_n)_{n=0}^\infty$ be a sequence of points $v_n \in f(U)$. For each index $n \in \mathbb{N}$, there is some point $u_n \in U$ so that $f(u_n) = v_n$. Since U is compact, the sequence $(u_n)_{n=0}^\infty$ has a convergent subsequence $(u_{n_k})_{k=0}^\infty$. Since f is continuous,

$$\lim_{k\to\infty} v_{n_k} = \lim_{k\to\infty} f(u_{n_k}) = f\left(\lim_{n\to\infty} u_{n_k}\right);$$

that is, the subsequence $(v_{n_k})_{k=0}^\infty$ of $(v_n)_{n=0}^\infty$ converges to a point in $f(U)$. Since every sequence in $f(U)$ has a subsequence which converges to a limit in $f(U)$, we conclude that $f(U)$ is compact. ■

The above result implies that sequential compactness is a topological invariant of metric spaces, since it is preserved under continuous maps between metric spaces. Intuitively, a compact metric space can be viewed as self-contained, since it is impossible for a divergent sequence to fully escape. We will end this section with an equivalent characterization of sequential compactness which may be surprising; a metric space is (sequentially) compact if and only if it is complete and totally bounded. In order to show this, we will require the following supporting lemma:

Lemma 3.6.21 Sequences in totally bounded subsets. *If a subset $U \subseteq X$ of a metric space (X, ρ) is totally bounded, then every sequence of points in U has a Cauchy subsequence.*

Proof. If $U = \emptyset$ is empty, then the desired result holds vacuously. We may therefore suppose that $U \neq \emptyset$ is non-empty. Let $(u_n)_{n=0}^{\infty}$ be a sequence of points $u_n \in U$. We aim to show that this sequence has a Cauchy subsequence. To that end, we first observe that since U is totally bounded in X, it is bounded by Proposition 3.4.24 Total boundedness implies boundedness, and so $\operatorname{diam}(U) < \infty$ is finite. let $(r_m)_{m=0}^{\infty}$ be a sequence of positive real numbers $r_n > 0$ so that $r_0 > \operatorname{diam}(U)$ and

$$\lim_{m\to\infty} r_m = 0.$$

We argue inductively that for each natural number $m \in \mathbb{N}$, there is a subsequence $(v_{m,n})_{n=0}^{\infty}$ of $(u_n)_{n=0}^{\infty}$ whose entries are all contained in an open ball of radius r_m, and moreover that if $m \leq p$ then $(v_{p,n})_{n=0}^{\infty}$ is a subsequence of $(v_{m,n})_{n=0}^{\infty}$.

Indeed, for the base case $m = 0$, we observe that U is contained in an open ball of radius $r_0 > \operatorname{diam}(U)$ around any of its points, so we may take $v_{0,n} = u_n$ for all indices $n \in \mathbb{N}$.

For the inductive step, suppose for the inductive hypothesis that for some natural number $m \in \mathbb{N}$ the sequence $(u_n)_{n=0}^{\infty}$ has a subsequence $(v_{m,n})_{n=0}^{\infty}$ whose entries are all contained in an open ball of radius r_m. Since U is totally bounded, it is contained in a union of finitely many open balls of radius r_{m+1}. The pigeonhole principle implies that infinitely many entries of $(v_{m,n})_{n=0}^{\infty}$ lie in one of these open balls. Removing all other entries, we obtain a further subsequence $(v_{m+1,n})_{n=0}^{\infty}$ whose entries all lie in a ball of radius r_{m+1}.

Consider the subsequence $(u_{n_k})_{k=0}^{\infty}$ of $(u_n)_{n=0}^{\infty}$ defined by the formula

$$u_{n_k} = v_{n,n}.$$

We will show that this subsequence is Cauchy. Indeed, let $\epsilon > 0$ be a positive real number. Since $(r_m)_{m=0}^{\infty}$ converges to 0, there is a natural number $m_\epsilon \in \mathbb{N}$ so that

$$r_m < \frac{\epsilon}{2}$$

for all indices $m \geq m_\epsilon$. Indeed, since $(u_{n_k})_{k=m_\epsilon}^{\infty}$ is a subsequence of $(v_{m_\epsilon,n})_{n=0}^{\infty}$, its entries all lie in an open ball of radius $r_m < \frac{\epsilon}{2}$. In particular, the entries of $(u_{n_k})_{k=m_\epsilon}^{\infty}$ have distance strictly less than ϵ from each other. Thus $(u_{n_k})_{k=0}^{\infty}$ is Cauchy. ■

Theorem 3.6.22 Heine-Borel Theorem. (Eduard Heine; Émile Borel) *A subset of a metric space is compact if and only if it is complete and totally bounded.*

Proof. Let $U \subseteq X$ be a subset of a metric space (X, ρ), and suppose first that U is compact. If $(u_n)_{n=0}^{\infty}$ is a Cauchy sequence of points $u_n \in U$, then compactness implies that $(u_n)_{n=0}^{\infty}$ has a subsequence which converges to a limit in U. Proposition 3.6.12 Cauchy sequences with convergent subsequences implies that $(u_n)_{n=0}^{\infty}$ converges to this limit. Since every Cauchy sequence in U converges to a limit in U, U is complete.

Now suppose for a contradiction that U is not totally bounded in X. Then there exists a positive real number $\epsilon > 0$ so that U is not contained in any finite union of open balls of radius ϵ. In particular, there is a sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U$ so that

$$u_n \not\subseteq \bigcup_{k=0}^{n-1} \mathrm{B}_\epsilon (u_k)$$

for all indices $n \in \mathbb{N}$. Since U is compact, this sequence has a subsequence $(u_{n_k})_{k=0}^{\infty}$ which converges to some point $u \in U$. In particular, there is a natural number $k_\epsilon \in \mathbb{N}$ so that

$$\rho (u_{n_k}, u) < \frac{\epsilon}{2}$$

for all indices $k \geq k_\epsilon$. We observe that

$$\begin{aligned}\rho\left(u_{n_{k_\epsilon+1}}, u_{n_{k_\epsilon}}\right) &\leq \rho\left(u_{n_{k_\epsilon+1}}, u\right) + \rho\left(u, u_{n_{k_\epsilon}}\right) = \rho\left(u_{n_{k_\epsilon+1}}, u\right) + \rho\left(u_{n_{k_\epsilon}}, u\right) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,\end{aligned}$$

so that

$$u_{n_{k_\epsilon+1}} \in \mathrm{B}_\epsilon\left(u_{n_{k_\epsilon}}\right) \subseteq \bigcup_{k=0}^{n_{k_\epsilon+1}-1} \mathrm{B}_\epsilon (u_k).$$

This contradiction implies that U is totally bounded in X.

Conversely, now suppose that U is complete and totally bounded in X. Then Lemma 3.6.21 Sequences in totally bounded subsets implies that every sequence of points in U has a Cauchy subsequence, and completeness implies that such a subsequence converges to a limit in U. Thus every sequence of points in U has a subsequence which converges to a limit in U; that is, U is compact. ■

We may use equivalent characterizations of completeness and total boundedness in Euclidean space to reduce the above theorem into a result which makes it easier to determine if a subset of Euclidean space is compact. Since Euclidean space is complete, a subset is complete if and only if it is closed. Moreover, as we have previously shown, a subset of Euclidean space is totally bounded if and only if it is bounded. Therefore, a subset of Euclidean space is sequentially compact if and only if it is closed and bounded. This is the content of the following corollary to the Heine-Borel Theorem, which is sometimes also called the Heine-Borel theorem.

Corollary 3.6.23 Compact subsets of Euclidean space. *A subset of Euclidean space is compact if and only if it is closed and bounded.*

Proof. The Heine-Borel Theorem implies that a subset of Euclidean space is compact if and only if it is complete and totally bounded. Since Euclidean space is complete by Corollary 3.6.16 Completeness of the p-norm, Proposition 3.6.7 Closed subsets of complete metric spaces implies that a subset of Euclidean space is complete if and only if it is closed. Moreover, Proposition 3.4.25 Total boundedness in Euclidean space implies that a subset of Euclidean space is totally bounded if and only if it is bounded. ■

Corollary 3.6.23 Compact subsets of Euclidean space allows our intuition about the geometry and point-set topology of Euclidean spaces to inform our understanding of compactness in arbitrary metric spaces.

Completeness and compactness are important geometric and topological properties for the study of metric geometry. While completeness is not easily generalizable to the broader spaces which we will study in Chapter 4 Topological Spaces, compactness remains a central idea in the study of topological spaces.

We will see that in the absence of a metric, we will rely on an equivalent characterization of compactness derived from a similar idea to total boundedness.

3.6.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Cauchy sequences. Determine whether or not the given sequence of points in a metric space is Cauchy.

1. Consider the sequence $(x_n)_{n=0}^{\infty}$ of real numbers x_n defined by the formula

$$x_n = 2 + 3^{-n}.$$

Determine whether or not the sequence $(x_n)_{n=0}^{\infty}$ is a Cauchy sequence in the real line $(\mathbb{R}, d)$.

2. Consider the sequence $(x_n)_{n=0}^{\infty}$ of real numbers x_n defined by the formula

$$x_n = 2 + 3^{-n}.$$

Determine whether or not the sequence $(x_n)_{n=0}^{\infty}$ is a Cauchy sequence in the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

3. Consider the sequence $(f_n)_{n=0}^{\infty}$ of bounded real-valued functions $f_n\colon [0,\infty) \to \mathbb{R}$ defined by the formula

$$f_n(x) = \frac{1}{x^n}.$$

Determine whether or not the sequence $(f_n)_{n=0}^{\infty}$ is a Cauchy sequence in the metric space $(\mathrm{B}([1,\infty)), d_\infty)$.

Complete subspaces of Euclidean space. Determine whether the given subset of Euclidean space is complete or incomplete.

4. Let U be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$U = \left\{\mathbf{x} \in \mathbb{R}^2 \;\middle|\; \mathbf{x} \neq \mathbf{0}\right\}.$$

Determine whether U is complete or incomplete in $\mathbb{R}^2$.

5. Let V be the subset of 3-dimensional Euclidean space $\mathbb{R}^3$ defined by

$$V = \left\{\mathbf{x} \in \mathbb{R}^3 \;\middle|\; \|\mathbf{x}\| = 2\right\}.$$

Determine whether V is complete or incomplete in $\mathbb{R}^3$.

6. Determine whether the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is complete or incomplete in the real line $\mathbb{R}$.

Compact subspaces of Euclidean space. Determine whether or not the given subset of Euclidean space is compact.

7. Let U be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$U = \left\{ \mathbf{x} \in \mathbb{R}^2 \;\middle|\; \mathbf{x} \neq \mathbf{0} \right\}.$$

Determine whether or not U is compact in $\mathbb{R}^2$.

8. Let V be the subset of 3-dimensional Euclidean space $\mathbb{R}^3$ defined by

$$V = \left\{ \mathbf{x} \in \mathbb{R}^3 \;\middle|\; \|\mathbf{x}\| = 2 \right\}.$$

Determine whether or not V is compact.

9. Determine whether or not the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are compact.

10. Let $a, b \in \mathbb{R}$ be real numbers, and suppose that $a \leq b$. Determine whether or not the closed interval $[a, b]$ is compact.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.6.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Completeness of discrete metric spaces.

(a) Prove that any discrete metric space is complete.

(b) Prove that any metric space with finite ground set is complete.

(c) Consider the metric subspace (X, d) of the real line $(\mathbb{R}, d)$ with ground set

$$X = \left\{ \frac{1}{n} \in \mathbb{R} \;\middle|\; n \in \mathbb{N} \setminus \{0\} \right\}.$$

Argue that (X, d) is a counterexample to the *false* claim that any metric space whose points are all isolated points is complete.

2. Incompleteness of the p-adic metric. Fix a prime number $p \in \mathbb{N}$, and recall that the **p-adic valuation** $|\cdot|_p : \mathbb{Q} \to \mathbb{Q}$ is a rational-valued function defined by the formula

$$|q|_p = \begin{cases} 0 & q = 0 \\ p^{-k_p(q)} & q \neq 0, \end{cases}$$

where $k_p(q) \in \mathbb{Z}$ is the unique integer so that $q = p^{k_p(q)} \left(\frac{a}{b}\right)$ for some integers $a, b \in \mathbb{Z}$ which are not divisible by p.

Moreover, recall from (b) of Problem 3.1.5.3 Ultrametrics that the p-adic valuation $|\cdot|_p$ induces an ultrametric $\rho_p \colon \mathbb{Q}^2 \to \mathbb{R}$ called the **p-adic metric** defined by the formula

$$\rho_p(q_1, q_2) = |q_1 - q_2|_p .$$

(a) Consider the sequence $(q_n)_{n=0}^{\infty}$ of rational numbers $q_n \in \mathbb{Q}$ defined by the formula

$$q_n = \sum_{k=0}^{n} p^k = 1 + p + p^2 + \cdots + p^n.$$

Show that $(q_n)_{n=0}^{\infty}$ is a Cauchy sequence in $(\mathbb{Q}, \rho_p)$.

(b) Consider the sequence $(r_n)_{n=0}^{\infty}$ of rational numbers $r_n \in \mathbb{Q}$ defined by the formula

$$r_n = \sum_{k=0}^{n} p^{k!} = p + p + p^2 + p^6 + \cdots + p^{n!}.$$

Show that $(r_n)_{n=0}^{\infty}$ is a Cauchy sequence in $(\mathbb{Q}, \rho_p)$.

While the Cauchy sequence $(q_n)_{n=0}^{\infty}$ converges in the p-adic metric to a rational number, the Cauchy sequence $(r_n)_{n=0}^{\infty}$ diverges and therefore is a witness to the incompleteness of $(\mathbb{Q}, \rho_p)$.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

In this chapter, we have seen that a metric allows us to easily develop a notion of closeness for points in a set based on the standard ordering on the nonnegative real numbers, and we have exploited this notion to explore analysis in metric spaces. In the next chapter, we will examine more abstract notions of closeness which will come from objects known as **topologies**.

Chapter 4

Topological Spaces

In the previous chapter, we examined several point-set topological concepts such as sequential and map convergence, openness and closedness, continuity, and completeness and compactness in the context of metric spaces. However, we also saw other instances of these topological notions, such as the pointwise convergence of a sequence of maps to a metric space, which in spite of the similarities with sequential convergence in a metric space, cannot be described as convergence with respect to a metric. What then, is the structure with respect to which a pointwise convergence sequence of maps converges? In this chapter, we introduce **topological spaces**, which provide a more general setting than metric spaces in which we may consider many (but not all) of the same analytical notions as in the previous chapter. Just as a metric space is a ground set considered with the additional structure of a metric, a topological space is a set considered with the additional structure of a **topology**, which designates certain privileged subsets of the ground set as **open**. In the previous chapter, we took the real-valued function which computed the distance between two points as fundamental, and we developed a sense of openness for subsets based on this measure of distance. Now, we will take this openness to be fundamental and generalize the notions of convergence, continuity, and compactness investigated in the previous chapter.

4.1 Introduction to Topology

In the previous chapter, we used the distance function on a metric space to provide a quantitative measure of how close two points were to each other. In this chapter, we will see that we can relax this to a more qualitative notion of closeness and still construct many of the point-set topological concepts defined for metric spaces. In particular, many of the definitions of the previous chapter can be equivalently defined in terms of open neighborhoods.

4.1.1 Examples of Topologies

A **topology** on a set is just a designation of **open** subsets, which are required to satisfy some of the important properties of open sets in metric spaces. Specifically, besides some non-degeneracy requirements, the collection of open sets in a topological space is required to be stable under arbitrary unions and finite intersections.

Definition 4.1.1 Topology. A collection $\tau \subseteq \mathcal{P}(X)$ of subsets of a set X, which is called the **ground set** or **underlying set** and whose elements are called **points**, is called a **topology** on $\mathcal{P}(X)$, and its elements are called **open**, if it satisfies the following conditions:

- *Non-degeneracy.*

  The empty set $\emptyset \in \tau$ and the ground set X are open.

- *$\cup$-stability.*

  For any indexed family $(U_j)_{j\in\mathcal{J}}$ of open sets $U_j \in \tau$, the union

  $$\bigcup_{j\in\mathcal{J}} U_j \in \tau$$

  is open.

- *Finite $\cap$-stability.*

  For any finite sequence $(U_j)_{j=1}^{m}$ of open sets $U_j \in \tau$, the intersection

  $$\bigcap_{j=1}^{m} U_j \in \tau$$

  is open.

In this case, the ordered pair (X, τ) is called a **topological space**. ◆

Example 4.1.2 Topological spaces. The following are examples of topological spaces:

- *Discrete topology.*

  For any set X, the power set $\mathcal{P}(X)$ of X is a topology on X. Considered in this context, $\mathcal{P}(X)$ is called the **discrete topology** on X and is denoted $\top_X$. The topological space $(X, \top_X)$ is called a **discrete (topological) space**.

- *Indiscrete topology.*

  For any set X, the set $\bot_X = \{\emptyset, X\}$ is a topology on X called the **discrete topology** on X. The topological space $(X, \bot_X)$ is called an **indiscrete (topological) space**.

- *Sierpiński space.*

  The **Sierpiński space** is the topological space (X, τ) with ground set $X = \{0, 1\}$ and topology $\tau = \{\emptyset, \{1\}, \{0, 1\}\}$. This topological space is a well-known simple counterexample to many false statements about topological spaces that feel true intuitively.

▲

Remark 4.1.3 Implied topologies. We will see that in general, there many topologies on a given set, no one of which can truly be considered canonical. However, sometimes if a topology τ on a set X is clear from context, previously specified, or otherwise implied, some mathematicians will refer to the topological space (X, τ) by only the underlying set X.

At the beginning of Chapter 3 Metric Spaces, we saw how some structures on a ground set induce others in a canonical way. For example, we saw that an inner product on a real or complex vector space induces a norm on that vector space, and that a norm on such a vector space induces a metric on the underlying set. In this way, we viewed an inner product space as a special kind of normed vector space and a normed vector space as a special kind of metric space. Similarly, we will now see that, in a metric space, the metric induces a topology on the ground set in a canonical way, so that we may view a metric space as a special kind of topological space.

Definition 4.1.4 Metric topology. The **metric topology** τ_ρ induced on a set X by a metric $\rho\colon X \times X \to \mathbb{R}$ on X is the collection of open subsets in X. That τ_ρ is a topology on X follows immediately from (1), (2), and (4) of Proposition 3.4.10 Topology of a metric space.

A topology τ on a set X is called **metrizable** if it is the metric topology induced by a metric on X, and **non-metrizable** otherwise. A topological space (X, τ) is called **metrizable** (resp. **non-metrizable**) if the topology is metrizable (resp. non-metrizable). ◆

Example 4.1.5 Metric topologies. The following are examples of metrizable and non-metrizable topologies:

- The **Euclidean topology** τ_d on n-dimensional real coordinate space $\mathbb{R}^n$ is the metric topology induced by the Euclidean metric $d\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$. This topology is also called the **standard topology** or the **usual topology** on $\mathbb{R}^n$.

- The discrete topology $\top_X$ on a set X is the metric topology induced by the discrete metric $\delta_X\colon X \times X \to \mathbb{R}$. In particular, any discrete topological space is metrizable.

- On the other hand, the indiscrete topology $\bot_X$ is non-metrizable unless $|X| \leq 1$, in which case $\bot_X = \top_X$.

- The Sierpiński space $(\{0,1\}, \{\emptyset, \{1\}, \{0,1\}\})$ is non-metrizable; that is, the topology $\{\emptyset, \{1\}, \{0,1\}\}$ is not induced on the ground set $\{0,1\}$ by any metric. More generally, a finite topological space is metrizable if and only if it is discrete.

▲

The above definition shows that we actually are already familiar with a special class of topological spaces; just as a metric space hides inside any normed vector space, so too does a topological space hide inside any metric space. However, as shown in the last two examples above, not every topological space arises in this manner. Moreover, there is in general no unique metric which induces a topology on a metrizable topological space; for example, any positive scalar multiple a metric induces the same topology on the underlying set.

In the same way that a metric induces a metric on any subset of its underlying set, a topology on a set induces a topology on any subset of its underlying set. For metric spaces, this subspace construction is a simple restriction of the metric to a subset of its domain. There is a similar, but slightly more intricate construction for topological spaces.

Lemma 4.1.6 Subspace topology. *Fix a subset $U \subseteq X$ of a topological space (X, τ). Then the collection*

$$\{U \cap V \in \mathcal{P}(X) \mid V \in \tau\}$$

is a topology on U.

Proof. Denote by τ' the collection

$$\tau' = \{U \cap V \in \mathcal{P}(X) \mid V \in \tau\}.$$

Since τ is non-degenerate, $\emptyset, X \in \tau$ are open in X, and so

$$\emptyset = U \cap \emptyset \in \tau'$$

and

$$U = U \cap X \in \tau';$$

that is, τ' is non-degenerate.

Now consider an indexed family $(W_j)_{j\in\mathcal{J}}$ of sets $W_j \in \tau'$. Then for each index $j \in \mathcal{J}$, $W_j = U \cap V_j$ for some open set $V_j \in \tau$ in X. Since τ is stable under unions,

$$\bigcup_{j\in\mathcal{J}} V_j \in \tau$$

is open in X, and so

$$\bigcup_{j\in\mathcal{J}} W_j = \bigcup_{j\in\mathcal{J}} (U \cap V_j) = U \cap \bigcup_{j\in\mathcal{J}} V_j \in \tau';$$

that is, τ' is stable under unions.

Finally, consider a finite sequence $(W_j)_{j=1}^m$ of sets $W_j \in \tau'$. Then for each index $1 \le j \le m$, $W_j = U \cap V_j$ for some open set $V_j \in \tau$ in X. Since τ is stable under finite intersections,

$$\bigcap_{j=1}^m V_j \in \tau$$

is open in X, and so

$$\bigcap_{j=1}^m W_j = \bigcap_{j=1}^m (U \cap V_j) = U \cap \bigcap_{j=1}^m V_j \in \tau';$$

that is, τ' is stable under finite intersections.

Since τ' is non-degenerate, stable under unions, and stable under finite intersections, it is a topology on U. ■

Definition 4.1.7 Subspace topology. The **subspace topology** $\tau|_U$ induced on a subset $U \subseteq X$ of a topological space (X, τ) is the collection

$$\tau|_U = \{U \cap V \in \mathcal{P}(X) \mid V \in \tau\}.$$

This collection $\tau|_U$ is a topology on U by Lemma 4.1.6 Subspace topology, and the topological space $(U, \tau|_U)$ is called a **(topological) subspace** of (X, τ). ◆

It turns out that this notion of the restriction of a topology to a subspace is compatible with the restriction of a metric to a subspace. That is, the restriction of a metric topology is precisely the metric topology induced by the restriction of that metric to that subset.

Lemma 4.1.8 Topological subspaces of metric spaces. *Let $U \subseteq X$ be a subset of a metric space (X, ρ). Then $\tau_\rho|_U = \tau_{\rho|_U}$.*

Proof. Part (a) of Problem 3.4.5.1 Open and closed sets in metric subspaces implies that a subset $V \subseteq U$ is open in U if and only if $V = U \cap W$ for some open subset $W \subseteq X$; that is, $V \in \tau_{\rho|_U}$ if and only if $V = U \cap W$ for some

$W \in \tau_\rho$. We conclude that $\tau_{\rho|_U} = \tau_\rho|_U$. ■

One consequence of Lemma 4.1.8 Topological subspaces of metric spaces is that metric subspaces are topological subspaces. Just as with metric subspaces, we will often abuse notation and refer to a subspace topology with the same notation as the original topology, so that we may speak of a topological subspace (U, τ) of a topological space (X, τ); here we really mean $(U, \tau|_U)$.

4.1.2 Algebra of Open and Closed Sets

The topology in a topological space elevates certain subsets of the ground set as **open**. In an analogous manner to our definition of closedness for subsets of a metric space, a subset of a topological space is called **closed** if and only if its complement is open.

Definition 4.1.9 Closed subset. A subset $U \subseteq X$ of a topological space (X, τ) is called **closed** in X if its complement $X \setminus U \in \tau$ is open in X. ◆

The defining properties of a topology impose constraints on the set of closed subsets of a topological space.

Lemma 4.1.10 Closed subsets of a topological space. *Let (X, τ) be a topological space.*

1. *The empty set $\emptyset$ and the ground set X are closed in X.*
2. *For any indexed family $(U_j)_{j\in\mathcal{J}}$ of closed sets U_j in X, the intersection*
$$\bigcap_{j\in\mathcal{J}} U_j$$
is closed in X.
3. *For any finite sequence $(U_j)_{j=1}^m$ of closed sets U_j in X, the union*
$$\bigcup_{j=1}^{m} U_j$$
is closed in X.

Proof.

1. The non-degeneracy of the topology τ implies that $\emptyset, X \in \tau$ are open in X, and so their complements $\emptyset = X \setminus X$ and $X = X \setminus \emptyset$ are closed in X.
2. For each index $j \in \mathcal{J}$, $U_j = X \setminus V_j$ for some open subset $V_j \in \tau$. The stability of the topology τ under unions implies that
$$\bigcup_{j\in\mathcal{J}} V_j \in \tau$$
is open in X, and so
$$\bigcap_{j\in\mathcal{J}} U_j = \bigcap_{j\in\mathcal{J}} (X \setminus V_j) = X \setminus \bigcup_{j\in\mathcal{J}} V_j$$
is closed in X.
3. For each index $1 \le j \le m$, $U_j = X \setminus V_j$ for some open subset $V_j \in \tau$.

The stability of the topology τ under finite intersections implies that

$$\bigcap_{j=1}^{m} V_j \in \tau$$

is open in X, and so

$$\bigcup_{j=1}^{m} U_j = \bigcup_{j=1}^{m} (X \setminus V_j) = X \setminus \bigcap_{j=1}^{m} V_j$$

is closed in X.

■

There is also a natural generalization of the notion of neighborhoods to the setting of topological spaces. Rather than requiring a **neighborhood** of a point in a topological space to contain an open ball centered around that point, we require that it contain some open subset which contains the point.

Definition 4.1.11 Neighborhood. A subset $N \subseteq X$ of a topological space (X, τ) is called a **neighborhood** of a point $x \in X$ in X if $x \in U \subseteq N$ for some open set $U \in \tau$ in X. The set of neighborhoods of x in X is variously denoted $\mathcal{N}(x)$, $\mathcal{N}_X(x)$, and/or $\mathcal{N}_{(X,\tau)}(x)$, depending on which information is clear from context. ◆

Example 4.1.12 Neighborhoods. Recall that the Sierpiński space (X, τ) is the topological space with ground set $X = \{0, 1\}$ and topology $\tau = \{\emptyset, \{1\}, \{0, 1\}\}$. Thus $X = \{0, 1\}$ is the only neighborhood of the point 0, but $\{1\}$ and $\{0, 1\}$ are neighborhoods of the point 1; that is,

$$\mathcal{N}(0) = \{\{0, 1\}\} \qquad \mathcal{N}(1) = \{\{1\}, \{0, 1\}\}$$

▲

Just as in a metric space, we will interpret the relationship of between a point in a topological space and a neighborhood to mean that the neighborhood contains all points which are "close enough" to that point. In a metric space, this notion of nearness for points can be *quantifiably* determined. In a topological space, however, we will use the concept of neighborhoods introduced above to obtain *qualitative* measures of closeness for points: two points are considered to be "close" in a topological space if many neighborhoods of one point also contain the other point.

The cognitive metaphor described above is called **locality**. For example, a point-set topological property can be said to hold locally if any point in the topological space has a neighborhood on which that property holds. Similarly, the **local properties** of a topological space are those which involve the existence of neighborhoods around any given point which satisfy certain conditions. These local properties of a topological space can be contrasted with its **global properties**, which hold on its entire ground set. For example, we will later see manifolds to witness strong local properties while having incredibly large variability in their global properties. We will see several examples of local properties, such as continuity, in the remainder of our investigation of topological spaces.

In many ways, the concept of topological spaces was created exactly to formalize the most general setting in which locality can be considered even in the absence of a metric to provide a quantitative measure of nearness for points. Because of the definition of the metric topology induced by a metric on its ground set, these notions of metric and topological locality coincide for metric topological spaces. That is, a metric space and its associated metric topological space have the same neighborhoods. This also implies that all notions in the context of metric spaces defined solely on neighborhoods can be reliably generalized to the context of topological spaces. In particular, we can use neighborhoods to define the **interior**, **closure**, and **boundary** of a subset of a topological space.

Definition 4.1.13 Interior; closure; boundary. Let $U \subseteq X$ be a subset of a topological space (X, τ).

- *Interior.*

  A point $x \in X$ is an **interior point** of U if $U \in \mathcal{N}_X(x)$ is a neighborhood of x in X. The set of interior points of U is called the **interior** of U and is denoted $\mathrm{Int}(U)$ or U°.

- *Closure.*

  A point $x \in X$ is a **point of closure** of U in X if every neighborhood of x in X meets U; that is, x is a point of closure of U in X if $N \cap U \neq \emptyset$ for any neighborhood $N \in \mathcal{N}_X(x)$ of x in X. The set of points of closure of U in X is called the **closure** of U in X and is denoted $\mathrm{Cl}(U)$ or $\overline{U}$.

- *Boundary.*

  A point $x \in X$ is a **boundary point** of U in X if every neighborhood of x in X meets both U and its complement $X \setminus U$; that is, x is a boundary point of U in X if $N \cap U \neq \emptyset$ and $N \setminus U \neq \emptyset$ for any neighborhood $N \in \mathcal{N}_X(x)$ of x in X. The set of boundary points of U in X is called the **boundary** of U in X and is denoted $\mathrm{Bd}(U)$ or ∂U.

◆

That the above definitions are phrased in terms of neighborhoods implies that many of the related results in the setting of metric spaces also hold for topological spaces in general. Even better, the proofs of these results for metric spaces can easily be altered to construct proofs of their generalizations.

Theorem 4.1.14 Interior; closure; boundary. *Let U be a subset of a metric space (X, τ).*

1. *Interior.*

   U is open in X if and only if $\mathrm{Int}(U) = U$. More generally, U has interior

   $$\mathrm{Int}(U) = \bigcup_{V \in \mathcal{O}_U} V = X \setminus \mathrm{Cl}(X \setminus U),$$

   where $\mathcal{O}_U$ is the collection of open sets in X which are contained in U. Finally, $\mathrm{Int}(\mathrm{Int}(U)) = \mathrm{Int}(U)$.

2. *Closure.*

   U is closed in X if and only if $\mathrm{Cl}(U) = U$. More generally, U has closure

   $$\mathrm{Cl}(U) = \bigcap_{V \in \mathcal{C}_U} V = X \setminus \mathrm{Int}(X \setminus U),$$

   where $\mathcal{C}_U$ is the collection of closed sets in X which contain U. Finally, $\mathrm{Cl}(\mathrm{Cl}(U)) = \mathrm{Cl}(U)$.

3. *Boundary.*

   U has boundary $\mathrm{Bd}(U) = \mathrm{Cl}(U) \cap \mathrm{Cl}(X \setminus U)$.

The proof of Theorem 4.1.14 Interior; closure; boundary is nearly identical to the proofs of Proposition 3.4.9 Interior/closure, Proposition 3.4.9 Interior/closure, Theorem 3.4.11 Interior/closure, and Corollary 3.4.12 Idempotence of interior/closure and the solution to Problem 3.4.5.2 Interior, closure, and boundary of complements, and so it is left to the reader.

Now that we have defined the central object of study, the remainder of this chapter will be dedicated to expanding our understanding of the point-set topological notions and methods developed in Chapter 3 Metric Spaces to the more general setting of topological spaces. In the next section, we will construct more examples of topologies on a set coming from the natural partial order on its power set.

4.1.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Examples of Topologies. Verify that the given collection of subsets comprise a topology.

1. A subset $U \subseteq X$ of a set X is called **cofinite** in X if the complement $X \setminus U$ is finite. The **cofinite topology**

$$\tau = \{U \in \mathcal{P}(X) \mid U = \emptyset \text{ or } X \setminus U \text{ is finite}\}$$

on X is the collection of all cofinite subsets of X along with the empty set $\emptyset$. Verify that τ is a topology on X.

2. A subset $U \subseteq X$ of a set X is called **cocountable** in X if the complement $X \setminus U$ is countable. The **cocountable topology**

$$\tau = \{U \in \mathcal{P}(X) \mid U = \emptyset \text{ or } X \setminus U \text{ is countable}\}$$

on X is the collection of all cocountable subsets of X along with the empty set $\emptyset$. Verify that τ is a topology on X.

3. Fix an element $x \in X$ of a set X. The **particular point topology**

$$\tau = \{U \in \mathcal{P}(X) \mid U = \emptyset \text{ or } x \in U\}$$

on X is the collection of all subsets of X which contain x along with the empty set $\emptyset$. Verify that τ is a topology on X.

4. Fix an element $x \in X$ of a set X. The **excluded point topology**

$$\tau = \{U \in \mathcal{P}(X) \mid U = X \text{ or } x \notin U\}$$

on X is the collection of all subsets of X which do not contain x along with X itself. Verify that τ is a topology on X.

Computing subspace topologies. Describe the subspace topology induced on the given subset of a topological space.

5. Let $U \subseteq X$ be a subset of a set X. Verify that if X is equipped with the cofinite topology, then the subspace topology induced on U is the cofinite topology on U.
6. Let $U \subseteq X$ be a subset of a set X. Verify that if X is equipped with the cocountable topology, then the subspace topology induced on U is the cocountable topology on U.
7. Let $U \subseteq X$ be a subset of a set X, and fix a point $x \in X$. Verify that if X is equipped with the particular point topology, then the subspace topology induced on U is either the particular point topology if $x \in U$ or the discrete topology if $x \notin U$.
8. Let $U \subseteq X$ be a subset of a set X, and fix a point $x \in X$. Verify that if X is equipped with the excluded point topology, then the subspace topology induced on U is either the excluded point topology if $x \in U$ or the indiscrete topology if $x \notin U$.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

4.1.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Topologies induced by p-metrics.** Show that for all $1 \leq p \leq \infty$, the p-metrics $d_p \colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ induce the same metric topology on n-dimensional real coordinate space $\mathbb{R}^n$.

   Hint. You may find (b) of Problem 3.1.5.2 Norm equivalence useful.
2. **Closed sets in subspaces.** Let $U \subseteq X$ be a subset of a topological space (X, τ). Prove that a subset $W \subseteq U$ is closed in U if and only if $W = U \cap V$ for some closed set $V \subseteq X$ in X.
3. **Monotonicity of interior, closure, and boundary.** Fix subsets $U, V \subseteq X$ of a topological space (X, τ).

   (a) Prove that if $U \subseteq V$, then $\operatorname{Int}(U) \subseteq \operatorname{Int}(V)$ and $\operatorname{Cl}(U) \subseteq \operatorname{Cl}(V)$.

   (b) Give concrete examples so that $A \subsetneq B$ but $\operatorname{Int}(A) = \operatorname{Int}(B)$ and $\operatorname{Cl}(A) = \operatorname{Cl}(B)$.

   (c) Give concrete examples so that $A \subseteq B$ but $\operatorname{Bd}(A) \nsubseteq \operatorname{Bd}(B)$.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

4.2 Comparison and Topology

In this section, we will explore the relationships between topology and order theory. We will explore the order-theoretic properties of the partial order on the set of topologies on a fixed set, and we will explore ways of generating topologies by specifying some (but not necessarily all) open sets. Finally we will explore some natural topologies inherited by any totally ordered set.

4.2.1 The Lattice of Topologies

We now investigate the structure of the collection of topologies on a given set. As a subcollection of the second power set, this collection of topologies inherits a partial order by containment $\subseteq$. Much of this section will be devoted to exploring the order-theoretic properties of this partially ordered set. In particular, we will see that any collection of topologies has a unique supremum and infimum in this partially ordered set, and therefore that any collection of subsets of a set generates a unique topology.

Definition 4.2.1 Comparison of topologies. Denote by $\mathcal{T}_X \subseteq \mathcal{P}(\mathcal{P}(X))$ the set of topologies on a set X. $\mathcal{T}_X$ is partially ordered by containment, and the partially ordered set $(\mathcal{T}_X, \subseteq)$ is called the **lattice of topologies** on X.

- *Coarsening.*

  For all topologies $\tau_1, \tau_2 \in \mathcal{T}_X$ on X, if $\tau_1 \subseteq \tau_2$, then we say that τ_1 is **coarser** than τ_2, or equivalently that τ_1 is a **coarsening** of τ_2.

  Similarly, if $\tau_1 \subsetneq \tau_2$, then we say that τ_1 is **strictly coarser** than τ_2, or equivalently that τ_1 is a **strict coarsening** of τ_2.

- *Refinement.*

  For all topologies $\tau_1, \tau_2 \in \mathcal{T}_X$ on X, if $\tau_1 \supseteq \tau_2$, then we say that τ_1 is **finer** than τ_2, or equivalently that τ_1 is a **refinement** of τ_2.

  Similarly, if $\tau_1 \supsetneq \tau_2$, then we say that τ_1 is **strictly finer** than τ_2, or equivalently that τ_1 is a **strict refinement** of τ_2.

◆

We will eventually see that any collection of topologies on a given set has a unique supremum and infimum in the partial order of topological comparison. More easy to see, however, is that the lattice of topologies on that set itself has a maximum and minimum; any topology is coarser than the discrete topology and finer than the indiscrete topology. This justifies our use of the “top” and “bottom” notation for the discrete topology $\top_X$ and the indiscrete topology $\bot_X$ on a set X; in the lattice $\mathcal{T}_X$ of topologies on X, $\top_X$ and $\bot_X$ are the maximum and minimum elements, respectively.

Lemma 4.2.2 Bounds on topologies. *The lattice of topologies on a set is bounded as a partially ordered set.*

Proof. Consider the lattice $\mathcal{T}_X$ of topologies on a set X. We observe that

$$\bot_X = \{\emptyset, X\} \subseteq \tau \subseteq \mathcal{P}(X) = \top_X$$

for all topologies $\tau \in \mathcal{T}_X$, and so $\mathcal{T}_X$ is bounded from above and below. ■

As stated previously, we will show that any collection of topologies on a given set has a supremum and a infimum with respect to the partial order by containment. Since the intersection of a collection of topologies on a set is also a topology on that set, the existence of the greatest lower bound, or meet, of such a collection of topologies is easy to show.

Lemma 4.2.3 Intersection of topologies. *The intersection of a collection of topologies on a set is a topology.*

Proof. Let $(\tau_j)_{j\in\mathcal{J}}$ be an indexed family of topologies $\tau_j \in \mathcal{T}_X$ on a set X, and let

$$\tau = \bigcap_{j\in\mathcal{J}} \tau_j$$

be the intersection. Since $\emptyset, X \in \tau_j$ for each index $j \in \mathcal{J}$, $\emptyset, X \in \tau$; that is, τ is non-degenerate.

Now let $(U_k)_{k\in\mathcal{K}}$ be an indexed family of sets $U_k \in \tau$. We observe that $U_k \in \tau_j$ for all indices $j \in \mathcal{J}$ and $k \in \mathcal{K}$, and so the stability of each τ_j under unions implies that

$$\bigcup_{k\in\mathcal{K}} U_k \in \tau_j$$

for each index $j \in \mathcal{J}$. Since this holds for all indices $j \in \mathcal{J}$,

$$\bigcup_{k\in\mathcal{K}} U_k \in \tau;$$

that is, τ is stable under unions.

Finally let $(U_k)_{k=1}^n$ be a finite sequence of sets $U_k \in \tau$. We observe that $U_k \in \tau_j$ for all indices $j \in \mathcal{J}$ and $1 \le k \le n$, and so the stability of each τ_j under finite intersection implies that

$$\bigcap_{k=1}^{n} U_k \in \tau_j$$

for each index $j \in \mathcal{J}$. Since this holds for all indices $j \in \mathcal{J}$,

$$\bigcup_{k=1}^{n} U_k \in \tau;$$

that is, τ is stable under finite intersections.

Since τ is non-degenerate, stable under unions, and stable under finite intersections, it is a topology on X. ■

Since the intersection of sets is the infimum in the partial order by containment, we will see that Lemma 4.2.3 Intersection of topologies implies that the intersection of topologies is also the infimum in the partial order of topological comparison. That is, the intersection of a collection of topologies on a set is the finest common coarsening.

It might seem plausible that the dual notion of union might provide us with a supremum of such a collection with respect to topological comparison. Unfortunately, this is not the case; the union of topologies on a set is not, in general, also a topology on that set. However, there is a unique coarsest common refinement of such a collection. In order to show that this is the case, we need to investigate the ways in which collections of subsets of a set which may not themselves be topologies may generate a topology on that set.

Definition 4.2.4 Generated topology. A collection $\mathcal{S} \subseteq \mathcal{P}(X)$ of subsets of a set X is called a **subbase** of a topology τ (and its elements are called **subbasic open sets**) on X if τ is the coarsest topology on X which contains

$\mathcal{S}$; that is, $\mathcal{S}$ is a subbase for τ if every topology on X which contains $\mathcal{S}$ is a refinement of τ.

Every collection $\mathcal{S} \subseteq \mathcal{P}(X)$ of subsets of a set X is a subbase of a unique topology

$$\langle \mathcal{S} \rangle = \bigcap_{\mathcal{S} \subseteq \tau \in \mathcal{T}_X} \tau$$

on X called the topology **generated** by $\mathcal{S}$. ◆

Example 4.2.5 Subbases. The following are examples of subbases of topologies:

- The metric topology τ_ρ induced on a set X by a metric $\rho\colon X \times X \to \mathbb{R}$ is generated by the collection of metric open balls; that is,

  $$\tau_\rho = \langle \{ \mathrm{B}_r(x) \in \mathcal{P}(X) \mid r > 0 \text{ and } x \in X \} \rangle .$$

- The indiscrete topology $\perp_X$ on a set X is generated by the empty set $\emptyset$; that is, $\perp_X = \langle \emptyset \rangle$.

- The discrete topology $\top_X$ on a set X is generated by the singleton sets $\{\{x\} \in \mathcal{P}(X) \mid x \in X\}$; that is,

  $$\top_X = \langle \{\{x\} \in \mathcal{P}(X) \mid x \in X\} \rangle .$$

- Let $((X_j, \tau_j))_{j \in \mathcal{J}}$ be an indexed family of topological spaces (X_j, τ_j). The **box topology** is the topology

  $$\tau = \left\langle \left\{ \prod_{j \in \mathcal{J}} U_j \in \mathcal{P}\left(\prod_{j \in \mathcal{J}} X_j \right) \,\middle|\, (U_j)_{j \in \mathcal{J}} \in \prod_{j \in \mathcal{J}} \tau_j \right\} \right\rangle$$

  on the Cartesian product $\prod_{j \in \mathcal{J}} X_j$ of the ground sets generated by products of open sets.

  We will see that if the index set $\mathcal{J}$ is infinite, then the box topology described above is too fine to be of use. In Section 4.4 Continuity, we will introduce the dual notions of **initial** and **final topologies** to give natural topologies on sets constructed from the ground sets of topological spaces, including via Cartesian products, disjoint unions, and quotients.

▲

The concept of generated topologies will now allow us to show that the lattice of topologies on a given set is a complete lattice. That is, any collection of topologies has a unique least upper bound and a unique greatest lower bound.

Theorem 4.2.6 Complete lattice of topologies. *The lattice $\mathcal{T}_X$ of topologies on a set X is a complete lattice; that is, any collection $\mathcal{S} \subseteq \mathcal{T}_X$ of topologies on X has a unique supremum and infimum.*

Proof. Let $\mathcal{S} \subseteq \mathcal{T}_X$ be a collection of topologies on a set X. Since the intersection

$$\bigcap_{\tau \in \mathcal{S}} \tau$$

is an infimum for $\mathcal{S}$ in the partially ordered set $(\mathcal{P}(\mathcal{P}(X)), \subseteq)$, it is an infimum for $\mathcal{S}$ in $(\mathcal{T}_X, \subseteq)$; that is, this intersection is the finest common coarsening of the topologies in $\mathcal{S}$. It therefore suffices to show that $\mathcal{S}$ has a supremum.

To that end, let

$$\sigma = \left\langle \bigcup_{\tau \in \mathcal{S}} \tau \right\rangle$$

be the topology on X generated by the union of all topologies in $\mathcal{S}$. Then σ is a common refinement of each topology in $\mathcal{S}$. Moreover, if $\sigma' \in \mathcal{T}_X$ is a common refinement of each topology in $\mathcal{S}$, then it contains the union of all such topologies, and therefore is a refinement of τ. Thus σ is an infimum for $\mathcal{S}$ in $(\mathcal{T}_X, \subseteq)$; that is, this intersection is the coarsest common refinement of the topologies in $\mathcal{S}$. ■

So the lattice of topologies on any given set is in fact a complete lattice under the partial order of containment, where infima are intersections and suprema are the topologies generated by unions. The following alternative characterization of subbases of topology will be useful for specifying certain nice subbases.

Lemma 4.2.7 Subbases of topology. *A collection $\mathcal{S} \subseteq \mathcal{P}(X)$ of subsets of a set X is a subbase of a topology τ if and only if for every point $x \in U$ in every open set $U \in \tau$,*

$$x \in \bigcap_{k=1}^{n} S_k \subseteq U$$

for some finite sequence $(S_k)_{k=1}^{n}$ of subbasic open sets $S_k \in \mathcal{S}$.

You will have an opportunity to prove Lemma 4.2.7 Subbases of topology in the problems for this section. For the rest of this section, we will explore the properties of certain nice subbases of topologies known as **bases**, which can be said to capture the point-set topology of a given topological space.

4.2.2 Base of Topology

Text before the first subdivision.

Text after the last subdivision.

4.2.3 Order Topologies

Text before the first subdivision.

Text after the last subdivision.

Text after the last subsection.

4.2.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

1. **Comparison of topologies on $\mathbb{R}$.** Compare the following topologies on the real numbers $\mathbb{R}$: the indiscrete topology $\perp_{\mathbb{R}}$; the discrete topology $\top_{\mathbb{R}}$; the Euclidean topology τ_d; the cofinite topology τ_1; the cocountable

topology τ_2; the particular point topology τ_3 specified with particular point $0 \in \mathbb{R}$; and the excluded point topology τ_4 specified with excluded point $0 \in \mathbb{R}$.

2. **Unions of topologies.** Give an example of two topologies on a set whose union is not a topology on that set.

3. **From subbase to topology.** Let $X = \{a, b, c, d\}$, and consider the collection $\mathcal{S} \subseteq \mathcal{P}(X)$ be the collection of subsets of X defined by

$$\mathcal{S} = \{\{a, b\}, \{b, c\}, \{c, d\}\}.$$

Compute explicitly the topology $\tau = \langle \mathcal{S} \rangle$ on X generated by $\mathcal{S}$.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

4.2.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Monotonicity of generated topologies.** Fix collections $\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}(X)$ of subsets of a set X. Show that if $\mathcal{A} \subseteq \mathcal{B}$, then $\langle \mathcal{A} \rangle \subseteq \langle \mathcal{B} \rangle$.

2. **Topological comparison and interior/closure.** Fix two topologies $\tau_1, \tau_2 \in \mathcal{T}_X$ on a set X, and suppose that $\tau_1 \subseteq \tau_2$.

   (a) For a subset U, compare its interior $\mathrm{Int}_{(X,\tau_1)}(U)$ in the topological space (X, τ_1) with its interior $\mathrm{Int}_{(X,\tau_2)}(U)$ in the topological space (X, τ_2).

   (b) For a subset U, compare its closure $\mathrm{Cl}_{(X,\tau_1)}(U)$ in the topological space (X, τ_1) with its closure $\mathrm{Cl}_{(X,\tau_2)}(U)$ in the topological space (X, τ_2).

3. **Subbase of the Euclidean topology.** Consider the collections $\mathcal{U}$ and $\mathcal{D}$ of upwards- and downwards pointing rays in the real numbers $\mathbb{R}$, respectively; that is,

$$\mathcal{U} = \{(a, \infty) \in \mathcal{P}(\mathbb{R}) \mid a \in \mathbb{R}\}$$
$$\mathcal{D} = \{(-\infty, b) \in \mathcal{P}(\mathbb{R}) \mid b \in \mathbb{R}\}.$$

Show that $\mathcal{U} \cup \mathcal{D}$ is a subbase of the Euclidean topology τ_d on $\mathbb{R}$.

Hint. First argue that every open set in the real line $(\mathbb{R}, \tau_d)$ is a union of bounded open intervals. What do finite intersections of subbasic open sets (elements of $\mathcal{U} \cap \mathcal{D}$) look like?

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

4.3 Convergence and Separation

We are now ready to reexamine limiting behavior such as convergence and divergence in the more general setting of topological spaces. In the previous chapter, we explored the convergence of objects such as sequences and maps in the context of metric spaces. In a metric space, we had the benefit of a metric granting us a quantitative measure of closeness upon which our formulation of convergence relied. In a general topological space, however, we don't explicitly have access to the same kind of data to measure how close points are to each other. Instead, we must rely on other qualitative measures of locality to formalize limiting behavior.

We will see that, in contrast to the relatively amenable case of convergence in a metric space, topological convergence can be far more outlandish. For example, some convergent sequences and maps in some topological spaces converge to more than point, and many of the reasonable results from the previous chapter don't hold in certain classes of topological spaces. In particular, many point-set topological notions such as continuity and compactness cannot in general be described in sequential terms as in the context of metric spaces. With this in mind, we will briefly introduce sequential convergence in topological spaces, but we will almost immediately generalize this to discuss the limiting behavior of a more general class of indexed families of points. Then we will discuss the separation axioms, which will allow us to consider certain classes of topological spaces in which limiting behavior is more well-behaved.

4.3.1 Sequential Convergence and Beyond

In the previous chapter, we defined convergence in a metric space in a way which relied explicitly on the metric in question. In topological spaces, however, we must rely only on what data is available from the topology. Instead of defining sequential convergence and divergence in terms of some distance function, we will define it in terms of the neighborhoods of points. A sequence of points in a topological space is said to converge to a point if the values of the sequence are eventually contained in any neighborhood of the putative limit.

Definition 4.3.1 Sequential convergence/divergence in a topological space. A sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a topological space (X, τ) **converges** to a point $x \in X$ if for every neighborhood N of x in X, there is a natural number n_N so that $x_n \in N$ for all indices $n \geq n_N$.

A sequence of points in a topological space is called **convergent** if it converges to some point and **divergent** otherwise if it does not converge to any point. ◆

We first observe that the above notion of sequential convergence coincides precisely with our previous notion of sequential convergence in metric spaces whenever a topological space is metrizable.

Lemma 4.3.2 Sequential convergence in metrizable spaces. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of points in a metric space (X, ρ). For any point $x \in X$, the sequence $(x_n)_{n=0}^{\infty}$ converges to x in the metric space (X, ρ) if and only if $(x_n)_{n=0}^{\infty}$ converges to x in the metrizable topological space (X, τ_ρ).*

Proof. First suppose that $(x_n)_{n=0}^{\infty}$ converges to x in the metric space (X, ρ), and let $N \subseteq X$ be a neighorhood of x in X. Then $\mathrm{B}_\epsilon(x) \subseteq N$ for some positive real number $\epsilon > 0$. There is some natural number $n_N \in \mathbb{N}$ so that

$$\rho(x_n, x) < \epsilon$$

for all indices $n \geq n_N$. In particular, we observe that $x_n \in \mathrm{B}_\epsilon(x) \subseteq N$ for all indices $n \geq n_N$. Thus $(x_n)_{n=0}^\infty$ converges to x in the metric topological space (X, τ_ρ).

Conversely, now suppose that $(x_n)_{n=0}^\infty$ converges to x in the metric topological space (X, τ_ρ), and let $\epsilon > 0$ be a positive real number. Since $\mathrm{B}_\epsilon(x)$ is a neighborhood of x in X, there is a natural number $n_\epsilon \in \mathbb{N}$ so that $x_n \in \mathrm{B}_\epsilon(x)$ for all indices $n \geq n_\epsilon$. In particular,

$$\rho(x_n, x) < \epsilon$$

for all indices $n \geq n_\epsilon$. Thus $(x_n)_{n=0}^\infty$ converges to x in the metric space (X, ρ). ■

One consequence of Lemma 4.3.2 Sequential convergence in metrizable spaces is that we have already explored plenty of examples of convergent and divergent sequences in metrizable topological spaces, and we are familiar with the properties of sequential convergence in metrizable topological spaces. For example, Lemma 4.3.2 Sequential convergence in metrizable spaces and Proposition 3.2.3 Uniqueness of sequential limits in a metric space implies that limits of convergent sequences in metrizable topological spaces are unique.

Corollary 4.3.3 Uniqueness of sequential limits in metrizable spaces. *Convergent sequences of points in metrizable topological spaces converge to unique points.*

One might hope that Corollary 4.3.3 Uniqueness of sequential limits in metrizable spaces could be extended to apply to sequential convergence in all topological spaces, metrizable and non-metrizable. However, our familiar intuition regarding such convergence can break down in non-metrizable spaces.

Example 4.3.4 Sequential convergence/divergence in non-metrizable spaces. The following are examples of convergent and divergent sequences in non-metrizable topological spaces:

- Let $(x_n)_{n=0}^\infty$ be a sequence of points $x_n \in X$ in an indiscrete topological space $(X, \perp_X)$. For any point $x \in X$, since the only neighborhood of x in X is X itself, the sequence $(x_n)_{n=0}^\infty$ converges to x.

- Recall that the Sierpiński space (X, τ) is the topological space with ground set $X = \{0, 1\}$ and topology $\tau = \{\emptyset, \{1\}, \{0, 1\}\}$. Let $(x_n)_{n=0}^\infty$ be the constant sequence of points $x_n \in X$ defined by the formula $x_n = 1$.

  The two neighborhoods of 1 in X are $\{1\}$ and $\{0, 1\}$, and these neighborhoods both contain every entry of the sequence $(x_n)_{n=0}^\infty$. In particular, the entries of $(x_n)_{n=0}^\infty$ eventually lie in any neighborhood of 1, so that $(x_n)_{n=0}^\infty$ converges to 1.

  However, the only neighborhood of 0 in X is $\{0, 1\}$, and this neighborhood contains every entry of the sequence $(x_n)_{n=0}^\infty$. In particular, the entries of $(x_n)_{n=0}^\infty$ eventually lie in any neighborhood of 0. So $(x_n)_{n=0}^\infty$ also converges to 0.

▲

The above example shows that in our generalization of metric spaces to topological spaces, we have lost at least one nice property of limiting behavior; sequences of points can converge to more than one limit. This by itself is not cause to abandon our study of the limiting behavior of sequences; we will see that many of the topological spaces encountered in practice are **Hausdorff**, a class of topological spaces in which convergent sequences converge to unique limits, even if these spaces are not metrizable.

The true reasons for our shift in focus from sequences to more general indexed families of points called **nets** are more subtle, and they are concerned with adapting the sequential characterizations of many of the point-set topological notions introduced in Chapter 3 Metric Spaces. For example, while we have seen characterizations of continuity and compactness in terms of the convergence of sequences, such characterizations are not extended to topological spaces in general. Indeed, there are topological spaces for which the limiting behavior of sequences alone is not sufficient data to fully describe central point-set topological properties such as continuity or compactness.

Example 4.3.5 The cocountable topology. The **cocountable topology** τ on a set X is the set

$$\tau = \{U \in \mathcal{P}(X) \mid U = \emptyset \text{ or } X \setminus U \text{ is countable}\}.$$

One can verify that τ is a topology on X for which all convergent sequences are eventually constant. We have seen in Problem 3.2.5.1 Eventually constant sequences that the discrete topology has the same property. Moreover, if X is uncountable, then τ is not the discrete topology on X.

In particular, if X is uncountable, then sequential convergence in the cocountable topology τ on X is precisely the same as sequential convergence in the discrete topology $\top_X$ on X, even though these topologies are quite different. ▲

The above example of a topological space whose point-set topological properties cannot be defined solely in terms of the convergence/divergence of sequences motivates the following broadening of our scope to include more general indexed families of points. Instead of only considering families indexed by the natural numbers $\mathbb{N}$, we allow the indexing sets to be particular types of preordered sets called **directed sets**.

Definition 4.3.6 Directed set. A preordered set $(A, \precsim)$ is called **upwards directed** (resp. **downwards directed**) if every finite set of elements in A has a common upper bound (resp. common lower bound). An family $(x_\alpha)_{\alpha \in A}$ indexed by a directed set A is called a **net**. ◆

Example 4.3.7 Directed sets. The following are examples of directed sets.

- Any totally ordered set is directed both upwards and downwards. Indeed, given two elements $x, y \in X$ in a totally ordered set $(X, \leq)$, either $x \leq y$, in which case $x \leq x, y \leq y$, or $y \leq x$, in which case $y \leq x, y \leq x$.

- The power set $\mathcal{P}(X)$ of a set X is directed both upwards and downwards. Indeed, given two subsets $U, V \subseteq X$,

$$U \cap V \subseteq U, V \subseteq U \cup V.$$

- Fix a point $c \in X$ in a metric space (X, ρ), and let $\precsim_c$ denote the binary relation on $X \setminus \{c\}$ defined so that $x \precsim_c y$ if and only if $\rho(x, c) \leq \rho(y, c)$. Then the preordered set $(X \setminus \{c\}, \precsim_c)$ is directed both upwards and downwards.

- Fix a point $x \in X$ in a topological space (X, τ). The set $\mathcal{N}(x)$ of neighborhoods of x in X is partial ordered by containment $\subseteq$, and $\mathcal{N}(x)$ is directed both upwards and downwards. Indeed, the intersection and union of any finite collection of neighborhoods of x in X is itself a neighborhood of x in X.

- Let $a, b \in \mathbb{R}$ be real numbers, and suppose that $a \leq b$. A **finite partition** of the closed interval $[a, b]$ is a finite sequence $(x_j)_{j=0}^m$ so that $x_0 = a$ and $x_m = b$. Given two such partitions P and Q, we write $P \preceq Q$ if P is a subsequence of Q. Considered with this binary relation, the set of finite partitions of $[a, b]$ is directed both upwards and downwards. This directed set will be an important part of the definition of the Riemann integral in Section 6.1 Riemann Sums and Integrals.

▲

Definition 4.3.8 Net convergence. Let $(x_\alpha)_{\alpha \in A}$ be a net of points $x_\alpha \in X$ in a topological space (X, τ) indexed by a directed set $(A, \preceq)$.

- *Upwards convergence.*

  Suppose that the indexing set A is upwards directed. A net $(x_\alpha)_{\alpha \in A}$ of points $x_\alpha \in X$ **converges upwards** to a point $x \in X$ if for all neighborhoods $N \in \mathcal{N}(x)$ of x in X, there is an index $\alpha_N \in A$ so that $x_\alpha \in N$ for all indices $\alpha \succeq \alpha_N$.

  The net $(x_\alpha)_{\alpha \in A}$ is called **upwards convergent** if it converges upward to a point in X, and $(x_\alpha)_{\alpha \in A}$ is called **upwards divergent** if it does not converge upwards to any point in X.

  If $(x_\alpha)_{\alpha \in A}$ converges upwards to a unique point $x \in X$, then this point is called the **upwards limit** of $(x_\alpha)_{\alpha \in A}$ and is denoted

  $$\lim_{\alpha \uparrow} x_\alpha = x.$$

- *Downwards convergence.*

  Suppose that the indexing set A is downwards directed. A net $(x_\alpha)_{\alpha \in A}$ of points $x_\alpha \in X$ **converges downwards** to a point $x \in X$ if for all neighborhoods $N \in \mathcal{N}(x)$ of x in X, there is an index $\alpha_N \in A$ so that $x_\alpha \in N$ for all indices $\alpha \preceq \alpha_N$.

  The net $(x_\alpha)_{\alpha \in A}$ is called **downwards convergent** if it converges downward to a point in X, and $(x_\alpha)_{\alpha \in A}$ is called **downwards divergent** if it does not converge downwards to any point in X.

  If $(x_\alpha)_{\alpha \in A}$ converges downwards to a unique point $x \in X$, then this point is called the **downwards limit** of $(x_\alpha)_{\alpha \in A}$ and is denoted

  $$\lim_{\alpha \downarrow} x_\alpha = x.$$

◆

Remark 4.3.9 Direction of convergence. If the indexing set A of a net $(x_\alpha)_{\alpha \in A}$ of points $x_\alpha \in X$ in a topological space (X, τ) is directed either only upwards or only downwards, then one may speak of the convergence or limit of said net without specifying the direction. Similarly, we may avoid specifying the direction of convergence explicitly if it is clear from context.

We will see that all interesting point-set topological notions can be defined in terms of the convergence of nets. For example, we now give such characterizations of limit points and isolated points.

Definition 4.3.10 Limit/isolated points. Consider a subset $U \subseteq X$ of a topological space (X, τ).

- *Limit point.*

We say that U **accumulates** at a point $x \in X$ if $N \cap U \not\subseteq \{x\}$ for all neighborhoods $N \in \mathcal{N}_X(x)$ of x in X. In this case, x is called a **limit point** (or **accumulation point**) of U in X. The set of such limit points of U in X is called the **derived set** of U and is denoted U'.

- *Isolated point.*

  We say that a point $x \in X$ is **isolated** from U if $N \cap U \subseteq \{x\}$ for some neighborhood $N \in \mathcal{N}_X(x)$ of x in X. If this condition holds for some point $x \in U$, then x is called an **isolated point** of U.

◆

Proposition 4.3.11 Net characterization of limit points. *Let $U \subseteq X$ be a subset of a topological space (X, τ). A point $x \in X$ is a limit point of U in X if and only if some net $(u_\alpha)_{\alpha \in A}$ of points $u_\alpha \in U \setminus \{x\}$ converges to x.*

Proof. First suppose that $x \in X$ is a limit point of U in X. Then for each neighborhood $N \in \mathcal{N}_X(x)$ of x in X, there is a point $u \in N \cap U \setminus \{x\}$. The Axiom of choice allows us to choose an indexed family $(u_N)_{N \in \mathcal{N}_X(x)}$ of points $u_N \in U \setminus \{x\}$ so that $u_N \in N$ for all indices $N \in \mathcal{N}_X(x)$.

We observe that for each neighborhood $N \in \mathcal{N}_X(x)$ of x in X,

$$u_M \in M \subseteq N$$

for all indices $M \subseteq N$. Thus $(u_N)_{N \in \mathcal{N}_X(x)}$ converges downwards to x.

Conversely, suppose that $x \in X$ is isolated from U. Then $N \cap U \subseteq \{x\}$ for some neighborhood $N \in \mathcal{N}_X(x)$ of x in X. Let $(u_\alpha)_{\alpha \in A}$ be a net of points $u_\alpha \in U \setminus \{x\}$. We observe that $u_\alpha \notin N$ for all indices $\alpha \in A$, and so $(u_\alpha)_{\alpha \in A}$ does not converge to x. ■

The main utility of net convergence is in generalizing the sequential characterizations of various point-set topological notions to general topological spaces. Indeed, all sequential characterizations of such properties for metric spaces have analogs in terms of the convergence of nets. However, we will see that additional hypotheses on the spaces in question will allow us to recover these sequential characterizations, even if they do not hold in the most general case. For example, first-countability is sufficient (although not necessary) to guarantee the equivalence of the general topological notions with their sequential counterparts.

4.3.2 The Separation Axioms

The non-degeneracy of a metric implies that any two distinct points in a metric space are separated by some positive distance, and hence are the centers of disjoint metric open balls of positive radius. These separation properties of a metric space have strong implications for the uniqueness of limits. For general topological spaces, there is a zoo of distinct yet related versions of these **separation axioms** which seek to compare how a given topology separates points relative to the discrete and indiscrete topologies.

The indiscrete topological spaces are the *least* separated among all topological spaces. If a topological space (X, τ) is indiscrete, then there is no point-set topological way to distinguish between the points of X. At the very least, all separation axioms must address this glaring pathology.

Definition 4.3.12 Topological distinguishability. Let (X, τ) be a topological space.

- *Topological indistinguishability.*

  Two points $x_1, x_2 \in X$ are called **topologically indistinguishable** if they have precisely the same neighborhoods

  $$\mathcal{N}(x_1) = \mathcal{N}(x_2);$$

  that is, x_1 and x_2 are topologically indistinguishable if for any subset $N \subseteq X$, N is a neighborhood of x_1 in X if and only if N is a neighborhood of x_2 in X.

- *Topological distinguishability.*

  Two points $x_1, x_2 \in X$ are called **topologically distinguishable** if

  $$\mathcal{N}(x_1) \neq \mathcal{N}(x_2);$$

  that is, x_1 and x_2 are topologically indistinguishable if there is some subset $N \subseteq X$ which is a neighborhood of one of x_1 and x_2 but not a neighborhood of the other.

A topological space (X, τ) is called T_0 or **Kolmogorov** if all pairs of distinct points in X are topologically distinguishable. ◆

Example 4.3.13 T_0 **spaces.** The following are examples and non-examples of T_0 spaces:

- Any two points $x_1, x_2 \in X$ in an indiscrete space $(X, \perp_X)$ are topologically indistinguishable. As such, any indiscrete space with at least two points is not T_0.

- The points 0 and 1 in the Sierpiński space (X, τ) with ground set $X = \{0, 1\}$ and topology $\tau = \{\emptyset, \{1\}, \{0, 1\}\}$ are topologically distinguishable, since $\{1\}$ is a neighborhood of 1 in X but not a neighborhood of 0 in X. In particular, (X, τ) is T_0.

▲

We have previously seen that sequential limits in the Sierpiński space are not unique, and so T_0 is not a strong enough separation axiom to impose unique limits. One issue of separability in the Sierpiński space is that every neighborhood of 0 contains 1. The following separation axiom aims to address this issue.

Definition 4.3.14 Separation. Two points $x_1, x_2 \in X$ in a topological space (X, τ) are called **separated** if there are neighborhoods $N_1 \subseteq X$ and $N_2 \subseteq X$ of x_1 and x_2 in X, respectively, so that $x_1 \notin N_2$ and $x_2 \notin N_1$.

A topological space (X, τ) is called T_1 or **Fréchet** if all pairs of distinct points in X are separated. ◆

Example 4.3.15 T_1 **spaces.** The following are examples and non-examples of T_1 spaces:

- The points 0 and 1 in the Sierpiński space (X, τ) with ground set $X = \{0, 1\}$ and topology $\tau = \{\emptyset, \{1\}, \{0, 1\}\}$ are not separated, since every neighborhood of 0 in X contains 1. Since two distinct points are not separated, the Sierpiński space is not T_1.

- A subset $U \subseteq X$ of a set X is called **cofinite** if its complement $X \setminus U$ is finite. The cofinite subsets of X comprise a topology τ on X called the **cofinite topology** on X.

If X is finite, then every subset of X is cofinite, and so the cofinite topology on X is discrete. However, if X is infinite, then for any distinct points $x_1, x_2 \in X$, $N_1 = X \setminus \{x_2\}$ and $N_2 = X \setminus \{x_1\}$ are neighborhoods of x_1 and x_2 in X, respectively, so that $x_1 \notin N_2$ and $x_2 \notin N_1$, and so x_1 and x_2 are separated. Since any pair of distinct points are separated, (X, τ) is T_1.

▲

Lemma 4.3.16 Topological distinguishability and separation. *Two separated points in a topological space are topologically distinguishable.*

Proof. Let $x_1, x_2 \in X$ be points in a topological space (X, τ). If x_1 and x_2 are separated, then there are neighborhoods $N_1 \in \mathcal{N}(x_1)$ and $N_2 \in \mathcal{N}(x_2)$ of x_1 and x_2 in X, respectively, so that $x_1 \notin N_2$ and $x_2 \notin N_1$. In particular, $N_1 \in \mathcal{N}(x_1) \setminus \mathcal{N}(x_2)$ is a neighborhood of x_1 but not a neighborhood of x_2, and so x_1 and x_2 are topologically distinguishable. ■

Proposition 4.3.17 T_0 and T_1 spaces. *Any T_1 space is T_0.*

Proof. Let (X, τ) be a T_1 space. Then any two distinct points $x_1, x_2 \in X$ are separated, and so Lemma 4.3.16 Topological distinguishability and separation implies that x_1 and x_2 are topologically distinguishable. Since any two distinct points are topologically distinguishable, (X, τ) is T_0. ■

Although T_1 is a much stronger separation axiom than T_0, it is still not strong enough to force the uniqueness of sequential/net limits. Indeed, we note that the sequence $(n)_{n=0}^{\infty}$ of natural numbers converges in the cofinite topology on $\mathbb{N}$ to every single natural number. One issue here is that while distinct points in a T_1 space have distinct neighborhoods, they might not have *disjoint* neighborhoods. The following and final separation axiom we will consider here aims to address this issue.

Definition 4.3.18 Separation by neighborhoods. Two points $x_1, x_2 \in X$ in a topological space (X, τ) are called **separated by neighborhoods** if there are neighborhoods $N_1 \in \mathcal{N}(x_1)$ and $N_2 \in \mathcal{N}(x_2)$ of x_1 and x_2 in X, respectively, so that $N_1 \cap N_2 = \emptyset$.

A topological space (X, τ) is called T_2 or **Hausdorff** if all pairs of distinct points in X are separated by neighborhoods. ◆

Example 4.3.19 T_2 spaces. The following are examples and non-examples of T_2 spaces:

- If X is an infinite set, then no two cofinite subsets of X are disjoint. Indeed, given two such cofinite subsets $U, V \subseteq X$,

  $$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$$

  is the union of two finite sets and is therefore finite. Since X is infinite and $U \cap V$ is cofinite, $U \cap V \neq \emptyset$ is non-empty.

  In particular, if X is an infinite set, then the above argument implies that the cofinite topology on X is not T_2, since any neighborhoods of any two distinct points $x_1, x_2 \in X$ intersect.

- Let (X, ρ) be a metric space, and consider two distinct points $x_1, x_2 \in X$. The non-degeneracy of the metric ρ implies that $\rho(x_1, x_2) > 0$. Note that for any point $x \in X$, if $\rho(x, x_1) < \frac{\rho(x_1, x_2)}{2}$, then Proposition 3.1.5 Reverse triangle inequality implies that

  $$\rho(x, x_2) \geq |\rho(x_1, x_2) - \rho(z, x_1)| \geq \rho(x_1, x_2) - \rho(z, x_1)$$

$$> \rho(x_1, x_2) - \frac{\rho(x_1, x_2)}{2} = \rho(x_1, x_2).$$

Similarly, if $\rho(x, x_2) < \frac{\rho(x_1,x_2)}{2}$, then $\rho(x, x_2) > \frac{\rho(x_1,x_2)}{2}$. In particular, let $N_i = \mathrm{B}_{\frac{\rho(x_1,x_2)}{2}}(x_i) \in \mathcal{N}(x_i)$ for each $i \in \{1,2\}$, and note that $N_1 \cap N_2 = \emptyset$. Thus every metrizable space is T_2.

▲

Lemma 4.3.20 Separation by neighborhoods and separation. *Two points in a topological space which are separated by neighborhoods are separated.*

Proof. Let $x_1, x_2 \in X$ be points in a topological space (X, τ). If x_1 and x_2 are separated by neighborhoods, then there are neighborhoods $N_1 \in \mathcal{N}(x_1)$ and $N_2 \in \mathcal{N}(x_2)$ of x_1 and x_2 in X, respectively, so that $N_1 \cap N_2 = \emptyset$. In particular, $x_1 \notin N_2$ and $x_2 \notin N_1$, and so x_1 and x_2 are separated. ■

Proposition 4.3.21 T_1 and T_2 spaces. *Any T_2 space is T_1.*

Proof. Let (X, τ) be a T_2 space. Then any two distinct points $x_1, x_2 \in X$ are separated by neighborhoods, and so Lemma 4.3.20 Separation by neighborhoods and separation implies that x_1 and x_2 are separated. Since any two distinct points are separated, (X, τ) is T_1. ■

The main attraction of the T_2 separation axiom is that it is the minimal required hypotheses on a topological space to guarantee the uniqueness of limits for convergent sequences and nets of points.

Theorem 4.3.22 Uniqueness of limits in T_2 spaces. *If a net of points in a T_2 space converges, then it converges to a unique limit.*

Proof. Let $(x_\alpha)_{\alpha \in A}$ be a net of points $x_\alpha \in X$ in a T_2 space (X, α) indexed by a directed set $(A, \precsim)$. We will prove the result in the case A is directed upwards; the downwards directed case is entirely analogous.

Suppose for a contradiction that the net $(x_\alpha)_{\alpha \in A}$ converges upwards to two distinct points $x_1, x_2 \in X$. Since (X, τ) is T_2, x_1 and x_2 are separated by neighborhoods; that is, there are neighborhoods $N_1 \in \mathcal{N}(x_1)$ and $N_2 \in \mathcal{N}(x_2)$ of x_1 and x_2 in X, respectively, so that $N_1 \cap N_2 = \emptyset$. Since $(x_\alpha)_{\alpha \in A}$ converges to both x_1 and x_2, there are indices $\alpha_{N_1}, \alpha_{N_2} \in A$ so that $x_\alpha \in N_1$ for all indices $\alpha \succsim \alpha_{N_1}$ and $x_\alpha \in N_2$ for all indices $\alpha \succsim \alpha_{N_2}$.

Since A is directed upwards, there is some index $\alpha \in A$ so that $\alpha_{N_1}, \alpha_{N_2} \preceq \alpha$. We conclude that $x_\alpha \in N_1$ and $x_\alpha \in N_2$, and so $x_\alpha \in N_1 \cap N_2 = \emptyset$. This contradiction implies that $(x_\alpha)_{\alpha \in A}$ converges to at most one point in X. ■

While there are ***many*** separation axioms beyond T_0, T_1, and T_2, their utility is highly context-dependent and field-specific. However, the T_2/Hausdorff property is ubiquitous throughout all of point-set topology and real analysis.

We will revisit Hausdorff spaces in the following section, wherein we will define and characterize map convergence and continuity for maps between topological spaces.

4.3.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Net convergence. Verify that the given net of points converges.

1. Let F be the collection of finite subsets of the natural numbers $\mathbb{N}$; that is,
$$F = \{S \in \mathcal{P}(N) \mid S \text{ is finite}\}.$$
F is partially ordered and directed both upwards and downwards by containment $\subseteq$. Consider the net $(x_S)_{S\in F}$ of real numbers $x_S \in \mathbb{R}$ defined by the formula
$$x_S = 2^{-\max(S)}.$$
Verify that $(x_S)_{S\in F}$ converges upwards to 0.

2. Let $(x_N)_{N\in\mathcal{N}(x)}$ be a net of points in a topological space (X, τ) indexed by the neighborhood system $\mathcal{N}(x)$. Verify that if $x_N \in N$ for all indices $N \in \mathcal{N}(x)$, then $(x_N)_{N\in\mathcal{N}(x)}$ converges downwards to x.

Separation axioms. Determine which separation axioms are satisfied by the given topological space.

3. Is an indiscrete topological space $(X, \bot_X)$ T_0? Is it T_1? Is it T_2?

4. Is discrete topological space $(X, \top_X)$ T_0? Is it T_1? Is it T_2?

5. Is Euclidean space $(\mathbb{R}^n, \tau_d)$ T_0? Is it T_1? Is it T_2?

6. Let τ be the cofinite topology on a set X. Is the topological space (X, τ) T_0? Is it T_1? Is it T_2?

7. Let τ be the cocountable topology on a set X. Is the topological space (X, τ) T_0? Is it T_1? Is it T_2?

8. Let τ be a particular point topology on a set X. Is the topological space (X, τ) T_0? Is it T_1? Is it T_2?

9. Let τ be an excluded point topology on a set X. Is the topological space (X, τ) T_0? Is it T_1? Is it T_2?

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

4.3.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Net convergence and subspaces. Let $U \subseteq X$ be a subset of a topological space (X, τ), and consider a net $(u_\alpha)_{\alpha\in A}$ of points $u_\alpha \in U$ indexed by an upwards directed set $(A, \precsim)$. Prove that for a point $u \in U$, $(u_\alpha)_{\alpha\in A}$ converges upwards to u as a net in X if and only if it converges upwards to u as a net in U.

Hint. What are the neighborhoods of u in the subspace $(U, \tau|_U)$?

2. Points of closure. Let $U \subseteq X$ be a subset of a topological space (X, τ).

(a) Prove that a point $x \in X$ is a point of closure of U in X if and only if a net of points in U converges to x.

(b) Prove that $\mathrm{Cl}\,(U) = U \cup U'$.

We remark that the union $\mathrm{Cl}\,(U) = U \cup U'$ need not be a disjoint union; that is, U and its derived set U' might have significant overlap.

3. Non-Hausdorff spaces. Let (X, τ) be a topological space. For two points $x, y \in X$, let $\preceq$ be the binary relation on the Cartesian product $\mathcal{N}\,(x) \times \mathcal{N}\,(y)$ defined by the following condition: $(M_1, V_1) \preceq (M_2, V_2)$ if and only if both $M_1 \subseteq M_2$ and $V_1 \subseteq V_2$.

(a) Show that $\mathcal{N}\,(x) \times \mathcal{N}\,(y)$ is partially ordered and directed both upwards and downwards by $\preceq$.

(b) Prove that if (X, τ) is not Hausdorff, then there is a convergent net of points in X which does not have a unique limit.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

4.4 Continuity

In this section, we extend the notion of continuity from the specific setting of maps between metric spaces to the broader setting of maps between topological spaces. As in metric geometry, a map between topological spaces is **continuous** at a point if the images of nearby points are nearby to the image of that point. We will see several characterizations of topological continuity which are analogous to the metric characterizations.

4.4.1 Map Convergence and Continuity at a Point

Other than the ϵ-δ characterization, all other characterizations of continuity for maps between metric spaces provided in Proposition 3.5.1 Characterizations of metric continuity may be generalized to maps between topological spaces. While the neighborhood characterization is identical, we have already seen that the point-set topology of general topological spaces requires us to consider nets of points rather than sequences. So the sequential characterization of continuity will become the net characterization. We begin with the convergence of maps between topological spaces.

Definition 4.4.1 Topological map convergence. A map $f\colon X \to Y$ from a topological space (X, τ_X) to a topological space (Y, τ_Y) **converges** at a point $c \in X$ to a point $l \in Y$ if for all neighborhoods $N \in \mathcal{N}_Y\,(l)$ of l in Y, there is a neighborhood $M \in \mathcal{N}_X\,(c)$ of c in X so that $f\,(x) \in N$ for all points $x \in M \setminus \{c\}$.

Such a map $f\colon X \to Y$ is called **convergent** at a point $c \in X$ if it converges at c to some point in Y and **divergent** at c otherwise if it does not converge at c to any point in Y. If such a map $f\colon X \to Y$ converges at a point $c \in X$ to a *unique* point $l \in Y$, then this unique point is called a **limit**, and is denoted

$$\lim_{x \to c} f(x) = l.$$

◆

As with previous generalizations of concepts introduced in Chapter 3 Metric Spaces, the convergence of a map between metric spaces is equivalent to its convergence as a map between the underlying metrizable topological spaces.

Lemma 4.4.2 Map convergence for metrizable spaces. *Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). For any points $c \in X$ and $l \in Y$, f converges at c to l as a map between metric spaces if and only if f converges at c to l as a map from the metrizable space (X, τ_{ρ_X}) to the metrizable space (Y, τ_{ρ_Y}).*

Proof. First suppose that f converges at c to l as a map between metric spaces, and let $N \in \mathcal{N}_Y(l)$ be a neighborhood of l in Y. Then $\mathrm{B}_\epsilon(l) \subseteq N$ for some positive real number $\epsilon > 0$. Since f converges at c to l, there is some positive real number $\delta > 0$ so that

$$\rho_Y(f(x), l) < \epsilon$$

for all points $x \in X$ so that $0 < \rho_X(x, c) < \delta$. Let $M = \mathrm{B}_\delta(c)$, and note that $M \in \mathcal{N}_X(c)$ is a neighborhood of c in X. Moreover, for all points $x \in X$, if

$$x \in M \setminus \{c\} = \mathrm{B}_\delta(c) \setminus \{c\} = \{x \in X \mid 0 < \rho_X(x, c) < \delta\},$$

then $\rho_Y(f(x), l) < \epsilon$, so that $x \in \mathrm{B}_\epsilon(l) \subseteq N$. We conclude that f converges at c to l as a map between metrizable spaces.

Conversely, now suppose that f converges at c to l as a map between metrizable spaces, and let $\epsilon > 0$ be a positive real number. Then $N = \mathrm{B}_\epsilon(l) \in \mathcal{N}_Y(l)$ is a neighborhood of l in Y, and so there is some neighborhood $M \in \mathcal{N}_X(c)$ so that $f(x) \in N$ for all points $x \in M \setminus \{c\}$.

Since M is a neighborhood of c, $\mathrm{B}_\delta(c)$ for some positive real number $\delta > 0$. In particular, for all points $x \in X$, if $0 < \rho_X(x, c) < \delta$, then

$$x \in \{x \in X \mid 0 < \rho_X(x, c) < \delta\} = \mathrm{B}_\delta(c) \setminus \{c\} = M \setminus \{c\},$$

and so $f(x) \in N = \mathrm{B}_\epsilon(l)$; that is, $\rho_Y(f(x), l) < \epsilon$. We conclude that f converges at c to l as a map between metric spaces. ■

Since the convergence of maps between metrizable topological spaces is precisely the convergence of maps between metric spaces, we already have extensive experience working with the convergence of such maps. For example, the convergence of maps between metrizable spaces may be characterized in terms of sequential convergence. However, only one of the implications relevant for such a characterization holds for topological spaces in general.

Proposition 4.4.3 Sequential "characterization" of map convergence. *Let $f\colon X \to Y$ be a map from a topological space (X, τ_X) to a topological space (Y, τ_Y). If f converges at a point $c \in X$ to a point $l \in Y$, then for all sequences $(x_n)_{n=0}^\infty$ of points $x_n \in X \setminus \{c\}$ converging to c, the sequence $(f(x_n))_{n=0}^\infty$ converges to l.*

Proof. Let $(x_n)_{n=0}^\infty$ be a sequence of points $x_n \in X \setminus \{c\}$ converging to c, and consider a neighborhood $N \in \mathcal{N}_Y(l)$ of l in Y. Since f converges at c to l, $f(M \setminus \{c\}) \subseteq N$ for some neighborhood $M \in \mathcal{N}_X(c)$ of c in X. Moreover, since $(x_n)_{n=0}^\infty$ converges to c, there is some natural number $n_M \in \mathbb{N}$ so that $x_n \in M$ for all indices $n \geq n_M$.

We observe that $x_n \in M \setminus \{c\}$ and so

$$f(x_n) \in f(M \setminus \{c\}) \subseteq N$$

for all indices $n \geq n_M$. Thus the sequence $(f_n)_{n=0}^{\infty}$ converges to l. ■

One might hope to find a converse to Proposition 4.4.3 Sequential "characterization" of map convergence as exists for metric spaces, but unfortunately no such result holds for topological spaces in general. However, we can obtain a full characterization of map convergence by considering the convergence of arbitrary nets instead of only sequences.

Theorem 4.4.4 Net characterization of map convergence. *Let $f\colon X \to Y$ be a map from a topological space (X, τ_X) to a topological space (Y, τ_Y), and fix points $c \in X$ and $l \in Y$. Then the following are equivalent:*

1. *The map f converges to c at l.*

2. *For every net $(x_\alpha)_{\alpha \in A}$ of points $x_\alpha \in X \setminus \{c\}$ indexed by an upwards directed set $(A, \precsim)$, if $(x_\alpha)_{\alpha \in A}$ converges upwards to c, then $(f(x_\alpha))_{\alpha \in A}$ converges upwards to l.*

3. *For every net $(x_\beta)_{\beta \in B}$ of points $x_\beta \in X \setminus \{c\}$ indexed by a downwards directed set $(B, \precsim)$, if $(x_\beta)_{\beta \in B}$ converges downwards to c, then $(f(x_\beta))_{\beta \in B}$ converges downwards to l.*

Proof. First suppose that f converges at c to l, and let $(x_\alpha)_{\alpha \in A}$ be a net of points $x_\alpha \in X \setminus \{c\}$ indexed by an upwards directed set $(A, \precsim)$ which converges upwards to c. Let $N \in \mathcal{N}_Y(l)$ be a neighborhood of l in Y. Since f converges at c to l, $f(M \setminus \{c\}) \subseteq N$ for some neighborhood $M \in \mathcal{N}_X(c)$ of c in X.

Moreover, since $(x_\alpha)_{\alpha \in A}$ converges upwards to c, there is an index $\alpha_{N_l} \in A$ so that $x_\alpha \in M$ for all indices $\alpha \succeq \alpha_{N_l}$. In particular, $x_\alpha \in M \setminus \{c\}$, so that $f(x_\alpha) \in N$ for all indices $\alpha \succeq \alpha_{N_l}$. We conclude that $(f(x_\alpha))_{\alpha \in A}$ converges upwards to l.

Now suppose that for every net $(x_\alpha)_{\alpha \in A}$ of points $x_\alpha \in X \setminus \{c\}$ indexed by an upwards directed set $(A, \precsim)$, if $(x_\alpha)_{\alpha \in A}$ converges upwards to c, then $(f(x_\alpha))_{\alpha \in A}$ converges upwards to l. Let $(x_\beta)_{\beta \in B}$ be a net of points $x_\beta \in X \setminus \{c\}$ indexed by a downwards directed set $(B, \precsim)$ which converges downwards to c.

Let $A = B$ and $\succsim = \precsim^{\mathrm{op}}$ be the opposite of $\precsim$, as described in Problem 1.5.5.3 Opposite of an order. Since $(B, \precsim)$ is downwards directed, $(A, \succsim)$ is upwards directed. Moreover, one can check that the downwards convergence of $(x_\beta)_{\beta \in B}$ as a net indexed by $(B, \precsim)$ implies the upwards convergence of $(x_\alpha)_{\alpha \in A}$ as a net indexed by $(A, \succsim)$. In particular, $(f(x_\alpha))_{\alpha \in A}$ converges upwards to l as a net indexed by $(A, \succsim)$, which again implies that $(f(x_\beta))_{\beta \in B}$ converges downwards to l as a net indexed by $(B, \precsim)$.

Finally, we wish to prove that if for every net $(x_\beta)_{\beta \in B}$ of points $x_\beta \in X \setminus \{c\}$ indexed by a downwards directed set $(B, \precsim)$, if $(x_\beta)_{\beta \in B}$ converges downwards to c, then $(f(x_\beta))_{\beta \in B}$ converges downwards to l, then f converges to c at l. We will prove the contrapositive implication.

To that end, suppose that f does not converge at c to l. Then there is a neighborhood $N \in \mathcal{N}_Y(l)$ of l in Y so that for every neighborhood $M \in \mathcal{N}_X(c)$ of c in X, there is some point $x \in M \setminus \{c\}$ so that $f(x) \notin N$. Recall that the set $\mathcal{N}_X(c)$ of neighborhoods of c in X is partially ordered and directed downwards by containment $\subseteq$. The Axiom of choice allows us to find an indexed family $(x_M)_{M \in \mathcal{N}_X(c)}$ of points $x_N \in X$ so that for each index $M \in \mathcal{N}_X(c)$, $x_M \in M \setminus \{c\}$ but $f(x_M) \notin N$.

Since $x_M \notin N$ for all indices $M \in \mathcal{N}_X(c)$, the net $(f(x_M))_{M \in \mathcal{N}_X(c)}$ cannot converge downwards to l. It therefore suffices to show that the net $(x_M)_{M \in \mathcal{N}_X(c)}$ converges downwards to c. To that end, let $M \in \mathcal{N}_X(c)$ be a neighborhood of c in X. Then

$$x_P \in P \subseteq M$$

for all indices $P \subseteq M$; that is, the net $(x_M)_{M \in \mathcal{N}_X(c)}$ converges downwards to c. ■

We will use Theorem 4.4.4 Net characterization of map convergence to construct an analogue for the sequential characterization of continuity at a point.

Theorem 4.4.5 Characterizations of topological continuity. *Let $f\colon X \to Y$ be a map from a topological space (X, τ_X) to a topological space (Y, τ_Y). For a point $c \in X$, the following are equivalent:*

1. *Convergence characterization of continuity.*

   The map f converges at c to $f(c)$.

2. *Neighborhood characterization of continuity.*

   The pre-image $f^{-1}(N)$ of any neighborhood $N \in \mathcal{N}_Y(f(c))$ of $f(c)$ in Y is a neighborhood of c in X.

3. *Net characterization of continuity.*

   For every net $(x_\alpha)_{\alpha \in A}$ of points $x_\alpha \in X$ indexed by an upwards directed set $(A, \precsim)$, if $(x_\alpha)_{\alpha \in A}$ converges upwards to c, then $(f(x_\alpha))_{\alpha \in A}$ converges upwards to $f(c)$.

Proof. First suppose that f converges at c to $f(c)$, and let $N \in \mathcal{N}_Y(f(c))$ be a neighborhood of $f(c)$ in Y. Then $f(M \setminus \{c\}) \subseteq N$ for some neighborhood $M \in \mathcal{N}_X(c)$ of c in X. We observe that $f(c) \in N$ as well, so that $f(M) \subseteq N$. Since M is a neighborhood of c in X, $c \in U \subseteq M$ for some open set $U \in \tau_X$ in X. Therefore,

$$x \in U \subseteq M \subseteq \{x \in X \mid f(x) \in N\} = f^{-1}(N),$$

and so the pre-image $f^{-1}(N) \in \mathcal{N}_X(c)$ is a neighborhood of c in X.

Now suppose that the pre-image $f^{-1}(N)$ of every neighborhood $M \in \mathcal{N}_Y(f(c))$ of $f(c)$ in Y is a neighborhood of c in X, and let $(x_\alpha)_{\alpha \in A}$ be a net of points $x_\alpha \in X$ indexed by an upwards directed set $(A, \precsim)$ which converges upwards to c. Let $N \in \mathcal{N}_Y(f(c))$ be a neighborhood of $f(c)$ in Y.

Since $f^{-1}(N)$ is a neighborhood of $f(c)$ in X, there is an index $\alpha_N \in A$ so that $x_\alpha \in f^{-1}(N)$ for all indices $\alpha \succeq \alpha_N$. In particular, $f(x_\alpha) \in N$ for all indices $\alpha \succeq \alpha_N$, and so the net $(f(x_\alpha))_{\alpha \in A}$ converges upwards to $f(c)$.

Finally, now suppose that for every net $(x_\alpha)_{\alpha \in A}$ of points $x_\alpha \in X$ indexed by an upwards directed set $(A, \precsim)$, if $(x_\alpha)_{\alpha \in A}$ converges upwards to c, then $(f(x_\alpha))_{\alpha \in A}$ converges upwards to $f(c)$. Then Theorem 4.4.4 Net characterization of map convergence implies that f converges at c to $f(c)$. ■

Definition 4.4.6 Continuity. Let $f\colon X \to Y$ be a map from a topological space (X, τ_X) to a topological space (Y, τ_Y).

- *Continuity.*

  f is **continuous** at a point $c \in X$ if it satisfies any (and therefore all) of the conditions in Theorem 4.4.5 Characterizations of topological continuity. f is **continuous** if it is continuous at every point $c \in X$.

- *Discontinuity.*

  f is **discontinuous** at a point $c \in X$ if it does not satisfies any of the conditions in Theorem 4.4.5 Characterizations of topological continuity. f is **discontinuous** if it is discontinuous at some point $c \in X$.

The set of continuous maps from X to Y is denoted $\mathrm{C}(X, Y)$. In the special case $(Y, \tau_Y) = (\mathbb{R}, \tau_d)$, the set of continuous, real-valued functions on a topological space (X, τ_X) is denoted $\mathrm{C}(X) = \mathrm{C}(X, \mathbb{R})$. ◆

Example 4.4.7 Continuity at a point. Let $f\colon \mathbb{R} \to \mathbb{R}$ be the real-valued function defined piecewise by the formula

$$f(x) = \begin{cases} 0 & x = 0 \\ 1 & x \neq 0. \end{cases}$$

While f is not continuous at 0 when the domain and codomain are equipped with the Euclidean topology τ_d, we can construct a topology τ on $\mathbb{R}$ so that f is a continuous map from $(\mathbb{R}, \tau)$ to $(\mathbb{R}, \tau_d)$.

Specifically, let

$$\tau = \{U \in \mathcal{P}\,\mathbb{R} \mid 0 \in U \text{ or } U = \emptyset\}.$$

One can check that τ is a topology on $\mathbb{R}$ called the **particular point topology**. We see that if $N \in \mathcal{N}_{(\mathbb{R},\tau_d)}(0)$ is a neighborhood of 0 in $(\mathbb{R}, \tau_d)$, then

$$f^{-1}(N) = \begin{cases} \mathbb{R} & 1 \in N \\ \{0\} & 1 \notin N. \end{cases}$$

In either case, $f^{-1}(N)$ is an open neighborhood of 0 in the particular point topology, and so f is continuous at 0 as a map from $(\mathbb{R}, \tau)$ to $(\mathbb{R}, \tau_d)$. ▲

Many of the standard properties of continuous maps between metric spaces also apply to continuous maps between topological spaces. For example, continuity is preserved by composition.

Proposition 4.4.8 Composition of continuous maps. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between topological spaces (X, τ_X), (Y, τ_Y), and (Z, τ_Z). If for some point $x \in X$, f is continuous at x and g is continuous at $f(x)$, then the composition $g \circ f$ is continuous at x.*

Proof. Let $W \in \mathcal{N}_Z((g \circ f)(x))$ be a neighborhood of $(g \circ f)(x) = g(f(x))$ in Z. Since g is continuous at $f(x)$, the pre-image $g^{-1}(W) \in \mathcal{N}_Y(f(x))$ is a neighborhood of $f(x)$ in Y. Moreover, since f is continuous at x, the pre-image

$$\begin{aligned} f^{-1}\left(g^{-1}(W)\right) &= \left\{x \in X \;\middle|\; f(x) \in g^{-1}(W)\right\} = \{x \in X \mid g(f(x)) \in W\} \\ &= \{x \in X \mid (g \circ f)(x) \in W\} = (g \circ f)^{-1}(W). \end{aligned}$$

Thus the neighborhood characterization of continuity in Theorem 4.4.5 Characterizations of topological continuity implies that $g \circ f$ is continuous at x. ■

Corollary 4.4.9 Composition of continuous maps. *The composition of continuous maps between topological spaces is continuous.*

Proof. ■

As previously mentioned, we can recover the sequential characterization of map convergence (and therefore the sequential characterization of continuity at a point) by imposing additional hypotheses on the spaces in question. For example, to establish a sequential characterization of convergence of maps from a topological space X to a topological space Y, it is sufficient to require that X be first-countable.

4.4.2 Continuous Maps between Topological Spaces

While we have seen characterizations of continuity at a point, we will be more interested in maps which are continuous everywhere. For the remainder of this section, we will explore the properties of such maps, and define maximal and minimal topologies with respect to continuity. To begin, we observe that Corollary 3.5.4 Open set characterization of metric continuity holds for maps between topological spaces.

Proposition 4.4.10 Open set characterization of continuity. *A map $f\colon X \to Y$ from a topological space (X, τ_X) to a topological space (Y, τ_Y) is continuous if and only if the pre-image $f^{-1}(V)$ of every open set $V \in \tau_Y$ in Y is open in X.*

Proof. Suppose first that f is continuous, and let $V \subseteq Y$ be an open subset of Y. Let $c \in f^{-1}(V)$. Since V is open, it is a neighborhood of $f(c)$, and so the neighborhood characterization of continuity in Theorem 4.4.5 Characterizations of topological continuity implies that $f^{-1}(V)$ is a neighborhood of c. Since this is true of every point $c \in f^{-1}(V)$, $f^{-1}(V) \in \tau_X$ is open in X.

Conversely, now suppose that the pre-image of every open set in Y is an open set in X, and consider a point $c \in X$ and a neighborhood $N \subseteq Y$ of $f(c)$ in Y. Then $f(c) \in V \subseteq N$ for some open set $V \in \tau_Y$, and so $f^{-1}(V) \in \tau_X$ is open in X. We observe that

$$c \in f^{-1}(V) = \{x \in X \mid f(x) \in V\} \subseteq \{x \in X \mid f(x) \in N\} = f^{-1}(N),$$

and so $f^{-1}(N)$ is a neighborhood of c. The neighborhood characterization of continuity in Theorem 4.4.5 Characterizations of topological continuity now implies that f is continuous at c. Since this holds for all points $c \in X$, f is continuous. ■

Example 4.4.11 Continuity. The following are examples of continuous and discontinuous maps between topological spaces:

- Any map $f\colon X \to Y$ from a discrete topological space $(X, \top_X)$ to a topological space (Y, τ_Y) is continuous.
- Any map $f\colon X \to Y$ from a topological space (X, τ_X) to an indiscrete topological space $(Y, \bot_Y)$ is continuous.
- Any constant map $f\colon X \to Y$ from a topological space (X, τ_X) to a topological space (Y, τ_Y) is continuous.
- Any non-constant map $f\colon X \to Y$ from an indiscrete topological space $(X, \bot_X)$ to a T_0 topological space (Y, τ_Y) with at least two points is discontinuous.

▲

We will use the observations of Example 4.4.11 Continuity together with Theorem 4.2.6 Complete lattice of topologies as a platform for building topologies in such a way to guarantee the continuity of certain maps. First, however, we will require the following corollary:

Corollary 4.4.12 Continuity, coarsening, and refinement. *Let $f\colon X \to Y$ be a continuous map from a topological space (X, τ_X) to a topological space (Y, τ_Y). Then for any refinement $\tau'_X \supseteq \tau_X$ of τ_X and any coarsening $\tau'_Y \subseteq \tau_Y$ of τ_Y, f is continuous as a map from (X, τ'_X) to (Y, τ'_Y).*

Proof. Let $V \in \tau'_Y$ be an open set in (Y, τ'_Y). Since $\tau'_Y \subseteq \tau_Y$, $V \in \tau_Y$, and so the continuity of f as a map from (X, τ_X) to (Y, τ_Y) implies that $f^{-1}(V) \in \tau_X$ is an open set in (X, τ_X). Since $\tau_X \subseteq \tau'_X$, $V \in \tau'_X$.

In summary, we have shown that for any open set $V \in \tau'_Y$, the pre-image $f^{-1}(V) \in \tau'_X$ is open. Thus f is a continuous map from (X, τ'_X) to (Y, τ'_Y). ■

To summarize Corollary 4.4.12 Continuity, coarsening, and refinement, we note that it is easy for a map to be continuous if the topology on its domain is very fine and the topology on its codomain is very coarse. For the same reason, it is very hard for a map to be continuous if the topology on its domain is very coarse and the topology on its codomain is very fine. We exploit this tension to define topologies which are just coarse enough or just fine enough to make certain maps continuous.

Definition 4.4.13 Final topology. The **final topology** on a set Y induced by an indexed family $(f_j)_{j \in \mathcal{J}}$ of maps $f_j\colon X_j \to Y$ from topological spaces (X_j, τ_{X_j}) to a set Y is the finest topology on Y so that each map $f_j\colon X_j \to Y$ is continuous. Concretely, the final topology τ induced on Y by $(f_j)_{j \in \mathcal{J}}$ is defined by

$$\tau = \bigcap_{j \in \mathcal{J}} \left\{ V \in \mathcal{P}(Y) \,\middle|\, f_j^{-1}(V) \in \tau_{X_j} \right\}.$$

◆

Example 4.4.14 Quotient topology. Let $q\colon X \to X/{\equiv}$ be the quotient map associated to an equivalence relation $\equiv$ on a topological space (X, τ). Concretely, q sends a point $x \in X$ to its equivalence class $[x]_{\equiv}$. The **quotient topology** on $X/{\equiv}$ is the final topology induced by q; that is, the quotient topology is the finest topology on $X/{\equiv}$ so that q is continuous. When $X/{\equiv}$ is equipped with the quotient topology, the resulting topological space is called a **quotient space**. ▲

The final topology on a set can also be characterized in terms of the continuity of outgoing maps.

Lemma 4.4.15 Continuous maps from final spaces. *Let τ_Y be the final topology on a set Y induced by an indexed family $(f_j)_{j \in \mathcal{J}}$ of maps $f_j\colon X_j \to Y$ from topological spaces (X_j, τ_{X_j}) to Y. Then a map $g\colon Y \to Z$ from (Y, τ_Y) to a topological space (X, τ_X) is continuous if and only if the composition $g \circ f_j\colon X_j \to Z$ is continuous for all indices $j \in \mathcal{J}$.*

Proof. If g is continuous, then $g \circ f_j$ is continuous for each index $j \in \mathcal{J}$ by Corollary 4.4.9 Composition of continuous maps. Thus it suffices to prove the converse implication. To that end, suppose that $g \circ f_j$ is continuous for each index $j \in \mathcal{J}$, and let $W \in \tau_Z$ be an open set in Z. We wish to show that $g^{-1}(W) \in \tau_Y$ is open in Y.

To that end, we note that

$$f_j^{-1}\left(g^{-1}(W)\right) = (g \circ f_j)^{-1}(W) \in \tau_X$$

is open in X for each index $j \in \mathcal{J}$, and so

$$g^{-1}(W) \in \bigcap_{j \in \mathcal{J}} \left\{ V \in \mathcal{P}(Y) \,\middle|\, f_j^{-1}(V) \in \tau_X \right\} = \tau_Y$$

is open in Y. We conclude by the Proposition 4.4.10 Open set characterization of continuity that g is continuous. ■

Dual to the notion of a final topology is that of an initial topology. While final topologies are as fine as possible while requiring the continuity of some incoming maps, initial topologies are as coarse as possible while requiring the continuity of some outgoing maps.

Definition 4.4.16 Initial topology. The **initial topology** on a set X induced by an indexed family $(f_j)_{j\in\mathcal{J}}$ of maps $f_j\colon X\to Y_j$ from a set X to topological spaces $\left(Y_j,\tau_{Y_j}\right)$ is the coarsest topology on X so that each map $f_j\colon X\to Y_j$ is continuous. Concretely, the initial topology τ induced on X by $(f_j)_{j\in\mathcal{J}}$ is

$$\tau=\left\langle\bigcup_{j\in\mathcal{J}}\left\{f_j^{-1}(V)\in\mathcal{P}(X)\,\middle|\,V\in\tau_{Y_j}\right\}\right\rangle.$$

◆

Example 4.4.17 Initial topologies. The following are examples of initial topologies:

- The subspace topology $\tau|_U$ on a subset $U\subseteq X$ of a topological space (X,τ) is the initial topology induced by the inclusion map $\iota\colon U\to X$.
- The **product topology** on a Cartesian product $X=\prod_{j\in\mathcal{J}}Y_j$ of topological spaces $\left(Y_j,\tau_{X_j}\right)$ is the initial topology induced by the indexed family $(\pi_j)_{j\in\mathcal{J}}$ of projection maps $\pi_j\colon X\to Y_j$.

▲

The initial topology on a set can also be characterized in terms of the continuity of incoming maps. The proof of this characterization is slightly harder than the analogous result for final topologies, since in this case we have a subbase for the initial topology rather than an explicit description.

Lemma 4.4.18 Continuous maps to initial spaces. *Let τ_Y be the initial topology on a set Y induced by an indexed family $(g_j)_{j\in\mathcal{J}}$ of maps $g_j\colon Y\to Z_j$ from Y to topological spaces $\left(Z_j,\tau_{Z_j}\right)$. Then a map $f\colon X\to Y$ from a topological space (X,τ_X) to (Y,τ_Y) is continuous if and only if the composition $g_j\circ f\colon X\to Z_j$ is continuous for all indices $j\in\mathcal{J}$.*

Proof. If f is continuous, then $g_j\circ f$ is continuous for each index $j\in\mathcal{J}$ by Corollary 4.4.9 Composition of continuous maps. Thus it suffices to prove the converse implication. To that end, suppose that $g_j\circ f$ is continuous for each index $j\in\mathcal{J}$, and let $V\in\tau_Y$ be an open set in Y. Then `[provisional cross-reference: problem-explicit-descriptions-of-the-generated-topology]` implies that

$$V=\bigcup_{k\in\mathcal{K}}\bigcap_{l=1}^{p_k}g_{j_{k,l}}^{-1}(W_{k,l})$$

for some indices $j_{k,l}\in\mathcal{J}$ and open sets $W_{k,l}\in\tau_{Z_{j_{k,l}}}$. Since topologies are stable under finite intersections and arbitrary unions,

$$\begin{aligned}f^{-1}(V)&=f^{-1}\left(\bigcup_{k\in\mathcal{K}}\bigcap_{l=1}^{p_k}g_{j_{k,l}}^{-1}(W_{k,l})\right)=\bigcup_{k\in\mathcal{K}}\bigcap_{l=1}^{p_k}f^{-1}\left(g_{j_{k,l}}^{-1}(W_{k,l})\right)\\&=\bigcup_{k\in\mathcal{K}}\bigcap_{l=1}^{p_k}\left(g_{j_{k,l}}\circ f\right)^{-1}(W_{k,l})\in\tau_X\end{aligned}$$

is open in X, and so f is continuous. ■

Typically, the above examples of final and initial topologies are implicitly assumed, so that unless otherwise specified, quotient maps, subset inclusions, and projection maps of Cartesian products are continuous.

In the next section, we will explore how compactness generalizes to topological spaces. In particular, we will use compactness to construct a subbase for a natural topology on the set $\mathrm{C}(X, Y)$ of continuous maps from a topological space X to a topological space Y.

4.4.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

4.4.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

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4.5 Compactness and Connectedness [SKIP]

Recall that a subset $U \subseteq X$ of a metric space (X, ρ) is (sequentially) compact if every sequence of points in U has a subsequence which converges to a limit in U. In this section, we investigate two analogues of this property for general topological spaces.

4.5.1 Open Covers

Text before the first subdivision.

Text after the last subdivision.

4.5.2 Compactness and Sequential Compactness

Text before the first subdivision.

Text after the last subdivision.

4.5.3 Connected and Disconnected Spaces

Text before the first subdivision.

Text after the last subdivision.

Text after the last subsection.

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Text after the last section.

Part III

Calculus

Chapter 5

Differentiation and Smoothness

In this chapter, we begin our study of calculus by exploring differentiation and differentiable functions. We will apply the point-set topological techniques developed in Part II Point-Set Topology to the well-known field of differential calculus, using limits to compute and analyze instantaneous rates of change of functions.

5.1 Differentiability

The fundamental object in differential calculus is the **differentiable function** and its **derivative**. In this section, we formally differentiability as a property of real-valued functions defined on subsets of the real line $\mathbb{R}$. We will develop techniques of differentiation, compile a list of differentiable functions, and compare the regularities of various functions.

5.1.1 Introduction to Differentiation

A real-valued function defined on a subset of the real line is called **differentiable** at a point if it can be well-approximated by a linear function in a neighborhood of that point. In this case, the **derivative** of the function gives the slope of this local linear approximation.

Definition 5.1.1 Differentiability. A real-valued function $f\colon X \to \mathbb{R}$ on a subset $X \subseteq \mathbb{R}$ is called **differentiable** at a point $c \in X \cap X'$ if there is a real number $D \in \mathbb{R}$ so that for all positive real numbers $\epsilon > 0$, there is a positive real number $\delta > 0$ so that

$$\left| \frac{f(x) - f(c)}{x - c} - D \right| < \epsilon$$

for all points $x \in X$ so that $0 < |x - c| < \delta$; that is, f is differentiable at c if c is a limit point of X in $\mathbb{R}$ and

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = D.$$

In this case, the above limit is called the **derivative** of f at c and is denoted $f'(c) = D$.

The **differentiability locus** $\mathrm{DL}(f)$ of f is the subset of $X \cap X'$ comprised by the points at which f is differentiable. The function f is called **differentiable** if $\mathrm{DL}(f) = X \cap X'$. The **derivative** of f is the real-valued function $f' \colon \mathrm{DL}(f) \to \mathbb{R}$. ◆

Example 5.1.2 Differentiability. The following are examples of differentiable functions:

- Let $f \colon \mathbb{R} \to \mathbb{R}$ be a constant function defined by the formula $f(x) = a$ for some real number $a \in \mathbb{R}$, and consider a point $c \in \mathbb{R}$. Since for any $\epsilon > 0$,

$$\begin{aligned}\left|\frac{f(x) - f(c)}{x - c} - 0\right| &= \left|\frac{a - a}{x - c}\right| \\ &= \left|\frac{0}{x - c}\right| = |0| = 0 < \epsilon\end{aligned}$$

  for all points $x \in \mathbb{R} \setminus \{c\}$, f is differentiable at c, with derivative $f'(c) = 0$. Since c was chosen arbitrarily, f has differentiability locus $\mathrm{DL}(f) = \mathbb{R} = \mathbb{R} \cap \mathbb{R}'$, so that f is differentiable.

- Let $g \colon \mathbb{R} \to \mathbb{R}$ be a linear function defined by the formula $g(x) = ax + b$ for some real numbers $a, b \in \mathbb{R}$, and consider a point $c \in \mathbb{R}$. Since for any positive real number $\epsilon > 0$,

$$\begin{aligned}\left|\frac{g(x) - g(c)}{x - c} - a\right| &= \left|\frac{(ax + b) - (ac + b)}{x - c} - a\right| = \left|\frac{ax + b - ac - b}{x - c} - a\right| \\ &= \left|\frac{ax - ac}{x - c} - a\right| = \left|\frac{a(x - c)}{x - c} - a\right| \\ &= |a - a| = |0| = 0 < \epsilon\end{aligned}$$

  for all points $x \in \mathbb{R} \setminus \{c\}$, f is differentiable at c, with derivative $f'(c) = 0$. Since c was chosen arbitrarily, g has differentiability locus $\mathrm{DL}(g) = \mathbb{R} = \mathbb{R} \cap \mathbb{R}'$, so that g is differentiable.

- Let $h \colon \mathbb{R} \to \mathbb{R}$ be the quadratic function defined by the formula $h(x) = x^2$, and consider a point $c \in \mathbb{R}$. For any positive real number $\epsilon > 0$, let $\delta = \epsilon > 0$. We observe that

$$\begin{aligned}\left|\frac{h(x) - h(c)}{x - c} - 2c\right| &= \left|\frac{x^2 - c^2}{x - c} - 2c\right| = \left|\frac{(x + c)(x - c)}{x - c} - 2c\right| \\ &= |x + c - 2c| = |x - c| < \delta = \epsilon\end{aligned}$$

  for all points $x \in \mathbb{R}$ so that $0 < |x - c| < \delta$. We conclude that h is differentiable at c, with derivative $h'(c) = 2c$. Since c was chosen arbitrarily, h has differentiability locus $\mathrm{DL}(h) = \mathbb{R} = \mathbb{R} \cap \mathbb{R}'$, so that h is differentiable.

▲

Extensions of the above examples can be used to compute the derivatives of a plethora of differentiable functions. Specifically, we will shortly see how to compute the derivative of a complicated function by breaking it down into smaller differentiable pieces. First, however, we observe that differentiability at a point is a stronger condition than continuity at that point.

Lemma 5.1.3 Differentiability implies continuity. *A real-valued function $f\colon X \to \mathbb{R}$ on a subset $X \subseteq \mathbb{R}$ is continuous at every point $c \in \mathrm{DL}(f)$.*

Proof. Let $f\colon X \to \mathbb{R}$ be a real-valued function on a subset $X \subseteq \mathbb{R}$, and suppose that f is differentiable at a point $c \in X$. Then there is a positive real number $\delta_f > 0$ so that

$$\left|\frac{f(x) - f(c)}{x - c} - f'(c)\right| < 1$$

for all points $x \in X$ so that $0 < |x - c| < \delta_f$. Let $\epsilon > 0$ be a positive real number, and let

$$\delta = \min\left(\left\{\delta_f, \frac{\epsilon}{1 + |f'(c)|}\right\}\right) > 0.$$

We observe that

$$\begin{aligned}
|f(x) - f(c)| = \left|\frac{f(x) - f(c)}{x - c}\right| \cdot |x - c| &= \left|\frac{f(x) - f(c)}{x - c} - f'(c) + f'(c)\right| \cdot |x - c| \\
&\leq \left(\left|\frac{f(x) - f(c)}{x - c} - f'(c)\right| + |f'(c)|\right) |x - c| \\
&< (1 + |f'(c)|)\,\delta < (1 + |f'(c)|)\left(\frac{\epsilon}{1 + |f'(c)|}\right) = \epsilon
\end{aligned}$$

for any point $x \in X$ so that $0 < |x - c| < \delta$, and so we conclude that f is continuous at c. ∎

While differentiable functions are continuous, continuous functions need not be differentiable. For example, the absolute value function $|x|$ is continuous, but $|x|$ is not differentiable at the point $x = 0$.

We now observe that differentiability is preserved under the pointwise addition, subtraction, and multiplication of functions.

Proposition 5.1.4 Arithmetic of differentiable functions.

1. *Addition of differentiable functions.*

   Let $f, g\colon X \to \mathbb{R}$ be real-valued functions on a subset $X \subseteq \mathbb{R}$. For a point $c \in X$, if f and g are differentiable at c, then $f + g$ is differentiable at c, and

   $$(f + g)'(c) = f'(c) + g'(c).$$

2. *Multiplication of differentiable functions.*

   Let $f, g\colon X \to \mathbb{R}$ be real-valued functions on a subset $X \subseteq \mathbb{R}$. For a point $c \in X$, if f and g are differentiable at c, then $f \cdot g$ is differentiable at c, and

   $$(f \cdot g)'(c) = f'(c)\,g(c) + f(c)\,g'(c).$$

Proof.

1. Let $\epsilon > 0$ be a positive real number. Since f and g are differentiable at c, there are positive real numbers $\delta_f, \delta_g > 0$ so that

   $$\left|\frac{f(x) - f(c)}{x - c} - f'(c)\right| < \frac{\epsilon}{2}$$

   for all points $x \in X$ so that $0 < |x - c| < \delta_f$, and

   $$\left|\frac{g(x) - g(c)}{x - c} - g'(c)\right| < \frac{\epsilon}{2}$$

for all points $x \in X$ so that $0 < |x - c| < \delta_g$. Let $\delta = \min(\{\delta_f, \delta_g\}) > 0$, and note that

$$\begin{aligned}\left|\frac{(f+g)(x)-(f+g)(c)}{x-c} - (f'(c)+g'(c))\right| &= \left|\frac{f(x)+g(x)-f(c)-g(c)}{x-c} - f'(c) - g'(c)\right| \\ &= \left|\frac{f(x)-f(c)}{x-c} - f'(c) + \frac{g(x)-g(c)}{x-c} - g'(c)\right| \\ &\leq \left|\frac{f(x)-f(c)}{x-c} - f'(c)\right| + \left|\frac{g(x)-g(c)}{x-c} - g'(c)\right| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon\end{aligned}$$

for all points $x \in X$ so that $0 < |x - c| < \delta$. Thus $f + g$ is differentiable at c, with derivative

$$(f+g)'(c) = f'(c) + g'(c).$$

2. Because utilizing the ϵ-δ characterization of convergence is slightly more complicated than it's worth, we compute the derivative directly. Since f and g are differentiable at c,

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c} = f'(c)$$

and

$$\lim_{x\to c}\frac{g(x)-g(c)}{x-c} = g'(c).$$

Moreover, Lemma 5.1.3 Differentiability implies continuity implies that g is continuous at c, so that

$$\lim_{x\to c} g(x) = g(c).$$

We conclude that

$$\begin{aligned}\lim_{x\to c}\frac{(f\cdot g)(x)-(f\cdot g)(c)}{x-c} &= \lim_{x\to c}\frac{f(x)g(x)-f(c)g(x)+f(c)g(x)-f(c)g(c)}{x-c} \\ &= \lim_{x\to c}\left(\left(\frac{f(x)-f(c)}{x-c}\right)g(x) + f(c)\left(\frac{g(x)-g(c)}{x-c}\right)\right) \\ &= f'(c)g(c) + f(c)g'(c).\end{aligned}$$

Thus $f \cdot g$ is differentiable at c, with derivative $(f\cdot g)'(c) = f'(c)g(c) + f(c)g'(c)$.

■

In fact, modest hypotheses on the denominator of a quotient of differentiable functions implies that such a quotient is also differentiable. You will have an opportunity to prove this **quotient rule** in the problems for this section. First, however, we observe the following immediate corollary of Proposition 5.1.4 Arithmetic of differentiable functions:

Corollary 5.1.5 Differentiability of polynomials. *Polynomial functions are differentiable at every real number.*

We can combine functions in more ways than just pointwise arithmetic. One of the fundamental such operations is composition. Here we see that the composition of two differentiable functions is also differentiable.

Theorem 5.1.6 Chain rule. *Let $f\colon X \to \mathbb{R}$ and $g\colon Y \to \mathbb{R}$ be real-valued functions on subsets $X, Y \subseteq \mathbb{R}$, and suppose that $f(X) \subseteq Y$. If f is differentiable at a point $c \in X$ and g is differentiable at $f(c)$, then $g \circ f\colon X \to \mathbb{R}$ is differentiable at c, with derivative*

$$(g \circ f)'(c) = g'(f(c))\, f'(c).$$

Proof. We first argue that

$$\lim_{x \to c} \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} = g'(f(c)).$$

Indeed, let $\epsilon > 0$ be a positive real number. Since g is differentiable at $f(c)$, there is a positive real number $\eta > 0$ so that

$$\left| \frac{g(y) - g(f(c))}{y - f(c)} - g'(f(c)) \right| < \epsilon$$

for all points $y \in Y$ so that $0 < |y - f(c)| < \eta$. Since f is differentiable at c, it is continuous at c by Lemma 5.1.3 Differentiability implies continuity, and so there is a positive real number $\delta > 0$ so that $|f(x) - f(c)| < \eta$ for all points $x \in X$ so that $|x - c| < \delta$. In particular, if $0 < |x - c| < \delta$, then

$$\left| \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} - g'(f(c)) \right| < \epsilon.$$

We conclude that

$$\lim_{x \to c} \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} = g'(f(c)).$$

We now observe that

$$\begin{aligned}
\lim_{x \to c} \frac{(g \circ f)(x) - (g \circ f)(c)}{x - c} &= \lim_{x \to c} \frac{g(f(x)) - g(f(c))}{x - c} \\
&= \lim_{x \to c} \left(\frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \right) \left(\frac{f(x) - f(c)}{x - c} \right) \\
&= \left(\lim_{x \to c} \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \right) \left(\lim_{x \to c} \frac{f(x) - f(c)}{x - c} \right) \\
&= g'(f(c))\, f'(c),
\end{aligned}$$

and so $g \circ f$ is differentiable at c, with derivative $(g \circ f)'(c) = g'(f(c))\, f'(c)$. ∎

The Chain rule is one of the most important foundational results in differential calculus, and has important generalizations to the differentiability of maps between higher-dimensional Euclidean spaces.

5.1.2 Regularity and Smoothness

For real-valued functions which are differentiable at many points in their domain, we can view the derivative not just as a number, but rather as a real-valued function which itself may be subject to analysis. A natural question in this setting is whether we can differentiate this derivative itself. This is not always possible; the derivative of a differentiable function need not itself be differentiable. In fact, such a derivative need not even be continuous. Differentiable functions whose derivative is itself differentiable are called **twice**

differentiable, and the derivative of the derivative of such a function is called its **second derivative**. Continuing in such a manner, we may speak of n-**times differentiable** functions and their n**th derivatives**.

Definition 5.1.7 Higher order differentiability. Fix a real-valued function $f\colon X \to \mathbb{R}$ on a subset $X \subseteq \mathbb{R}$. The higher order differentiability properties of f are defined recursively as follows:

- f is 0-times differentiable, its 0**-times differentiability locus** is $\mathrm{DL}^0(f) = X$, and its 0**th derivative** is $f^{(0)} = f\colon \mathrm{DL}^0(f) \to \mathbb{R}$.

- Suppose that f has k-times differentiability locus $\mathrm{DL}^k(f)$ and kth derivative $f^{(k)}\colon \mathrm{DL}^k(f) \to \mathbb{R}$. The $(k+1)$-times differentiability locus $\mathrm{DL}^{k+1}(f)$ is the differentiability locus

$$\begin{aligned}\mathrm{DL}^{k+1}(f) = \mathrm{DL}\left(f^{(k)}\right) &= \left\{x \in \mathrm{DL}^k(f) \,\middle|\, f^{(k)} \text{ is differentiable at } x\right\}\\ &\subseteq \mathrm{DL}^k(f) \cap \left(\mathrm{DL}^k(f)\right)'.\end{aligned}$$

  The $(k+1)$**st derivative** $f^{(k+1)}$ of f is the derivative

$$f^{(k+1)} = \left(f^{(k)}\right)'\colon \mathrm{DL}^{k+1}(f) \to \mathbb{R}.$$

  Then f is $(k+1)$**-times differentiable** if $f^{(k)}$ is differentiable at every point of $\mathrm{DL}^k(f) \cap \left(\mathrm{DL}^k(f)\right)'$; that is, f is $(k+1)$-times differentiable if

$$\mathrm{DL}^{k+1}(f) = \mathrm{DL}^k(f) \cap \left(\mathrm{DL}^k(f)\right)'.$$

◆

Example 5.1.8 Higher order differentiability. The following are examples of higher order derivatives of functions:

- Since polynomial functions are differentiable, and their derivatives are polynomials (which are also differentiable), every polynomial function is k-times differentiable for all $k \in \mathbb{N}$.

- Fix a natural number $k \in \mathbb{N}$, and consider the power function $p\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $p(x) = x^{k+\frac{1}{3}}$. Once we have explored the calculus of power series in Section 7.3 Power Series, we will see that p is k-times differentiable but not $(k+1)$-times differentiable.

▲

As previously mentioned, the derivative of a differentiable function need not be continuous. Requiring continuity in the derivatives of a function yield the concept of **regularity**. Intuitively, the regularity of a function determines how many of its derivatives are continuous.

Definition 5.1.9 Regularity. A real-valued function $f\colon X \to \mathbb{R}$ on a subset $X \subseteq \mathbb{R}$ is called k-times continuously differentiable if it is k-times differentiable and its kth derivative $f^{(k)}\colon \mathrm{DL}^k(f) \to \mathbb{R}$ is continuous. The set of k-times continuously differentiable functions on X is denoted $\mathrm{C}^k(X)$. ◆

Example 5.1.10 Regularity. The following are examples of functions of different regularity:

- Since polynomial functions are continuous and differentiable, and their derivatives are polynomials (which are also continuous and differentiable), every polynomial function is k-times continuously differentiable for all $k \in \mathbb{N}$.

- Fix a natural number $k \in \mathbb{N}$, and consider the power function $p \colon \mathbb{R} \to \mathbb{R}$ defined by the formula $p(x) = x^{k+\frac{1}{3}}$. Once we have explored the calculus of power series in Section 7.3 Power Series, we will see that p is k-times continuously differentiable but not $(k+1)$-times differentiable; that is $p \in \mathrm{C}^k(\mathbb{R}) \setminus \mathrm{C}^{k+1}(\mathbb{R})$.

- Let $f \colon \mathbb{R} \to \mathbb{R}$ be the real-valued function defined piecewise by the formula
$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0. \end{cases}$$
Then f is differentiable but not continuously differentiable.

▲

Lemma 5.1.11 Monotonicity of regularity classes. *For any subset $X \subseteq \mathbb{R}$,*
$$\mathrm{C}^0(X) \supseteq \mathrm{C}^1(X) \supseteq \mathrm{C}^2(X) \supseteq \mathrm{C}^3(X) \supseteq \cdots.$$
Moreover, the intersection $\bigcap_{k=0}^{\infty} \mathrm{C}^k(X)$ is non-empty.

Proof. Let $k \in \mathbb{N}$ be a natural number, and suppose that $f \in \mathrm{C}^{k+1}(X)$; that is, $f \colon X \to \mathbb{R}$ is a $(k+1)$-times continuously differentiable real-valued function on X. Since f is $(k+1)$-times differentiable, it is k-times differentiable, and its kth derivative $f^{(k)} \colon \mathrm{DL}^k(f) \to \mathbb{R}$ is itself differentiable. Lemma 5.1.3 Differentiability implies continuity implies that $f^{(k)}$ is continuous at every point $x \in \mathrm{DL}^k(f) \cap \left(\mathrm{DL}^k(f)\right)'$.

On the other hand, if $x \in \mathrm{DL}^k(f) \setminus \left(\mathrm{DL}^k(f)\right)'$, then x is an isolated point of $\mathrm{DL}^k(f)$, and therefore $f^{(k)}$ is automatically continuous at x. Thus $f^{(k)}$ is continuous, and so $f \in \mathrm{C}^k(X)$ is k-times continuously differentiable.

We have already seen that constant functions are continuously differentiable, and their derivatives are also constant functions. Thus any constant function is k-times continuously differentiable for all $k \in \mathbb{N}$, and so
$$\bigcap_{k=0}^{\infty} \mathrm{C}^k(X) \neq \emptyset.$$

■

Definition 5.1.12 Smoothness. A real-valued function $f \colon X \to \mathbb{R}$ on a subset $X \subseteq \mathbb{R}$ is called **smooth** if it is k-times continuously differentiable for all $k \in \mathbb{N}$. The set of smooth functions on X is denoted
$$\mathrm{C}^{\infty}(X) = \bigcup_{k=0}^{\infty} \mathrm{C}^k(X).$$

◆

There are several interesting topologies on the space $\mathrm{C}^k(X)$ of k-times differentiable functions on a subset $X \subseteq \mathbb{R}$ In general, the uniform convergence of a sequence of continuously differentiable functions does not imply the uniform convergence of its derivatives. However, if the derivative converge uniformly, then the uniform limit of the functions is differentiable, and its derivative is the uniform limit of the derivatives.

Text after the last subsection.

5.1.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Locus of differentiability. Determine the differentiability locus of the given function.

1. Determine the differentiability locus $\mathrm{DL}(f)$ of the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = |x|^3$.

2. Determine the differentiability locus $\mathrm{DL}(g)$ of the real-valued function $g\colon \mathbb{R} \to \mathbb{R}$ defined piecewise by the formula

$$g(x) = \begin{cases} x^2 & x < 0 \\ 2x^2 & x \geq 0. \end{cases}$$

Computing derivatives. Compute formulas for the derivatives of the given differentiable functions.

3. Compute a formula for the derivative of the differentiable function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$f(x) = \sin^2(x) + \cos^2(x).$$

4. Compute a formula for the derivative of the differentiable function $g\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$g(y) = \frac{1}{1+y^2}.$$

5. Compute a formula for the derivative of the differentiable function $h\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$h(z) = \sqrt{\sin(z) + 2}.$$

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

5.1.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be

less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **The quotient rule.** In this problem, we complete Proposition 5.1.4 Arithmetic of differentiable functions to address the quotient of two differentiable functions.

   (a) Let $r\colon \mathbb{R}\setminus\{0\} \to \mathbb{R}\setminus\{0\}$ be the function defined by the formula

   $$r(x) = \frac{1}{x}.$$

   Show that r is differentiable, with derivative

   $$r'(x) = -\frac{1}{x^2}$$

   (b) Let $g\colon X \to \mathbb{R}\setminus\{0\}$ be a non-zero real-valued function defined on a subset $X \subseteq \mathbb{R}$. Prove that if g is differentiable at a point $c \in X \cap X'$, then $\frac{1}{g}$ is differentiable at c, with derivative

   $$\left(\frac{1}{g}\right)'(c) = -\frac{g'(c)}{g(c)^2}.$$

   (c) Let $f, g\colon X \to \mathbb{R}$ be real-valued functions defined on a subset $X \subseteq \mathbb{R}$. Prove that if $g(x) \neq 0$ for all points $x \in X$ and f and g are differentiable at a point $c \in X \cap X'$, then $\frac{f}{g}$ is differentiable at c, with derivative

   $$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)\,g(c) - f(c)\,g'(c)}{g(c)^2}.$$

   The result proven in (c) above is usually called the **quotient rule**.

2. **The power rule.** For each real number $p \in \mathbb{R}$, consider the power function $f_p\colon (0,\infty) \to \mathbb{R}$ defined by the formula $f_p(x) = x^p$.

   (a) Prove that for each natural number $n \in \mathbb{N}$, f_n is differentiable, with derivative

   $$f_n'(x) = nx^{n-1}.$$

   Hint. Either prove the desired result by induction on $n \in \mathbb{N}$ or use the binomial theorem.

   (b) Show that for any integer $n \in \mathbb{Z}$, f_n is differentiable, with derivative $f_n'(x) = nx^{n-1}$.

   (c) Show that for any rational number $p \in \mathbb{Q}$ that if f_p is differentiable, then its derivative is $f_p'(x) = px^{p-1}$.

   Hint. Write $p = \frac{m}{n}$ for some integers $m, n \in \mathbb{Z}$ with $n \neq 0$, and differentiate the equation $f_p(x)^n = x^m$.

   The rules proven above also hold for irrational powers, although we will require additional machinery in order to prove this final extension. In general, this is called the **power rule**.

3. Derivatives of trigonometric functions. Text before the parts.

(a) Compute the limits
$$\lim_{\theta\to 0}\frac{\sin(\theta)}{\theta}$$
and
$$\lim_{\theta\to 0}\frac{\cos(\theta)-1}{\theta}.$$

Hint. You may wish to use the Pythagorean theorem $\sin^2(\theta)+\cos^2(\theta)=1$, which holds for all $\theta\in\mathbb{R}$, and/or the bounds
$$\cos(\theta)\leq\frac{\sin(\theta)}{\theta}\leq\frac{1}{\cos(\theta)},$$
which hold for all $\theta\in\mathbb{R}$ so that $0<|\theta|<\frac{\pi}{2}$.

(b) Show that the trigonometric function $\sin(x)$ is differentiable, with derivative $\cos(x)$.

Hint. You may wish to use the sum of angles formula
$$\sin(\alpha+\beta)=\sin(\alpha)\cos(\beta)+\cos(\alpha)\sin(\beta),$$
which holds for all real numbers $\alpha,\beta\in\mathbb{R}$.

(c) Show that the trigonometric function $\cos(x)$ is differentiable, with derivative $-\sin(x)$.

Hint. You may wish to use the trigonometric identities $\sin(x)=\cos\left(x-\frac{\pi}{2}\right)$ and $\cos(x)=-\sin\left(x-\frac{\pi}{2}\right)$, which holds for all $x\in\mathbb{R}$.

(d) Show that the trigonometric function $\tan(x)$ is differentiable, with derivative $\frac{1}{\cos^2(x)}$.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

5.2 Analysis of Critical Points

In this section, we analyze the local behavior of a real-valued function based on differentiability criteria and/or the value of the derivative. We will make a distinction between **critical points** and **regular points** and begin to develop tools to analyze the local behavior near each. Finally, we will also classify certain kinds of critical points based on several derivative tests.

5.2.1 Local Behavior Near Critical Points

The local behavior of a differentiable real-valued function near a point can be very different depending on the sign of the derivative at that point. We will make the distinction between **critical points**, where the derivative vanishes (or does not exist), and **regular points**, where the derivative is non-zero.

Definition 5.2.1 Critical/regular point/value. Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on an interval $X \subseteq \mathbb{R}$.

- *Critical point/value.*

  A point $x \in X$ is called a **critical point** of f if either $f'(x) = 0$, in which case x is called a **stationary (critical) point** of f, or f is not differentiable at x.

  A real number $y \in \mathbb{R}$ is called a **critical value** of f if $f(x) = y$ for some critical point $x \in X$.

- *Regular point/value.*

  A point $x \in X$ is called a **regular point** of f if it is not a critical point of f; that is, x is a regular point of f if f is differentiable at x and $f'(x) \neq 0$.

  A real number $y \in \mathbb{R}$ is called a **regular value** of f if it is not a critical value of f; that is, y is a regular value of f if the pre-image

  $$f^{-1}(y) = \{x \in X \mid f(x) = y\}$$

  consists only of regular points of f.

◆

Example 5.2.2 Critical/regular points/values. The following are examples of critical points, regular points, critical values, and regular values of functions:

- Let $f\colon \mathbb{R} \to \mathbb{R}$ be the function defined piecewise by the formula

  $$f(x) = \operatorname{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0. \end{cases}$$

  Then every point $x \in \mathbb{R}$ is a critical point of f. If $x \neq 0$, then x is a stationary critical point of f, and if $x = 0$, then x is a point of non-differentiability. Thus 1, 0, and -1 are critical values of f, and all other real numbers are regular values.

- Let $g\colon \mathbb{R} \to \mathbb{R}$ be the function defined piecewise by the formula

  $$g(y) = |y| = \begin{cases} y & y \geq 0 \\ -y & y < 0. \end{cases}$$

  Then 0 is the only critical point of g, and it is a point of non-differentiability. Thus 0 is a critical value of g, and all other real numbers are regular values.

- Let $h\colon \mathbb{R} \to \mathbb{R}$ be the function defined by the formula $h(z) = z^2$. Then 0 is the only critical point of h, and it is a stationary critical point. Thus 0 is a critical value of g, and all other real numbers are regular values.

▲

One use of the critical points of a real-valued function is in finding **local extrema**, which occur when the restriction of a function to a neighborhood attains an extreme value on that neighborhood.

Definition 5.2.3 Local extrema. Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on a subset $X \subseteq \mathbb{R}$.

- *Local maximum.*

  f attains a **local maximum** at a point $x \in X$ if there is a neighborhood $N \in \mathcal{N}_X(x)$ of x in X so that

  $$f(y) \leq f(x)$$

  for all points $y \in N$.

- *Local minimum.*

  f attains a **local minimum** at a point $x \in X$ if there is a neighborhood $N \in \mathcal{N}_X(x)$ of x in X so that

  $$f(x) \leq f(y)$$

  for all points $y \in N$.

Local extremum is a generic term which encompasses both local maxima and local minima. ◆

In fact, local extrema may only be attained at critical points. If a local extremum is attained at a point of non-differentiability, then this is clear. However, if the local extremum is attained at a point at which the function is differentiable, then this point is a stationary critical point. This is the content of the following lemma:

Lemma 5.2.4 Local extrema are stationary points. *Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on a subset $X \subseteq \mathbb{R}$ which attains a local extremum at a point $c \in X$ so that every neighborhood of c contains points strictly greater than c and points strictly less than c. If f is differentiable at c, then $f'(c) = 0$; that is, c is a stationary point of f.*

Proof. By contraposition. Suppose first that $f'(c) > 0$ is positive. Then since $\frac{f'(c)}{2} > 0$ is positive, there is some positive real number $\delta > 0$ so that

$$\left|\frac{f(x) - f(c)}{x - c} - f'(c)\right| < \frac{f'(c)}{2}$$

for all points $x \in X$ so that $0 < |x - c| < \delta$. In particular,

$$\frac{f(x) - f(c)}{x - c} > f'(c) - \frac{f'(c)}{2} = \frac{f'(c)}{2} > 0$$

for all points $x \in X$ so that $0 < |x - c| < \delta$, so that $f(x) - f(c)$ and $x - c$ have the same sign; that is, so long as $x \in [a, b]$ and $0 < |x - c| < \delta$, then $f(x) < f(c)$ if $x < c$ and $f(x) > f(c)$ if $x > c$. In particular, since the neighborhood $\mathrm{B}_\delta(c) = (c - \delta, c + \delta) \cap X$ contains points strictly less than c and contains points strictly larger than c, we conclude that f does not achieve a maximum or minimum at c.

Similarly, now suppose that $f'(c) < 0$ is negative. Then since $-\frac{f'(c)}{2} > 0$ is positive, there is some positive real number $\delta > 0$ so that

$$\left|\frac{f(x) - f(c)}{x - c} - f'(c)\right| < -\frac{f'(c)}{2}$$

for all points $x \in X$ so that $0 < |x - c| < \delta$. In particular,

$$\frac{f(x) - f(c)}{x - c} < -f'(c) + \frac{f'(c)}{2} = -\frac{f'(c)}{2} < 0$$

for all points $x \in X$ so that $0 < |x - c| < \delta$, so that $f(x) - f(c)$ and $x - c$ have opposite sign; that is, so long as $x \in [a, b]$ and $0 < |x - c| < \delta$, then $f(x) > f(c)$ if $x < c$ and $f(x) < f(c)$ if $x > c$. In particular, since the neighborhood $\mathrm{B}_\delta(c) = (c - \delta, c + \delta) \cap X$ contains points strictly less than c and contains points strictly larger than c, we conclude that f does not achieve a maximum or minimum at c. ■

A famous consequence of Lemma 5.2.4 Local extrema are stationary points due to Fermat occurs when we suppose the domain of the function in question is an interval. In this case, interior points automatically satisfy the hypothesis of Lemma 5.2.4 Local extrema are stationary points.

Theorem 5.2.5 Interior extremum theorem. (Pierre de Fermat) *Let $f: X \to \mathbb{R}$ be a real-valued function defined on an interval $X \subseteq \mathbb{R}$ which attains a local extremum at an interior point $c \in \mathrm{Int}(X)$. If f is differentiable at c, then $f'(c) = 0$; that is, c is a stationary point of f.*

Proof. Any neighborhood $N \in \mathcal{N}_X(c)$ of an interior point $c \in \mathrm{Int}(X)$ contains an open interval containing c, and therefore contains points strictly greater than c and points strictly less than c. The desired conclusion now follows immediately from Lemma 5.2.4 Local extrema are stationary points. ■

One use of the Interior extremum theorem is to prove the following theorem of Darboux, which establishes an intermediate value property for the derivative of a differentiable function ***even if said function is not continuously differentiable***.

Theorem 5.2.6 Darboux's theorem. (Jean-Gaston Darboux) *Let $f: X \to \mathbb{R}$ be a real-valued function defined on an interval $X \subseteq \mathbb{R}$. If f is differentiable, then $\mathrm{im}(f') = f'(X)$ is an interval.*

Proof. Let $a, b \in X$, and suppose that $a < b$. There are three cases; either $f'(a) = f'(b)$, $f'(a) < f'(b)$, or $f'(a) > f'(b)$. If $f(a) = f(b)$, then $\mathrm{im}(f')$ vacuously contains all points between $f'(a)$ and $f'(b)$.

Suppose that $f'(a) < f'(b)$, let $f'(a) < r < f'(b)$, and consider the function $g: X \to \mathbb{R}$ defined by the formula

$$g(x) = f(x) - rx.$$

We observe that $g'(a) = f'(a) - r < 0$ and $g'(b) = f'(b) - r > 0$. In particular, the argument in the proof of Lemma 5.2.4 Local extrema are stationary points implies that g does not attain a minimum value at a or b.

However, since g is continuous on the compact interval $[a, b]$, it attains a minimum value on $[a, b]$. Thus g attains a minimum value at some interior point $c \in (a, b)$, and so the Interior extremum theorem implies that $g'(c) = 0$; that is,

$$g'(c) = f'(c) - r,$$

so that $f'(c) = r$.

Now suppose that $f'(b) < f'(a)$, let $f'(b) < r < f'(a)$, and consider the function $g: X \to \mathbb{R}$ defined by the formula

$$g(x) = f(x) - rx.$$

We observe that $g'(a) = f'(a) - r > 0$ and $g'(b) = f'(b) - r < 0$. In particular, the argument in the proof of Lemma 5.2.4 Local extrema are stationary points implies that g does not attain a maximum value at a or b.

However, since g is continuous on the compact interval $[a, b]$, it attains a maximum value on $[a, b]$. Thus g attains a maximum value at some interior point $c \in (a, b)$, and so the Interior extremum theorem implies that $g'(c) = 0$; that is,

$$g'(c) = f'(c) - r,$$

so that $f'(c) = r$. ■

We have thus far only explored the local behavior of functions near critical points, but there is also much to say of the behavior of a function near regular points. For example, if a differentiable function has only regular points, then it is strictly monotone.

Corollary 5.2.7 Monotonicity in the absence of critical points. *Let $f: X \to \mathbb{R}$ be a differentiable function defined on an interval $X \subseteq \mathbb{R}$. If f does not have any critical points in X, then f is strictly monotone.*

Proof. If f' takes both positive and negative values on X, then Darboux's theorem implies that $f'(x) = 0$ for some point $x \in X$. This contradiction implies that f' either takes only positive values on X or takes only negative values on X. We will prove the result in the case $f'(x) > 0$ for all points $x \in X$; the case $f'(x) < 0$ for all $x \in X$ is similar.

Let $a, b \in X$, and suppose that $a < b$. Since f is continuous on the compact interval $[a, b] \subseteq X$, it attains a maximum value on $[a, b]$. If f attains a maximum value at an interior point $c \in (a, b)$, then the Interior extremum theorem implies that $f'(c) = 0$. This contradiction implies that f attains a maximum value at one of the endpoints a or b.

Since $f'(a) > 0$ and any neighborhood of a in $[a, b]$ contains points which are strictly larger than a, the argument in the proof of Lemma 5.2.4 Local extrema are stationary points implies that f does not attain a maximum value at a. Thus we conclude that $f(a) < f(b)$. Since a and b were chosen arbitrarily, f is increasing. ■

Stronger versions of Corollary 5.2.7 Monotonicity in the absence of critical points are usually called the **inverse function theorem**, and establish the differentiability of the left inverse of any such function. You will have an opportunity to explore these additional conclusions in the problems for this section.

5.2.2 Classification of Critical Points

While the previous results of this section are quite strong, we wish to establish even stronger characterizations of real-valued functions based on local analyses near critical points. However, these analyses will require us to consider critical points which are not themselves limits of critical points.

Definition 5.2.8 Isolated critical point. Let $f: X \to \mathbb{R}$ be a real-valued function defined on a subset $X \subseteq \mathbb{R}$. A critical point $x \in X$ of f is called **isolated** if there is a neighborhood $N \in \mathcal{N}_X(x)$ so that $N \setminus \{x\}$ consists only of regular points of f. ◆

Example 5.2.9 Isolated critical points. The following are examples and non-examples of isolated critical points of real-valued functions:

- Let $f: \mathbb{R} \to \mathbb{R}$ be the continuous real-valued function defined piecewise by the formula

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

  Then f has infinitely many non-zero isolated critical points which accumulate at the critical point 0.

- Let $g\colon \mathbb{R} \to \mathbb{R}$ be the continuous real-valued function defined by the formula $g(y) = |y|$. Then g has one critical point at 0, which is isolated.
- Let $h\colon [-1,1] \to \mathbb{R}$ be the continuous real-valued function defined by the formula
$$h(z) = \sqrt{1 - z^2}.$$
Then h has three critical points at -1, 0, and 1, each of which is isolated.

▲

If c is an isolated critical point of a continuous real-valued function defined on an interval, then continuity and Corollary 5.2.7 Monotonicity in the absence of critical points implies that the restrictions of f to some intervals $(c-\delta, c]$ and $[c, c+\delta)$ are strictly monotone. This can be used to detect the presence of a local extremum.

Theorem 5.2.10 First derivative test. *Let $f\colon X \to \mathbb{R}$ be a continuous real-valued function defined on an interval $X \subseteq \mathbb{R}$. If $c \in \operatorname{Int}(X)$ is an interior point of X in $\mathbb{R}$ and an isolated critical point of f, then f attains a local extremum at c if and only if the derivative f' changes sign at c.*

Proof. Under these assumptions, the previous discussion applies; there is some positive real number $\delta > 0$ so that $(c-\delta, c+\delta) \subseteq X$ so that f' does not change sign on the intervals $(c-\delta, c)$ and $(c, c+\delta)$.

If f' takes only positive values on both $(c-\delta, c)$ and $(c, c+\delta)$, then the continuity of f and Corollary 5.2.7 Monotonicity in the absence of critical points implies that the restrictions of f to the intervals $(c-\delta, c]$ and $[c, c+\delta)$ are increasing. We conclude that the restriction of f to $(c-\delta, c+\delta)$ is increasing, so that
$$f(c-\eta) < f(c) < f(c+\eta)$$
for all sufficiently small positive real numbers $0 < \eta < \delta$. In particular, this implies that f cannot attain a local extremum at c.

If f' takes only negative values on both $(c-\delta, c)$ and $(c, c+\delta)$, then the continuity of f and Corollary 5.2.7 Monotonicity in the absence of critical points implies that the restrictions of f to the intervals $(c-\delta, c]$ and $[c, c+\delta)$ are decreasing. We conclude that the restriction of f to $(c-\delta, c+\delta)$ is decreasing, so that
$$f(c-\eta) > f(c) > f(c+\eta)$$
for all sufficiently small positive real numbers $0 < \eta < \delta$. In particular, this implies that f cannot attain a local extremum at c.

If f' takes only positive values on $(c-\delta, c)$ and only negative values on $(c, c+\delta)$, then the continuity of f and Corollary 5.2.7 Monotonicity in the absence of critical points implies that the restrictions of f to the intervals $(c-\delta, c]$ and $[c, c+\delta)$ are increasing and decreasing, respectively. We conclude that
$$f(c-\eta) < f(c) > f(c+\eta)$$
for all sufficiently small positive real numbers $0 < \eta < \delta$. In particular, this implies that f attains a local maximum at c.

Finally, if f' takes only negative values on $(c-\delta, c)$ and only positive values on $(c, c+\delta)$, then the continuity of f and Corollary 5.2.7 Monotonicity in the absence of critical points implies that the restrictions of f to the intervals $(c-\delta, c]$ and $[c, c+\delta)$ are decreasing and increasing, respectively. We conclude that
$$f(c-\eta) > f(c) < f(c+\eta)$$

for all sufficiently small positive real numbers $0 < \eta < \delta$. In particular, this implies that f attains a local minimum at c. ■

Example 5.2.11 Using the first derivative test. Let $n \geq 1$ be a positive integer, and consider the real-valued function $f : \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x^n$. Recall that f is differentiable, with derivative $f'(x) = nx^{n-1}$. If $n = 1$, then

$$f'(x) = 1x^0 = 1 \neq 0$$

for all points $x \neq 0$, and so f has no critical points. However, if $n \geq 2$, then

$$f'(x) = nx^{n-1}$$

vanishes precisely when $x = 0$; that is, 0 is the only critical point of f, and it is a stationary critical point.

We observe that if $n \geq 2$ is even, then $f'(x) < 0$ when $x < 0$ and $f'(x) > 0$ when $x > 0$. The First derivative test implies that f attains a local minimum at 0 in this case.

On the other hand, if $n \geq 2$ is odd, then $f'(x) > 0$ when $x \neq 0$. The First derivative test implies that f does not attain a local extremum at 0 in this case. ▲

When a real-valued function is twice-differentiable at an isolated stationary critical point, its second derivative can also sometimes be used to detect the presence of local extrema at critical points.

Corollary 5.2.12 Second derivative test. *Let $f : X \to \mathbb{R}$ be a differentiable real-valued function defined on an interval $X \subseteq \mathbb{R}$. Suppose that $c \in \operatorname{Int}(X)$ is an interior point of X in $\mathbb{R}$ and an isolated stationary critical point of f at which f is twice-differentiable. If $f''(c) > 0$, then f attains a local minimum at 0, and if $f''(c) < 0$, then f attains a local maximum at 0.*

Proof. Since c is a stationary critical point of f, $f'(c) = 0$. Since $\frac{|f''(c)|}{2} > 0$ is a positive real number, there is a positive real number $\delta > 0$ so that $(c - \delta, c + \delta) \subseteq X$ and

$$\left| \frac{f'(x) - f'(c)}{x - c} - f''(c) \right| < \frac{|f''(c)|}{2}$$

for all points $x \in (c - \delta, c + \delta)$. If $f''(c) > 0$, then

$$\frac{f'(x)}{x - c} = \frac{f'(x) - f'(c)}{x - c} > \frac{f''(c)}{2} > 0,$$

and so $f'(x)$ and $x - c$ have the same sign; that is, f' takes only negative values on $(c - \delta, c)$ and takes only positive values on $(c, c + \delta)$. The First derivative test now implies that f attains a local minimum at c.

On the other hand, if $f''(c) > 0$, then

$$\frac{f'(x)}{x - c} = \frac{f'(x) - f'(c)}{x - c} < \frac{f''(c)}{2} < 0,$$

and so $f'(x)$ and $x - c$ have opposite signs; that is, f' takes only positive values on $(c - \delta, c)$ and takes only negative values on $(c, c + \delta)$. The First derivative test now implies that f attains a local maximum at c. ■

Example 5.2.13 Using the second derivative test. Let $n \geq 2$, and consider the real-valued function $f : \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x^n$. Recall that f is differentiable, with derivative $f'(x) = nx^{n-1}$ and second derivative $f''(x) = n(n-1)x^{n-2}$. We have seen in Example 5.2.11 Using the first derivative test that f has a single stationary critical point at 0.

We observe that if $n = 2$, then $f''(0) = 2 \cdot 1 = 2 > 0$, and so the Second derivative test implies that f attains a local minimum at 0 in this case. However, if $n \geq 3$, then

$$f''(0) = n(n-1)0^{n-2} = 0,$$

and so the Second derivative test is inconclusive. ▲

So we see that the Second derivative test may not be conclusive even for functions whose isolated critical points may be analyzed by the First derivative test.

In the next section, we will relate the instantaneous rate of change of a differentiable function (that is, its derivative) to its average rate of change via the **mean value theorem**.

5.2.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Examples of critical points. Give examples of differentiable functions which satisfy the given conditions.

1. Give an example of a differentiable function which has a single critical point which is not a local extremum.
2. Give an example of a differentiable function which attains a local minimum but no global minima.
3. Give an example of a differentiable function which has no critical points.
4. Give an example of a differentiable function which has exactly one critical point but no global extrema.
5. Give an example of a differentiable function which attains infinitely many global extrema.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

5.2.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be

less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Misconceptions about critical points.** Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on an interval $X \subseteq \mathbb{R}$. ***The following statements are false!*** Explain why the statement is incorrect, and disprove them by providing a counterexample.

   (a) If f attains a local extremum at a point $c \in X$, then c is a critical point of f.

   (b) If a point $c \in X$ is a critical point of f, then f attains a local extremum at c.

   (c) If a point $c \in X$ is a regular point of f, then c has a neighborhood in X on which f is monotone.

   (d) f is strictly monotone if and only if f has no critical points.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

5.3 Applications of the Mean Value Theorem

In this section, we introduce one of the fundamental results of differential calculus. The **mean value theorem** relates the instantaneous rate of change of a differentiable function given by its derivative and its average rate of change given by a difference quotient. While not immediately obvious, the mean value theorem has several deep consequences for the study of differentiable functions. We will derive the mean value theorem and exhibit several of these corollaries.

5.3.1 Average and Instantaneous Rates of Change

The **average rate of change** of a real-valued function $f\colon X \to \mathbb{R}$ defined on some subset $X \subseteq \mathbb{R}$ which contains a closed interval $[a, b] \subseteq X$ is the difference quotient

$$\frac{f(b) - f(a)}{b - a}.$$

In Section 5.1 Differentiability, we saw that the derivative of a differentiable function is a limit of such difference quotients; as such, the derivative is often interpreted as an **instantaneous rate of change**. The **mean value theorem** relates these two rates of change under some mild hypotheses on f.

Before we explore the Mean value theorem in its full generality, we prove a special case. The proof of the Mean value theorem will rely on the following supporting lemma:

Lemma 5.3.1 Rolle's lemma. (Michel Rolle) *Let $f\colon [a, b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a, b] \subsetneq \mathbb{R}$. If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) and $f(a) = f(b)$, then $f'(c) = 0$ for some point $a < c < b$.*

Proof. Let $a < c < b$, and suppose that $f'(c) > 0$. Then since $\frac{f'(c)}{2} > 0$ is positive, there is some positive real number $\delta > 0$ so that

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < \frac{f'(c)}{2}$$

for all points $x \in [a, b]$ so that $0 < |x - c| < \delta$. In particular,

$$\frac{f(x) - f(c)}{x - c} > f'(c) - \frac{f'(c)}{2} = \frac{f'(c)}{2} > 0$$

for all points $x \in [a, b]$ so that $0 < |x - c| < \delta$, so that $f(x) - f(c)$ and $x - c$ have the same sign; that is, so long as $x \in [a, b]$ and $0 < |x - c| < \delta$, then $f(x) < f(c)$ if $x < c$ and $f(x) > f(c)$ if $x > c$. In particular, we conclude that f does not achieve a maximum or minimum at c.

Similarly, now suppose that $f'(c) < 0$. Then since $-\frac{f'(c)}{2} > 0$ is positive, there is some positive real number $\delta > 0$ so that

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < -\frac{f'(c)}{2}$$

for all points $x \in [a, b]$ so that $0 < |x - c| < \delta$. In particular,

$$\frac{f(x) - f(c)}{x - c} < -f'(c) + \frac{f'(c)}{2} = -\frac{f'(c)}{2} < 0$$

for all points $x \in [a, b]$ so that $0 < |x - c| < \delta$, so that $f(x) - f(c)$ and $x - c$ have opposite sign; that is, so long as $x \in [a, b]$ and $0 < |x - c| < \delta$, then $f(x) > f(c)$ if $x < c$ and $f(x) < f(c)$ if $x > c$. In particular, we conclude that f does not achieve a maximum or minimum at c.

However, $[a, b]$ is closed and bounded, and hence compact by Corollary 3.6.23 Compact subsets of Euclidean space. Since f is continuous and $[a, b]$ is compact, Proposition 3.6.20 Continuous images of compact subsets implies that

$$\operatorname{im}(f) = f([a, b]) = \{f(x) \in \mathbb{R} \mid a \leq x \leq b\}$$

is compact and hence also closed and bounded. In particular, f attains a maximum $\max_{a \leq x \leq b} f(x) = \max(\operatorname{im}(f))$ and a minimum $\min_{a \leq x \leq b} f(x) = \min(\operatorname{im}(f))$.

To summarize our argument so far, we have shown that f does not achieve a maximum or minimum value at any point where its derivative is non-zero, but f must achieve a maximum and minimum value somewhere in $[a, b]$. If f achieves a maximum or a minimum value at an interior point $a < c < b$, then the above argument shows that $f'(c) = 0$. Otherwise, f attains both its maximum and minimum values at the endpoints a and b. In this case, f must be constant because $f(a) = f(b)$. In particular, this implies that $f'(c) = 0$ for all points $a < c < b$. ■

We remark that the details of the above proof of Rolle's lemma are necessary even if we have an understanding of the analysis of critical points and local extrema. Indeed, the result often taught in calculus classes that for a function $f\colon X \to \mathbb{R}$, if f is differentiable at a point $c \in X$ and $f'(c) > 0$, then f is increasing near c is false; such a conclusion actually requires continuous differentiability near c.

Theorem 5.3.2 Mean value theorem. (Joseph-Louis Lagrange) *Let $f\colon [a, b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a, b] \subsetneq \mathbb{R}$. If f is continuous on the closed interval $[a, b]$ and differentiable on the open* *interval* (a, b)*, then*

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some point $a < c < b$*.*

Proof. Let $g\colon [a, b] \to \mathbb{R}$ be the real-valued function defined by the formula

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a}\right) x.$$

Then g is continuous by Proposition 3.5.6 Vector arithmetic of continuous maps and differentiable on the open interval (a, b) by Proposition 5.1.4 Arithmetic of differentiable functions, with derivative

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Moreover,

$$\begin{aligned} g(b) &= f(b) - \left(\frac{f(b) - f(a)}{b - a}\right) b \\ &= f(b) - \left(\frac{f(b) - f(a)}{b - a}\right)(b - a) - \left(\frac{f(b) - f(a)}{b - a}\right) a \\ &= f(b) - (f(b) - f(a)) - \left(\frac{f(b) - f(a)}{b - a}\right) a \\ &= f(b) - \left(\frac{f(b) - f(a)}{b - a}\right) a = g(a). \end{aligned}$$

Lemma 5.3.1 Rolle's lemma now implies that $g'(c) = 0$ for some interior point $a < c < b$; that is,

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

■

Example 5.3.3 Hypotheses of the Mean value theorem. The following examples demonstrate that each of the hypotheses of the Mean value theorem is necessary:

1. Let $f\colon [0, 1] \cup [2, 3] \to \mathbb{R}$ be the real-valued function defined piecewise by the formula

$$f(x) = \begin{cases} 0 & 0 \leq x \leq 1 \\ 1 & 2 \leq x \leq 3. \end{cases}$$

   Then f is continuous on $[0, 1] \cup [2, 3]$ and differentiable on the interior $(0, 1) \cup (2, 3)$, but

$$f'(x) = 0 \neq \frac{1}{3} = \frac{1 - 0}{3 - 0} = \frac{f(3) - f(0)}{3 - 0}$$

   for all interior points $x \in (0, 1) \cup (2, 3)$. This example shows that f must be defined on a closed interval in order to guarantee the conclusions of the Mean value theorem.

2. Let $f\colon [0, 1] \to \mathbb{R}$ be the real-valued function defined piecewise by the formula

$$f(x) = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1. \end{cases}$$

Then f is continuous on $[0,1)$ and differentiable on the interior $(0,1)$, but

$$f'(x) = 0 \neq 1 = \frac{1-0}{1-0} = \frac{f(1) - f(0)}{1-0}$$

for all interior points $x \in (0,1)$. This example shows that f must be continuous on the closed interval $[0,1]$ in order to guarantee the conclusions of the Mean value theorem.

3. Let $f\colon [-1,1] \to \mathbb{R}$ be the real-valued function defined by the formula $f(x) = |x|$. Then f is continuous on $[0,1]$ and differentiable at every interior point except $x = 0$, but

$$f'(x) = \pm 1 \neq 0 = \frac{1-1}{1-(-1)} = \frac{f(1) - f(-1)}{1-(-1)}$$

for all interior points $x \in (-1,1) \setminus \{0\}$. This example shows that f must be differentiable on the open interval $(-1,1)$ in order to guarantee the conclusions of the Mean value theorem.

▲

While all of the hypotheses of the Mean value theorem are necessary in order to guarantee its conclusion, they are sufficient as written. Indeed, a function $f\colon [a,b] \to \mathbb{R}$ need not be differentiable at the endpoints a or b in order to apply the Mean value theorem; for an example, see the function defined by the formula

$$f(x) = \sqrt{\left(\frac{a-b}{2}\right)^2 - \left(x - \frac{a+b}{2}\right)^2},$$

which is differentiable on the open interval (a,b) but not at the endpoints and yet satisfies the conclusions of the Mean value theorem.

5.3.2 Beyond the Mean Value Theorem

Rolle's lemma and the Mean value theorem are two of a constellation of similar results relating the derivatives of some differentiable functions to the average rates of change of said functions. Particularly useful among these results is the following lemma, from which the Mean value theorem and more can be derived:

Lemma 5.3.4 Generalized mean value theorem. *Let $f, g, h\colon [a,b] \to \mathbb{R}$ be real-valued functions defined on a closed interval $[a,b]$, and consider the real-valued function $D\colon [a,b] \to \mathbb{R}$ defined by the formula*

$$\begin{aligned} D(x) &= \det \begin{pmatrix} f(x) & g(x) & h(x) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{pmatrix} \\ &= f(x)\left(g(a)\,h(b) - g(b)\,h(b)\right) - g(x)\left(f(a)\,h(b) - f(b)\,h(a)\right) \\ &\quad + h(x)\left(f(a)\,g(b) - f(b)\,g(a)\right). \end{aligned}$$

If f, g, and h are continuous on the closed interval $[a,b]$ and differentiable on the interior (a,b), then D is differentiable on (a,b), and $D'(c) = 0$ for some interior point $a < c < b$.

Proof. We observe that D is a linear combination of f, g, and h, and so is continuous on $[a,b]$ by Proposition 3.5.6 Vector arithmetic of continuous maps and is differentiable on (a,b) by Proposition 5.1.4 Arithmetic of differentiable functions. Moreover,

$$D(a) = \det\begin{pmatrix} f(a) & g(a) & h(a) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{pmatrix}$$

and

$$D(b) = \det\begin{pmatrix} f(b) & g(b) & h(b) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{pmatrix}$$

are determinants of 3×3 matrices with repeated rows, and so $D(a) = D(b) = 0$ by a standard result of linear algebra. Alternatively, one can verify that $D(a) = D(b) = 0$ by direct computation. In any case, Rolle's lemma now implies that $D'(c) = 0$ for some interior point $a < c < b$. ■

One can use Lemma 5.3.4 Generalized mean value theorem to construct another proof of the Mean value theorem, but its real strength is in its generality. We include the following alternative proof of Lemma 5.3.4 Generalized mean value theorem for completeness:

Alternative proof of the Mean value theorem. Let $g, h\colon [a,b] \to \mathbb{R}$ be the real-valued functions defined by the formulas $g(x) = x$ and $h(x) = 1$, and consider the function $D\colon [a,b] \to \mathbb{R}$ defined as in Lemma 5.3.4 Generalized mean value theorem. Then

$$\begin{aligned} D(x) = \det\begin{pmatrix} f(x) & g(x) & h(x) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{pmatrix} &= \det\begin{pmatrix} f(x) & x & 1 \\ f(a) & a & 1 \\ f(b) & b & 1 \end{pmatrix} \\ &= f(x)(a-b) - x(f(a) - f(b)) + 1(f(a)b - f(b)a), \end{aligned}$$

and so

$$D'(x) = f'(x)(a-b) - (f(a) - f(b)).$$

Lemma 5.3.4 Generalized mean value theorem implies that there is some interior point $a < c < b$ so that $D'(c) = 0$. We observe that

$$\begin{aligned} 0 = D'(c) &= f'(c)(a-b) - (f(a) - f(b)) \\ f(a) - f(b) &= f'(c)(a-b) \\ \frac{f(b) - f(a)}{b-a} = \frac{f(a) - f(b)}{a-b} &= f'(c). \end{aligned}$$

■

Lemma 5.3.4 Generalized mean value theorem relates the derivatives of three differentiable functions, and the Theorem 5.3.2 Mean value theorem deals with the derivative of just one differentiable function. Intermediate between these results is the following result, which relates the derivatives of two differentiable functions:

Theorem 5.3.5 Extended mean value theorem. (Augustin-Louis Cauchy) *Let $f, g\colon [a,b] \to \mathbb{R}$ be real-valued functions defined on a closed interval $[a,b] \subsetneq \mathbb{R}$. If f and g are continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b), then*

$$f'(c)(g(b) - g(a)) = (f(b) - f(a))g'(c)$$

for some point $a < c < b$.

Proof. Let $h\colon [a,b] \to \mathbb{R}$ be the constant real-valued function defined by the formula $h(x) = 1$, and consider the function $D\colon [a,b] \to \mathbb{R}$ defined as in Lemma 5.3.4 Generalized mean value theorem. Then

$$\begin{aligned} D(x) &= \det \begin{pmatrix} f(x) & g(x) & h(x) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{pmatrix} = \det \begin{pmatrix} f(x) & g(x) & 1 \\ f(a) & g(a) & 1 \\ f(b) & g(b) & 1 \end{pmatrix} \\ &= f(x)(g(a) - g(b)) - g(x)(f(a) - f(b)) + (f(a)g(b) - f(b)g(a)), \end{aligned}$$

and so

$$D'(x) = f'(x)(g(a) - g(b)) - g'(x)(f(a) - f(b)).$$

Lemma 5.3.4 Generalized mean value theorem implies that there is some interior point $a < c < b$ so that $D'(c) = 0$. We observe that

$$\begin{aligned} 0 = D'(c) &= f'(c)(g(a) - g(b)) - g'(c)(f(a) - f(b)) \\ f'(c)(g(b) - g(a)) &= g'(c)(f(b) - f(a)). \end{aligned}$$

■

We will use the Extended mean value theorem to compute certain limits taking indeterminate form.

Corollary 5.3.6 L'Hôpital's rule. (Guillaume de l'Hôpital) *Let $f, g\colon (a,b) \to \mathbb{R}$ be real-valued functions on an open interval $(a,b) \subseteq \mathbb{R}$, and consider an interior point $a < c < b$. Suppose that either*

$$\lim_{x \to c} f(x) = \lim_{x \to c} g(x) = 0$$

or

$$\lim_{x \to c} |f(x)| = \lim_{x \to c} |g(x)| = \infty.$$

If f and g are differentiable on $(a,b) \setminus c$, $g'(x) \neq 0$ for all $x \in (a,b) \setminus \{c\}$, and

$$\lim_{x \to c} \frac{f'(x)}{g'(x)} = L$$

for some extended real number $L \in [-\infty, \infty]$, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = L$$

Proof. Since g is continuous on (a,b) and $g'(x) \neq 0$ for all $x \in (a,b) \setminus \{c\}$, $g(x)$ attains no local extrema on $(a,b) \setminus \{c\}$. In particular, $g(x)$ is strictly monotone on (a,c) and on (b,c). Without loss of generality, we may shrink the interval (a,b) so that $g(x) \neq 0$ for all points $x \in (a,b) \setminus \{c\}$.

For each point $x \in (a,b) \setminus \{c\}$, let I_x denote the interval with endpoints x and c, including x but not c; that is,

$$I_x = \begin{cases} [x,c) & a < x < c < b \\ (c,x] & a < c < x < b. \end{cases}$$

Moreover, let $i, s\colon (a,b) \setminus \{c\} \to [-\infty, \infty]$ be the extended real-valued functions defined by the formulas

$$i(x) = \inf_{y \in I_x} \frac{f'(y)}{g'(y)} \qquad s(x) = \sup_{y \in I_x} \frac{f'(y)}{g'(y)}.$$

We observe that since $\lim_{x\to c}\frac{f'(x)}{g'(x)} = L$,

$$\lim_{x\to c} i(x) = \lim_{x\to c} s(x) = L.$$

Consider distinct points $y, z \in I_x$. Since the restriction of g to either $(a, c]$ or $[c, b)$ is strictly monotone and therefore injective, $g(y) \neq g(z)$. Moreover, the restrictions of f and g to the closed interval between y and z satisfy the hypotheses of Theorem 5.3.5 Extended mean value theorem, and so

$$\frac{f(z) - f(y)}{g(z) - g(y)} = \frac{f'(t)}{g'(t)}$$

for some point $t \in I_x$ lying strictly between y and z. In particular, we conclude that

$$i(x) \leq \frac{f(z) - f(y)}{g(z) - g(y)} \leq s(x)$$

for all $x \in (a, b) \setminus \{c\}$ and distinct points $y, z \in I_x$.

We first suppose that

$$\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = 0.$$

Then the above argument implies that

$$i(x) \leq \frac{\frac{f(x)}{g(x)} - \frac{f(y)}{g(x)}}{1 - \frac{g(y)}{g(x)}} \leq s(x)$$

for all $x \in (a, b)$ and $y \in I_x \setminus \{x\}$. Taking a limit as $y \to c$, we obtain

$$i(x) \leq \frac{f(x)}{g(x)} \leq s(x).$$

The Squeeze theorem and Theorem 3.2.19 Sequential characterization of map convergence now implies that

$$\lim_{x\to c} \frac{f(x)}{g(x)} = L.$$

We now suppose that

$$\lim_{x\to c} |f(x)| = \lim_{x\to c} |g(x)| = \infty.$$

Then the above argument implies that

$$i(x) \leq \frac{\frac{f(y)}{g(y)} - \frac{f(x)}{g(y)}}{1 - \frac{g(x)}{g(y)}} \leq s(x)$$

for all $x \in (a, b)$ and $y \in I_x \setminus \{x\}$. Taking limits as $y \to c$, we observe that

$$\lim_{y\to c} \frac{f(x)}{g(y)} = \lim_{y\to c} \frac{g(x)}{g(y)} = 0.$$

We conclude that

$$i(x) \leq \liminf_{y\in I_x} \frac{f(y)}{g(y)} \leq \limsup_{y\in I_x} \frac{f(y)}{g(y)} \leq s(x)$$

for all $x \in (a, b) \setminus \{c\}$. The Squeeze theorem and Theorem 3.2.19 Sequential characterization of map convergence now implies that

$$\lim_{x\to c} \frac{f(x)}{g(x)} = L.$$

■

While different in formulation than the other Mean value theorem-like results covered in this section, L'Hôpital's rule also relates the values of some differentiable functions with the values of their derivatives.

In the next section, we will explore approximations of multiply-differentiable functions by higher order polynomials.

5.3.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Hypotheses and conclusions of the Mean value theorem. Determine whether the given function satisfies the hypotheses and conclusions of the Mean value theorem.

1. Let $f\colon [0,1] \to \mathbb{R}$ be the real-valued function defined on the interval $[0,1]$ by the formula

$$f(x) = \sqrt[3]{x^2 - x}.$$

(a) Determine whether or not f satisfies the hypotheses of the Mean value theorem.

(b) Determine whether or not f satisfies the conclusions of the Mean value theorem. If so, find a point $0 < c < 1$ for which

$$f'(c) = \frac{f(1) - f(0)}{1 - 0}.$$

2. Let $g\colon [-1,1] \to \mathbb{R}$ be the real-valued function defined on the interval $[0,1]$ by the formula

$$g(x) = x^{\frac{3}{5}}.$$

(a) Determine whether or not g satisfies the hypotheses of the Mean value theorem.

(b) Determine whether or not g satisfies the conclusions of the Mean value theorem. If so, find a point $-1 < c < 1$ for which

$$g'(c) = \frac{g(1) - g(-1)}{1 - (-1)}.$$

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

5.3.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Functions with the same derivative.**

   (a) Let $f\colon X \to \mathbb{R}$ be a differentiable real-valued function defined on an interval X. Prove that if $f'(x) = 0$ for all $x \in X$, then f is constant.

   (b) Let $f, g\colon X \to \mathbb{R}$ be differentiable real-valued functions defined on an interval X. Prove that if $f'(x) = g'(x)$ for all $x \in X$, then f and g differ by a constant.

   The above result will be important in Chapter 6 Riemann Integration, where we will use it to compare two anti-derivatives of the same continuous function on a closed interval.

2. **Differentiability and Lipschitz continuity.** Let $f\colon X \to \mathbb{R}$ be a differentiable real-valued function defined on an interval $X \subseteq \mathbb{R}$. Prove that f is Lipschitz continuous if and only if its derivative f' is bounded.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

5.4 Higher Order Approximations

In this section, we explore the properties of the **Taylor approximations** of a multiply differentiable function. In a sense which we will make precise, these polynomials are the best possible approximations for such a function near a given point. We will also compute the error in such approximations by means similar to the Mean value theorem.

5.4.1 Dominance and Asymptotics

How can we compare the speed at which two functions converge to 0 or diverge to infinity? We may be able to see graphically that while both x^2 and x^4 converge at 0 to 0, x^4 converges far more quickly. One attempt to quantify observations of this type is the family of related binary relations comprising the notion of **asymptotic comparison**.

Definition 5.4.1 Asymptotic comparison. Let $f, g\colon X \to \mathbb{R}$ be real-valued functions defined on a subset $X \subseteq \mathbb{R}$, and fix a limit point $c \in X'$.

- *Asymptotic dominance.*

  We say that f is **asymptotically dominated** by g near c if

  $$|f(x)| = \lambda(x)\,|g(x)|$$

  for some non-negative real-valued function $\lambda\colon X \to [0, \infty)$ and non-negative real number $0 \le L < \infty$ so that

  $$\lim_{x \to c} \lambda(x) = L.$$

  In this case, we write $f \preccurlyeq g$ as $x \to c$.

- *Asymptotic comparability.*

  We say that f and g are **asymptotically comparable** near c if

  $$|f(x)| = \lambda(x)\,|g(x)|$$

  for some non-negative real-valued function $\lambda\colon X \to [0,\infty)$ and positive real number $0 < L < \infty$ so that

  $$\lim_{x\to c} \lambda(x) = L.$$

  In this case, we write $f \asymp g$ as $x \to c$.

- *Asymptotic.*

  We say that f is **asymptotic** to g near c if

  $$|f(x)| = \lambda(x)\,|g(x)|$$

  for some non-negative real-valued function $\lambda\colon X \to [0,\infty)$ so that

  $$\lim_{x\to c} \lambda(x) = 1.$$

  In this case, we write $f \sim g$ as $x \to c$.

◆

Example 5.4.2 Asymptotic comparison. The following are examples of asymptotic dominance, asymptotic comparability, and asymptotics:

- Consider two power functions $f, g\colon (0,\infty) \to \mathbb{R}$ defined by the formulas $f(x) = ax^p$ and $g(x) = bx^q$ for some real numbers $a, b, p, q \in \mathbb{R}$ so that $a, b \neq 0$. Suppose that $p \leq q$, and consider the real-valued function $\lambda\colon (0,\infty) \to \mathbb{R}$ defined by the formula

  $$\lambda(x) = \left|\frac{a}{b}\right| x^{p-q}.$$

  We observe that

  $$\begin{aligned} |f(x)| &= |ax^p| = |a| \cdot |x|^p = |a|\,x^p \\ &= \frac{|a|}{|b|} x^{p-q} \cdot |b|\,x^q = \left|\frac{a}{b}\right| x^{p-q} \cdot |bx^q| = \lambda(x)\,|g(x)| \end{aligned}$$

  for all points $0 < x < \infty$, and

  $$\begin{aligned} \lim_{x\to\infty} \lambda(x) &= \lim_{x\to\infty} \left|\frac{a}{b}\right| x^{p-q} = \left|\frac{a}{b}\right| \lim_{x\to\infty} x^{p-q} \\ &= \left|\frac{a}{b}\right| \cdot 0 = 0, \end{aligned}$$

  since $p - q < 0$. We conclude that $f \preccurlyeq g$ as $x \to \infty$.

  On the other hand, $|g(x)| = \left(\frac{1}{\lambda(x)}\right) |f(x)|$ for all points $0 < x < \infty$, and

  $$\begin{aligned} \lim_{x\to 0} \frac{1}{\lambda(x)} &= \lim_{x\to 0} \frac{1}{\left|\frac{a}{b}\right| x^{p-q}} = \lim_{x\to 0} \left|\frac{b}{a}\right| x^{q-p} \\ &= \left|\frac{b}{a}\right| \lim_{x\to 0} x^{q-p} = \left|\frac{b}{a}\right| \cdot 0 = 0, \end{aligned}$$

  since $q - p > 0$. We conclude that $g \preccurlyeq f$ as $x \to 0$.

- Let $f\colon \mathbb{R}\setminus\{x_1,\dots,x_n\}\to\mathbb{R}$ be a rational function with formula

$$f(x)=\frac{a_m x^m+a_{m-1}x^{m-1}+\cdots+a_2x^2+a_1x+a_0}{b_nx^n+b_{n-1}x^{n-1}+\cdots+b_2x^2+b_1x+b_0}$$

in standard form, and consider the power functions $g,h\colon (0,\infty)\to\mathbb{R}$ defined by the formulas

$$g(x)=x^{m-n} \qquad\qquad h(x)=\left(\frac{a_m}{b_n}\right)x^{m-n}.$$

Let $\lambda\colon (0,\infty)\setminus\{x_1,\dots,x_n\}\to\mathbb{R}$ be the real-valued function defined by the formula

$$\lambda(x)=\frac{a_m+a_{m-1}x^{-1}+\cdots+a_2x^{2-m}a_1x^{1-m}+a_0x^{-m}}{b_n+b_{n-1}x^{-1}+\cdots+b_2x^{2-n}+b_1x^{1-n}+b_0x^{-n}}.$$

We observe that

$$\begin{aligned}\lambda(x)\left|g(x)\right| &= x^{m-n}\left|\frac{a_m+a_{m-1}x^{-1}+\cdots+a_2x^{2-m}a_1x^{1-m}+a_0x^{-m}}{b_n+b_{n-1}x^{-1}+\cdots+b_2x^{2-n}+b_1x^{1-n}+b_0x^{-n}}\right|\\ &= \left(\frac{x^m}{x^n}\right)\left|\frac{a_m+a_{m-1}x^{-1}+\cdots+a_2x^{2-m}a_1x^{1-m}+a_0x^{-m}}{b_n+b_{n-1}x^{-1}+\cdots+b_2x^{2-n}+b_1x^{1-n}+b_0x^{-n}}\right|\\ &= \left|\frac{a_mx^m+a_{m-1}x^{m-1}+\cdots+a_2x^2+a_1x+a_0}{b_nx^n+b_{n-1}x^{n-1}+\cdots+b_2x^2+b_1x+b_0}\right| = \left|f(x)\right|,\end{aligned}$$

and

$$\begin{aligned}\lim_{x\to\infty}\lambda(x) &= \lim_{x\to\infty}\frac{a_m+a_{m-1}x^{-1}+\cdots+a_2x^{2-m}a_1x^{1-m}+a_0x^{-m}}{b_n+b_{n-1}x^{-1}+\cdots+b_2x^{2-n}+b_1x^{1-n}+b_0x^{-n}}\\ &= \frac{a_m}{b_n}\in(0,\infty).\end{aligned}$$

We conclude that $f\asymp g$ as $x\to\infty$.

Moreover,

$$\begin{aligned}\left|\frac{b_n}{a_m}\right|\lambda(x)\left|h(x)\right| &= \left|\frac{b_n}{a_m}\right|\lambda(x)\left|\left(\frac{a_m}{b_n}\right)x^{m-n}\right|\\ &= \lambda(x)\left|g(x)\right| = \left|f(x)\right|,\end{aligned}$$

and

$$\lim_{x\to\infty}\left|\frac{b_n}{a_m}\right|\lambda(x)=\left|\frac{b_n}{a_m}\right|\lim_{x\to\infty}\lambda(x)=\left|\frac{b_n}{a_m}\right|\left|\frac{a_m}{b_n}\right|=1.$$

We conclude that $f\sim h$ as $x\to\infty$.

▲

One can see from the definitions that if two functions are asymptotic, then they are asymptotically comparable. Moreover, if two functions are asymptotically comparable, then each asymptotically dominates the other. In fact, the converse implication also holds.

Lemma 5.4.3 Asymptotic comparison. *Let $f,g\colon X\to\mathbb{R}$ be real-valued functions defined on a subset $X\subseteq\mathbb{R}$, and fix a limit point $c\in X'$. Then $f\asymp g$ as $x\to c$ if and only if both $f\preccurlyeq g$ and $g\preccurlyeq f$ as $x\to c$.*

Proof. First suppose that $f \asymp g$ as $x \to c$. Then there is a non-negative real-valued function $\lambda \colon X \to [0, \infty)$ and a positive real number $0 < L < \infty$ so that $|f(x)| = \lambda(x) |g(x)|$ for all $x \in X$ and

$$\lim_{x \to c} \lambda(x) = L.$$

Since $0 \leq L < \infty$, we conclude that $f \preccurlyeq g$ as $x \to c$. Moreover, we observe that

$$|g(x)| = |g(x)| \left(\frac{\lambda(x)}{\lambda(x)} \right) = \left(\frac{1}{\lambda(x)} \right) |f(x)|,$$

and

$$\lim_{x \to c} \frac{1}{\lambda(x)} = \frac{1}{\lim_{x \to c} \lambda(x)} = \frac{1}{L} \in (0, \infty),$$

so that $g \preccurlyeq f$ as $x \to c$.

Now suppose that both $f \preccurlyeq g$ and $g \preccurlyeq f$ as $x \to c$. Then there are non-negative real-valued functions $\lambda, \mu \colon X \to [0, \infty)$ and non-negative real numbers $L, M \in [0, \infty)$ so that $|f(x)| = \lambda(x) |g(x)|$ and $|g(x)| = \mu(x) |f(x)|$ for all $x \in X$ and

$$\lim_{x \to c} \lambda(x) = L \qquad \lim_{x \to c} \mu(x) = M.$$

We first observe that for any point $x \in X$, if any one of $\lambda(x)$, $\mu(x)$, $f(x)$, or $g(x)$ is 0, then $f(x) = g(x) = 0$.

There are two cases: either there is a sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X \backslash \{c\}$ converging to c so that $f(x_n), g(x_n) \neq 0$ for all $n \in \mathbb{N}$, or no such sequence exists. If there is such a sequence, we observe that

$$\begin{aligned} \lambda(x_n) \mu(x_n) = \lambda(x_n) \mu(x_n) \left| \frac{f(x_n) g(x_n)}{f(x_n) g(x_n)} \right| &= \left| \frac{\lambda(x_n) f(x_n)}{g(x_n)} \right| \cdot \left| \frac{\mu(x_n) g(x_n)}{f(x_n)} \right| \\ = \left| \frac{g(x_n)}{g(x_n)} \right| \cdot \left| \frac{f(x_n)}{f(x_n)} \right| = 1 \end{aligned}$$

for all $n \in \mathbb{N}$, and so

$$\begin{aligned} LM &= \left(\lim_{n \to \infty} \lambda(x_n) \right) \left(\lim_{n \to \infty} \mu(x_n) \right) \\ &= \lim_{n \to \infty} \lambda(x_n) \mu(x_n) = \lim_{n \to \infty} 1 = 1. \end{aligned}$$

In particular, we conclude that $L, M \neq 0$ are non-zero, so that $f \asymp g$.

On the other hand, if no such sequence exists, then $f(x) = g(x) = 0$ for all $x \in X$ sufficiently close to c. Let $\nu \colon X \to \mathbb{R}$ be the real-valued function defined piecewise by the formula

$$\nu(x) = \begin{cases} \lambda(x) & f(x) \neq 0 \\ 1 & f(x) = 0. \end{cases}$$

Then $\nu(x) |f(x)| = |g(x)|$ and

$$\lim_{x \to c} \nu(x) = 1,$$

and so $f \asymp g$. ■

Definition 5.4.4 Strict asymptotic dominance. Let $f, g \colon X \to \mathbb{R}$ be real-valued functions defined on a subset $X \subseteq \mathbb{R}$, and fix a limit point $c \in X'$. We say that f is **strictly asymptotically dominated** by g near c if $f \preccurlyeq g$

but $f \not\asymp g$ as $x \to c$. In this case, we write $f \prec g$ as $x \to c$. ◆

We will shortly find the following characterization of strict asymptotic dominance useful:

Proposition 5.4.5 Strict asymptotic dominance. *Let $f, g\colon X \to \mathbb{R}$ be real-valued functions defined on a subset $X \subseteq \mathbb{R}$, and fix a limit point $c \in X'$. If $g(x) \neq 0$ for all $x \in X \setminus \{c\}$, then $f \prec g$ if and only if*

$$\lim_{x \to c} \frac{f(x)}{g(x)} = 0.$$

Proof. Let $\lambda\colon X \to [0, \infty)$ be the non-negative real-valued function defined by the formula

$$\lambda(x) = \left| \frac{f(x)}{g(x)} \right|.$$

Then $|f(x)| = \lambda(x)\,|g(x)|$ for all $x \in X$, and

$$\lim_{x \to c} \lambda(x) = \lim_{x \to c} \left| \frac{f(x)}{g(x)} \right| = \left| \lim_{x \to c} \frac{f(x)}{g(x)} \right|.$$

We conclude that $f \prec g$ if and only if

$$\lim_{x \to c} \frac{f(x)}{g(x)} = 0.$$

■

While we will mainly utilize the language of asymptotic comparison to describe the error in a Taylor approximation, these ideas are ubiquitous throughout mathematical analysis and applied mathematics.

5.4.2 Polynomial Approximations of Differentiable Functions

We now arrive at one of the fundamental results in differential calculus. **Taylor's theorem** establishes a best possible polynomial approximation p for a multiply differentiable function f near a given point. We quantify the quality of this approximation by asymptotically comparing the difference $f - p$ to well-known power functions vanishing at c at high order.

Theorem 5.4.6 Taylor's theorem. (Brook Taylor) *Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on a subset $X \subseteq \mathbb{R}$. If f is k-times differentiable at a point $c \in X$, then there is a unique polynomial function p of degree at most k so that $f(x) - p(x) \prec (x - c)^k$.*

Proof. For uniqueness, let $p, q\colon \mathbb{R} \to \mathbb{R}$ be polynomial functions of degree at most k, and suppose that $f(x) - p(x) \prec (x - c)^k$ and $f(x) - q(x) \prec (x - c)^k$. Then

$$0 \leq \left| \frac{p(x) - q(x)}{(x - c)^k} \right| \leq \left| \frac{f(x) - p(x)}{(x - c)^k} \right| + \left| \frac{f(x) - q(x)}{(x - c)^k} \right|$$

for all $x \in X$. The Theorem 3.3.3 Squeeze theorem now implies that

$$\lim_{x \to c} \left| \frac{p(x) - q(x)}{(x - c)^k} \right| = 0,$$

so that $p(x) - q(x) \prec (x - c)^k$ as $x \to c$. This polynomial $p(x) - q(x)$ can be written in the form

$$p(x) - q(x) = a_0 + a_1(x-c) + a_2(x-c)^2 + \cdots + a_k(x-c)^k$$

for some real numbers $a_0, \ldots, a_k \in \mathbb{R}$. For each index $0 \leq j \leq k$, we observe that

$$\lim_{x \to c} \left| \frac{p(x) - q(x)}{(x-c)^j} \right| = \left(\lim_{x \to c} \left| \frac{p(x) - q(x)}{(x-c)^k} \right| \right) \left(\lim_{x \to c} \left| (x-c)^{k-j} \right| \right) = 0 \cdot 0 = 0.$$

If $j = 0$, then

$$\begin{aligned} 0 &= \lim_{x \to c} \left| \frac{p(x) - q(x)}{(x-c)^0} \right| = \lim_{x \to c} |p(x) - q(x)| \\ &= \lim_{x \to c} \left| a_0 + a_1(x-c) + a_2(x-c)^2 + \cdots + a_k(x-c)^k \right| = |a_0|. \end{aligned}$$

If $j = 1$, then

$$\begin{aligned} 0 &= \lim_{x \to c} \left| \frac{p(x) - q(x)}{(x-c)^1} \right| = \lim_{x \to c} \left| \frac{p(x) - q(x)}{x-c} \right| \\ &= \lim_{x \to c} \left| a_1 + a_2(x-c)^1 + \cdots + a_k(x-c)^{k-1} \right| = |a_1|. \end{aligned}$$

Continuing in this manner, we conclude that $a_0 = a_1 = \cdots = a_k = 0$, so that $p = q$.

For existence, let $p \colon \mathbb{R} \to \mathbb{R}$ be the polynomial function defined by the formula

$$p(x) = f(c) + f'(c)(x-c) + \frac{f''(c)(x-c)^2}{2} + \cdots + \frac{f^{(k)}(c)(x-c)^k}{k!}.$$

We observe that $p^{(j)}(c) = f^{(j)}(c)$ for all $0 \leq j \leq c$. We will show that $f(x) - p(x) \prec (x-c)^k$ by repeated applications of L'Hôpital's rule. Indeed,

$$\begin{aligned} \lim_{x \to c} \frac{f(x) - p(x)}{(x-c)^k} &= \lim_{x \to c} \frac{f'(x) - p'(x)}{k(x-c)^{k-1}} = \lim_{x \to c} \frac{f''(x) - p''(x)}{k(k-1)(x-c)^{k-2}} \\ &= \cdots = \lim_{x \to c} \frac{f^{(k-1)}(x) - p^{(k-1)}(x)}{k!(x-c)} \\ &= \left(\frac{1}{k!} \right) \left(\lim_{x \to c} \frac{f^{(k-1)}(x) - f^{(k-1)}(c)}{x-c} - \lim_{x \to c} \frac{p^{(k-1)}(x) - p^{(k-1)}(c)}{x-c} \right) \\ &= \frac{f^{(k)}(c) - p^{(k)}(c)}{k!} = \frac{0}{k!} = 0, \end{aligned}$$

so that $f(x) - p(x) \prec (x-c)^k$. ■

In summary, if a real-valued function f is k-times differentiable at a point c, then the unique polynomial of degree at most k with the same value, and derivatives as f up to order k is the best possible polynomial approximation of f near c. This polynomial is called the **Taylor approximation** of f at c.

Definition 5.4.7 Taylor approximation/remainder. Let $f \colon X \to \mathbb{R}$ be a real-valued function defined on a subset $X \subseteq \mathbb{R}$. If f is k-times differentiable at a point $c \in X$, then the unique polynomial $P_c^k(f, x)$ of degree at most k so that $f(x) - P_c^k(f, x) \prec (x-c)^k$ is called the order k **Taylor approximation** of f

at c, and $R_c^k(f,x) = f(x) - P_c^k(f,x)$ is called the order k **Taylor remainder** of f at c. ◆

The content of Taylor's theorem is not in finding the Taylor approximation, but rather in bounding the Taylor remainder. Under stronger hypotheses on f, one can find explicit formulas for the Taylor remainder. For example, the following result utilizes Theorem 5.3.5 Extended mean value theorem to compute a family of such an explicit formulas.

Theorem 5.4.8 Mean value form of the Taylor remainder. *Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on an interval $X \subseteq \mathbb{R}$, and suppose that f is k-times differentiable at an interior point $c \in \operatorname{Int}(X)$, with order k Taylor approximation $P_c^k(f,x)$ and order k Taylor remainder $R_c^k(f,x)$.*

Fix a point $x \in X$, and a real-valued function $G\colon I_x \to \mathbb{R}$ defined on the closed interval I_x between c and x. If both $f^{(k)}$ and G are continuous on I_x and differentiable on the open interval $\operatorname{Int}(I_x)$, and if G has no critical points on $\operatorname{Int}(I_x)$, then

$$R_c^k(f,x) = \left(\frac{f^{(k+1)}(y)(x-y)^k}{k!}\right)\left(\frac{G(x)-G(c)}{G'(y)}\right)$$

for some interior point $y \in \operatorname{Int}(I_x)$.

Proof. Consider the real-valued function $F\colon X \to \mathbb{R}$ defined by the formula

$$F(y) = f(y) + f'(y)(x-y) + \frac{f''(y)(x-y)^2}{2} + \cdots + \frac{f^{(k)}(y)(x-y)^k}{k!}.$$

We observe that F is continuous on the closed interval I_x and differentiable on the open interval $\operatorname{Int}(I_x)$, with derivative

$$\begin{aligned}F'(y) = f'(y) + (f''(y)(x-y) - f'(y)) + \left(\frac{f^{(3)}(y)(x-y)^2}{2} - f''(y)(x-y)\right) \\ + \cdots + \left(\frac{f^{(k+1)}(y)(x-y)^k}{k!} - \frac{f^{(k)}(y)(x-y)^{k-1}}{(k-1)!}\right) \\ = \frac{f^{(k+1)}(y)(x-y)^k}{k!}.\end{aligned}$$

By the Theorem 5.3.5 Extended mean value theorem, there is some interior point $y \in \operatorname{Int}(I_x)$ so that

$$F'(y)(G(x)-G(c)) = (F(x)-F(c))G'(y).$$

We observe that $F(x) = f(x)$ and $F(c) = P_c^k(f,x)$, so that

$$\begin{aligned}R_c^k(f,x) &= f(x) - P_c^k(f,x) = F(x) - F(c) \\ &= F'(y)\left(\frac{G(x)-G(c)}{G'(y)}\right) \\ &= \left(\frac{f^{(k+1)}(y)(x-y)^k}{k!}\right)\left(\frac{G(x)-G(c)}{G'(y)}\right).\end{aligned}$$

■

The utility of Theorem 5.4.8 Mean value form of the Taylor remainder is in choosing specific functions G in order to obtain particularly nice explicit formulas for the Taylor remainder. For example, we obtain the **Lagrange form** of the Taylor remainder, which is a higher order generalization of the Mean value theorem.

Corollary 5.4.9 Lagrange form of the Taylor remainder. (Joseph-Louis Lagrange) *Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on an interval $X \subseteq \mathbb{R}$, and suppose that f is k-times differentiable at an interior point $c \in \operatorname{Int}(X)$, with order k Taylor approximation $P_c^k(f,x)$ and order k Taylor remainder $R_c^k(f,x)$.*

Fix a point $x \in X$. If $f^{(k)}$ is continuous on the closed interval I_x between c and x and differentiable on the open interval $\operatorname{Int}(I_x)$, then

$$R_c^k(f,x) = \frac{f^{(k+1)}(y)(x-c)^{k+1}}{(k+1)!}$$

for some interior point $y \in \operatorname{Int}(I_x)$.

Proof. This follows immediately from Theorem 5.4.8 Mean value form of the Taylor remainder by taking $G(y) = (x-y)^{k+1}$. Indeed, in this case,

$$\begin{aligned} R_c^k(f,x) &= \left(\frac{f^{(k+1)}(y)(x-y)^k}{k!}\right)\left(\frac{G(x)-G(c)}{G'(y)}\right) \\ &= \left(\frac{f^{(k+1)}(y)(x-y)^k}{k!}\right)\left(\frac{-(x-c)^{k+1}}{(k+1)(x-y)^k(-1)}\right) \\ &= \frac{f^{(k+1)}(y)(x-c)^{k+1}}{(k+1)!} \end{aligned}$$

for some interior point $y \in \operatorname{Int}(I_x)$. ■

We observe that Corollary 5.4.9 Lagrange form of the Taylor remainder is precisely the Mean value theorem when $k = 0$. Other important forms of the Taylor remainder can be found by choosing different functions G. You will have an opportunity to explore the **Cauchy form** of the Taylor remainder in the problems for this section.

In Chapter 7 Infinite Series, we will explore **Taylor series**, which are related approximations which utilize all derivatives of an infinitely differentiable function.

5.4.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

5.4.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Taylor approximations of non-analytic functions. Let $f\colon \mathbb{R} \to \mathbb{R}$ be the real-valued function defined piecewise by the formula

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x > 0 \\ 0 & x \leq 0. \end{cases}$$

(a) Prove that $u^{\frac{n}{2}} \prec e^u$ as $u \to \infty$ for all natural numbers $n \in \mathbb{N}$.

(b) Prove that $f(x) \prec x^n$ as $x \to 0$ for all natural numbers $n \in \mathbb{N}$.

Hint. Make the substitution $u = \frac{1}{x^2}$.

(c) Compute all Taylor approximations $P_0^n(f, x)$ of f near 0.

(d) Let $g \in \mathrm{C}^{\infty}(X)$ be a smooth function defined on an interval $X \subseteq \mathbb{R}$. If there is a point $c \in X$ so that $g^{(n)}(c) = 0$ for all positive integers $n \geq 1$, can we conclude that g is constant on a neighborhood of c?

f is our first example of a smooth function which is not **analytic**. We will explore these functions more in Chapter 7 Infinite Series.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

Text after the last section.

Chapter 6

Riemann Integration

In this chapter, we initiate our study of integral calculus by constructing the Riemann integral and examining its properties. Both because the Riemann integral does not allow for the integration of a wide array of functions and because it does not behave well under limits, we will also begin to prosecute a case for a stronger theory of integration which will be explored further in `[provisional cross-reference: part-measure-and-integration]`.

6.1 Riemann Sums and Integrals

In this section, we construct the **Riemann integral** as a limit of **Riemann sums**. We prove several equivalent characterizations of integrability and explore some preliminary properties of integration.

6.1.1 Finite Integration Methods

Before we introduce the **Riemann integral** of a function, we must develop some preliminary results about **Riemann sums**. These sums are associated to a function and a **tagged partition** of a closed interval, which both decomposes the interval into a union of closed subintervals and specifies a point in each subinterval.

Definition 6.1.1 Partitions. Fix real numbers $a, b \in \mathbb{R}$ so that $a \leq b$.

- *Partition.*

  A **partition** P of the closed interval $[a, b]$ is an increasing finite sequence $P = (x_k)_{k=0}^{n}$ whose first entry is $x_0 = a$ and whose last entry is $x_n = b$; that is,

  $$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b.$$

  The **mesh** $\operatorname{mesh}(P)$ of such a partition $P = (x_k)_{k=0}^{n}$ is the maximal distance between any adjacent entries of P; that is,

  $$\operatorname{mesh}(P) = \max_{0 \leq k < n} |x_{k+1} - x_k| = \max_{0 \leq k < n} (x_{k+1} - x_k).$$

- *Tagged partition.*

  A **tagged partition** of the closed interval $[a, b]$ is an ordered pair (P, t), where $P = (x_k)_{k=0}^{n}$ is a partition of $[a, b]$ and $t = (t_k)_{k=0}^{n-1}$ is a finite sequence of points $x_k \leq t_k \leq x_{k+1}$.

◆

Example 6.1.2 Tagged partitions. The following are examples of tagged partitions:

- *Left and right tagged partitions.*

  Let $P = (x_k)_{k=0}^{n}$ be a partition of a closed interval $[a, b]$. The associated **left tagged** partition (P, l) is defined by the formula $l_k = x_k$ for all indices $0 \leq k < n$. Similarly, the associated **right tagged** partition (P, r) is defined by the formula $r_k = x_{k+1}$ for all indices $0 \leq k < n$.

  For example, let $n \geq 1$ be a positive integer, and consider the partition $P = (x_k)_{k=0}^{n}$ of the closed interval $[0, 1]$ defined by the formula $x_k = \frac{k}{n}$. The associated left tagged partition is (P, l), where $l = (l_k)_{k=0}^{n-1}$ is given by the formula
  $$l_k = x_k = \frac{k}{n},$$
  and the associated right tagged partition is (P, r), where $r = (r_k)_{k=0}^{n-1}$ is given by the formula
  $$r_k = x_{k+1} = \frac{k+1}{n}.$$
  We note that the mesh $\operatorname{mesh}(P)$ does not depend on the tagging, and is $\operatorname{mesh}(P) = \frac{1}{n}$ in this example.

- *Centrally tagged partitions.*

  Let $P = (x_k)_{k=0}^{n}$ be a partition of a closed interval $[a, b]$. The associated **centrally tagged** partition (P, c) is defined by the formula
  $$c_k = \frac{x_k + x_{k+1}}{2}$$
  for all indices $0 \leq k < n$.

  For example, let $n \geq 1$ be a positive integer, and consider the partition $P = (x_k)_{k=0}^{2^n}$ of the closed interval $[0, 1]$ defined by the formula $x_k = k2^{-n}$. The associated left tagged partition is (P, c), where $c = (c_k)_{k=0}^{2^n-1}$ is given by the formula
  $$l_k = \frac{x_k + x_{k+1}}{2} = \frac{k2^{-n} + (k+1)\,2^{-k}}{2}.$$
  We note that the mesh $\operatorname{mesh}(P)$ does not depend on the tagging, and is $\operatorname{mesh}(P) = 2^{-n}$ in this example.

▲

We remark that it is not required for the subintervals determined by a partition to be of equal length, although on occasion such a requirement makes it easier to compute associated quantities. Moreover, it is not required that the tagging of each subinterval be done in a coherent way.

Definition 6.1.3 Comparison of partitions. Let $P = (x_j)_{j=0}^{m}$ and $Q = (y_k)_{k=0}^{n}$ be partitions of a closed interval $[a, b]$. If $\{x_j\}_{j=0}^{m} \subseteq \{y_k\}_{k=0}^{n}$, then we write $P \preceq Q$ and say that P is **coarser** than Q; that Q is **finer** than P; Q is a **refinement** of P; and/or that P is a coarsening of Q.

Let (P, s) and (Q, t) be tagged partitions of the closed interval $[a, b]$, where $P = (x_j)_{j=0}^{m}$, $Q = (y_k)_{k=0}^{n}$, $s = (s_j)_{j=0}^{m-1}$, and $t = (t_k)_{k=0}^{n-1}$. If both $\{x_j\}_{j=0}^{m} \subseteq \{y_k\}_{k=0}^{n}$ and $\{s_j\}_{j=0}^{m-1} \subseteq \{t_k\}_{k=0}^{n-1}$, then we write $(P, s) \preceq (Q, t)$ and say that (P, s) is **coarser** than (Q, t); that (Q, t) is **finer** than (P, s); (Q, t) is a **refinement** of (P, s); and/or that (P, s) is a coarsening of (Q, t). ◆

Both partitions and tagged partitions are partially ordered by containment. In fact, the sets of partitions and tagged partitions of a closed interval form an upwards directed set, in that any pair of such (tagged) partitions has a common refinement.

Lemma 6.1.4 Partitions are directed. *The sets of partitions and tagged partitions of a closed interval are directed upwards by the partial order* $\preceq$*.*

Proof. Let $P = (x_j)_{j=0}^m$ and $Q = (y_k)_{k=0}^n$ be partitions of a closed interval. Let $R = (z_l)_{l=0}^p$ be an increasing sequence which enumerates

$$\{z_l\}_{l=0}^p = \{x_j\}_{j=0}^m \cup \{y_k\}_{k=0}^n .$$

Since

$$\begin{aligned}\min_{0\le l\le p} z_l &= \min\left(\left\{\min_{0\le j\le m} x_j, \min_{0\le k\le n} y_k\right\}\right)\\ &= \min\left(\{a, a\}\right) = a\end{aligned}$$

and

$$\begin{aligned}\max_{0\le l\le p} z_l &= \max\left(\left\{\max_{0\le j\le m} x_j, \max_{0\le k\le n} y_k\right\}\right)\\ &= \max\left(\{b, b\}\right) = b,\end{aligned}$$

R is a partition of $[a, b]$. Moreover, R was constructed to be a common refinement of P and Q.

If P and Q were tagged, then we note that each subinterval determined by R is contained in a subinterval determined by P and a subinterval determined by Q. If each such subinterval contains at most one tag from either P or Q, we may choose arbitrary tags in other subintervals. Otherwise, we may subdivide further so that this process produces a tagging of R which is a refinement of those on P and Q. ■

Associated to each real-valued function and tagged partition is a **Riemann sum**. This sum is meant to approximate the signed area enclosed by the graph of said function, and does so with varying success depending on the integrability of the function and the **mesh** of the partition.

Definition 6.1.5 Riemann sum. Let $f\colon [a, b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a, b]$. The **Riemann sum** $\mathcal{R}_f(P, t)$ of f associated to a tagged partition (P, t) of $[a, b]$ is the real number

$$\mathcal{R}_f(P, t) = \sum_{k=0}^{n-1} f(t_k)(x_{k+1} - x_k),$$

where $P = (x_k)_{k=0}^n$ and $t = (t_k)_{k=0}^{n-1}$. ◆

Example 6.1.6 Riemann sums. The following are examples of Riemann sums:

- *Left and right Riemann sums.*

  The **left Riemann sum** of a function f associated to a partition P of a closed interval $[a, b]$ is the Riemann sum $\mathcal{R}_f(P, l)$ of f associated to the associated left tagged partition (P, l) of $[a, b]$. Similarly, the **right Riemann sum** of f associated P is the Riemann sum $\mathcal{R}_f(P, r)$ of f

associated to the associated right tagged partition (P, r) of $[a, b]$.

For example, consider the function $f\colon [0, 16] \to \mathbb{R}$ defined by the formula $f(x) = \sqrt{x}$ and the partition $P = (0, 1, 4, 9, 16)$ of the closed interval $[0, 16]$. The left Riemann sum of f associated to P is

$$\begin{aligned}\mathcal{R}_f(P, l) &= f(0)(1-0) + f(1)(4-1) + f(4)(9-4) + f(9)(16-9)\\ &= 0 \cdot 1 + 1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 = 0 + 3 + 10 + 21 = 34,\end{aligned}$$

while the right Riemann sum of f associated to P is

$$\begin{aligned}\mathcal{R}_f(P, r) &= f(1)(1-0) + f(4)(4-1) + f(9)(9-4) + f(16)(16-9)\\ &= 1 \cdot 1 + 2 \cdot 3 + 3 \cdot 5 + 4 \cdot 7 = 1 + 6 + 15 + 28 = 50.\end{aligned}$$

- *Central Riemann sums.*

  The **central Riemann sum** of a function f associated to a partition P of a closed interval $[a, b]$ is the Riemann sum $\mathcal{R}_f(P, c)$ of f associated to the associated central tagged partition (P, c) of $[a, b]$.

  For example, consider the function $g\colon [0, 100] \to \mathbb{R}$ defined by the formula $g(x) = 3x - 25$ and the partition $P = (0, 10, 50, 70, 100)$ of the closed interval $[0, 16]$. The central Riemann sum of g associated to P is

$$\begin{aligned}\mathcal{R}_g(P, c) &= g(5)(10-0) + g(30)(50-10) + g(60)(70-50) + g(85)(100-70)\\ &= (-10) \cdot 10 + 65 \cdot 40 + 155 \cdot 20 + 230 \cdot 30 = 12{,}500.\end{aligned}$$

▲

Related to the notion of Riemann sums are the **Darboux sums**. These sums are associated to an untagged partition. Rather than using the value of the function at a chosen point, the height of each term of a Darboux sum is determined by the supremum or infimum of the function over the given subinterval.

Definition 6.1.7 Darboux sum. Let $f\colon [a, b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a, b]$.

- *Upper Darboux sum.*

  The **upper Darboux sum** $\mathcal{U}_f(P)$ of f associated to a partition $P = (x_k)_{k=0}^n$ of $[a, b]$ is the extended real number

$$\mathcal{U}_f(P) = \sum_{k=0}^{n-1} \sup_{x_k \le t \le x_{k+1}} f(t)(x_{k+1} - x_k)$$

- *Lower Darboux sum.*

  The **upper Darboux sum** $\mathcal{L}_f(P)$ of f associated to a partition $P = (x_k)_{k=0}^n$ of $[a, b]$ is the extended real number

$$\mathcal{L}_f(P) = \sum_{k=0}^{n-1} \inf_{x_k \le t \le x_{k+1}} f(t)(x_{k+1} - x_k)$$

◆

We note that if the integrand is continuous, then Darboux sums are Riemann sums. However, Darboux sums need not be Riemann sums in general. We will show that the Riemann integral is also a limit of Darboux sums. In order to prove such a result, we will require the following properties:

Proposition 6.1.8 Properties of Darboux sums. *Let $f\colon [a,b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a,b]$.*

1. *Monotonicity.*

   For any partitions P and Q of $[a,b]$, if $P \preceq Q$, then $\mathcal{U}_f(P) \geq \mathcal{U}_f(Q)$ and $\mathcal{L}_f(P) \leq \mathcal{L}_f(Q)$.

2. *Bounds on Riemann sums.*

   For any tagged partition (P,t) of $[a,b]$,

$$\mathcal{L}_f(P) \leq \mathcal{R}_f(P,t) \leq \mathcal{U}_f(P).$$

Proof.

1. You will have an opportunity to prove this in the problems for this section.

2. Let $P = (x_k)_{k=0}^{n}$ and $t = (t_k)_{k=0}^{n-1}$. We observe that

$$\begin{aligned}\inf_{x_k \leq x \leq x_{k+1}} f(x)(x_{k+1}-x_k) &\leq f(t_k)(x_{k+1}-x_k) \\ &\leq \sup_{x_k \leq x \leq x_{k+1}} f(x)(x_{k+1}-x_k)\end{aligned}$$

   for all indices $0 \leq k < n$, and so

$$\begin{aligned}\sum_{k=0}^{n-1} \inf_{x_k \leq x \leq x_{k+1}} f(x)(x_{k+1}-x_k) &\leq \sum_{k=0}^{n-1} f(t_k)(x_{k+1}-x_k) \\ &\leq\leq \sum_{k=0}^{n-1} \sup_{x_k x \leq x_{k+1}} f(x)(x_{k+1}-x_k);\end{aligned}$$

   that is, $\mathcal{L}_f(P) \leq \mathcal{R}_f(P,t) \leq \mathcal{U}_f(P)$.

■

While Riemann sums and Darboux sums may give decent approximations of area computations, we will need to involve limits in order to perform precise computations.

6.1.2 Riemann Integrability

We now explore the Riemann integrability of various functions and introduce the Riemann integral. We will see several equivalent characterizations of this property, expressed as conditions on limits of certain nets of Riemann sums and Darboux sums.

Theorem 6.1.9 Riemann integrability. *Let $f\colon [a,b] \to \mathbb{R}$ be a bounded real-valued function defined on a closed interval $[a,b]$, and let $I \in \mathbb{R}$ be a real number. The following are equivalent:*

1. *For any positive real number $\epsilon > 0$, there is a positive real number $\delta > 0$ so that*

$$|\mathcal{R}_f(P,t) - I| < \epsilon$$

   for all tagged partitions (P,t) of $[a,b]$ of mesh $\operatorname{mesh}(P) < \delta$.

2. *The net $\mathcal{R}_f = (\mathcal{R}_f(P,t))_{(P,t)}$ of real numbers which associates to each*

tagged partition (P, t) of $[a, b]$ the corresponding Riemann sum of f converges upwards to I.

3. *The nets $\mathcal{U}_f = (\mathcal{U}_f(P))_P$ and $\mathcal{L}_f = (\mathcal{L}_f(P))_P$ of real numbers which associate to each partition P of $[a, b]$ the corresponding upper and lower Darboux sums of f converge upwards to I.*

Proof. Let $B = \sup_{a \leq x \leq b} |f(x)|$. If $B = 0$, then $f(x) = 0$ for all $a \leq x \leq b$, and so each of the three above properties holds. Thus we may suppose for the remainder of the proof that $B > 0$. First suppose that for any positive real number $\epsilon > 0$, there is a positive real number $\delta > 0$ so that

$$|\mathcal{R}_f(P, t) - I| < \epsilon$$

for all tagged partitions (P, t) of $[a, b]$ of mesh $\mathrm{mesh}(P) < \delta$. Let $N \in \mathcal{N}_{\mathbb{R}}(I)$ be a neighborhood of I in $\mathbb{R}$, and note that $\mathrm{B}_\epsilon(I) \subseteq N$ for some positive real number $\epsilon > 0$. Choose a tagged partition (P_ϵ, t_ϵ) of mesh $\mathrm{mesh}(P_\epsilon) < \delta$. Then for all refinements $(P, t) \succeq (P_\epsilon, t_\epsilon)$ of (P_ϵ, t_ϵ), $\mathrm{mesh}(P) \leq \mathrm{mesh}(P_\epsilon) < \delta$, and so

$$|\mathcal{R}_f(P, t) - I| < \epsilon.$$

We conclude that the net $\mathcal{R}_f = (\mathcal{R}_f(P, t))_{(P,t)}$ converges upwards to I.

Now suppose that the net $\mathcal{R}_f = (\mathcal{R}_f(P, t))_{(P,t)}$ converges upwards to I. Since the nets $\mathcal{U}_f = (\mathcal{U}_f(P))_P$ and $\mathcal{L}_f = (\mathcal{L}_f(P))_P$ are monotone and bounded by Proposition 6.1.8 Properties of Darboux sums, they converge to real numbers

$$U = \lim_{P\uparrow} \mathcal{U}_f(P) = \inf_P \mathcal{U}_f(P)$$
$$L = \lim_{P\uparrow} \mathcal{L}_f(P) = \sup_P \mathcal{U}_f(P)$$

by a net analogue of the Monotone convergence theorem. So it suffices to show $I = U = L$. We first observe that for any partition P of $[a, b]$ and positive real number $\epsilon > 0$, there is some tagged partition (Q, t) so that $P \preceq Q$ and $|\mathcal{R}_f(Q, t) - I| < \epsilon$. In particular,

$$\mathcal{L}_f(P) \leq \mathcal{L}_f(Q) \leq \mathcal{R}_f(Q, t) < I + \epsilon,$$

and

$$\mathcal{U}_f(P) \geq \mathcal{U}_f(Q) \geq \mathcal{R}_f(Q, t) > I - \epsilon.$$

Since this holds for all partitions P and all positive real numbers $\epsilon > 0$,

$$L = \sup_P \mathcal{L}_f(P) \leq I \leq \inf_P \mathcal{U}_f(P) = U.$$

It remains to show that $U \leq I \leq L$. We will show that $U \leq I$; the proof that $I \leq L$ is similar.

To that end, let $\epsilon > 0$ be a positive real number. Then there is some tagged partition (P_ϵ, t_ϵ) of $[a, b]$ so that

$$|\mathcal{R}_f(P, t) - I| < \frac{\epsilon}{3}$$

for all refinements $(P, t) \succeq (P_\epsilon, s_\epsilon)$. In particular, by further refining (P_ϵ, s_ϵ) we may assume without loss of generality that it is left tagged. Let $P_\epsilon = (x_k)_{k=0}^n$. We aim to define a further refinement of P_ϵ.

For each index $0 \leq k < n$, there is some point $x_k \leq u_k \leq x_{k+1}$ so that

$$f(u_k) + \frac{\epsilon}{3(b-a)} > \sup_{x_k \leq x \leq x_{k+1}} f(x).$$

If we may choose $u_k = x_k$, do so. Otherwise, we subdivide $[x_k, x_{k+1}]$ into the union of two closed intervals, one of length

$$L_k \leq \frac{\epsilon}{6Bn}$$

containing x_k and the other of length $x_{k+1} - x_k - L_k$ containing u_k. Tagging these intervals with x_k and u_k, respectively, we obtain a further refinement $(Q_\epsilon, t_\epsilon) \succeq (P_\epsilon, s_\epsilon)$. Moreover, we may replace the tagging s_ϵ with $u = (u_k)_{k=0}^{n-1}$ to obtain a tagged partition (P_ϵ, u). We observe that

$$\begin{aligned}
\mathcal{R}_f(Q_\epsilon, t_\epsilon) - \mathcal{R}_f(P_\epsilon, u) &= \sum_{k=0}^{n-1} f(x_k) L_k + f(u_k)(x_{k+1} - x_k - L_k) \\
&\quad - \sum_{k=0}^{n-1} f(u_k)(x_{k+1} - x_k) \\
&= \sum_{k=0}^{n-1} (f(x_k) - f(u_k)) L_k.
\end{aligned}$$

In particular,

$$\begin{aligned}
|\mathcal{R}_f(Q_\epsilon, t_\epsilon) - \mathcal{R}_f(P_\epsilon, u)| &= \left| \sum_{k=0}^{n-1} (f(x_k) - f(u_k)) L_k \right| \\
&\leq \sum_{k=0}^{n-1} |f(x_k) - f(u_k)| L_k \\
&\leq \sum_{k=0}^{n-1} (|f(x_k)| + |f(u_k)|) L_k \\
&\leq \sum_{k=0}^{n-1} \frac{2B\epsilon}{6Bn} = \frac{\epsilon}{3}.
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathcal{R}_f(P_\epsilon, u) &= \sum_{k=0}^{n-1} f(u_k)(x_{k+1} - x_k) \\
&> \sum_{k=0}^{n-1} \left(\sup_{x_k \leq x \leq x_{k+1}} f(x) - \frac{\epsilon}{3(b-a)} \right) (x_{k+1} - x_k) \\
&= \sum_{k=0}^{n-1} \left(\sup_{x_k \leq x \leq x_{k+1}} f(x)(x_{k+1} - x_k) - \left(\frac{\epsilon}{3(b-a)} \right) (x_{k+1} - x_k) \right) \\
&= \mathcal{U}_f(P_\epsilon) - \frac{\epsilon}{3}
\end{aligned}$$

so that $|\mathcal{U}_f(P_\epsilon) - \mathcal{R}_f(P_\epsilon, u)| < \frac{\epsilon}{3}$. We conclude that

$$\begin{aligned}
|\mathcal{U}_f(P_\epsilon) - I| &\leq |\mathcal{U}_f(P_\epsilon) - \mathcal{R}_f(P_\epsilon, u)| + |\mathcal{R}_f(P_\epsilon, u) - \mathcal{R}_f(Q_\epsilon, t_\epsilon)| \\
&\quad + |\mathcal{R}_f(Q_\epsilon, t_\epsilon) - I| \\
&< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon.
\end{aligned}$$

In particular,

$$U = \inf_P \mathcal{U}_f(P) \leq \mathcal{U}_f(\mathcal{P}_\epsilon) < I + \epsilon;$$

since ϵ was an arbitrary positive real number, $U \leq I$.

Finally, now suppose that the nets $\mathcal{U}_f = (\mathcal{U}_f(P))_P$ and $\mathcal{L}_f = (\mathcal{L}_f(P))_P$ converge upwards to I, and let $\epsilon > 0$ be a positive real number. Then there is a partition $P_\epsilon = (y_k)_{k=0}^n$ of $[a,b]$ so that

$$\max\left(\{|\mathcal{U}_f(P) - I|, |\mathcal{L}_f(P) - I|\}\right) < \frac{\epsilon}{2}$$

for all refinements $P \succeq P_\epsilon$. Without loss of generality, we may assume that $n \geq 2$. Let

$$\delta = \min\left(\{y_{k+1} - y_k\}_{k=0}^{n-1} \cup \left\{\frac{\epsilon}{4(n-1)B}, 1\right\}\right).$$

Suppose that (P,t) is a tagged partition of $[a,b]$ of mesh $\operatorname{mesh}(P) < \delta$, where $P = (x_j)_{j=0}^m$ and $t = (t_j)_{j=0}^{m-1}$. For each index $0 \leq j < m$, the corresponding subinterval $[x_j, x_{j+1}]$ either lies entirely inside a subinterval $[y_k, y_{k+1}]$ determined by P_ϵ or lies in the union $[y_k, y_{k+2}] = [y_k, y_{k+1}] \cup [y_{k+1}, y_{k+2}]$ of two adjacent such subintervals. In particular,

$$x_{j+1} - x_j \leq \operatorname{mesh}(P) < \delta \leq \min\left(\{y_{k+1} - y_k\}_{k=0}^{n-1}\right),$$

so that $[x_j, x_{j+1}]$ cannot span the union of three or more such intervals. Moreover, it cannot occur more than $n-1$ times that a subinterval determined by P spans two adjacent subintervals determined by P_ϵ.

We observe that if $[x_j, x_{j+1}] \subseteq [y_k, y_{k+1}]$ for some index $0 \leq k < n$, then

$$\inf_{y_k \leq x \leq y_{k+1}} f(x)(x_{j+1} - x_j) \leq f(t_j)(x_{j+1} - x_j) \leq \sup_{y_k \leq x \leq y_{k+1}} f(x)(x_{j+1} - x_j).$$

On the other hand, if otherwise $[x_j, x_{j+1}] \subseteq [y_k, y_{k+2}]$ for some index $0 \leq k < n-1$, then

$$y_k < x_j < y_{k+1} < x_{j+1} < y_{k+2}.$$

In this case, if $x_j \leq t_j \leq y_{k+1}$, then

$$\inf_{y_k \leq x \leq y_{k+1}} f(x)(y_{k+1} - x_j) \leq f(t_j)(y_{k+1} - x_j) \leq \sup_{y_k \leq x \leq y_{k+1}} f(x)(y_{k+1} - x_j),$$

and

$$\left|f(t_j) - \sup_{y_{k+1} \leq x \leq y_{k+2}} f(x)\right|(x_{j+1} - y_{k+1}) < 2B\delta \leq \frac{\epsilon}{2(n-1)}$$
$$\left|f(t_j) - \inf_{y_{k+1} \leq x \leq y_{k+2}} f(x)\right|(x_{j+1} - y_{k+1}) < 2B\delta \leq \frac{\epsilon}{2(n-1)}.$$

Since this can occur a maximum of $n-1$ times, we conclude that

$$I - \epsilon < \mathcal{L}_f(P_\epsilon) - \frac{\epsilon}{2} < \mathcal{R}_f(P,t) < \mathcal{U}_f(P_\epsilon) + \frac{\epsilon}{2} < I + \epsilon,$$

so that $|\mathcal{R}_f(P,t) - I| < \epsilon$. ■

The limits of upper and lower Darboux sums are called the **upper** and **lower Darboux integrals**. The third condition in Theorem 6.1.9 Riemann integrability, that the upper and lower Darboux integrals agree, is sometimes called **Darboux integrability**. Theorem 6.1.9 Riemann integrability shows that Darboux integrability is equivalent to other characterizations of what we call **Riemann integrability**.

Definition 6.1.10 Riemann integrability/integral. Let $f\colon [a,b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a,b]$. f is **Riemann integrable** if there is a real number $I \in \mathbb{R}$ satisfying any (and therefore all) of the equivalent conditions in Theorem 6.1.9 Riemann integrability. In this case, that $\mathbb{R}$ is Hausdorff implies that I is uniquely defined by these properties. We call this number

$$I = \lim_{(P,t)\uparrow} \mathcal{R}_f(P,t) = \lim_{P\uparrow} \mathcal{U}_f(P,t) = \lim_{P\uparrow} \mathcal{L}_f(P,t) = \inf_P \mathcal{U}_f(P) = \sup_P \mathcal{L}_f(P)$$

the **(definite) Riemann integral** of f from a to b, and we denote it by

$$\int_a^b f(x)\,\mathrm{d}x.$$

◆

Example 6.1.11 Riemann integrability. The following are examples and non-examples of Riemann integrable functions and their Riemann integrals:

- Let $c \in \mathbb{R}$ be a real number, and consider the constant function $f\colon [a,b] \to \mathbb{R}$ defined by the formula $f(x) = c$. We observe that

$$\begin{aligned}\mathcal{R}_f(P,t) &= \sum_{k=0}^{n-1} f(t_k)(x_{k+1}-x_k) = \sum_{k=0}^{n-1} c\,(x_{k+1}-x_k)\\ &= c\sum_{k=0}^{n-1}(x_{k+1}-x_k) = c\,(b-a)\end{aligned}$$

  for all tagged partitions (P,t) of $[a,b]$, where $P = (x_k)_{k=0}^{n}$ and $t = (t_k)_{k=0}^{n-1}$. This implies that

$$\lim_{(P,t)\uparrow} \mathcal{R}_f(P,t) = \lim_{(P,t)\uparrow} c\,(b-a) = c\,(b-a);$$

  that is, f is Riemann integrable, with Riemann integral

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b c\,\mathrm{d}x = c\,(b-a).$$

- Let $g\colon [0,2] \to \mathbb{R}$ be the real-valued function defined piecewise on the closed interval $[0,2]$ by the formula

$$g(y) = \begin{cases} 1 & 0 \le y \le 1 \\ 2 & 1 < y \le 2. \end{cases}$$

  Let $P = (y_k)_{k=0}^{n}$ be a partition of $[0,2]$. Then there is a unique index $0 \le j < n$ so that $y_j \le 1 < y_{j+1}$. We observe that the restriction of g to every other subinterval of the partition is constant, so that

$$\begin{aligned}\mathcal{U}_g(P) - \mathcal{L}_g(P) &= \sum_{k=0}^{n-1} \sup_{y_k \le y \le y_{k+1}} g(y)\,(y_{k+1}-y_k) - \sum_{k=0}^{n-1} \inf_{y_k \le y \le y_{k+1}} g(y)\,(y_{k+1}-y_k)\\ &= \sum_{k=0}^{n-1}\left(\sup_{y_k \le y \le y_{k+1}} g(y) - \inf_{y_k \le y \le y_{k+1}} g(y)\right)(y_{k+1}-y_k)\end{aligned}$$

$$= \left(\sup_{y_j \le y \le y_{j+1}} g(y) - \inf_{y_j \le y \le y_{j+1}} g(y) \right) (y_{j+1} - y_j)$$
$$= 1 (y_{j+1} - y_j) \le \operatorname{mesh}(P).$$

This implies that g is Riemann integrable, although this method does not necessarily directly compute the Riemann integral $\int_0^2 g(y)\, \mathrm{d}y$.

- More generally, let $h\colon [a,b] \to \mathbb{R}$ be a non-decreasing real-valued function defined on a closed interval $[a,b]$. We observe that

$$\begin{aligned}
\mathcal{U}_g(P) - \mathcal{L}_g(P) &= \sum_{k=0}^{n-1} \sup_{z_k \le z \le z_{k+1}} h(z)(z_{k+1} - z_k) - \sum_{k=0}^{n-1} \inf_{z_k \le z \le z_{k+1}} h(z)(z_{k+1} - z_k) \\
&= \sum_{k=0}^{n-1} h(z_{k+1})(z_{k+1} - z_k) - \sum_{k=0}^{n-1} h(z_k)(z_{k+1} - z_k) \\
&= \sum_{k=0}^{n-1} (h(z_{k+1}) - h(z_k))(z_{k+1} - z_k) \le \sum_{k=0}^{n-1} (h(z_{k+1}) - h(z_k)) \operatorname{mesh}(P) \\
&= (h(b) - h(a)) \operatorname{mesh}(P)
\end{aligned}$$

for any partition $P = (z_k)_{k=0}^n$ of $[a,b]$. This implies that h is Riemann integrable, but the above computation doesn't necessarily aid in the computation of the Riemann integral $\int_a^b h(z)\, \mathrm{d}z$.

- Let $\chi\colon [0,1] \to \mathbb{R}$ be the real-valued function defined piecewise by the formula

$$\chi(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q}. \end{cases}$$

Since any closed interval contains both rational and irrational numbers,

$$\begin{aligned}
\mathcal{U}_\chi(P) &= \sum_{k=0}^{n-1} \sup_{x_k \le x \le x_{k+1}} \chi(x)(x_{k+1} - x_k) \\
&= \sum_{k=0}^{n-1} 1 (x_{k+1} - x_k) = 1 - 0 = 1
\end{aligned}$$

and

$$\begin{aligned}
\mathcal{L}_\chi(P) &= \sum_{k=0}^{n-1} \inf_{x_k \le x \le x_{k+1}} \chi(x)(x_{k+1} - x_k) \\
&= \sum_{k=0}^{n-1} 0 (x_{k+1} - x_k) = 0
\end{aligned}$$

for any partition $P = (x_k)_{k=0}^n$ of $[0,1]$. We conclude that χ is not Riemann integrable.

▲

In general, it can be quite difficult to compute Riemann integrals without additional tools. We will see some such tools in Section 6.2 The Fundamental Theorems of Calculus, but even without differential calculus, we may still expand our inventory of Riemann integrable functions significantly. For example, any continuous function defined on a closed interval is Riemann integrable.

Proposition 6.1.12 Continuity implies integrability. *Let $f\colon [a,b]\to\mathbb{R}$ be a real-valued function defined on a closed interval $[a,b]$. If f is continuous, then it is Riemann integrable.*

Proof. Let

$$U = \lim_{P\uparrow} \mathcal{U}_f(P) \qquad\qquad L = \lim_{P\uparrow} \mathcal{L}_f(P).$$

We will show that $U \le L$. To that end, the Heine-Cantor theorem implies that f is uniformly continuous on $[a,b]$. Let $\epsilon > 0$ be a positive real number. Then there is a positive real number $\delta > 0$ so that

$$|f(x_1) - f(x_2)| < \frac{\epsilon}{b-a}$$

for all points $x_1, x_2 \in [a,b]$ so that $|x_1 - x_2| < \delta$. We observe that

$$\begin{aligned}
|\mathcal{U}_f(P) - \mathcal{L}_f(P)| &= \left|\sum_{k=0}^{n-1} \sup_{x_k\le x\le x_{k+1}} f(x)(x_{k+1}-x_k) - \sum_{k=0}^{n-1} \inf_{x_k\le x\le x_{k+1}} f(x)(x_{k+1}-x_k)\right| \\
&= \left|\sum_{k=0}^{n-1} \left(\sup_{x_k\le x\le x_{k+1}} f(x) - \inf_{x_k\le x\le x_{k+1}} f(x)\right)(x_{k+1}-x_k)\right| \\
&\le \sum_{k=0}^{n-1} \left|\sup_{x_k\le x\le x_{k+1}} f(x) - \inf_{x_k\le x\le x_{k+1}} f(x)\right|(x_{k+1}-x_k) \\
&\le \sum_{k=0}^{n-1} \left(\frac{\epsilon}{b-a}\right)(x_{k+1}-x_k) = \left(\frac{\epsilon}{b-a}\right)(b-a) = \epsilon
\end{aligned}$$

for any partition $P = (x_k)_{k=0}^{n}$ of $[a,b]$ of mesh $\operatorname{mesh}(P) < \delta$. In particular, $|U - L| \le \epsilon$ for all positive real numbers $\epsilon > 0$, and so $U = L$. Theorem 6.1.9 Riemann integrability implies that f is Riemann integrable. ■

We end this section with a few results about the interplay between Riemann integrals and arithmetic, comparison, and limits. First, the set of Riemann integrable functions is a vector space, and the Riemann integral is a linear functional on this vector space.

Proposition 6.1.13 Linearity of Riemann integration. *Let*

$$f = \sum_{j=1}^{m} a_j f_j$$

be a real linear combination of real-valued functions $f_j\colon [a,b]\to\mathbb{R}$ defined on a closed interval $[a,b]$. If each f_j is Riemann integrable, then so too is f, and

$$\int_a^b f(x)\,\mathrm{d}x = \sum_{j=1}^{m} a_j \int_a^b f_j(x)\,\mathrm{d}x.$$

Proof. We observe that for each tagged partition (P,t) of $[a,b]$,

$$\begin{aligned}
\mathcal{R}_f(P,t) &= \sum_{k=0}^{n-1} f(t_k)(x_{k+1}-x_k) = \sum_{k=0}^{n-1}\left(\sum_{j=1}^{m} a_j f_j(t_k)\right)(x_{k+1}-x_k) \\
&= \sum_{k=0}^{n-1}\sum_{j=1}^{m} a_j f_j(t_k)(x_{k+1}-x_k) = \sum_{j=1}^{m} a_j \sum_{k=0}^{n-1} f_j(t_k)(x_{k+1}-x_k)
\end{aligned}$$

$$= \sum_{j=1}^{m} a_j \mathcal{R}_{f_j}(P,t),$$

where $P = (x_k)_{k=0}^{n}$ and $t = (t_k)_{k=0}^{n-1}$. Thus

$$\begin{aligned}\int_a^b f(x)\,\mathrm{d}x &= \lim_{(P,t)\uparrow} \mathcal{R}_f(P,t) = \lim_{(P,t)\uparrow} \sum_{j=1}^{m} a_j \mathcal{R}_{f_j}(P,t) \\ &= \sum_{j=1}^{m} a_j \lim_{(P,t)\uparrow} \mathcal{R}_{f_j}(P,t) = \sum_{j=1}^{m} a_j \int_a^b f_j(x)\,\mathrm{d}x.\end{aligned}$$

■

We emphasize that while it is true that the product of Riemann integrable functions is also Riemann integrable, it is not in general true that the Riemann integral of such a product is the product of the Riemann integrals of the factors; that is,

$$\int_a^b f(x)\,g(x)\,\mathrm{d}x \neq \left(\int_a^b f(x)\,\mathrm{d}x\right)\left(\int_a^b g(x)\,\mathrm{d}x\right).$$

You will have an opportunity to explore this further in the problems for this section. First, however, we note that definite integrals preserve comparison of functions.

Lemma 6.1.14 Riemann integration and comparison. *Let $f, g\colon [a,b] \to \mathbb{R}$ be real-valued functions defined on a closed interval $[a,b]$. If f and g are Riemann integrable and $f(x) \leq g(x)$ for all points $a \leq x \leq b$, then*

$$\int_a^b f(x)\,\mathrm{d}x \leq \int_a^b g(x)\,\mathrm{d}x.$$

Proof. We observe that for all tagged partitions (P,t) of $[a,b]$,

$$\begin{aligned}\mathcal{R}_f(P,t) &= \sum_{k=0}^{n-1} f(t_k)(x_{k+1} - x_k) \\ &\leq \sum_{k=0}^{n-1} g(t_k)(x_{k+1} - x_k) = \mathcal{R}_g(P,t),\end{aligned}$$

where $P = (x_k)_{k=0}^{n}$ and $t = (t_k)_{k=0}^{n-1}$. Thus

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{(P,t)\uparrow} \mathcal{R}_f(P,t) \leq \lim_{(P,t)\uparrow} \mathcal{R}_g(P,t) = \int_a^b g(x)\,\mathrm{d}x.$$

■

While the behavior of Riemann integrals under limits can unfortunately be quite subtle, we can establish some preliminary properties. The following theorem establishes the continuity of the Riemann integral on the vector space of Riemann integrable functions on a closed interval, taken with the compact open topology:

Theorem 6.1.15 Uniform convergence and Riemann integrability. *Let $(f_n)_{n=0}^{\infty}$ be a uniformly convergent sequence of real-valued functions $f_n\colon [a,b] \to \mathbb{R}$ defined on a closed interval $[a,b]$. If each f_n is Riemann integrable, then the* *uniform limit* $\operatorname{unif\,lim}_{n\to\infty} f_n$ *is Riemann integrable, with Riemann integral*

$$\int_a^b \operatorname*{unif\,lim}_{n\to\infty} f_n(x)\,\mathrm{d}x = \lim_{n\to\infty}\int_a^b f_n(x)\,\mathrm{d}x.$$

Proof. We first argue that the sequence $(I_n)_{n=0}^{\infty}$ of Riemann integrals

$$I_n = \int_a^b f_n(x)\,\mathrm{d}x$$

is Cauchy. To that end, let $f = \operatorname{unif\,lim}_{n\to\infty} f_n$. Since $(f_n)_{n=0}^{\infty}$ converges uniformly to f, there is some natural number $n_\epsilon \in \mathbb{N}$ so that

$$|f_n(x) - f(x)| < \frac{\epsilon}{6(b-a)}$$

for all indices $n \ge n_\epsilon$ and points $a \le x \le b$. In particular,

$$\begin{aligned} |f_m(x) - f_n(x)| &\le |f_m(x) - f(x)| + |f_n(x) - f(x)| \\ &< \frac{\epsilon}{6(b-a)} + \frac{\epsilon}{6(b-a)} = \frac{\epsilon}{3(b-a)} \end{aligned}$$

for all indices $m, n \ge n_\epsilon$ and points $a \le x \le b$, and so

$$\begin{aligned} |\mathcal{R}_{f_m}(P,t) - \mathcal{R}_{f_n}(P,t)| &= \left|\sum_{i=0}^{l-1} f_m(t_i)(x_{i+1}-x_i) - \sum_{i=0}^{l-1} f_n(t_i)(x_{i+1}-x_i)\right| \\ &\le \sum_{i=0}^{l-1} |f_m(t_i) - f_n(t_i)|(x_{i+1}-x_i) \\ &< \sum_{i=0}^{l-1}\left(\frac{\epsilon}{3(b-a)}\right)(x_{i+1}-x_i) \\ &= \left(\frac{\epsilon}{3(b-a)}\right)(b-a) = \frac{\epsilon}{3} \end{aligned}$$

for all indices $m, n \ge n_\epsilon$ and tagged partitions (P,t) of $[a,b]$, where $P = (x_i)_{i=0}^{l}$ and $t = (t_i)_{i=0}^{l-1}$.

Let (P,t) be a tagged partition of $[a,b]$ so that

$$\begin{aligned} |\mathcal{R}_{f_m}(P,t) - I_m| &< \frac{\epsilon}{3} \\ |\mathcal{R}_{f_n}(P,t) - I_n| &< \frac{\epsilon}{3}. \end{aligned}$$

Thus

$$\begin{aligned} |I_m - I_n| &\le |I_m - \mathcal{R}_{f_m}(P,t)| + |\mathcal{R}_{f_m}(P,t) - \mathcal{R}_{f_n}(P,t)| + |\mathcal{R}_{f_n}(P,t) - I_n| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned}$$

We conclude that $(I_n)_{n=0}^{\infty}$ is Cauchy.

Corollary 3.6.15 Completeness of the real line implies that this sequence converges to a real number $I \in \mathbb{R}$. We will show that

$$\lim_{(P,t)\uparrow} \mathcal{R}_f(P,t) = I.$$

To that end, let $\epsilon > 0$ be a positive real number. Since $(f_n)_{n=0}^{\infty}$ converges uniformly to f, there is some natural number $n_\epsilon \in \mathbb{N}$ so that

$$|f_n(x) - f(x)| < \frac{\epsilon}{3(b-a)}$$

for all indices $n \geq n_\epsilon$ and points $a \leq x \leq b$. In particular,

$$\begin{aligned}
|\mathcal{R}_f(P,t) - \mathcal{R}_{f_n}(P,t)| &= \left| \sum_{i=0}^{l-1} f(t_i)(x_{i+1} - x_i) - \sum_{i=0}^{l-1} f_n(t_i)(x_{i+1} - x_i) \right| \\
&\leq \sum_{i=0}^{l-1} |f_n(t_i) - f(t_i)| (x_{i+1} - x_i) \\
&< \sum_{i=0}^{l-1} \left(\frac{\epsilon}{3(b-a)} \right) (x_{i+1} - x_i) \\
&= \left(\frac{\epsilon}{3(b-a)} \right) (b-a) = \frac{\epsilon}{3}
\end{aligned}$$

for all indices $n \geq n_\epsilon$ and tagged partitions (P,t) of $[a,b]$. Since $(I_n)_{n=0}^{\infty}$ converges to I, there is a natural number $m_\epsilon \in \mathbb{N}$ so that

$$|I_n - I| < \frac{\epsilon}{3}$$

for all indices $n \geq m_\epsilon$. Fix $n \geq \max(\{m_\epsilon, n_\epsilon\})$. Then there is some tagged partition (P_ϵ, t_ϵ) of $[a,b]$ so that

$$|\mathcal{R}_{f_n}(P,t) - I_n| < \frac{\epsilon}{3}$$

for all refinements $(P,t) \succeq (P_\epsilon, t_\epsilon)$. We observe that

$$\begin{aligned}
|\mathcal{R}_f(P,t) - I| &\leq |\mathcal{R}_f(P,t) - \mathcal{R}_{f_n}(P,t)| + |\mathcal{R}_{f_n}(P,t) - I_n| + |I_n - I| \\
&< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon
\end{aligned}$$

for all refinements $(P,t) \succeq (P_\epsilon, t_\epsilon)$. Thus the net $(\mathcal{R}_f(P,t))_{(P,t)}$ converges to I, and so f is Riemann integrable, with Riemann integral

$$\int_a^b f(x)\,\mathrm{d}x = I = \lim_{n\to\infty} I_n = \lim_{n\to\infty} \int_a^b f_n(x)\,\mathrm{d}x.$$

■

We will see in Section 6.3 Improper Integration that an analogue Theorem 6.1.15 Uniform convergence and Riemann integrability does not hold for improper integration over non-compact intervals.

In the next section, we will relate the processes of differentiation and integration. Along the way, we will find ways to compute Riemann integrals without computing Riemann sums and discover anti-derivatives of continuous functions.

6.1.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

6.1.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **More on integrability and continuity.** Let $f\colon [a,b] \to \mathbb{R}$ be a Riemann integrable real-valued function defined on a closed interval. Prove that $g \circ f$ is Riemann integrable for any continuous real-valued function $g\colon \mathbb{R} \to \mathbb{R}$.

   Hint. Use the Heine-Cantor theorem on an appropriate restriction of g.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

6.2 The Fundamental Theorems of Calculus

In this section, we develop the two **fundamental theorems of calculus**. These theorems bridge the two halves of calculus: differential calculus and integral calculus. We will see that differentiation and integration are related operations, and in a sense inverse to each other.

6.2.1 Differentiating Riemann Integrals

Before we establish the **first fundamental theorem of calculus**, we will require two further properties of the Riemann integral. The first establishes a relation between integrals of the same function on adjacent closed intervals, and the second is an integral analogue of the triangle inequality.

Lemma 6.2.1 Chasles' relation. (Michel Chasles) *Let $f\colon [a,c] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a,c]$. If f is Riemann integrable, then for any interior point $a < b < c$, the restrictions of f to the closed intervals $[a,b]$ and $[b,c]$ are Riemann integrable, and*

$$\int_a^c f(x)\, \mathrm{d}x = \int_a^b f(x)\, \mathrm{d}x + \int_b^c f(x)\, \mathrm{d}x.$$

Proof. Since f is Riemann integrable, there is a partition $P_\epsilon = (x_k)_{k=0}^n$ of $[a,c]$ so that

$$\mathcal{U}_f(P) - \mathcal{L}_f(P) < \epsilon$$

for all refinements $P \succeq P_\epsilon$. Without loss of generality, we may suppose that $b = x_j$ for some index $0 < j < n$. Consider the partitions $Q_\epsilon = (x_k)_{k=0}^j$ and $R_\epsilon = (k=j)^n$ of $[a,b]$ and $[b,c]$. We will show that the existence of Q_ϵ and R_ϵ implies the Riemann integrability of the restrictions of f to $[a,b]$ and $[b,c]$, respectively.

Indeed, note that for any refinement $Q \succeq Q_\epsilon$, we may form a refinement $Q' \succeq P_\epsilon$ by appending $(x_k)_{k=j+1}^n$ as a suffix. We observe that

$$\begin{aligned}
\mathcal{U}_f(Q) - \mathcal{L}_f(Q) &= \sum_{k=0}^{j-1}\left(\sup_{x_k \le x \le x_{k+1}} f(x) - \inf_{x_k \le x \le x_{k+1}} f(x)\right)(x_{k+1} - x_k) \\
&\le \sum_{k=0}^{n-1}\left(\sup_{x_k \le x \le x_{k+1}} f(x) - \inf_{x_k \le x \le x_{k+1}} f(x)\right)(x_{k+1} - x_k) \\
&= \mathcal{U}_f(Q') - \mathcal{L}_f(Q') < \epsilon.
\end{aligned}$$

Similarly, for any refinement $R \succeq R_\epsilon$, we may form a refinement $R' \succeq P_\epsilon$ by appending $(x_k)_{k=0}^{j-1}$ as a prefix. We observe that

$$\begin{aligned}
\mathcal{U}_f(R) - \mathcal{L}_f(R) &= \sum_{k=j}^{n-1}\left(\sup_{x_k \le x \le x_{k+1}} f(x) - \inf_{x_k \le x \le x_{k+1}} f(x)\right)(x_{k+1} - x_k) \\
&\le \sum_{k=0}^{n-1}\left(\sup_{x_k \le x \le x_{k+1}} f(x) - \inf_{x_k \le x \le x_{k+1}} f(x)\right)(x_{k+1} - x_k) \\
&= \mathcal{U}_f(R') - \mathcal{L}_f(R') < \epsilon.
\end{aligned}$$

Thus the restrictions of f to $[a,b]$ and $[b,c]$ are Riemann integrable. It remains only to show that $\int_a^c f(x)\, \mathrm{d}x = I$, where

$$I = \int_a^b f(x)\, \mathrm{d}x + \int_b^c f(x)\, \mathrm{d}x.$$

To that end, we observe that since f is bounded, there is a positive real number $B > 0$ so that $|f(x)| \le B$ for all points $a \le x \le b$. Note that for any positive real number $\epsilon > 0$, there is a positive real number $\delta > 0$ so that

$$\left|\mathcal{R}_f(Q,u) - \int_a^b f(x)\, \mathrm{d}x\right| < \frac{\epsilon}{3}$$

for all tagged partitions (Q,u) of $[a,b]$ of mesh $\operatorname{mesh}(Q) < \delta$,

$$\left|\mathcal{R}_f(R,v) - \int_b^c f(x)\, \mathrm{d}x\right| < \frac{\epsilon}{3}$$

for all tagged partitions (R,v) of $[b,c]$ of mesh $\operatorname{mesh}(R) < \delta$, and

$$\delta \le \frac{\epsilon}{6B}.$$

Let (P,t) be a tagged partition of $[a,c]$ of mesh $\operatorname{mesh}(P) < \delta$, where $P = (x_k)_{k=0}^n$ and $t = (t_k)_{k=0}^{n-1}$. If $x_j = b$ for some index $0 < j < n$, then consider the tagged partitions (Q,u) and (R,v) of $[a,b]$ and $[b,c]$, respectively, where

$$Q = (x_k)_{k=0}^j \qquad\qquad u = (t_k)_{k=0}^{j-1}$$

$$R = (x_k)_{k=j}^{n} \qquad\qquad v = (t_k)_{k=j}^{n-1}.$$

These tagged partitions both have mesh less than δ, and so

$$\begin{aligned}
|\mathcal{R}_f(P,t) - I| &= \left|\sum_{k=0}^{n-1} f(t_k)(x_{k+1} - x_k) - I\right| \\
&\leq \left|\sum_{k=0}^{j-1} f(t_k)(x_{k+1} - x_k) - \int_a^b f(x)\,\mathrm{d}x\right| \\
&\quad + \left|\sum_{k=j}^{n-1} f(t_k)(x_{k+1} - x_k) - \int_b^c f(x)\,\mathrm{d}x\right| \\
&= \left|\mathcal{R}_f(Q) - \int_a^b f(x)\,\mathrm{d}x\right| + \left|\mathcal{R}_f(Q) - \int_a^b f(x)\,\mathrm{d}x\right| \\
&< \frac{\epsilon}{3} + \frac{\epsilon}{3} = \frac{2\epsilon}{3} < \epsilon.
\end{aligned}$$

On the other hand, if $x_j \neq b$ for all indices $0 < j < n$, then there is a unique index $0 \leq j < n$ so that $x_j < b < x_{j+1}$. We may find a refinement (P', t') of (P, t) solely by adding in b and choosing tags y and z in the new subintervals $[x_j, b]$ and $[b, x_{j+1}]$. We observe that the above computation applies to (P', t'), and so

$$\begin{aligned}
|\mathcal{R}_f(P,t) - I| &\leq |\mathcal{R}_f(P,t) - \mathcal{R}_f(P',t')| + |\mathcal{R}_f(P',t') - I| \\
&< |f(t_j)(x_{j+1} - x_j) - f(y)(b - x_j) - f(z)(x_{j+1} - b)| + \frac{2\epsilon}{3} \\
&< 2B\delta + \frac{2\epsilon}{3} \leq \frac{\epsilon}{3} + \frac{2\epsilon}{3} = \epsilon.
\end{aligned}$$

In summary, we have shown that for all positive real numbers $\epsilon > 0$, there is some positive real number $\delta > 0$ so that $|\mathcal{R}_f(P,t) - I| < \epsilon$ for all tagged partitions (P, t) of $[a, c]$ of mesh $\operatorname{mesh}(P) < \delta$. We conclude that

$$\int_a^c f(x)\,\mathrm{d}x = I = \int_a^b f(x)\,\mathrm{d}x + \int_b^c f(x)\,\mathrm{d}x.$$

■

So far, we have only allowed the notation

$$\int_a^b f(x)\,\mathrm{d}x$$

when $a < b$. Chasles' relation provides a way to extend this to allow for any real numbers a and b. For example, if $a > b$, then we define

$$\int_a^b f(x)\,\mathrm{d}x = -\int_b^a f(x)\,,\mathrm{d}x.$$

Similarly, if $a = b$, then we define

$$\int_a^b f(x)\,\mathrm{d}x = 0.$$

You will have an opportunity in the problems for this section to prove that with these extensions of notation for Riemann integrals, Chasles' relation still holds.

While the following result may seem innocuous at first glance, it is a deeply useful estimate which reflects the geometric structure of the vector space of Riemann integrable functions.

Proposition 6.2.2 Triangle inequality for Riemann integrals. *Let $f\colon [a,b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a,b]$. If f is Riemann integrable, then $|f|$ is Riemann integrable, and*

$$\left|\int_a^b f(x)\,\mathrm{d}x\right| \leq \int_a^b |f(x)|\,\mathrm{d}x.$$

Proof. Since the function $g\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $g(x) = |x|$, the composition $g \circ f = |f|$ is Riemann integrable by Problem 6.1.4.1 More on integrability and continuity. Since $0 \leq |f(x)|$ for all points $a \leq x \leq b$,

$$0 = \int_a^b 0\,\mathrm{d}x \leq \int_a^b |f(x)|\,\mathrm{d}x$$

by Lemma 6.1.14 Riemann integration and comparison. Similarly, $-|f(x)| \leq f(x) \leq |f(x)|$ for all points $a \leq x \leq b$, and so

$$-\int_a^b |f(x)|\,\mathrm{d}x = \int_a^b -|f(x)|\,\mathrm{d}x \leq \int_a^b f(x)\,\mathrm{d}x \leq \int_a^b |f(x)|\,\mathrm{d}x$$

by Lemma 6.1.14 Riemann integration and comparison. Thus

$$\left|\int_a^b f(x)\,\mathrm{d}x\right| \leq \int_a^b |f(x)|\,\mathrm{d}x.$$

■

We are now ready to prove the **first fundamental theorem of calculus**. This foundational result establishes a relationship between differentiation integration: by integrating a continuous function f, we may obtain a differentiable function whose derivative is f.

Theorem 6.2.3 First fundamental theorem of calculus. *Let $f\colon [a,b] \to \mathbb{R}$ be a continuous real-valued function defined on a closed interval $[a,b]$, and consider the real-valued function $F\colon [a,b] \to \mathbb{R}$ defined by the formula*

$$F(x) = \int_a^x f(x)\,\mathrm{d}x.$$

Then F is differentiable, with derivative $F'(c) = f(c)$ for all points $a \leq c \leq b$.

Proof. Let $\epsilon > 0$ be a positive real number. Since f is continuous at c, there is a positive real number $\delta > 0$ so that $|f(x) - f(c)| < \frac{\epsilon}{2}$ for all points $x \in [a,b]$ so that $0 < |x - c| < \delta$. We observe that Chasles' relation and the Triangle inequality for Riemann integrals together imply that

$$\begin{aligned}
\left|\frac{F(x) - F(c)}{x - c} - f(c)\right| &= \left|\frac{1}{x-c}\left(\int_a^x f(x)\,\mathrm{d}x - \int_a^c f(x)\,\mathrm{d}x - f(c)(x-c)\right)\right| \\
&= \left|\frac{1}{x-c}\left(\int_c^x f(x)\,\mathrm{d}x - \int_c^x f(c)\,\mathrm{d}x\right)\right| \\
&= \frac{1}{|x-c|}\left|\int_c^x f(x) - f(c)\,\mathrm{d}x\right|
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{|x-c|} \int_{\min(\{c,x\})}^{\max(\{c,x\})} |f(x) - f(c)| \, \mathrm{d}x \\
&\leq \frac{1}{|x-c|} \int_{\min(\{c,x\})}^{\max(\{c,x\})} \frac{\epsilon}{2} \, \mathrm{d}x \\
&= \frac{\epsilon \left(\max\left(\{c,x\}\right) - \min\left(\{c,x\}\right)\right)}{2\,|x-c|} \\
&= \frac{\epsilon\,|x-c|}{2\,|x-c|} = \frac{\epsilon}{2} < \epsilon
\end{aligned}
$$

for all points $x \in [a,b]$ so that $0 < |x-c| < \delta$. Thus F is differentiable at c, with derivative $F'(c) = f(c)$. ■

Thus integrating a continuous function yields a differentiable function whose derivative is the integrand; we call such a function an **anti-derivative** (or **primitive**) of the integrand.

6.2.2 Integrating Derivatives

The First fundamental theorem of calculus establishes a relationship between differentiation and Riemann integration: the function obtained by integrating a continuous function is differentiable, with derivative the integrand. We now seek to establish a relationship in the other direction; that is, we explore Riemann integrals of the derivative of a differentiable function. In this case, the moral is similar; integrating the derivative of a function yields information about that function itself.

Theorem 6.2.4 Second fundamental theorem of calculus. *Let $f\colon [a,b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a,b]$. If f is continuously differentiable, then*

$$\int_a^b f'(x)\, \mathrm{d}x = f(b) - f(a).$$

Proof. Let $F\colon [a,b] \to \mathbb{R}$ be the real-valued function defined on $[a,b]$ by the formula

$$F(x) = \int_a^x f'(x)\, \mathrm{d}x.$$

Then the Theorem 6.2.3 First fundamental theorem of calculus implies that F is differentiable, with derivative $F'(x) = f'(x)$ for all points $a \leq x \leq b$.

Part (b) of Problem 5.3.4.1 Functions with the same derivative now implies that there is some real number $c \in \mathbb{R}$ so that $F(x) = f(x) + c$ for all points $a \leq x \leq b$. By evaluating at a, we observe that

$$0 = \int_a^a f'(x)\, \mathrm{d}x = F(a) = f(a) + c,$$

so that $c = -f(a)$. Thus

$$\int_a^b f'(x)\, \mathrm{d}x = F(b) = f(b) + c = f(b) - f(a).$$

■

There are stronger versions of the Second fundamental theorem of calculus which do not require continuity of the integrand, but rather only Riemann integrability. However, we will only use the result as stated.

Example 6.2.5 Computing Riemann integrals using the Second fundamental theorem of calculus. The following are examples of how to use the Second fundamental theorem of calculus to compute a Riemann integral:

- Consider the Riemann integral

$$\int_0^4 x^3 \,\mathrm{d}x.$$

Instead of computing this Riemann integral as a limit of Riemann sums, we instead recognize that $x^3 = f'(x)$, where $f\colon \mathbb{R} \to \mathbb{R}$ is the continuously differentiable real-valued function defined by the formula $f(x) = \frac{x^4}{4}$. Thus the Theorem 6.2.4 Second fundamental theorem of calculus implies that

$$\begin{aligned}\int_0^4 x^3 \,\mathrm{d}x &= \int_0^4 f'(x) \,\mathrm{d}x = f(4) - f(0)\\ &= \frac{4^4}{4} - \frac{0^4}{4} = 64 - 0 = 64.\end{aligned}$$

- Consider the Riemann integral

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(y) \,\mathrm{d}y.$$

Instead of computing this Riemann integral as a limit of Riemann sums, we instead recognize that $\cos(y) = g'(y)$, where $g\colon \mathbb{R} \to \mathbb{R}$ is the continuously differentiable real-valued function defined by the formula $g(y) = \sin(y)$. Thus the Theorem 6.2.4 Second fundamental theorem of calculus implies that

$$\begin{aligned}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(y) \,\mathrm{d}y &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} g'(y) \,\mathrm{d}y = g\left(\frac{\pi}{2}\right) - g\left(-\frac{\pi}{2}\right)\\ &= \sin\left(\frac{\pi}{2}\right) - \sin\left(-\frac{\pi}{2}\right) = 1 - (-1) = 2.\end{aligned}$$

- Consider the Riemann integral

$$\int_0^{\pi} 2z\sin(3z) + 3z^2\cos(3z) \,\mathrm{d}z.$$

Instead of computing this Riemann integral as a limit of Riemann sums, we instead recognize that $2z\sin(3z) + 3z^2\cos(3z) = h'(z)$, where $h\colon \mathbb{R} \to \mathbb{R}$ is the continuously differentiable real-valued function defined by the formula $h(z) = z^2\sin(3z)$. Thus the Theorem 6.2.4 Second fundamental theorem of calculus implies that

$$\begin{aligned}\int_0^{\pi} 2z\sin(3z) + 3z^2\cos(3z) \,\mathrm{d}z &= \int_0^{\pi} h'(z) \,\mathrm{d}z = h(\pi) - h(0)\\ &= \pi^2\sin(3\pi) - 0^2\sin(3\cdot 0) = 0 - 0 = 0.\end{aligned}$$

▲

The Second fundamental theorem of calculus not only exposes another link between differential calculus and integral calculus, but also provides an invaluable tool for computing Riemann integrals without resorting to limits of nets of

Riemann sums. Specifically, Second fundamental theorem of calculus implies that for continuous functions, Riemann integration is only as hard as computing anti-derivatives. However, we remark that computing anti-derivatives can still be quite difficult, even for some relatively simple integrands.

In the next section, we will try to expand our definition of integrals to include both unbounded functions and non-compact intervals.

6.2.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

6.2.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

6.3 Improper Integration

In this section, we expand the kinds of functions we consider as the integrands of Riemann integrals. Thus far, we have only allowed Riemann integrals of the form

$$\int_a^b f(x)\, \mathrm{d}x,$$

where $f\colon [a,b] \to \mathbb{R}$ is a Riemann integrable (and therefore bounded) function defined on a closed interval. In this section, we relax both of these conditions while maintaining a grasp on some of the nice properties of the theory of Riemann integration.

6.3.1 Exhaustion by Compact Intervals

We begin with an example that illustrates the dangers of integrating real-valued functions on larger and larger compact intervals: we may lose coherent interpretations of such integrals in the limit.

Example 6.3.1 Subtlety of improper integration. Consider the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x$. Then the Second fundamental theorem of calculus implies that

$$\int_a^b f(x)\, \mathrm{d}x = \frac{b^2}{2} - \frac{a^2}{2} = \frac{b^2 - a^2}{2}$$

for all real numbers $-\infty < a \leq b < \infty$. In particular, we observe that

$$\lim_{b\to\infty} \int_{-b}^{b} f(x)\, \mathrm{d}x = \lim_{b\to\infty} \frac{b^2 - (-b)^2}{2} = \lim_{b\to\infty} 0 = 0$$

but

$$\begin{aligned}\lim_{b\to\infty} \int_{1-b}^{1+b} f(x)\, \mathrm{d}x &= \lim_{b\to\infty} \frac{(1+b)^2 - (1-b)^2}{2} \\ &= \lim_{b\to\infty} \frac{1 + 2b + b^2 - 1 + 2b - b^2}{2} = \lim_{b\to\infty} \frac{4b}{2} \\ &= \lim_{b\to\infty} 2b = \infty,\end{aligned}$$

and

$$\begin{aligned}\lim_{b\to\infty} \int_{-1-b}^{-1+b} f(x)\, \mathrm{d}x &= \lim_{b\to\infty} \frac{(-1+b)^2 - (-1-b)^2}{2} \\ &= \lim_{b\to\infty} \frac{1 - 2b + b^2 - 1 - 2b - b^2}{2} = \lim_{b\to\infty} -\frac{4b}{2} \\ &= \lim_{b\to\infty} -2b = -\infty.\end{aligned}$$

This example serves to complicate what precisely is meant by the improper integral $\int_{-\infty}^{\infty} f(x)\, \mathrm{d}x$. ▲

Specifically, performing a Riemann integration on a non-compact interval will require additional care to ensure that said integral converges in a specific way. As in the case of Riemann integrals on compact intervals, we use nets to define the relevant mode of convergence.

Lemma 6.3.2 Compact subintervals are directed. *For any interval $X \subseteq \mathbb{R}$, the set $\mathcal{K}_X$ of all compact subintervals of the interior* $\operatorname{Int}(X)$ *is directed upwards.*

Proof. Let $U, V \in \mathcal{K}_X$ be two compact subintervals of the interior $\operatorname{Int}(X)$. Then $U = [a, b]$ and $V = [c, d]$ for some real numbers $a, b, c, d \in \mathbb{R}$ so that $a \leq b$ and $c \leq d$. Let

$$W = [\min(\{a, c\}), \max(\{b, d\})].$$

We observe that $\min(\{a, c\}) \in \operatorname{Int}(X)$ and $\max(\{b, d\}) \in \operatorname{Int}(X)$. Since X is an interval, so too is $\operatorname{Int}(X)$. We conclude that $W \subseteq \operatorname{Int}(X)$, and so $W \in \mathcal{K}_X$. Of course $U, V \subseteq W$. Since every pair of elements of $\mathcal{K}_X$ has a common upper bound, $\mathcal{K}_X$ is directed upwards. ■

The improper Riemann integrability of a real-valued function will be defined in terms of the convergence of certain nets of Riemann integrals defined on compact intervals. These nets will factor through the above directed set, and will be called **interior exhaustions** by compact subintervals.

Definition 6.3.3 Interior exhaustion by compact subintervals. Let $\mathcal{K}_X$ be the directed set of compact subintervals of the interior $\operatorname{Int}(X)$ of an interval $X \subseteq \mathbb{R}$. A family $(K_\alpha)_{\alpha\in A}$ of such compact subintervals $K_\alpha \in \mathcal{K}_X$ indexed by an upwards directed set $(A, \preceq)$ is called an **interior exhaustion** of X by compact intervals if it is order-preserving and

$$\operatorname{Int}(X) = \bigcup_{\alpha\in A} K_\alpha.$$

◆

Lemma 6.3.4 Interior exhaustion by compact subintervals. *Let $(K_\alpha)_{\alpha\in A}$ be a family of compact subintervals $K_\alpha \in \mathcal{K}_X$ of an interval $X \subseteq \mathbb{R}$ indexed by an upwards directed set $(A, \preceq)$. For each index $\alpha \in A$, write $K_\alpha = [a_\alpha, b_\alpha]$ for some real numbers $-\infty < a_\alpha \le b_\alpha < \infty$.*

Then $(K_\alpha)_{\alpha\in A}$ is an interior exhaustion of X by compact subintervals if and only if it satisfies the following conditions:

1. *For all indices $\alpha, \beta \in A$, if $\alpha \preceq \beta$, then $a_\alpha \ge a_\beta$ and $b_\alpha \le b_\beta$.*
2. *The nets $(a_\alpha)_{\alpha\in A}$ and $(b_\alpha)_{\alpha\in A}$ have upwards limits*

$$\lim_{\alpha\uparrow} a_\alpha = \inf(X) \qquad\qquad \lim_{\alpha\uparrow} b_\alpha = \sup(X).$$

You will have an opportunity to prove Lemma 6.3.4 Interior exhaustion by compact subintervals in the problems for this section. One consequence is that Riemann integrals of integrable functions defined on compact intervals are limits of Riemann integrals on exhaustions by compact subintervals.

Proposition 6.3.5 Improper integrals on compact intervals. *Let $f\colon [a,b] \to \mathbb{R}$ be a real-valued function defined on a closed interval $[a,b]$. If f is integrable, then*

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{\alpha\uparrow} \int_{a_\alpha}^{b_\alpha} f(x)\,\mathrm{d}x$$

for any interior exhaustion $([a_\alpha, b_\alpha])_{\alpha\in A}$ of $[a,b]$ by compact subintervals $[a_\alpha, b_\alpha] \in \mathcal{K}_{[a,b]}$.

Proof. Since f is integrable, it is bounded, and so there is some positive real number $B > 0$ so that $|f(x)| \le B$ for all points $a \le x \le b$. Let $\epsilon > 0$ be a positive real number. Then Lemma 6.3.4 Interior exhaustion by compact subintervals implies that there is an index $\alpha_\epsilon \in A$ so that $a_{\alpha_\epsilon} < a + \frac{\epsilon}{2B}$ and $b_{\alpha_\epsilon} > b - \frac{\epsilon}{2B}$. We observe that

$$\begin{aligned}
\left| \int_a^b f(x)\,\mathrm{d}x - \int_{a_\alpha}^{b_\alpha} f(x)\,\mathrm{d}x \right| &= \left| \int_a^{a_\alpha} f(x)\,\mathrm{d}x + \int_{b_\alpha}^{b} f(x)\,\mathrm{d}x \right| \\
&\le \left| \int_a^{a_\alpha} f(x)\,\mathrm{d}x \right| + \left| \int_{b_\alpha}^{b} f(x)\,\mathrm{d}x \right| \\
&\le \int_a^{a_\alpha} |f(x)|\,\mathrm{d}x + \int_{b_\alpha}^{b} |f(x)|\,\mathrm{d}x \\
&\le \int_a^{a_\alpha} B\,\mathrm{d}x + \int_{b_\alpha}^{b} B\,\mathrm{d}x
\end{aligned}$$

$$\begin{aligned}&\leq B\left(a_{\alpha}-a\right)+B\left(b-b_{\alpha}\right)\\&\leq B\left(a_{\alpha_{\epsilon}}-a\right)+B\left(b-b_{\alpha_{\epsilon}}\right)\\&<B\left(\frac{\epsilon}{2B}\right)+B\left(\frac{\epsilon}{2B}\right)=\frac{\epsilon}{2}+\frac{\epsilon}{2}=\epsilon\end{aligned}$$

for all indices $\alpha \succeq \alpha_{\epsilon}$. Thus

$$\lim_{\alpha\uparrow}\int_{a_{\alpha}}^{b_{\alpha}} f\left(x\right)\,\mathrm{d}x=\int_{a}^{b} f\left(x\right)\,\mathrm{d}x.$$

■

Proposition 6.3.6 Riemann integrals and interior exhaustions. *Let $f\colon X\to\mathbb{R}$ be a real-valued function defined on an interval X, and suppose that the restriction of f to any compact subinterval of the interior* $\mathrm{Int}\left(X\right)$ *is integrable. Then for any real number $I\in\mathbb{R}$, the following are equivalent:*

1. *For any interior exhaustion $\left(\left[a_{\alpha},b_{\alpha}\right]\right)_{\alpha\in A}$ of X by compact subintervals,*

$$\lim_{\alpha\uparrow}\int_{a_{\alpha}}^{b_{\alpha}} f\left(x\right)\,\mathrm{d}x=I$$

2. *For any positive real number ϵ, there is a compact subinterval $\left[a_{\epsilon},b_{\epsilon}\right]\in\mathcal{K}_X$ of* $\mathrm{Int}\left(X\right)$ *so that*

$$\left|\int_{a}^{b} f\left(x\right)\,\mathrm{d}x-I\right|<\epsilon$$

for all compact subintervals $\left[a_{\epsilon},b_{\epsilon}\right]\subseteq\left[a,b\right]\subseteq\mathrm{Int}\left(X\right)$.

Proof. First suppose that

$$\lim_{\alpha\uparrow}\int_{a_{\alpha}}^{b_{\alpha}} f\left(x\right)\,\mathrm{d}x=I$$

for any interior exhaustion $\left(\left[a_{\alpha},b_{\alpha}\right]\right)_{\alpha\in A}$ of X by compact subintervals. Then in particular we may choose the maximal such interior exhaustion $\left(K\right)_{K\in\mathcal{K}_X}$, obtaining the convergence

$$\lim_{\left[a,b\right]\uparrow}\int_{a}^{b} f\left(x\right)\,\mathrm{d}x=I.$$

Thus for any positive real number $\epsilon>0$, there is a compact subinterval $\left[a_{\epsilon},b_{\epsilon}\right]\in\mathcal{K}_X$ so that

$$\left|\int_{a}^{b} f\left(x\right)\,\mathrm{d}x-I\right|<\epsilon$$

for all compact subintervals $\left[a_{\epsilon},b_{\epsilon}\right]\subseteq\left[a,b\right]\subseteq\mathrm{Int}\left(X\right)$.

Conversely, now suppose that for any positive real number $\epsilon>0$, there is a compact subinterval $\left[a_{\epsilon},b_{\epsilon}\right]\in\mathcal{K}_X$ so that

$$\left|\int_{a}^{b} f\left(x\right)\,\mathrm{d}x-I\right|<\epsilon$$

for all compact subintervals $\left[a_{\epsilon},b_{\epsilon}\right]\subseteq\left[a,b\right]\subseteq\mathrm{Int}\left(X\right)$, and let $\left(\left[a_{\alpha},b_{\alpha}\right]\right)_{\alpha\in A}$ be an interior exhaustion of X by compact intervals $\left[a_{\alpha},b_{\alpha}\right]$. Lemma 6.3.4 Interior exhaustion by compact subintervals implies that there is an index $\alpha_{\epsilon}\in A$ so that

$$\left[a_{\epsilon},b_{\epsilon}\right]\subseteq\left[a_{\alpha_{\epsilon}},b_{\alpha_{\epsilon}}\right]\subseteq\mathrm{Int}\left(X\right).$$

We observe that

$$\left|\int_a^b f(x)\,\mathrm{d}x - I\right| < \epsilon$$

for all indices $\alpha \succeq \alpha_\epsilon$, since in this case

$$[a_\epsilon, b_\epsilon] \subseteq [a_{\alpha_\epsilon}, b_{\alpha_\epsilon}] \subseteq [a_\alpha, b_\alpha] \subseteq \mathrm{Int}(X).$$

We conclude that

$$\lim_{\alpha\uparrow} \int_{a_\alpha}^{b_\alpha} f(x)\,\mathrm{d}x = I.$$

■

Definition 6.3.7 Improper Riemann integral. Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on an interval X. f is called **improperly integrable** if the restriction of f to any compact subinterval of the interior $\mathrm{Int}(X)$ is integrable, and there is a real number $I \in \mathbb{R}$ which satisfies either (and therefore both) of the conditions in Proposition 6.3.6 Riemann integrals and interior exhaustions. In this case, that $\mathbb{R}$ is Hausdorff implies that the number I is uniquely defined by these properties. This number I is called the **improper Riemann integral** of f, and is denoted

$$\int_{\inf(X)}^{\sup(X)} f(x)\,\mathrm{d}x.$$

◆

Example 6.3.8 Improper Riemann integration. The following are examples and non-examples of improperly integrable functions and their improper Riemann integrals:

- Let $p \in (0,\infty)\setminus\{1\}$ be a real number, and consider the function $f\colon [1,\infty)$ defined by the formula $f(x) = \frac{1}{x^p}$. We observe that

  $$\int_a^b f(x)\,\mathrm{d}x = \frac{b^{1-p}}{1-p} - \frac{a^{1-p}}{1-p} = \frac{b^{1-p} - a^{1-p}}{1-p}$$

  for any compact subinterval $[a,b] \subseteq (1,\infty)$. We first note that if $p < 1$, then $1 - p > 0$, and so

  $$\lim_{b\to\infty} \int_a^b f(x)\,\mathrm{d}x = \lim_{b\to\infty} \frac{b^{1-p} - a^{1-p}}{1-p} = \infty.$$

  This implies that f is not improperly integrable in this case.

  On the other hand, now suppose that $p > 1$, and let $\epsilon > 0$ be a positive real number. Fix real numbers $1 < a_\epsilon \leq b_\epsilon < \infty$ so that

  $$a_\epsilon < \left(1 - \frac{\epsilon(p-1)}{2}\right)^{\frac{1}{1-p}} \qquad b_\epsilon > \left(\frac{\epsilon(p-1)}{2}\right)^{\frac{1}{1-p}}.$$

  We observe that

  $$\begin{aligned}\left|\int_a^b f(x)\,\mathrm{d}x - \frac{1}{p-1}\right| &= \left|\frac{b^{1-p} - a^{1-p}}{1-p} + \frac{1}{1-p}\right| \leq \frac{\left|b^{1-p}\right|}{p-1} + \frac{\left|1 - a^{1-p}\right|}{p-1} \\ &= \frac{b^{1-p}}{p-1} + \frac{1 - a^{1-p}}{p-1} \leq \frac{b_\epsilon^{1-p}}{p-1} + \frac{1 - a_\epsilon^{1-p}}{p-1}\end{aligned}$$

$$< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all compact subintervals $[a_\epsilon, b_\epsilon] \subseteq [a, b] \subsetneq (0, \infty)$. Thus f is improperly integrable, with improper Riemann integral

$$\int_1^\infty f(x)\ \mathrm{d}x = \frac{1}{p-1}.$$

- Let $p \in (0, \infty) \setminus \{1\}$ be a real number, and consider the function $f\colon (0, 1]$ defined by the formula $f(x) = \frac{1}{x^p}$. We observe that

  $$\int_a^b f(x)\ \mathrm{d}x = \frac{b^{1-p}}{1-p} - \frac{a^{1-p}}{1-p} = \frac{b^{1-p} - a^{1-p}}{1-p}$$

  for any compact subinterval $[a, b] \subseteq (0, 1)$. If We first note that if $p > 1$, then $1 - p < 0$, and so

  $$\lim_{a \to 0} \int_a^b f(x)\ \mathrm{d}x = \lim_{a \to 0} \frac{b^{1-p} - a^{1-p}}{1-p} = \lim_{a \to 0} \frac{a^{1-p} - b^{1-p}}{p-1} = \infty.$$

  This implies that f is not improperly integrable in this case.

  On the other hand, now suppose that $0 < p < 1$, and let $\epsilon > 0$ be a positive real number. Fix real numbers $1 < a_\epsilon \leq b_\epsilon < \infty$ so that

  $$a_\epsilon < \left(\frac{\epsilon(1-p)}{2}\right)^{\frac{1}{1-p}} \qquad b_\epsilon > \left(1 - \frac{\epsilon(1-p)}{2}\right)^{\frac{1}{1-p}}.$$

  We observe that

  $$\begin{aligned}
  \left|\int_a^b f(x)\ \mathrm{d}x - \frac{1}{1-p}\right| &= \left|\frac{b^{1-p} - a^{1-p}}{1-p} - \frac{1}{1-p}\right| \leq \frac{\left|b^{1-p} - 1\right|}{1-p} + \frac{\left|a^{1-p}\right|}{1-p} \\
  &= \frac{1 - b^{1-p}}{1-p} + \frac{a^{1-p}}{1-p} \leq \frac{1 - b_\epsilon^{1-p}}{1-p} + \frac{a_\epsilon^{1-p}}{1-p} \\
  &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon
  \end{aligned}$$

  for all compact subintervals $[a_\epsilon, b_\epsilon] \subseteq [a, b] \subsetneq (0, \infty)$. Thus f is improperly integrable, with improper Riemann integral

  $$\int_0^1 f(x)\ \mathrm{d}x = \frac{1}{1-p}.$$

- One question raised by the above examples is the nature of the case $p = 1$; that is, consider the function $f\colon (0, \infty)$ defined by the formula $f(x) = \frac{1}{x}$. We will see that f is not improperly integrable on $(0, 1]$ or on $[1, \infty)$.

  To that end, we first note that

  $$\int_a^b f(x)\ \mathrm{d}x = \ln(b) - \ln(a) = \ln\left(\frac{b}{a}\right)$$

  for any compact subinterval $[a, b] \subseteq (0, \infty)$. In particular, we note that

  $$\lim_{a \to 0} \int_a^b f(x)\ \mathrm{d}x = \lim_{a \to 0} \ln\left(\frac{b}{a}\right) = \infty$$

  and

  $$\lim_{b \to \infty} \int_a^b f(x)\ \mathrm{d}x = \lim_{b \to \infty} \ln\left(\frac{b}{a}\right) = \infty.$$

  These computations imply that the restrictions of f to the intervals $(0, 1]$ and $[1, \infty)$ are not improperly integrable.

▲

Proposition 6.3.9 Arithmetic of improperly integrable functions.

1. *Addition of improperly integrable functions.*

   Let $f, g\colon X \to \mathbb{R}$ be real-valued functions defined on an interval X. If f and g are improperly integrable, then so is $f+g$, and

$$\int_{\inf(X)}^{\sup(X)} (f+g)(x)\ \mathrm{d}x = \int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x + \int_{\inf(X)}^{\sup(X)} g(x)\ \mathrm{d}x.$$

2. *Scalar multiplication of improperly integrable functions.*

   Let $f\colon X \to \mathbb{R}$ be real-valued functions defined on an interval X. If f is improperly integrable, then so is cf for any real number $c \in \mathbb{R}$, and

$$\int_{\inf(X)}^{\sup(X)} (cf)(x)\ \mathrm{d}x = c\int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x.$$

Proof.

1. Let

$$I_f = \int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x \qquad\qquad I_g = \int_{\inf(X)}^{\sup(X)} g(x)\ \mathrm{d}x.$$

   Let $\epsilon > 0$ be a positive real number. Since f and g are improperly integrable, there is a compact subinterval $[a_\epsilon, b_\epsilon] \in \mathcal{K}_X$ of the interior $\operatorname{Int}(X)$ so that

$$\left|\int_a^b f(x)\ \mathrm{d}x - I_f\right| < \frac{\epsilon}{2}$$

   and

$$\left|\int_a^b g(x)\ \mathrm{d}x - I_g\right| < \frac{\epsilon}{2}$$

   for all compact subintervals $[a_\epsilon, b_\epsilon] \subseteq [a,b] \subseteq \operatorname{Int}(X)$. We observe that

$$\begin{aligned}\left|\int_a^b (f+g)(x)\,\mathrm{d}x - (I_f + I_g)\right| &= \left|\int_a^b f(x)\ \mathrm{d}x + \int_a^b g(x)\ \mathrm{d}x - I_f - I_g\right| \\ &\le \left|\int_a^b f(x)\ \mathrm{d}x - I_f\right| + \left|\int_a^b g(x)\ \mathrm{d}x - I_g\right| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,\end{aligned}$$

   for all compact subintervals $[a_\epsilon, b_\epsilon] \subseteq [a,b] \subseteq \operatorname{Int}(X)$. We conclude that $f+g$ is improperly integrable, with improper integral

$$\int_{\inf(X)}^{\sup(X)} (f+g)(x)\ \mathrm{d}x = I_f + I_g = \int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x + \int_{\inf(X)}^{\sup(X)} g(x)\ \mathrm{d}x.$$

2. If $c = 0$, then $cf = 0f = 0$ is improperly integrable, with

$$\begin{aligned}\int_{\inf(X)}^{\sup(X)} (cf)(x)\ \mathrm{d}x &= \int_{\inf(X)}^{\sup(X)} 0\,\mathrm{d}x = 0 \\ &= 0\int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x = c\int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x.\end{aligned}$$

We may therefore suppose that $c \neq 0$. Let

$$I = \int_{\inf(X)}^{\sup(X)} f(x) \, \mathrm{d}x.$$

Let $\epsilon > 0$ be a positive real number. Since f is improperly integrable, there is a compact subinterval $[a_\epsilon, b_\epsilon] \in \mathcal{K}_X$ of the interior $\operatorname{Int}(X)$ so that

$$\left| \int_a^b f(x) \, \mathrm{d}x - I \right| < \frac{\epsilon}{|c|}$$

for all compact subintervals $[a_\epsilon, b_\epsilon] \subseteq [a, b] \subseteq \operatorname{Int}(X)$. We observe that

$$\begin{aligned} \left| \int_a^b (cf)(x) \, \mathrm{d}x - cI \right| &= \left| c \int_a^b f(x) \, \mathrm{d}x - cI \right| \\ &= |c| \left| \int_a^b f(x) \, \mathrm{d}x - I_f \right| \\ &< \frac{|c| \, \epsilon}{|c|} = \epsilon, \end{aligned}$$

for all compact subintervals $[a_\epsilon, b_\epsilon] \subseteq [a, b] \subseteq \operatorname{Int}(X)$. We conclude that cf is improperly integrable, with improper integral

$$\int_{\inf(X)}^{\sup(X)} (cf)(x) \, \mathrm{d}x = cI = c \int_{\inf(X)}^{\sup(X)} f(x) \, \mathrm{d}x.$$

■

To summarize, a function is improperly integrable if the proper Riemann integral of a restriction of said function to a compact subinterval of the interior of the domain can be made arbitrarily close to a given real number just by choosing the compact subinterval to be wide enough.

6.3.2 Tests for Integral Convergence

We now explore several sufficient criteria for establishing improper Riemann integrability. We first observe that while a real-valued function f defined on compact intervals is Riemann integrable if and only if $|f|$ is Riemann integrable, the same cannot be said of functions defined on non-compact intervals.

Example 6.3.10 Conditional convergence of improper integrals. Let $f\colon (0, \infty) \to \mathbb{R}$ be the real-valued function defined by the formula

$$f(x) = \frac{\sin(x)}{x}.$$

We don't yet have the tools to verify this, but f is improperly Riemann integrable, while $|f|$ is not. We might call the convergence of the corresponding improper integral of f **conditional**, in contrast with **absolutely convergent** improper integrals. ▲

Definition 6.3.11 Absolute/conditional Riemann integrability. Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on an interval $X \subseteq \mathbb{R}$.

- *Absolute Riemann integrability.*

f is called **absolutely Riemann integrable** if $|f|$ is improperly Riemann integrable. One might also describe the improper Riemann integral

$$\int_{\inf(X)}^{\sup(X)} f(x)\,\mathrm{d}x$$

as **absolutely convergent** in this case.

- *Conditional Riemann integrability.*

  f is called **conditionally Riemann integrable** if it is improperly Riemann integrable but not absolutely Riemann integrable; that is f is conditionally Riemann integrable if f is improperly Riemann integrable but $|f|$ is not. One might also describe the improper Riemann integral

  $$\int_{\inf(X)}^{\sup(X)} f(x)\,\mathrm{d}x$$

  as **conditionally convergent** in this case.

◆

We will show that absolute integrability implies improper integrability. This will follow from the comparison test for improper integrability, of which we first establish the following special case:

Lemma 6.3.12 Comparison of non-negative improper integrals. *Let $f, g\colon X \to [0, \infty)$ be non-negative real-valued functions defined on an interval $X \subseteq \mathbb{R}$, and suppose that $0 \leq f(x) \leq g(x)$ for all points $x \in X$. If the restriction of f to any compact subinterval of the interior $\operatorname{Int}(X)$ is integrable and g is improperly integrable, then f is improperly integrable, and*

$$0 \leq \int_{\inf(X)}^{\sup(X)} f(x)\,\mathrm{d}x \leq \int_{\inf(X)}^{\sup(X)} g(x)\,\mathrm{d}x.$$

Proof. Consider the nets $\left(F_{[a,b]}\right)_{[a,b]\in\mathcal{K}_X}$ and $\left(G_{[a,b]}\right)_{[a,b]\in\mathcal{K}_X}$ defined by the formulas

$$F_{[a,b]} = \int_a^b f(x)\,\mathrm{d}x \qquad\qquad G_{[a,b]} = \int_a^b g(x)\,\mathrm{d}x$$

for all compact subintervals $[a, b] \in \mathcal{K}_X$ of the interior $\operatorname{Int}(X)$. We first observe that these nets are order-preserving. Indeed, for any compact subintervals $[a, b], [c, d] \in \mathcal{K}_X$, if $[a, b] \subseteq [c, d]$, then Chasles' relation and Lemma 6.1.14 Riemann integration and comparison imply that

$$\begin{aligned} F_{[a,b]} &= \int_a^b f(x)\,\mathrm{d}x \\ &\leq \int_c^a f(x)\,\mathrm{d}x + \int_a^b f(x)\,\mathrm{d}x + \int_b^d f(x)\,\mathrm{d}x \\ &= \int_c^d f(x)\,\mathrm{d}x = F_{[c,d]}, \end{aligned}$$

and similarly that $G_{[a,b]} \leq G_{[c,d]}$. The net analogue of the Monotone convergence theorem implies that

$$\int_{\inf(X)}^{\sup(X)} g(x)\,\mathrm{d}x = \lim_{[a,b]\uparrow} G_{[a,b]} = \sup_{[a,b]\in\mathcal{K}_X} G_{[a,b]}.$$

We observe that for any compact subinterval $[a,b] \in \mathcal{K}_X$ of the interior $\operatorname{Int}(X)$,

$$\begin{aligned} F_{[a,b]} &= \int_a^b f(x)\ \mathrm{d}x \le \int_a^b g(x)\ \mathrm{d}x \\ &\le \sup_{[c,d]\in\mathcal{K}_X} G([c,d]) = \int_{\inf(X)}^{\sup(X)} g(x)\ \mathrm{d}x \end{aligned}$$

by Lemma 6.1.14 Riemann integration and comparison, and so another application of the net analogue of the Monotone convergence theorem implies that $\left(F_{[a,b]}\right)_{[a,b]\in\mathcal{K}_X}$ converges upwards to

$$\lim_{[a,b]\uparrow} F_{[a,b]} = \sup_{[a,b]\in\mathcal{K}_X} F_{[a,b]} \le \int_{\inf(X)}^{\sup(X)} g(x)\ \mathrm{d}x.$$

That is, f is improperly integrable, with improper integral

$$0 \le \int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x \le \int_{\inf(X)}^{\sup(X)} g(x)\ \mathrm{d}x.$$

■

Theorem 6.3.13 Comparison test for integral convergence. *Let $f,g,h\colon X \to \mathbb{R}$ be real-valued functions defined on an interval X, and suppose that $f(x) \le g(x) \le h(x)$ for all points $x \in X$. If the restriction of g to any compact subinterval of the interior $\operatorname{Int}(X)$ is integrable and f and h are improperly integrable, then g is improperly integrable, and*

$$\int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x \le \int_{\inf(X)}^{\sup(X)} g(x)\ \mathrm{d}x \le \int_{\inf(X)}^{\sup(X)} h(x)\ \mathrm{d}x.$$

Proof. We note that $g-f$ and $h-f$ satisfy the hypotheses of Lemma 6.3.12 Comparison of non-negative improper integrals, and so the desired result now follows immediately from Lemma 6.3.12 Comparison of non-negative improper integrals and Proposition 6.3.9 Arithmetic of improperly integrable functions. ■

Corollary 6.3.14 Absolute integrability implies improper integrability. *Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on an interval X. If the restriction of f to every compact subinterval of the interior $\operatorname{Int}(X)$ is integrable and f is absolutely integrable, then f is improperly integrable, and*

$$\left|\int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x\right| \le \int_{\inf(X)}^{\sup(X)} |f(x)|\ \mathrm{d}x.$$

Proof. Since f is absolutely integrable, $|f|$ is improperly integrable. We observe that (2) of Proposition 6.3.9 Arithmetic of improperly integrable functions implies that $-|f|$ is also improperly integrable. Since $-|f(x)| \le f(x) \le |f(x)|$, the Comparison test for integral convergence implies that f is improperly integrable, with improper integral

$$-\int_{\inf(X)}^{\sup(X)} |f(x)|\ \mathrm{d}x \le \int_{\inf(X)}^{\sup(X)} f(x)\ \mathrm{d}x \le \int_{\inf(X)}^{\sup(X)} |f(x)|\ \mathrm{d}x.$$

Taking absolute values, we obtain

$$0 \leq \left| \int_{\inf(X)}^{\sup(X)} f(x) \, \mathrm{d}x \right| \leq \int_{\inf(X)}^{\sup(X)} |f(x)| \, \mathrm{d}x.$$

■

Example 6.3.15 Comparison of improper integrals. Let $f\colon [1, \infty) \to \mathbb{R}$ be the real-valued function defined by the formula

$$f(x) = \frac{a}{x^2 + b}$$

for some non-negative real numbers $a, b \geq 0$. Since f is continuous, the restriction of f to any compact subinterval of $(1, \infty)$ is integrable. We observe that

$$0 \leq f(x) = \frac{a}{x^2 + b} < \frac{a}{x^2}$$

for all points $1 \leq x < \infty$. Since $\frac{a}{x^2}$ is improperly integrable, the Comparison test for integral convergence implies that f is improperly integrable, with improper integral

$$0 \leq \int_1^\infty f(x) \, \mathrm{d}x \leq \int_1^\infty \frac{a}{x^2} \, \mathrm{d}x = \frac{a}{2-1} = a.$$

▲

The hypothesis that $b \geq 0$ is non-negative in Example 6.3.15 Comparison of improper integrals was necessary in order to obtain the given comparison. However, the same conclusion holds for negative $b < 0$. One way to see this improper integrability is to apply the following result:

Theorem 6.3.16 Limit comparison test for integral convergence. *Let $f, g\colon X \to \mathbb{R}$ be real-valued functions defined on an interval X, and suppose that $f \asymp g$ as $x \to \inf(X)$ and $x \to \sup(X)$. If the restriction of f to any compact subinterval of the interior $\operatorname{Int}(X)$ is integrable and g is improperly integrable, then f is improperly integrable.*

We will see in Chapter 7 Infinite Series that several of these tests for improper integrability correspond to analogous tests for series convergence. This is one example of a deep relationship between integrals and series which will be explored further in `[provisional cross-reference: chapter-lebesgue-integration]`.

In the next section, we will explore some of the properties of the theory of Riemann integration which make it less suitable as a vehicle for exploring the analysis of real-valued functions.

6.3.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

6.3.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these problems are available in Appendix A Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

6.4 Deficiencies of the Riemann Integral (SKIP)

Text before the first subsection.

Text after the last subsection.

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In the next chapter, we will explore measure theory, a branch of mathematics invented in order to construct a more robust theory of integration free from the issues mentioned in Section 6.4 Deficiencies of the Riemann Integral (SKIP).

Chapter 7

Infinite Series

In this chapter, we explore the convergence and divergence of infinite series of real numbers and real-valued functions. After establishing several sufficient criteria for convergence, we will investigate **power series**. The most productive source of power series is Taylor's theorem, which when applied to smooth functions produces the partial sums of a **Taylor series**. We will use such Taylor series to analyze **analytic functions**, which are locally equal to their Taylor series.

7.1 Partial Sums and Infinite Series

We begin by developing a notion of convergence and divergence for series of real numbers. These ideas build directly on the convergence and divergence of sequences of real numbers. In fact, a series converges/diverges if and only if an associated sequence of real numbers (although *not* the sequence of terms) does.

7.1.1 Infinitely Iterated Sums

Because the sum of two real numbers is itself a real number (that is, addition is a binary relation on $\mathbb{R}$), the sum of three (or four, or five, or six) real numbers is also a real number. In fact, we can add together any finite sequence of real numbers and obtain a real number. If we instead start with an infinite sequence of real numbers, we can thus form a sequence whose nth entry is the sum of the first n numbers.

Definition 7.1.1 Series. The **series** associated to a sequence $(x_n)_{n=0}^{\infty}$ of real numbers $x_n \in \mathbb{R}$ called **terms** is a formal expression of the form

$$\sum_{n=0}^{\infty} x_n.$$

For each index $n \in \mathbb{N}$, the nth **partial sum** S_n is the finite sum

$$S_n = \sum_{k=0}^{n} x_k,$$

and the nth **tail** T_n is the series

$$T_n = \sum_{k=n+1}^{\infty} x_k = \sum_{k=0}^{\infty} x_{k+n+1}.$$

◆

Example 7.1.2 Geometric series. A **geometric series** of **initial value** $a \in \mathbb{R}$ and **ratio** $r \in \mathbb{R}$ is the series of the form

$$\sum_{n=0}^{\infty} ar^n.$$

If $r = 1$, then the partial sums of such a geometric series are

$$\begin{aligned} S_n &= \sum_{k=0}^{n} ar^k = a + ar + ar^2 + \cdots + ar^n \\ &= a + a + a + \cdots + a = a\,(n+1) \end{aligned}$$

On the other hand, if $r \neq 1$, then we claim that the partial sums of such a geometric series are

$$\begin{aligned} S_n &= \sum_{k=0}^{n} ar^k = a + ar + ar^2 + \cdots + ar^n \\ &= \frac{a\left(1 - r^{n+1}\right)}{1 - r}. \end{aligned}$$

We can verify this computation by induction on $n \in \mathbb{N}$. Indeed, for the base case $n = 0$, we see that

$$\begin{aligned} S_n = S_0 &= \sum_{k=0}^{0} ar^k = ar^0 = a \\ &= a\left(\frac{1-r}{1-r}\right) = \frac{a\left(1 - r^1\right)}{1 - r} = \frac{a\left(1 - r^{n+1}\right)}{1 - r}. \end{aligned}$$

For the inductive step, suppose that

$$S_n = \frac{a\left(1 - r^{n+1}\right)}{1 - r}$$

for some index $n \in \mathbb{N}$. We now observe that

$$\begin{aligned} S_{n+1} &= \sum_{k=0}^{n+1} ar^k = \sum_{k=0}^{n} ar^k + ar^{n+1} = \frac{a\left(1 - r^{n+1}\right)}{1 - r} + ar^{n+1} \\ &= \frac{a\left(1 - r^{n+1}\right) + ar^{n+1}\,(1 - r)}{1 - r} \\ &= \frac{a - ar^{n+1} + ar^{n+1} - ar^{n+2}}{1 - r} \\ &= \frac{a - ar^{n+2}}{1 - r} = \frac{a\left(1 - r^{(n+1)+1}\right)}{1 - r}. \end{aligned}$$

▲

Definition 7.1.3 Series convergence/divergence. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_m \in \mathbb{R}$, and consider the associated series $\sum_{n=0}^{\infty} x_n$.

- *Series convergence.*

  The series $\sum_{n=0}^{\infty} x_n$ **converges** to a real number $S \in \mathbb{R}$ if the sequence $(S_n)_{n=0}^{\infty}$ of partial sums

  $$S_n = \sum_{k=0}^{n} x_k$$

  converges to S. In this case, we call S the **value** of the series, and write

  $$\sum_{n=0}^{\infty} x_n = S = \lim_{n\to\infty} \sum_{k=0}^{n} x_k.$$

- *Series divergence.*

  The series $\sum_{n=0}^{\infty} x_n$ **diverges** if the sequence $(S_n)_{n=0}^{\infty}$ of partial sums

  $$S_n = \sum_{k=0}^{n} x_k$$

  diverges.

  A special case of series divergence is series **divergence to infinity**. If the sequence $(S_n)_{n=0}^{\infty}$ diverges to $\pm\infty$, then we write

  $$\sum_{n=0}^{\infty} x_n = \pm\infty = \lim_{n\to\infty} \sum_{k=0}^{n} x_k.$$

◆

Example 7.1.4 Series convergence/divergence. Consider the geometric series $\sum_{n=0}^{\infty} ar^n$ of initial value $a \in \mathbb{R}$ and ratio $r \in \mathbb{R}$, and let $(S_n)_{n=0}^{\infty}$ be the sequence of partial sums

$$S_n = \sum_{k=0}^{n} ar^k.$$

We will see that the convergence or divergence of this geometric series depends on both a and r.

- We first observe that if $a = 0$, then

  $$S_n = \sum_{k=0}^{n} ar^k = \sum_{k=0}^{n} 0 = 0$$

  for all indices $n \in \mathbb{N}$, and so

  $$\sum_{n=0}^{\infty} ar^n = \lim_{n\to\infty} S_n = \lim_{n\to\infty} 0 = 0.$$

- Now suppose that $a \neq 0$ and $r = 0$. Then

  $$S_n = a$$

  for all indices $n \in \mathbb{N}$, and so

  $$\sum_{n=0}^{\infty} ar^n = \lim_{n\to\infty} S_n = \lim_{n\to\infty} a = a.$$

- Now suppose that $a \neq 0$ and $0 < |r| < 1$. Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \frac{\ln(\epsilon) + \ln(1-r)}{\ln(|r|)} - 1$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned}\left|S_n - \frac{a}{1-r}\right| &= \left|\frac{a\left(1-r^{n+1}\right)}{1-r} - \frac{a}{1-r}\right| = \frac{\left|-r^{n+1}\right|}{|1-r|} \\ &= \frac{|r|^{n+1}}{1-r} \leq \frac{|r|^{n_\epsilon+1}}{1-r} < \epsilon\end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus the geometric series converges to

$$\sum_{n=0}^{\infty} ar^n = \lim_{n\to\infty} S_n = \frac{a}{1-r}.$$

- Now suppose that $a > 0$ and $r = 1$. Let $B \in \mathbb{R}$ be a real number. The Archimedean principle implies that

$$n_B > \frac{B}{a} - 1$$

for some natural number $n_B \in \mathbb{N}$. We observe

$$S_n = (n+1)\,a \geq (n_B + 1)\,a > B$$

for all indices $n \geq n_B$. Thus the geometric series diverges to ∞; that is,

$$\sum_{n=0}^{\infty} ar^n = \lim_{n\to\infty} S_n = \infty.$$

- Now suppose that $a < 0$ and $r = 1$. Let $B \in \mathbb{R}$ be a real number. The Archimedean principle implies that

$$n_B > \frac{B}{a} - 1$$

for some natural number $n_B \in \mathbb{N}$. We observe

$$S_n = (n+1)\,a \leq (n_B + 1)\,a < B$$

for all indices $n \geq n_B$. Thus the geometric series diverges to $-\infty$; that is,

$$\sum_{n=0}^{\infty} ar^n = \lim_{n\to\infty} S_n = -\infty.$$

- Now suppose that $a \neq 0$ and $r = -1$. Then the geometric series has partial sums

$$\begin{aligned}S_n &= \frac{a\left(1-r^{n+1}\right)}{1-r} = \frac{a\left(1-(-1)^{n+1}\right)}{1-(-1)} = \frac{a\left(1+(-1)^{n+1}\right)}{2} \\ &= \begin{cases} a & n \text{ is even} \\ 0 & n \text{ is odd} \end{cases}\end{aligned}$$

- Now suppose that $a > 0$ and $r > 1$. Let $B \in \mathbb{R}$ be a real number. The Archimedean principle implies that

$$n_B > \frac{\ln(a - B(1-r)) - \ln(a)}{\ln(r)} - 1$$

for some natural number $n_B \in \mathbb{N}$. We observe that

$$S_n = \frac{a\left(1 - r^{n+1}\right)}{1-r} \geq \frac{a\left(1 - r^{n_B+1}\right)}{1-r} > B$$

for all indices $n \geq n_B$. Thus the geometric series diverges to ∞; that is,

$$\sum_{n=0}^{\infty} ar^n = \lim_{n\to\infty} S_n = \infty.$$

- Now suppose that $a < 0$ and $r > 1$. Let $B \in \mathbb{R}$ be a real number. The Archimedean principle implies that

$$n_B > \frac{\ln(a - B(1-r)) - \ln(a)}{\ln(r)} - 1$$

for some natural number $n_B \in \mathbb{N}$. We observe that

$$S_n = \frac{a\left(1 - r^{n+1}\right)}{1-r} \leq \frac{a\left(1 - r^{n_B+1}\right)}{1-r} < B$$

for all indices $n \geq n_B$. Thus the geometric series diverges to ∞; that is,

$$\sum_{n=0}^{\infty} ar^n = \lim_{n\to\infty} S_n = -\infty.$$

- Finally, suppose that $a \neq 0$ and $r < -1$. We observe that

$$|S_{n+1} - S_n| = \left|\sum_{k=0}^{n+1} ar^k - \sum_{k=0}^{n} ar^k\right| = \left|ar^{n+1}\right| = |a|\,|r|^{n+1} \geq 1$$

for any index $n \in \mathbb{N}$ so that

$$n \geq -\frac{\ln(|a|)}{\ln(|r|)} - 1.$$

In particular, this implies that the sequence $(S_n)_{n=0}^{\infty}$ cannot be Cauchy, and therefore cannot converge. So the geometric series diverges.

To summarize, the geometric series $\sum_{n=0}^{\infty} ar^n$ converges if and only if $a = 0$ or $|r| < 1$. ▲

Several of the ideas presented above in Example 7.1.4 Series convergence/divergence can be generalized to help determine the convergence or divergence behavior of more general series. For example, the sequence of partial sums cannot be Cauchy unless the sequence of terms converges to 0.

Proposition 7.1.5 Convergent series have convergent terms. *Let*

$(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$. If the series

$$\sum_{n=0}^{\infty} x_n$$

converges, then $(x_n)_{n=0}^{\infty}$ converges to 0.

Proof. By contraposition. Suppose that $(x_n)_{n=0}^{\infty}$. Then there is a positive real number $\epsilon > 0$ so that $|x_n - 0| \geq \epsilon$ for infinitely many natural numbers $n \in \mathbb{N}$. Let $(S_n)_{n=0}^{\infty}$ be the sequence of partial sums

$$S_n = \sum_{k=0}^{n} x_n.$$

For any positive integer $n \geq 1$ so that $|x_n - 0| \geq \epsilon$, we observe that

$$\begin{aligned} |S_n - S_{n-1}| &= \left| \sum_{k=0}^{n} x_k - \sum_{k=0}^{n-1} x_k \right| = |x_n| \\ &= |x_n - 0| \geq \epsilon. \end{aligned}$$

Since this occurs for infinitely many positive integers $n \geq 1$, we conclude that the sequence $(S_n)_{n=0}^{\infty}$ cannot be Cauchy. Proposition 3.6.3 Convergent sequences are Cauchy now implies that $(S_n)_{n=0}^{\infty}$, and therefore the series $\sum_{n=0}^{\infty} x_n$, diverges. ■

Series convergence behaves well under termwise arithmetic. In particular, we may add convergent series term by term or scale a convergent series by a real number, and the convergent of the series is preserved.

Proposition 7.1.6 Arithmetic of convergent series.

1. *Addition of convergent series.*

   Let $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ be sequences of real numbers $x_n, y_n \in \mathbb{R}$. If the series $\sum_{n=0}^{\infty} x_n$ and $\sum_{n=0}^{\infty} y_n$ converge, then so too does the series $\sum_{n=0}^{\infty} x_n + y_n$, and

   $$\sum_{n=0}^{\infty} x_n + y_n = \sum_{n=0}^{\infty} x_n + \sum_{n=0}^{\infty} y_n.$$

2. *Scalar multiplication of convergent series.*

   Let $c \in \mathbb{R}$ be a real number and $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$. If the series $\sum_{n=0}^{\infty} x_n$ converges, then so too does the series $\sum_{n=0}^{\infty} cx_n$, and

   $$\sum_{n=0}^{\infty} cx_n = c \sum_{n=0}^{\infty} x_n.$$

Proof.

1. Let $(S_n)_{n=0}^{\infty}$ and $(T_n)_{n=0}^{\infty}$ be the sequences of partial sums

   $$S_n = \sum_{k=0}^{n} x_k \qquad\qquad T_n = \sum_{k=0}^{n} y_k,$$

and denote by X and Y the corresponding values of the series

$$X = \sum_{n=0}^{\infty} x_n = \lim_{n\to\infty} S_n$$
$$Y = \sum_{n=0}^{\infty} y_n = \lim_{n\to\infty} T_n.$$

Let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$|S_n - X| < \frac{\epsilon}{2}$$

and

$$|T_n - Y| < \frac{\epsilon}{2}$$

for all indices $n \geq n_\epsilon$. We observe that

$$\begin{aligned} \left| \sum_{k=0}^{n} x_k + y_k - (X+Y) \right| &= |S_n + T_n - X - Y| \leq |S_n - X| + |T_n - Y| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. We conclude that the series $\sum_{n=0}^{\infty} x_n + y_n$ converges to

$$\begin{aligned} \sum_{n=0}^{\infty} x_n + y_n = \lim_{n\to\infty} \sum_{k=0}^{n} x_n + y_n = X + Y \\ = \sum_{n=0}^{\infty} x_n + \sum_{n=0}^{\infty} y_n. \end{aligned}$$

2. If $c = 0$, then

$$\sum_{k=0}^{n} cx_k = \sum_{k=0}^{n} 0 = 0$$

for all indices $n \in \mathbb{N}$, and so

$$\begin{aligned} \sum_{n=0}^{\infty} cx_n = \lim_{n\to\infty} \sum_{k=0}^{n} cx_k &= \lim_{n\to\infty} 0 = 0 \\ &= 0 \sum_{n=0}^{\infty} x_n = c \sum_{n=0}^{\infty} x_n. \end{aligned}$$

We may therefore assume that $c \neq 0$. Let $(S_n)_{n=0}^{\infty}$ be the sequence of partial sums

$$S_n = \sum_{k=0}^{n} x_k,$$

and denote by X the corresponding value of the series

$$X = \sum_{n=0}^{\infty} x_n = \lim_{n\to\infty} S_n.$$

Let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$|S_n - X| < \frac{\epsilon}{|c|}$$

for all indices $n \geq n_\epsilon$. We observe that

$$\begin{aligned} \left| \sum_{k=0}^{n} cx_k - (cX) \right| &= |cS_n - cX| = |c|\,|S_n - X| \\ &< |c| \left(\frac{\epsilon}{|c|} \right) = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. We conclude that the series $\sum_{n=0}^{\infty} cx_n$ converges to

$$\begin{aligned} \sum_{n=0}^{\infty} cx_n = \lim_{n\to\infty} \sum_{k=0}^{n} cx_n = cX \\ = c \sum_{n=0}^{\infty} x_n. \end{aligned}$$

■

We will also see that termwise multiplication of convergent series produces convergent series. In order to prove this, we will require additional sufficient criteria for series convergence developed in Section 7.2 Tests for Series Convergence. We can, however, compare convergent series by comparing their terms.

Lemma 7.1.7 Comparison of convergent series. *Let $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ be sequences of real numbers $x_n, y_n \in \mathbb{R}$. If $x_n \leq y_n$ for all indices $n \in \mathbb{N}$ and the series $\sum_{n=0}^{\infty} x_n$ and $\sum_{n=0}^{\infty} y_n$ converge, then*

$$\sum_{n=0}^{\infty} x_n \leq \sum_{n=0}^{\infty} y_n.$$

Proof. Let $(S_n)_{n=0}^{\infty}$ and $(T_n)_{n=0}^{\infty}$ be the sequences of partial sums

$$S_n = \sum_{k=0}^{n} x_k \qquad T_n = \sum_{k=0}^{n} y_k.$$

Since $x_k \leq y_k$ for each index $0 \leq k \leq n$, $S_n \leq T_n$. We conclude that

$$\sum_{n=0}^{\infty} x_n = \lim_{n\to\infty} S_n \leq \lim_{n\to\infty} T_n = \sum_{n=0}^{\infty} y_n$$

by Proposition 3.3.2 Comparison of limits. ■

In the next section, we will build on Lemma 7.1.7 Comparison of convergent series significantly, establishing an analogue of the Comparison test for integral convergence for infinite series.

7.1.2 Rearrangements of Infinite Series [SKIP]

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7.1.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

7.1.4 Reading Questions

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these reading questions are available. However, these are provided in order to check your work. If you are unable to solve the above reading questions, discuss the problem with classmates and/or instructor rather than spoiling the answers.

7.2 Tests for Series Convergence

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7.3 Power Series

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7.4 Analytic Functions

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Exercises

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Appendices

Appendix A

Selected Solutions

I · Foundations of Analysis
1 · Introductory Set Theory
1.1 · What Is a Set?
1.1.4 · Exercises

Elements of sets. Determine whether or not the given object is an element of the given set. Write your answer using the notation $\in$ or $\notin$.

1.1.4.1. Is 3 an element of the set $\{1, 2, 3\}$?

Answer. Yes, $3 \in \{1, 2, 3\}$.

1.1.4.2. Is 5 an element of the set $\{3, 9, 12\}$?

Answer. No, $3 \notin \{1, 2, 3\}$.

1.1.4.3. Is 8 an element of the set $\{0, 2, 4, \ldots, 12\}$?

Answer. If we interpret the roster notation as

$$\{0, 2, 4, \ldots, 12\} = \{0, 2, 4, 6, 8, 10, 12\},$$

then $8 \in \{0, 2, 4, \ldots, 12\}$.

1.1.4.4. Is 50 an element of the set $\{0, 1, 4, 9, 16, \ldots\}$?

Answer. If we interpret the roster notation as

$$\{0, 1, 4, 9, 16, \ldots\} = \left\{n^2 \;\middle|\; n \in \{0, 1, 2, 3, \ldots\}\right\},$$

then $50 \notin \{0, 1, 4, 9, 16, \ldots\}$.

Roster notation. Write down the specified set in roster notation.

1.1.4.5. Write down the set E of even numbers between 1 and 15 in roster notation.

Answer. $E = \{2, 4, 6, 8, 10, 12, 14\}$.

1.1.4.6. Write down the set L of letters in your full name in roster notation.

Answer. The correct answer will differ depending on your full name. My full name is “Max Lahn”, and so I would answer

$$L = \{\mathrm{m}, \mathrm{a}, \mathrm{x}, \mathrm{l}, \mathrm{h}, \mathrm{n}\}.$$

1.1.4.7. Write down the set C of colors in the rainbow in roster notation.

Answer. The correct answer will differ depending on what you consider to be separate colors. I would answer

$$C = \{\text{red}, \text{orange}, \text{yellow}, \text{green}, \text{blue}, \text{indigo}, \text{violet}\}$$

Set containment. Let S, T, and U be the sets defined by

$$S = \{0, 1, 2\} \qquad T = \{3, 4\} \qquad U = \{1, 2, 3\}.$$

Determine whether or not the stated set containment is true or false.

1.1.4.8. Is $S \subseteq U$?

Answer. No, $S \not\subseteq U$.

1.1.4.9. Is $S \cap T \subsetneq U$?

Answer. Yes, $S \cap T \subsetneq U$.

Solution. Yes,

$$S \cap T = \{0, 1, 2\} \cap \{3, 4\} = \emptyset \subsetneq \{1, 2, 3\} = U$$

1.1.4.10. Is $U \subseteq S \cup T$?

Answer. Yes, $U \subseteq S \cup T$.

Solution. Yes,

$$\begin{aligned} U &= \{1, 2, 3\} \subseteq \{0, 1, 2, 3, 4\} \\ &= \{0, 1, 2\} \cup \{3, 4\} = S \cup T \end{aligned}$$

Power sets. Write down the power sets of the given sets in roster notation.

1.1.4.11. Write down the power set $\mathcal{P}(S)$ of the set $S = \{1\}$ in roster notation.

Answer. The power set $\mathcal{P}(S)$ of the set $S = \{1\}$ is the set

$$\mathcal{P}(S) = \{\emptyset, \{1\}\}.$$

Solution. Choosing a subset of the set $S = \{1\}$ is equivalent to making a unique binary choices of whether or not to include the element 1. We see that S has the following 2 subsets: $\emptyset$, and $\{1\}$. Therefore, the power set $\mathcal{P}(S)$ of the set $S = \{1\}$ is the set

$$\mathcal{P}(S) = \{\emptyset, \{1\}\}.$$

1.1.4.12. Write down the power set $\mathcal{P}(T)$ of the set $T = \{1, 2\}$ in roster notation.

Answer. The power set $\mathcal{P}(T)$ of the set $T = \{1, 2\}$ is the set

$$\mathcal{P}(T) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Solution. Choosing a subset of the set $T = \{1, 2\}$ is equivalent to making a unique sequence of two binary choices of whether to include each element 1 and 2. We see that T has the following 4 subsets: $\emptyset$, $\{1\}$, $\{2\}$, and $\{1, 2\}$. Therefore, the power set $\mathcal{P}(T)$ of the set $S = \{1, 2\}$ is the set

$$\mathcal{P}(T) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

1.1.4.13. Write down the power set $\mathcal{P}(U)$ of the set $U = \{1, 2, 3\}$ in roster notation.

Answer. The power set $\mathcal{P}(U)$ of the set $U = \{1, 2, 3\}$ is the set

$$\mathcal{P}(U) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}.$$

Solution. Choosing a subset of the set $U = \{1, 2, 3\}$ is equivalent to making a unique sequence of three binary choices of whether to include each element 1, 2, and 3. We see that U has the following 8 subsets: $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$, and $\{1, 2, 3\}$. Therefore, the power set $\mathcal{P}(U)$ of the set $U = \{1, 2, 3\}$ is the set

$$\mathcal{P}(U) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}.$$

Set Operations. Compute the given set operations. Write your answer in roster or set-builder notation, or as a well-known set.

1.1.4.14. Compute $S \cap T$, where

$$S = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \ldots\}$$

is the set of even natural numbers and

$$T = \{n \in \mathbb{N} \mid n = 2k + 1 \text{ for some } k \in \mathbb{N}\} = \{1, 3, 5, 7, \ldots\}$$

is the set of odd integers.

Answer. $S \cap T = \emptyset$.

Solution. No natural number is both even and odd, and so $S \cap T = \emptyset$.

1.1.4.15. Compute $S \cup T$, where $S = \{0, 3, 4, 9\}$ and $T = \{1, 3, 6\}$.

Answer. $S \cup T = \{0, 1, 3, 4, 6, 9\}$.

Solution. $S \cup T$ contains all the elements of S and all the elements of T, so that $S \cup T = \{0, 1, 3, 4, 6, 9\}$.

1.1.4.16. Compute $S \setminus T$, where

$$S = \{n \in \mathbb{N} \mid n = 5k \text{ for some } k \in \mathbb{N}\} = \{0, 5, 10, 15, \ldots\}$$

is the set of natural numbers divisible by 5, and

$$T = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \ldots\}$$

is the set of even natural numbers.

Answer. $S \setminus T = \{n \in \mathbb{N} \mid n = 10k + 5 \text{ for some } k \in \mathbb{N}\} = \{5, 15, 25, 35, \ldots\}$.

Solution. $S \setminus T$ contains all the odd multiples of 5, so that

$$\begin{aligned} S \setminus T &= \{n \in \mathbb{N} \mid n = 10k + 5 \text{ for some } k \in \mathbb{N}\} \\ &= \{5, 15, 25, 35, \ldots\}. \end{aligned}$$

1.1.5 · Problems

1.1.5.1. Distributivity of set operations.

(a) Prove that $S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$ for all sets S, T, and U.

Solution. Let $x \in S \cap (T \cup U)$. Then $x \in S$ and $x \in T \cup U$, and so

$x \in T$ or $x \in U$. If $x \in T$, then $x \in S \cap T$, and so $x \in (S \cap T) \cup (S \cap U)$. On the other hand, if $x \in U$, then $x \in S \cap U$, and so $x \in (S \cap T) \cup (S \cap U)$.

In summary, we have shown that $x \in (S \cap T) \cup (S \cap U)$ for all elements $x \in S \cap (T \cup U)$, and so

$$S \cap (T \cup U) \subseteq (S \cap T) \cup (S \cap U).$$

Conversely, now let $x \in (S \cap T) \cup (S \cap U)$. Then $x \in S \cap T$ or $x \in S \cap U$. If $x \in S \cap T$, then $x \in S$ and $x \in T$. In particular, $x \in T \cup U$, and so $x \in S \cap (T \cup U)$.

In summary, we have shown that $x \in S \cap (T \cup U)$ for all $x \in (S \cap T) \cup (S \cap U)$, and so

$$(S \cap T) \cup (S \cap U) \subseteq S \cap (T \cup U);$$

we now conclude that $S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$.

(b) Prove that $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$ for all sets S, T, and U.

Solution. Let $x \in S \cup (T \cap U)$. Then $x \in S$ or $x \in T \cap U$. If $x \in S$, then $x \in S \cup T$ and $x \in S \cup U$, and so $x \in (S \cup T) \cap (S \cup U)$. On the other hand, if $x \in T \cap U$, then $x \in T$ and $x \in U$. In this case, $x \in S \cup T$ and $x \in S \cup U$, and so $x \in (S \cup T) \cap (S \cup U)$.

In summary, we have shown that $x \in (S \cup T) \cap (S \cup U)$ for all elements $x \in S \cup (T \cap U)$, and so

$$S \cup (T \cap U) \subseteq (S \cup T) \cap (S \cup U).$$

Conversely, now let $x \in (S \cup T) \cap (S \cup U)$. Then $x \in S \cup T$ and $x \in S \cup U$. In particular, if $x \notin S$, then $x \in T$ and $x \in U$, and so $x \in T \cap U$. We conclude that $x \in S \cup (T \cap U)$.

In summary, we have shown that $x \in S \cup (T \cap U)$ for all $x \in (S \cup T) \cap (S \cup U)$, and so

$$(S \cup T) \cap (S \cup U) \subseteq S \cup (T \cap U);$$

we now conclude that $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$.

Taken together, these results are called the **distributive properties** of the set operations.

1.1.5.2. De Morgan's laws.

(a) Prove that $S \setminus (T \cap U) = (S \setminus T) \cup (S \setminus U)$ for all sets S, T, and U.

Solution. First let $x \in S \setminus (T \cap U)$. Then $x \in S$ and $x \notin T \cap U$. We want to show that $x \in S \setminus T$ or $x \in S \setminus U$. To that end, suppose that $x \notin S \setminus T$. Since $x \in S$, we must have $x \in T$.

But $x \notin T \cap U$, and so $x \notin U$; that is, $x \in S \setminus U$. Since if $x \notin S \setminus T$, then $x \in S \setminus U$, we conclude that $x \in (S \setminus T) \cup (S \setminus U)$.

Conversely, now let $x \in (S \setminus T) \cup (S \setminus U)$, so that $x \in S \setminus T$ or $x \in S \setminus U$. If $x \in S \setminus T$, then $x \in S$ and $x \notin T$, so that $x \notin T \cap U$. We conclude that $x \in S \setminus (T \cap U)$.

In summary, we have shown that $x \in S \setminus (T \cap U)$ if and only if $x \in (S \setminus T) \cup (S \setminus U)$. We conclude that $S \setminus (T \cap U) = (S \setminus T) \cup (S \setminus U)$.

(b) Prove that $S \setminus (T \cup U) = (S \setminus T) \cap (S \setminus U)$.

Solution. First let $x \in S \setminus (T \cup U)$. Then $x \in S$ and $x \notin T \cup U$. If $x \in T$, then $x \in T \cup U$, and so $x \notin T$; we conclude that $x \in S \setminus T$. Similarly, if $x \in U$, then $x \in T \cup U$, and so $x \notin U$; we conclude that $x \in S \setminus U$. Thus $x \in (S \setminus T) \cap (S \setminus U)$.

Conversely, now let $x \in (S \setminus T) \cap (S \setminus U)$, so that $x \in S \setminus T$ and $x \in S \setminus U$. Thus $x \in S$, $x \notin T$, and $x \notin U$. If $x \in T \cup U$, then $x \in T$ or $x \in U$, and so $x \notin T \cup U$; we conclude that $x \in S \setminus (T \cup U)$.

In summary, we have shown that $x \in S \setminus (T \cup U)$ if and only if $x \in (S \setminus T) \cap (S \setminus U)$. We conclude that $S \setminus (T \cup U) = (S \setminus T) \cap (S \setminus U)$.

Taken together, these results are called **De Morgan's laws**.

1.1.5.3. Power sets. Determine whether the following statement is true or false:

The power set of a set contains at least two distinct elements.

If the statement is true, prove it. If the statement is false, disprove it by providing a counterexample.

Solution. The statement is false. For a counterexample, we consider the empty set $\emptyset$. The only subset of the empty set $\emptyset$ is $\emptyset$ itself.

1.1.5.4. Set operations and containment. Let S and T be sets.

(a) Prove that $S \cap T = S$ if and only if $S \subseteq T$.

Solution. Since $S \cap T \subseteq S$, it suffices to show that $S \subseteq S \cap T$ if and only if $S \subseteq T$. To that end, note that if $S \subseteq S \cap T$, then (1) of Proposition 1.1.14 Set operations and containment implies that $S \subseteq T$.

Conversely, we note that $S \subseteq S$, so that if $S \subseteq T$, then (1) of Proposition 1.1.14 Set operations and containment implies that $S \subseteq S \cap T$.

(b) Prove that $S \cup T = S$ if and only if $T \subseteq S$.

Solution. Since $S \cup S \cup T$, it suffices to show that $S \cup T \subseteq S$ if and only if $T \subseteq S$. To that end, note that if $S \cup T \subseteq S$, then (2) of Proposition 1.1.14 Set operations and containment implies that $T \subseteq S$.

Conversely, we note that $S \subseteq S$, so that if $T \subseteq S$, then (2) of Proposition 1.1.14 Set operations and containment implies that $S \cup T \subseteq S$.

(c) Prove that $S \setminus T = S$ if and only if $S \cap T = \emptyset$.

Solution. First suppose that $S \cap T = \emptyset$. Then $x \notin T$ for all $x \in S$, and so

$$S \setminus T = \{x \mid x \in S \text{ and } x \notin T\} = \{x \mid x \in S\} = S.$$

Conversely, now suppose that $S \cap T \neq \emptyset$. Then $x \in S \cap T$, and so $x \in S$ and $x \in T$. Since $x \in T$, $x \notin S \setminus T$. So x is an element of S but not $S \setminus T$; we conclude that $S \setminus T \neq S$.

1.2 · Maps and Functions
1.2.4 · Exercises

Cartesian products. Write down the Cartesian products of the given sets in roster notation.

1.2.4.1. Write down the Cartesian product $S \times T$ of the sets $S = \{0,1\}$ and $T = \{0,1,2\}$ in roster notation.

Answer. The Cartesian product $S \times T$ of the sets $S = \{0,1\}$ and $T = \{0,1,2\}$ is

$$S \times T = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2)\}.$$

1.2.4.2. Write down the Cartesian product $U \times V$ of the sets $U = \{0\}$ and $V = \{1,2,3,4\}$ in roster notation.

Answer. The Cartesian product $U \times V$ of the sets $U = \{0\}$ and $V = \{1,2,3,4\}$ is

$$U \times V = \{(0,1), (0,2), (0,3), (0,4)\}.$$

1.2.4.3. Write down the Cartesian product $X \times Y$ of the sets $X = \{0,1,2,3,4,5\}$ and $Y = \emptyset$ in roster notation.

Answer. The Cartesian product $X \times Y$ of the sets $X = \{0,1,2,3,4,5\}$ and $Y = \emptyset$ is $X \times Y = \emptyset$.

Is it a map? Consider the sets $X = \{0,2,4,6,8\}$ and $Y = \{0,1,2,3,4\}$. Determine whether or not the given object is a map from X to Y.

1.2.4.4. Determine whether or not

$$f = \{(0,0), (2,1), (4,2), (6,3), (8,4)\}$$

is a map from X to Y.

Answer. Yes, $f \colon X \to Y$ is a map from X to Y.

1.2.4.5. Determine whether or not

$$g = \{(0,1), (2,1), (4,4), (8,1)\}$$

is a map from X to Y.

Answer. No, g is not a map from X to Y.

Solution. There is no output $y \in Y$ so that $(6, y) \in g$, and so g does not satisfy the totality condition. Thus g is not a map from X to Y.

1.2.4.6. Determine whether or not

$$h = \{(0,1), (2,1), (4,4), (4,3), (6,2), (8,1)\}$$

is a map from X to Y.

Answer. No, h is not a map from X to Y.

Solution. Note that $(4,4), (4,3) \in h$, and so h does not satisfy the uniqueness condition. Thus h is not a map from X to Y.

1.2.4.7. Determine whether or not

$$j = \{(0,0), (2,0), (4,0), (6,0), (8,0)\}$$

is a map from X to Y.

Answer. Yes, $j \colon X \to Y$ is a map from X to Y.

1.2.4.8. Images and pre-images. Let $f_4 \colon \{1,2\} \to \{\text{red}, \text{blue}\}$ be the map from $\{1,2\}$ to $\{\text{red}, \text{blue}\}$ defined in Example 1.2.4 Maps; that is,

$$f_4 = \{(1, \text{blue}), (2, \text{blue})\}.$$

Compute the given images and pre-images. Express your answer in roster notation if necessary.

(a) Compute the images $f_4(1)$ and $f_4(2)$.

Answer. $f_4(1) = f_4(2) = \text{blue}$.

Solution. Since $(1, \text{blue}) \in f_4$, the image $f_4(1)$ of 1 under f_4 is

$$f_4(1) = \text{blue}.$$

Similarly, since $(2, \text{blue}) \in f_4$, the image $f_4(2)$ of 2 under f_4 is

$$f_4(2) = \text{blue}.$$

(b) Compute the image $\operatorname{im}(f_4)$.

Answer. $\operatorname{im}(f_4) = \{\text{blue}\}$.

Solution. Since $f_4(x) = \text{blue}$ for all $x \in \{1, 2\}$, the image $\operatorname{im}(f_4)$ of f_4 is

$$\operatorname{im}(f_4) = f_4(\{1, 2\}) = \{\text{blue}\}.$$

(c) Compute the pre-images $f_4^{-1}(\text{red})$ and $f_4^{-1}(\text{blue})$.

Answer. $f_4^{-1}(\text{red}) = \emptyset$ and $f_4^{-1}(\text{blue}) = \{1, 2\}$.

Solution. Since $f_4(x) = \text{blue}$ for all $x \in \{1, 2\}$, the pre-images $f_4^{-1}(\text{red})$ and $f_4^{-1}(\text{blue})$ of red and blue under f_4 are

$$f_4^{-1}(\text{red}) = \{x \in \{1, 2\} \mid f_4(x) = \text{red}\} = \emptyset$$

and

$$f_4^{-1}(\text{blue}) = \{x \in \{1, 2\} \mid f_4(x) = \text{blue}\} = \{1, 2\}.$$

Determining injectivity/surjectivity/bijectivity. Determine whether or not the given map is injective, surjective, and/or bijective.

1.2.4.9. Let $X = \{0, 1\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f\colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 0)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

Answer. f is neither injective, surjective, nor bijective.

1.2.4.10. Let $X = \{0, 1\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f\colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 2)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

Answer. f is injective, but it is neither surjective nor bijective.

1.2.4.11. Let $X = \{0, 1, 2\}$ and $Y = \{0, 1\}$, and consider the following map $f\colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 1), (2, 1)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

Answer. f is surjective, but it is neither injective nor bijective.

1.2.4.12. Let $X = \{0, 1, 2\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f : X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 2), (2, 1)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

Answer. f is injective, surjective, and bijective.

1.2.5 · Problems

1.2.5.1. Non-commutativity of Cartesian products. Give examples of sets X and Y so that $X \times Y \neq Y \times X$.

Solution. If $X = \{0\}$ and $Y = \{1\}$, then

$$\begin{aligned} X \times Y &= \{0\} \times \{1\} = \{(0, 1)\} \\ &\neq \{(1, 0)\} = \{1\} \times \{0\} = Y \times X. \end{aligned}$$

1.2.5.2. Existence of maps. Determine whether the following statement is true or false:

> For any sets X and Y, there is a map from X to Y.

If the statement is true, prove it. If the statement is false, disprove it by providing a counterexample.

Solution. Let $X = \{0, 1, 2, 3, 4, 5\}$ and $Y = \emptyset$. As shown in Exercise Group 1.2.4.1–3 Cartesian products, $X \times Y = \emptyset$. In particular, no subset of $X \times Y$ can satisfy the totality condition, and so there are no maps from X to Y.

1.2.5.3. Non-commutativity of composition.

(a) Give examples of maps f and g so that the composition $g \circ f$ is well-defined but the composition $f \circ g$ is not.

Solution. Let $X = \{0\}$, $Y = \{1\}$, and $Z = \{2\}$, and consider the maps $f : X \to Y$ and $g : Y \to Z$ defined by $f(0) = 1$ and $g(1) = 2$.

Since the codomain Y of f is the domain of g, the composition $g \circ f$ is well-defined. However, since the codomain Z of g is not the domain X of f, the composition $f \circ g$ is not well-defined.

(b) Give examples of maps f and g so that the compositions $g \circ f$ and $f \circ g$ are well-defined but $g \circ f \neq f \circ g$.

Solution. Let $X = \{0, 1, 2\}$, and consider the maps $f, g : X \to X$ from X to itself defined by

$$\begin{array}{lll} f(0) = 1 & f(1) = 2 & f(2) = 0 \\ g(0) = 1 & g(1) = 0 & g(2) = 2 \end{array}$$

Then

$$\begin{aligned} (g \circ f)(0) &= g(f(0)) = g(1) = 0 \\ &\neq 2 = f(1) = f(g(0)) = (f \circ g)(0), \end{aligned}$$

and so $g \circ f \neq f \circ g$.

1.2.5.4. Composition and injectivity/surjectivity/bijectivity. Let $f : X \to Y$ and $g : Y \to Z$ be maps between sets X, Y, and Z.

(a) Prove that if f and g are injective, then so too is their composition $g \circ f\colon X \to Z$.

Solution. Let $x_1, x_2 \in X$, and suppose that $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then

$$g(f(x_1)) = (g \circ f)(x_1) = (g \circ f)(x_2) = g(f(x_2)),$$

and so the injectivity of g implies that $f(x_1) = f(x_2)$. Moreover, the injectivity of f implies that $x_1 = x_2$. In summary, we have shown that for all inputs $x_1, x_2 \in X$, if $(g \circ f)(x_1) = (g \circ f)(x_2)$, then $x_1 = x_2$; that is, $g \circ f$ is injective.

(b) Prove that if f and g are surjective, then so too is their composition $g \circ f\colon X \to Z$.

Solution. Let $z \in Z$. Since g is surjective, $z = g(y)$ for some $y \in Y$. Moreover, since f is surjective, $y = f(x)$ for some $x \in X$. Thus

$$z = g(y) = g(f(x)) = (g \circ f)(x).$$

In summary, we have shown that for every output $z \in Z$, there is some input $x \in X$ so that $z = (g \circ f)(x)$; that is, $g \circ f$ is surjective.

(c) Prove that if f and g are bijective, then so too is their composition $g \circ f\colon X \to Z$.

Solution. Since f and g are bijective, they are both injective and surjective. (1) and (2) above not imply that their composition $g \circ f$ is both injective and surjective; that is, $g \circ f$ is bijective.

(d) Prove that if the composition $g \circ f\colon X \to Z$ is injective, then so too is f.

Solution. Suppose that f is not injective. Then $f(x_1) = f(x_2)$ for some distinct inputs $x_1, x_2 \in X$. Note that

$$(g \circ f)(x_1) = g(f(x_1)) = g(f(x_2)) = (g \circ f)(x_2).$$

Since $(g \circ f)(x_1) = (g \circ f)(x_2)$ but $x_1 \neq x_2$, $g \circ f$ is not injective.

In summary, we have shown that if f is not injective, then $g \circ f$ is not injective. The contrapositive of this implication is the desired result: if $g \circ f$ is injective, then so is f.

(e) Prove that if the composition $g \circ f\colon X \to Z$ is surjective, then so too is g.

Solution. Suppose that g is not surjective. Then there is some $z \in Z$ so that $z \neq g(y)$ for all $y \in Y$. Note that

$$(g \circ f)(x) = g(f(x)) \neq z$$

for all $x \in X$, and so $g \circ f$ is not surjective.

In summary, we have shown that if g is not surjective, then $g \circ f$ is not surjective. The contrapositive of this implication is the desired result: if $g \circ f$ is surjective, then so is g.

Note that the above results are used in the proof of Theorem 1.2.15 Invertibility. In order to avoid circular reasoning, you should prove the above statements by using only the definitions of injectivity, surjectivity, and bijectivity (and not the alternative characterizations given by Theorem 1.2.15 Invertibility and Corollary 1.2.16 Invertibility).

1.2.5.5. Indexed intersections and unions. Let $(U_j)_{j\in\mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$.

(a) Prove that for any subset $V \subseteq X$,

$$V \subseteq \bigcap_{j\in\mathcal{J}} U_j$$

if and only if $V \subseteq U_j$ for all indices $j \in \mathcal{J}$, and

$$\bigcup_{j\in\mathcal{J}} U_j \subseteq V$$

if and only if $U_j \subseteq V$ for all indices $j \in \mathcal{J}$.

Solution. First suppose that $V \subseteq \bigcap_{j\in\mathcal{J}} U_j$. Then for all $x \in V$,

$$x \in \bigcap_{j\in\mathcal{J}} U_j = \{x \in X \mid x \in U_j \text{ for all } j \in \mathcal{J}\},$$

and so $x \in U_j$ for all indices $j \in \mathcal{J}$. Thus $V \subseteq U_j$ for all indices $j \in \mathcal{J}$.

Conversely, now suppose that $V \subseteq U_j$ for all indices $j \in \mathcal{J}$, and let $x \in V$. Then $x \in U_j$ for all indices $j \in \mathcal{J}$, and so

$$x \in \{x \in X \mid x \in U_j \text{ for all } j \in \mathcal{J}\} = \bigcap_{j\in\mathcal{J}} U_j.$$

We conclude that $V \subseteq \bigcap_{j\in\mathcal{J}} U_j$.

Now suppose that $\bigcup_{j\in\mathcal{J}} U_j \subseteq V$. Fix an index $k \in \mathcal{J}$, and let $x \in U_k$. Then

$$x \in \{x \in X \mid x \in U_j \text{ for some } j \in \mathcal{J}\} = \bigcup_{j\in\mathcal{J}} U_j,$$

and so $x \in V$. We conclude that $U_k \subseteq V$, and note that this holds for all indices $k \in \mathcal{J}$.

Conversely, now suppose that $U_j \subseteq V$ for all indices $j \in \mathcal{J}$, and let $x \in \bigcup_{j\in\mathcal{J}} U_j$. Then

$$x \in \bigcup_{j\in\mathcal{J}} U_j = \{x \in X \mid x \in U_j \text{ for some } j \in \mathcal{J}\},$$

and so $x \in U_j$ for some index $j \in \mathcal{J}$. Since $U_j \subseteq V$, we have $x \in V$. We conclude that $\bigcup_{j\in\mathcal{J}} U_j \subseteq V$.

(b) Prove that

$$\bigcap_{k\in\mathcal{K}} U_{f(k)} = \bigcap_{j\in\mathcal{J}} U_j$$

and

$$\bigcup_{k\in\mathcal{K}} U_{f(k)} = \bigcup_{j\in\mathcal{J}} U_j$$

for any set $\mathcal{K}$ and surjection $f\colon \mathcal{K} \to \mathcal{J}$.

Solution. Let $x \in X$, and suppose first that $x \in U_j$ for all indices $j \in \mathcal{J}$. Then $f(k) \in \mathcal{J}$ for all indices $k \in \mathcal{K}$, and so $x \in U_{f(k)}$ for all such indices. Thus

$$\bigcap_{j\in\mathcal{J}} U_j \subseteq \bigcap_{k\in\mathcal{K}} U_{f(k)}.$$

Conversely, now suppose that $x \in U_{f(k)}$ for all indices $k \in \mathcal{K}$. Since $f\colon \mathcal{K} \to \mathcal{J}$ is surjective, this implies that $x \in U_j$ for all indices $j \in \mathcal{J}$. Thus

$$\bigcap_{k \in \mathcal{K}} U_{f(k)} \subseteq \bigcap_{j \in \mathcal{J}} U_j,$$

and so

$$\bigcap_{k \in \mathcal{K}} U_{f(k)} = \bigcap_{j \in \mathcal{J}} U_j.$$

Now suppose that $x \in U_j$ for some index $j \in \mathcal{J}$. Since $f\colon \mathcal{K} \to \mathcal{J}$ is surjective, $f(k) = j$ for some index $k \in \mathcal{K}$, and so $x \in U_j = U_{f(k)}$. In particular, $x \in U_{f(k)}$ for some index $k \in \mathcal{K}$, and so

$$\bigcup_{j \in \mathcal{J}} U_j \subseteq \bigcup_{k \in \mathcal{K}} U_{f(k)}.$$

Conversely, now suppose that $x \in U_{f(k)}$ for some index $k \in \mathcal{K}$. Since $f(k) \in \mathcal{J}$ is surjective, this implies that $x \in U_j$ for some index $j = f(k) \in \mathcal{J}$. Thus

$$\bigcup_{k \in \mathcal{K}} U_{f(k)} \subseteq \bigcup_{j \in \mathcal{J}} U_j,$$

and so

$$\bigcup_{k \in \mathcal{K}} U_{f(k)} = \bigcup_{j \in \mathcal{J}} U_j.$$

(c) Prove that

$$V \cup \bigcap_{j \in \mathcal{J}} U_j = \bigcap_{j \in \mathcal{J}} (V \cup U_j)$$

and

$$V \cap \bigcup_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \cap U_j)$$

for any subset $V \subseteq X$.

Solution. Let $x \in X$, and suppose that $x \in V \cup \bigcap_{j \in \mathcal{J}} U_j$. Then $x \in V$ or $x \in U_j$ for all indices $j \in \mathcal{J}$. If $x \in V$, then $x \in V \cup U_j$ for all indices $j \in \mathcal{J}$, and so

$$x \in \bigcap_{j \in \mathcal{J}} V \cup U_j.$$

On the other hand, if $x \notin V$, then $x \in U_j$ for all indices $j \in \mathcal{J}$, and so $x \in V \cup U_j$ for all indices $j \in \mathcal{J}$; that is,

$$x \in \bigcap_{j \in \mathcal{J}} V \cup U_j.$$

Conversely, now suppose that $x \in \bigcap_{j \in \mathcal{J}} V \cup U_j$. Then $x \in V \cup U_j$ for all indices $j \in \mathcal{J}$. If $x \notin V$, then $x \in U_j$ for all indices $j \in \mathcal{J}$, and so $x \in \bigcap_{j \in \mathcal{J}} U_j$. We conclude that

$$x \in V \cup \bigcap_{j \in \mathcal{J}} U_j.$$

In summary, we have shown that $x \in V \cup \bigcap_{j \in \mathcal{J}} U_j$ if and only if $x \in \bigcap_{j \in \mathcal{J}} V \cup U_j$, and so

$$V \cup \bigcap_{j \in \mathcal{J}} U_j = \bigcap_{k \in \mathcal{J}} V \cup U_k.$$

Now suppose that $x \in V \cap \bigcup_{j\in\mathcal{J}} U_j$. Then $x \in V$, and $x \in U_j$ for some index $j \in \mathcal{J}$. In particular, $x \in V \cap U_j$ for some index $j \in \mathcal{J}$, and so

$$x \in \bigcup_{j\in\mathcal{J}} V \cap U_j.$$

Conversely, now suppose that $x \in \bigcup_{j\in\mathcal{J}} V \cap U_j$. Then $x \in V \cap U_j$ for some index $j \in \mathcal{J}$, and so $x \in V$ and $x \in U_j$ for some index $j \in \mathcal{J}$. We conclude that

$$x \in V \cap \bigcup_{j\in\mathcal{J}} U_j.$$

In summary, we have shown that $x \in V \cap \bigcup_{j\in\mathcal{J}} U_j$ if and only if $x \in \bigcup_{j\in\mathcal{J}} V \cap U_j$, and so

$$V \cap \bigcup_{j\in\mathcal{J}} U_j = \bigcup_{j\in\mathcal{J}} V \cap U_j.$$

(d) Prove that

$$V \setminus \bigcap_{j\in\mathcal{J}} U_j = \bigcup_{j\in\mathcal{J}} (V \setminus U_j)$$

and

$$V \setminus \bigcup_{j\in\mathcal{J}} U_j = \bigcup_{j\in\mathcal{J}} (V \setminus U_j)$$

for any subset $V \subseteq X$.

Solution. Let $x \in X$, and suppose that $x \in V \setminus \bigcap_{j\in\mathcal{J}} U_j$. Then $x \in V$ and $x \notin \bigcap_{j\in\mathcal{J}} U_j$. So $x \notin U_j$ for some index $j \in \mathcal{J}$, and so $x \in V \setminus U_j$ for some index $j \in \mathcal{J}$; that is,

$$x \in \bigcup_{j\in\mathcal{J}} V \setminus U_j.$$

Conversely, now suppose that $x \in \bigcup_{j\in\mathcal{J}} V \setminus U_j$. Then $x \in V \setminus U_j$ for some index $j \in \mathcal{J}$, and so $x \in V$ and $x \notin U_j$ for some index $j \in \mathcal{J}$. In particular, $x \notin \bigcap_{j\in\mathcal{J}} U_j$, and so

$$x \in V \setminus \bigcap_{j\in\mathcal{J}} U_j.$$

In summary, we have shown that $x \in V \setminus \bigcap_{j\in\mathcal{J}} U_j$ if and only if $x \in \bigcup_{j\in\mathcal{J}} V \setminus U_j$, and so

$$V \setminus \bigcap_{j\in\mathcal{J}} U_j = \bigcup_{k\in\mathcal{J}} V \setminus U_k.$$

Let $x \in X$, and suppose that $x \in V \setminus \bigcup_{j\in\mathcal{J}} U_j$. Then $x \in V$ and $x \notin \bigcup_{j\in\mathcal{J}} U_j$. So $x \notin U_j$ for all indices $j \in \mathcal{J}$, and so $x \in V \setminus U_j$ for all indices $j \in \mathcal{J}$; that is,

$$x \in \bigcap_{j\in\mathcal{J}} V \setminus U_j.$$

Conversely, now suppose that $x \in \bigcap_{j\in\mathcal{J}} V \setminus U_j$. Then $x \in V \setminus U_j$ for all indices $j \in \mathcal{J}$, and so $x \in V$ and $x \notin U_j$ for all indices $j \in \mathcal{J}$. In particular, $x \notin \bigcup_{j\in\mathcal{J}} U_j$, and so

$$x \in V \setminus \bigcup_{j\in\mathcal{J}} U_j.$$

In summary, we have shown that $x \in V \setminus \bigcup_{j \in \mathcal{J}} U_j$ if and only if $x \in \bigcap_{j \in \mathcal{J}} V \setminus U_j$, and so

$$V \setminus \bigcup_{j \in \mathcal{J}} U_j = \bigcap_{k \in \mathcal{J}} V \setminus U_k.$$

1.4 · Finitary Operations and Relations
1.4.4 · Exercises

1.4.4.1. Indexed Cartesian products. Let $U_1 = \{1,2\}$, $U_2 = \{1,2,3\}$, and $U_3 = \{0,1\}$, and consider the Cartesian product $U = \prod_{k=1}^{3} U_k$.

(a) Write down the Cartesian product $U = \prod_{k=1}^{3} U_k$ in roster notation.

Answer. We have

$$\begin{aligned} U = \{&(1,1,0), (1,1,1), (1,2,0), (1,2,1), (1,3,0), (1,3,1), \\ &(2,1,0), (2,1,1), (2,2,0), (2,2,1), (2,3,0), (2,3,1)\}. \end{aligned}$$

(b) For each index $k \in \{1,2,3\}$, let $\pi_k \colon U \to U_k$ be the projection map onto the kth factor. Compute $\pi_1(2,3,0)$, $\pi_2(1,1,1)$, and $\pi_3(2,2,0)$.

Answer. We have

$$\pi_1(2,3,0) = 2 \qquad \pi_2(1,1,1) = 1 \qquad \pi_3(2,2,0) = 0.$$

1.4.4.2. Exponentiation of natural numbers. Let $e \colon \mathbb{N}^2 \to \mathbb{N}$ be the **exponentiation** operation defined by the formula

$$e(m,n) = m^n;$$

that is, e is defined recursively by the formulae $e(m,0) = 1$ and $e(m,n+1) = e^{m,n} \cdot m$.

(a) Compute $e(1,2)$ and $e(2,1)$. Is exponentiation a commutative binary operation?

Answer. $e(1,2) = 1^2 = 1$ and $e(2,1) = 2^1 = 2$. Since $e(1,2) \neq e(2,1)$, exponentiation is non-commutative.

(b) Compute $e(e(2,1),2)$ and $e(2,e(1,2))$. Is exponentiation an associative binary operation?

Answer. We compute

$$e(e(2,1),2) = e\left(2^1,2\right) = e(2,2) = 2^2 = 4$$

and

$$e(2,e(1,2)) = e\left(2,1^2\right) = e(2,1) = 2^1 = 2.$$

Since $e(e(2,1),2) \neq e(2,e(1,2))$, exponentiation is non-associative.

Equivalence classes. Let $X = \{0,1,2,3,\ldots,10\}$. Write down all the equivalence classes for the given equivalence relation on X in roster notation.

1.4.4.3. Consider the equivalence relation F on X defined by the condition that for all $x,y \in X$, $x \; F \; y$ if and only if the English words for x and y start with the same letter. Write down the equivalence classes in roster notation.

Answer. The equivalence classes are

$$\begin{aligned}
[0]_F &= \{0\} \\
[1]_F &= \{1\} \\
[2]_F = [3]_F = [10]_F &= \{2, 3, 10\} \\
[4]_F = [5]_F &= \{4, 5\} \\
[6]_F = [7]_F &= \{6, 7\} \\
[8]_F &= \{8\} \\
[8]_F &= \{8\}\,.
\end{aligned}$$

Solution. 0 is the only element of X beginning with Z, and so

$$[0]_F = \{0\}\,.$$

1 is the only element of X beginning with O, and so

$$[1]_F = \{1\}\,.$$

2, 3, and 10 are the only elements of X beginning with T, and so

$$[2]_F = [3]_F = [10]_F = \{2, 3, 10\}\,.$$

4 and 5 are the only elements of X beginning with F, and so

$$[4]_F = [5]_F = \{4, 5\}\,.$$

6 and 7 are the only elements of X beginning with S, and so

$$[6]_F = [7]_F = \{6, 7\}\,.$$

8 is the only element of X beginning with E, and so

$$[8]_F = \{8\}\,.$$

Finally, 9 is the only element of X beginning with N, and so

$$[9]_F = \{9\}\,.$$

1.4.4.4. Consider the equivalence relation L on X defined by the condition that for all $x, y \in X$, $x\ L\ y$ if and only if the English words for x and y end with the same letter. Write down all the equivalence classes in roster notation.

Answer. The equivalence classes are

$$\begin{aligned}
[0]_L = [2]_L &= \{0, 2\} \\
[1]_L = [3]_L = [5]_L = [9]_L &= \{1, 3, 5, 9\} \\
[4]_L = &= \{4\} \\
[6]_L = &= \{6\} \\
[7]_L = [10]_L &= \{7, 10\} \\
[8]_L &= \{8\}\,.
\end{aligned}$$

Solution. 0 and 2 are the only elements of X ending with O, and so

$$[0]_L = [2]_L = \{0, 2\}\,.$$

1, 3, 5, and 9 are the only elements of X ending with E, and so

$$[1]_L = [3]_L = [5]_L = [9]_L = \{1, 3, 5, 9\}.$$

4 is the only element of X ending with R, and so

$$[4]_L = \{4\}.$$

6 is the only element of X ending with X, and so

$$[6]_L = \{6\}.$$

7 and 10 are the only elements of X ending with N, and so

$$[7]_L = [10]_L = \{7, 10\}.$$

8 is the only element of X ending with T, and so

$$[8]_L = \{8\}.$$

1.4.5 · Problems

1.4.5.1. Revisiting commutativity/associativity of Cartesian products. Recall from Problem 1.2.5.1 Non-commutativity of Cartesian products that there are sets X and Y so that $X \times Y \neq Y \times X$. Similarly, there are sets X, Y, and Z so that $(X \times Y) \times Z \neq X \times (Y \times Z)$.

(a) Prove that there is a bijection between $X \times Y$ and $Y \times X$.

Solution. Consider the maps $f\colon X \times Y \to Y \times X$ and $g\colon Y \times X \to X \times Y$ defined by the formulae $f(x, y) = (y, x)$ and $g(y, x) = (x, y)$. We observe that

$$(g \circ f)(x, y) = g(f(x, y)) = g(y, x) = (x, y) = \mathrm{id}_{X \times Y}(x, y)$$

and

$$(f \circ g)(y, x) = f(g(y, x)) = f(x, y) = (y, x) = \mathrm{id}_{Y \times X}(y, x)$$

for all elements $x \in X$ and $y \in Y$, and so $g \circ f = \mathrm{id}_{X \times Y}$ and $f \circ g = \mathrm{id}_{Y \times X}$; that is, f and g are inverses, and therefore bijections by Corollary 1.2.16 Invertibility.

(b) Prove that there is a bijection between $(X \times Y) \times Z$ and $X \times (Y \times Z)$.

Solution. Consider the maps $f\colon (X \times Y) \times Z \to X \times (Y \times Z)$ and $g\colon X \times (Y \times Z) \to (X \times Y) \times Z$ defined by the formulae $f((x, y), z) = (x, (y, z))$ and $g(x, (y, z)) = ((x, y), z)$. We observe that

$$\begin{aligned}(g \circ f)((x, y), z) &= g(f((x, y), z)) = g(x, (y, z)) \\ &= ((x, y), z) = \mathrm{id}_{(X \times Y) \times Z}((x, y), z)\end{aligned}$$

and

$$\begin{aligned}(f \circ g)(x, (y, z)) &= f(g(x, (y, z))) = f((x, y), z) \\ &= (x, (y, z)) = \mathrm{id}_{X \times (Y \times Z)}(x, (y, z))\end{aligned}$$

for all elements $x \in X$, $y \in Y$, and $z \in Z$, and so $g \circ f = \mathrm{id}_{(X \times Y) \times Z}$ and $f \circ g = \mathrm{id}_{X \times (Y \times Z)}$; that is, f and g are inverses, and therefore bijections by Corollary 1.2.16 Invertibility.

While the sets $X \times Y$ and $Y \times X$ and the sets $(X \times Y) \times Z$ and $X \times (Y \times Z)$ are not equal, they may safely be identified via these bijections.

1.4.5.2. Cross products. The **cross product** $\times\colon \left(\mathbb{R}^3\right)^2 \to \mathbb{R}^3$ is the binary operation on 3-dimensional real coordinate space

$$\mathbb{R}^3 = \{(x_1, x_2, x_3) \mid x_1, x_2, x_3 \in \mathbb{R}\}$$

defined by the formula

$$(x_1, x_2, x_3) \times (y_1, y_2, y_3) = (x_2y_3 - x_3y_2, x_3y_1 - x_1y_3, x_1y_2 - x_2y_1)$$

For this problem, you may assume all standard properties of the arithmetic of real numbers, even though we will not formally define $\mathbb{R}$ until Section 2.2 Constructing the Real Numbers.

(a) Determine whether or not the cross product $\times$ on $\mathbb{R}^3$ is commutative. Justify your answer.

Solution. Consider the vectors $\mathbf{e}_1 = (1,0,0)$, $\mathbf{e}_2 = (0,1,0)$, and $\mathbf{e}_3 = (0,0,1)$. We note that

$$\begin{aligned}\mathbf{e}_1 \times \mathbf{e}_2 &= (1,0,0) \times (0,1,0)\\ &= (0\cdot 0 - 0\cdot 1, 0\cdot 0 - 1\cdot 0, 1\cdot 1 - 0\cdot 0) = (0,0,1) = \mathbf{e}_3\end{aligned}$$

and

$$\begin{aligned}\mathbf{e}_2 \times \mathbf{e}_1 &= (0,1,0) \times (1,0,0)\\ &= (1\cdot 0 - 0\cdot 0, 0\cdot 1 - 0\cdot 0, 0\cdot 0 - 1\cdot 1) = (0,0,-1) = -\mathbf{e}_3.\end{aligned}$$

Since

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3 \neq -\mathbf{e}_3 = \mathbf{e}_2 \times \mathbf{e}_1,$$

the cross product $\times$ is non-commutative.

(b) Determine whether or not the cross product $\times$ on $\mathbb{R}^3$ is associative. Justify your answer.

Solution. Consider the vectors $\mathbf{e}_1 = (1,0,0)$, $\mathbf{e}_2 = (0,1,0)$, and $\mathbf{e}_3 = (0,0,1)$. We note that

$$\begin{aligned}\mathbf{e}_1 \times \mathbf{e}_1 &= (1,0,0) \times (1,0,0)\\ &= (0\cdot 0 - 0\cdot 0, 0\cdot 1 - 1\cdot 0, 1\cdot 0 - 0\cdot 1) = (0,0,0) = \mathbf{0},\end{aligned}$$

$$\begin{aligned}\mathbf{0} \times \mathbf{e}_2 &= (0,0,0) \times (0,1,0)\\ &= (0\cdot 0 - 0\cdot 1, 0\cdot 0 - 0\cdot 0, 0\cdot 1 - 0\cdot 0) = (0,0,0) = \mathbf{0},\end{aligned}$$

and

$$\begin{aligned}\mathbf{e}_1 \times \mathbf{e}_3 &= (1,0,0) \times (0,0,1)\\ &= (0\cdot 1 - 0\cdot 0, 0\cdot 0 - 1\cdot 1, 1\cdot 0 - 0\cdot 0) = (0,-1,0) = -\mathbf{e}_2.\end{aligned}$$

Together with the computations above in (a), we see that

$$\begin{aligned}(\mathbf{e}_1 \times \mathbf{e}_1) \times \mathbf{e}_2 = \mathbf{0} \times \mathbf{e}_2 = \mathbf{0}&\\ \neq -\mathbf{e}_2 = \mathbf{e}_1 \times \mathbf{e}_3 = \mathbf{e}_1 \times (\mathbf{e}_1 \times \mathbf{e}_2)&,\end{aligned}$$

and so the cross product $\times$ is non-associative.

1.4.5.3. Generating equivalence relations. Let X be a set.

(a) Prove that the intersection $R = \bigcap_{j \in \mathcal{J}} R_j$ of any indexed family $(R_j)_{j \in \mathcal{J}}$ of equivalence relations R_j on X is an equivalence relation on X.

Solution. Let $x \in X$. Since for each index $j \in \mathcal{J}$, the equivalence relation R_j is reflexive, we have $(x, x) \in R_j$. Thus

$$(x, x) \in \bigcap_{j \in \mathcal{J}} R_j = R,$$

and so R is reflexive.

Now let $x_1, x_2 \in X$, and suppose that $(x_1, x_2) \in R$. Then $(x_1, x_2) \in R_j$ for all indices $j \in \mathcal{J}$, and so $(x_2, x_1) \in R_j$ for all such indices by symmetry. Thus

$$(x_2, x_1) \in \bigcap_{j \in \mathcal{J}} R_j = R,$$

and so R is symmetric.

Finally, let $x_1, x_2, x_3 \in X$, and suppose that $(x_1, x_2), (x_2, x_3) \in R$. Then $(x_1, x_2), (x_2, x_3) \in R_j$ for all indices $j \in \mathcal{J}$, and so $(x_1, x_3) \in R_j$ for all such indices by transitivity. Thus

$$(x_1, x_3) \in \bigcap_{j \in \mathcal{J}} R_j = R,$$

and so R is transitive. Since R is reflexive, symmetric, and transitive, it is an equivalence relation on X.

(b) Prove that for any binary relation R on X, there is a unique equivalence relation S on X which satisfies the following conditions:

(a) S is a **refinement** of R; that is, $R \subseteq S$.

(b) S is **coarser** than any equivalence relation which **refines** R; that is, $S \subseteq T$ for any equivalence relation T on X so that $R \subseteq T$.

Solution. Denote by $\mathcal{T}$ the set of equivalence relations on X which contain R, and let

$$S = \bigcap_{T \in \mathcal{T}} T.$$

By (a) above, S is an equivalence relation on X. Since $R \subseteq T$ for all $T \in \mathcal{T}$, (a) of Problem 1.2.5.5 Indexed intersections and unions implies that $R \subseteq S$; that is, S satisfies condition (1). On the other hand, $S \subseteq T$ for all $T \in \mathcal{T}$, and so S satisfies condition (2).

Uniqueness of S follows from the fact that $S \in \mathcal{T}$; specifically, if an equivalence T satisfies condition (1), then $T \in \mathcal{T}$ and so $S \subseteq T$. If T also satisfies condition (2), then $T \subseteq S$, and so $T = S$.

The unique equivalence relation on X constructed in (b) above is called the equivalence relation **generated** by R.

1.4.5.4. Universal property of quotients. Let $\equiv$ be an equivalence relation on a set X. A map $f \colon X \to Y$ from X to a set Y is called **$\equiv$-invariant** if for all elements $x_1, x_2 \in X$, if $x_1 \equiv x_2$, then $f(x_1) = f(x_2)$.

Prove that for every $\equiv$-invariant map $f \colon X \to Y$, there is a map $\overline{f} \colon X/{\equiv} \to Y$ so that $\overline{f} \circ \pi_{\equiv} = f$.

Solution. Let $x_1, x_2 \in X$, and suppose that $[x_1]_\equiv = [x_2]_\equiv$. Then $x_1 \equiv x_2$, and so $f(x_1) = f(x_2)$. We conclude that the map $\overline{f}\colon X/{\equiv} \to Y$ defined by the formula

$$\overline{f}([x]_\equiv) = f(x)$$

is well-defined. Moreover, we note that

$$\left(\overline{f} \circ \pi_\equiv\right)(x) = \overline{f}(\pi_\equiv(x)) = \overline{f}([x]_\equiv) = f(x)$$

for all inputs $x \in X$, and so $\overline{f} \circ \pi_\equiv = f$.

1.5 · Introduction to Order Theory
1.5.4 · Exercises

Identifying partial and total orders. Determine whether or not the given preorder is a partial order. If the preorder is a partial order, determine whether or not it is total.

1.5.4.1. Let R be the preorder on the set $\{1,2,3,4\}$ defined by

$$\{(4,2),(4,3),(2,1),(3,1),(4,1),(1,3),(2,3),(4,4),(3,3),(2,2),(1,1)\}.$$

Determine whether or not R is a partial order on $\{1,2,3,4\}$. If so, determine whether or not it is total.

Answer. R is not a partial order.

Solution. We observe that $(1,3),(3,1) \in R$, but $1 \neq 3$. Thus R is not anti-symmetric, and so R is not a partial order.

1.5.4.2. Let $\precsim$ be the preorder on the set P of all people (living or dead) defined so that for all people $p, q \in P$, $p \precsim q$ if and only if either p is either q or an ancestor of q. Determine whether or not $\precsim$ is a partial order on P. If so, determine whether or not it is total.

Answer. $\precsim$ is a partial order but not a total order.

Solution. Let $p, q \in P$ be people, and suppose that $p \preceq q$ and $q \preceq p$. If p and q are different people, then it is not possible for each to be the ancestor of the other (since ancestors are born strictly before their descendants). Thus p and q are the same person. We conclude that the preorder $\precsim$ is anti-symmetric and therefore a partial order.

However, $\precsim$ is not a total order, since for example my siblings and I are not comparable. For example, my sister is not my ancestor, and I am not my sister's ancestor.

1.5.4.3. Let $\precsim$ be the preorder on the set C of all circles in the Cartesian plane so that for all circles $c, d \in C$, $c \precsim d$ if and only if the diameter of c is at most the diameter of d. Determine whether or not $\precsim$ is a partial order on C. If so, determine whether or not it is total.

Answer. $\precsim$ is not a partial order.

Solution. Since there are distinct circles with the same diameter, $\precsim$ is not anti-symmetric, and therefore is not a partial order.

Bounds. Determine whether or not the following subsets of the partially ordered set $\mathcal{P}(\mathbb{N})$ are bounds for the given element.

1.5.4.4. Which of the following are upper bounds for the set $\{1,2,3\}$ in the partially ordered set $\mathcal{P}(\mathbb{N})$?

A. $\emptyset$

B. $\{1, 2\}$

C. $\{1, 2, 3\}$

D. $\mathbb{N} \setminus \{1, 2\}$

E. $\mathbb{N} \setminus \{0\}$

F. $\mathbb{N}$

Answer. C, E, F.

Solution.

A. *Incorrect.*

B. *Incorrect.*

C. *Correct.*

D. *Incorrect.*

E. *Correct.*

F. *Correct.*

1.5.4.5. Which of the following are lower bounds for the set $\{1, 2, 3\}$ in the partially ordered set $\mathcal{P}(\mathbb{N})$?

A. $\emptyset$

B. $\{1, 2\}$

C. $\{1, 2, 3\}$

D. $\mathbb{N} \setminus \{1, 2\}$

E. $\mathbb{N} \setminus \{0\}$

F. $\mathbb{N}$

Answer. A, B, C.

Solution.

A. *Correct.*

B. *Correct.*

C. *Correct.*

D. *Incorrect.*

E. *Incorrect.*

F. *Incorrect.*

Boundedness. Determine the boundedness of the following subsets of the totally ordered set $\mathbb{N}$.

1.5.4.6. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded from below?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

Answer. A, B, C, D, E.

Solution.

A. *Correct.*

B. *Correct.*

C. *Correct.*

D. *Correct.*

E. *Correct.*

1.5.4.7. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded from above?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

Answer. A, B.

Solution.

A. *Correct.*

B. *Correct.*

C. *Incorrect.*

D. *Incorrect.*

E. *Incorrect.*

1.5.4.8. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

Answer. A, B.

Solution.

A. *Correct.*

B. *Correct.*

C. *Incorrect.*

D. *Incorrect.*

E. *Incorrect.*

1.5.5 · Problems

1.5.5.1. Order by divisibility. Consider the binary relation $\mid$ on the natural numbers $\mathbb{N}$ defined so that for all natural numbers $m, n \in \mathbb{N}$, $m \mid n$ if and only if $mk = n$ for some natural number $k \in \mathbb{N}$; that is, $m \mid n$ if and only if n is a multiple of m.

(a) Prove that $\mid$ is a partial order on the set $\mathbb{N}$ of natural numbers.

Solution. We must prove that the binary relation $\mid$ is reflexive, anti-symmetric, and transitive. To that end, first let $n \in \mathbb{N}$ be a natural number, and note that $n \cdot 1 = n$. Thus $n \mid n$, and so the binary relation $\mid$ is reflexive.

Now let $m, n \in \mathbb{N}$ be natural numbers, and suppose that $m \mid n$ and $n \mid m$. Then $mk = n$ and $nl = m$ for some natural numbers $k, l \in \mathbb{N}$. If $m = 0$, then

$$n = mk = 0k = 0 = m.$$

On the other hand, if $m \neq 0$, then we observe that

$$m \cdot 1 = m = nl = (mk)\, l = m\,(kl)\,,,$$

and so Corollary 1.3.16 Cancellation of multiplication implies that $kl = 1$. Since $k, l \in \mathbb{N}$ are natural numbers, we must have $k = l = 1$, so that

$$m = nl = n \cdot 1 = n.$$

In summary, we have shown that if $m \mid n$ and $n \mid m$, then $m = n$. Thus the binary relation $\mid$ is anti-symmetric

Finally, let $m, n, p \in \mathbb{N}$ be natural numbers, and suppose that $m \mid n$ and $n \mid p$. Then $mk = n$ and $nl = p$ for some natural numbers $k, l \in \mathbb{N}$, and so

$$m\,(kl) = (mk)\, l = nl = p.$$

Thus the binary relation $\mid$ is transitive.

Since the binary relation $\mid$ is reflexive, anti-symmetric, and transitive, it is a partial order on $\mathbb{N}$.

(b) Is the binary relation $\mid$ a total order on $\mathbb{N}$? Justify your answer.

Solution. No, the binary relation $\mid$ is not a total order on $\mathbb{N}$, since for example $2 \not| 3$ and $3 \not| 2$.

(c) What are the extrema of the partially ordered set $(\mathbb{N}, \mid)$?

Solution. For all natural numbers $n \in \mathbb{N}$, $1n = n$ and $0n = 0$, so that $1 \mid n \mid 0$. Thus 1 and 0 are the minimum and maximum of the partially ordered set $(\mathbb{N}, \mid)$, respectively.

(d) We will abuse notation and denote the restriction of $|$ to the set $\mathbb{N} \setminus \{0,1\}$ by $|$ as well. What are the extremal elements of the partially ordered set $(\mathbb{N} \setminus \{0,1\}, |)$? Are these extremal elements extrema?

Solution. Let $n \in \mathbb{N} \setminus \{0,1\}$, and note that $n \mid 2n$ but $n \neq 2n$. Thus there are no maximal elements of the partially ordered set $(\mathbb{N} \setminus \{0,1\}, |)$.

Conversely, let $p \in \mathbb{N}$ be a prime number. Since the only natural number divisors of p are 1 and p, p is a minimal element of the partially ordered set $(\mathbb{N} \setminus \{0,1\}, |)$. On the other hand, any composite natural number has proper divisors, and so is not minimal.

1.5.5.2. Extremal subsets. Find the extremal elements of the partially ordered set $\mathcal{P}(\mathbb{N}) \setminus \{\emptyset, \mathbb{N}\}$.

Solution. Denote by X the partially ordered set $\mathcal{P}(\mathbb{N}) \setminus \{\emptyset, \mathbb{N}\}$. For all natural numbers $n \in \mathbb{N}$, the only subsets of the set $\{n\}$ are $\emptyset$ and $\{n\}$ itself. Thus $\{n\}$ is minimal in X. On the other hand, any proper subset of $\mathbb{N}$ which contains at least two natural numbers has a proper non-empty subset, and so is not minimal in X.

For all natural numbers $n \in \mathbb{N}$, the only subsets of $\mathbb{N}$ which contain $\mathbb{N} \setminus \{n\}$ are $\mathbb{N}$ and $\mathbb{N} \setminus \{n\}$ itself. Thus $\mathbb{N} \setminus \{n\}$ is maximal in X. On the other hand, any non-empty subset of $\mathbb{N}$ which misses at least two natural numbers is contained in a strictly larger proper subset, and so is not maximal in X.

1.5.5.3. Opposite of an order. The **opposite** R^{op} of a binary relation R on a set X is the binary relation defined by

$$R^{\mathrm{op}} = \left\{ (x_2, x_1) \in X^2 \;\middle|\; (x_1, x_2) \in R \right\}.$$

(a) Prove that the opposite of a preorder is a preorder.

Solution. Let $\precsim$ be a preorder on a set X, and denote by $\succsim$ the opposite. We will show that the binary operation $\succsim$ is reflexive and transitive. To that end, first let $x \in X$. The reflexivity of $\precsim$ implies that $x \precsim x$, and so $x \succsim x$. Thus the binary relation $\succsim$ is reflexive.

Now let $x, y, z \in X$, and suppose that $x \succsim y$ and $y \succsim z$. Then $z \precsim y$ and $y \precsim x$, and so the transitivity of $\precsim$ implies that $z \precsim x$. Thus $x \succsim z$, and so the binary relation $\succsim$ is transitive.

Since the binary relation $\succsim$ is reflexive and transitive, it is a preorder on X.

(b) Prove that the opposite of a partial order is a partial order.

Solution. Let $\preceq$ be a partial order on a set X, and denote by $\succeq$ the opposite. Since $\preceq$ is a preorder, $\succeq$ is a preorder on X by (a) above. Thus $\succeq$ is reflexive and transitive, and so it remains to show that $\succeq$ is anti-symmetric.

To that end, let $x, y \in X$, and suppose that both $x \succeq y$ and $y \succeq x$. Then $y \preceq x$ and $x \preceq y$, and so the anti-symmetry of $\preceq$ implies that $x = y$. Thus $\succeq$ is anti-symmetric.

Since the binary relation $\succeq$ is reflexive, anti-symmetric, and transitive, it is a partial order on X.

(c) Prove that the opposite of a total order is a total order.

Solution. Let $\leq$ be a total order on a set X, and denote by $\geq$ the opposite. Since $\leq$ is a partial order, $\geq$ is a partial order on X by (b) above. Thus $\geq$ is reflexive, anti-symmetric, and transitive, and so it remains to show that $\geq$ is total. To that end, let $x, y \in X$. The totality of $\leq$ implies that either $x \leq y$ or $y \leq x$. Thus $y \geq x$ and $x \geq y$, and so the binary relation $\geq$ is total.

Since $\geq$ is anti-symmetric, transitive, and total, it is a total order on X.

1.5.5.4. Orders from pullbacks.

(a) Prove that the pullback of a preorder is a preorder.

Solution. Consider the pullback $\precsim_f$ of a preorder $\precsim$ on a set Y along a map f from a set X to Y. For all elements $x \in X$, the reflexivity of $\precsim$ implies that $f(x) \precsim f(x)$, and so $x \precsim_f x$; that is, $\precsim_f$ is reflexive.

For all elements $x_1, x_2, x_3 \in X$, suppose that $x_1 \precsim_f x_2$ and $x_2 \precsim_f x_3$. Then $f(x_1) \precsim f(x_2)$ and $f(x_2) \precsim f(x_3)$, and so the transitivity of $\precsim$ implies that $f(x_1) \precsim f(x_3)$; that is, $x_1 \precsim_f x_3$, and so $\precsim_f$ is transitive.

Since the binary relation $\precsim_f$ is reflexive and transitive, it is a preorder on X.

(b) Prove that the pullback $\preceq_f$ of a partial order $\preceq$ on a set Y along a map $f\colon X \to Y$ is a partial order on X if and only if f is injective.

Solution. Since $\preceq$ is a partial order on Y, it is a preorder on Y, and thus $\preceq_f$ is a preorder on X by (a) above. Thus it suffices to show that $\preceq_f$ is anti-symmetric if and only if f is injective. To that end, suppose first that $\preceq_f$ is not anti-symmetric. Then $x_1 \preceq_f x_2$ and $x_2 \preceq_f x_1$ for some elements $x_1, x_2 \in X$ so that $x_1 \neq x_2$. We observe that $f(x_1) \preceq f(x_2)$ and $f(x_2) \preceq f(x_1)$, and so the anti-symmetry of $\preceq$ implies that $f(x_1) = f(x_2)$. Since $x_1 \neq x_2$ but $f(x_1) = f(x_2)$, f is not injective.

Conversely, suppose that f is not injective. Then $f(x_1) = f(x_2)$ for some elements $x_1, x_2 \in X$ so that $x_1 \neq x_2$. We see that $f(x_1) \preceq f(x_2)$ and $f(x_2) \preceq f(x_1)$, so that $x_1 \preceq_f x_2$ and $x_2 \preceq_f x_1$. That $x_1 \neq x_2$ now implies that $\preceq_f$ is not anti-symmetric.

(c) Is the pullback of a total order along an injective map necessarily a total order? If so, prove it. If not, provide a counterexample.

Solution. Yes, the pullback of a total order along an injective map is necessarily a total order. Indeed, consider the pullback $\leq_f$ of a total order $\leq$ on a set Y along an injective map $f\colon X \to Y$ from a set X to Y. Since $\leq$ is a total order, Lemma 1.5.14 Total orders are partial orders implies that it is a partial order, and so $\leq_f$ is a partial order by (b) above. Thus it suffices to show that $\leq_f$ must be total.

To that end, let $x_1, x_2 \in X$, and note that the totality of $\leq$ implies that either $f(x_1) \leq f(x_2)$ or $f(x_2) \leq f(x_1)$ (or both); that is, either $x_1 \leq_f x_2$ or $x_2 \leq_f x_1$ (or both). Thus $\leq_f$ is total, and hence a total order on X.

2 · Constructing the Number Systems
2.2 · Constructing the Real Numbers
2.2.6 · Problems

2.2.6.1. Cancellation of addition. Prove that for all Dedekind cuts $X, Y, Z \in \mathbb{R}$, if $X + Z \subseteq Y + Z$, then $X \subseteq Y$.

Solution. If $X + Z \subseteq Y + Z$, then

$$X = X + X_0 = X + Z + (-Z) \leq Y + Z + (-Z) = Y + X_0 = Y$$

by Theorem 2.2.15 Arithmetic of Dedekind cuts.

2.3 · The Complete Ordered Field

2.3.3 · Exercises

Algebraic structures other than fields. Determine which of the ordered field axioms are satisfied by the given algebraic structure.

2.3.3.1. Which of the ordered field axioms are not satisfied by the set $\mathbb{Z}$ of integers?

Solution. The integers $\mathbb{Z}$ satisfy all the ordered field axioms except the existence of multiplicative inverses. In fact, only 1 and -1 have multiplicative inverses.

2.3.3.2. Which of the ordered field axioms are not satisfied by the set $\mathbb{N}$ of natural numbers?

Solution. The natural numbers $\mathbb{N}$ satisfy all the ordered field axioms except the existence of additive and multiplicative inverses. In fact, only 0 has an additive inverse, and only 1 has a multiplicative inverse.

2.3.3.3. Let X be a set. Which of the ordered field axioms are not satisfied by the ordered quadruple $(\mathcal{P}(X), \cup, \cap, \subseteq)$?

Solution. The ordered quadruple $(\mathcal{P}(X), \cap, \cup, \subseteq)$ satisfies all the ordered field axioms except the existence additive inverses, the existence of multiplicative inverses, and totality.

2.3.4 · Problems

2.3.4.1. Uniqueness in a field. Let $(F, +, \cdot)$ be a field

(a) Show that the additive identity element $0 \in F$ is unique.

Solution. Let $x \in F$, and suppose that x is an additive identity element; that is, suppose that $x + y = y + x = y$ for all $y \in F$. Then in particular,

$$x = x + 0 = 0.$$

(b) Show that the multiplicative identity element $1 \in F$ is unique.

Solution. Let $x \in F$, and suppose that x is a multiplicative identity element; that is, suppose that $x \cdot y = y \cdot x = y$ for all $y \in F$. Then in particular,

$$x = x \cdot 1 = 1.$$

(c) Show that additive inverses are unique; that is, given elements $x, y, z \in F$, show that if

$$x + y = x + z = 0,$$

then $y = z$.

Solution. Suppose that $x+y = x+z = 0$ for some elements $x, y, z \in F$, and note that

$$\begin{aligned} y &= y + 0 = y + x + z \\ &= x + y + z = 0 + z = z. \end{aligned}$$

(d) Show that multiplicative inverses are unique; that is, given elements $x, y, z \in F$, show that if

$$x \cdot y = x \cdot z = 1,$$

then $y = z$.

Solution. Suppose that $x \cdot y = x \cdot z = 1$ for some elements $x, y, z \in F$, and note that

$$\begin{aligned} y &= y \cdot 1 = y \cdot x \cdot z \\ &= x \cdot y \cdot z = 1 \cdot z = z. \end{aligned}$$

2.3.4.2. Sign in an ordered field. Let $(F, +, \cdot, \leq)$ be an ordered field. As in the case of the rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$, we can define the **sign** of an element of F by comparison with the additive identity element 0:

- *Positivity.*

  A non-zero element $x \in F \setminus \{0\}$ is called **positive** if $0 \leq x$.

- *Negativity.*

  A non-zero element $x \in F \setminus \{0\}$ is called **negative** if $x \leq 0$.

- *Zero.*

  The additive identity element $0 \in F$ is not positive or negative.

(a) Let $x \in F$. Show that if $x \neq 0$, then x and its additive inverse $-x$ have opposite sign.

Solution. We first note that if $-x = 0$, then

$$x = x + 0 = x + (-x) = 0.$$

Thus if $x \neq 0$, then $-x \neq 0$. If x is positive, then $0 \leq x$, and so

$$-x = 0 + (-x) \leq x + (-x) = 0,$$

and so $-x$ is negative. On the other hand, if x is negative, then $x \leq 0$, and so

$$0 = x + (-x) \leq 0 + (-x) = -x,$$

and so $-x$ is positive.

(b) More generally, show that for all elements $x, y \in F$, if $x \leq y$ then $-y \leq -x$.

Solution. We note that

$$\begin{aligned} -y &= 0 + (-y) = x + (-x) + (-y) \\ &\leq y + (-x) + (-y) = y + (-y) + (-x) = 0 + (-x) = -x. \end{aligned}$$

(c) Show that $(-1) \cdot x = -x$ for all elements $x \in F$.

Solution. We observe that

$$x + ((-1) \cdot x) = (1 \cdot x) + ((-1) \cdot x) = (1 + (-1)) \cdot x = 0 \cdot x = 0,$$

and so $(-1) \cdot x$ is an additive inverse for x. Thus (c) in Problem 2.3.4.1 Uniqueness in a field implies that $-1 \cdot x = -x$.

(d) Show that the product of two negative elements is positive.

Solution. Let $x, y \in F$ be negative elements. Then $x \neq 0$ and $y \neq 0$; we first argue that $x \cdot y \neq 0$. Indeed, if $x \cdot y = 0$, then

$$\begin{aligned} 1 = 1 \cdot 1 = x \cdot x^{-1} \cdot y \cdot y^{-1} &= x \cdot x^{-1} \cdot y \cdot y^{-1} \\ = x \cdot y \cdot x^{-1} \cdot y^{-1} &= 0 \cdot x^{-1} \cdot y^{-1} = 0. \end{aligned}$$

This contradiction implies that $x \cdot y \neq 0$. We now observe that (a) implies that $-x$ and $-y$ are positive. Moreover, (c) implies that

$$\begin{aligned} x \cdot y &= 1 \cdot x \cdot y = (-1) \cdot (-1) \cdot x \cdot y \\ &= (-1) \cdot x \cdot (-1) \cdot y = (-x) \cdot (-y) \geq 0 \end{aligned}$$

is positive.

2.3.4.3. Infima and Dedekind completeness. Let $(F, +, \cdot, \leq)$ be an ordered field, and consider the map $n \colon F \to F$ defined by the formula $n(x) = -x$; that is, n sends an input $x \in F$ to its additive inverse $-x$.

(a) Show that for a subset $U \subseteq F$, U is bounded from above if and only if $n(U)$ is bounded from below, and U is bounded from below if and only if $n(U)$ is bounded from above.

Solution. If U is bounded from above, then there is some element $x \in F$ so that $u \leq x$ for all $u \in U$. We observe that (b) in Problem 2.3.4.2 Sign in an ordered field implies that

$$n(x) = -x \leq -u = n(u)$$

for all $u \in U$, and so $n(x)$ is a lower bound for $n(U)$.

Similarly, if U is bounded from below, then there is some element $x \in F$ so that $x \leq u$ for all $u \in U$. We observe that (b) in Problem 2.3.4.2 Sign in an ordered field implies that

$$n(u) = -u \leq -x = n(x)$$

for all $u \in U$, and so $n(x)$ is an upper bound for $n(U)$.

The reverse implications follow from the observation that n is its own inverse and then applying the above arguments to $n(U)$: if $n(U)$ is bounded from above (resp. below), then $U = n(n(U))$ is bounded from below (resp. above).

(b) Show that a subset $U \subseteq F$ has a supremum if and only if $n(U)$ has an infimum, and in this case

$$\inf(n(U)) = n(\sup(U)).$$

Similarly, show that U has an infimum if and only if $n(U)$ has a supremum, and in this case

$$\sup(n(U)) = n(\inf(U))$$

Solution. The argument in the above proof of (a) shows that n interchanges upper and lower bounds. Specifically, n sends the set of upper bounds (resp. lower bounds) of a set U to the set of lower bounds (resp. upper bounds) of $n(U)$. Moreover, by (b) of Problem 2.3.4.2 Sign in an ordered field, if $x \leq y$, then $n(y) \leq n(x)$. As such, n sends a least upper bound (resp. greatest lower bound) for U to a greatest lower bound (resp. least upper bound) for $n(U)$. In particular, if U has a supremum (resp. infimum), then $n(U)$ has an infimum (resp. supremum). The converse holds by applying the same result to $n(U)$ and observing that n is an involution; that is, that $n \circ n = \mathrm{id}_F$.

(c) Show that the ordered field $(F, +, \cdot, \leq)$ is Dedekind complete if and only if every non-empty subset of F which is bounded from below has an infimum.

Solution. If F is Dedekind complete, then every non-empty subset of F which is bounded from above has a supremum. Let U be a non-empty subset of F which is bounded from below. By (a) above, $n(U)$ is a non-empty subset of F which is bounded from above and hence has a supremum. Now (b) implies that U has an infimum.

Conversely, now suppose that every non-empty subset of F which is bounded from below has an infimum, and let U be a non-empty subset of F which is bounded from above. By (a) above, $n(U)$ is a non-empty subset of F which is bounded from below and hence has a supremum. Now (b) implies that U has a supremum. Thus F is Dedekind complete.

II · Point-Set Topology
3 · Metric Spaces
3.1 · Introduction to Metric Geometry
3.1.4 · Exercises

Computing distances in metric spaces. Compute the distance between the given points in the indicated metric space.

3.1.4.1. Let d be the Euclidean metric on the real coordinate plane $\mathbb{R}^2$.

(a) Compute $d(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.

Answer. $d(\mathbf{x}, \mathbf{y}) = \sqrt{20}$.

Solution. We compute

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= \sqrt{(3-(-1))^2 + (2-4)^2} = \sqrt{4^2 + (-2)^2} \\ &= \sqrt{16+4} = \sqrt{20}. \end{aligned}$$

(b) Compute $d(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

Answer. $d(\mathbf{x}, \mathbf{y}) = 5$.

Solution. We compute

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{3^2 + 4^2} \\ &= \sqrt{9+16} = \sqrt{25} = 5. \end{aligned}$$

3.1.4.2. Let d_3 be the 3-metric on the real coordinate plane $\mathbb{R}^2$.

(a) Compute $d_3(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.

Answer. $d_3(\mathbf{x}, \mathbf{y}) = \sqrt[3]{72}$.

Solution. We compute

$$\begin{aligned} d_3(\mathbf{x}, \mathbf{y}) &= \sqrt[3]{|3-(-1)|^3 + |2-4|^3} = \sqrt[3]{4^3 + 2^3} \\ &= \sqrt[3]{64+8} = \sqrt[3]{72}. \end{aligned}$$

3.1.4.3. Let d_∞ be the ∞-metric on the real coordinate plane $\mathbb{R}^2$.

(a) Compute $d_\infty(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.

Answer. $d_\infty(\mathbf{x}, \mathbf{y}) = 4$.

Solution. We compute

$$\begin{aligned} d_\infty(\mathbf{x}, \mathbf{y}) &= \max(\{|3-(-1)|, |2-4|\}) \\ &= \max(\{|4|, |-2|\}) = \max(\{4, 2\}) = 4. \end{aligned}$$

(b) Compute $d_\infty(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

Answer. $d_3(\mathbf{x}, \mathbf{y}) = 4$.

Solution. We compute

$$\begin{aligned} d_\infty(\mathbf{x}, \mathbf{y}) &= \max(\{|3-0|, |4-0|\}) = \max(\{|3|, |4|\}) \\ &= max(\{3, 4\}) = 4. \end{aligned}$$

3.1.4.4. Let $\delta_{\mathbb{R}^2}$ be the discrete metric on the real coordinate plane $\mathbb{R}^2$.

(a) Compute $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.

Answer. $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y}) = 1$.

Solution. Since $\mathbf{x} \neq \mathbf{y}$, $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y}) = 1$.

(b) Compute $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

Answer. $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y}) = 1$.

Solution. Since $\mathbf{x} \neq \mathbf{y}$, $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y}) = 1$.

Verifying metric properties. Determine whether or not the given function is a metric.

3.1.4.5. Let $d\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be the Euclidean metric on the real numbers $\mathbb{R}$; that is,

$$d(x_1, x_2) = |x_1 - x_2|$$

for all real numbers $x_1, x_2 \in \mathbb{R}$. Verify that d is a metric on $\mathbb{R}$.

Solution. For all real numbers $x_1, x_2 \in \mathbb{R}$, if $x_1 = x_2$, then

$$|x_1 - x_2| = |0| = 0.$$

Conversely, if $x_1 \neq x_2$, then

$$d(x_1, x_2) = |x_1 - x_2| = \max(\{x_1, x_2\}) - \min(\{x_1, x_2\}) > 0.$$

Since $d(x_1, x_2) = 0$ if and only if $x_1 = x_2$, d is non-degenerate.

Since

$$d(x_1, x_2) = |x_1 - x_2| = |x_2 - x_1| = d(x_2, x_1)$$

for all real numbers $x_1, x_2 \in X$, d is symmetric.

Finally, consider three points $x_1, x_2, x_3 \in \mathbb{R}$. If $x_1 \geq x_2 \geq x_3$, then

$$\begin{aligned} |x_1 - x_3| &= x_1 - x_3 = x_1 - x_2 + x_2 - x_3 \\ &= |x_1 - x_2| + |x_2 - x_3| = d\left(x_1, x_2\right) + d\left(x_2, x_3\right). \end{aligned}$$

If $x_1 \geq x_2 \geq x_3$, then

$$\begin{aligned} |x_1 - x_3| &= x_1 - x_3 = x_1 - x_2 + x_2 - x_3 \\ &= |x_1 - x_2| - |x_2 - x_3| \leq d\left(x_1, x_2\right) + d\left(x_2, x_3\right). \end{aligned}$$

If $x_2 \geq x_1 \geq x_3$, then

$$\begin{aligned} |x_1 - x_3| &= x_1 - x_3 = x_1 - x_2 + x_2 - x_3 \\ &= -|x_1 - x_2| + |x_2 - x_3| \leq d\left(x_1, x_2\right) + d\left(x_2, x_3\right). \end{aligned}$$

If $x_2 \geq x_3 \geq x_1$, then

$$\begin{aligned} |x_1 - x_3| &= x_3 - x_1 = x_3 - x_2 + x_2 - x_1 \\ &= |x_1 - x_2| - |x_2 - x_3| \leq d\left(x_1, x_2\right) + d\left(x_2, x_3\right). \end{aligned}$$

If $x_3 \geq x_1 \geq x_2$, then

$$\begin{aligned} |x_1 - x_3| &= x_3 - x_1 = x_3 - x_2 + x_2 - x_1 \\ &= -|x_1 - x_2| + |x_2 - x_3| \leq d\left(x_1, x_2\right) + d\left(x_2, x_3\right). \end{aligned}$$

Finally, if $x_3 \geq x_2 \geq x_1$, then

$$\begin{aligned} |x_1 - x_3| &= x_3 - x_1 = x_3 - x_2 + x_2 - x_1 \\ &= |x_1 - x_2| + |x_2 - x_3| = d\left(x_1, x_2\right) + d\left(x_2, x_3\right). \end{aligned}$$

In any case, since $d\left(x_1, x_3\right) \leq d\left(x_1, x_2\right) + d\left(x_2, x_3\right)$, d satisfies the triangle inequality.

Since d is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on the real numbers $\mathbb{R}$.

3.1.4.6. Let $\delta_X \colon X \times X \to \mathbb{R}$ be the discrete metric on a set X; that is,

$$\delta_X\left(x_1, x_2\right) = \begin{cases} 0 & x_1 = x_2 \\ 1 & x_1 \neq x_2 \end{cases}$$

for all points $x_1, x_2 \in X$. Verify that δ_X is a metric on X.

Solution. For all points $x_1, x_2 \in X$, if $x_1 = x_2$, then

$$\delta_X\left(x_1, x_2\right) = 0.$$

Conversely, if $x_1 \neq x_2$, then

$$\delta_X\left(x_1, x_2\right) = 1 \neq 0.$$

Since $\delta_X\left(x_1, x_2\right) = 0$ if and only if $x_1 = x_2$, δ_X is non-degenerate.

Consider two points $x_1, x_2 \in X$. If $x_1 = x_2$, then $x_2 = x_1$, and so

$$\delta_X\left(x_1, x_2\right) = 0 = \delta_X\left(x_2, x_1\right).$$

Conversely, if $x_1 \neq x_2$, then $x_2 \neq x_1$, and so

$$\delta_X\left(x_1, x_2\right) = 1 = \delta_X\left(x_2, x_1\right).$$

In either case, since $\delta_X(x_1, x_2) = \delta_X(x_2, x_1)$, δ_X is symmetric.

Finally, consider three points $x_1, x_2, x_3 \in X$. If $x_1 = x_3$, then

$$\delta_X(x_1, x_3) = 0 \leq \delta_X(x_1, x_2) + \delta_X(x_2, x_3),$$

since δ_X takes only non-negative values. On the other hand, suppose that $x_1 \neq x_3$. Then it is not the case that $x_1 = x_2$ and $x_2 = x_3$, and so at least one of $\delta_X(x_1, x_2)$ and $\delta_X(x_2, x_3)$ is 1. We conclude that

$$\delta_X(x_1, x_3) = 1 \leq \delta_X(x_1, x_2) + \delta_X(x_2, x_3).$$

In either case, since $\delta_X(x_1, x_3) \leq \delta_X(x_1, x_2) + \delta_X(x_2, x_3)$, δ_X satisfies the triangle inequality.

Since δ_X is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on X.

3.1.4.7. Let $\rho\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be the real-valued function defined by the formula

$$\rho(x_1, x_2) = x_1 - x_2.$$

Determine whether or not ρ is a metric on the real numbers $\mathbb{R}$.

Solution. We note that

$$\rho(0, 1) = 0 - 1 = -1 \neq 1 = 1 - 0 = \rho(1, 0),$$

and so ρ is not symmetric. Since ρ is not symmetric, it is not a metric on the real numbers $\mathbb{R}$.

3.1.4.8. Subspaces of discrete metric spaces. Let $U \subseteq X$ be a subset of a discrete metric space (X, δ_X). Verify that the metric $\delta_X|_U$ induced on U by δ_X is the discrete metric δ_U on U.

Solution. Let $\iota\colon U \to X$ be the inclusion map of U into X, and consider points $u_1, u_2 \in U$. If $u_1 = u_2$, then $\iota(u_1) = \iota(u_2)$, and so

$$\delta_X|_U(u_1, u_2) = \delta_X(\iota(u_1), \iota(u_2)) = 0 = \delta_U(u_1, u_2).$$

On the other hand, since ι is injective, if $u_1 \neq u_2$, then $\iota(u_1) \neq \iota(u_2)$, and so

$$\delta_X|_U(u_1, u_2) = \delta_X(\iota(u_1), \iota(u_2)) = 1 = \delta_U(u_1, u_2).$$

In either case, $\delta_X|_U(u_1, u_2) = \delta_U(u_1, u_2)$, and so $\delta_X|_U = \delta_U$.

3.1.4.9. Scalar multiples of metrics. Give an example of a real number $\alpha \in \mathbb{R}$ and a metric $\rho\colon X \times X \to \mathbb{R}$ so that $\alpha\rho$ is not a metric.

Answer. Any non-positive real number $\alpha \leq 0$ and any metric $\rho\colon X \times X \to \mathbb{R}$ on a set X with at least two distinct points are correct answers.

Solution. Since Lemma 3.1.4 Metrics are non-negative implies that metrics take only non-negative values, any non-positive real number $\alpha \leq 0$ and any metric $\rho\colon X \times X \to \mathbb{R}$ on a set X with at least two distinct points are correct answers.

3.1.5 · Problems

3.1.5.1. From pseudometrics to metrics. A real-valued function $\rho\colon X^2 \to \mathbb{R}$ on the Cartesian square X^2 of a set X, which is called the **ground set** and whose elements are called **points**, is a **pseudometric** on X if it has the following properties:

(a) *Identity.*

$\rho(x, x) = 0$ for any point $x \in X$.

(b) *Symmetry.*

$\rho(x_1, x_2) = \rho(x_2, x_1)$ for any points $x_1, x_2 \in X$.

(c) *Triangle inequality.*

$\rho(x_1, x_3) \leq \rho(x_1, x_2) + \rho(x_2, x_3)$ for any points $x_1, x_2, x_3 \in X$.

In this case, the ordered pair (X, ρ) is called a **pseudometric space**.

(a) Let ρ be a pseudometric on a set X, and consider the binary relation $\equiv_\rho$ on X defined so that $x_1 \equiv_\rho x_2$ if and only if $\rho(x_1, x_2) = 0$. Prove that $\equiv_\rho$ is an equivalence relation on X.

Solution. Let $x \in X$, and note that $\rho(x, x) = 0$, so that $x \equiv_\rho x$. Since this holds for all points $x \in X$, $\equiv_\rho$ is reflexive.

Now let $x_1, x_2 \in X$, and suppose that $x_1 \equiv_\rho x_2$. Then $\rho(x_1, x_2) = 0$, and so

$$\rho(x_2, x_1) = \rho(x_1, x_2) = 0;$$

that is, $x_2 \equiv_\rho x_1$. Since this holds for all points $x_1, x_2 \in X$, $\equiv_\rho$ is symmetric.

Finally, let $x_1, x_2, x_3 \in X$, and suppose that $x_1 \equiv_\rho x_2$ and $x_2 \equiv_\rho x_3$. Then

$$\rho(x_1, x_2) = \rho(x_2, x_3) = 0.$$

Lemma 3.1.4 Metrics are non-negative and the triangle inequality together imply that

$$0 \leq \rho(x_1, x_3) \leq \rho(x_1, x_2) + \rho(x_2, x_3) = 0 + 0 = 0,$$

so that $\rho(x_1, x_3) = 0$; that is, $x_1 \equiv_\rho x_3$. Since this holds for all points $x_1, x_2, x_3 \in X$, $\equiv_\rho$ is transitive.

Since the binary relation $\equiv_\rho$ is reflexive, symmetric, and transitive, it is an equivalence relation on X.

(b) Prove that a pseudometric ρ descends to a well-defined metric $\overline{\rho}$ on the quotient set $X/{\equiv_\rho}$ defined by the formula

$$\overline{\rho}\left([x_1]_{\equiv_\rho}, [x_2]_{\equiv_\rho}\right) = \rho(x_1, x_2).$$

Solution. Let $x_1, x_2, x_3, x_4 \in X$, and suppose that $x_1 \equiv_\rho x_3$ and $x_2 \equiv_\rho x_4$. Then

$$\rho(x_1, x_3) = \rho(x_2, x_4) = 0,$$

and so

$$\begin{aligned}
\rho(x_1, x_2) &\leq \rho(x_1, x_3) + \rho(x_3, x_4) + \rho(x_4, x_2) \\
&= \rho(x_1, x_3) + \rho(x_3, x_4) + \rho(x_2, x_4) \\
&= 0 + \rho(x_3, x_4) + 0 = \rho(x_3, x_4),
\end{aligned}$$

and similarly

$$\begin{aligned}
\rho(x_3, x_4) &\leq \rho(x_3, x_1) + \rho(x_1, x_2) + \rho(x_2, x_4) \\
&= \rho(x_1, x_3) + \rho(x_1, x_2) + \rho(x_2, x_4) \\
&= 0 + \rho(x_1, x_2) + 0 = \rho(x_1, x_2).
\end{aligned}$$

We conclude that $\rho(x_1, x_2) = \rho(x_3, x_4)$, and so $\overline{\rho}$ is well-defined. It remains to show that $\overline{\rho}$ is a metric on $X/\equiv_\rho$.

To that end, first let $x_1, x_2 \in X$, and note that

$$\overline{\rho}\left([x_1]_{\equiv_\rho}, [x_2]_{\equiv_\rho}\right) = 0$$

if and only if $\rho(x_1, x_2) = 0$, which occurs precisely when $x_1 \equiv_\rho x_2$; that is, $\overline{\rho}\left([x_1]_{\equiv_\rho}, [x_2]_{\equiv_\rho}\right) = 0$ if and only if $[x_1]_{\equiv_\rho} = [x_2]_{\equiv_\rho}$, and so $\overline{\rho}$ is non-degenerate.

Now let $x_1, x_2 \in X$, and note that

$$\begin{aligned}\overline{\rho}\left([x_1]_{\equiv_\rho}, [x_2]_{\equiv_\rho}\right) &= \rho(x_1, x_2) = \rho(x_2, x_1)\\ &= \overline{\rho}\left([x_2]_{\equiv_\rho}, [x_1]_{\equiv_\rho}\right),\end{aligned}$$

so that $\overline{\rho}$ is symmetric.

Finally, let $x_1, x_2, x_3 \in X$, and note that

$$\begin{aligned}\overline{\rho}\left([x_1]_{\equiv_\rho}, [x_3]_{\equiv_\rho}\right) &= \rho(x_1, x_3) \leq \rho(x_1, x_2) + \rho(x_2, x_3)\\ &= \overline{\rho}\left([x_1]_{\equiv_\rho}, [x_2]_{\equiv_\rho}\right) + \overline{\rho}\left([x_2]_{\equiv_\rho}, [x_3]_{\equiv_\rho}\right),\end{aligned}$$

so that $\overline{\rho}$ satisfies the triangle inequality.

Since $\overline{\rho}$ is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on $X/\equiv_\rho$.

3.1.5.2. Norm equivalence. Two norms $\|\cdot\|_1, \|\cdot\|_2 : V \to \mathbb{R}$ on a real or complex vector space V are called **equivalent** if there are positive real numbers $A, B \in (0, \infty)$ so that

$$A\|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1 \leq B\|\mathbf{v}\|_2$$

for all vectors $\mathbf{v} \in V$.

(a) Prove that norm equivalence is an equivalence relation on the set of norms on V.

Solution. First let $\|\cdot\| : V \to \mathbb{R}$ be a norm on V, and note that

$$1\|\mathbf{v}\| \leq \|\mathbf{v}\| \leq 1\|\mathbf{v}\|$$

for all vectors $\mathbf{v} \in V$. Thus $\|\cdot\|$ and $\|\cdot\|$ are equivalent, and so norm equivalence is reflexive.

Now let $\|\cdot\|_1, \|\cdot\|_2 : V \to \mathbb{R}$ be norms on V, and suppose that $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent. Then there are positive real numbers $A, B \in (0, \infty)$ so that

$$A\|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1 \leq B\|\mathbf{v}\|_2$$

for all vectors $\mathbf{v} \in V$, and so

$$\frac{1}{B}\|\mathbf{v}\|_1 \leq \|\mathbf{v}\|_2 \leq \frac{1}{A}\|\mathbf{v}\|_1$$

for all vectors $\mathbf{v} \in V$. Thus $\|\cdot\|_2$ and $\|\cdot\|_1$ are equivalent, and so norm equivalence is symmetric.

Finally, let $\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_3 : V \to \mathbb{R}$ be norms on V, and suppose that $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent and that $\|\cdot\|_2$ and $\|\cdot\|_3$ are equivalent. Then there are positive real numbers $A, B, C, D \in (0, \infty)$ so that

$$A \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1 \leq B \|\mathbf{v}\|_2$$

and

$$C \|\mathbf{v}\|_3 \leq \|\mathbf{v}\|_2 \leq D \|\mathbf{v}\|_3$$

for all vectors $\mathbf{v} \in V$, and so

$$\begin{aligned} AC \|\mathbf{v}\|_3 &\leq A \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1 \\ &\leq B \|\mathbf{v}\|_2 \leq BD \|\mathbf{v}\|_3 \end{aligned}$$

for all vectors $\mathbf{v} \in V$. Thus $\|\cdot\|_1$ and $\|\cdot\|_3$ are equivalent, and so norm equivalence is transitive.

Since norm equivalence is reflexive, symmetric, and transitive, it is an equivalence relation on the set of norms on V.

(b) Prove that for a given natural number $n \in \mathbb{N}$, any two p-norms $\|\cdot\|_p$ on n-dimensional real coordinate space $\mathbb{R}^n$ for $1 \leq p \leq \infty$ are all equivalent.

Solution. We will show that any p-norm on $\mathbb{R}^n$ with $1 \leq p < \infty$ is equivalent to the ∞-norm; (a) above implies that this suffices. To that end, let $\mathbf{v} = (v_k)_{k=1}^n \in \mathbb{R}^n$, and first note that

$$|v_j| = \left(|v_j|^p\right)^{\frac{1}{p}} \leq \left(\sum_{k=1}^m |v_k|^p\right)^{\frac{1}{p}} = \|\mathbf{v}\|_p$$

for all indices $1 \leq j \leq m$, and so

$$\|\mathbf{v}\|_\infty = \max_{1 \leq j \leq m} |v_j| \leq \|\mathbf{v}\|_p .$$

On the other hand, we note that

$$\begin{aligned} \|\mathbf{v}\|_p = \left(\sum_{j=1}^m |v_j|^p\right)^{\frac{1}{p}} &\leq \left(\sum_{j=1}^m \max_{1 \leq k \leq m} |v_k|^p\right)^{\frac{1}{p}} \\ &= \left(m \|\mathbf{v}\|_\infty^p\right)^{\frac{1}{p}} = m^{\frac{1}{p}} \|\mathbf{v}\|_\infty . \end{aligned}$$

Since

$$1 \|\mathbf{v}\|_\infty \|\mathbf{v}\|_p \leq m^{\frac{1}{p}} \|\mathbf{v}\|_\infty$$

for all vectors $\mathbf{v} \in V$, $\|\cdot\|_p$ and $\|\cdot\|_\infty$ are equivalent.

3.1.5.3. Ultrametrics. A real-valued function $\rho \colon X^2 \to \mathbb{R}$ on the Cartesian square X^2 of a set X is called an **ultrametric** on X, which is called the **ground set** and whose elements are called **points** if it satisfies the following conditions:

- *Non-degeneracy.*

  For any points $x_1, x_2 \in X$, $\rho(x_1, x_2) = 0$ if and only if $x_1 = x_2$.

- *Symmetry.*

  $\rho(x_1, x_2) = \rho(x_2, x_1)$ for any points $x_1, x_2 \in X$.

- *Strong triangle inequality.*

  $\rho(x_1, x_3) \leq \max(\{\rho(x_1, x_2), \rho(x_2, x_3)\})$ for any points $x_1, x_2, x_3 \in X$.

If a real-valued function $\rho\colon X^2 \to \mathbb{R}$ is an ultrametric on X, then the ordered pair (X, ρ) is called an ultrametric space.

(a) Prove that any ultrametric is a metric.

Solution. An ultrametric is non-degenerate and symmetric, and so it remains only to show that it satisfies the triangle inequality. We first observe that given points $x_1, x_2 \in X$ in an ultrametric space (X, ρ),

$$\rho(x_1, x_2) = \max(\{\rho(x_1, x_2), \rho(x_2, x_1)\}) \geq \rho(x_1, x_1) = 0;$$

that is, ultrametrics take only positive values. This implies that

$$\rho(x_1, x_3) \leq \max(\{\rho(x_1, x_2), \rho(x_2, x_3)\}) \leq \rho(x_1, x_2) + \rho(x_2, x_3)$$

for all points $x_1, x_2, x_3 \in X$, so that ρ satisfies the triangle inequality and is therefore a metric on X.

(b) Fix a prime number $p \in \mathbb{N}$. The ***p*-adic valuation** $|\cdot|_p : \mathbb{Q} \to \mathbb{Q}$ is a rational-valued function defined by the formula

$$|q|_p = \begin{cases} 0 & q = 0 \\ p^{-k_p(q)} & q \neq 0, \end{cases}$$

where $k_p(q) \in \mathbb{Z}$ is the unique integer so that $q = p^{k_p(q)}\left(\frac{a}{b}\right)$ for some integers $a, b \in \mathbb{Z}$ which are not divisible by p.

Prove that the p-adic valuation $|\cdot|_p$ induces an ultrametric $\rho_p\colon \mathbb{Q}^2 \to \mathbb{R}$ defined by the formula

$$\rho_p(q_1, q_2) = |q_1 - q_2|_p.$$

This ultrametric is called the ***p*-adic metric** on $\mathbb{Q}$.

Solution. First let $q_1, q_2 \in \mathbb{Q}$ be rational numbers, and suppose that $q_1 = q_2$. Then

$$\rho_p(q_1, q_2) = |q_1 - q_2|_p = |0|_p = 0.$$

Conversely, now suppose that $q_1 \neq q_2$. Then $q_1 - q_2 \neq 0$, and so

$$\rho_p(q_1, q_2) = |q_1 - q_2|_p = p^{-k_p(q_1 - q_2)} \neq 0.$$

Since $\rho_p(q_1, q_2) = 0$ if and only if $q_1 = q_2$, ρ_p is non-degenerate.

Now let $q_1, q_2 \in \mathbb{Q}$, and observe that if

$$q_1 - q_2 = p^k\left(\frac{a}{b}\right)$$

for some integers $a, b \in \mathbb{Z}$ which are not divisible by p, then

$$q_2 - q_1 = p^k\left(\frac{-a}{b}\right),$$

and p does not divide $-a$ or b. Thus $k_p(q_1 - q_2) = k_p(q_2 - q_1)$, and so

$$\rho_p(q_1, q_2) = |q_1 - q_2|_p = p^{-k_p(q_1 - q_2)}$$

$$= p^{-k_p(q_2-q_1)} = |q_2 - q_1|_p = \rho_p(q_2, q_1).$$

We conclude that ρ_p is symmetric.

Finally, let $q_1, q_2, q_3 \in \mathbb{Q}$, and write

$$q_1 - q_2 = p^j\left(\frac{a}{b}\right) \qquad\qquad q_2 - q_3 = p^k\left(\frac{c}{d}\right)$$

for some integers $j, k \in \mathbb{Z}$ and some integers $a, b, c, d \in \mathbb{Z}$ which are not divisible by p. We observe that if $j \leq k$, then

$$\begin{aligned} q_1 - q_3 = q_1 - q_2 + q_2 - q_3 &= p^j\left(\frac{a}{b}\right) + p^k\left(\frac{c}{d}\right) \\ = p^j\left(\frac{a}{b} + p^{k-j}\left(\frac{c}{d}\right)\right) &= p^j\left(\frac{ad + p^{k-j}bc}{bd}\right), \end{aligned}$$

where p does not divide $ad + p^{k-j}bc$ or bd. On the other hand, if $k \leq j$, then

$$\begin{aligned} q_1 - q_3 = q_1 - q_2 + q_2 - q_3 &= p^j\left(\frac{a}{b}\right) + p^k\left(\frac{c}{d}\right) \\ = p^k\left(p^{j-k}\left(\frac{a}{b}\right) + \frac{c}{d}\right) &= p^j\left(\frac{p^{j-k}ad + bc}{bd}\right), \end{aligned}$$

where p does not divide $p^{j-k}ad + bc$ or bd.

In summary, we have shown that

$$k_p(q_1 - q_3) = \min(\{k_p(q_1 - q_2), k_p(q_2 - q_3)\}),$$

so that

$$\begin{aligned} \rho_p(q_1, q_3) = |q_1 - q_3|_p &= p^{-k_p(q_1-q_3)} = p^{-\min(\{k_p(q_1-q_2), k_p(q_2-q_3)\})} \\ &= p^{\max(\{-k_p(q_1-q_2), -k_p(q_2-q_3)\})} = \max\left(\left\{p^{-k_p(q_1-q_2)}, p^{-k_p(q_2-q_3)}\right\}\right) \\ &= \max\left(\left\{|q_1 - q_2|_p, |q_2 - q_3|_p\right\}\right) = \max(\{\rho_p(q_1, q_2), \rho_p(q_2, q_3)\}); \end{aligned}$$

that is, ρ_p satisfies the strong triangle inequality.

Since the p-adic metric $\rho_p\colon \mathbb{Q} \times \mathbb{Q} \to \mathbb{R}$ is non-degenerate, symmetric, and satisfies the strong triangle inequality, it is an ultrametric on $\mathbb{Q}$.

(c) Prove that for all points $x_1, x_2, x_3 \in X$ in an ultrametric space (X, ρ), at least two of the distances $\rho(x_1, x_2)$, $\rho(x_2, x_3)$, and $\rho(x_1, x_3)$ are equal.

Solution. This follows from applications of the strong triangle inequality. In particular, suppose that $\rho(x_1, x_2) \neq \rho(x_2, x_3)$. Up to renaming x_1, x_2, and x_3, we may suppose that $\rho(x_1, x_2) < \rho(x_2, x_3)$. Then

$$\rho(x_1, x_3) \leq \max(\{\rho(x_1, x_2), \rho(x_2, x_3)\}) = \rho(x_2, x_3).$$

Both $\rho(x_1, x_2) < \rho(x_2, x_3)$ and $\rho(x_1, x_3) \leq \rho(x_2, x_3)$, and so

$$\max(\{\rho(x_1, x_2), \rho(x_1, x_3)\}) \leq \rho(x_2, x_3).$$

On the other hand,

$$\rho(x_2, x_3) \leq \max(\{\rho(x_2, x_1), \rho(x_1, x_3)\}) = \max(\{\rho(x_1, x_2), \rho(x_1, x_3)\}),$$

and so $\rho(x_2, x_3) = \max(\{\rho(x_1, x_2), \rho(x_1, x_3)\})$. But we assumed that $\rho(x_1, x_2) \neq \rho(x_2, x_3)$, and so $\rho(x_2, x_3) = \rho(x_1, x_3)$.

The geometry of non-Archimedean metric spaces differs in several ways from our intuitions about Euclidean geometry informed by our experience of distance in the physical world. We will see several of these differences over the course of this chapter.

3.1.5.4. Reverse triangle inequality for norms. Prove that

$$|\|\mathbf{v}_1\| - \|\mathbf{v}_2\|| \leq \|\mathbf{v}_1 - \mathbf{v}_2\|$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$ in a normed vector space $(V, \|\cdot\|)$.

Solution. The absolute homogeneity and subadditivity of the norm $\|\cdot\|$ on V implies that

$$\begin{aligned}\|\mathbf{v}_1\| - \|\mathbf{v}_2\| &\leq \|\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_2\| - \|\mathbf{v}_2\| \\ &= \|\mathbf{v}_1 - \mathbf{v}_2\| + \|\mathbf{v}_2\| - \|\mathbf{v}_2\| = \|\mathbf{v}_1 - \mathbf{v}_2\|\end{aligned}$$

and

$$\begin{aligned}\|\mathbf{v}_2\| - \|\mathbf{v}_1\| &\leq \|\mathbf{v}_2 - \mathbf{v}_1 + \mathbf{v}_1\| - \|\mathbf{v}_1\| \\ &= \|\mathbf{v}_2 - \mathbf{v}_1\| + \|\mathbf{v}_1\| - \|\mathbf{v}_1\| = \|\mathbf{v}_2 - \mathbf{v}_1\| \\ &= |-1| \|\mathbf{v}_2 - \mathbf{v}_1\| = \|\mathbf{v}_1 - \mathbf{v}_2\|.\end{aligned}$$

Thus

$$|\|\mathbf{v}_1\| - \|\mathbf{v}_2\|| = \max\left(\{\|\mathbf{v}_1\| - \|\mathbf{v}_2\|, \|\mathbf{v}_2\| - \|\mathbf{v}_1\|\}\right) \leq \|\mathbf{v}_1 - \mathbf{v}_2\|.$$

3.2 · Convergence and Limits
3.2.4 · Exercises

Convergence in your own words. Describe *in your own words* the given mode of convergence. ***Do not*** just copy the definition.

3.2.4.1. Describe *in your own words* what it means for a sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) to converge to a limit $x \in X$.

3.2.4.2. Describe *in your own words* what it means for a sequence $(f_n)_{n=0}^{\infty}$ of maps $f\colon X \to Y$ from a set X to a metric space (Y, ρ) to converge pointwise to a limit $f\colon X \to Y$.

3.2.4.3. Describe *in your own words* what it means for a sequence $(f_n)_{n=0}^{\infty}$ of maps $f\colon X \to Y$ from a set X to a metric space (Y, ρ) to converge uniformly to a limit $f\colon X \to Y$.

3.2.4.4. Describe *in your own words* what it means for a map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) to converge at a point $c \in X$ to a limit $l \in Y$.

Sequential convergence. Determine whether the following sequences of real numbers converge or diverge in the Euclidean metric d on the real numbers $\mathbb{R}$. If the given sequence converges, find the limit.

3.2.4.5. Consider the sequence $(a_n)_{n=1}^{\infty}$ of real numbers $a_n \in \mathbb{R}$ defined by the formula

$$a_n = \frac{\cos(n\pi)}{n}.$$

Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Answer. The sequence $(a_n)_{n=1}^{\infty}$ converges to 0; that is,

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{\cos(n\pi)}{n} = 0.$$

Solution. Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that $n_\epsilon > \frac{1}{\epsilon}$ for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d(a_n, 0) &= \left| \frac{\cos(n\pi)}{n} - 0 \right| = \left| \frac{(-1)^n}{n} \right| \\ &= \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(a_n)_{n=1}^{\infty}$ converges to 0; that is,

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{\cos(n\pi)}{n} = 0.$$

3.2.4.6. Consider the sequence $(b_n)_{n=0}^{\infty}$ of real numbers $b_n \in \mathbb{R}$ defined by the formula

$$b_n = n\cos(n\pi).$$

Determine whether the sequence $(b_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Answer. The sequence $(b_n)_{n=0}^{\infty}$ diverges.

Solution. Let $l \in \mathbb{R}$ be a real number. Then

$$\begin{aligned} d(b_n, l) &= |n\cos(n\pi) - l| = |(-1)^n n - l| \\ &= |n - l| = n - l \geq 1 \end{aligned}$$

for all even indices $n \geq \lceil l \rceil + 1$, and so $(b_n)_{n=0}^{\infty}$ does not converge to l. Since l was chosen to be an arbitrary real number, the sequence $(b_n)_{n=0}^{\infty}$ diverges.

3.2.4.7. Consider the sequence $(c_n)_{n=0}^{\infty}$ of real numbers $c_n \in \mathbb{R}$ defined by the formula

$$c_n = n\sin(n\pi).$$

Determine whether the sequence $(c_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Answer. The sequence $(c_n)_{n=0}^{\infty}$ converges to 0; that is,

$$\lim_{n\to\infty} c_n = \lim_{n\to\infty} n\sin(n\pi) = 0.$$

Solution. Let $\epsilon > 0$ be a positive real number, and observe that

$$\begin{aligned} d(c_n, 0) = |n\sin(n\pi) - 0| = |0 - 0| \\ = |0| = 0 < \epsilon \end{aligned}$$

for all indices $n \in \mathbb{N}$. Thus the sequence $(c_n)_{n=0}^{\infty}$ converges to 0; that is,

$$\lim_{n\to\infty} c_n = \lim_{n\to\infty} n\sin(n\pi) = 0.$$

3.2.4.8. Consider the sequence $(d_n)_{n=0}^{\infty}$ of real numbers $d_n \in \mathbb{R}$ defined by the formula

$$d_n = \frac{2}{n+1}.$$

Determine whether the sequence $(d_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Answer. The sequence $(d_n)_{n=0}^{\infty}$ converges to 0; that is,

$$\lim_{n\to\infty} d_n = \lim_{n\to\infty} \frac{2}{n+1} = 0.$$

Solution. Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \frac{2-\epsilon}{\epsilon}$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d\left(d_n, 0\right) &= \left|\frac{2}{n+1} - 0\right| = \left|\frac{2}{n+1}\right| = \frac{2}{n+1} \\ &\leq \frac{2}{n_\epsilon + 1} \leq \frac{2}{\left(\frac{2-\epsilon}{\epsilon}\right) + 1} = \frac{2}{\left(\frac{2}{\epsilon}\right)} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(d_n)_{n=0}^{\infty}$ converges to 0; that is,

$$\lim_{n\to\infty} d_n = \lim_{n\to\infty} \frac{2}{n+1} = 0.$$

3.2.4.9. Consider the sequence $(e_n)_{n=0}^{\infty}$ of real numbers $e_n \in \mathbb{R}$ defined by the formula

$$e_n = \frac{2n}{n+1}.$$

Determine whether the sequence $(e_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Answer. The sequence $(e_n)_{n=0}^{\infty}$ converges to 2; that is,

$$\lim_{n\to\infty} e_n = \lim_{n\to\infty} \frac{2n}{n+1} = 2.$$

Solution. Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \frac{2-\epsilon}{\epsilon}$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d\left(e_n, 2\right) &= \left|\frac{2n}{n+1} - 2\right| = \left|-\frac{2}{n+1}\right| = \frac{2}{n+1} \\ &\leq \frac{2}{n_\epsilon + 1} \leq \frac{2}{\frac{2-\epsilon}{\epsilon} + 1} = \frac{2}{\left(\frac{2}{\epsilon}\right)} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(e_n)_{n=0}^{\infty}$ converges to 2; that is,

$$\lim_{n\to\infty} e_n = \lim_{n\to\infty} \frac{2n}{n+1} = 2.$$

3.2.4.10. Consider the sequence $(f_n)_{n=0}^{\infty}$ of real numbers $f_n \in \mathbb{R}$ defined by the formula

$$f_n = \sqrt{\frac{2n}{n+1}}.$$

Determine whether the sequence $(f_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Answer. The sequence $(f_n)_{n=0}^{\infty}$ converges to $\sqrt{2}$; that is,

$$\lim_{n\to\infty} f_n = \lim_{n\to\infty} \sqrt{\frac{2n}{n+1}} = \sqrt{2}.$$

Solution. Let $\epsilon > 0$ be a positive real number. The argument above in (5) implies that there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$d\left(\frac{2n}{n+1}, 2\right) < 2\epsilon\sqrt{2}$$

for all indices $n \geq n_\epsilon$. We observe that

$$\left|\sqrt{\frac{2n}{n+1}} + \sqrt{2}\right| \geq \sqrt{\frac{2n}{n+1}} + \sqrt{2} \geq \sqrt{2 - 2\epsilon\sqrt{2}} + \sqrt{2} \geq 2\sqrt{2}$$

for all indices $n \geq n_\epsilon$, and so

$$\begin{aligned} d\left(f_n, \sqrt{2}\right) &= \left|\sqrt{\frac{2n}{n+1}} - \sqrt{2}\right| = \frac{\left|\frac{2n}{n+1} - 2\right|}{\left|\sqrt{\frac{2n}{n+1}} + \sqrt{2}\right|} \\ &= \frac{d\left(\frac{2n}{n+1}, 2\right)}{\left|\sqrt{\frac{2n}{n+1}} + \sqrt{2}\right|} < \frac{2\epsilon\sqrt{2}}{2\sqrt{2}} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(f_n)_{n=0}^{\infty}$ converges to $\sqrt{2}$; that is,

$$\lim_{n\to\infty} f_n = \lim_{n\to\infty} \sqrt{\frac{2n}{n+1}} = \sqrt{2}.$$

3.2.4.11. Consider the sequence $(g_n)_{n=0}^{\infty}$ of real numbers $g_n \in \mathbb{R}$ defined by the formula

$$g_n = \frac{3^n}{n!}.$$

Determine whether the sequence $(g_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Answer. The sequence $(g_n)_{n=0}^{\infty}$ converges to 0; that is,

$$\lim_{n\to\infty} g_n = \lim_{n\to\infty} \frac{3^n}{n!} = 0.$$

Solution. We observe that an inductive argument shows that

$$\frac{3^{n-1}}{(n-1)!} < 1$$

whenever $n \geq 8$. Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon \geq \max\left(\left\{8, \frac{3}{\epsilon}\right\}\right)$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$d\left(g_n, 0\right) = \left|\frac{3^n}{n!} - 0\right| = \left|\frac{3^n}{n!}\right| = \frac{3^n}{n!}$$

$$= \left(\frac{3}{n}\right)\left(\frac{3^{n-1}}{(n-1)!}\right) < \frac{3}{n_\epsilon} < \epsilon$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(g_n)_{n=0}^{\infty}$ converges to 0; that is,

$$\lim_{n\to\infty} g_n = \lim_{n\to\infty} \frac{3^n}{n!} = 0.$$

Pointwise and uniform convergence. Determine whether the following sequences of real-valued functions converge or diverge both pointwise and uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If the given sequence converges pointwise, find the pointwise limit. Express your answer using limit notation. If the given sequence converges uniformly, find the uniform limit.

3.2.4.12. Consider the sequence $(a_n)_{n=1}^{\infty}$ of real-valued functions $a_n : \mathbb{R} \to \mathbb{R}$ defined by the formulae

$$a_n(x) = \left(x - \frac{1}{n}\right)^2.$$

(a) Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

Answer. Let $a : \mathbb{R} \to \mathbb{R}$ be the function defined by the formula $a(x) = x^2$. Then the sequence $(a_n)_{n=1}^{\infty}$ converges pointwise to a; that is,

$$\lim_{n\to\infty} a_n = a.$$

Solution. Let $a : \mathbb{R} \to \mathbb{R}$ be the function defined by the formula $a(x) = x^2$. Fix a point $x \in \mathbb{R}$, and let $\epsilon > 0$ be a positive real number. We observe that if $n > \frac{4|x|}{\epsilon}$, then

$$\frac{2|x|}{n} < \frac{2|x|}{\left(\frac{4|x|}{\epsilon}\right)} = \frac{\epsilon}{2}.$$

Similarly, we observe that if $n > \sqrt{\frac{2}{\epsilon}}$, then

$$\frac{1}{n^2} < \frac{1}{\left(\sqrt{\frac{2}{\epsilon}}\right)^2} = \frac{\epsilon}{2}.$$

The Archimedean principle implies that

$$n_\epsilon > \max\left(\left\{\frac{4|x|}{\epsilon}, \sqrt{\frac{2}{\epsilon}}\right\}\right)$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$d(a_n(x), a(x)) = \left|\left(x - \frac{1}{n}\right)^2 - x^2\right| = \left|x^2 - \left(\frac{2}{n}\right)x + \frac{1}{n^2} - x^2\right|$$
$$= \left|-\frac{2}{n}x + \frac{1}{n^2}\right| \leq \frac{2|x|}{n} + \frac{1}{n^2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(a_n(x))_{n=1}^{\infty}$ converges to $a(x)$. Since this holds for all points $x \in \mathbb{R}$, the sequence $(a_n)_{n=1}^{\infty}$ converges pointwise to a; that is,

$$\lim_{n\to\infty} a_n = a.$$

(b) Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

Answer. The sequence $(a_n)_{n=1}^{\infty}$ does not converge uniformly.

Solution. The dependence of n_ϵ on x in the above solution for (a) is a good sign that the sequence $(a_n)_{n=1}^{\infty}$ does not converge uniformly. Indeed, we observe that

$$\begin{aligned} d(a_n(n), a(n)) &= \left| \left(n - \frac{1}{n}\right)^2 - n^2 \right| = \left| n^2 - \left(\frac{2}{n}\right) n + \frac{1}{n^2} - n^2 \right| \\ &= \left| -2 + \frac{1}{n^2} \right| \geq 2 - \frac{1}{n} \geq 1 \end{aligned}$$

for all positive integers $n \geq 1$. Thus the sequence $(a_n)_{n=1}^{\infty}$ does not converge uniformly to a. Since $(a_n)_{n=1}^{\infty}$ converges pointwise to a, this implies that the sequence $(a_n)_{n=1}^{\infty}$ does not converge uniformly.

3.2.4.13. Consider the sequence $(b_n)_{n=1}^{\infty}$ of real-valued functions $b_n\colon [-10, 10] \to \mathbb{R}$ defined by the formulae

$$b_n(x) = \frac{x^2}{n}.$$

(a) Determine whether the sequence $(b_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

Answer. Let $b\colon [-10, 10] \to \mathbb{R}$ be the constant function defined by the formula $b(x) = 0$. Then the sequence $(b_n)_{n=1}^{\infty}$ converges pointwise to b; that is,

$$\lim_{n\to\infty} b_n = b.$$

Solution. Let $b\colon [-10, 10] \to \mathbb{R}$ be the constant function defined by the formula $b(x) = 0$. Fix a point $x \in [-10, 10]$, and let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \frac{|x|^2}{\epsilon}$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d(b_n(x), b(x)) &= \left| \frac{x^2}{n} - 0 \right| = \left| \frac{x^2}{n} \right| \\ &= \frac{|x|^2}{n} \leq \frac{|x|^2}{n_\epsilon} < \frac{|x|^2}{\left(\frac{|x|^2}{\epsilon}\right)} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(b_n(x))_{n=1}^{\infty}$ converges to $b(x)$. Since this holds for all points $x \in [-10, 10]$, the sequence $(b_n)_{n=1}^{\infty}$ converges pointwise to b; that is,

$$\lim_{n\to\infty} b_n = b.$$

(b) Determine whether the sequence $(b_n)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

Answer. Let $b\colon [-10, 10] \to \mathbb{R}$ be the constant function defined by the formula $b(x) = 0$. Then the sequence $(b_n)_{n=1}^{\infty}$ converges uniformly to b; that is,

$$\operatorname*{unif\,lim}_{n\to\infty} b_n = b.$$

Solution. The dependence of n_ϵ on x in the above solution for (a) is a good sign that the sequence $(b_n)_{n=1}^{\infty}$ does not converge uniformly. However, the bounds on the input $-10 \leq x \leq 10$ allow us to obtain uniform convergence.

Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \frac{100}{\epsilon}$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d(b_n(x), b(x)) &= \left| \frac{x^2}{n} - 0 \right| = \left| \frac{x^2}{n} \right| \\ &= \frac{|x|^2}{n} \leq \frac{10^2}{n_\epsilon} < \frac{100}{\left(\frac{100}{\epsilon}\right)} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$ and points $x \in [-10, 10]$. Thus the sequence $(b_n)_{n=1}^{\infty}$ converges uniformly to b; that is,

$$\operatorname*{unif\,lim}_{n\to\infty} b_n = b.$$

3.2.4.14. Consider the sequence $(c_n)_{n=1}^{\infty}$ of real-valued functions $c_n\colon \mathbb{R} \to \mathbb{R}$ defined by the formulae

$$c_n(x) = \frac{x^2}{n}.$$

(a) Determine whether the sequence $(c_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

Answer. Let $c\colon \mathbb{R} \to \mathbb{R}$ be the constant function defined by the formula $c(x) = 0$. Then the sequence $(c_n)_{n=1}^{\infty}$ converges pointwise to c; that is,

$$\lim_{n\to\infty} c_n = c.$$

Solution. Let $c\colon \mathbb{R} \to \mathbb{R}$ be the constant function defined by the formula $c(x) = 0$. Fix a point $x \in \mathbb{R}$, and let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \frac{|x|^2}{\epsilon}$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d\left(c_n\left(x\right), c\left(x\right)\right) &= \left|\frac{x^2}{n} - 0\right| = \left|\frac{x^2}{n}\right| \\ &= \frac{|x|^2}{n} \leq \frac{|x|^2}{n_\epsilon} < \frac{|x|^2}{\left(\frac{|x|^2}{\epsilon}\right)} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus the sequence $\left(c_n\left(x\right)\right)_{n=1}^{\infty}$ converges to $c\left(x\right)$. Since this holds for all points $x \in \mathbb{R}$, the sequence $\left(c_n\right)_{n=1}^{\infty}$ converges pointwise to c; that is,

$$\lim_{n\to\infty} c_n = c.$$

(b) Determine whether the sequence $\left(c_n\right)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

Answer. The sequence $\left(c_n\right)_{n=1}^{\infty}$ does not converge uniformly.

Solution. The dependence of n_ϵ on x in the above solution for (a) is a good sign that the sequence $\left(c_n\right)_{n=1}^{\infty}$ does not converge uniformly. Indeed, we observe that

$$\begin{aligned} d\left(c_n\left(\sqrt{n}\right), c\left(\sqrt{n}\right)\right) &= \left|\frac{\left(\sqrt{n}\right)^2}{n} - 0\right| = \left|\frac{n}{n}\right| \\ &= |1| = 1 \end{aligned}$$

for all positive integers $n \geq 1$. Thus the sequence $\left(c_n\right)_{n=1}^{\infty}$ does not converge uniformly to c. Since $\left(c_n\right)_{n=1}^{\infty}$ converges pointwise to c, this implies that the sequence $\left(c_n\right)_{n=1}^{\infty}$ does not converge uniformly.

3.2.4.15. Consider the sequence $\left(d_n\right)_{n=0}^{\infty}$ of real-valued functions $d_n\colon [-10, 10] \to \mathbb{R}$ defined by the formulae

$$d_n\left(x\right) = \frac{1}{3^{x+n}}.$$

(a) Determine whether the sequence $\left(d_n\right)_{n=0}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

Answer. Let $d\colon [-10, 10] \to \mathbb{R}$ be the constant function defined by the formula $d\left(x\right) = 0$. Then the sequence $\left(d_n\right)_{n=0}^{\infty}$ converges pointwise to d; that is,

$$\lim_{n\to\infty} d_n = d.$$

Solution. Let $d\colon [-10, 10] \to \mathbb{R}$ be the constant function defined by the formula $d\left(x\right) = 0$. Fix a point $x \in [-10, 10]$, and let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \log_3\left(\frac{1}{\epsilon}\right) - x$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d\left(d_n\left(x\right), d\left(x\right)\right) &= \left|\frac{1}{3^{x+n}} - 0\right| = \left|\frac{1}{3^{x+n}}\right| = \frac{1}{3^{x+n}} \\ &\leq \frac{1}{3^{x+n_\epsilon}} < \frac{1}{3^{x+\log_3\left(\frac{1}{\epsilon}\right)-x}} = \frac{1}{3^{\log_3\left(\frac{1}{\epsilon}\right)}} \\ &= \frac{1}{\left(\frac{1}{\epsilon}\right)} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus the sequence $\left(d_n\left(x\right)\right)_{n=0}^{\infty}$ converges to $d\left(x\right)$. Since this holds for all points $x \in [-10, 10]$, the sequence $\left(d_n\right)_{n=1}^{\infty}$ converges pointwise to d; that is,

$$\lim_{n\to\infty} d_n = d.$$

(b) Determine whether the sequence $\left(d_n\right)_{n=0}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

Answer. Let $d\colon [-10, 10] \to \mathbb{R}$ be the constant function defined by the formula $d\left(x\right) = 0$. Then the sequence $\left(d_n\right)_{n=0}^{\infty}$ converges uniformly to b; that is,

$$\operatorname*{unif\,lim}_{n\to\infty} d_n = d.$$

Solution. The dependence of n_ϵ on x in the above solution for (a) is a good sign that the sequence $\left(d_n\right)_{n=0}^{\infty}$ does not converge uniformly. However, the bounds on the input $-10 \leq x \leq 10$ allow us to obtain uniform convergence.

Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \ln\left(\frac{1}{\epsilon}\right) + 10$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d\left(d_n\left(x\right), d\left(x\right)\right) &= \left|\frac{1}{3^{x+n}} - 0\right| = \left|\frac{1}{3^{x+n}}\right| = \frac{1}{3^{x+n}} \\ &\leq \frac{1}{3^{-10+n_\epsilon}} < \frac{1}{3^{-10+\log_3\left(\frac{1}{\epsilon}\right)+10}} = \frac{1}{3^{\log_3\left(\frac{1}{\epsilon}\right)}} \\ &= \frac{1}{\left(\frac{1}{\epsilon}\right)} = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$ and points $x \in [-10, 10]$. Thus the sequence $\left(d_n\right)_{n=0}^{\infty}$ converges uniformly to d; that is,

$$\operatorname*{unif\,lim}_{n\to\infty} d_n = d.$$

Limit points and isolated points. Find all limit points and isolated points of the given subset of the Euclidean plane $\mathbb{R}^2$.

3.2.4.16. Let U be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$U = \left\{ \mathbf{x} \in \mathbb{R}^2 \,\middle|\, 1 \leq \|\mathbf{x}\| < 2 \right\}.$$

Find all limit points and isolated points of U in $\mathbb{R}^2$.

Answer. Every point $\mathbf{x} \in \mathbb{R}^2$ so that $1 \leq \|\mathbf{x}\| \leq 2$ is a limit point of U in $\mathbb{R}^2$, and U has no isolated points.

3.2.4.17. Let V be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$V = \left\{ \mathbf{x} \in \mathbb{R}^2 \,\middle|\, \|\mathbf{x}\| \in \mathbb{N} \right\}.$$

Find all limit points and isolated points of V in $\mathbb{R}^2$.

Answer. Every point $\mathbf{x} \in \mathbb{R}^2$ so that $\|\mathbf{x}\| \in \mathbb{N} \setminus \{0\}$ is a positive integer is a limit point of U in $\mathbb{R}^2$, and $\mathbf{0}$ is the only isolated point of U.

3.2.4.18. Let W be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$W = \left\{ (x, y) \in \mathbb{R}^2 \,\middle|\, x, y \in \mathbb{Z} \right\}.$$

Find all limit points and isolated points of W in $\mathbb{R}^2$.

Answer. W has no limit points in $\mathbb{R}^2$, and every point in W is an isolated point.

3.2.4.19. Let X be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$X = \left\{ (x, y) \in \mathbb{R}^2 \,\middle|\, x, y \in \mathbb{Q} \right\}.$$

Find all limit points and isolated points of X in $\mathbb{R}^2$.

Answer. Every point in $\mathbb{R}^2$ is a limit point of X in $\mathbb{R}^2$, and X has no isolated points.

3.2.4.20. Let Y be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$Y = \left\{ (x, y) \in \mathbb{R}^2 \,\middle|\, x, y \notin \mathbb{Q} \right\}.$$

Find all limit points and isolated points of Y in $\mathbb{R}^2$.

Answer. Every point in $\mathbb{R}^2$ is a limit point of Y in $\mathbb{R}^2$, and Y has no isolated points.

3.2.4.21. Let Z be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$Z = \left\{ \left(\frac{1}{m}, \frac{1}{n} \right) \in \mathbb{R}^2 \,\middle|\, m, n \in \mathbb{N} \setminus \{0\} \right\}.$$

Find all limit points and isolated points of Z in $\mathbb{R}^2$.

Answer. The set of limit points of Z in $\mathbb{R}^2$ is

$$Z' = \left\{ \left(\frac{1}{m}, 0 \right) \in \mathbb{R}^2 \,\middle|\, m \in \mathbb{N} \setminus \{0\} \right\} \cup \left\{ \left(0, \frac{1}{n} \right) \in \mathbb{R}^2 \,\middle|\, n \in \mathbb{N} \setminus \{0\} \right\},$$

and every point of Z is an isolated point.

Map convergence. Determine whether or not the given real-valued function converges at the given input. If the function converges, find the limit and express your answer using limit notation.

3.2.4.22. Consider the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined piecewise by the formula

$$f(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0. \end{cases}$$

Determine whether or not f converges at 0. If f converges at 0, find the limit and express your answer using limit notation.

Answer. f does not converge at 0.

Solution. We observe that

$$\lim_{n\to\infty} f\left(\frac{1}{n}\right) = \lim_{n\to\infty} 1 = 1$$

and

$$\lim_{n\to\infty} f\left(-\frac{1}{n}\right) = \lim_{n\to\infty} -1 = -1.$$

If f converged at 0, then the above limits to have the same value. Since they do not have the same value, f does not converge at 0.

3.2.4.23. Consider the real-valued function $g\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$g(y) = y^2 - 2y + 1.$$

Determine whether or not g converges at 0. If g converges at 0, find the limit and express your answer using limit notation.

Answer. g converges at 0 to 1; that is,

$$\lim_{y\to 0} g(y) = 1.$$

Solution. Let $\epsilon > 0$ be a positive real number, and let $\delta = \min\left(\left\{1, \frac{\epsilon}{3}\right\}\right) > 0$. We observe that

$$\begin{aligned} d(g(y), 1) &= \left|y^2 - 2y + 1 - 1\right| = \left|y^2 - 2y\right| = |y| \cdot |y - 2| \\ &< |y - 0| \cdot 3 = 3d(y, 0) < 3\delta \leq 3\left(\frac{\epsilon}{3}\right) = \epsilon \end{aligned}$$

for all $y \in \mathbb{R}$ so that $0 < d(y, 0) < \delta$, and so g converges to 1 at 0. Since 0 is a limit point of $\mathbb{R}$, this limit is unique; that is,

$$\lim_{y\to 0} g(y) = 1.$$

3.2.4.24. Consider the real-valued function $h\colon [0, \infty) \to \mathbb{R}$ defined by the formula

$$h(z) = \sqrt{z}.$$

Determine whether or not h converges at 4. If h converges at 4, find the limit and express your answer using limit notation.

Answer. $\lim_{x\to 4} h(z) = 2$.

Solution. Let $\epsilon > 0$ be a positive real number, and let $\delta = \min(\{3, 3\epsilon\}) > 0$. We observe that if $d(z, 4) < 3$, then $|z - 4| < 3$, and so

$$\begin{aligned} z &\geq 1 \\ \sqrt{z} &\geq 1 \end{aligned}$$

$$\sqrt{z} + 2 \geq 3$$
$$\frac{1}{\sqrt{z} + 2} \leq \frac{1}{3}.$$

We conclude that

$$\begin{aligned} d\left(h\left(z\right), 2\right) = \left|\sqrt{z} - 2\right| &= \frac{|z-4|}{\sqrt{z}+2} \\ &< \frac{d\left(z, 4\right)}{3} < \frac{\delta}{3} \leq \frac{3\epsilon}{3} = \epsilon \end{aligned}$$

for all $z \in \mathbb{R}$ so that $0 < d\left(z, 4\right) < \delta$, and so h converges to 2 at 4. Since 4 is a limit point of $\mathbb{R}$, this limit is unique; that is,

$$\lim_{z \to 4} h\left(z\right) = 2.$$

3.2.5 · Problems

3.2.5.1. Eventually constant sequences. A sequence $\left(x_n\right)_{n=0}^{\infty}$ of elements $x_n \in X$ of a set X is called **eventually constant** if there is a point $x \in X$ and a natural number $n_0 \in \mathbb{N}$ so that $x_n = x$ for all indices $n \geq n_0$.

(a) Prove that any eventually constant sequence of points in a metric space is convergent.

Solution. Let $\left(x_n\right)_{n=0}^{\infty}$ be an eventually constant sequence of points $x_n \in X$ in a metric space (X, ρ). Then there is a point $x \in X$ and a natural number $n_0 \in \mathbb{N}$ so that $x_n = x$ for all indices $n \geq n_0$. Let $\epsilon > 0$ be a positive real number, and observe that

$$\rho\left(x_n, x\right) = \rho\left(x, x\right) = 0 < \epsilon$$

for all indices $n \geq n_0$. We conclude that the sequence $\left(x_n\right)_{n=0}^{\infty}$ converges to x.

(b) Prove that any sequence of points in a discrete metric space is convergent if and only if it is eventually constant.

Solution. Let $\left(x_n\right)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a discrete metric space (X, δ_X). If $\left(x_n\right)_{n=0}^{\infty}$ is eventually constant, then it is convergent by (a) above. Thus we may suppose that $\left(x_n\right)_{n=0}^{\infty}$ converges to some point $x \in X$.

Let $\epsilon = \frac{1}{2}$. Since $\epsilon > 0$ is positive, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\delta_X\left(x_n, x\right) < \epsilon = \frac{1}{2}$$

for all indices $n \geq n_\epsilon$. Since the values of δ_X are 0 and 1, we conclude that

$$\delta_X\left(x_n, x\right) = 0$$

and hence $x_n = x$ for all indices $n \geq n_\epsilon$. Thus $\left(x_n\right)_{n=0}^{\infty}$ is eventually constant.

(c) Give an example of a sequence of real numbers which converges in the Euclidean metric d on $\mathbb{R}$ but diverges in the discrete metric $\delta_{\mathbb{R}}$ on $\mathbb{R}$.

Solution. $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ converges to 0 in the Euclidean metric, but this sequence is not eventually constant and so does not converge in the discrete metric.

3.2.5.2. Sequential convergence in the infinity metric. Let $m \in \mathbb{N}$ be a natural number, and consider a sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ of points $\mathbf{x}_n = (x_{1,n}, \ldots, x_{m,n}) \in \mathbb{R}^m$. Prove that for any vector $\mathbf{x} = (x_1, \ldots, x_m) \in \mathbb{R}$, the following are equivalent:

(a) The sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the ∞-metric d_∞ on $\mathbb{R}^m$.

(b) For all indices $1 \leq j \leq m$, the sequence $(x_{j,n})_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$.

Solution. First suppose that the sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the ∞-metric d_∞ on $\mathbb{R}^m$. Fix an index $1 \leq j \leq m$, and let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$d_\infty(\mathbf{x}_n, \mathbf{x}) < \epsilon$$

for all indices $n \geq n_\epsilon$. We observe that

$$\begin{aligned} d(x_{j,n}, x_j) = |x_{j,n} - x_j| &\leq \max_{1 \leq k \leq m} |x_{k,n} - x_k| \\ &= \|\mathbf{x}_n - \mathbf{x}\|_\infty = d_\infty(\mathbf{x}_n, \mathbf{x}) < \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$, and so $(x_{j,n})_{n=0}^{\infty}$ converges to x_j.

Conversely, now suppose that for all indices $1 \leq j \leq m$, the sequence $(x_{j,n})_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$. Let $\epsilon > 0$ be a positive real number. Then for each index $1 \leq j \leq m$, there is a natural number $n_{j,\epsilon} \in \mathbb{N}$ so that

$$d(x_{j,n}, x_j) < \epsilon$$

for all indices $n \geq n_{j,\epsilon}$. Let $n_\epsilon = \max_{1 \leq j \leq m} n_{j,\epsilon} \in \mathbb{N}$, and note that

$$\begin{aligned} d_\infty(\mathbf{x}_n, \mathbf{x}) = \|\mathbf{x}_n - \mathbf{x}\|_\infty &= \max_{1 \leq j \leq m} |x_{j,n} - x_j| \\ &= \max_{1 \leq j \leq m} d(x_{j,n}, x_j) < \max_{1 \leq j \leq m} \epsilon = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$, and so the sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the ∞-metric d_∞ on $\mathbb{R}^m$.

3.2.5.3. Alternative characterization of uniform convergence. Let $(f_n)_{n=0}^{\infty}$ be a sequence of maps $f_n \colon X \to Y$ from a non-empty set $X \neq \emptyset$ to a metric space (Y, ρ). Prove that $(f_n)_{n=0}^{\infty}$ converges to a map $f \colon X \to Y$ from X to Y if and only if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$0 \leq \sup_{x \in X} \rho(f_n(x), f(x)) < \epsilon$$

for all indices $n \geq n_\epsilon$.

Solution. We observe that for a given index $n \in \mathbb{N}$,

$$0 \leq \sup_{x \in X} \rho(f_n(x), f(x)) < \epsilon$$

if and only if

$$\rho(f_n(x), f(x)) < \epsilon$$

for all inputs $x \in X$. Thus the sequence $(f_n)_{n=0}^{\infty}$ converges uniformly to f if and only if there is a natural number $n_\epsilon \in \mathbb{N}$ so that either (and therefore both) of the above conditions holds for all indices $n \geq n_\epsilon$.

3.2.5.4. Uniqueness of map limits. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) which has at least two distinct points. Prove the following partial converse to Corollary 3.2.24 Uniqueness of map limits: If f converges at a point $c \in X$ to a unique point $l \in Y$, then c is a limit point of X.

Solution. By contraposition. Suppose that c is an isolated point of Y. Then there are no sequences $(x_n)_{n=0}^{\infty}$ of points $x_n \in X \setminus \{c\}$ which converge to c, and so for each point $y \in Y$, Theorem 3.2.19 Sequential characterization of map convergence implies that f converges at x to y. Since Y has at least two distinct points, then f does not converge to a unique limit at c.

In summary, we have shown that if c is an isolated point, then f does not converge to a unique limit at c. Equivalently, if f converges at c to a unique limit, then c is a limit point of X.

3.3 · Limits in the Euclidean Metric
3.3.4 · Exercises

Limits and arithmetic. Compute the limits of the given sequences of real numbers.

3.3.4.1. Compute $\lim_{n\to\infty} \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3}$.

Answer. $\lim_{n\to\infty} \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3} = 0$.

Solution. We compute

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3} &= \lim_{n\to\infty} \frac{1}{n} + \lim_{n\to\infty} \frac{1}{n^2} + \lim_{n\to\infty} \frac{1}{n^3} \\ &= 0 + 0 + 0 = 0.\end{aligned}$$

3.3.4.2. Compute $\lim_{n\to\infty} \frac{2n+1}{3n}$.

Answer. $\lim_{n\to\infty} \frac{2n+1}{3n} = \frac{2}{3}$.

Solution. We compute

$$\begin{aligned}\lim_{n\to\infty} \frac{2n+1}{3n} &= \lim_{n\to\infty} \frac{2n}{3n} + \lim_{n\to\infty} \frac{1}{3n} \\ &= \lim_{n\to\infty} \frac{2}{3} + \frac{1}{3} \lim_{n\to\infty} \frac{1}{n} \\ &= \frac{2}{3} + 0 = \frac{2}{3}.\end{aligned}$$

3.3.4.3. Compute $\lim_{n\to\infty} \frac{1-n^2}{n}$.

Answer. $\lim_{n\to\infty} \frac{1-n^2}{n} = -\infty$.

Solution. We compute

$$\begin{aligned}\lim_{n\to\infty} \frac{1-n^2}{n} &= \lim_{n\to\infty} \left(\frac{1}{n} - n\right) = \lim_{n\to\infty} \frac{1}{n} - \lim_{n\to\infty} n \\ &= 0 - \infty = -\infty.\end{aligned}$$

3.3.4.4. Limits and strict comparison. Give examples of convergent sequences $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ of real numbers $x_n, y_n \in \mathbb{R}$ so that $x_n < y_n$ for all indices $n \in \mathbb{N}$ but

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n.$$

Solution. For each natural number $n \in \mathbb{N}$, let $x_n = -2^{-n}$ and $y_n = 2^{-n}$.

Then $x_n < y_n$ for all indices $n \in \mathbb{N}$, but

$$\lim_{n\to\infty} x_n = 0 = \lim_{n\to\infty} y_n.$$

Computing upper/lower limits. Compute the upper and lower limits of the given sequences of real numbers.

3.3.4.5. Compute $\limsup_{n\to\infty} 2^n$ and $\liminf_{n\to\infty} 2^n$.

Answer. $\limsup_{n\to\infty} 2^n = \liminf_{n\to\infty} 2^n = \infty$.

3.3.4.6. Compute $\limsup_{n\to\infty} 2^{-n}$ and $\liminf_{n\to\infty} 2^{-n}$.

Answer. $\limsup_{n\to\infty} 2^{-n} = \liminf_{n\to\infty} 2^{-n} = 0$.

3.3.4.7. Compute $\limsup_{n\to\infty} (-2)^n$ and $\liminf_{n\to\infty} (-2)^n$.

Answer. $\limsup_{n\to\infty} (-2)^n = \infty$ and $\liminf_{n\to\infty} (-2)^n = -\infty$.

3.3.4.8. Compute $\limsup_{n\to\infty} (-2)^{-n}$ and $\liminf_{n\to\infty} (-2)^{-n}$.

Answer. $\limsup_{n\to\infty} (-2)^n = \liminf_{n\to\infty} (-2)^n = 0$.

3.3.5 · Problems

3.3.5.1. Limits and scalar multiplication. Let $\|\cdot\|$ be a norm on a real or complex vector space V. Prove that

$$\lim_{n\to\infty} \alpha_n \mathbf{v}_n = \lim_{n\to\infty} \alpha_n \lim_{n\to\infty} \mathbf{v}_n$$

for any convergent sequences $(\alpha_n)_{n=0}^{\infty}$ and $(\mathbf{v}_n)_{n=0}^{\infty}$ of scalars α_n and vectors $\mathbf{v}_n \in V$.

Solution. Let $\alpha = \lim_{n\to\infty} \alpha_n$ and $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$. Let $\epsilon > 0$ be a positive real number.

$$\begin{aligned} 0 \le \|\alpha_n \mathbf{v}_n - \alpha \mathbf{v}\| &= \|\alpha_n \mathbf{v}_n - \alpha \mathbf{v}_n + \alpha \mathbf{v}_n - \alpha \mathbf{v}\| \le \|\alpha_n \mathbf{v}_n - \alpha \mathbf{v}_n\| + \|\alpha \mathbf{v}_n - \alpha \mathbf{v}\| \\ &= \|(\alpha_n - \alpha)\mathbf{v}_n\| + \|\alpha(\mathbf{v}_n - \mathbf{v})\| = |\alpha_n - \alpha| \|\mathbf{v}_n\| + |\alpha| \|\mathbf{v}_n - \mathbf{v}\|. \end{aligned}$$

Proposition 3.2.6 Sequential convergence and Proposition 3.3.1 Limits and vector arithmetic together imply that

$$\begin{aligned} \lim_{n\to\infty} \left(|\alpha_n - \alpha| \|\mathbf{v}_n\| + |\alpha| \|\mathbf{v}_n - \mathbf{v}\|\right) &= \left(\lim_{n\to\infty} |\alpha_n - \alpha|\right)\left(\lim_{n\to\infty} \|\mathbf{v}_n\|\right) + |\alpha| \lim_{n\to\infty} \|\mathbf{v}_n - \mathbf{v}\| \\ &= 0 \cdot \|\mathbf{v}\| + |\alpha| \cdot 0 = 0 + 0 = 0. \end{aligned}$$

In summary, we have shown that

$$0 \le \|\alpha_n \mathbf{v}_n - \alpha \mathbf{v}\| \le |\alpha_n - \alpha| \|\mathbf{v}_n\| + |\alpha| \|\mathbf{v}_n - \mathbf{v}\|$$

and that

$$\lim_{n\to\infty} \left(|\alpha_n - \alpha| \|\mathbf{v}_n\| + |\alpha| \|\mathbf{v}_n - \mathbf{v}\|\right) = \lim_{n\to\infty} 0 = 0.$$

Theorem 3.3.3 Squeeze theorem implies that

$$\lim_{n\to\infty} \|\alpha_n \mathbf{v}_n - \alpha \mathbf{v}\| = 0,$$

and so $\lim_{n\to\infty} \alpha_n \mathbf{v}_n = \alpha \mathbf{v}$ by Proposition 3.2.6 Sequential convergence.

3.3.5.2. Limits and division. Prove that

$$\lim_{n\to\infty} \frac{1}{x_n} = \frac{1}{\lim_{n\to\infty} x_n}$$

for every sequence $(x_n)_{n=0}^{\infty}$ of non-zero real numbers $x_n \in \mathbb{R} \setminus \{0\}$ which converges to a non-zero real number limit

$$\lim_{n\to\infty} x_n \neq 0.$$

Solution. Let $x = \lim_{n\to\infty} x_n$. Since $(x_n)_{n=0}^{\infty}$ converges to x and $x \neq 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$|x_n - x| < \min\left(\left\{\frac{|x|}{2}, \frac{\epsilon\,|x|^2}{2}\right\}\right)$$

for all indices $n \geq n_\epsilon$. We observe that if $|x_n - x| < \frac{|x|}{2}$, then

$$\frac{|x|}{2} = |x| - \frac{|x|}{2} \leq |x - x_n| + |x_n| - \frac{|x|}{2}$$

$$< \frac{|x|}{2} + |x_n| - \frac{|x|}{2} = |x_n|\,.$$

In particular,

$$\begin{aligned}\left|\frac{1}{x_n} - \frac{1}{x}\right| &= \left|\frac{x - x_n}{x_n x}\right| = \frac{|x_n - x|}{|x_n| \cdot |x|}\\ &\leq \frac{\left(\frac{\epsilon|x|^2}{2}\right)}{\left(\frac{|x|}{2}\right)|x|} = \frac{2\epsilon\,|x|^2}{2\,|x|^2} = \epsilon\end{aligned}$$

for all indices $n \geq n_\epsilon$. Thus

$$\lim_{n\to\infty} \frac{1}{x_n} = \frac{1}{x} = \frac{1}{\lim_{n\to\infty} x_n}.$$

3.3.5.3. Squeezing to infinity. Let $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ be sequences of real numbers $x_n, y_n \in \mathbb{R}$.

(a) Prove that if $x_n \leq y_n$ for all indices $n \in \mathbb{N}$ and $(x_n)_{n=0}^{\infty}$ diverges to ∞, then $(y_n)_{n=0}^{\infty}$ diverges to ∞.

Solution. Let $B \in \mathbb{R}$ be a real number. Since $(x_n)_{n=0}^{\infty}$ diverges to ∞, there is a natural number $n_B \in \mathbb{N}$ so that $x_n > B$ for all indices $n \geq n_B$. We observe that

$$y_n \geq x_n > B$$

for all indices $n \geq n_B$. Thus the sequence $(y_n)_{n=0}^{\infty}$ diverges to ∞.

(b) Prove that if $x_n \leq y_n$ for all indices $n \in \mathbb{N}$ and $(y_n)_{n=0}^{\infty}$ diverges to $-\infty$, then $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$.

Solution. Let $B \in \mathbb{R}$ be a real number. Since $(y_n)_{n=0}^{\infty}$ diverges to $-\infty$, there is a natural number $n_B \in \mathbb{N}$ so that $y_n < B$ for all indices $n \geq n_B$. We observe that

$$x_n \leq y_n < B$$

for all indices $n \geq n_B$. Thus the sequence $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$.

We can interpret the above results as versions of the Squeeze theorem for divergence to infinity.

3.3.5.4. Reciprocals of infinite limits. Prove that a sequence $(x_n)_{n=0}^{\infty}$ of negative real numbers $x_n < 0$ diverges to $-\infty$ if and only if

$$\lim_{n\to\infty} \frac{1}{x_n} = 0.$$

Solution. First suppose that $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$, and let $\epsilon > 0$. Then $-\frac{1}{\epsilon} \in \mathbb{R}$ is a real number, and so there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$x_n < -\frac{1}{\epsilon}$$

for all indices $n \geq n_\epsilon$. In particular,

$$d\left(\frac{1}{x_n}, 0\right) = \left|\frac{1}{x_n} - 0\right| = \left|\frac{1}{x_n}\right| = -\frac{1}{x_n} < \epsilon$$

for all indices $n \geq n_\epsilon$, and so $\left(\frac{1}{x_n}\right)_{n=0}^{\infty}$ converges to 0.

Conversely, now suppose that $\left(\frac{1}{x_n}\right)_{n=0}^{\infty}$ converges to 0, and let $B \in \mathbb{R}$ be a real number. If $B \leq 0$ is non-negative, then

$$x_n < 0 \leq B$$

for all indices $n \in \mathbb{N}$. On the other hand, if $B < 0$ is negative, then $-\frac{1}{B} > 0$ is a positive real number, and so there is a natural number n_B so that

$$d\left(\frac{1}{x_n}, 0\right) < -\frac{1}{B}$$

for all indices $n \geq n_B$. In particular,

$$x_n = -|x_n| = -\frac{1}{\left|\frac{1}{x_n} - 0\right|} = -\frac{1}{d\left(\frac{1}{x_n}, 0\right)} < B$$

for all indices $n \geq n_B$, and so $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$.

3.3.5.5. Divergence and upper/lower limits. In this problem, you will complete the proof of Theorem 3.3.20 Convergence and upper and lower limits. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$.

(a) Prove that $(x_n)_{n=0}^{\infty}$ diverges to ∞ if and only if

$$\liminf_{n\to\infty} x_n = \infty.$$

Solution. We note that for a natural number $m \in \mathbb{N}$ and a real number $B \in \mathbb{R}$, $x_n \geq B$ for all indices $n \geq m$ if and only if

$$\inf_{n\geq m} x_n \geq B.$$

First suppose that the sequence $(x_n)_{n=0}^{\infty}$ diverges to ∞. Then for all real numbers $B \in \mathbb{R}$, there is a natural number $m_B \in \mathbb{N}$ so that $x_n > B + 1$ for all indices $n \geq m_B$. We observe that

$$\inf_{n\geq m} x_n \geq \inf_{n\geq m_B} x_n \geq B + 1 > B$$

for all indices $m \geq m_B$, and so the sequence $(\inf_{n\geq m} x_n)_{m=0}^{\infty}$ diverges to ∞; that is,

$$\liminf_{n\to\infty} x_n = \lim_{m\to\infty} \inf_{n\geq m} x_n = \infty.$$

Conversely, now suppose that $\liminf_{n\to\infty} x_n = \infty$, and let $B \in \mathbb{R}$ be a real number. Then there is a natural number $n_B \in \mathbb{N}$ so that

$$\inf_{n\geq m} x_n > B$$

for all indices $m \geq n_B$. In particular,

$$x_n \geq \inf_{k\geq n_B} x_k > B$$

for all indices $n \geq n_B$. Thus the sequence $(x_n)_{n=0}^{\infty}$ diverges to ∞.

(b) Prove that $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$ if and only if

$$\limsup_{n\to\infty} x_n = -\infty.$$

Solution. We note that for a natural number $m \in \mathbb{N}$ and a real number $B \in \mathbb{R}$, $x_n \leq B$ for all indices $n \geq m$ if and only if

$$\sup_{n\geq m} x_n \leq B.$$

First suppose that the sequence $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$. Then for all real numbers $B \in \mathbb{R}$, there is a natural number $m_B \in \mathbb{N}$ so that $x_n < B-1$ for all indices $n \geq m_B$. We observe that

$$\sup_{n\geq m} x_n \leq \sup_{n\geq m_B} x_n \leq B - 1 < B$$

for all indices $m \geq m_B$, and so the sequence $\left(\sup_{n\geq m} x_n\right)_{m=0}^{\infty}$ diverges to $-\infty$; that is,

$$\limsup_{n\to\infty} x_n = \lim_{m\to\infty} \sup_{n\geq m} x_n = -\infty.$$

Conversely, now suppose that $\limsup_{n\to\infty} x_n = -\infty$, and let $B \in \mathbb{R}$ be a real number. Then there is a natural number $n_B \in \mathbb{N}$ so that

$$\sup_{n\geq m} x_n < B$$

for all indices $m \geq n_B$. In particular,

$$x_n \leq \sup_{k\geq n_B} x_k < B$$

for all indices $n \geq n_B$. Thus the sequence $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$.

3.4 · Open and Closed Sets

3.4.4 · Exercises

Neighborhoods. Determine whether or not the given set is a neighborhood of the given point.

3.4.4.1. Let
$$A = (0, 2) = \{x \in \mathbb{R} \mid 0 < x < 2\}$$
be the open interval between 0 and 2. Determine whether or not A is a neighborhood of the point 1 in the metric space $(\mathbb{R}, d)$.

Answer. A is a neighborhood of 1 in $\mathbb{R}$.

Solution. Since A contains the open ball
$$\begin{aligned} \mathrm{B}_1(1) &= \{x \in \mathbb{R} \mid |x-1| < 1\} = \{x \in \mathbb{R} \mid 0 < x < 2\} \\ &= (0, 2) = A \end{aligned}$$
of radius 1 and center 1, A is a neighborhood of 1 in $\mathbb{R}$.

3.4.4.2. Let
$$B = [0, 2] = \{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$$
be the closed interval between 0 and 2. Determine whether or not B is a neighborhood of the point 1 in the metric space $(\mathbb{R}, d)$.

Answer. B is a neighborhood of 1 in $\mathbb{R}$.

Solution. Since B contains the open ball
$$\begin{aligned} \mathrm{B}_1(1) &= \{x \in \mathbb{R} \mid |x-1| < 1\} = \{x \in \mathbb{R} \mid 0 < x < 2\} \\ &= (0, 2) \subsetneq [0, 2] = B \end{aligned}$$
of radius 1 and center 1, B is a neighborhood of 1 in $\mathbb{R}$.

3.4.4.3. Let
$$C = [0, 2) = \{x \in \mathbb{R} \mid 0 \leq x < 2\}$$
be the half-closed, half-open interval between 0 and 2. Determine whether or not C is a neighborhood of the point 0 in the metric space $(\mathbb{R}, d)$.

Answer. C is not a neighborhood of 0 in $\mathbb{R}$.

Solution. We note that for any positive real number $r > 0$, the open ball
$$\mathrm{B}_r(0) = \{x \in \mathbb{R} \mid |x - 0| < r\} = \{x \in \mathbb{R} \mid -r < x < r\} = (-r, r)$$
contains negative numbers. Thus any neighborhood of 0 in $\mathbb{R}$ contains negative numbers. Since every element of C is non-negative, C is not a neighborhood of 1 in $\mathbb{R}$.

3.4.4.4. Let
$$D = 2\mathbb{Z} = \{2k \in \mathbb{Z} \mid k \in \mathbb{Z}\} = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$$
be the set of even integers. Determine whether or not D is a neighborhood of the point 0 in the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

Answer. D is a neighborhood of 0 in $(\mathbb{R}, \delta_{\mathbb{R}})$.

Solution. We note that since the discrete metric $\delta_{\mathbb{R}}$ takes only the values 0 and 1,
$$\mathrm{B}_{\frac{1}{2}}(0) = \left\{x \in \mathbb{R} \,\middle|\, \delta_{\mathbb{R}}(x, 0) < \frac{1}{2}\right\} = \{x \in \mathbb{R} \mid \delta_{\mathbb{R}}(x, 0) = 0\} = \{0\}.$$
Since $0 \in D$, $\mathrm{B}_{\frac{1}{2}}(0) \subseteq D$, and so D is a neighborhood of 0 in $(\mathbb{R}, \delta_{\mathbb{R}})$.

3.4.4.5. Let
$$E = [0, 1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

be the closed interval between 0 and 1. Determine whether or not E is a neighborhood of the point 0 in the metric subspace $([0,2], d)$ of the real line $(\mathbb{R}, d)$.

Answer. E is a neighborhood of 0 in $[0,2]$.

Solution. Since E contains the open ball

$$\begin{aligned} \mathrm{B}_1(0) &= \{x \in [0,2] \mid |x-0| < 1\} = \{x \in \mathbb{R} \mid 0 \leq x \leq 2 \text{ and } -1 < x < 1\} \\ &= \{x \in \mathbb{R} \mid 0 \leq x < 1\} = [0,1) \subseteq [0,1] = E \end{aligned}$$

of radius 1 and center 1, E is a neighborhood of 1 in $[0,2]$.

Open and closed sets. Determine whether the given subset of a metric space is open, closed, both open and closed, or neither open nor closed.

3.4.4.6. Let

$$A = (0,2) = \{x \in \mathbb{R} \mid 0 < x < 2\}$$

be the open interval between 0 and 2. Determine whether or not A is open or closed in the metric space $(\mathbb{R}, d)$.

Answer. A is open in $\mathbb{R}$ but not closed.

3.4.4.7. Let

$$B = [0,2] = \{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$$

be the closed interval between 0 and 2. Determine whether or not B is open or closed in the metric space $(\mathbb{R}, d)$.

Answer. B is closed in $\mathbb{R}$ but not open.

3.4.4.8. Let

$$C = [0,2) = \{x \in \mathbb{R} \mid 0 \leq x < 2\}$$

be the half-closed, half-open interval between 0 and 2. Determine whether or not C is open or closed in the metric space $(\mathbb{R}, d)$.

Answer. C is neither open nor closed in $\mathbb{R}$.

3.4.4.9. Let

$$D = 2\mathbb{Z} = \{2k \in \mathbb{Z} \mid k \in \mathbb{Z}\} = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$$

be the set of even integers. Determine whether or not D is open or closed in the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

Answer. D both open and closed in $(\mathbb{R}, \delta_{\mathbb{R}})$.

3.4.4.10. Let

$$E = [0,1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

be the closed interval between 0 and 1. Determine whether or not E is open or closed in the metric subspace $([0,2], d)$ of the real line $(\mathbb{R}, d)$.

Answer. E is closed in $[0,2]$ but not open.

3.4.4.11. Let

$$F = [0,1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

be the closed interval between 0 and 1. Determine whether or not F is open or closed in the metric subspace $([0,1] \cup [2,3], d)$ of the real line $(\mathbb{R}, d)$.

Answer. F is both open and closed in $[0,1] \cup [2,3]$.

Diameter. Compute the diameter of the given subsets of metric spaces.

3.4.4.12. Let $A = \{0, 1, 2\}$. Compute the diameter of A as a subset of the metric space $(\mathbb{R}, d)$.

Answer. $\operatorname{diam}(A) = 2$.

3.4.4.13. Let $B = \{0, 1, 2, 5\}$. Compute the diameter of B as a subset of the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

Answer. $\operatorname{diam}(B) = 1$.

3.4.4.14. Let

$$C = \{0, 1, 2\} \cup \left\{3 - \frac{1}{n}\right\}_{n=1}^{\infty}.$$

Compute the diameter of C as a subset of the metric space $(\mathbb{R}, d)$.

Answer. $\operatorname{diam}(C) = 4$.

3.4.4.15. Let

$$D = 3\mathbb{Z} = \{3k \in \mathbb{Z}\}\, k \in \mathbb{Z}$$

be the set of integers which are divisible by 3. Compute the diameter of D as a subset of the metric space $(\mathbb{R}, d)$.

Answer. $\operatorname{diam}(D) = \infty$.

3.4.4.16. Let $E = \{\pi\}$. Compute the diameter of E as a subset of the metric space $(\mathbb{R}, d)$.

Answer. $\operatorname{diam}(E) = 0$.

3.4.4.17. Let $F = \{\pi\}$. Compute the diameter of F as a subset of the discrete metric space $(\mathbb{R}, \delta_X)$.

Answer. $\operatorname{diam}(F) = 0$.

Bounded and unbounded subsets of metric spaces. Let

$$U = (0, 1) = \{x \in \mathbb{R} \mid 0 < x < 1\}$$

be the open interval between 0 and 1. Determine whether U is a bounded or unbounded subset of the given metric space.

3.4.4.18. Determine whether U is bounded or unbounded in the metric space $(\mathbb{R}, d)$.

Answer. U is bounded in $\mathbb{R}$

3.4.4.19. Determine whether U is bounded or unbounded in the subspace $((0, \infty), d)$ of the metric space $(\mathbb{R}, d)$.

Answer. U is bounded in $(0, \infty)$

3.4.4.20. Determine whether U is bounded or unbounded in the metric space $((0, \infty), \rho)$, where ρ is the metric on $(0, \infty)$ defined by the formula

$$\rho(x, y) = \left|\ln\left(\frac{x}{y}\right)\right|.$$

Answer. U is unbounded in $((0, \infty), \rho)$.

3.4.4.21. Determine whether U is bounded or unbounded in the discrete metric space $\left((0, \infty), \delta_{(0,\infty)}\right)$.

Answer. U is bounded in $\left((0, \infty), \delta_{(0,\infty)}\right)$

The uniform metric. Compute the distance between the given bounded maps in the uniform metric.

3.4.4.22. Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be the constant real-valued functions defined by the formulae $f(x) = 2$ and $g(x) = \pi$. Compute the uniform distance $d_\infty(f, g)$.

Answer. $d_\infty(f, g) = \pi - 2$.

Solution. We compute

$$\begin{aligned} d_\infty(f,g) &= \sup_{x\in\mathbb{R}} d(f(x), g(x)) = \sup_{x\in\mathbb{R}} |2-\pi| \\ &= \sup_{x\in\mathbb{R}} (\pi - 2) = \pi - 2. \end{aligned}$$

3.4.4.23. Let $f, g\colon [-1, 1] \to \mathbb{R}$ be the real-valued functions defined by the formulae $f(x) = 2$ and $g(x) = x^2$. Compute the uniform distance $d_\infty(f, g)$.

Answer. $d_\infty(f, g) = 2$.

Solution. We compute

$$\begin{aligned} d_\infty(f,g) &= \sup_{-1\le x\le 1} d(f(x), g(x)) = \sup_{-1\le x\le 1} \left|2-x^2\right| \\ &= \sup_{-1\le x\le 1} 2 - x^2 = 2 - 0 = 2. \end{aligned}$$

3.4.4.24. Let $f, g\colon (0, 2) \to \mathbb{R}$ be the real-valued functions defined by the formulae $f(x) = 2 - x$ and $g(x) = 1 + 3x$. Compute the uniform distance $d_\infty(f, g)$.

Answer. $d_\infty(f, g) = 2$.

Solution. We compute

$$\begin{aligned} d_\infty(f,g) &= \sup_{0<x<2} d(f(x), g(x)) = \sup_{0<x<2} |2 - x - (1+3x)| \\ &= \sup_{0<x<2} |1 - 4x| = 7. \end{aligned}$$

3.4.5 · Problems

3.4.5.1. Open and closed sets in metric subspaces. Let $U \subseteq X$ be a subset of a metric space (X, ρ).

- **(a)** Prove that a subset $V \subseteq U$ is open in the metric subspace (U, ρ) if and only if $V = U \cap W$ for some open set $W \subseteq X$ in X.
- **(b)** Prove that if a subset $V \subseteq U$ is open in X, then it is open in U.
- **(c)** Construct an explicit counterexample to the converse of (b) above.

3.4.5.2. Interior, closure, and boundary of complements. Let $U \subseteq X$ be a subset of a metric space (X, ρ).

- **(a)** Prove that $\operatorname{Int}(X \setminus U) = X \setminus \operatorname{Cl}(U)$.
- **(b)** Prove that $\operatorname{Cl}(X \setminus U) = X \setminus \operatorname{Int}(U)$.
- **(c)** Prove that $\operatorname{Bd}(X \setminus U) = \operatorname{Bd}(U) = \operatorname{Cl}(U) \cap \operatorname{Cl}(X \setminus U)$.
- **(d)** Prove that $\operatorname{Bd}(U) = \operatorname{Cl}(U) \setminus \operatorname{Int}(U)$.

3.4.5.3. Closure and limit points. Let $U \subseteq X$ be a subset of a metric space (X, ρ). Prove that $\operatorname{Cl}(U) = U \cup U'$.

3.4.5.4. Open balls in ultrametric spaces. Prove that any point in an open ball in an ultrametric space is its center.

3.4.5.5. Diameter in the real line. Let $\emptyset \subsetneq U \subseteq \mathbb{R}$ be a non-empty subset of the real line $(\mathbb{R}, d)$. Prove that U has diameter

$$\operatorname{diam}(U) = \sup(U) - \inf(U)$$

3.5 · Notions of Metric Continuity
3.5.4 · Exercises

Continuity at a point. Determine whether or not the given map between metric spaces is continuous at the given point.

3.5.4.1. Let $f\colon \mathbb{R} \to \mathbb{R}$ be the linear function defined by the formula $f(x) = 3x - 1$. Determine whether or not f is continuous at 2.

Answer. f is continuous at 2.

Solution. Let $\epsilon > 0$ be a positive real number, and let $\delta = \frac{\epsilon}{3} > 0$. We observe that

$$\begin{aligned} d(f(x), f(2)) &= |(3x-1) - (3(2)-1)| \\ &= |3x-6| = 3|x-2| = 3d(x,2) < 3\delta = \epsilon \end{aligned}$$

for all points $x \in \mathbb{R}$ so that $d(x,2) < \delta$. Thus f is continuous at 2.

3.5.4.2. Let $g\colon \mathbb{R} \to \mathbb{R}$ be the linear function defined by the formula $f(y) = 2y^2 - 1$. Determine whether or not g is continuous at 4.

Answer. g is continuous at 4.

Solution. Let $\epsilon > 0$ be a positive real number, and let

$$\delta = \max\left(\left\{1, \frac{\epsilon}{18}\right\}\right) > 0.$$

We observe that

$$\begin{aligned} d(g(y), g(4)) &= \left|\left(2y^2 - 1\right) - \left(2(4)^2 - 1\right)\right| = \left|2y^2 - 32\right| = 2|y+4| \cdot |y-4| \\ &= 2|y+4|\, d(y,4) < 2 \cdot 9\delta < \epsilon \end{aligned}$$

for all points $y \in \mathbb{R}$ so that $d(y,4) < \delta$. Thus g is continuous at 4.

3.5.4.3. Let $h\colon [0,\infty) \to \mathbb{R}$ be the linear function defined by the formula $h(z) = \sqrt{z}$. Determine whether or not h is continuous at 1.

Answer. h is continuous at 4.

Solution. Let $\epsilon > 0$ be a positive real number, and let $\delta = \epsilon$. We observe that

$$d(h(z), h(1)) = \left|\sqrt{z} - \sqrt{1}\right| = \left|\frac{z-1}{\sqrt{z}+1}\right| < \frac{\delta}{1} = \epsilon$$

for all points $z \in \mathbb{R}$ so that $d(z,1) < \delta$. Thus h is continuous at 1.

3.5.4.4. Uniform and Lipschitz continuity. Let $-\infty < a < b < \infty$ be real numbers, and consider the real-valued function $f\colon [a,b] \to \mathbb{R}$ defined by the formula

$$f(x) = x^3.$$

(a) Verify that f is not uniformly continuous.

Solution. We observe that for any positive real number $\delta > 0$,

$$d\left(f\left(x+\frac{\delta}{2}\right), f(x)\right) = \left|\left(x+\frac{\delta}{2}\right)^3 - x^3\right| = \left|\frac{\delta}{2}x^2 + \frac{\delta^2}{4}x + \frac{\delta^3}{8}\right|$$

is the absolute value of a quadratic function of x and is therefore unbounded from above. This implies that for any positive real number $\delta > 0$, there is some real number $x \in \mathbb{R}$ so that $d\left(f\left(x+\frac{\delta}{2}\right), f(x)\right) \geq 1$, and so f is not uniformly continuous.

(b) Without appealing to Proposition 3.5.20 Lipschitz continuity implies uniform continuity, verify that f is not Lipschitz continuous.

Solution. We observe that

$$d(f(x_1), f(x_2)) = \left|x_1^3 - x_2^3\right| = |x_1 - x_2| \cdot \left|x_1^2 + x_1x_2 + x_2^2\right| = \left|x_1^2 + x_1x_2 + x_2^2\right| \cdot d(x_1, x_2)$$

for any points $x_1, x_2 \in \mathbb{R}$. Since $x_1^2 + x_1x_2 + x_2^2$ is a quadratic polynomial in x_1 and x_2, it is not bounded in absolute value, and so f cannot be Lipschitz continuous.

(c) Verify that the restriction of f to a closed interval $[a, b] \subsetneq \mathbb{R}$ is uniformly continuous.

Solution. Let $\epsilon > 0$ be a positive real number, and consider the positive real number

$$\delta = \frac{\epsilon}{3\max\left(\left\{|a|^2, |b|^2\right\}\right)}.$$

We observe that

$$\left|x_1^3 - x_2^3\right| = |x_1 - x_2| \cdot \left|x_1^2 + x_1x_2 + x_2^2\right| \leq d(x_1, x_2) \cdot 3\max\left(\left\{|a|^2, |b|^2\right\}\right) < 3\delta\max\left(\left\{|a|^2, |b|^2\right\}\right) = \epsilon$$

for all points $x_1, x_2 \in [a, b]$ so that $d(x_1, x_2) < \delta$. Thus the restriction of f to $[a, b]$ is uniformly continuous.

(d) Verify that the restriction of f to a closed interval $[a, b] \subsetneq \mathbb{R}$ is Lipschitz continuous.

Solution. We observe that

$$\left|x_1^3 - x_2^3\right| = |x_1 - x_2| \cdot \left|x_1^2 + x_1x_2 + x_2^2\right| \leq d(x_1, x_2) \cdot 3\max\left(\left\{|a|^2, |b|^2\right\}\right)$$

for all points $x_1, x_2 \in [a, b]$. Thus the restriction of f to $[a, b]$ is Lipschitz continuous, with Lipschitz constant

$$L_f \leq 3\max\left(\left\{|a|^2, |b|^2\right\}\right).$$

3.5.5 · Problems

3.5.5.1. Continuity of rational functions.

(a) Prove using one or more characterizations of continuity that the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x$ is continuous.

(b) Argue that any polynomial function is continuous.

(c) Argue that any rational function is continuous on its implicit domain.

3.5.5.2. Uniform limits of uniformly continuous maps. Let $(f_n)_{n=0}^{\infty}$ be a sequence of uniformly continuous maps $f_n\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). Prove that if $(f_n)_{n=0}^{\infty}$ converges uniformly, the uniform limit is also uniformly continuous.

3.5.5.3. Continuous pointwise limits of discontinuous maps. We have seen in Theorem 3.5.9 Uniform limits of continuous maps and Problem 3.5.5.2 Uniform limits of uniformly continuous maps that uniform limits preserve continuity and uniform continuity. In this problem, we will construct counterexamples to the converses of these results.

(a) Give an example of a sequence $(f_n)_{n=0}^{\infty}$ of continuous maps $f_n\colon X \to Y$ from a metric space X to a metric space Y which converges pointwise but not uniformly to a continuous map $f\colon X \to Y$ from X to Y.

(b) Give an example of a sequence $(g_n)_{n=0}^{\infty}$ of discontinuous maps $g_n\colon X \to Y$ from a metric space X to a metric space Y which converges uniformly to a continuous map $g\colon X \to Y$ from X to Y.

3.5.5.4. Uniform limits of Lipschitz continuous maps. Consider the sequence $(f_n)_{n=1}^{\infty}$ of real-valued functions $f_n\colon\ [0, \infty) \to \mathbb{R}$ defined by the formula

$$f_n(x) = \sqrt{x + \frac{1}{n}}.$$

(a) Show that f_n is Lipschitz continuous for each index $n \geq 1$.

(b) Show that the sequence $(f_n)_{n=0}^{\infty}$ converges uniformly, but the uniform limit is not Lipschitz continuous.

(c) Prove that the limit of a uniformly convergent sequence $(g_n)_{n=0}^{\infty}$ of Lipschitz continuous maps $g_n\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is Lipschitz continuous if the sequence $(L_{g_n})_{n=0}^{\infty}$ of Lipschitz constants L_{g_n} is bounded from above.

3.6 · Completeness and Compactness
3.6.4 · Exercises

Cauchy sequences. Determine whether or not the given sequence of points in a metric space is Cauchy.

3.6.4.1. Consider the sequence $(x_n)_{n=0}^{\infty}$ of real numbers x_n defined by the formula

$$x_n = 2 + 3^{-n}.$$

Determine whether or not the sequence $(x_n)_{n=0}^{\infty}$ is a Cauchy sequence in the real line $(\mathbb{R}, d)$.

Answer. The sequence $(x_n)_{n=0}^{\infty}$ is a Cauchy sequence in the real line $(\mathbb{R}, d)$.

Solution. Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \log_3\left(\frac{1}{\epsilon}\right)$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d(x_m, x_n) &= \left|\left(2 + 3^{-m}\right) - \left(2 + 3^{-n}\right)\right| = 3^{-\min(\{m,n\})} - 3^{-\max(\{m,n\})} \\ &= 3^{-\min(\{m,n\})}\left(1 - 3^{\min(\{m,n\})-\max(\{m,n\})}\right) \leq 3^{-n_\epsilon} < \epsilon \end{aligned}$$

for all indices $m, n \geq n_\epsilon$, and so the sequence $(x_n)_{n=0}^\infty$ is Cauchy.

3.6.4.2. Consider the sequence $(x_n)_{n=0}^\infty$ of real numbers x_n defined by the formula

$$x_n = 2 + 3^{-n}.$$

Determine whether or not the sequence $(x_n)_{n=0}^\infty$ is a Cauchy sequence in the discrete metric space $(\mathbb{R}, \delta_\mathbb{R})$.

Answer. The sequence $(x_n)_{n=0}^\infty$ is not a Cauchy sequence in the discrete metric space $(\mathbb{R}, \delta_\mathbb{R})$.

Solution. Since

$$x_n = 3^{-n} > 3^{-(n+1)} = x_{n+1}$$

for all indices $n \in \mathbb{N}$,

$$\delta_\mathbb{R}(x_n, x_{n+1}) = 1$$

for all such indices. In particular, there is no natural number $n_1 \in \mathbb{N}$ so that $\delta_\mathbb{R}(x_m, x_n) < 1$ for all indices $m, n \geq n_1$. Thus the sequence $(x_n)_{n=0}^\infty$ is not a Cauchy sequence in the discrete metric space $(\mathbb{R}, \delta_\mathbb{R})$.

3.6.4.3. Consider the sequence $(f_n)_{n=0}^\infty$ of bounded real-valued functions $f_n\colon [0,\infty) \to \mathbb{R}$ defined by the formula

$$f_n(x) = \frac{1}{x^n}.$$

Determine whether or not the sequence $(f_n)_{n=0}^\infty$ is a Cauchy sequence in the metric space $(\mathrm{B}([1,\infty)), d_\infty)$.

Answer. The sequence $(f_n)_{n=0}^\infty$ is a Cauchy sequence in the metric space $(\mathrm{B}([1,\infty)), d_\infty)$.

Complete subspaces of Euclidean space. Determine whether the given subset of Euclidean space is complete or incomplete.

3.6.4.4. Let U be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$U = \left\{\mathbf{x} \in \mathbb{R}^2 \,\middle|\, \mathbf{x} \neq \mathbf{0}\right\}.$$

Determine whether U is complete or incomplete in $\mathbb{R}^2$.

Answer. U is incomplete in $\mathbb{R}^2$.

Solution. For any vector $\mathbf{u} \in U$, the sequence $(\mathbf{u}_n)_{n=1}^\infty$ defined by the formula

$$\mathbf{u}_n = \frac{\mathbf{u}}{n}$$

is a Cauchy sequence which does not converge to a limit in U, and so U is not complete.

3.6.4.5. Let V be the subset of 3-dimensional Euclidean space $\mathbb{R}^3$ defined by

$$V = \left\{ \mathbf{x} \in \mathbb{R}^3 \,\middle|\, \|\mathbf{x}\| = 2 \right\}.$$

Determine whether V is complete or incomplete in $\mathbb{R}^3$.

Answer. V is complete in $\mathbb{R}^3$.

Solution. V is the pre-image of the closed set $\{2\}$ of $\mathbb{R}$ under the continuous map $\|\cdot\| : \mathbb{R}^3 \to \mathbb{R}$, and so V is closed. Since V is a closed subset of a complete metric space, it is complete.

3.6.4.6. Determine whether the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is complete or incomplete in the real line $\mathbb{R}$.

Answer. The irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are incomplete in the real line $\mathbb{R}$.

Solution. For any irrational number $x \in \mathbb{R} \setminus \mathbb{Q}$, the sequence $(x_n)_{n=1}^{\infty}$ defined by the formula

$$x_n = \frac{x}{n}$$

is a Cauchy sequence of irrational numbers which does not converge to a limit in $\mathbb{R} \setminus \mathbb{Q}$, and so the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are incomplete in the real line $\mathbb{R}$.

Compact subspaces of Euclidean space. Determine whether or not the given subset of Euclidean space is compact.

3.6.4.7. Let U be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$U = \left\{ \mathbf{x} \in \mathbb{R}^2 \,\middle|\, \mathbf{x} \neq \mathbf{0} \right\}.$$

Determine whether or not U is compact in $\mathbb{R}^2$.

Answer. U is not compact.

Solution. As shown in Exercise Group 3.6.4.7–10 Compact subspaces of Euclidean space, U is not complete in $\mathbb{R}^2$. Thus the Heine-Borel Theorem implies that U is not compact.

3.6.4.8. Let V be the subset of 3-dimensional Euclidean space $\mathbb{R}^3$ defined by

$$V = \left\{ \mathbf{x} \in \mathbb{R}^3 \,\middle|\, \|\mathbf{x}\| = 2 \right\}.$$

Determine whether or not V is compact.

Answer. V is compact.

Solution. As shown in Exercise Group 3.6.4.7–10 Compact subspaces of Euclidean space, V is complete in $\mathbb{R}^2$. Moreover, $V \subseteq \mathrm{B}_3(\mathbf{0})$ is bounded. Thus Corollary 3.6.23 Compact subsets of Euclidean space implies that V is compact.

3.6.4.9. Determine whether or not the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are compact.

Answer. The irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are not compact.

Solution. As shown in Exercise Group 3.6.4.7–10 Compact subspaces of Euclidean space, the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are not complete in the real line $\mathbb{R}$. Thus the Heine-Borel Theorem implies that $\mathbb{R} \setminus \mathbb{Q}$ is not compact.

3.6.4.10. Let $a, b \in \mathbb{R}$ be real numbers, and suppose that $a \leq b$. Determine whether or not the closed interval $[a, b]$ is compact.

Answer. The closed interval $[a, b]$ is compact.

Solution. Since the interval $[a, b]$ is closed in the real line $\mathbb{R}$, Proposition 3.6.7 Closed subsets of complete metric spaces implies that $[a, b]$ is complete in $\mathbb{R}^2$. Moreover, $[a, b] \subsetneq \mathrm{B}_{b-a+1}(a)$ is bounded. Thus Corollary 3.6.23 Compact subsets of Euclidean space implies that $[a, b]$ is compact.

3.6.5 · Problems

3.6.5.1. Completeness of discrete metric spaces.

(a) Prove that any discrete metric space is complete.

(b) Prove that any metric space with finite ground set is complete.

(c) Consider the metric subspace (X, d) of the real line $(\mathbb{R}, d)$ with ground set

$$X = \left\{ \frac{1}{n} \in \mathbb{R} \,\middle|\, n \in \mathbb{N} \setminus \{0\} \right\}.$$

Argue that (X, d) is a counterexample to the *false* claim that any metric space whose points are all isolated points is complete.

3.6.5.2. Incompleteness of the p-adic metric. Fix a prime number $p \in \mathbb{N}$, and recall that the **p-adic valuation** $|\cdot|_p : \mathbb{Q} \to \mathbb{Q}$ is a rational-valued function defined by the formula

$$|q|_p = \begin{cases} 0 & q = 0 \\ p^{-k_p(q)} & q \neq 0, \end{cases}$$

where $k_p(q) \in \mathbb{Z}$ is the unique integer so that $q = p^{k_p(q)} \left(\frac{a}{b}\right)$ for some integers $a, b \in \mathbb{Z}$ which are not divisible by p.

Moreover, recall from (b) of Problem 3.1.5.3 Ultrametrics that the p-adic valuation $|\cdot|_p$ induces an ultrametric $\rho_p \colon \mathbb{Q}^2 \to \mathbb{R}$ called the **p-adic metric** defined by the formula

$$\rho_p(q_1, q_2) = |q_1 - q_2|_p .$$

(a) Consider the sequence $(q_n)_{n=0}^{\infty}$ of rational numbers $q_n \in \mathbb{Q}$ defined by the formula

$$q_n = \sum_{k=0}^{n} p^k = 1 + p + p^2 + \cdots + p^n.$$

Show that $(q_n)_{n=0}^{\infty}$ is a Cauchy sequence in $(\mathbb{Q}, \rho_p)$.

(b) Consider the sequence $(r_n)_{n=0}^{\infty}$ of rational numbers $r_n \in \mathbb{Q}$ defined by the formula

$$r_n = \sum_{k=0}^{n} p^{k!} = p + p + p^2 + p^6 + \cdots + p^{n!}.$$

Show that $(r_n)_{n=0}^{\infty}$ is a Cauchy sequence in $(\mathbb{Q}, \rho_p)$.

While the Cauchy sequence $(q_n)_{n=0}^{\infty}$ converges in the p-adic metric to a rational number, the Cauchy sequence $(r_n)_{n=0}^{\infty}$ diverges and therefore is a witness to the incompleteness of $(\mathbb{Q}, \rho_p)$.

4 · Topological Spaces

4.1 · Introduction to Topology

4.1.3 · Exercises

Examples of Topologies. Verify that the given collection of subsets comprise a topology.

4.1.3.1. A subset $U \subseteq X$ of a set X is called **cofinite** in X if the complement $X \setminus U$ is finite. The **cofinite topology**

$$\tau = \{U \in \mathcal{P}(X) \mid U = \emptyset \text{ or } X \setminus U \text{ is finite}\}$$

on X is the collection of all cofinite subsets of X along with the empty set $\emptyset$. Verify that τ is a topology on X.

Solution. We see that $\emptyset \in \tau$ and $X \setminus X = \emptyset$ is finite, so that X is cofinite in X. Thus $\emptyset, X \in \tau$, and so τ is non-degenerate.

Now let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of subsets $U_j \in \tau$. Since unions with the empty set do not affect the value of the union, we may suppose that $U_j \neq \emptyset$ for all indices $j \in \mathcal{J}$. So $X \setminus U_j$ is finite for each index $j \in \mathcal{J}$. (4) of Theorem 1.2.19 Analogous properties of indexed intersections and unions implies that

$$X \setminus \bigcup_{j \in \mathcal{J}} U_j = \bigcap_{j \in \mathcal{J}} (X \setminus U_j)$$

is an intersection of finite sets, and so is itself finite. Thus $\bigcup_{j \in \mathcal{J}} U_j \in \tau$ is cofinite in X, and so τ is stable under unions.

Finally, let $(U_j)_{j=1}^m$ be a finite sequence of subsets $U_j \in \tau$. If $U_j = \emptyset$ for any index $j \in \mathcal{J}$, then

$$\bigcap_{j \in \mathcal{J}} U_j = \emptyset \in \tau.$$

So we may suppose that $U_j \neq \emptyset$ for all indices $j \in \mathcal{J}$. In particular, $X \setminus U_j$ is finite for each index $j \in \mathcal{J}$. (4) of Theorem 1.2.19 Analogous properties of indexed intersections and unions implies that

$$X \setminus \bigcap_{j=1}^{m} U_j = \bigcup_{j=1}^{m} (X \setminus U_j)$$

is a finite union of finite sets, and so is itself finite. Thus $\bigcap_{j=1}^m U_j \in \tau$ is cofinite in X, and so τ is stable under finite intersections.

Since τ is non-degenerate, stable under unions, and stable under finite intersections, it is a topology on X.

4.1.3.2. A subset $U \subseteq X$ of a set X is called **cocountable** in X if the complement $X \setminus U$ is countable. The **cocountable topology**

$$\tau = \{U \in \mathcal{P}(X) \mid U = \emptyset \text{ or } X \setminus U \text{ is countable}\}$$

on X is the collection of all cocountable subsets of X along with the empty set $\emptyset$. Verify that τ is a topology on X.

Solution. We see that $\emptyset \in \tau$ and $X \setminus X = \emptyset$ is countable, so that X is cocountable in X. Thus $\emptyset, X \in \tau$, and so τ is non-degenerate.

Now let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of subsets $U_j \in \tau$. Since unions with the empty set do not affect the value of the union, we may suppose that $U_j \neq \emptyset$ for all indices $j \in \mathcal{J}$. So $X \setminus U_j$ is countable for each index $j \in \mathcal{J}$. (4) of Theorem 1.2.19 Analogous properties of indexed intersections and unions implies that

$$X \setminus \bigcup_{j \in \mathcal{J}} U_j = \bigcap_{j \in \mathcal{J}} (X \setminus U_j)$$

is an intersection of countable sets, and so is itself countable. Thus $\bigcup_{j\in\mathcal{J}} U_j \in \tau$ is cocountable in X, and so τ is stable under unions.

Finally, let $(U_j)_{j=1}^m$ be a finite sequence of subsets $U_j \in \tau$. If $U_j = \emptyset$ for any index $j \in \mathcal{J}$, then

$$\bigcap_{j\in\mathcal{J}} U_j = \emptyset \in \tau.$$

So we may suppose that $U_j \neq \emptyset$ for all indices $j \in \mathcal{J}$. In particular, $X \setminus U_j$ is countable for each index $j \in \mathcal{J}$. (4) of Theorem 1.2.19 Analogous properties of indexed intersections and unions implies that

$$X \setminus \bigcap_{j=1}^{m} U_j = \bigcup_{j=1}^{m} (X \setminus U_j)$$

is a finite union of countable sets, and so is itself countable. Thus $\bigcap_{j=1}^m U_j \in \tau$ is cocountable in X, and so τ is stable under finite intersections.

Since τ is non-degenerate, stable under unions, and stable under finite intersections, it is a topology on X.

4.1.3.3. Fix an element $x \in X$ of a set X. The **particular point topology**

$$\tau = \{U \in \mathcal{P}(X) \mid U = \emptyset \text{ or } x \in U\}$$

on X is the collection of all subsets of X which contain x along with the empty set $\emptyset$. Verify that τ is a topology on X.

Solution. We see that $\emptyset \in \tau$ and $x \in X$, so that $X \in \tau$. Since $\emptyset, X \in \tau$, τ is non-degenerate.

Now let $(U_j)_{j\in\mathcal{J}}$ be an indexed family of subsets $U_j \in \tau$. If $U_j = \emptyset$ for all indices $j \in \mathcal{J}$, then

$$\bigcup_{j\in\mathcal{J}} U_j = \bigcup_{j\in\mathcal{J}} \emptyset = \emptyset \in \tau.$$

Otherwise, if $U_j \neq \emptyset$ for some index $j \in \mathcal{J}$, then $x \in U_j$ for some index $j \in \mathcal{J}$, and so

$$x \in \bigcup_{j\in\mathcal{J}} U_j \in \tau.$$

Thus τ is stable under unions.

Finally, let $(U_j)_{j=1}^m$ be a finite sequence of subsets $U_j \in \tau$. If $U_j = \emptyset$ for any index $1 \leq j \leq m$, then

$$\bigcap_{j=1}^{m} U_j = \emptyset \in \tau.$$

So we may suppose that $U_j \neq \emptyset$ for all indices $1 \leq j \leq m$. In particular, $x \in U_j$ for all indices $1 \leq j \leq m$, and so

$$x \in \bigcap_{j=1}^{m} U_j \in \tau.$$

Thus τ is stable under finite intersections.

Since τ is non-degenerate, stable under unions, and stable under finite intersections, it is a topology on X.

4.1.3.4. Fix an element $x \in X$ of a set X. The **excluded point topology**

$$\tau = \{U \in \mathcal{P}(X) \mid U = X \text{ or } x \notin U\}$$

on X is the collection of all subsets of X which do not contain x along with X itself. Verify that τ is a topology on X.

Solution. We see that $X \in \tau$ and $x \notin \emptyset$, so that $\emptyset \in \tau$. Since $\emptyset, X \in \tau$, τ is non-degenerate.

Now let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of subsets $U_j \in \tau$. If $U_j = X$ for some index $j \in \mathcal{J}$, then

$$\bigcup_{j \in \mathcal{J}} U_j = X \in \tau.$$

Otherwise, if $U_j \neq X$ for all indices $j \in \mathcal{J}$, then $x \notin U_j$ for all indices $j \in \mathcal{J}$, and so

$$x \in \bigcup_{j \in \mathcal{J}} U_j \in \tau.$$

Thus τ is stable under unions.

Finally, let $(U_j)_{j=1}^{m}$ be a finite sequence of subsets $U_j \in \tau$. If $U_j = X$ for all indices $1 \leq j \leq m$, then

$$\bigcap_{j=1}^{m} U_j = \bigcap_{j=1}^{m} X = X \in \tau.$$

So we may suppose that $U_j \neq X$ for all indices $1 \leq j \leq m$. In particular, $x \notin U_j$ for some index $1 \leq j \leq m$, and so

$$x \notin \bigcap_{j=1}^{m} U_j \in \tau.$$

Thus τ is stable under finite intersections.

Since τ is non-degenerate, stable under unions, and stable under finite intersections, it is a topology on X.

Computing subspace topologies. Describe the subspace topology induced on the given subset of a topological space.

4.1.3.5. Let $U \subseteq X$ be a subset of a set X. Verify that if X is equipped with the cofinite topology, then the subspace topology induced on U is the cofinite topology on U.

Solution. Denote by τ_X and τ_U the cofinite topologies on X and U; we aim to show that $\tau_X|_U = \tau_U$.

To that end, let $W \in \mathcal{P}(X)$ be a subset of X, and suppose first that $W \in \tau_U$. If $W = \emptyset$, then $W = U \cap \emptyset \in \tau_X|_U$. On the other hand, if $W \neq \emptyset$, then $U \setminus W$ is finite. We observe that $X \setminus (U \setminus W) \in \tau_X$ is cofinite in X, and so

$$\begin{aligned} W &= U \setminus (U \setminus W) = \{u \in U \mid u \notin U \setminus W\} \\ &= \{x \in X \mid x \in U \text{ and } x \notin U \setminus W\} = U \cap \{x \in X \mid x \notin U \setminus W\} \\ &= U \cap (X \setminus (U \setminus W)) \in \tau_X|_U . \end{aligned}$$

Conversely, now suppose that $W \in \tau_X|_U$. Then $W = U \cap V$ for some $V \in \tau_X$. If $V = \emptyset$, then

$$W = U \cap V = U \cap \emptyset = \emptyset \in \tau_U.$$

On the other hand, if $V \neq \emptyset$, then $X \setminus V$ is finite, and so (a) of Problem 1.1.5.2 De Morgan's laws implies that

$$\begin{aligned} U \setminus W &= U \setminus (U \cap V) = (U \setminus U) \cup (U \setminus V) \\ &= \emptyset \cup (U \setminus V) = U \setminus V \subseteq X \setminus V \end{aligned}$$

is finite, so that $W \in \tau_U$.

In summary, we have shown that $W \in \tau_U$ if and only if $W \in \tau_X|_U$. We conclude that $\tau_X|_U = \tau_U$.

4.1.3.6. Let $U \subseteq X$ be a subset of a set X. Verify that if X is equipped with the cocountable topology, then the subspace topology induced on U is the cocountable topology on U.

Solution. Denote by τ_X and τ_U the cocountable topologies on X and U; we aim to show that $\tau_X|_U = \tau_U$.

To that end, let $W \in \mathcal{P}(X)$ be a subset of X, and suppose first that $W \in \tau_U$. If $W = \emptyset$, then $W = U \cap \emptyset \in \tau_X|_U$. On the other hand, if $W \neq \emptyset$, then $U \setminus W$ is countable. We observe that $X \setminus (U \setminus W) \in \tau_X$ is cocountable in X, and so

$$\begin{aligned} W &= U \setminus (U \setminus W) = \{u \in U \mid u \notin U \setminus W\} \\ &= \{x \in X \mid x \in U \text{ and } x \notin U \setminus W\} = U \cap \{x \in X \mid x \notin U \setminus W\} \\ &= U \cap (X \setminus (U \setminus W)) \in \tau_X|_U . \end{aligned}$$

Conversely, now suppose that $W \in \tau_X|_U$. Then $W = U \cap V$ for some $V \in \tau_X$. If $V = \emptyset$, then

$$W = U \cap V = U \cap \emptyset = \emptyset \in \tau_U .$$

On the other hand, if $V \neq \emptyset$, then $X \setminus V$ is countable, and so (a) of Problem 1.1.5.2 De Morgan's laws implies that

$$\begin{aligned} U \setminus W &= U \setminus (U \cap V) = (U \setminus U) \cup (U \setminus V) \\ &= \emptyset \cup (U \setminus V) = U \setminus V \subseteq X \setminus V \end{aligned}$$

is countable, so that $W \in \tau_U$.

In summary, we have shown that $W \in \tau_U$ if and only if $W \in \tau_X|_U$. We conclude that $\tau_X|_U = \tau_U$.

4.1.3.7. Let $U \subseteq X$ be a subset of a set X, and fix a point $x \in X$. Verify that if X is equipped with the particular point topology, then the subspace topology induced on U is either the particular point topology if $x \in U$ or the discrete topology if $x \notin U$.

Solution. Denote by τ_X the particular point topology. First suppose that $x \in U$, and denote by τ_U the particular point topology; we aim to show that $\tau_X|_U = \tau_U$.

To that end, let $W \in \mathcal{P}(U)$ be a subset of U, and suppose first that $W \in \tau_U$. If $W = \emptyset$, then $W = U \cap \emptyset \in \tau_X|_U$. On the other hand, if $W \neq \emptyset$, then $x \in W \in \tau_X$, and so

$$W = U \cap W \in \tau_X|_U .$$

Now suppose that $W \in \tau_X|_U$. Then $W = U \cap V$ for some $V \in \tau_X$. If $V = \emptyset$, then

$$W = U \cap V = U \cap \emptyset = \emptyset \in \tau_U .$$

On the other hand, if $V \neq \emptyset$, then $x \in V$, and so

$$x \in U \cap V = W \in \tau_U.$$

In summary, we have shown that $W \in \tau_X|_U$ if and only if $W \in \tau_U$. We conclude that $\tau_X|_U = \tau_U$.

Now suppose that $x \notin U$. Then for any subset $W \subseteq U$, (a) of Problem 1.1.5.1 Distributivity of set operations implies that

$$\begin{aligned} W &= U \cap W = (U \cap W) \cup \emptyset \\ &= (U \cap W) \cup (U \cap \{x\}) = U \cap (W \cup \{x\}) \in \tau_X|_U. \end{aligned}$$

We conclude that $\tau_X|_U = \mathcal{P}(U) = \top_U$ is the discrete topology on U.

4.1.3.8. Let $U \subseteq X$ be a subset of a set X, and fix a point $x \in X$. Verify that if X is equipped with the excluded point topology, then the subspace topology induced on U is either the excluded point topology if $x \in U$ or the indiscrete topology if $x \notin U$.

Solution. Denote by τ_X the excluded point topology. First suppose that $x \in U$, and denote by τ_U the excluded point topology; we aim to show that $\tau_X|_U = \tau_U$.

To that end, let $W \in \mathcal{P}(U)$ be a subset of U, and suppose first that $W \in \tau_U$. If $W = U$, then $W = U \cap X \in \tau_X|_U$. On the other hand, if $W \neq U$, then $x \notin W \in \tau_X$, and so

$$W = U \cap W \in \tau_X|_U.$$

Now suppose that $W \in \tau_X|_U$. Then $W = U \cap V$ for some $V \in \tau_X$. If $V = X$, then

$$W = U \cap X = U \in \tau_U.$$

On the other hand, if $V \neq X$, then $x \notin V$, and so

$$x \notin U \cap V = W \in \tau_U.$$

In summary, we have shown that $W \in \tau_X|_U$ if and only if $W \in \tau_U$. We conclude that $\tau_X|_U = \tau_U$.

Now suppose that $x \notin U$. Then $x \notin W \in \tau_X$ for any subset $W \subseteq U$, and so

$$W = U \cap W \in \tau_X|_U.$$

We conclude that $\tau_X|_U = \mathcal{P}(U) = \top_U$ is the discrete topology on U.

4.1.4 · Problems

4.1.4.1. Topologies induced by p-metrics. Show that for all $1 \leq p \leq \infty$, the p-metrics $d_p \colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ induce the same metric topology on n-dimensional real coordinate space $\mathbb{R}^n$.

4.1.4.2. Closed sets in subspaces. Let $U \subseteq X$ be a subset of a topological space (X, τ). Prove that a subset $W \subseteq U$ is closed in U if and only if $W = U \cap V$ for some closed set $V \subseteq X$ in X.

4.1.4.3. Monotonicity of interior, closure, and boundary. Fix subsets $U, V \subseteq X$ of a topological space (X, τ).

(a) Prove that if $U \subseteq V$, then $\operatorname{Int}(U) \subseteq \operatorname{Int}(V)$ and $\operatorname{Cl}(U) \subseteq \operatorname{Cl}(V)$.

(b) Give concrete examples so that $A \subsetneq B$ but $\operatorname{Int}(A) = \operatorname{Int}(B)$ and $\operatorname{Cl}(A) = \operatorname{Cl}(B)$.

(c) Give concrete examples so that $A \subseteq B$ but $\mathrm{Bd}(A) \not\subseteq \mathrm{Bd}(B)$.

4.2 · Comparison and Topology

4.2.4 · Exercises

4.2.4.1. Comparison of topologies on $\mathbb{R}$. Compare the following topologies on the real numbers $\mathbb{R}$: the indiscrete topology $\perp_{\mathbb{R}}$; the discrete topology $\top_{\mathbb{R}}$; the Euclidean topology τ_d; the cofinite topology τ_1; the cocountable topology τ_2; the particular point topology τ_3 specified with particular point $0 \in \mathbb{R}$; and the excluded point topology τ_4 specified with excluded point $0 \in \mathbb{R}$.

Answer. We observe that all comparison relationships can be summarized by

$$\begin{aligned}
&\perp_{\mathbb{R}} \subseteq \tau_1 \subseteq \tau_2 \subseteq \top_{\mathbb{R}} \\
&\perp_{\mathbb{R}} \subseteq \tau_1 \subseteq \tau_d \subseteq \top_{\mathbb{R}} \\
&\perp_{\mathbb{R}} \subseteq \tau_3 \subseteq \top_{\mathbb{R}} \\
&\perp_{\mathbb{R}} \subseteq \tau_4 \subseteq \top_{\mathbb{R}}.
\end{aligned}$$

4.2.4.2. Unions of topologies. Give an example of two topologies on a set whose union is not a topology on that set.

Solution. Let $X = \{a, b, c\}$, and consider the topologies

$$\tau_1 = \{\emptyset, \{a\}, X\} \qquad\qquad \tau_2 = \{\emptyset, \{b\}, X\}.$$

Then $\tau_1 \cup \tau_2 = \{\emptyset, \{a\}, \{b\}, X\}$ is not a topology on X, since it is not stable under unions. Indeed, $\{a\}, \{b\} \in \tau_1 \cup \tau_2$, but $\{a\} \cup \{b\} = \{a, b\} \notin \tau_1 \cup \tau_2$.

4.2.4.3. From subbase to topology. Let $X = \{a, b, c, d\}$, and consider the collection $\mathcal{S} \subseteq \mathcal{P}(X)$ be the collection of subsets of X defined by

$$\mathcal{S} = \{\{a, b\}, \{b, c\}, \{c, d\}\}.$$

Compute explicitly the topology $\tau = \langle \mathcal{S} \rangle$ on X generated by $\mathcal{S}$.

Answer. The topology $\tau = \langle \mathcal{S} \rangle$ on X generated by $\mathcal{S}$ is

$$\tau = \{\emptyset, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, d\}, \{a, b, c\}, \{b, c, d\}, \{a, b, c, d\}\}.$$

4.2.5 · Problems

4.2.5.1. Monotonicity of generated topologies. Fix collections $\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}(X)$ of subsets of a set X. Show that if $\mathcal{A} \subseteq \mathcal{B}$, then $\langle \mathcal{A} \rangle \subseteq \langle \mathcal{B} \rangle$.

4.2.5.2. Topological comparison and interior/closure. Fix two topologies $\tau_1, \tau_2 \in \mathcal{T}_X$ on a set X, and suppose that $\tau_1 \subseteq \tau_2$.

(a) For a subset U, compare its interior $\mathrm{Int}_{(X,\tau_1)}(U)$ in the topological space (X, τ_1) with its interior $\mathrm{Int}_{(X,\tau_2)}(U)$ in the topological space (X, τ_2).

(b) For a subset U, compare its closure $\mathrm{Cl}_{(X,\tau_1)}(U)$ in the topological space (X, τ_1) with its closure $\mathrm{Cl}_{(X,\tau_2)}(U)$ in the topological space (X, τ_2).

4.2.5.3. Subbase of the Euclidean topology. Consider the collections $\mathcal{U}$ and $\mathcal{D}$ of upwards- and downwards pointing rays in the real numbers $\mathbb{R}$, respectively; that is,

$$\begin{aligned}
\mathcal{U} &= \{(a, \infty) \in \mathcal{P}(\mathbb{R}) \mid a \in \mathbb{R}\} \\
\mathcal{D} &= \{(-\infty, b) \in \mathcal{P}(\mathbb{R}) \mid b \in \mathbb{R}\}.
\end{aligned}$$

Show that $\mathcal{U} \cup \mathcal{D}$ is a subbase of the Euclidean topology τ_d on $\mathbb{R}$.

4.3 · Convergence and Separation
4.3.3 · Exercises

Net convergence. Verify that the given net of points converges.

4.3.3.1. Let F be the collection of finite subsets of the natural numbers $\mathbb{N}$; that is,

$$F = \{S \in \mathcal{P}(N) \mid S \text{ is finite}\}.$$

F is partially ordered and directed both upwards and downwards by containment $\subseteq$. Consider the net $(x_S)_{S\in F}$ of real numbers $x_S \in \mathbb{R}$ defined by the formula

$$x_S = 2^{-\max(S)}.$$

Verify that $(x_S)_{S\in F}$ converges upwards to 0.

Solution. Let $N \in \mathcal{N}(0)$ be a neighborhood of 0 in $\mathbb{R}$. Then $\mathrm{B}_\epsilon(0) \subseteq N$ for some positive real number $\epsilon > 0$. The Archimedean principle implies that

$$n > \log_2\left(\frac{1}{\epsilon}\right)$$

for some natural number $n_\epsilon \in \mathbb{N}$. Let $S_N = \{n_\epsilon\} \in F$. We observe that for all indices $S \supseteq S_N$,

$$\begin{aligned} d(x_S, 0) &= \left|2^{-\max(S)} - 0\right| = \left|2^{-\max(S)}\right| = 2^{-\max(S)} \\ &\leq 2^{-\max(S_N)} = 2^{-n_\epsilon} < \epsilon, \end{aligned}$$

and so $x_S \in \mathrm{B}_\epsilon(0) \subseteq N$. We conclude that the net $(x_S)_{S\in F}$ converges upwards to 0.

4.3.3.2. Let $(x_N)_{N\in\mathcal{N}(x)}$ be a net of points in a topological space (X,τ) indexed by the neighborhood system $\mathcal{N}(x)$. Verify that if $x_N \in N$ for all indices $N \in \mathcal{N}(x)$, then $(x_N)_{N\in\mathcal{N}(x)}$ converges downwards to x.

Solution. Let $N \in \mathcal{N}(x)$ be a neighborhood of x in X. Since

$$x_M \in M \subseteq N$$

for all indices $M \subseteq N$, the net $(x_N)_{N\in\mathcal{N}(x)}$ converges downwards to x.

Separation axioms. Determine which separation axioms are satisfied by the given topological space.

4.3.3.3. Is an indiscrete topological space $(X, \perp_X)$ T_0? Is it T_1? Is it T_2?

Answer. If X has at most one element, then the indiscrete topological space $(X, \perp_X)$ is T_0, T_1, and T_2. However, if X has at least two elements, then the indiscrete topological space $(X, \perp_X)$ is neither T_0, T_1, nor T_2.

4.3.3.4. Is discrete topological space $(X, \top_X)$ T_0? Is it T_1? Is it T_2?

Answer. The discrete topological space $(X, \top_X)$ is T_0, T_1, and T_2.

4.3.3.5. Is Euclidean space $(\mathbb{R}^n, \tau_d)$ T_0? Is it T_1? Is it T_2?

Answer. Euclidean space $(\mathbb{R}^n, \tau_d)$ is T_0, T_1, and T_2.

4.3.3.6. Let τ be the cofinite topology on a set X. Is the topological space (X,τ) T_0? Is it T_1? Is it T_2?

Answer. The topological space (X,τ) is T_0, T_1, and T_2 if X is finite. However, if X is infinite, then the topological space (X,τ) is only T_0 and

T_1.

4.3.3.7. Let τ be the cocountable topology on a set X. Is the topological space (X, τ) T_0? Is it T_1? Is it T_2?

Answer. The topological space (X, τ) is T_0, T_1, and T_2 if X is countable. However, if X is uncountable, then the topological space (X, τ) is only T_0 and T_1.

4.3.3.8. Let τ be a particular point topology on a set X. Is the topological space (X, τ) T_0? Is it T_1? Is it T_2?

Answer. The topological space (X, τ) is T_0, T_1, and T_2 if X has at most one element. However, if X has at least two elements, then the topological space (X, τ) is only T_0.

4.3.3.9. Let τ be an excluded point topology on a set X. Is the topological space (X, τ) T_0? Is it T_1? Is it T_2?

Answer. The topological space (X, τ) is T_0, T_1, and T_2 if X has at most one element. However, if X has at least two elements, then the topological space (X, τ) is only T_0.

4.3.4 · Problems

4.3.4.1. Net convergence and subspaces. Let $U \subseteq X$ be a subset of a topological space (X, τ), and consider a net $(u_\alpha)_{\alpha \in A}$ of points $u_\alpha \in U$ indexed by an upwards directed set $(A, \precsim)$. Prove that for a point $u \in U$, $(u_\alpha)_{\alpha \in A}$ converges upwards to u as a net in X if and only if it converges upwards to u as a net in U.

4.3.4.2. Points of closure. Let $U \subseteq X$ be a subset of a topological space (X, τ).

(a) Prove that a point $x \in X$ is a point of closure of U in X if and only if a net of points in U converges to x.

(b) Prove that $\mathrm{Cl}(U) = U \cup U'$.

We remark that the union $\mathrm{Cl}(U) = U \cup U'$ need not be a disjoint union; that is, U and its derived set U' might have significant overlap.

4.3.4.3. Non-Hausdorff spaces. Let (X, τ) be a topological space. For two points $x, y \in X$, let $\preceq$ be the binary relation on the Cartesian product $\mathcal{N}(x) \times \mathcal{N}(y)$ defined by the following condition: $(M_1, V_1) \preceq (M_2, V_2)$ if and only if both $M_1 \subseteq M_2$ and $V_1 \subseteq V_2$.

(a) Show that $\mathcal{N}(x) \times \mathcal{N}(y)$ is partially ordered and directed both upwards and downwards by $\preceq$.

(b) Prove that if (X, τ) is not Hausdorff, then there is a convergent net of points in X which does not have a unique limit.

III · Calculus
5 · Differentiation and Smoothness
5.1 · Differentiability
5.1.3 · Exercises

Locus of differentiability. Determine the differentiability locus of the given function.

5.1.3.1. Determine the differentiability locus $\mathrm{DL}(f)$ of the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = |x|^3$.

Answer. f has differentiability locus $\mathrm{DL}(f) = \mathbb{R}$; that is, f is differentiable.

Solution. We first show that f is differentiable at 0. Let $\epsilon > 0$ be a positive real number, and consider the positive real number $\delta = \sqrt{\epsilon}$. We observe that

$$\begin{aligned}\left|\frac{f(x) - f(0)}{x - 0} - 0\right| &= \left|\frac{|x|^3 - |0|^3}{x - 0}\right| = \left|\frac{|x|^3}{x}\right| \\ &= \frac{|x|^3}{|x|} = |x|^2 < \delta^2 = \epsilon\end{aligned}$$

for all $x \in \mathbb{R}$ so that $0 < |x - 0| < \delta$. Thus f is differentiable at 0, with derivative $f'(0) = 0$.

At any point $c \in \mathbb{R} \setminus \{0\}$, f is a composition of differentiable functions $x \mapsto |x|$ and $x \mapsto x^3$, and so is also differentiable. We conclude that f has differentiability locus $\mathrm{DL}(f) = \mathbb{R}$; that is, f is differentiable.

5.1.3.2. Determine the differentiability locus $\mathrm{DL}(g)$ of the real-valued function $g\colon \mathbb{R} \to \mathbb{R}$ defined piecewise by the formula

$$g(x) = \begin{cases} x^2 & x < 0 \\ 2x^2 & x \geq 0. \end{cases}$$

Answer. g has differentiability locus $\mathrm{DL}(g) = \mathbb{R}$; that is, g is differentiable.

Solution. We first show that g is differentiable at 0. Let $\epsilon > 0$ be a positive real number, and consider the positive real number $\delta = \frac{\epsilon}{2} > 0$. We observe that

$$\begin{aligned}\left|\frac{g(x) - g(0)}{x - 0} - 0\right| &= \left|\frac{x^2 - 2(0)^2}{x - 0}\right| = \left|\frac{x^2}{x}\right| \\ &= |x| = |x - 0| < \delta - \frac{\epsilon}{2} < \epsilon\end{aligned}$$

for all points $-\delta < x < 0$. Similarly,

$$\begin{aligned}\left|\frac{g(x) - g(0)}{x - 0} - 0\right| &= \left|\frac{2x^2 - 2(0)^2}{x - 0}\right| = \left|\frac{2x^2}{x}\right| \\ &= |2x| = 2|x - 0| < 2\delta = \epsilon\end{aligned}$$

for all points $0 < x \leq \delta$. Thus g is differentiable at 0, with derivative $g'(0) = 0$.

At any point $c \in \mathbb{R} \setminus \{0\}$, g is locally equal to a polynomial and therefore differentiable. We conclude that g has differentiability locus $\mathrm{DL}(g) = \mathbb{R}$; that is, g is differentiable.

Computing derivatives. Compute formulas for the derivatives of the given differentiable functions.

5.1.3.3. Compute a formula for the derivative of the differentiable function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$f(x) = \sin^2(x) + \cos^2(x).$$

Answer. f has derivative given by the formula $f'(x) = 0$.

5.1.3.4. Compute a formula for the derivative of the differentiable function $g\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$g(y) = \frac{1}{1+y^2}.$$

Answer. g has derivative given by the formula

$$g'(y) = -\frac{2y}{(1+y^2)^2}.$$

5.1.3.5. Compute a formula for the derivative of the differentiable function $h\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$h(z) = \sqrt{\sin(z) + 2}.$$

Answer. h has derivative given by the formula

$$h'(z) = \frac{\cos(z)}{\sqrt{\sin(z) + 2}}.$$

5.1.4 · Problems

5.1.4.1. The quotient rule. In this problem, we complete Proposition 5.1.4 Arithmetic of differentiable functions to address the quotient of two differentiable functions.

(a) Let $r\colon \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$ be the function defined by the formula

$$r(x) = \frac{1}{x}.$$

Show that r is differentiable, with derivative

$$r'(x) = -\frac{1}{x^2}$$

(b) Let $g\colon X \to \mathbb{R}\setminus\{0\}$ be a non-zero real-valued function defined on a subset $X \subseteq \mathbb{R}$. Prove that if g is differentiable at a point $c \in X \cap X'$, then $\frac{1}{g}$ is differentiable at c, with derivative

$$\left(\frac{1}{g}\right)'(c) = -\frac{g'(c)}{g(c)^2}.$$

(c) Let $f, g\colon X \to \mathbb{R}$ be real-valued functions defined on a subset $X \subseteq \mathbb{R}$. Prove that if $g(x) \neq 0$ for all points $x \in X$ and f and g are differentiable at a point $c \in X \cap X'$, then $\frac{f}{g}$ is differentiable at c, with derivative

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)\, g(c) - f(c)\, g'(c)}{g(c)^2}.$$

The result proven in (c) above is usually called the **quotient rule**.

5.1.4.2. The power rule. For each real number $p \in \mathbb{R}$, consider the power function $f_p\colon (0, \infty) \to \mathbb{R}$ defined by the formula $f_p(x) = x^p$.

(a) Prove that for each natural number $n \in \mathbb{N}$, f_n is differentiable, with derivative

$$f_n'(x) = nx^{n-1}.$$

(b) Show that for any integer $n \in \mathbb{Z}$, f_n is differentiable, with derivative $f_n'(x) = nx^{n-1}$.

(c) Show that for any rational number $p \in \mathbb{Q}$ that if f_p is differentiable, then its derivative is $f_p'(x) = px^{p-1}$.

The rules proven above also hold for irrational powers, although we will require additional machinery in order to prove this final extension. In general, this is called the **power rule**.

5.1.4.3. Derivatives of trigonometric functions. Text before the parts.

(a) Compute the limits

$$\lim_{\theta \to 0} \frac{\sin(\theta)}{\theta}$$

and

$$\lim_{\theta \to 0} \frac{\cos(\theta) - 1}{\theta}.$$

(b) Show that the trigonometric function $\sin(x)$ is differentiable, with derivative $\cos(x)$.

(c) Show that the trigonometric function $\cos(x)$ is differentiable, with derivative $-\sin(x)$.

(d) Show that the trigonometric function $\tan(x)$ is differentiable, with derivative $\frac{1}{\cos^2(x)}$.

5.2 · Analysis of Critical Points
5.2.3 · Exercises

Examples of critical points. Give examples of differentiable functions which satisfy the given conditions.

5.2.3.1. Give an example of a differentiable function which has a single critical point which is not a local extremum.

Answer. The function $f(x) = x^3$ has a single critical point which is not a local extremum.

5.2.3.2. Give an example of a differentiable function which attains a local minimum but no global minima.

Answer. The function $f(x) = x\sin(x)$ attains a local minimum at 0 but no global minima.

5.2.3.3. Give an example of a differentiable function which has no critical points.

Answer. The function $f(x) = x$ has no critical points

5.2.3.4. Give an example of a differentiable function which has exactly one critical point but no global extrema.

Answer. The function $f(x) = x^3$ has a critical point at 0 but no global extrema.

5.2.3.5. Give an example of a differentiable function which attains infinitely many global extrema.

Answer. The function $f(x) = \sin(x)$ attains a global extremum at each integer multiple of π.

5.2.4 · Problems

5.2.4.1. Misconceptions about critical points. Let $f\colon X \to \mathbb{R}$ be a real-valued function defined on an interval $X \subseteq \mathbb{R}$. ***The following statements are false!*** Explain why the statement is incorrect, and disprove them by providing a counterexample.

(a) If f attains a local extremum at a point $c \in X$, then c is a critical point of f.

(b) If a point $c \in X$ is a critical point of f, then f attains a local extremum at c.

(c) If a point $c \in X$ is a regular point of f, then c has a neighborhood in X on which f is monotone.

(d) f is strictly monotone if and only if f has no critical points.

5.3 · Applications of the Mean Value Theorem

5.3.3 · Exercises

Hypotheses and conclusions of the Mean value theorem. Determine whether the given function satisfies the hypotheses and conclusions of the Mean value theorem.

5.3.3.1. Let $f\colon [0,1] \to \mathbb{R}$ be the real-valued function defined on the interval $[0,1]$ by the formula

$$f(x) = \sqrt[3]{x^2 - x}.$$

(a) Determine whether or not f satisfies the hypotheses of the Mean value theorem.

Answer. Yes, f satisfies the hypotheses of the Mean value theorem.

(b) Determine whether or not f satisfies the conclusions of the Mean value theorem. If so, find a point $0 < c < 1$ for which

$$f'(c) = \frac{f(1) - f(0)}{1 - 0}.$$

Answer. Yes, f satisfies the conclusions of the Mean value theorem. We may choose $c = \frac{1}{2}$.

5.3.3.2. Let $g\colon [-1,1] \to \mathbb{R}$ be the real-valued function defined on the interval $[0,1]$ by the formula

$$g(x) = x^{\frac{3}{5}}.$$

(a) Determine whether or not g satisfies the hypotheses of the Mean value theorem.

Answer. No, g does not satisfy the hypotheses of the Mean value theorem.

(b) Determine whether or not g satisfies the conclusions of the Mean value theorem. If so, find a point $-1 < c < 1$ for which

$$g'(c) = \frac{g(1) - g(-1)}{1 - (-1)}.$$

Answer. Yes, g satisfies the conclusions of the Mean value theorem. We may choose $c = \pm \left(\frac{3}{5}\right)^{\frac{5}{2}}$.

5.3.4 · Problems

5.3.4.1. Functions with the same derivative.

(a) Let $f\colon X \to \mathbb{R}$ be a differentiable real-valued function defined on an interval X. Prove that if $f'(x) = 0$ for all $x \in X$, then f is constant.

(b) Let $f, g\colon X \to \mathbb{R}$ be differentiable real-valued functions defined on an interval X. Prove that if $f'(x) = g'(x)$ for all $x \in X$, then f and g differ by a constant.

The above result will be important in Chapter 6 Riemann Integration, where we will use it to compare two anti-derivatives of the same continuous function on a closed interval.

5.3.4.2. Differentiability and Lipschitz continuity. Let $f\colon X \to \mathbb{R}$ be a differentiable real-valued function defined on an interval $X \subseteq \mathbb{R}$. Prove that f is Lipschitz continuous if and only if its derivative f' is bounded.

5.4 · Higher Order Approximations

5.4.4 · Problems

5.4.4.1. Taylor approximations of non-analytic functions. Let $f\colon \mathbb{R} \to \mathbb{R}$ be the real-valued function defined piecewise by the formula

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x > 0 \\ 0 & x \leq 0. \end{cases}$$

(a) Prove that $u^{\frac{n}{2}} \prec e^u$ as $u \to \infty$ for all natural numbers $n \in \mathbb{N}$.

(b) Prove that $f(x) \prec x^n$ as $x \to 0$ for all natural numbers $n \in \mathbb{N}$.

(c) Compute all Taylor approximations $P_0^n(f, x)$ of f near 0.

(d) Let $g \in \mathrm{C}^\infty(X)$ be a smooth function defined on an interval $X \subseteq \mathbb{R}$. If there is a point $c \in X$ so that $g^{(n)}(c) = 0$ for all positive integers $n \geq 1$, can we conclude that g is constant on a neighborhood of c?

f is our first example of a smooth function which is not **analytic**. We will explore these functions more in Chapter 7 Infinite Series.

6 · Riemann Integration

6.1 · Riemann Sums and Integrals

6.1.4 · Problems

6.1.4.1. More on integrability and continuity. Let $f\colon [a, b] \to \mathbb{R}$ be a Riemann integrable real-valued function defined on a closed interval. Prove that $g \circ f$ is Riemann integrable for any continuous real-valued function $g\colon \mathbb{R} \to \mathbb{R}$.

Appendix B

List of Symbols

(Continued on next page)

Symbol	Description	Page
L	the standard ordering	22
$\prod_{j\in\mathcal{J}} U_j$	the Cartesian product of a family $(U_j)_{j\in\mathcal{J}}$ of sets U	33
π_k	projection onto the kth factor	33
X^n	the nth Cartesian power of a set X	35
$\sum_{j=1}^{m} n_j$	recursive summation notation	37
$\prod_{j=1}^{m} n_j$	recursive product notation	37
R_f	the pullback of a finitary relation R along a map f	37
$R\|_U$	the restriction of a finitary relation R to a subset U	??
$[x]_\equiv$	the equivalence class of x under $\equiv$	39
$X/{\equiv}$	the quotient of X by $\equiv$	39
$\max(U)$	the maximum of U	44
$\min(U)$	the minimum of U	44
$\|X\|$	the cardinality of X	51
$\mathbb{R}$	the set of real numbers	54
X_q	the rational Dedekind cut associated to q	??
$\mathbb{R}(x)$	the field of rational functions	65
$\sup(U)$	the supremum of U	66
$\inf(U)$	the infimum of U	66
		70
$\overline{z}$	the conjugate of z	71
$\|z\|$	the magnitude of z	71
$\overline{\mathbb{R}}$	the set of extended real numbers	71
∞	positive infinity	71
$-\infty$	negative infinity	71
		71
δ_X	the discrete metric on X	76
$\\|\mathbf{v}\\|$	The norm of a vector $\mathbf{v}$	79
$\\|\cdot\\|_p$	the p-norm	80
$\\|\cdot\\|_p$	the p-norm	80
d_p	the p-metric	??
$\lim_{n\to\infty} x_n$	the limit of $(x_n)_{n=0}^{\infty}$	87
$\lim_{n\to\infty} f_n$	pointwise limit of the sequence $(f_n)_{n=0}^{\infty}$	90
$\operatorname{unif}\lim_{n\to\infty} f_n$	uniform limit of the sequence $(f_n)_{n=0}^{\infty}$	92
U'	the derived set of U	96
$\lim_{x\to c} f(x)$	the limit of a map f at a limit point c	97
		110
$\limsup_{n\to\infty} x_n$	the upper limit of $(x_n)_{n=0}^{\infty}$	114
$\liminf_{n\to\infty} x_n$	the lower limit of $(x_n)_{n=0}^{\infty}$	114
$\mathrm{B}_r(c)$	the open metric ball of radius r and center c	120
$\mathrm{D}_r(c)$	the closed metric disk of radius r and center c	120
$\mathrm{S}_r(c)$	the metric sphere of radius r and center c	120
$\mathrm{Int}(U)$	the interior of U	122
$\mathrm{Cl}(U)$	the closure of U	122
$\mathrm{Bd}(U)$	the boundary of U	122
$\mathrm{diam}(U)$	the diameter of U	128
		132
ρ_∞	the uniform metric	134
$\mathrm{C}(X,Y)$	the space of continuous maps from X to Y	139

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Symbol	Description	Page
$\mathrm{C}(X)$	the space of continuous real-valued functions on X	139
$\mathrm{UC}(X,Y)$	the set of uniformly continuous maps from X to Y	144
$\mathrm{UC}(X)$	the set of uniformly continuous real-valued functions on X	144
$\mathrm{LC}(X,Y)$	the set of Lipschitz continuous maps from X to Y	148
$\mathrm{LC}(X)$	the set of Lipschitz continuous real-valued functions on X	148
L_f	the Lipschitz constant of f	148
$(x_{n_k})_{k=0}^{\infty}$	a subsequence of $(x_n)_{n=0}^{\infty}$	156
$\top_X$	the discrete topology on X	169
$\bot_X$	the indiscrete topology on X	169
τ_ρ	the metric topology induced by ρ	170
$\tau\|_U$	the subspace topology induced on U by τ.	171
$\mathcal{N}(x)$	the set of neighborhoods of x	173
$\mathcal{N}_X(x)$	the set of neighborhoods of x in X	173
$\mathcal{N}_{(X,\tau)}(x)$	the set of neighborhoods of x in (X,τ)	173
$\mathrm{Int}(U)$	the interior of U	174
U°	the interior of U	174
$\mathrm{Cl}(U)$	the closure of U	174
$\overline{U}$	the closure of U	174
$\mathrm{Bd}(U)$	the boundary of U	174
∂U	the boundary of U	174
$\mathcal{T}_X$	the lattice of topologies on X	177
$\langle \mathcal{S} \rangle$	the topology generated by $\mathcal{S}$.	178
$\mathrm{C}(X,Y)$	the space of continuous maps from X to Y	194
$\mathrm{C}(X)$	the space of continuous real-valued functions on X	194
$f'(c)$	the derivative of f at c	202
f'	the derivative of f	202
$\mathrm{DL}^k(f)$	the k-times differentiability locus of f	207
$f^{(k)}$	the kth derivative of f	207
$\mathrm{C}^k(X)$	the set of k-times continuously differentiable functions on X	207
$\mathrm{C}^\infty(X)$	the set of smooth functions on X	208
$\preccurlyeq$	asymptotic dominance	227
$\asymp$	asymptotic comparability	227
$\sim$	asymptotic	227
$\prec$	strict asymptotic dominance	230
$P_c^k(f,x)$	the order k Taylor approximation of f at c	232
$R_c^k(f,x)$	the order k Taylor remainder of f at c	232
$\mathrm{mesh}(P)$	the mesh of P	236
$\mathcal{R}_f(P,t)$	the Riemann sum of f associated to (P,t).	238
$\mathcal{U}_f(P)$	the upper Darboux sum of f associated to P	239
$\mathcal{L}_f(P)$	the lower Darboux sum of f associated to P	239
$\int_a^b f(x)\,\mathrm{d}x$	the definite (Riemann) integral of f from a to b	244
$\mathcal{K}_X$	the set of compact subintervals of $\mathrm{Int}(X)$	257
$\sum_{n=0}^{\infty} x_n$	the series associated to the sequence $(x_n)_{n=0}^{\infty}$	269

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Colophon

This book was authored in PreTeXt.